Arithmetic and Observation: Classifications, Recovery, and Transcendence Barriers

Unified Edition of Twelve Successor Manuscripts

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Abstract

This unified manuscript brings together twelve successive studies of arithmetic observation, optimization, inverse recovery, and the transfer of analytic constructions to questions of irrationality and transcendence. The development begins with observation boundaries and optimal values, passes through structural recovery and effective logical classification, and ends with arithmetic certificates, entropy thresholds, primitive integer forms, and exact quadratic exceptions. The twelve parts retain their substantive arguments, hypotheses, proofs, examples, target registers, and limitations. A common introduction, continuous numbering, linked cross-references, and one consolidated bibliography provide a single reading structure. An editorial roadmap identifies later strengthenings and distinguishes model classifications from unresolved transfers to named constants. The earlier 223-page treatise and the subsequent 447-page compilation are not reproduced. This edition consolidates the twelve manuscripts written after those volumes; it does not assert a new resolution of their outstanding named-constant conjectures.

Scope of this edition. The source sequence begins with *Arithmetic of Observation Boundaries* and ends with *Arithmetic Rigidity, Quadratic Exceptions, and Constant Transfer*. All twelve successor manuscripts are represented. The two earlier volumes remain cited where they supply mathematical inputs, but neither is included as a part of this document. The TEX source embeds the entire present text and bibliography and needs no external chapter files.

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Editorial synthesis and reading map

The common question is how much mathematical structure can be recovered from observations, and which additional information is required before that recovery decides an arithmetic property. The first parts develop arithmetic classifications for boundaries and optimal values. The middle parts test how those mechanisms interact with inverse problems, logic, and exact finite certificates. The last parts separate analytic convergence, arithmetic cancellation, and genuine value classifications. These developments are interdependent, so the original chronological order is retained. Moving isolated proofs into new thematic chapters would obscure their local definitions and inherited assumptions.

Twelve parts, one argument sequence

Part	Main subject	Role in the combined development
I	Observation boundaries	Arithmetic tests for extraction, stopping, and spectral boundaries; the first kernel and height obstructions.
II	Optimal values	Transfer from boundary locations to objective values; mixed-sign spectra and stronger positive-kernel obstructions.
III	Independence and entropy optima	Cubic spectra, families of stopping values, integer extraction rates, entropy optimization, and genuinely reduced denominators.
IV	Rigidity of optimization	The full planar stopping family, entropy-game ranges, differential complexity, and finite algebraic covers.
V	Structural and adaptive recovery	Unrestricted mixture identification, finite spectral complexity, robust certificates, and exact adaptive thresholds.
VI	Logic and effective transfer	The fifty-seven requested logical problems, with distinctions between solved subcases, external announcements, and failed reductions.
VII	Arithmetic beyond convergence	Effective logarithmic-root and quantile classification, finite cyclotomic certificates, and limits of arithmetic transfer.
VIII	Arithmetic certificates	Inverse realization, moment matching, entropy tests, Mahler relation ideals, and growing arithmetic kernels.
IX	Thresholds and primitive forms	Sharp alphabet and degree costs, exact Laguerre content, Wilson approximants, and formal Gamma isolation.
X	Sparse certificates	Information loss, a four-atom entropy example, rare cancellation, curvature counts, and nonlinear Lambert independence.
XI	Spectral entropy and transfer	Totally-real entropy thresholds, a third Lambert jet, full-degree positive pencils, and parameter laws.
XII	Quadratic exceptions	Exact relative and conditional entropy classifications, the cubic ternary obstruction, actual beta-period separation, fixed-parameter content bounds, and repaired Catalan convergence.

How later parts change the earlier frontier

Several earlier conclusions remain valid while losing their status as the strongest statement available in the sequence. The following crosswalk records the main changes; it does not count each intermediate formulation as a separate discovery.

Spectra, stopping, and extraction. The one-sided rank-two spectral classification in Part I is extended to all sign patterns in Part II; the real-cubic branch becomes unconditional in Part III. The unrestricted higher-rank conditional framework retains its hypotheses. The stopping branch progresses from boundary arithmetic to objective values and then to independence of the full planar even-payoff family in Part IV. An optimizer and its objective value are different numbers. Likewise, the extraction branch's geometric results for real shape parameters do not automatically extend arithmetic conclusions proved only for rational or integer parameters.

Entropy thresholds answer different questions. In this roadmap, natural entropy is $H(p) = -\sum_i p_i \log p_i$, bit entropy is $H_2(p) = H(p)/\log 2$, and effective rank is $E(p) = e^{H(p)}$. Their arithmetic classifications must not be interchanged; retained parts specify their local notation. Part IX proves a three-symbol minimum for algebraic irrational bit entropy among positive algebraic probability vectors. Part X constructs four irrational algebraic probabilities with rational bit entropy, and Part XI makes four the exact minimum when all entries are irrational and totally real. The unrestricted ternary question is still open within these arguments. Part XII excludes every cubic probability field, leaving degree at least four; when the entropy denominator is odd, the remaining degree is odd and at least five. Its separate three-symbol threshold concerns algebraic exponential relative entropy for nonrational totally positive algebraic laws, with a positive rational reference and a nonzero real algebraic exponent. It is not a revision of the totally-real threshold four.

Positivity of conjugates matters. Totally real means that every conjugate is real. Totally positive means that every conjugate is positive. A displayed positive probability need not be totally positive. The relative-entropy classification in Part XII uses positivity of the conjugate laws and has genuine quadratic exceptions. This is compatible with the effective-rank classification in Part XI; the two exponential expressions have different factors and permit different cancellations.

Primitive content and rare indices. Fixed-kernel estimates, standard denominator estimates, and estimates after actual primitive reduction are different assertions. The primitive normalization discussion in Part III prevents an overbroad reading of earlier positivity obstructions. For Laguerre forms, the $O(\log X)$ exceptional count in Part X concerns $\log g_n \geq \varepsilon n \log n$. The later $O_{A,\varepsilon}((\log X)^4)$ count in Part XII concerns the smaller threshold $\log g_n(A) \geq \varepsilon n$ at fixed A. Both statements are retained. The parameter-residue law of Part XI and the fixed-parameter index law of Part XII average different variables. The latter distribution is for a prime-support upper bound, not for the primitive content itself. None of these density or counting statements removes every possible infinite exceptional subsequence.

Catalan approximation spaces. The determinant constructions develop through indefinite mixed pencils, positive full-degree pencils, polynomial-invariant subspaces, and full paired polynomial spaces. Their conclusions are not interchangeable. Part XII proves a uniform root gap for the restricted invariant space and recovers convergence by restoring the omitted sector.

The repaired convergence still lacks the primitive-height estimate needed for an arithmetic conclusion. Earlier positive interval constructions are distinct families and are not contradicted by the invariant-space gap.

Lambert functions and Gamma periods. Part XI extends the earlier Lambert independence results to three jets together with a prime series, under its stated derivative condition. It does not prove independence for every finite jet family or specialize functional independence to a particular numerical value. The final determinant-budget optimization concerns a specified sufficient clearing bound, not the actual arithmetic degree of a Lambert constant. The Gamma branch moves from a formal identity calculus to an actual fourteen-dimensional beta-period span in Part XII using classical transcendence theory. The twelve displayed transcendental ratios are classical-theorem consequences, not twelve newly solved open conjectures. Linear independence of that span is not algebraic independence of its entries.

Editorial conventions and retained limitations

The mathematical text is preserved at the source-part level. Individual title pages, abstracts, tables of contents, and bibliographies have been replaced by common frontmatter and a consolidated bibliography. Labels and citations have unique internal namespaces, and verification appendices continue the global section numbering. Repeated definitions and short supporting arguments are retained when they keep a proof intelligible. Only explicitly recorded typographic corrections and a terminology clarification are made; this is not a new full proof audit.

Statements such as “new”, “previous”, and “unresolved here” inside a source part refer to its place in the development. The forward notes identify subsequent progress. Target counts in different registers overlap and must not be added as counts of discoveries. External announcements retain their source dates and qualifications; this compilation does not independently update every literature-status statement or replay external formal proof projects. Earlier-volume inputs and other cited theorems remain mathematical dependencies even though the present TEX is a single independently compilable file.

The final part records zero newly closed previously open named-constant conjectures among its sixteen targets. The exact model classifications, inverse constructions, and obstruction theorems elsewhere in the volume have their stated scopes. Historical priority and journal-level significance are separate questions from assembling these arguments.

Part I
Arithmetic of Observation Boundaries

Editorial integration note. This is the first of the twelve included successors. The rank-two spectral theorem is strengthened in Part II. The positivity obstructions here retain their stated denominator normalization; the primitive-reduction distinction is developed in Part III.

1 Scope, provenance, and the arithmetic change of viewpoint

The starting point is the pair of working manuscripts *Observation and Recovery in Arithmetic and Probability: Prime Flows, Rare Events, and Infinite Mixtures* (223 pages) [1] and *Observation and Recovery: Geometry, Mixtures, and Spectral Inversion* (447 pages) [4]. Their recovery theorems identify parameters from observations, while their boundary theorems define distinguished constants by positivity, contact, or optimization. The present manuscript asks for the arithmetic nature of those constants. It does not concatenate the earlier manuscripts, and it does not treat their recovery statements as transcendence theorems.

There are three logically different tasks. An inverse theorem proves that observations determine a number. A numerical method approximates that number with certified error. A Diophantine argument proves that it cannot be rational or algebraic. The third task needs arithmetic information, such as denominator growth, a differential specialization theorem, or a transcendence theorem for exponentiation. Neither uniqueness nor fast numerical convergence supplies that information by itself.

1.1 What is proved, what is inherited, and what is not claimed

The proofs below establish transfer statements under their displayed hypotheses. The deep transcendence inputs are classical and are cited explicitly. A new application to a boundary from the preceding working manuscripts is described as a *transfer result*, not as a resolution of a historically documented conjecture. A theorem about a newly formulated class of positive kernels or spectra is described as a *class theorem*. A previously known classification is used only as a calibration case. A numerical value is illustrative and never used as an arithmetic proof.

In particular, proving the irrationality of an entire family of newly studied capacities is different from proving the irrationality of a famous pre-existing constant. Both can be mathematically useful, but they should not be counted together as independent famous problems closed. Rational-affine variants of a single boundary constant are also counted as one family.

The statements here are supported by written proofs and cross-checks within this research workflow. They are not presented as externally refereed results, and no claim of Annals-level novelty follows from the length of this manuscript or the number of examples.

1.2 The two transfer mechanisms

The first mechanism starts with a boundary equation

$$U(\xi)f(\xi) + V(\xi)f'(\xi) = 0,$$

where f is an E-function and U, V are algebraic rational functions. If f and f' are linearly independent over $\overline{\mathbb{Q}}(z)$, Beukers' theorem excludes every nonzero algebraic ordinary point ξ

satisfying this equation. The arithmetic of the *parameters* is crucial: the standard Kummer E-function input used here requires rational hypergeometric parameters. This explains why some contact locations are proved transcendental, while an implicitly varying capacity parameter is proved only irrational.

The second mechanism starts with a critical value of a finite positive spectrum. The critical-point and critical-value equations give two algebraic equations on a two-dimensional torus. A one-sided sign condition rules out repeated-factor degeneracy. An irrational-weight derivation has no invariant nonconstant Laurent irreducible factors, so the two equations force algebraic torus coordinates. Gelfond–Schneider then excludes all positive points except the explicitly identifiable point $x = 1$.

1.3 Notation and foundational inputs

We write $\overline{\mathbb{Q}}$ for the field of algebraic numbers, trdeg for transcendence degree, and $M(a; b; z) = {}_1F_1(a; b; z)$ for Kummer's function. All real powers of positive numbers use the real logarithm. Algebraic independence of an infinite family means algebraic independence of every finite subfamily. Boundary existence and uniqueness are proved below where needed, or tied explicitly to a displayed analytic characterization from the earlier manuscripts. A literature status is dated to the searches completed for this manuscript; an unverified recent proof claim is not counted as a resolution.

2 An arithmetic obstruction on the gamma-extraction boundary

The gamma-extraction curve in the preceding manuscript produces a family of implicitly defined constants. Its arithmetic nature was not determined there. We prove that every rational point of its parameter axis strictly above $1/2$ gives an irrational boundary constant. The argument uses a classical specialization theorem for E-functions, not a new transcendence theorem. The new application is to the probabilistic boundary and its exact contact relation. In particular, the result does not settle the irrationality of any famous classical constant whose status was previously open.

2.1 The inherited boundary and the precise arithmetic question

For $d > 1/2$, the previous manuscript [4] constructed a unique number

$$A(d) \in \left(\max(0, 1 - d), \frac{1}{4d}\right). \tag{2.1}$$

Put $\eta = a + d - 1$ and, for $\eta > 0$, define

$$B_{a,d}(z) = {}_1F_1(a - 1; \eta; z) + \frac{z}{2\eta}{}_1F_1(a; \eta + 1; z). \tag{2.2}$$

The defining equation, with uniqueness in (2.1), is

$$B_{A(d),d}(2(d - A(d))) = 0. \tag{2.3}$$

This equation comes from the inverse Laplace transform of $(u + 1)u^{-a}(u + 2)^{-d}$. At the positivity boundary its density has a unique double zero at

$$x_*(d) = d - A(d). \tag{2.4}$$

The exact formula for the contact point is essential to the arithmetic argument. A generic unspecified zero of an E-function would not supply it.

On the first excess-power McKay line with rational parameter $\kappa > 1/2$, the maximum endpoint-rate gamma shape is $h_*(\kappa) = \kappa - A(\kappa)$. Thus $h_*(d)$ and $x_*(d)$ are the same number in this normalization. The ID-remainder capacity is $d - 1/(4d)$, and its deficit from the unrestricted capacity is $1/(4d) - A(d)$. These rational-affine variants must not be counted as independent arithmetic discoveries.

2.2 The classical specialization input, with its hypotheses checked

We use the following precise consequence of Beukers' refinement of the Siegel–Shidlovskii theorem [2, Corollary 1.4]. If a vector of E-functions F satisfies $F' = M(z)F$, with $M \in \overline{\mathbb{Q}}(z)^{r\times r}$, and its coordinates are linearly independent over $\overline{\mathbb{Q}}(z)$, then their values at a nonzero algebraic point ξ are linearly independent over $\overline{\mathbb{Q}}$ whenever ξ is not a pole of M. In particular this is a linear-independence theorem over the full field of algebraic numbers, not just over the rationals.

Lemma 2.1 (Rational Kummer parameters give strict E-functions)**.** *Let $\alpha, \beta \in \mathbb{Q}$, with $\beta \notin \{0, -1, -2, \ldots\}$. The function $y(z) = {}_1F_1(\alpha; \beta; z)$ and its derivative are E-functions. Their first-order system is*

$$\frac{d}{dz}\begin{pmatrix} y \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \alpha/z & 1 - \beta/z \end{pmatrix}\begin{pmatrix} y \\ y' \end{pmatrix}. \tag{2.5}$$

Its only finite possible singularity is zero.

Proof. The polynomial case is immediate, so suppose that no numerator Pochhammer symbol vanishes. Write

$$y(z) = \sum_{n\geq 0} u_n \frac{z^n}{n!}, \qquad u_n = \frac{(\alpha)_n}{(\beta)_n}.$$

The recurrence for u_n gives $|u_n| \leq C^{n+1}$ for a fixed C, so the series is entire of exponential type. We give the denominator check explicitly. Write $\alpha = U/V$, $\beta = W/Z$ in lowest terms with positive V, Z. For a prime p not dividing VZ, and for each power p^k, the counts of indices $0 \leq j < n$ for which p^k divides $U + Vj$ and $W + Zj$ differ by at most one: each is the count of one residue class modulo p^k in an interval of length n. All relevant nonzero factors have absolute value at most $C_1(n+1)$. Therefore the denominator valuation of every u_j, $j \leq n$, at such a prime is at most $\lfloor \log_p(C_1(n+1)) \rfloor$.

For the finitely many primes dividing VZ, the denominator valuation is $O(n)$: the explicit factor $(Z/V)^j$ contributes linearly, and summing the residue-class counts of divisibility over p^k also contributes $O(n)$. Consequently one common denominator of $u_0, \ldots, u_n$ divides a fixed-prime factor of size $\exp(O(n))$ times $\operatorname{lcm}(1, \ldots, \lfloor C_1(n+1) \rfloor)$. The elementary Chebyshev bound for this least common multiple is $\exp(O(n))$. Thus the coefficients satisfy the strict exponential height and denominator bounds in the definition of an E-function. The coefficients of y' are u_{n+1} and satisfy the same bounds. Finally Kummer's equation $zy'' + (\beta - z)y' - \alpha y = 0$ proves (2.5); see also [3, Section 13.2]. □

The next check avoids any irreducibility assumption about the whole Kummer differential equation. That distinction matters when d is a positive integer.

Lemma 2.2 (The analytic solution is not hyperexponential)**.** *Suppose $\beta \notin \{0, -1, -2, \ldots\}$ and neither α nor $\beta - \alpha$ is a nonpositive integer. Then ${}_1F_1(\alpha; \beta; z)$ and its derivative are linearly independent over $\mathbb{C}(z)$.*

Proof. Otherwise y'/y would be rational. Since y is entire and $y(0) = 1$, its finite zeros would be exactly the finitely many poles of this rational logarithmic derivative, with positive integer residues. Let P be the polynomial containing those zeros with their multiplicities. The logarithmic derivative of y/P is an entire rational function, hence a polynomial. It follows that $y = e^{Q(z)}P(z)$ for a polynomial Q, after absorbing a nonzero constant into P.

The coefficient ratio gives exponential-type growth of y, also for the complex parameters of this lemma. Thus Q has degree at most one, and we may write $y = e^{cz}P$ with $n = \deg P \geq 0$. Substitute this expression into Kummer's equation. The coefficient of degree $n+1$, after removing e^{cz}, is proportional to $c^2 - c$, so $c = 0$ or $c = 1$. If $c = 0$, the coefficient of degree n forces $\alpha = -n$. If $c = 1$, the transformed equation is

$$zP'' + (\beta + z)P' + (\beta - \alpha)P = 0,$$

whose leading coefficient forces $\beta - \alpha = -n$. Both alternatives contradict the assumptions. □

Proposition 2.3 (A standard E-function consequence for Robin zeros)**.** *Let $\alpha, \beta \in \mathbb{Q}$ satisfy the hypotheses of Lemma 2.2. At every nonzero algebraic ξ, the two numbers*

$$ {}_1F_1(\alpha;\beta;\xi), \qquad \frac{d}{dz}{}_1F_1(\alpha;\beta;z)\Big|_{z=\xi}$$

are linearly independent over $\overline{\mathbb{Q}}$. Their quotient is therefore transcendental. If $P, Q \in \overline{\mathbb{Q}}[z]$ are relatively prime and not both zero, every nonzero zero of $P(z)y(z)+Q(z)y'(z)$ is transcendental. This proposition is a classical-theorem application, not a claim of a new general E-function theorem.

Proof. Apply the stated Beukers theorem to (2.5), using the two preceding lemmas. Every nonzero algebraic point is ordinary for this system. Linear independence also ensures that neither value is zero. An algebraic quotient would give a forbidden linear relation. At an algebraic zero of $Py + Qy'$, coprimality ensures that $P(\xi), Q(\xi)$ are not both zero, giving the same contradiction. □

2.3 Irrationality of every rational-parameter curved boundary

Theorem 2.4 (The gamma boundary contains no rational point)**.** *For every rational $d > 1/2$, the number $A(d)$ is irrational. Consequently the maximum endpoint-rate gamma shape $h_*(d) = x_*(d) = d - A(d)$ and the capacity gain $1/(4d) - A(d)$ are irrational as well. More precisely,*

$$\mathbb{Q}(A(d)) = \mathbb{Q}(h_*(d)) = \mathbb{Q}(1/(4d) - A(d)).$$

The curved boundary $a = A(d)$ has no point with both coordinates rational. In contrast, the linear boundary $A(d) = 1 - d$ for $d \leq 1/2$ has rational value at every rational d.

Proof. Suppose $a = A(d)$ were rational. By (2.1), $0 < a < 1/2$, $\eta = a + d - 1 > 0$, and $d - a > 0$. Set

$$\alpha = a - 1, \qquad \beta = \eta, \qquad p = 1 - a.$$

Then $\alpha \in (-1, -1/2)$ is not a nonpositive integer, $\beta > 0$, and $\beta - \alpha = d > 0$ is not a nonpositive integer. The derivative identity for Kummer's function gives

$$B_{a,d}(z) = y(z) - \frac{z}{2p}y'(z). \tag{2.6}$$

The coefficients in this linear combination are relatively prime polynomials with rational coefficients. The contact relation (2.3) says that it vanishes at $z_* = 2(d - a) > 0$, a rational number. This contradicts Proposition 2.3. The remaining assertions are immediate rational-affine changes of this one irrational number, and the inherited formula on the linear face. □

For example, the theorem settles the following three members of the newly defined family, with exact inherited enclosures:

d	$A(d)$	$h_*(d) = x_*(d)$
1	irrational in $(0, 1/4)$	irrational in $(3/4, 1)$
3/2	irrational in $(0, 1/6)$	irrational in $(4/3, 3/2)$
2	irrational in $(0, 1/8)$	irrational in $(15/8, 2)$

These rows illustrate one uniform theorem; they are not three independent open-problem closures.

For integer $d = m \geq 1$ the Kummer equation can indeed be reducible. Its second Frobenius solution is

$$z^{1-\eta}{}_1F_1(1-m; 2-\eta; z),$$

a power times a polynomial of degree $m-1$. When a is rational in the interval used above, $\eta = a + m - 1$ is not an integer, so this solution is not the analytic normalized solution y. Lemma 2.2 concerns that particular analytic solution and remains valid. Thus $d = 1$ and $d = 2$ are not omitted exceptional cases. The excluded endpoint $a = 0$ also matters: for $d > 1$, (2.2) becomes $B_{0,d}(z) = 1 - z/\{2(d-1)\}$, whose positive zero is algebraic when d is rational. The strict interior assumptions cannot be discarded.

2.4 An exact rational-input decision procedure

The arithmetic obstruction has an effective consequence for the factorization problem. This is grouped with Theorem 2.4, not counted as a second independent irrationality theorem.

Corollary 2.5 (Finite rational arithmetic decides rational factor pairs)**.** *For rational $\kappa > 1/2$ and nonnegative rational shapes h_-, h_+, simultaneous gamma extraction at the two original rates can be decided by a terminating algorithm using rational arithmetic. No equality oracle for a special-function value is needed. The algorithm makes no uniform complexity or irrationality-measure claim.*

Proof. Put $a = \kappa - h_-$ and $d = \kappa - h_+$. The previous exact diagram says that feasibility is $a \geq A(d)$. Reject if $a \leq 0$ or $a + d < 1$. If $d \leq 1/2$, accept exactly when $a + d \geq 1$. For $d > 1/2$, reject $a + d = 1$ and accept $a \geq 1/(4d)$. The only remaining case has

$$\max(0, 1-d) < a < 1/(4d) < 1/2.$$

Set $p = 1 - a \in (1/2, 1)$, $z = 2(d-a) > 0$, and $\eta = a + d - 1 > 0$. The one-point criterion reduces feasibility to $B_{a,d}(z) \geq 0$. Here all parameters are rational, and

$$B_{a,d}(z) = \sum_{n \geq 0} b_n, \qquad b_n = \left(1 - \frac{n}{2p}\right) \frac{(-p)_n}{(\eta)_n} \frac{z^n}{n!}. \tag{2.7}$$

All terms are rational; $b_n > 0$ for every $n \geq 2$. For $n \geq 3$,

$$\frac{b_{n+1}}{b_n} = \frac{n+1-2p}{n-2p} \frac{n-p}{n+\eta} \frac{z}{n+1} \leq \frac{2z}{n+1}.$$

Therefore, for $N \geq \max(3, \lceil 4z \rceil)$,

$$\sum_{n=0}^{N} b_n \;\leq B_{a,d}(z) \;\leq \sum_{n=0}^{N} b_n + 2b_{N+1}. \tag{2.8}$$

These rational intervals shrink to the value. Proposition 2.3 ensures that the value is not zero. Increasing N therefore eventually certifies its sign. A positive sign accepts the factor pair and a negative sign rejects it. This proves termination and exact correctness, including the cases of negative d and atomic remainders already handled by rational inequalities. □

The same sign certificates give arbitrarily tight rational brackets for $A(d)$ when $d > 1/2$ is rational. Start with the rational upper bound $1/(4d)$ and enumerate rational points decreasing to $\max(0, 1-d)$ until a negative one-point value is certified; then bisect. None of the rational midpoints is a boundary point, so every sign test terminates. This is an exact computability statement; it gives no effective bound on how close $A(d)$ can be to rationals of a given denominator.

What this does not prove. Transcendence of $A(1)$, $A(3/2)$, $A(2)$, or of the general rational-parameter boundary remains unresolved here. If one merely assumes $A(d)$ algebraic, the Kummer parameters need not be rational; Lemma 2.1 no longer applies. No assertion that such a function is an E-function has been justified. Also $A(d)$ is a root in a parameter of a family, not a value of one fixed rational- parameter E-function at a rational point, so irrationality-exponent theorems for those values cannot simply be applied to $A(d)$. The same obstruction prevents importing this proof to the entire intermediate-rate capacity curve, whose density contact location is not known to be algebraic under a rational-shape assumption. No result here concerns Euler's constant, Catalan's constant, or an individual odd zeta value beyond the results already known for them.

3 Arithmetic of stopping boundaries and Bessel obstruction constants

This section investigates three targets. The first is the arithmetic classification of the power-payoff stopping boundary for a Bessel bridge, including the reflected Brownian bridge and Brownian excursion. The second concerns the locations and values of the Bessel-ratio extrema that occurred in the earlier HCM obstruction analysis. Both admit complete classifications at rational parameters. The third is the signed Brownian-bridge stopping constant; the method does not decide its arithmetic nature. The stopping problems themselves are established in the literature [5, 6]. We have not located a primary source posing the arithmetic classifications below as open conjectures. They are therefore recorded as arithmetic transfers to existing constants, not as closures of named historical conjectures. The transcendence input is classical; no new version of the Siegel–Shidlovskii theorem is claimed.

3.1 The arithmetic input and a logarithmic-derivative mechanism

We use the ordinary-point linear specialization theorem stated in Section 2, in the precise form of Beukers' Corollary 1.4 [2]. In particular, there is no unspecified finite exceptional set at ordinary algebraic points.

Here an E-function is an entire function $f(z) = \sum_{n\geq 0} a_n z^n/n!$ with algebraic coefficients, satisfying a linear differential equation over $\overline{\mathbb{Q}}(z)$, whose coefficient heights and common denominators grow at most exponentially. All functions used below have rational coefficients. For completeness, the elementary denominator estimate needed here is supplied next.

Lemma 3.1 (Rational hypergeometric coefficients)**.** *For positive rational a, b, the numbers $r_n = (a)_n/(b)_n$ have at most exponential absolute growth and a common denominator for*

$r_0, \dots, r_n$ bounded by C^{n+1}, with C depending only on a, b. Consequently

$$M(a; b; z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}$$

and its derivative are E-functions. For rational $\mu > -1$, the normalized modified Bessel function

$$H_\mu(z) = \sum_{n=0}^{\infty} \frac{(z^2/4)^n}{n!(\mu+1)_n}$$

and its derivative are also E-functions.

Proof. The Kummer assertions and their denominator bounds are Lemma 2.1. Their equation is

$$zf'' + (b - z)f' - af = 0. \tag{3.1}$$

For the Bessel function the odd exponential coefficients vanish, whereas the even ones are

$$\frac{(2n)!}{4^n n!(\mu+1)_n} = \frac{(1/2)_n}{(\mu+1)_n}.$$

The same rational Pochhammer-ratio estimate applies, and

$$H_\mu'' + \frac{2\mu+1}{z} H_\mu' - H_\mu = 0 \tag{3.2}$$

supplies the differential equation. Derivatives shift the exponential coefficients and preserve the bounds. □

Lemma 3.2 (A classical specialization mechanism)**.** *Suppose f, f' are E-functions satisfying a two-dimensional first-order system over $\overline{\mathbb{Q}}(z)$, and f'/f is not rational. At every nonzero algebraic point ξ where the system is regular, $f(\xi) \neq 0$ and $f'(\xi)/f(\xi)$ is transcendental. Hence if*

$$Q(x, f'(x)/f(x)) = 0, \qquad Q \in \overline{\mathbb{Q}}[X, Y],$$

and $Q(x, Y)$ is not the zero polynomial, then such a nonzero nonsingular real x cannot be algebraic.

Proof. A rational linear relation between f and f' would make f'/f rational. Thus Beukers' corollary applies to the pair. A zero of $f(\xi)$ or an algebraic value of the quotient would give a nontrivial algebraic linear relation between the two values. In the last assertion, an algebraic x would make the quotient a root of a nonzero polynomial with algebraic coefficients, a contradiction. □

We will repeatedly use an elementary analytic observation. If an entire function of exponential type has rational logarithmic derivative, it has the form $e^{\lambda z}P(z)$ for a polynomial P. To verify this, the poles of the logarithmic derivative are exactly the finitely many zeros, with positive integral residues. After removing these polynomial factors, integration of the remaining polynomial logarithmic derivative gives $e^{Q(z)}$. Exponential type forces $\deg Q \leq 1$. This observation concerns the particular entire solution, and does not assume irreducibility of its differential equation.

3.2 Target I: an exact arithmetic classification of bridge boundaries

For $d, q > 0$ put $a = q/2$, $b = d/2$ and define $z_{d,q} > 0$ by

$$z_{d,q} M'(a; b; z_{d,q}) = aM(a; b; z_{d,q}), \qquad C_{d,q} = \sqrt{2z_{d,q}}. \tag{3.3}$$

The next proof includes existence and uniqueness, rather than treating a numerical solution as a defined constant.

Theorem 3.3 (Bridge-boundary classification)**.** *For every $d, q > 0$, equation (3.3) has exactly one positive solution. For rational $d, q > 0$,*

$$C_{d,q} \text{ is algebraic} \quad \Longleftrightarrow \quad q - d \in 2\mathbb{Z}_{\geq 0}. \tag{3.4}$$

In every other rational-parameter case $C_{d,q}$ is transcendental. If $q - d = 2N$ with $N \geq 0$, set $P_N(z) = M(-N; d/2; -z)$. The algebraic boundary is specified without special functions by the unique positive root of

$$zP_N'(z) + (z - q/2)P_N(z) = 0. \tag{3.5}$$

Proof. Write $f(z) = \sum_{n\geq 0} c_n z^n$ with $c_n > 0$. For $z > 0$, the weights $c_n z^n / f(z)$ define a probability distribution on the nonnegative integers. Its mean is $L(z) = zf'(z)/f(z)$ and

$$\frac{dL}{d\log z} = \mathrm{Var}_z(n) > 0.$$

The variance is strictly positive since at least two weights are positive. Also $L(z) \to 0$ as $z \downarrow 0$. For every fixed m, the probability of $n \leq m$ tends to zero as $z \to \infty$: the numerator has degree at most m, while the denominator is at least its positive term of degree $m + 1$. Thus $L(z) \to \infty$. This proves existence and uniqueness of $L(z) = a$.

We classify when f'/f can be rational. The coefficient bounds in Lemma 3.1 imply exponential type. If its logarithmic derivative is rational, write $f = e^{\lambda z} P(z)$ and let $N = \deg P$. Substitution into (3.1) first forces $\lambda(\lambda - 1) = 0$ by its highest-degree term. If $\lambda = 0$, the next highest-degree term gives $a = -N$, impossible for $a > 0$. If $\lambda = 1$, substitution gives

$$zP'' + (z + b)P' + (b - a)P = 0,$$

whose highest coefficient forces $a - b = N$. Conversely, when $a - b = N \in \mathbb{Z}_{\geq 0}$, Kummer's transformation gives $f(z) = e^z M(-N; b; -z)$. The latter polynomial has rational coefficients when a, b are rational, and is positive for $z > 0$.

Outside this exceptional family, the pair (f, f') is linearly independent and solves

$$\begin{pmatrix} f \\ f' \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ a/z & 1 - b/z \end{pmatrix} \begin{pmatrix} f \\ f' \end{pmatrix}.$$

Its only possible finite singular point is zero. If $z_{d,q}$ were algebraic, equation (3.3) would contradict Lemma 3.2. In the exceptional family the boundary equation is precisely (3.5). This is a nonzero polynomial, since its highest term has degree $N + 1$, and hence its positive root is algebraic. Finally $C_{d,q}$ is algebraic if and only if $z_{d,q} = C_{d,q}^2/2$ is algebraic. □

The identification with stochastic boundaries is as follows. The squared Bessel bridge Q of dimension $d > 0$ has local generator

$$2u\partial_{uu} + \left(d - \frac{2u}{1 - t}\right)\partial_u.$$

For payoff $Q_\tau^{q/2}$, the ansatz $V(t,u) = (1-t)^{q/2} g(u/(1-t))$ gives

$$4yg'' + 2(d-y)g' - qg = 0.$$

The solution regular at zero is a multiple of $M(q/2; d/2; y/2)$. Value matching and smooth fit at $y = C_{d,q}^2$ give exactly (3.3). Hobson and Liu's verification theorem establishes that the resulting boundary $\sqrt{Q_t} = C_{d,q}\sqrt{1-t}$ is optimal; that stochastic verification is an external input, not a new claim here [6, Theorem 3.1]. The same boundary for $d = 1$ and payoff $|X_\tau|^q$ is the reflected Brownian-bridge result of Ekström and Wanntorp [5, Theorem 3.2].

Three examples make the exceptional set concrete:

- For the reflected Brownian bridge, rational $q > 0$ gives an algebraic boundary exactly when $q = 1, 3, 5, \ldots$. In particular, $C_{1,1} = 1$ and $C_{1,3} = 3^{1/4}$. These elementary cases are not being presented as new constants.
- The quadratic-payoff boundary $C_{1,2}$ is transcendental.
- The Brownian-excursion linear-payoff boundary $C_{3,1} \simeq 1.50339538$ is transcendental. In this case the regular solution is proportional to $x^{-1}\int_0^x e^{t^2/2}\,dt$, so its equation can also be written

$$C_{3,1} e^{C_{3,1}^2/2} = 2\int_0^{C_{3,1}} e^{t^2/2}\,dt.$$

These examples constitute one classification theorem, not three independent conjecture closures. The distinction between reflected and signed Brownian payoffs is essential.

3.3 Target II: arithmetic of the Bessel obstruction extrema

For $\mu > -1$, define

$$R_\mu(y) = \frac{I_{\mu+1}(y)}{I_\mu(y)}, \qquad D_\mu(y) = R_\mu(y) + yR'_\mu(y), \qquad y > 0.$$

These are the real obstruction functions in the earlier HCM analysis. We reproduce their shape argument so that the constants whose arithmetic is asserted have fully specified existence and uniqueness.

Lemma 3.4 (Shape, with both half-integer endpoints)**.** *If $\mu \geq 1/2$, D_μ increases strictly from zero to one; its supremum is the unattained algebraic number 1. If $-1 < \mu < 1/2$, it has a unique global maximum at $y_\mu > 0$, with $M_\mu = D_\mu(y_\mu) > 1$. For $-1/2 \leq \mu < 1/2$ this is its only stationary point. For $-1 < \mu < -1/2$ there is exactly one subsequent stationary point, a minimum with value in $(0,1)$. At every stationary point y,*

$$D''_\mu(y) = -\frac{2(D_\mu(y)^2 - 1)}{y}. \tag{3.6}$$

Proof. Put $\alpha = \mu + 1/2$, $v = R_\mu + \alpha/y$, and $D = D_\mu$. The Bessel equation and recurrence give

$$R' = 1 - \frac{2\alpha}{y}R - R^2, \qquad v' = 1 - v^2 + \frac{\alpha(\alpha-1)}{y^2} = \frac{D-v}{y}, \qquad D' = 2(1-vD). \tag{3.7}$$

For clarity, positivity needed in this argument does not follow just from the Riccati equation. Write $H_\mu(y) = \sum_{n\geq 0} c_n y^{2n}$, where $c_n > 0$ for $\mu > -1$. Under the probability weights $c_n y^{2n}/H_\mu(y)$, one has $yR(y) = 2\mathbb{E}_y n$ and hence

$$D(y) = (yR(y))' = \frac{4}{y}\,\mathrm{Var}_y(n) > 0.$$

All differentiations are justified by convergence of the entire series. The standard differentiated large-argument expansions used below follow from [3, §10.40(i)].

At a zero of v' one has $v'' = -2\alpha(\alpha-1)/y^3$. At a zero of D', positivity gives $v = 1/D > 0$, and differentiation gives (3.6). Such a stationary point cannot have $D = 1$. Otherwise $v = 1$ and $v' = 0$, so $\alpha(\alpha-1) = 0$. At these two exceptional parameters $v = \tanh y$ or $v = \coth y$, respectively, neither of which equals one at a finite positive point.

The endpoint expansions, also valid after differentiation, are

$$\begin{aligned} D(y) &= \frac{y}{\mu+1} + O(y^3) && (y \downarrow 0), \\ v(y) &= 1 + \frac{\alpha(\alpha-1)}{2y^2} + O(y^{-3}) && (y \to \infty), \\ D(y) &= 1 - \frac{\alpha(\alpha-1)}{2y^2} + O(y^{-3}) && (y \to \infty). \end{aligned}$$

If $0 < \alpha < 1$, v' starts negative, ends positive, and every zero of v' is a strict upward crossing. Thus v has one minimum. Before that minimum $D < v$, so any stationary point of D would have $D < 1$ and be a strict minimum. This cannot be the first zero of D', which starts positive. After that minimum $D > v$, so every stationary point of D is a strict maximum above one. Since D' ends negative, there is exactly one.

If $\alpha < 0$, v' starts positive, ends negative, and each zero is a strict downward crossing. Before its unique maximum, every stationary point of D must be a maximum above one; after it, every stationary point must be a minimum below one. There must be a value $D > 1$: otherwise $[y(1-R(y))]' = 1 - D \geq 0$, whereas $y(1-R(y))$ starts at zero and tends to $\alpha < 0$. Since D eventually lies below one and increases to one, the maximum and the later minimum both exist and are unique. For $\alpha = 0$, $R = \tanh y$, and $D' = 2\operatorname{sech}^2 y(1 - y\tanh y)$; $y\tanh y$ increases from zero to infinity, giving one maximum.

For $\alpha > 1$, v' starts negative and any zero would have to be a downward crossing. There are no such zeros. Therefore every possible stationary point of D would be a minimum below one, which again cannot be the first zero of D'. Thus $D' > 0$. For $\alpha = 1$ the conclusion follows from $v = \coth y$ and the same argument, or direct differentiation. The endpoint limits finish the proof. □

Theorem 3.5 (Transcendental extremum locations and values)**.** *Let $\mu > -1$ be rational. Every positive stationary point of D_μ, and its value, is transcendental. In particular, for rational $-1 < \mu < 1/2$, both y_μ and M_μ in Lemma 3.4 are transcendental. For rational $-1 < \mu < -1/2$, the later minimum has transcendental location and value as well. The endpoint $\mu = -1/2$ is included: $y_{-1/2}\tanh y_{-1/2} = 1$ and $M_{-1/2} = y_{-1/2}$.*

Proof. The entire even function H_μ of Lemma 3.1 is of exponential type, and its logarithmic derivative is R_μ. Suppose this derivative were rational. The analytic observation above would give $H_\mu = e^{\lambda z}P(z)$. Evenness implies $e^{2\lambda z} = P(-z)/P(z)$, forcing $\lambda = 0$; otherwise the exponential has an essential singularity at infinity, whereas the rational function does not. Thus H_μ would be a polynomial, contrary to its strictly positive infinitely many even Taylor coefficients. The system associated with (3.2) has no finite singular point except possibly zero. Lemma 3.2 shows that $R_\mu(y)$ is transcendental for every nonzero algebraic y.

At a stationary point, substitute $D = y(1-R^2) - 2\mu R$ into $D' = 0$ in (3.7). Writing $\alpha = \mu + 1/2$ gives the polynomial equation

$$y - (yR + \alpha)\{y(1-R^2) - 2\mu R\} = 0. \tag{3.8}$$

For a fixed nonzero algebraic y this is a nonzero polynomial in R, since its cubic coefficient is y^2. It would force R to be algebraic. Therefore a stationary location y is transcendental.

The value requires a further elimination; transcendence of the location alone does not suffice. Write $m = D(y)$. At a stationary point, $v = 1/m$ and

$$D = v + y(1 - v^2) + \frac{\alpha(\alpha - 1)}{y}.$$

Eliminating v yields

$$(m^2 - 1)y^2 - m(m^2 - 1)y + (\mu^2 - 1/4)m^2 = 0. \tag{3.9}$$

Lemma 3.4 showed $m > 0$ and $m \neq 1$. If m were algebraic, this would be a nonzero quadratic equation over $\overline{\mathbb{Q}}$ for y, contradicting the preceding paragraph. Thus m is transcendental. The statements about which extrema exist follow from Lemma 3.4. □

Corollary 3.6 (The normalized critical scale)**.** *For rational $-1 < \mu < 1/2$, the positive number*

$$S_\mu = \sqrt{\frac{y_\mu(M_\mu^2 - 1)}{3}}$$

is transcendental. Thus the coefficient πS_μ in the earlier near-threshold HCM asymptotic has a transcendental ratio to π. This statement does not assert that πS_μ itself is transcendental.

Proof. Set $K = 3S_\mu^2 = y_\mu(M_\mu^2 - 1) > 0$ and $A = \mu^2 - 1/4$. Equation (3.9) gives $K(M_\mu - y_\mu) = AM_\mu^2$. If K were algebraic, substitution $y_\mu = M_\mu - AM_\mu^2/K$ into its definition would give

$$(KM_\mu - AM_\mu^2)(M_\mu^2 - 1) - K^2 = 0.$$

This is a nonzero polynomial over $\overline{\mathbb{Q}}$ (its constant term is $-K^2$), contradicting Theorem 3.5. Hence K, and therefore S_μ, is transcendental. □

The new step relative to the earlier analytic obstruction analysis is arithmetic: the Riccati stationary relation passes transcendence from a logarithmic derivative to the extremum location, and the finite-fiber algebraic relation (3.9) passes it to the extremal value. The analytic shape lemma is repeated for self-containment and is not counted as a new result. The maximum, minimum, and normalized scale are grouped as one target family.

3.4 Target III: the signed Brownian boundary remains undecided

For signed payoff X_τ the Brownian-bridge boundary is $B\sqrt{1-t}$ with $B \simeq 0.8399236756$, not the reflected boundary $C_{1,1} = 1$. Its defining equation is

$$(1 - B^2)F(B) = B, \qquad F(x) = \int_0^\infty e^{xu - u^2/2}\, du = e^{x^2/2}\left(\sqrt{\pi/2} + \int_0^x e^{-u^2/2}\, du\right). \tag{3.10}$$

This is the established stopping constant in [5, Theorem 2.1]. Its existence and uniqueness on $(0, 1)$ follow directly: $F' = 1 + xF$, so $h(x) = (1-x^2)F(x) - x$ has $h'(x) = -x(1+x^2)F(x) - x^2 < 0$ for $x > 0$, while $h(0) > 0$ and $h(1) = -1$.

The naive E-function argument fails because $F(0) = \sqrt{\pi/2}$ is not algebraic. Dividing by this value does not repair the issue: the new derivative at zero is then $\sqrt{2/\pi}$. More precisely, if B were algebraic and $z = B^2/2$, equation (3.10) would yield

$$\sqrt{\pi/2} = \frac{B}{1 - B^2}e^{-z} - BM(1/2; 3/2; -z). \tag{3.11}$$

The right side combines E-function values at an algebraic argument, but the specialization theorem used here concerns relations with algebraic coefficients; it gives no contradiction to (3.11). No irrationality or transcendence assertion for B is proved in this section. We also do not claim that the literature search establishes its global open status. This is an explicitly recorded failed route, preventing a normalization constant from being silently treated as algebraic.

Proposition 3.7 (A conditional link to the E–G intersection problem)**.** *Let* $\mathbf{E}$ *be the ring of values of* E*-functions at algebraic points. If* $\pi \notin \mathbf{E}$*, the signed stopping constant* B *is transcendental. In particular this follows from the conjecture* $\mathbf{E} \cap \mathbf{G} = \overline{\mathbb{Q}}$*, where* $\mathbf{G}$ *is the ring of values of analytic continuations of* G*-functions at algebraic points, discussed in [7, Introduction].*

Proof. If B were algebraic, the right side of (3.11) would belong to $\mathbf{E}$. The ring is closed under sums, products, and multiplication by algebraic numbers; these are the usual closure properties of E-functions, with an algebraic rescaling of the argument if needed. Squaring gives $\pi \in \mathbf{E}$. On the other hand, $\pi = 6 \arcsin(1/2)$ belongs to $\mathbf{G}$: the series

$$\arcsin z = \sum_{n \geq 0} \frac{\binom{2n}{n}}{4^n (2n+1)} z^{2n+1}$$

is a G-function, with $(1-z^2)g'' - zg' = 0$ and exponential coefficient and common-denominator bounds. Its value at $1/2$ is inside its disk of convergence, so no branch choice is needed. The intersection conjecture would then force π to be algebraic, a contradiction. This proves the conditional assertion only. □

Accounting. There are two unconditionally proved arithmetic target families and one target with a conditional reduction but no unconditional decision. The bridge classification uses established optimal-stopping formulas; the Bessel classification adds arithmetic information to the earlier manuscript's obstruction constants. Neither family has been verified as a historically stated open arithmetic conjecture. Numerical approximations in the examples serve only to identify the constants; every arithmetic conclusion rests on an exact equation and a proof.

4 Arithmetic of critical spectral thresholds and frozen edges

The inverse-observation theory supplies constants as minima and as right boundaries of zero sets. Their uniqueness does not determine their arithmetic nature. This section classifies a substantial class of finite spectral minima, then contrasts it with finite-prime and infinite-prime boundary equations. The main reduction uses the Gelfond–Schneider theorem: a positive algebraic number other than one, raised to an algebraic irrational power, is transcendental. This is a classical theorem, not a new transcendence principle; a modern primary proof source is [8]. The elementary frozen-edge consequences of Lindemann–Weierstrass are explicitly separated from the critical-value argument.

4.1 A complete classification for a one-sided rank-two spectrum

Let

$$S(x) = \sum_{j=1}^{m} w_j x^{r_j}, \qquad T(C) = \min_{x>0} \frac{1+Cx}{S(x)}, \qquad 0 < r_j < 1, \quad w_j > 0, \quad C > 0. \tag{4.1}$$

Weights need not sum to one. Put

$$W = \sum_j w_j, \qquad M = \sum_j w_j r_j.$$

All positive real powers in this section use the real logarithm.

Theorem 4.1 (The unique algebraic critical value)**.** *Suppose the weights and C are algebraic. Suppose there are an algebraic irrational number θ, a positive integer q, and integers m_j, n_j such that*

$$r_j = \frac{m_j + n_j\theta}{q}, \qquad n_j \geq 0 \text{ for every } j, \qquad n_j > 0 \text{ for at least one } j. \tag{4.2}$$

The same conclusion holds with all $n_j \leq 0$ and at least one strict inequality. Then the following conditions are equivalent:

1. *$T(C)$ is algebraic;*
2. *the minimizer in (4.1) is $x = 1$;*
3. *$C = M/(W - M)$.*

At this exceptional setting,

$$T(C) = \frac{1 + C}{W} = \frac{1}{W - M}. \tag{4.3}$$

At every other positive algebraic setting C, the critical value is transcendental.

If all w_j and C are positive rational numbers, the exceptional setting is impossible, and every such critical value is transcendental.

The one-sided condition is a condition on the coefficients of the irrational basis element, not on the signs of the physical exponents: the latter always lie in $(0, 1)$. It permits any finite number of distinct irrational powers and arbitrarily many rational powers. It is not merely a two-term formula. Its role is to prevent a repeated algebraic component at the critical point; this issue cannot be omitted from a general algebraic-elimination argument.

Lemma 4.2 (Contact geometry)**.** *The minimum in (4.1) has a unique minimizer x_C. It is characterized by*

$$C = \frac{S'(x_C)}{S(x_C) - x_C S'(x_C)}. \tag{4.4}$$

In particular $x_C = 1$ if and only if $C = M/(W - M)$.

Proof. The objective tends to infinity as x tends to zero or infinity, because all exponents lie strictly between zero and one. Write $H = S - xS' > 0$. Strict concavity gives

$$\frac{d}{dx}\frac{S'(x)}{H(x)} = \frac{S''(x)S(x)}{H(x)^2} < 0.$$

Moreover xS'/H is bounded above and below by positive constants, using the smallest and largest exponents. Thus S'/H decreases continuously from infinity to zero. The derivative of the objective vanishes exactly at (4.4), which proves existence and uniqueness. Evaluation at one gives the last assertion. □

The following elementary algebraic lemma supplies the step that a formal resultant calculation can otherwise hide. Let $\overline{\mathbb{Q}}$ denote the algebraic numbers.

Lemma 4.3 (An irrational diagonal derivation has no invariant curve)**.** *On* $\overline{\mathbb{Q}}[u^{\pm 1}, v^{\pm 1}]$*, put*

$$D = u\partial_u + \theta v\partial_v, \qquad \theta \in \overline{\mathbb{Q}} \setminus \mathbb{Q}.$$

If H *is a nonunit irreducible Laurent polynomial, then* H *and* DH *are coprime. Every common zero of* H *and* DH *in* $(\mathbb{C}^*)^2$ *consequently has both coordinates algebraic.*

Proof. If $DH = 0$, the coefficient of each monomial $u^a v^b$ is multiplied by $a + b\theta$. Irrationality forces every nonconstant coefficient to vanish, so H would be a constant, contrary to the hypothesis. Suppose instead that $DH = AH$ for a Laurent polynomial A. For a nonzero Laurent polynomial P, denote the convex hull of its exponent vectors by $\mathcal{N}(P)$. Products satisfy

$$\mathcal{N}(PQ) = \mathcal{N}(P) + \mathcal{N}(Q).$$

One direct proof is to maximize a generic real linear functional: each factor has a unique maximizing exponent, so the corresponding coefficient in the product cannot cancel. Continuity gives equality of support functions in every direction. Differentiation deletes monomials but introduces none. Hence

$$\mathcal{N}(H) + \mathcal{N}(A) = \mathcal{N}(DH) \subseteq \mathcal{N}(H).$$

Taking support functions in all directions forces $\mathcal{N}(A) = \{(0,0)\}$, so A is constant. The equation $DH = cH$ then says that all nonzero monomials of H have the same weight $a + b\theta$. These weights are distinct for distinct integer pairs, so H is a single monomial, a unit. This contradiction proves coprimality.

Here are details of the algebraicity assertion. Multiplying H by a Laurent monomial does not change the common torus zeros of H and DH, by the product rule. We therefore normalize H to an ordinary polynomial divisible by neither u nor v; it remains irreducible. If H depends only on u, irreducibility over the algebraically closed field $\overline{\mathbb{Q}}$ makes it an associate of $u - a$ for a nonzero algebraic a; it has no common torus zero with DH. Otherwise view the two polynomial equations as equations in v over $\overline{\mathbb{Q}}(u)$. Their coprimality gives a resultant which is a nonzero polynomial in u, vanishing at any common torus zero. Thus its u coordinate is algebraic. The specialization of H at this coordinate cannot vanish identically as a polynomial in v: otherwise $u - a$ would divide H, against irreducibility and dependence on v. The v coordinate is therefore algebraic as well. □

Proof of Theorem 4.1. Only the implication from algebraicity to $x_C = 1$ needs proof. Suppose $\tau = T(C)$ is algebraic and put

$$u_* = x_C^{1/q}, \qquad v_* = x_C^{\theta/q}, \qquad P(u,v) = \sum_j w_j u^{m_j} v^{n_j}, \qquad F(u,v) = P(u,v) - \frac{1 + Cu^q}{\tau}.$$

The value and stationary equations give

$$F(u_*, v_*) = 0, \qquad DF(u_*, v_*) = 0. \tag{4.5}$$

Indeed, along $(u,v) = (x^{1/q}, x^{\theta/q})$, logarithmic differentiation is D/q. Under (4.2),

$$F_v(u_*, v_*) = \sum_j w_j n_j u_*^{m_j} v_*^{n_j - 1} > 0.$$

For nonpositive n_j the same expression is strictly negative. Factor F over the algebraic numbers. Since its v derivative does not vanish at the point, exactly one irreducible factor H vanishes there, and its multiplicity in F is one. Thus $F = HR$ with $R(u_*, v_*) \neq 0$. Equation (4.5) now gives $DH(u_*, v_*) = 0$. Lemma 4.3 proves that u_*, v_* are algebraic. But $v_* = u_*^{\theta}$.

Gelfond–Schneider forces $u_* = 1$, because $u_* > 0$ and the exponent is algebraic irrational. Hence $x_C = 1$.

The contact lemma and direct evaluation give the other implications and (4.3). Finally, if the weights are rational, then $W \in \mathbb{Q}$ and

$$M = \frac{1}{q}\sum_j w_j m_j + \frac{\theta}{q}\sum_j w_j n_j$$

is irrational, since the second coefficient is nonzero. If $C = M/(W - M)$ were rational, then $M = CW/(1 + C)$ would be rational, a contradiction. □

Remark 4.4 (Why one cannot simply assume a squarefree level set). Without the sign condition, an algebraic level polynomial F can in principle have a repeated factor through the contact point. Then $F = DF = 0$ may hold on a whole curve, and a resultant can vanish identically. The preceding proof establishes a nonvanishing partial derivative before selecting its factor; it does not assume away that difficulty. The same proof works for any rank-two spectrum whose algebraic critical level is nonsingular at the contact point. No unconditional theorem for all mixed-sign irrational coefficients is claimed here.

4.2 Critical locations and three specific threshold constants

Proposition 4.5 (Rational spectra and a single irrational coset)**.** *Assume the weights and C are positive algebraic.*

1. *If all r_j are rational, both x_C and $T(C)$ are algebraic.*
2. *If $r_j = \alpha + k_j/q$ for a fixed algebraic irrational α, integers k_j, and a positive integer q, then x_C is algebraic. If $x_C \neq 1$, both $T(C)$ and $\log x_C$ are transcendental.*

Proof. For the first assertion put $u = x^{1/q}$ with q a common denominator. The stationary equation becomes a nonzero polynomial equation over the algebraic numbers after clearing powers of u. Its positive solution is algebraic, and the objective at that solution is algebraic. The polynomial is nonzero, for otherwise the objective would be constant, contrary to its divergence at both ends.

For the second assertion put $u = x^{1/q}$ and $P(u) = \sum_j w_j u^{k_j}$, a positive Laurent polynomial on $(0, \infty)$. Then $S(x) = u^{q\alpha}P(u)$, and the stationary equation is

$$qCu^q P(u) - (1 + Cu^q)\{q\alpha P(u) + uP'(u)\} = 0. \tag{4.6}$$

It is nonzero: for the largest index k occurring in P, its highest coefficient is a positive multiple of $q - q\alpha - k = q(1 - r_{\max}) > 0$. Consequently the positive root u_C, and hence x_C, is algebraic. If $u_C \neq 1$, Gelfond–Schneider makes $u_C^{q\alpha}$ transcendental. The remaining factors in $(1 + Cu_C^q)/(u_C^{q\alpha}P(u_C))$ are nonzero algebraic, so $T(C)$ is transcendental. Finally $\log x_C$ cannot be nonzero algebraic, by the Hermite–Lindemann consequence of Lindemann–Weierstrass [9]. □

The following are three concrete constants arising from the same critical observation mechanism. Their proofs are not three independent transcendence breakthroughs; the first is a direct classical evaluation, while the second and third illustrate the critical-value classification.

1. For $S(x) = x^{1/\sqrt{2}}$ and $C = 1$, the contact is $x_* = 1 + \sqrt{2}$, and

$$\tau_1 = \min_{x>0} \frac{1 + x}{x^{1/\sqrt{2}}} = \frac{2 + \sqrt{2}}{(1 + \sqrt{2})^{1/\sqrt{2}}} \tag{4.7}$$

is transcendental by Gelfond–Schneider. This explicit corollary is included as a calibration example, not as a newly solved historical problem.

2. The mixed rational–irrational threshold

$$\tau_2 = \min_{x>0} \frac{2(1+x)}{x^{1/3} + x^{1/\sqrt{2}}} \tag{4.8}$$

is transcendental. Use $q = 6$, $\theta = \sqrt{2}$ and exponent pairs $(2, 0), (0, 3)$ in Theorem 4.1. Both weights and the setting are rational, so no exceptional contact is possible. This proves a statement about the minimum itself without first finding an explicit closed form for its minimizer.

3. The two-irrational-frequency threshold

$$\tau_3 = \min_{x>0} \frac{2(1+x)}{x^{(\sqrt{2}-1)/2} + x^{\sqrt{2}/2}} \tag{4.9}$$

is also transcendental, and its contact is algebraic. More explicitly, put $\theta = \sqrt{2} - 1$. If u_* is the unique positive root of

$$(1-\theta)u^3 + (2-\theta)u^2 - (1+\theta)u - \theta = 0, \tag{4.10}$$

then

$$x_* = u_*^2, \qquad \tau_3 = \frac{2(1+u_*^2)}{u_*^\theta(1+u_*)}.$$

Equation (4.10) is the stationary equation for $2(1+u^2)/(u^\theta(1+u))$; uniqueness follows from Lemma 4.2. Substitution of $u = 1$ gives $2 - 4\theta \neq 0$, so Gelfond–Schneider applies.

There is a necessary exception even with an algebraic irrational frequency. For any such $r \in (0, 1)$, the one-atom spectrum and the algebraic setting $C = r/(1-r)$ have $x_C = 1$ and $T(C) = 1/(1-r)$ algebraic. Even rational C does not suffice if the weights are only assumed algebraic: take

$$r_1 = \frac{1}{4}, \quad r_2 = \frac{1}{\sqrt{2}}, \quad w_1 = \frac{r_2 - 1/2}{r_2 - 1/4}, \quad w_2 = \frac{1/4}{r_2 - 1/4}.$$

The weights are positive algebraic, sum to one, and have weighted mean $1/2$. At $C = 1$ the threshold is exactly 2. These examples rule out an unqualified claim that an irrational frequency automatically makes every critical value transcendental.

4.3 Frozen edges: what algebraic frequencies do prove

For positive distinct frequencies and magnitudes define the frozen edge by

$$\sum_{j=1}^{m} a_j e^{-\lambda_j S} = 1, \qquad a_j, \lambda_j > 0. \tag{4.11}$$

There is a unique real solution, since the left side is strictly decreasing from infinity to zero. In independent finite-sum models this is the rightmost zero boundary; the arithmetic statements below concern the explicitly defined constant S.

Proposition 4.6 (Classical consequences for algebraic frequencies)**.** *If every a_j and λ_j is algebraic, then $S = 0$ if $\sum a_j = 1$, and otherwise S is transcendental. More generally, every nonzero complex zero of*

$$1 - \sum_j c_j e^{-\lambda_j z}, \qquad c_j \in \overline{\mathbb{Q}} \setminus \{0\},$$

is transcendental when the distinct positive λ_j are algebraic.

In particular let $S_{1,\sqrt{2}}$ solve

$$e^{-S_{1,\sqrt{2}}} + e^{-\sqrt{2}S_{1,\sqrt{2}}} = 1. \tag{4.12}$$

Both $S_{1,\sqrt{2}}$ and $e^{-S_{1,\sqrt{2}}}$ are transcendental.

Proof. The zero statement follows immediately from Lindemann–Weierstrass: if $z \neq 0$ were algebraic, the distinct algebraic exponents $0, -\lambda_1 z, \ldots, -\lambda_m z$ would give a nontrivial algebraic linear relation among their exponentials. The frozen edge is zero exactly when $\sum a_j = 1$, proving the first claim.

For (4.12), the edge is positive and hence transcendental by the first claim. Set $X = e^{-S_{1,\sqrt{2}}}$, so $0 < X < 1$ and $X + X^{\sqrt{2}} = 1$. If X were algebraic, then $X^{\sqrt{2}} = 1 - X$ would be algebraic, contradicting Gelfond–Schneider. □

These are transfers of classical transcendence theorems, not new solutions of long-standing logarithm problems. The zero statement also does not classify the real and imaginary parts separately: a transcendental complex number can have an algebraic real part. The condition on complex coefficients is algebraicity of c_j, not algebraicity of a chosen angle in $c_j = a_j e^{i\theta_j}$.

4.4 Finite prime edges and the barrier at infinitely many primes

Prime frequencies are logarithms of algebraic numbers, not algebraic numbers themselves. Therefore Proposition 4.6 does not apply to them. One can nevertheless obtain a different, elementary classification at the level of rationality.

Proposition 4.7 (Every finite unit-weight prime edge is irrational)**.** *Let $\mathcal{P}$ be a finite set of at least two primes. The unique positive solution of*

$$\sum_{p \in \mathcal{P}} p^{-\sigma} = 1 \tag{4.13}$$

is irrational.

Proof. Existence and positivity follow from strict decrease, the value $|\mathcal{P}| > 1$ at zero, and the limit zero at infinity. Suppose $\sigma = a/b$ with positive integers a, b. Work in a number field containing all positive radicals $p^{1/b}$ and choose a nonarchimedean valuation v above any one prime $p_0 \in \mathcal{P}$. Normalize it so that $v(p_0) > 0$. For every other rational prime p, $v(p) = 0$. Thus

$$v(p_0^{-a/b}) = -\frac{a}{b} v(p_0) < 0, \qquad v(p^{-a/b}) = 0 \quad (p \neq p_0).$$

In a sum with a unique term of strictly smallest valuation, the valuation of the sum equals that smallest valuation. This follows from the ultrametric inequality applied in both directions to that term and the sum of the others. The left side of (4.13) therefore has negative valuation, whereas $v(1) = 0$, a contradiction. □

For example the natural two-prime constant

$$2^{-\sigma_{2,3}} + 3^{-\sigma_{2,3}} = 1 \tag{4.14}$$

is irrational. The corresponding exponential polynomial is a standard nonlattice self-similar-string model studied in [11]. The valuation proof is an elementary finite radical argument; no historical priority is claimed for this irrationality observation. We do not prove that $\sigma_{2,3}$ is transcendental. If it were algebraic irrational, Gelfond–Schneider would make each summand transcendental, but that does not prevent their sum from being one. Baker's theorem concerns algebraic linear combinations of finitely many logarithms [10]; the displayed additive relation between two unknown powers is not such a relation.

For comparison, the commensurable edge $2^{-S} + 4^{-S} = 1$ has

$$S = \frac{\log((1+\sqrt{5})/2)}{\log 2}.$$

It is transcendental: if $S = a/b$ were rational, then $((1+\sqrt{5})/2)^b = 2^a$, impossible because the golden ratio is an algebraic unit whereas a nonzero power of two is not. If S were algebraic irrational, Gelfond–Schneider would contradict the algebraicity of 2^S. This familiar logarithm corollary is a useful comparison, not a new difficult constant.

Two infinite constants resist the preceding arguments:

$$P(\sigma_P) = 1, \qquad P(s) = \sum_{p \text{ prime}} p^{-s}, \qquad \zeta(\rho) = 2. \tag{4.15}$$

The second constant is the exponent in the classical ordered factorization problem; see [12] for its role. Both equations have unique real solutions greater than one, by positivity and strict monotonicity. For P, divergence at one follows from divergence of the sum of prime reciprocals; for ζ it is the usual pole. Both functions tend to zero and one, respectively, at positive infinity.

No irrationality or transcendence conclusion for σ_P or ρ is obtained here. The obstruction is specific:

1. The critical-value theorem is a statement about two Laurent variables and a finite algebraic level polynomial. Neither infinite equation supplies such a polynomial or the associated critical-point equation.

2. The finite-prime valuation proof concerns equality of a *finite* sum in a number field. Real convergence of an infinite radical series does not preserve that valuation argument. Limits of irrational, or even transcendental, numbers may be rational.

3. Lindemann–Weierstrass requires algebraic exponents. The products $s \log p$ are not algebraic merely because s is assumed algebraic. The usual E-function value theorems likewise require their algebraic-coefficient differential setting; writing a prime Dirichlet series does not provide it.

4. Recovering the full spectrum or locating its boundary analytically does not provide the Diophantine estimates needed to exclude rational or algebraic boundary locations.

The checked primary sources supply no theorem that resolves these two arithmetic classifications by the reductions used here. This is a limitation of the present work and search, not an assertion of an exhaustive priority survey. In particular nothing in this section classifies the ordinates of the nontrivial Riemann-zeta zeros or resolves the Riemann hypothesis.

The substantive finite-spectrum result is Theorem 4.1: algebraicity of a variational critical value forces the unique normalized contact $x = 1$, and positive rational data exclude that exception. The explicit constants, frozen-edge corollaries, and finite-prime irrationality proof are kept separate from the unresolved infinite-prime targets.

5 Exponential-integral values beyond the algebraic exponential field

This section records unconditional consequences of classical transcendence theory. Its main tool is the Siegel–Shidlovskii theorem, not an inference from statistical identifiability. The consequences include the transcendence of $\log \mathrm{Ein}(1)$, a simultaneous statement for logarithmic moments of truncated exponential laws, and transcendence of an extremum location. We make no priority claim for these consequences and no claim to resolve the arithmetic nature of Euler's constant or the classical elementary constants discussed below.

Define the entire exponential integral by

$$\mathrm{Ein}(z) = \int_0^z \frac{1 - e^{-t}}{t}\, dt = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n\, n!},$$

and let

$$\mathcal{F}_{\exp} = \overline{\mathbb{Q}}(e^b : b \in \overline{\mathbb{Q}}).$$

This is a field generated by exponentials of algebraic numbers; it is not a field closed under taking exponentials of all its own elements.

Theorem 5.1 (A simultaneous classical-theory consequence)**.** *For every finite collection of distinct nonzero algebraic numbers $a_1, \dots, a_s$, the numbers*

$$\mathrm{Ein}(a_1), \dots, \mathrm{Ein}(a_s)$$

are algebraically independent over $\mathcal{F}_{\exp}$. In particular, each is transcendental over that entire field, not only over $\overline{\mathbb{Q}}(e)$.

The arithmetic properties of exponential-integral and polyexponential values are classical applications of E-function theory; see [15, 2]. Related probabilistic interpretations and higher moment constants also occur in [16]. We give the finite-field reduction and differential argument explicitly because independence from every finite collection of algebraic exponentials is what the logarithmic consequence needs. Our literature check does not certify novelty of this formulation.

5.1 The differential and arithmetic steps

We use the following classical antiderivative criterion [13, 14]. If $K \subset L$ are differential fields of characteristic zero with the same algebraically closed constant field C, and $f_1', \dots, f_s' \in K$, then algebraic dependence of the f_j over K implies

$$\sum_{j=1}^{s} c_j f_j \in K$$

for some $c_j \in C$ not all zero. Here the fields will be subfields of meromorphic functions on $\mathbb{C}$, containing all complex constants, so the common constant field is exactly $\mathbb{C}$.

Lemma 5.2 (The formal simple-pole obstruction)**.** *Let $b_1, \dots, b_r \in \mathbb{C}$ be linearly independent over $\mathbb{Q}$, and put $K = \mathbb{C}(z, e^{b_1 z}, \dots, e^{b_r z})$. Suppose that distinct nonzero $a_1, \dots, a_s$ are integer linear combinations of the b_i. Then $\mathrm{Ein}(a_1 z), \dots, \mathrm{Ein}(a_s z)$ are algebraically independent over K.*

Proof. The functions $w_i = e^{b_i z}$ are algebraically independent over $\mathbb{C}(z)$. Indeed, a polynomial relation becomes a finite exponential sum $\sum_\lambda p_\lambda(z)e^{\lambda z} = 0$ with distinct λ after clearing rational denominators. On a ray on which the real parts of all λz are distinct, its largest exponential cannot be canceled by the others with polynomial coefficients. Thus K is the rational function field $\mathbb{C}(w_1, \dots, w_r)(z)$, with derivation

$$D = \frac{\partial}{\partial z} + \sum_{i=1}^{r} b_i w_i \frac{\partial}{\partial w_i}.$$

No rational function $R \in K$ has derivative $H(w)/z$ for a nonzero $H \in \mathbb{C}(w_1, \dots, w_r)$. To prove this, take the formal Laurent expansion of R in z with coefficient field $\mathbb{C}(w_1, \dots, w_r)$. If R has a pole of order $m \geq 1$, its derivative has a pole of order $m + 1$: the derivative of the leading $A(w)z^{-m}$ contributes $-mA(w)z^{-m-1}$, while the w-derivations have pole order at most m. If R is regular at this formal divisor, then so is DR. A simple pole is impossible in either case. This is a valuation in the abstract rational function field. It does not treat the analytic values $w_i(0) = 1$ as independent coordinates at a point, and it does not contradict analyticity of Ein at the complex number zero.

Write $f_j(z) = \mathrm{Ein}(a_j z)$. We have

$$f_j'(z) = \frac{1 - e^{-a_j z}}{z}.$$

A nonzero constant linear combination of these derivatives is $H(w)/z$ with $H \neq 0$: the distinct nonzero a_j yield distinct nonconstant Laurent monomials $e^{-a_j z}$, which cannot cancel each other or the constant monomial. The simple-pole obstruction shows that no such combination has a primitive in K. The antiderivative criterion therefore proves the lemma. □

Proof of Theorem 5.1. A putative polynomial relation over $\mathcal{F}_{\exp}$ involves only finitely many exponentials $e^{c_1}, \dots, e^{c_u}$ with algebraic c_i. Take a $\mathbb{Q}$-basis for the span of the c_i and a_j, and divide its elements by a common positive integer so that all c_i, a_j are integer linear combinations of the resulting basis $b_1, \dots, b_r$.

Apply the Siegel–Shidlovskii transcendence-degree theorem in the form of [2, Theorem 1.1] to the vector consisting of

$$1, \quad e^{b_i z}\ (1 \leq i \leq r), \quad e^{-a_j z},\ \mathrm{Ein}(a_j z)\ (1 \leq j \leq s).$$

These are E-functions over a fixed number field. For the integral functions the coefficients in exponential-series notation are $(-1)^{n-1}a_j^n/n$. Their conjugates and common denominators grow at most exponentially: one uses a fixed denominator for a_j and $\mathrm{lcm}(1, \dots, n) \leq C^n$. The displayed vector satisfies a homogeneous first-order linear differential system over $\overline{\mathbb{Q}}(z)$, since the derivatives of exponentials are constant multiples of themselves and $f_j' = (1 - e^{-a_j z})/z$. Its only possible finite singularity is zero. The inclusion of 1 makes the system homogeneous.

Lemma 5.2 shows that the generated function field has transcendence degree $r + s$ over $\mathbb{C}(z)$, and hence also over $\overline{\mathbb{Q}}(z)$. The additional exponentials $e^{-a_j z}$ are Laurent monomials in the $e^{b_i z}$ and do not increase this degree. The specialization theorem at the nonsingular algebraic point $z = 1$ therefore gives

$$\mathrm{trdeg}_{\overline{\mathbb{Q}}}\, \overline{\mathbb{Q}}(e^{b_1}, \dots, e^{b_r}, \mathrm{Ein}(a_1), \dots, \mathrm{Ein}(a_s)) = r + s.$$

There are exactly $r + s$ displayed generators. They are consequently algebraically independent. This rules out the proposed relation, because its coefficients lie in $\overline{\mathbb{Q}}(e^{b_1}, \dots, e^{b_r})$. □

5.2 Logarithms and logarithmic statistical moments

Corollary 5.3 (A logarithmic transference consequence)**.** *Let $a_1, \dots, a_s$ be distinct nonzero algebraic numbers and choose any complex logarithm L_j of $\mathrm{Ein}(a_j)$. For every nonzero vector $(q_1, \dots, q_s) \in \mathbb{Q}^s$,*

$$\sum_{j=1}^{s} q_j L_j$$

is transcendental. In particular, the real number $\log \mathrm{Ein}(1)$ is transcendental.

Proof. Theorem 5.1 ensures the values are nonzero. If the sum were an algebraic number β, clear the denominators of the q_j to obtain integers n_j, not all zero, and a positive integer d with

$$\prod_j \mathrm{Ein}(a_j)^{n_j} = e^{d\beta} \in \mathcal{F}_{\exp}.$$

Multiplying out negative powers gives a nonzero polynomial relation over $\mathcal{F}_{\exp}$, a contradiction. For $a = 1$ the defining integral is positive, so its real logarithm is defined. □

This proves transcendence of every nonzero rational linear combination; it does not assert algebraic independence of the logarithms or their linear independence over all algebraic coefficients.

Corollary 5.4 (Logarithmic moments of truncated exponential laws)**.** *For $a > 0$, let X_a have density*

$$p_a(x) = \frac{ae^{-ax}}{1 - e^{-a}}, \qquad 0 < x < 1,$$

and put $m(a) = \mathbb{E}[-\log X_a]$. If $a_1, \dots, a_s$ are distinct positive algebraic numbers, then $m(a_1), \dots, m(a_s)$ are algebraically independent over $\mathcal{F}_{\exp}$. Every nonzero rational linear combination of their real logarithms is transcendental as well.

Proof. Integration by parts, with primitive $(1 - e^{-ax})/a$ for e^{-ax}, gives

$$\int_0^1 (-\log x) e^{-ax}\, dx = \frac{1}{a} \int_0^1 \frac{1 - e^{-ax}}{x}\, dx = \frac{\mathrm{Ein}(a)}{a}.$$

The boundary term at zero vanishes because $x \log x \to 0$. Consequently

$$m(a) = \frac{\mathrm{Ein}(a)}{1 - e^{-a}}.$$

The denominators are nonzero elements of $\mathcal{F}_{\exp}$, so Theorem 5.1 proves the independence claim. The same integer-power argument as in Corollary 5.3 proves the logarithmic statement. □

This is a transfer to a precisely specified statistical functional, not a statement that ordinary exponential-family identifiability proves transcendence. The arithmetic input is the E-function theorem.

5.3 A transcendental extremum location

Let N_x be Poisson with mean $x > 0$, and conditional on N_x take that many independent exponential random variables of mean one. Let M_x be their maximum, with $M_x = 0$ when $N_x = 0$. For $y \geq 0$,

$$\mathbb{P}(M_x \leq y) = \exp(-xe^{-y}), \qquad \mathbb{E}M_x = \mathrm{Ein}(x).$$

The second identity follows by integrating the survival function and substituting $t = xe^{-y}$. This connects Ein with a Poisson sample-size extreme-value statistic rather than introducing an unrelated named constant.

Proposition 5.5 (Unique power-cost optimum)**.** *For every algebraic $0 < \theta < 1$, the function*

$$x \longmapsto \frac{\mathbb{E}M_x}{x^\theta} = \frac{\mathrm{Ein}(x)}{x^\theta}$$

has a unique maximizer $x_\theta \in (0, \infty)$, and x_θ is transcendental. It is characterized by

$$1 - e^{-x_\theta} = \theta \,\mathrm{Ein}(x_\theta). \tag{5.1}$$

Proof. Write $f(x) = 1 - e^{-x}$, $A(x) = \mathrm{Ein}(x)$, and $h(x) = x/(e^x - 1)$. The function h strictly decreases from one to zero. Moreover

$$\frac{f(x)}{A(x)} = \frac{\int_0^x h(t)f(t)\,dt/t}{\int_0^x f(t)\,dt/t} > h(x).$$

For $\eta(x) = f(x)/A(x)$, differentiation therefore gives

$$\eta'(x) = \frac{\eta(x)}{x}\{h(x) - \eta(x)\} < 0.$$

Since $A(x) \sim x$ at zero and $A(x) \to \infty$ at infinity, η decreases continuously from one to zero. The derivative of $x^{-\theta}A(x)$ has the sign of $\eta(x) - \theta$, proving a unique maximum and (5.1). The endpoint limits are zero: at zero use $A(x) \sim x$, and at infinity use $A(x) \le 1 + \log x$ for $x \ge 1$.

If x_θ were algebraic, (5.1) would put $\mathrm{Ein}(x_\theta)$ in $\mathcal{F}_{\exp}$, contradicting Theorem 5.1. No arithmetic assertion about the maximal objective value itself is implied by this argument. □

5.4 The barrier at the classical elementary constants

The E-function theorem specializes at algebraic points and requires arithmetic control of Taylor coefficients. Neither hypothesis can be replaced by observable injectivity or by functional independence alone. For instance

$$F(z) = e^z - e, \qquad H(z) = e^{\sqrt{2}z} - e^{\sqrt{2}}$$

are algebraically independent over $\mathbb{C}(z)$, by independence of the two exponentials, but both vanish at the algebraic point $z = 1$. Their constant Taylor coefficients are not algebraic, so the arithmetic specialization theorem does not apply. Likewise, the observable family $f_c(z) = e^z + c$ identifies an arbitrary complex parameter c exactly from $f_c(0)$, regardless of whether c is rational, algebraic irrational, or transcendental.

For the classical targets $e + \pi$, $e\pi$, e^e, and π^π, no unconditional irrationality proof is obtained here. These are among the standard open arithmetic targets discussed in [17]. The product $e\pi$ must not be confused with e^π, whose transcendence is classical. One unconditional observation is that $e + \pi$ and $e\pi$ cannot both be algebraic: otherwise e and π would be roots of a quadratic polynomial with algebraic coefficients.

For precision, the usual conditional implications are as follows. Schanuel's conjecture asserts

$$\mathrm{trdeg}_{\mathbb{Q}}\, \mathbb{Q}(z_1, \ldots, z_n, e^{z_1}, \ldots, e^{z_n}) \ge n$$

when the z_i are linearly independent over $\mathbb{Q}$. Applying it to $1, i\pi$ would make e, π algebraically independent, and hence also make $e + \pi, e\pi$ algebraically independent, since $\mathbb{Q}(e, \pi)$ is algebraic

over $\mathbb{Q}(e+\pi, e\pi)$. Applying it to $1, e$ would make e, e^e algebraically independent. Finally $i\pi, \log\pi, \pi\log\pi$ are $\mathbb{Q}$-linearly independent: imaginary parts eliminate the first coefficient, and irrationality of π eliminates the other two. Schanuel's conjecture would therefore make $\pi, \log\pi, \pi^\pi$ algebraically independent. These are conditional deductions, not proofs of the conjecture or of the open individual targets. No result above controls E-function values at arbitrary transcendental arguments, and no such extension is assumed.

6 Arithmetic kernels for Euler, Gompertz, Catalan, and zeta constants

This contribution tests four named arithmetic targets: the irrationality of Euler's constant γ, the irrationality of the Euler–Gompertz constant δ, the irrationality of Catalan's constant G, and the irrationality of $\zeta(5)$. The positive result concerns a related, explicitly specified family of exponential integrals. It is a consequence of classical E-function theory and does not identify δ itself as transcendental. The negative results below are exact statements about specified approximation constructions; they do not assert that all approximation methods fail.

6.1 What the arithmetic transfer must preserve

Put

$$\gamma = -\int_0^\infty e^{-t}\log t\,dt, \qquad \delta_x = \int_0^\infty \frac{e^{-t}}{t+x}\,dt \quad (x>0), \qquad \delta = \delta_1.$$

For integers $s \geq 2$, write

$$H_n^{(s)} = \sum_{j=1}^n j^{-s}, \quad H_n = \sum_{j=1}^n j^{-1}, \quad D_n = \operatorname{lcm}(1,\dots,n), \quad H_0^{(s)} = H_0 = 0.$$

A recoverable integral is not automatically an arithmetic certificate. The elementary certificate needed here is the following: if $A_n, B_n \in \mathbb{Z}$ and $0 < |B_n\xi - A_n| \longrightarrow 0$, then ξ is irrational. Indeed, $\xi = a/b$ would make every nonzero form have absolute value at least $1/b$. Consequently, a construction must retain a nonzero coefficient of the intended constant, prove that the form is nonzero, and control its value *after* clearing denominators. These three requirements are separate from the injectivity and stability of an observation transform.

The primary references provide a useful calibration. Van Assche and Wolfs construct improved approximants to both γ and δ, but their published denominator/error analysis still does not establish either irrationality [18]. Rivoal proves disjunctive transcendence for γ and δ [15]. The relation $\gamma + \delta/e = \mathrm{Ein}(1)$ and its cancellation mechanism have also been studied explicitly by Powers [19]. None of those facts is counted here as a new individual-constant result.

6.2 A positive transfer: a geometric family has at most two exceptions

Define

$$\mathrm{Ein}(z) = \int_0^z \frac{1-e^{-u}}{u}\,du = \sum_{n\geq 1}\frac{(-1)^{n+1}z^n}{n\,n!}, \qquad \mathcal{E} = \overline{\mathbb{Q}}(e^a : a \in \overline{\mathbb{Q}}).$$

The multi-point E-function theorem proved in Theorem 5.1 states that distinct nonzero algebraic arguments $a_1,\dots,a_m$ give values $\mathrm{Ein}(a_1),\dots,\mathrm{Ein}(a_m)$ algebraically independent over $\mathcal{E}$. Its external inputs are the Kolchin–Ostrowski theorem and the Siegel–Shidlovskii specialization theorem, in the form recorded by Beukers [2, 14]. The argument below is a transfer of that result, not another proof or another count of the same independence theorem.

Lemma 6.1 (The exact boundary split)**.** *For $x > 0$,*

$$\mathrm{Ein}(x) = \gamma + \log x + e^{-x}\delta_x. \tag{6.1}$$

Proof. Split the integral of $e^{-t}\log t$ at x. Integration by parts, using $1 - e^{-t}$ on the first interval, gives

$$\int_0^x e^{-t}\log t\,dt = (1 - e^{-x})\log x - \mathrm{Ein}(x).$$

The endpoint at zero vanishes because $(1 - e^{-t})\log t \to 0$. On the second interval,

$$\int_x^\infty e^{-t}\log t\,dt = e^{-x}\log x + \int_x^\infty \frac{e^{-t}}{t}\,dt = e^{-x}\log x + e^{-x}\delta_x.$$

Adding and using the definition of γ proves the identity. □

Theorem 6.2 (A finite defect in a geometric family)**.** *Fix an integer $b \geq 2$. If $j_1, \ldots, j_m$ are distinct nonnegative integers, then*

$$\mathrm{trdeg}_{\mathcal{E}}\,\mathcal{E}(\delta_{b^{j_1}}, \ldots, \delta_{b^{j_m}}) \geq m - 2. \tag{6.2}$$

Consequently, at most two members of $\{\delta_{b^j} : j \geq 0\}$ are algebraic over $\mathcal{E}$, and in particular at most two are algebraic over $\mathbb{Q}$. There is a subset of at most two indices whose removal leaves a family algebraically independent over $\mathcal{E}$.

Proof. Write $K = \mathcal{E}(\delta_{b^{j_1}}, \ldots, \delta_{b^{j_m}})$. Since $e^{-b^{j_i}} \in \mathcal{E}$ and $\log(b^{j_i}) = j_i \log b$, Lemma 6.1 implies

$$\mathcal{E}\big(\mathrm{Ein}(b^{j_1}), \ldots, \mathrm{Ein}(b^{j_m})\big) \subset K(\gamma, \log b).$$

The field on the left has transcendence degree m over $\mathcal{E}$ by Theorem 5.1. Adjoining two elements raises transcendence degree by at most two. This proves (6.2). If three members were algebraic over $\mathcal{E}$, apply the bound to those three.

For the last assertion, choose a maximal algebraically independent subfamily B of the countable family, over $\mathcal{E}$. Every omitted member is algebraic over $\mathcal{E}(B)$ and hence over a field generated by a finite subset of B. If three members were omitted, take the union B_0 of three finite witness sets. The finite family consisting of B_0 and those three members would have transcendence degree $|B_0|$, contrary to (6.2), which requires at least $|B_0| + 1$. □

For example, the twelve explicit integrals

$$\delta_1, \delta_2, \delta_4, \delta_8, \delta_{16}, \delta_{32}, \delta_{64}, \delta_{128}, \delta_{256}, \delta_{512}, \delta_{1024}, \delta_{2048} \tag{6.3}$$

contain at least ten algebraically independent numbers. The theorem neither specifies the two possible exceptional indices nor proves that the first member δ_1 is irrational. In particular, twelve entries in (6.3) are not twelve individually resolved conjectures.

Provenance and exact gain. This is a self-posed family transfer from the multi-point E-function input. The one-point split and the associated disjunctive results are classical [15]. The source search did not establish that the particular geometric-family formulation is new; no priority claim is made. The substantive gain over a single disjunction is the uniform deficit of at most two, for every finite subset, including the infinite-family consequence. The two lost dimensions are precisely the common terms γ and $\log b$; observing arbitrarily many scales does not assign those dimensions to individual members.

6.3 Target δ: an exact optimization over all integer square kernels

The integral representation makes positive polynomial weights a natural first transfer from positive observation kernels. Here that entire class can be optimized exactly.

Lemma 6.3 (A Laguerre bound with an arithmetic consequence)**.** *Let $P \in \mathbb{R}[t]$ have degree n and leading coefficient $a_n \neq 0$. For every $x > 0$,*

$$I_x(P) := \int_0^\infty \frac{e^{-t}P(t)^2}{t+x}\,dt \geq \frac{a_n^2(n!)^2}{x+2n+1}. \tag{6.4}$$

If x is a positive integer and $P \in \mathbb{Z}[t]$, then

$$I_x(P) = P(-x)^2\delta_x + A_x(P), \qquad A_x(P) \in \mathbb{Z}. \tag{6.5}$$

Proof. Let $\pi_n(t) = (-1)^n n! L_n(t)$ be the monic Laguerre polynomial, where

$$L_n(t) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{t^k}{k!}.$$

Rodrigues' formula and n integrations by parts give orthogonality to all polynomials of degree below n in $L^2(e^{-t}dt)$ and

$$\int_0^\infty \pi_n(t)^2 e^{-t}dt = (n!)^2.$$

For completeness, Rodrigues' formula is $e^{-t}\pi_n(t) = (-1)^n(d/dt)^n(e^{-t}t^n)$; the boundary terms vanish, and $\pi_n^{(n)} = n!$ gives the displayed norm. The coefficient of t^{n-1} in π_n is $-n^2$. Comparing the two leading coefficients of $t\pi_n$ and π_{n+1}, and then using orthogonality, yields

$$\int_0^\infty t\pi_n(t)^2 e^{-t}dt = (2n+1)(n!)^2.$$

Also $\int P\pi_n e^{-t}dt = a_n(n!)^2$. Apply Cauchy–Schwarz to $P/\sqrt{t+x}$ and $\sqrt{t+x}\pi_n$ in $L^2(e^{-t}dt)$:

$$a_n^2(n!)^4 \leq I_x(P)(x+2n+1)(n!)^2.$$

This proves (6.4). For the arithmetic assertion, division by the monic polynomial $t+x$ gives $P(t)^2 = P(-x)^2 + (t+x)Q(t)$ with $Q \in \mathbb{Z}[t]$. Every moment $\int_0^\infty t^k e^{-t}dt = k!$ is an integer. □

Theorem 6.4 (The exact positive-square obstruction for δ)**.** *Over all nonzero integer polynomials of arbitrary degree,*

$$\inf_{0 \neq P \in \mathbb{Z}[t]} I_1(P) = 4\delta - 2 > 0. \tag{6.6}$$

Equality holds exactly for $P(t) = t-1$ and $P(t) = 1-t$. Thus no sequence of these integral integer linear forms can tend to zero.

Proof. Polynomial division and factorial moments give

$$I_1(t-1) = 4\delta - 2 > 0, \qquad \int_0^\infty \frac{e^{-t}t(t-2)^2}{1+t}\,dt = 6 - 9\delta > 0.$$

Both integrands are positive off a finite set, so $1/2 < \delta < 2/3$ and $0 < 4\delta - 2 < 2/3$. If $\deg P = n \geq 2$, Lemma 6.3 gives $I_1(P) \geq (n!)^2/(2n+2) \geq 2/3$; the last expression is increasing for $n \geq 2$. A nonzero constant integer polynomial gives $I_1(P) \geq \delta$, which exceeds $4\delta - 2$ because $\delta < 2/3$.

It remains to consider $P(t) = at + c$, $a, c \in \mathbb{Z}$, $a \neq 0$. If $|a| \geq 2$, the same bound gives $I_1(P) \geq a^2/4 \geq 1$. Changing the sign of P reduces the remaining case to $a = 1$. Direct integration gives

$$I_1(t + c) = 2c + (c - 1)^2\delta, \quad I_1(t + c) - I_1(t - 1) = (c + 1)\big(2 + (c - 3)\delta\big).$$

For $c \geq 0$ both factors are positive, using $\delta < 2/3$ when $c = 0$. For $c \leq -2$ both factors are negative, using $\delta > 1/2$. Equality therefore occurs only at $c = -1$. □

The result is an exact solution of a restricted, self-posed optimization problem. It is not a lower bound for arbitrary integer forms $q\delta - p$. The restriction matters: signed kernels, rational kernels, and Hermite–Padé constructions are not covered. For a rational polynomial $P = Q/d$, the integer form obtained by the natural square-denominator clearance is $d^2 I_1(P) = I_1(Q) \geq 4\delta - 2$; merely normalizing the kernel to have a small integral cannot evade the obstruction.

6.4 Target γ: a vanishing observation can erase the constant

Proposition 6.5 (The exact coefficient of γ)**.** *If $P(t) = \sum_{k=0}^{d} a_k t^k \in \mathbb{Z}[t]$, then*

$$\int_0^\infty e^{-t} P(t) \log t \, dt = A(P) - B(P)\gamma, \quad A(P) = \sum_{k=0}^{d} a_k k! H_k \in \mathbb{Z}, \quad B(P) = \sum_{k=0}^{d} a_k k! \in \mathbb{Z}. \tag{6.7}$$

In particular, for $n \geq 1$,

$$\int_0^\infty e^{-t} L_n(t) \, dt = 0, \qquad \int_0^\infty e^{-t} L_n(t) \log t \, dt = -\frac{1}{n}. \tag{6.8}$$

The small second integral has zero coefficient of γ. After multiplying L_n by $n!$ to obtain an integer polynomial, that integral equals $-(n-1)!$.

Proof. Differentiating $\Gamma(u + 1) = u\Gamma(u)$ successively at nonnegative integers, starting with $\Gamma'(1) = -\gamma$, gives $\Gamma'(k + 1) = k!(H_k - \gamma)$. Differentiation under the integral is justified by exponential decay at infinity and integrability of $t^k|\log t|$ at zero. Since $k!/j$ is an integer for $1 \leq j \leq k$, (6.7) follows. The coefficient $B(L_n)$ is $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$. Finally,

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} H_k = -\int_0^1 \frac{(1-u)^n}{1-u} \, du = -\frac{1}{n},$$

where $H_k = \int_0^1 (1 - u^k)/(1 - u) \, du$. This proves all assertions. □

This calculation identifies two failures of this particular attempt: Laguerre orthogonality removes the target coefficient, and the natural integer normalization reverses the apparent decay. A successful replacement must keep $B(P_n) \neq 0$ and make $0 < |A(P_n) - B(P_n)\gamma| \to 0$ simultaneously. The observation-theoretic statement that a family of transforms determines the exponential law does not provide either inequality.

6.5 Why adding finitely many scales does not create a canonical convergent series for γ

The following finite-scale version of the canonical cancellation test is another restricted obstruction. Its one-scale case is already present in Powers' analysis [19]; the proof here uses the locations and residues of the Borel singularities.

For a formal factorial series $\sum_{n\geq 0} a_n$, use the Borel convention $\widehat{A}(t) = \sum_{n\geq 0} a_n t^n/n!$. Define the two canonical Borel transforms

$$\widehat{\gamma}(t) = \frac{e^t/(1+t) - 1}{t}, \qquad \widehat{\delta}_x(t) = \frac{1}{x+t}.$$

The apparent singularity of $\widehat{\gamma}$ at zero is removable. Their Laplace integrals are γ and δ_x:

$$\int_0^\infty e^{-t}\widehat{\gamma}(t)dt = \int_0^\infty \frac{(1+t)^{-1} - e^{-t}}{t}\, dt = \gamma.$$

For example, the last identity follows by applying the logarithm identity $\log y = \int_0^\infty (e^{-t} - e^{-yt})dt/t$ to an exponential random variable y and taking expectation. Truncation at both endpoints, followed by dominated convergence, justifies the operation; the resulting integrand is bounded at zero and $O(t^{-2})$ at infinity.

Proposition 6.6 (Finite canonical pole cancellation)**.** *Let $x_1, \ldots, x_m > 0$ be distinct. The function*

$$c\widehat{\gamma}(t) + \sum_{j=1}^{m} c_j \widehat{\delta}_{x_j}(t)$$

is entire if and only if every coefficient with $x_j \neq 1$ vanishes and, if $x_j = 1$ is present, its coefficient equals c/e; if 1 is absent one must also have $c = 0$. Hence no nonzero linear combination with algebraic coefficients $c, c_1, \ldots, c_m$ cancels all the canonical poles. For such a nonzero algebraic combination, the associated factorial series fails even the necessary term-to-zero condition for ordinary convergence.

Proof. At $t = -1$, the residue of $\widehat{\gamma}$ is $-e^{-1}$. The residue of $\widehat{\delta}_{x_j}$ at $-x_j$ is 1. All the poles are simple, and their locations are distinct except for the possible match $x_j = 1$. This proves the stated cancellation conditions. After cancellation, the remaining formula has only removable finite singularities and is entire. If all coefficients are algebraic and $c \neq 0$, the required equality $c_j = c/e$ is impossible by the transcendence of e; if $c = 0$, distinct poles force every $c_j = 0$.

To prove the final statement, subtract the principal parts of the finitely many uncancelled poles. The remainder is entire. The uncancelled pole nearest zero is unique, since all x_j and 1 are positive. If its location is $-r$ and its nonzero residue is C, the Taylor coefficient b_n of the Borel transform satisfies $b_n = C(-1)^n r^{-n-1} + o(r^{-n})$ by Cauchy's estimate on a larger circle containing no other uncancelled pole. Thus the factorial-series term $n!b_n$ does not tend to zero. □

This theorem concerns the specified canonical representations. Divergence of such a representation proves no arithmetic classification of its Borel sum. In particular, it cannot upgrade the known transcendence of $\gamma + \delta/e$ to the transcendence of γ or δ individually. It does show exactly why algebraic finite-scale coefficients do not replicate the exceptional cancellation coefficient $1/e$.

6.6 Target G: localization versus the denominator ledger

The exact integral

$$G = \sum_{k\geq 0} \frac{(-1)^k}{(2k+1)^2} = \int_0^1 \frac{-\log x}{1+x^2}\, dx$$

follows by dominated convergence applied to geometric partial sums. For $P \in \mathbb{Z}[z]$, polynomial division gives

$$\int_0^1 \frac{-\log x}{1+x^2} P(x^2)\, dx = P(-1)G + \sum_{k=0}^{d-1} \frac{q_k}{(2k+1)^2}, \qquad q_k \in \mathbb{Z}, \quad d = \deg P. \tag{6.9}$$

Thus the coefficient of G is an evaluation outside the observation interval, at $z = -1$.

Proposition 6.7 (A complete ledger for beta localization)**.** *Fix positive integers a, b. Set $d_n = (a+b)n$ and*

$$P_n(z) = z^{an}(1-z)^{bn}, \quad I_n = \int_0^1 \frac{-\log x}{1+x^2} P_n(x^2)dx, \quad O_d = \operatorname{lcm}(1, 3, \ldots, 2d-1).$$

Then $O_{d_n}^2 I_n \in \mathbb{Z}G + \mathbb{Z}$, its coefficient of G is $O_{d_n}^2(-1)^{an}2^{bn} \neq 0$, and

$$\lim_{n\to\infty} \frac{1}{n} \log(O_{d_n}^2 I_n) = 4(a+b) + a\log a + b\log b - (a+b)\log(a+b) > 0. \tag{6.10}$$

Consequently this specified denominator clearance produces positive forms tending to infinity, not forms tending to zero.

Proof. The arithmetic assertions follow from (6.9) and $P_n(-1) = (-1)^{an}2^{bn}$. Substitute $u = x^2$:

$$I_n = \int_0^1 w(u)e^{n\phi(u)}du, \quad w(u) = \frac{-\log u}{4\sqrt{u}(1+u)}, \quad \phi(u) = a\log u + b\log(1-u).$$

The unique maximizer is $u_0 = a/(a+b)$, with $-\phi''(u_0) = (a+b)^3/(ab)$. Laplace's method, whose hypotheses hold because w is positive and smooth near this interior maximizer and the endpoints are exponentially negligible, gives

$$I_n \sim w(u_0)\sqrt{\frac{2\pi ab}{(a+b)^3 n}} \left(\frac{a^a b^b}{(a+b)^{a+b}}\right)^n.$$

To check endpoint negligibility directly, choose small $\varepsilon > 0$ so that $u^a(1-u)^b$ is uniformly below its maximum outside $(\varepsilon, 1-\varepsilon)$; the integrable factor w then gives an exponentially smaller bound. The usual Taylor expansion and a Gaussian dominating bound give the displayed interior asymptotic.

The prime number theorem in the form $\log D_N = N + o(N)$ implies $\log O_d = 2d + o(d)$, since removing the highest power of 2 changes the logarithm by only $O(\log d)$. Combining the two limits proves the equality in (6.10). Finally, with $\theta = a/(a+b)$,

$$a\log a + b\log b - (a+b)\log(a+b) = (a+b)\{\theta\log\theta + (1-\theta)\log(1-\theta)\} \geq -(a+b)\log 2,$$

which proves strict positivity. □

The proposition does not rule out cancellations that substantially reduce the actual common denominator of the rational part. It identifies the missing requirement precisely: a smaller denominator must be proved, prime by prime, rather than inferred from the analytic decay of I_n. This distinction is especially relevant to the contemporary Catalan claim: Sun's September 2026 preprint asserts irrationality, while Wachs' subsequent primary-source analysis reports an incomplete denominator ledger, including a missing contribution of the prime 2 [20, 21]. We use neither that claim nor its negation as a theorem about G; the result established here is Proposition 6.7.

6.7 Target $\zeta(5)$: the boundary that carries the constant

Proposition 6.8 (The endpoint coefficient)**.** *For $s \geq 2$ and $P(x) = \sum_{k=0}^d a_k x^k \in \mathbb{Z}[x]$,*

$$\frac{1}{(s-1)!}\int_0^1 \frac{(-\log x)^{s-1}P(x)}{1-x}\,dx = P(1)\zeta(s) - \sum_{k=0}^d a_k H_k^{(s)}. \tag{6.11}$$

The rational part has denominator dividing D_d^s. In particular, the positive localization $P_n(x) = x^n(1-x)^n$ removes the coefficient of $\zeta(5)$ exactly.

Proof. For a monomial, expand $(1-x)^{-1}$ and use Tonelli's theorem to obtain

$$\frac{1}{(s-1)!}\int_0^1 \frac{x^k(-\log x)^{s-1}}{1-x}dx = \sum_{j\geq 0}\frac{1}{(k+j+1)^s} = \zeta(s) - H_k^{(s)}.$$

Linearity proves the formula for an arbitrary polynomial, and every integer $1 \leq j \leq d$ divides D_d. Finally $P_n(1) = 0$. □

For example, at $n = 1$, the supposedly small $\zeta(5)$ form is simply

$$\frac{1}{24}\int_0^1 x(-\log x)^4 dx = \frac{1}{32};$$

there is no $\zeta(5)$ coefficient. Keeping the endpoint by taking $P_n(x) = x^n$ instead gives

$$\zeta(5) - H_n^{(5)} \sim \frac{1}{4n^4}, \qquad D_n^5(\zeta(5) - H_n^{(5)}) \longrightarrow \infty.$$

The first asymptotic follows by integral comparison and the second from $\log D_n = n + o(n)$. Thus both natural variants are completely accounted for: one erases the target and the other loses to denominator growth. Brown and Zudilin's cellular approximants are a genuinely different and far stronger construction [22]; these elementary calculations do not claim to improve them. No proof of the individual irrationality of $\zeta(5)$ is obtained in this contribution.

6.8 The surviving research requirements

The four target attempts leave different, explicit requirements. For γ, the logarithmic observation must retain its factorial-moment coefficient while producing a small integer form. For δ, any new positive-kernel construction must leave the integer-polynomial-square class optimized in Theorem 6.4, or introduce a quantitatively justified arithmetic normalization outside it. For G, the beta-localization forms require a much smaller denominator than the stated least-common-multiple clearance. For $\zeta(5)$, boundary localization must preserve a nonzero endpoint coefficient and simultaneously beat its denominator growth. The family result (6.2) succeeds for a different reason: it uses algebraic independence of E-function values and explicitly pays for the two shared nuisance constants. It supplies a proved family statement without assigning unproved arithmetic properties to any one of the four named constants.

7 Arithmetic height after exact observation: three constant targets

The three transcendence targets considered here are the transcendence of

$$P_2 = \sum_{p\ mprime} 2^{-p}, \qquad L_{\mathbb{P}} = \sum_{p\ mprime}\frac{1}{2^p-1}, \qquad E = \sum_{n\geq 1}\frac{1}{2^n-1}.$$

All sums converge absolutely. None of their transcendence questions is settled in this section. In particular, irrationality of $L_{\mathbb{P}}$ is not a new target: it was established by Tao and Teräväinen [24, Theorem 1.3]. Irrationality of E is classical [28, 29], and irrationality of P_2 has an elementary proof below. Our contribution to this investigation is a rigorous audit of two proposed transfers from exact observation to arithmetic: height-controlled rational truncation and finite Mahler systems. The first has sharply different costs for these constants; the second is impossible for the natural generating functions of P_2 and E. These are obstruction and calibration

results, not new solutions of historical transcendence problems. The no-Mahler conclusions are corollaries of existing structural theorems, with proofs supplied here.

For calibration, put

$$K = \sum_{j\geq 0} 2^{-2^j}, \qquad F = \sum_{j\geq 0} \frac{1}{2^{2^j}+1}.$$

Kempner's number K is already transcendental [26]. The reciprocal Fermat constant F is also already transcendental, and Coons proved the stronger sharp statement $\mu(F) = 2$ [25, Theorem 1]. Thus neither is counted as a newly resolved constant. We use them to identify exactly which arithmetic information a fast observation error does, and does not, provide.

7.1 Reduced height and the rational-grid correction

For a reduced rational approximation A_N/Q_N to a real number ξ, with $Q_N \to \infty$, define its approximation exponent along this particular sequence by

$$\beta_N(\xi; A_N/Q_N) = \frac{-\log|\xi - A_N/Q_N|}{\log Q_N}.$$

This is not the irrationality exponent $\mu(\xi)$, which optimizes over all rational approximations. A limit below two therefore says nothing by itself about the value of $\mu(\xi)$.

Lemma 7.1 (Rational-grid correction)**.** *Let $A_N \in \mathbb{Z}$, $Q_N \in \mathbb{N}$, $Q_N \to \infty$, and let $c \in \mathbb{Q}$. Suppose*

$$0 < |Q_N\xi - A_N - c| \longrightarrow 0.$$

Then ξ is irrational. Moreover, if $|Q_N\xi - A_N - c| = Q_N^{-\sigma+o(1)}$ with $\sigma > 0$, the corrected rational approximations $(A_N + c)/Q_N$ have error $Q_N^{-1-\sigma+o(1)}$. If their reduced denominators q_N satisfy $\log q_N/\log Q_N \to 1$, their approximation exponents tend to $1+\sigma$.

Proof. If $\xi = a/b$ and $c = u/v$ are rational, then $Q_N\xi - A_N - c$ belongs to $(bv)^{-1}\mathbb{Z}$. Its nonzero absolute value is at least $1/(bv)$, a contradiction. The remaining assertions follow by dividing by Q_N and then taking logarithms. □

The nonzero rational center c matters. A normalized truncation error converging to one can prove irrationality after correction, even though it does not tend to zero. The following exact calculation records the height cost of this correction for the reciprocal Fermat constant.

Theorem 7.2 (One carry and an exact denominator formula)**.** *Write $f_j = 2^{2^j}+1$ and*

$$F_N = \sum_{j=0}^{N} \frac{1}{f_j} = \frac{A_N}{Q_N}, \qquad Q_N = \prod_{j=0}^{N} f_j = 2^{2^{N+1}} - 1.$$

The displayed fraction is reduced. The raw truncation and its one-unit numerator correction satisfy

$$Q_N(F - F_N) \longrightarrow 1, \qquad 0 < Q_N(F - F_N) < 1, \tag{7.1}$$

$$\frac{A_N + 1}{Q_N} - F = Q_N^{-2}(1 + o(1)). \tag{7.2}$$

Furthermore,

$$\gcd(A_N + 1, Q_N) = \prod_{\substack{0\leq i\leq N \\ 2^{i+1}\mid N-i+1}} f_i. \tag{7.3}$$

Consequently the reduced denominator q_N of $(A_N+1)/Q_N$ obeys $Q_N/2^{N+1} \le q_N \le Q_N$. The raw approximation exponent tends to one, whereas the corrected exponent tends to two. In particular, (7.1) and Lemma 7.1 give an elementary proof of the already known irrationality of F.

Proof. The product identity follows by repeatedly factoring x^2-1. The f_j are pairwise coprime: if $i<j$, then $f_j \equiv 2 \pmod{f_i}$, while f_i is odd. Modulo f_i, the numerator $A_N = \sum_{j=0}^{N} Q_N/f_j$ is Q_N/f_i, which is a unit modulo f_i. This proves reduction.

Set $x = 2^{2^{N+1}} = Q_N + 1$. The remaining tail is

$$R_N = F - F_N = \sum_{j\ge 0} \frac{1}{x^{2^j}+1}.$$

It is positive and strictly smaller than $\sum_{m\ge 1} x^{-m} = 1/(x-1)$, since 2^j are distinct positive integers and each summand is strictly smaller than x^{-2^j}. Also,

$$R_N = \frac{1}{x+1} + \frac{1}{x^2+1} + O(x^{-4}) = x^{-1} + O(x^{-3}).$$

Thus $(x-1)R_N \to 1$, and

$$\frac{1}{Q_N} - R_N = \frac{1}{x-1} - R_N = x^{-2} + O(x^{-3}) = Q_N^{-2}(1+o(1)).$$

This proves (7.1)–(7.2).

For the exact cancellation, reduction modulo f_i gives

$$\frac{Q_N}{f_i} = \left(\prod_{j<i} f_j\right)\left(\prod_{i<j\le N} f_j\right) \equiv -2\cdot 2^{N-i} = -2^{N-i+1} \pmod{f_i}.$$

(The empty first product for $i=0$ is also -2 modulo $f_0 = 3$.) Hence $A_N + 1 \equiv 1 - 2^{N-i+1} \pmod{f_i}$. For every prime divisor ℓ of f_i, the order of 2 modulo ℓ is exactly 2^{i+1}: it divides 2^{i+1}, and $2^{2^i} \equiv -1 \not\equiv 1 \pmod{\ell}$. If $2^{i+1} \nmid N-i+1$, no prime divisor of f_i divides A_N+1. If $2^{i+1} \mid N-i+1$, the congruence $2^{N-i+1} \equiv 1 \pmod{f_i}$ holds for the whole integer f_i. Pairwise coprimality proves (7.3).

Every index contributing to that product satisfies $2^{i+1} \le N+1$. With $L = \lfloor \log_2(N+1) \rfloor$, the product is at most $\prod_{i=0}^{L-1} f_i = 2^{2^L} - 1 < 2^{N+1}$ (the empty product is one). Therefore $\log q_N = \log Q_N + O(N)$ and $\log q_N / \log Q_N \to 1$. The exponent statements follow. □

This calculation supplies a denominator formula and an explicit correction mechanism; it does not replace Coons's upper bound $\mu(F) \le 2$, nor the classical transcendence theorem. Fast convergence alone had exponent one. Arithmetic control of a normalized remainder and its cancellation supplied exponent two.

7.2 Prime supports: exact denominators and a vanishing height exponent

Let $p_1 < p_2 < \cdots$ be the primes. For binary observations supported on an increasing integer sequence a_j, denominators have a different geometry from those of reciprocal Mersenne observations.

Proposition 7.3 (Two exact height laws)**.** *Let $(a_j)_{j\ge1}$ be strictly increasing positive integers and put $X=\sum_{j\ge1}2^{-a_j}$. Its Nth truncation X_N has reduced denominator 2^{a_N}, and*

$$2^{-a_{N+1}}\le X-X_N\le 2^{1-a_{N+1}},\qquad \beta_N(X;X_N)=\frac{a_{N+1}}{a_N}+O(a_N^{-1}). \tag{7.4}$$

For the prime Lambert constant, the Nth truncation

$$Y_N=\sum_{j=1}^{N}\frac{1}{2^{p_j}-1}=\frac{B_N}{D_N}$$

is reduced with

$$D_N=\prod_{j=1}^{N}(2^{p_j}-1),\qquad \beta_N(L_{\mathbb{P}};Y_N)=\frac{p_{N+1}+O(1)}{\sum_{j\le N}p_j+O(1)}. \tag{7.5}$$

By the prime number theorem, as $x=p_N\to\infty$,

$$\beta_N(P_2;\textstyle\sum_{j\le N}2^{-p_j})\longrightarrow 1, \tag{7.6}$$

$$\beta_N(L_{\mathbb{P}};Y_N)\sim\frac{2\log x}{x}\longrightarrow 0. \tag{7.7}$$

Proof. The numerator $2^{a_N}X_N$ is odd, since its final summand is one and all previous summands are even. Summing over every integer exponent at least a_{N+1} gives the tail bound, and taking logarithms proves (7.4).

The Euclidean algorithm for exponents gives $\gcd(2^u-1,2^v-1)=2^{\gcd(u,v)}-1$. Thus the denominators $2^{p_j}-1$ are pairwise coprime. Reduction of B_N modulo any one of them leaves the unit $D_N/(2^{p_j}-1)$, proving that B_N/D_N is reduced. Also,

$$\log_2 D_N=\sum_{j\le N}p_j+\sum_{j\le N}\log_2(1-2^{-p_j})=\sum_{j\le N}p_j+O(1).$$

For $m=p_{N+1}$,

$$2^{-m}<L_{\mathbb{P}}-Y_N\le\frac{1}{1-2^{-m}}\sum_{r=m}^{\infty}2^{-r}\le\frac{2^{1-m}}{1-2^{-m}}.$$

This proves (7.5). Finally, the prime number theorem gives $p_{N+1}/p_N\to1$ and, by partial summation, $\sum_{p\le x}p\sim x^2/(2\log x)$. Substitution gives the two limits. □

Corollary 7.4 (Raw truncations are not continued-fraction approximants)**.** *For each of P_2, $L_{\mathbb{P}}$, and F, the raw truncations above are eventually not continued-fraction convergents. For K, none of its raw truncations is a convergent.*

Proof. A convergent a/q to an irrational real number satisfies $|\xi-a/q|<q^{-2}$. For P_2, $L_{\mathbb{P}}$, and F, the proved limits of the raw exponents are less than two. Their irrationality is known, and for P_2 is proved in the next paragraph. For the Nth Kempner truncation, $q=2^{2^N}$ and

$$K-K_N=\sum_{j\ge1}q^{-2^j}>q^{-2}.$$

Thus the necessary inequality fails in every case. □

For completeness, the binary expansion of P_2 has digit one exactly at prime positions. It is neither finite nor eventually all ones. If it were eventually periodic with period $h \geq 1$, choose a prime p beyond the preperiod. The position $p + ph = p(1 + h)$ would have the same digit as p but would be composite, a contradiction. Hence P_2 is irrational. This elementary identification of every prime from an exact binary expansion does not establish transcendence.

Equation (7.7) is particularly restrictive for the proposed observation transfer. A truncation giving roughly x binary digits of absolute accuracy has denominator bit length asymptotic to $x^2/(2 \log x)$. The whole prime support is identifiable, yet these specific rational approximations spend an increasing number of height bits per recovered value bit. This does not rule out other rational approximants: the irrationality theorem of Tao–Teräväinen already uses additional arithmetic structure.

7.3 The precise finite-Mahler obstruction

A power series $f(z) \in \overline{\mathbb{Q}}[[z]]$ is k-Mahler if there exist polynomials $r_0, \ldots, r_s \in \overline{\mathbb{Q}}[z]$, not all zero, such that

$$\sum_{j=0}^{s} r_j(z) f(z^{k^j}) = 0, \qquad k \geq 2.$$

An inhomogeneous relation with rational-function right side also implies a homogeneous one: its successive dilates span a finite-dimensional space over $\overline{\mathbb{Q}}(z)$, including the rational function 1. We use two established results of Adamczewski, Bell, and Smertnig [23, Theorems 1.1–1.2]: a Mahler series with a finite coefficient alphabet is automatic; and if a Mahler coefficient sequence has logarithmic Weil heights $h(a_n) = o(\log n)$, then $h(a_n) = O(\log \log n)$. These are substantial external inputs.

Theorem 7.5 (Two natural generating functions cannot enter a finite linear Mahler system)**.** *For every integer $k \geq 2$, neither*

$$P(z) = \sum_{p \text{ prime}} z^p \quad \textit{nor} \quad D(z) = \sum_{n \geq 1} d(n) z^n = \sum_{n \geq 1} \frac{z^n}{1 - z^n}$$

is k-Mahler. Here $d(n)$ counts positive divisors. In particular, neither function can be a component of a finite vector $\mathbf{f}(z)$ satisfying

$$\mathbf{f}(z) = A(z)\mathbf{f}(z^k), \qquad A(z) \in \mathrm{GL}_r(\overline{\mathbb{Q}}(z)).$$

The assertion excludes finite linear dilation systems for these natural functions; it is not an assertion about arbitrary representations of the individual numbers $P(1/2)$ or $D(1/2)$.

Proof. If P were k-Mahler, its coefficients, lying in $\{0, 1\}$, would form a k-automatic sequence. Thus a finite automaton would recognize the base-k representations of primes, with no leading zeroes. Here is a complete pumping argument excluding this.

Let the automaton have s states. Choose a prime $p > k^s$ with $p > k$, and write its digit word as uvw, where v is a nonempty loop among the first s transitions. Then $uv^t w$ is accepted for every integer $t \geq 0$. Let a be the integer represented by u, let c be the integer represented by v, and let $\ell = |v|$. Appending v acts on a prefix residue modulo p by the invertible affine map

$$T(x) = k^\ell x + c \quad \text{on } \mathbb{F}_p.$$

This map has finite order, say $r \geq 1$. Therefore $T^{1+r}(a) = T(a)$ modulo p. Appending the unchanged suffix w preserves this congruence, so the integer represented by $uv^{1+r}w$ is divisible

by p. It is strictly larger than p and still has no leading zero. The automaton accepts a composite, a contradiction. The nonregularity of prime representations is classical; see [27] for historical attribution and stronger results.

For D, we first establish $\log d(n) = o(\log n)$ elementarily. Fix $\varepsilon > 0$. For primes $p \geq 2^{1/\varepsilon}$ and exponents $a \geq 1$, $a + 1 \leq 2^a \leq p^{\varepsilon a}$. For the finitely many smaller primes, each supremum $\sup_{a \geq 0}(a+1)p^{-\varepsilon a}$ is finite. Multiplication gives $d(n) \leq C_\varepsilon n^\varepsilon$, and the asserted little-oh bound follows because ε was arbitrary. On the other hand, take $n_m = \prod_{j=1}^{m} p_j$. Then $d(n_m) = 2^m$. Bertrand's postulate gives $p_j \leq 2^j$, hence

$$\log n_m \leq \tfrac{1}{2}m(m+1)\log 2, \qquad \log\log n_m = O(\log m), \qquad \log d(n_m) = m \log 2.$$

Thus $\log d(n)$ is not $O(\log\log n)$. Since the logarithmic Weil height of the positive integer $d(n)$ is precisely $\log d(n)$, the quoted height-gap theorem excludes a Mahler equation.

Finally, the displayed vector equation implies $\mathbf{f}(z^{k^j}) = B_j(z)\mathbf{f}(z)$ with invertible rational-function matrices B_j. For any chosen component, $r+1$ such scalar dilates lie in a vector space of dimension at most r over $\overline{\mathbb{Q}}(z)$. They satisfy a nonzero linear relation; clearing denominators gives a Mahler relation. The two scalar obstructions therefore exclude the vector systems as well. □

The Mahler mechanisms for the calibration constants really do have the missing finite closure. Their natural functions obey

$$\mathcal{K}(z) = z + \mathcal{K}(z^2), \qquad \mathcal{F}(z) = \frac{z}{1+z} + \mathcal{F}(z^2).$$

By contrast, the no-Mahler theorem rules out obtaining such finite linear closure for P or D merely by adding finitely many auxiliary functions. It neither proves their special values algebraic nor proves them transcendental. For $\sum \omega(n) z^n$, the height-gap argument used for $d(n)$ does not apply: $\log \omega(n) = O(\log\log n)$, which is an allowed height range. No no-Mahler assertion about that function is made here.

7.4 Infinite inversion remains possible, but does not transfer heights

The prime support has a natural exact inversion, closely analogous to recovering an entire frequency family from its observations. Put

$$\mathcal{L}(z) = \sum_p \frac{z^p}{1 - z^p}, \qquad |z| < 1.$$

Then

$$\mathcal{L}(z) = \sum_{j \geq 1} P(z^j), \qquad P(z) = \sum_{j \geq 1} \mu(j)\mathcal{L}(z^j), \tag{7.8}$$

where μ is the Möbius function. This exact infinite inversion has an explicit finite observation error.

Proposition 7.6 (A controlled infinite observation identity)**.** *For $0 < \rho < 1$, $|z| \leq \rho$, and $M \geq 1$,*

$$\left| P(z) - \sum_{j=1}^{M} \mu(j)\mathcal{L}(z^j) \right| \leq \frac{\rho^{2(M+1)}}{(1-\rho)(1-\rho^2)^2}.$$

Proof. For $|w| \leq \rho$,

$$|\mathcal{L}(w)| \leq \sum_{n \geq 2} \frac{|w|^n}{1 - |w|^n} \leq \frac{|w|^2}{(1-\rho)(1-\rho^2)}.$$

Consequently all sums in (7.8) converge absolutely. Expanding its right side as a double sum, the coefficient of $P(z^m)$ is $\sum_{j|m} \mu(j)$, which is one for $m = 1$ and zero otherwise. Summing the preceding bound over $j > M$ proves the claim. □

This inversion does not preserve algebraicity under an infinite limit, and it supplies no bounds on degrees or heights of hypothetical algebraic values $\mathcal{L}(z^j)$. The distinction is mathematical, not merely computational: even a series of rational summands can converge exponentially or faster to a transcendental number, as the Kempner and reciprocal Fermat examples already demonstrate. The no-Mahler theorem explains why replacing this infinite observation identity by a finite linear dilation system for P cannot work.

What remains for the three targets. For P_2, the present proof supplies irrationality, a raw height exponent of one, and exclusion of every finite Mahler system for its prime-support generating function. It supplies neither an irrationality exponent greater than two nor a different transcendence criterion. For $L_{\mathbb{P}}$, irrationality is already due to Tao–Teräväinen; our exact denominator formula shows that prime truncations have height exponent tending to zero. Controlling a suitable corrected remainder or constructing arithmetic auxiliary forms is still missing. For E, the divisor generating function is excluded from every finite linear Mahler system by the height gap, and its known irrationality does not become a transcendence proof. The exact gcd formula and rational-grid correction above are proved statements within this investigation, while historical priority for those elementary refinements has not been established.

8 Target register and the remaining arithmetic gaps

The following register contains twenty-three explicitly specified targets. Rows are questions, not a count of independent discoveries. In particular, the two bridge examples belong to one theorem, the Bessel extremum coordinates belong to one family, and affine versions of a capacity are not counted again. "Not resolved here" is a statement about this manuscript; for less standard boundary constants it is not a claim that an exhaustive literature search has established a historically recognized open problem.

No.	Target and arithmetic question	Outcome and precise scope
1	Euler's constant γ: irrationality	Not resolved here. A relation involving Ein(1) supplies a disjunctive statement, not identification of the arithmetic nature of γ.
2	Gompertz's constant δ: irrationality	Not resolved here. The positive squared-integer-polynomial integral method has a sharp positive minimum; signed or more general auxiliary forms remain possible.
3	Catalan's constant G: irrationality	Not resolved here. The attempted elementary truncation route fails its denominator test. Recent claimed proofs are not accepted as verified inputs.
4	$\zeta(5)$: irrationality	Not resolved here. Simultaneous statements about several odd zeta values cannot be converted into an individual conclusion without an additional elimination argument.
5	$e + \pi$: irrationality or transcendence	Not resolved here. The elementary quadratic argument gives only that this number and $e\pi$ cannot both be algebraic.
6	$e\pi$: irrationality or transcendence	Not resolved here. This product is different from e^{π}. Schanuel's conjecture would give more, but is not proved.

No.	Target and arithmetic question	Outcome and precise scope
7	e^e: irrationality or transcendence	Not resolved here. The exponent is transcendental; the hypotheses of Gelfond–Schneider and algebraic-point E-function specialization do not apply.
8	π^π: irrationality or transcendence	Not resolved here. The conditional Schanuel deduction is recorded separately from unconditional theorems.
9	$P_2 = \sum_p 2^{-p}$: transcendence	Not resolved here. Its known elementary irrationality is proved for calibration. Its natural prime-support generating function is not Mahler for any integer base.
10	$L_{\mathbb{P}} = \sum_p (2^p - 1)^{-1}$: transcendence	Not resolved here. Irrationality was already proved by Tao–Teräväinen. The raw rational truncation exponent tends to zero.
11	Erdős–Borwein constant $E = \sum_{n\geq 1}(2^n - 1)^{-1}$: transcendence	Not resolved here. Irrationality is classical. The divisor generating function is excluded from every finite linear Mahler system by the coefficient-height gap.
12	Gamma capacity $A(1)$: irrationality	Proved, uniformly for $A(d)$ at every rational $d > 1/2$ (Theorem 2.4). Transcendence of these parameter roots is not proved.
13	Excursion boundary $C_{3,1}$: transcendence	Proved as part of the complete rational-parameter bridge classification (Theorem 3.3). No historically stated arithmetic conjecture is claimed closed.
14	Quadratic reflected-bridge boundary $C_{1,2}$: transcendence	Proved by the same classification. This is a second example of one theorem, not an independent discovery.
15	Bessel maximum (y_0, M_0): transcendence	Both coordinates are proved transcendental, within the uniform rational-parameter extremum theorem; the normalized critical scale is covered too (Theorem 3.5).
16	Mixed threshold τ_2 in (4.8): transcendence	Proved within the one-sided rank-two spectral classification (Theorem 4.1). The proof does not need an explicit minimizer.
17	Two-prime edge $\sigma_{2,3}$: irrationality	Proved by a finite-place valuation argument, uniformly for every finite unit-weight prime set of size at least two. Transcendence is not proved; no historical priority claim is made.
18	Prime-zeta edge $P(\sigma_P) = 1$: irrationality	Not resolved here. The finite radical valuation argument does not pass through a limit defined only by real convergence.
19	Ordered-factorization exponent $\zeta(\rho) = 2$: irrationality	Not resolved here. No finite algebraic differential system or sufficiently strong arithmetic approximation is obtained.
20	$\log \mathrm{Ein}(1)$: transcendence	Proved as a classical-theory consequence. The proof uses independence of exponential-integral values over the entire algebraic exponential field (Theorem 5.1).
21	Poisson extreme-value optimizer $x_{1/2}$: transcendence	Proved, with existence and uniqueness, as a member of the algebraic-power-cost family (Proposition 5.5). The optimum value itself is not classified.
22	Signed Brownian-bridge boundary B: irrationality	Not resolved unconditionally. The factor $\sqrt{\pi/2}$ blocks the proposed E-function argument. Transcendence follows conditionally from the E–G value-intersection conjecture (Proposition 3.7).
23	Dyadic Gompertz family $\delta_1, \delta_2, \ldots, \delta_{2^{m-1}}$: simultaneous transcendence	The family has transcendence degree at least $m - 2$. This is a simultaneous transfer from the exponential-integral theorem; it does not select which individual constants are transcendental.

8.1 Which extensions would be substantive next steps?

For the gamma capacity, proving transcendence would require control of a Kummer family at algebraic, potentially nonrational, parameters or a different arithmetic representation of its parameter root. The present rational-parameter E-function proof cannot supply that extension merely by changing the word "rational" to "algebraic." A quantitative irrationality measure is a separate problem as well.

For the spectral classification, a natural next question is to remove the one-sided condition on the irrational exponent coordinates. The precise issue is whether an algebraic critical level can be singular at the positive contact point. The existing proof applies whenever nonsingularity is established by another argument. For spectra of rational rank greater than two, two contact equations need not have a zero-dimensional algebraic intersection, so a higher-dimensional theorem needs additional structure.

For the famous individual constants, the missing step is arithmetic isolation. In the Gompertz family, a lower bound on transcendence degree leaves the identity of the exceptional members unspecified. For a single δ, positive squared-polynomial kernels are ruled out by the sharp bound proved above, but signed Padé constructions are not. For γ, Catalan's constant, and $\zeta(5)$, one needs controlled nonzero integer linear forms or a new specialization theorem with the relevant hypotheses actually verified.

For prime-supported constants, exact recovery of the prime support and uniform tail estimates are already available. The denominator calculations show why this is insufficient: absolute accuracy and arithmetic height can improve at very different rates. The finite-Mahler obstruction rules out one concrete attempt to import the lacunary method, while leaving nonlinear identities, new auxiliary forms, and other arithmetic structures open.

8.2 Research assessment

The strongest outcomes here are uniform statements about pre-existing analytic boundaries: a rational-point exclusion on the gamma capacity curve, an exact algebraic-versus-transcendental bridge classification, an arithmetic classification of Bessel extrema, and the rank-two spectral critical-value theorem with its sharp algebraic exception. The remaining positive results are carefully identified consequences of classical transcendence theory, finite radical arithmetic, or exact denominator bookkeeping. No new individual solution of a documented famous constant conjecture has been established in this investigation. The next assessment should focus on correctness, prior overlap, and the scope of these uniform theorems rather than counting their numerical instances.

9 Verification record and reproducibility

This file is a standalone LaTeX source: the bibliography and all proofs are embedded, and there are no external figure, bibliography, or input-file dependencies. Compilation uses standard pdfLaTeX packages. The foundational specialization and transcendence theorems remain cited external inputs; their consequences are proved explicitly.

The gamma-capacity proof and the two probability boundary families were cross-read independently. The exponential-integral proof was checked independently for its common-constant hypothesis, its formal pole argument, and the specialization point. The rank-two spectral proof was checked for repeated factors, the Newton-polytope argument, and the algebraic exceptional setting. The positive-square Gompertz optimization was independently checked, including all polynomial degrees and the equality cases. These are internal mathematical

reviews, not formal proof-assistant certification or external peer review.

The substantive assumptions that must not be weakened are rational Kummer parameters, ordinary nonzero algebraic specialization points, positivity and one-sided irrational coordinates in the spectral classification, and the integer-square restriction in the sharp Gompertz obstruction. The conditional signed-bridge result is kept distinct from the unconditional results.

Part II
Arithmetic of Optimal Values

Editorial integration note. The stopping-value and extraction results are extended in Parts III and IV. Keep the distinction between the boundary location, the attained value, and the chosen denominator normalization.

Original part framing.

Status. Working manuscript with complete proofs for the statements marked unconditional, cited classical inputs, and explicitly marked conditional and unresolved statements. Internal independent cross-reading is not external peer review or formal certification.

10 Research question and advances over the preceding paper

The objective of this investigation is to turn analytic information about observation and recovery into arithmetic information about individual constants. The starting sources are the 223-page foundational manuscript [1], the 447-page sequel compilation [30], and the subsequent 41-page arithmetic manuscript [31]. This paper extends the last of those sources. Their lengths are source identifiers, not evidence for the strength or correctness of a theorem. Each inherited statement used below is identified as such.

The earlier paper established arithmetic information about several boundary *locations*, a rank-two spectral classification with a sign condition, and family-level transcendence statements. These do not automatically classify an optimized *value*, remove a geometric hypothesis, or identify one particular member of a family. Those three gaps motivate the present work.

Direction	Previously established	Advance in this paper
Finite spectra	One-sided irrational coordinates in rank two	All sign patterns; exact exceptional setting retained (Theorem 11.1)
Stopping payoff	Arithmetic of the optimal boundary	All even-power optimal values at rational dimensions are transcendental (Theorem 12.2)
Joint payoff arithmetic	No exact joint classification of objectives	Full relation lattice and an explicit infinite independent family (Theorems 12.3 and 12.4)
Gamma extraction	Rational-input endpoint shape is irrational	Fixed-shape critical rate and contact are transcendental (Theorem 13.2)
Boundary tables	Large-order asymptotic	No algebraic-coefficient polynomial recurrence on a rational mesh (Theorem 13.3)
Classical constants	Selected positive-kernel failures	Bounds for whole nonnegative integer-polynomial classes, with denominator scope stated

The most direct individual-constant result is the even-payoff theorem: transcendence of a stopping boundary alone cannot imply transcendence of a function evaluated there. A separate elimination identity supplies that missing implication. The joint-value theorem provides a different kind of information: the ten displayed values are individually specified, and *all ten* are algebraically independent. This is stronger than saying that some unspecified members of a larger list are transcendental. Nevertheless, their independence is a transfer of classical

Lindemann–Weierstrass to a deliberately selected stochastic family, not the resolution of ten independent historical conjectures.

The spectral result uses another mechanism. At an algebraic putative optimal level, negative curvature along the physical trajectory and positive curvature in an auxiliary direction exclude a repeated irreducible component. Two coprime equations then force algebraicity in two variables. This converts a geometric sign issue into an exact arithmetic classification. In higher rank the same equations no longer force algebraicity; the needed additional conjectural input is stated without being hidden inside the proof.

Throughout, $\overline{\mathbb{Q}}$ denotes the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Independence of an infinite collection means independence of every finite subcollection. "Unresolved here" means that this manuscript has no proof; it does not by itself certify global historical open status. Examples obtained by changing a parameter in one theorem are counted as one target family. In particular, the classical-constant investigations do not supply a new proof of irrationality or transcendence of Euler's constant, the Euler–Gompertz constant, Catalan's constant, or an individual odd zeta value beyond previously known results.

11 Opposite curvatures remove the spectral sign restriction

The preceding spectral classification required all coefficients of one irrational basis element to have the same sign. That restriction ensured that an algebraic level set was nonsingular at its point of contact. Here we remove the restriction completely. A level set may be singular, but two opposite curvatures prevent the one type of repeated component that would invalidate the elimination argument. The result applies to every finite spectrum in a two-dimensional rational space containing one. An additional theorem identifies the exact transcendence-degree input needed in higher dimension. Its higher-rank classification is explicitly conditional, and none of these results is presented as the resolution of a previously named arithmetic conjecture.

11.1 The unconditional classification

Let

$$S(x)=\sum_{j=1}^{m} w_j x^{r_j}, \qquad T(C)=\min_{x>0}\frac{1+Cx}{S(x)}, \qquad W=\sum_j w_j, \quad M=\sum_j w_j r_j. \tag{11.1}$$

Throughout, $0<r_j<1$, all weights are positive, and real powers use the real logarithm. The weights need not sum to one.

Theorem 11.1 (All rank-two spectra, including mixed signs)**.** *Suppose $w_1,\dots,w_m,C$ are positive algebraic numbers and*

$$r_j=\frac{m_j+n_j\theta}{q}, \qquad m_j,n_j\in\mathbb{Z}, \quad q\in\mathbb{Z}_{>0}, \quad \theta\in(\overline{\mathbb{Q}}\cap\mathbb{R})\setminus\mathbb{Q}, \tag{11.2}$$

where at least one n_j is nonzero. There is no sign restriction on the n_j. The minimum has a unique contact x_C, and

$$T(C)\in\overline{\mathbb{Q}} \iff x_C=1 \iff C=C_*:=\frac{M}{W-M}. \tag{11.3}$$

At the unique exceptional setting,

$$T(C_*)=\frac{1}{W-M}. \tag{11.4}$$

Every other positive algebraic setting gives a transcendental value.

The theorem covers, in particular, arbitrary finite subsets of $\mathbb{Q}(\sqrt{d}) \cap (0,1)$ containing an irrational element, for any positive nonsquare integer d. It concerns the optimized value, whose contact need not have a known closed form. Gelfond–Schneider is the classical arithmetic input: if $a > 0$ is algebraic and $a \neq 1$, then a^θ is transcendental for algebraic irrational θ; see the primary proof account [32]. The work here is the reduction of the variational constant to that input in the presence of singular algebraic levels.

Lemma 11.2 (Contact and curvature)**.** *The contact in* (11.1) *is unique and satisfies*

$$C = \frac{S'(x_C)}{S(x_C) - x_C S'(x_C)}. \tag{11.5}$$

If $\tau = T(C)$*, then*

$$S(x_C) = \frac{1 + C x_C}{\tau}, \qquad S'(x_C) = \frac{C}{\tau}, \qquad S''(x_C) < 0. \tag{11.6}$$

Proof. The objective diverges at both ends of $(0, \infty)$ because all exponents are strictly between zero and one. Put $H = S - xS' > 0$. Then

$$\left(\frac{S'}{H}\right)' = \frac{S''S}{H^2} < 0, \qquad \frac{xS'}{H} = \frac{\sum_j w_j r_j x^{r_j}}{\sum_j w_j (1 - r_j) x^{r_j}}.$$

The last ratio is bounded above and below by positive constants. Consequently S'/H decreases from infinity to zero, and the unique stationary point is the unique minimum. Its value equation and (11.5) imply (11.6). □

We give the elimination step in a form that also exposes the higher-dimensional obstruction. For $z \in (0,\infty)^d$ and a Laurent polynomial F, its logarithmic Hessian is the ordinary real Hessian at $y = 0$ of the function

$$y \longmapsto F(z_1 e^{y_1}, \ldots, z_d e^{y_d}).$$

It is indefinite if its quadratic form takes both positive and negative values. This definition applies whether or not the ordinary gradient of F vanishes.

Lemma 11.3 (Coprime equations impose two algebraic constraints)**.** *If two nonzero Laurent polynomials over* $\overline{\mathbb{Q}}$ *in* $d \geq 2$ *variables are coprime, every common torus zero* z *satisfies*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(z_1, \ldots, z_d) \leq d - 2.$$

Proof. A nonzero equation excludes transcendence degree d. If the degree were $d-1$, choose $d-1$ coordinates forming a transcendence basis and relabel them $z_1, \ldots, z_{d-1}$. Evaluation embeds the rational function field in these variables into $\mathbb{C}$. The remaining coordinate has a minimal polynomial over that field. Every polynomial equation satisfied by the coordinates is divisible by this minimal polynomial in the one-variable polynomial ring over the rational function field. Clearing denominators and taking its primitive part, Gauss' lemma gives a common nonconstant irreducible factor of the two original polynomials after their Laurent monomials have been cleared. This factor is not a coordinate variable, since all coordinates are nonzero. It is therefore a common nonunit Laurent factor, a contradiction. □

Lemma 11.4 (No invariant Laurent hypersurface)**.** *Let* $b_1, \ldots, b_d$ *be rationally independent real algebraic numbers, and set*

$$D = \sum_{i=1}^{d} b_i z_i \partial_{z_i}.$$

For every nonunit irreducible Laurent polynomial H *over* $\overline{\mathbb{Q}}$*, the polynomials* H *and* DH *are coprime.*

Proof. Every monomial z^k is an eigenvector of D, with eigenvalue $b \cdot k$. Rational independence makes these eigenvalues distinct, and only the constant monomial has eigenvalue zero. Thus $DH = 0$ would force H to be a constant, which is excluded. If H divides DH, write $DH = AH$. Newton polytopes satisfy

$$\mathcal{N}(A) + \mathcal{N}(H) = \mathcal{N}(DH) \subseteq \mathcal{N}(H).$$

For completeness, the product identity follows by maximizing a generic linear functional: the unique maximizing monomial in each factor gives a noncancelling maximizing product. Continuity gives the identity for all support directions. Taking support functions in the displayed inclusion forces every exponent of A to be zero, so A is constant. Distinct eigenvalues then force H to have only one monomial. Such a Laurent polynomial is a unit, again a contradiction. □

Theorem 11.5 (An indefinite contact has codimension at least two)**.** *Let F be a nonzero Laurent polynomial with real algebraic coefficients, let D be as in Lemma 11.4, and let $z \in (0,\infty)^d$. Suppose*

$$F(z) = DF(z) = 0 \tag{11.7}$$

and the logarithmic Hessian of F at z is indefinite. Then

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(z_1, \dots, z_d) \leq d - 2. \tag{11.8}$$

For $d = 2$, both coordinates are algebraic.

Proof. Factor F in the Laurent polynomial ring over $\overline{\mathbb{Q}}$. If two distinct irreducible factors vanish at z, they are coprime and Lemma 11.3 applies. Otherwise exactly one irreducible factor H vanishes. Write $F = H^e R$ with $R(z) \neq 0$.

If $e = 1$, the equation $DF(z) = 0$ gives $DH(z) = 0$. Lemmas 11.4 and 11.3 again apply. It remains to exclude $e \geq 2$. Pass to logarithmic coordinates and write $h(y) = H(z_1 e^{y_1}, \dots, z_d e^{y_d})$ and similarly $r(y)$. Since $h(0) = 0$, the Hessian of $h^e r$ at zero is zero for $e \geq 3$. For $e = 2$, it is

$$2r(0)\, \nabla h(0) \nabla h(0)^{\mathsf{T}}. \tag{11.9}$$

This matrix has complex rank at most one. It is also the Hessian of a real-valued function on real coordinates, because F has real coefficients. Therefore it is a real symmetric matrix of real rank at most one. A real symmetric matrix of rank at most one is positive semidefinite or negative semidefinite. It cannot be indefinite. This contradicts the hypothesis in all cases $e \geq 2$. No choice of a real normalization for the complex factor H is needed. □

Proof of Theorem 11.1. Suppose $\tau = T(C)$ is algebraic. Put

$$u = x_C^{1/q}, \quad v = x_C^{\theta/q}, \quad P(U,V) = \sum_j w_j U^{m_j} V^{n_j}, \quad F(U,V) = P(U,V) - \frac{1 + CU^q}{\tau}.$$

Use $D = U\partial_U + \theta V \partial_V$ and $E = V\partial_V$. Along $(U,V) = (x^{1/q}, x^{\theta/q})$, the operator D is $qx\, d/dx$. The contact equations yield

$$\begin{aligned} F(u,v) &= DF(u,v) = 0, \\ D^2F(u,v) &= q^2 x_C^2 S''(x_C) < 0, \\ E^2F(u,v) &= \sum_j w_j n_j^2 u^{m_j} v^{n_j} > 0. \end{aligned} \tag{11.10}$$

In the second line, the additional first-derivative term is $q^2 x_C(S'(x_C) - C/\tau) = 0$. The last line is strictly positive because at least one $n_j \neq 0$. The operators D and E are constant directions in logarithmic coordinates, so the last two lines imply that the logarithmic Hessian is indefinite. In particular F is nonzero. Theorem 11.5 proves that u and v are algebraic. Since $v = u^\theta$ and $u > 0$, Gelfond–Schneider forces $u = 1$. Thus $x_C = 1$.

Conversely, evaluating (11.5) at one gives $C = M/(W - M)$, and at this setting the value is $(1 + C)/W = 1/(W - M)$, an algebraic number. This proves every implication. □

The sign removal is essential: the old proof used $F_V \neq 0$, whereas the new proof only needs $E^2 F > 0$. At a mixed-sign contact, F_V may vanish. The argument above includes that case and all intersections of distinct irreducible components; it never assumes that the entire level polynomial is squarefree.

11.2 Rational settings and genuinely mixed-sign examples

Corollary 11.6 (The rational cancellation criterion)**.** *Under Theorem 11.1, assume all weights are rational. Write*

$$A = \frac{1}{q}\sum_j w_j m_j, \qquad B = \frac{1}{q}\sum_j w_j n_j, \qquad M = A + B\theta.$$

If $B \neq 0$, every positive rational setting has transcendental value. If $B = 0$, the unique algebraic value among positive rational settings occurs at the rational setting $C_* = A/(W - A)$.

Proof. If $B \neq 0$, then M is irrational. A rational exceptional setting would give $M = C_* W/(1 + C_*) \in \mathbb{Q}$, a contradiction. If $B = 0$, the exceptional setting is rational, and the main theorem excludes all other positive rational settings. □

For a concrete family outside the old one-sided hypothesis, put

$$a = \frac{1}{\sqrt{2}}, \qquad \Theta_k = \min_{x>0} \frac{2(1 + kx)}{x^a + x^{1-a}} \quad (k > 0). \tag{11.11}$$

Both exponents lie in $(0, 1)$, and their coefficients of $\sqrt{2}$ have opposite signs. Equal weights give $W = 1$ and $M = 1/2$. Consequently

$$\Theta_1 = 2, \qquad \Theta_2, \Theta_3, \Theta_4, \Theta_5, \Theta_6, \Theta_7, \Theta_8, \Theta_9, \Theta_{10}, \Theta_{11} \quad \text{are transcendental.} \tag{11.12}$$

These ten constants are distinct. If $k_2 > k_1$ and x_2 is the minimizer for k_2, then

$$\Theta_{k_2} > \frac{2(1 + k_1 x_2)}{x_2^a + x_2^{1-a}} \geq \Theta_{k_1}.$$

Their common proof is one classification theorem, not ten independent solutions of historical conjectures. The value Θ_1 also supplies a necessary correction to any attempted extension of the old rational corollary: rational weights and a rational setting can give an algebraic threshold when opposite irrational coefficients cancel.

An asymmetric example is

$$\Psi = \min_{x>0} \frac{3(1 + x)}{2x^{1/\sqrt{2}} + x^{1-1/\sqrt{2}}}. \tag{11.13}$$

Here $M = (1 + 1/\sqrt{2})/3$ is irrational. Thus Ψ is transcendental, as is the corresponding value at every positive rational setting. In contrast with the symmetric family, there is no rational exceptional setting.

11.3 Higher rational rank: an exact conditional extension

The curvature theorem remains valid in any dimension. What fails unconditionally is the inference from two algebraic constraints to algebraicity of all coordinates. The following statement isolates the missing input without silently invoking a conjecture.

Theorem 11.7 (A sufficient exponential-line bound)**.** *Suppose w_j, C are positive algebraic, all $r_j \in (0,1)$ are algebraic, and at least one r_j is irrational. Let*

$$d = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{1, r_1, \dots, r_m\} \geq 2.$$

Choose a real algebraic rational basis $1, \beta_2, \dots, \beta_d$ and an integer $q > 0$ so that

$$r_j = \frac{1}{q}\left(m_{j1} + \sum_{i=2}^{d} m_{ji}\beta_i\right), \qquad m_{ji} \in \mathbb{Z}.$$

Set $b_1 = 1/q$ and $b_i = \beta_i/q$ for $i \geq 2$. Suppose the following bound holds at every real $t \neq 0$:

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{b_1 t}, \dots, e^{b_d t}) \geq d - 1. \tag{11.14}$$

Then the classification (11.3) and exceptional value (11.4) hold for this spectrum. It is enough to assume (11.14) only at the nonzero contact logarithm, if such a contact occurs.

Proof. Assume $\tau = T(C)$ is algebraic and put $t = \log x_C$ and $z_i = e^{b_i t}$. With multi-index notation define

$$P(Z) = \sum_j w_j \prod_{i=1}^{d} Z_i^{m_{ji}}, \qquad F(Z) = P(Z) - \frac{1 + CZ_1^q}{\tau}, \qquad D = \sum_{i=1}^{d} b_i Z_i \partial_{Z_i}.$$

Along the exponential line, $D = x\, d/dx$. Thus $F(z) = DF(z) = 0$ and $D^2F(z) = x_C^2 S''(x_C) < 0$. Some index $i \geq 2$ has $m_{ji} \neq 0$ for at least one j. For that index,

$$(Z_i\partial_{Z_i})^2 F(z) = \sum_j w_j m_{ji}^2 \prod_{\ell=1}^{d} z_\ell^{m_{j\ell}} > 0.$$

Theorem 11.5 gives transcendence degree at most $d-2$. If $t \neq 0$, this contradicts (11.14). Therefore $t = 0$ and $x_C = 1$. The converse and exceptional value follow from the contact calculation exactly as before. □

Corollary 11.8 (All finite algebraic spectra, conditional on Schanuel)**.** *Assume Schanuel's conjecture. For every finite algebraic spectrum in $(0,1)$ containing an irrational exponent, and every choice of positive algebraic weights and setting, the equivalence (11.3) holds.*

Proof. Schanuel's conjecture asserts that rationally independent complex numbers $s_1, \dots, s_d$ satisfy

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(s_1, \dots, s_d, e^{s_1}, \dots, e^{s_d}) \geq d;$$

see [33]. For $t \neq 0$, the numbers $b_i t$ are rationally independent. Extending the base from $\mathbb{Q}$ to its algebraic closure does not change transcendence degree. Since every b_i is algebraic and $b_1 \neq 0$, their field over $\overline{\mathbb{Q}}$ is generated by t. Adjoining t increases transcendence degree by at most one. Schanuel therefore implies (11.14), and Theorem 11.7 applies. □

For $d = 2$, bound (11.14) follows unconditionally from Gelfond–Schneider: two algebraic coordinates on the line would be related by an algebraic irrational power. In higher dimension we do not prove the bound. The conditional corollary is a precise statement of the remaining arithmetic input, not an unconditional extension. For example, the value

$$\min_{x>0} \frac{2(1+x)}{x^{\sqrt{2}/2} + x^{\sqrt{3}/3}} \tag{11.15}$$

is transcendental under the corollary: its mean exponent is not $1/2$ and the rational span with one has dimension three. This manuscript does not prove its transcendence unconditionally.

Proposition 11.9 (Curvature alone does not eliminate dimension three)**.** *There is a polynomial F with real algebraic coefficients in three variables, irreducible over $\overline{\mathbb{Q}}$, for which the conditions $F = DF = 0$ and opposite logarithmic curvatures hold on a positive real curve containing points with transcendental coordinates. The diagonal weights of D can be rationally independent algebraic numbers.*

Proof. Take

$$D = u\partial_u + \sqrt{2}\, v\partial_v + \sqrt{3}\, z\partial_z, \qquad E = v\partial_v,$$
$$F = z - 1 - \sqrt{3}(u-1) - 5(u-1)^2 + (v-1)^2.$$

The polynomial is irreducible: it is monic and linear in z, with coefficient one. At $p = (1, 1, 1)$, direct differentiation gives

$$F(p) = DF(p) = 0, \qquad D^2F(p) = -3 - \sqrt{3} < 0, \qquad E^2F(p) = 2 > 0.$$

Moreover

$$\nabla F(p) = (-\sqrt{3}, 0, 1), \qquad \nabla(DF)(p) = (-\sqrt{3} - 10, 2\sqrt{2}, \sqrt{3})$$

are independent. The real implicit function theorem gives a smooth one-dimensional common zero set near p. In a sufficiently small neighborhood its coordinates remain positive and the two strict curvature inequalities persist. This curve has uncountably many points, whereas algebraic triples form a countable set. Some of its points, in fact uncountably many, therefore have a transcendental coordinate. □

The proposition is an obstruction to an ambient elimination argument. It does not say that those curve points lie on the specific exponential line prescribed by a spectral model, and it is not a counterexample to the conditional classification. It locates the gap exactly: coprime equations cut curves in dimension three, while in dimension two they cut only algebraic points. No conclusion about prime-zeta edge constants, $\zeta(\rho) = 2$, or Riemann-zeta zero ordinates follows from this finite algebraic-spectrum result.

Provenance and count. The unconditional gain is one classification: removing the sign restriction from the rank-two spectral threshold theorem. Its proof uses a general curvature-and-factorization statement, which also yields the conditional higher-rank reduction. The displayed mixed-sign constants are applications of that classification. Classical Gelfond–Schneider and the conjectural Schanuel input are credited separately. The literature check did not establish historical priority for this exact variational statement; no such priority is asserted.

12 From stopping boundaries to optimal objective values

The preceding arithmetic analysis classified certain stopping *locations*; it did not classify the optimal expected payoffs. This distinction matters because the value of an analytic function at a transcendental boundary need not be transcendental. We investigate three objective-value targets here. For every rational dimension and every positive even integral payoff exponent, the optimal value is transcendental. On the algebraic-boundary family we determine all algebraic relations between finite collections of normalized optimal values, and construct an infinite algebraically independent collection with integer dimensions and integer payoff exponents. The linear Brownian-excursion value remains undecided by these arguments.

The stochastic optimal-stopping formulas are established results of Ekström–Wanntorp and Hobson–Liu [5, 34]. The new work here is their arithmetic analysis, using classical transcendence theorems and an explicit Wronskian elimination. No primary source located in our search states these arithmetic questions as historical open conjectures, and no such closure is claimed. The ten examples below illustrate one joint-independence theorem; they are not ten independent problem closures.

12.1 The objective and the inherited boundary input

Let R be a Bessel bridge of dimension $d > 0$, starting at zero and returning to zero at time one. For $q > 0$ define its optimal origin value by

$$\mathcal{V}_{d,q} = \sup_{0 \le \tau \le 1} \mathbb{E}[R_\tau^q]. \tag{12.1}$$

For integer d, this is the radial part of a d-dimensional Brownian bridge. For arbitrary positive d it is understood through the usual squared Bessel bridge, including its standard behavior at zero. Put $a = q/2$, $b = d/2$, and

$$f_{a,b}(z) = M(a; b; z) = \sum_{n \ge 0} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}.$$

The established stopping formula [34] is

$$z_{d,q} f'_{a,b}(z_{d,q}) = a f_{a,b}(z_{d,q}), \qquad C_{d,q} = \sqrt{2 z_{d,q}}, \qquad \mathcal{V}_{d,q} = \frac{(2 z_{d,q})^a}{f_{a,b}(z_{d,q})}. \tag{12.2}$$

The optimal rule first crosses $C_{d,q}\sqrt{1-t}$. For $d = 1$ this is the problem with payoff $|X_\tau|^q$ for a Brownian bridge, not the signed-payoff stopping problem.

We recall the exact arithmetic input from the preceding manuscript [31], without counting it as a new result:

$$\begin{gathered} d, q \in \mathbb{Q}_{>0}: \qquad z_{d,q} \text{ is algebraic} \iff a - b \in \mathbb{Z}_{\ge 0}; \\ \text{otherwise } z_{d,q} \text{ is transcendental.} \end{gathered} \tag{12.3}$$

For clarity, both the existence assertion and the arithmetic mechanism can be recalled briefly. The positive Taylor coefficients of f give probability weights proportional to $c_n z^n$. Then zf'/f is their mean, its derivative with respect to $\log z$ is their strictly positive variance, and this mean increases from zero to infinity. There is exactly one positive solution in (12.2).

For rational $a, b > 0$, f and f' are E-functions satisfying the first-order system

$$\begin{pmatrix} f \\ f' \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ a/z & 1 - b/z \end{pmatrix} \begin{pmatrix} f \\ f' \end{pmatrix}.$$

The denominator estimate for $(a)_n/(b)_n$ follows by counting residues in arithmetic progressions prime by prime; their common denominators grow exponentially. The only finite singular point of the system is zero. A rational logarithmic derivative of the entire exponential-type solution forces $f = e^{\lambda z}P(z)$ with P polynomial. Kummer's equation then forces either $\lambda = 0$, $a = -\deg P$, or $\lambda = 1$, $a - b = \deg P$. The first alternative is impossible. Outside the second family, Beukers' linear-specialization theorem [35, Corollary 1.4] excludes the boundary relation at a nonzero algebraic point. Inside the second family,

$$f_{a,b}(z) = e^z P_N(z), \qquad P_N(z) = M(-N; b; -z), \qquad N = a - b \in \mathbb{Z}_{\geq 0}, \tag{12.4}$$

and the boundary equation is a nonzero polynomial equation. This proves the recalled input and explains why it concerns the location rather than the objective.

We also use the classical Lindemann–Weierstrass theorem: exponentials of distinct algebraic numbers are linearly independent over $\overline{\mathbb{Q}}$. It follows here directly from the same Beukers theorem. The functions $e^{\lambda_j z}$ form a system with constant algebraic diagonal coefficients and are linearly independent over $\mathbb{C}(z)$ when the λ_j are distinct. Indeed, after clearing denominators, a polynomial-coefficient relation is impossible along a ray on which the real parts of $\lambda_j z$ are all distinct: its largest exponential dominates. Specializing at $z = 1$ gives the stated arithmetic conclusion.

12.2 Target I: every even-payoff optimum is transcendental

The essential new identity eliminates the special function from the optimal value, even when the boundary itself is transcendental.

Lemma 12.1 (Contiguous reduction and its constant determinant)**.** *Let $b > 0$ and $m \geq 1$ be an integer. Define polynomials $A_m, B_m \in \mathbb{Q}[b, z]$ by $A_1 = 1$, $B_1 = 0$, and*

$$A_{m+1} = \frac{zA'_m + (z + m + 1 - b)A_m}{m}, \tag{12.5}$$

$$B_{m+1} = \frac{(b-1)A_m + zB'_m + mB_m}{m}. \tag{12.6}$$

Thus rational numerical b gives rational-coefficient polynomials. Put

$$U_m = zA'_m + (z + 1 - b - m)A_m, \tag{12.7}$$

$$T_m = (b-1)A_m + zB'_m - mB_m, \tag{12.8}$$

$$K_m(b) = \frac{\prod_{j=1}^{m}(j-b)}{(m-1)!}. \tag{12.9}$$

Then

$$M(m; b; z) = A_m(z)M(1; b; z) + B_m(z), \qquad B_m U_m - A_m T_m = K_m(b). \tag{12.10}$$

Moreover $\deg A_m = m - 1$ *with leading coefficient* $1/(m-1)!$*, and* $\deg U_m = m$ *with the same leading coefficient.*

Proof. Write $\theta = z\,d/dz$ and $Y = M(1; b; z)$. Comparison of Taylor coefficients gives

$$\theta Y = (z + 1 - b)Y + (b - 1), \qquad M(m + 1; b; z) = \frac{(\theta + m)M(m; b; z)}{m}. \tag{12.11}$$

These are special cases of the standard contiguous relations [39, §13.3], but the coefficient check proves them directly. Substitution of the first identity into the second gives exactly

(12.5)–(12.6), proving the first formula in (12.10). The degree and leading coefficient claims follow by induction. The constant term is

$$A_m(0) = \frac{\prod_{j=2}^{m}(j-b)}{(m-1)!}, \qquad A_m(0) + B_m(0) = 1. \tag{12.12}$$

Here is a complete derivation of the determinant identity, including the potentially vanishing exceptional constants. On $z > 0$ let $H(z) = e^z z^{1-b}$. The equation

$$\mathcal{L}_m f = 0, \qquad \mathcal{L}_m = \theta(\theta + b - 1) - z(\theta + m),$$

is Kummer's equation multiplied by z. Direct substitution gives $\mathcal{L}_1 H = 0$, and the operator identity

$$\mathcal{L}_{m+1}(\theta + m) = (\theta + m)\mathcal{L}_m$$

follows from $\theta z = z(\theta + 1)$. Thus repeated application of the contiguous operator shows that both $F_m = M(m; b; z) = A_m Y + B_m$ and $G_m = A_m H$ solve Kummer's equation with upper parameter m.

The Wronskian in the convention $W(F, G) = FG' - F'G$ therefore obeys $W' = (1 - b/z)W$. It is a constant times $e^z z^{-b}$. Since $H'/H = 1 + (1 - b)/z$, direct multiplication using (12.11) gives

$$\begin{aligned} W(F_m, G_m) &= \frac{H}{z}\left[z(B_m A_m' - A_m B_m') + (z + 1 - b)A_m B_m - (b - 1)A_m^2\right] \\ &= e^z z^{-b}(B_m U_m - A_m T_m). \end{aligned}$$

It follows that the polynomial in the last parentheses is constant. At zero, (12.12) evaluates it as $(1 - b)A_m(0) = K_m(b)$. The calculation remains valid when this constant is zero; linear independence of F_m, G_m was not assumed. □

Theorem 12.2 (Transcendence of all rational-dimensional even-payoff values)**.** *For every $d \in \mathbb{Q}_{>0}$ and every integer $m \geq 1$, $\mathcal{V}_{d,2m}$ is transcendental. More explicitly, put $b = d/2$ and $z = z_{d,2m}$. If $b \notin \{1, \ldots, m\}$,*

$$\mathcal{V}_{d,2m} = \frac{(2z)^m U_m(z)}{K_m(b)}, \tag{12.13}$$

a nonconstant rational-coefficient polynomial of degree $2m$ in the transcendental boundary variable z. If $b \in \{1, \ldots, m\}$, then z is positive algebraic and $\mathcal{V}_{d,2m}$ is a nonzero algebraic multiple of e^{-z}.

Proof. At the optimal boundary the smooth-fit equation is $(\theta - m)F_m = 0$. By the first identity of (12.10) and (12.11), this is precisely $U_m(z)Y(z) + T_m(z) = 0$. Multiplying $F_m = A_m Y + B_m$ by U_m and using the constant determinant gives

$$F_m(z)U_m(z) = K_m(b). \tag{12.14}$$

If $b \notin \{1, \ldots, m\}$, the right side is nonzero. In particular $U_m(z) \neq 0$, so no division by a potentially vanishing polynomial is hidden in the argument. Formula (12.2) now gives (12.13). Its degree is $2m$ by Lemma 12.1. The boundary classification (12.3) says that z is transcendental. Were $\mathcal{V}_{d,2m}$ algebraic, (12.13) would be a nonzero polynomial equation over $\overline{\mathbb{Q}}$ for z, a contradiction.

If $b \in \{1, \ldots, m\}$, set $N = m - b \geq 0$. Equations (12.3) and (12.4) give

$$\mathcal{V}_{d,2m} = \frac{(2z)^m}{P_N(z)} e^{-z}.$$

The prefactor is nonzero algebraic because $z > 0$ is algebraic and $P_N(z) > 0$. Lindemann–Weierstrass makes e^{-z} transcendental. This covers every positive rational d. □

For the quadratic payoff the new polynomial identity is especially short. Here $U_1 = z - b$, $K_1 = 1 - b$, so for $d \neq 2$,

$$\mathcal{V}_{d,2} = \frac{C_{d,2}^2(C_{d,2}^2 - d)}{2 - d}. \tag{12.15}$$

Thus the reflected Brownian-bridge value is $C_{1,2}^4 - C_{1,2}^2$, while the three-dimensional quadratic value is $C_{3,2}^2(3 - C_{3,2}^2)$; both are transcendental. The exceptional value is $\mathcal{V}_{2,2} = 2/e$. At the next payoff exponent,

$$U_2(z) = z^2 + 2(1 - b)z + (b - 2)(b + 1), \qquad K_2 = (1 - b)(2 - b),$$

which gives an explicit quartic polynomial in z whenever $b \neq 1, 2$. These are instances of one uniform theorem. The polynomial relation also shows why no algebraic independence between a nonexceptional even-payoff value and its own boundary is asserted: they are algebraically dependent.

12.3 Target II: all algebraic relations on the algebraic-boundary family

Take any finite list of rational pairs (d_i, q_i) satisfying $q_i - d_i = 2N_i$ with $N_i \in \mathbb{Z}_{\geq 0}$, and let $z_i = z_{d_i,q_i}$. Define the positive algebraic number

$$a_i^* = \frac{(2z_i)^{q_i/2}}{M(-N_i; d_i/2; -z_i)}, \qquad W_i = \frac{\mathcal{V}_{d_i,q_i}}{a_i^*} = e^{-z_i}. \tag{12.16}$$

Rational powers here denote their positive real branches, which are algebraic at positive algebraic arguments.

Theorem 12.3 (Exact transcendence degree and relation lattice)**.** *For every such finite list,*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\mathcal{V}_{d_1,q_1}, \ldots, \mathcal{V}_{d_s,q_s}) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{z_1, \ldots, z_s\}. \tag{12.17}$$

More precisely, the kernel of the evaluation map from $\overline{\mathbb{Q}}[X_1^{\pm 1}, \ldots, X_s^{\pm 1}]$ to $\mathbb{C}$, $X_i \mapsto W_i$, is generated by the binomials

$$X_1^{n_1} \cdots X_s^{n_s} - 1, \qquad n \in \mathbb{Z}^s, \quad \sum_i n_i z_i = 0. \tag{12.18}$$

Thus the theorem determines all algebraic relations, after the explicit algebraic rescaling (12.16)*.*

Proof. Expand any Laurent polynomial at $W_i = e^{-z_i}$ and group its monomials by the algebraic exponent $\sum_i n_i z_i$. Lindemann–Weierstrass says that the sum vanishes exactly when the coefficients in each group add to zero. Within a group, subtract one fixed monomial from the others; their exponent differences lie in the integer relation lattice. This proves the kernel statement and shows that it is an exact description, not merely a list of some relations.

Let the rational rank of the z_i be r. Select r of them as a basis. Their exponentials are algebraically independent: different polynomial monomials give distinct algebraic exponents, again by Lindemann–Weierstrass. Every remaining exponential has an integral positive power equal to a Laurent monomial in the selected exponentials, because its exponent is a rational linear combination of the basis. It is therefore algebraic over the field they generate. The nonzero algebraic prefactors do not change transcendence degree, proving (12.17). □

The theorem includes dependent examples. On the diagonal $q = d$ one has $z = d/2$, so every finite collection has transcendence degree one, regardless of how many rational dimensions are used. Consequently merely listing many transcendental optimal values would not demonstrate independence. The following construction does produce independence with actual integer dimensions.

Theorem 12.4 (Infinitely many algebraically independent integer-dimensional optima)**.** *For each prime p, choose any positive integer solution*

$$x_p^2 - py_p^2 = 1, \qquad x_p > 1,$$

and put $d_p = x_p - 1$, $q_p = x_p + 1$. Then the collection

$$\left\{\mathcal{V}_{d_p,q_p} : p \text{ prime}\right\} \tag{12.19}$$

is algebraically independent over $\overline{\mathbb{Q}}$, meaning that every finite subcollection is algebraically independent. Every d_p and q_p is a positive integer, so these are optimal expected radial-power payoffs for ordinary multidimensional Brownian bridges. Explicitly,

$$\mathcal{V}_{d_p,q_p} = \frac{(y_p\sqrt{p})^{(x_p+1)/2}}{1 + y_p\sqrt{p}/(x_p-1)} \exp\left(-\frac{y_p\sqrt{p}}{2}\right). \tag{12.20}$$

Proof. Here $q_p = d_p + 2$, so $N = 1$ and $P_1(z) = 1 + 2z/d_p$. The boundary polynomial reduces to $z^2 = d_p(d_p+2)/4$. Hence

$$z_p = \frac{\sqrt{d_p(d_p+2)}}{2} = \frac{y_p\sqrt{p}}{2} > 0.$$

Formula (12.20) follows from (12.16). The numbers $\sqrt{p}$ for distinct primes are rationally linearly independent. One way to see this is to use the sign-changing automorphisms of the multiquadratic field generated by these square roots: prime factorization makes their square classes independent, and changing the sign of just one square root in a putative linear relation isolates its coefficient. Nonzero rational factors $y_p/2$ preserve this independence. Theorem 12.3 now proves the claim.

For completeness, the positive integral Pell solutions used here exist without any arithmetic conjecture. Dirichlet's elementary pigeonhole approximation gives infinitely many positive integers r, s with $|r - s\sqrt{p}| < 1/s$. Their nonzero integral norms $r^2 - ps^2$ lie in a fixed bounded set. Infinitely many pairs therefore have the same nonzero norm k, and an infinite subcollection has the same residues of r, s modulo $|k|$. For two distinct members $(r_1, s_1), (r_2, s_2)$ of this subcollection, chosen with $r_1 + s_1\sqrt{p} > r_2 + s_2\sqrt{p}$, the quotient

$$\frac{r_1 + s_1\sqrt{p}}{r_2 + s_2\sqrt{p}} = \frac{r_1r_2 - ps_1s_2}{k} + \frac{s_1r_2 - r_1s_2}{k}\sqrt{p} = x + y\sqrt{p}$$

has integral coefficients, norm one, and value greater than one. Its conjugate is its positive reciprocal, so $x > 1$ and $y > 0$. This supplies a solution for each prime. □

Here are ten explicit choices, all satisfying the displayed Pell equation exactly. They exhibit one joint theorem, not ten unrelated arithmetic claims.

p	x_p	y_p	d_p	q_p	z_p
2	3	2	2	4	$\sqrt{2}$
3	2	1	1	3	$\sqrt{3}/2$
5	9	4	8	10	$2\sqrt{5}$
7	8	3	7	9	$3\sqrt{7}/2$
11	10	3	9	11	$3\sqrt{11}/2$
13	649	180	648	650	$90\sqrt{13}$
17	33	8	32	34	$4\sqrt{17}$
19	170	39	169	171	$39\sqrt{19}/2$
23	24	5	23	25	$5\sqrt{23}/2$
29	9801	1820	9800	9802	$910\sqrt{29}$

For example, the optimal values for $(d, q) = (2, 4)$ and $(1, 3)$ are algebraically independent, not just individually transcendental. The infinite family is deliberately selected to make its arithmetic accessible. Its probabilistic meaning is standard, while the selection and the arithmetic question are posed here; historical novelty is not inferred from the absence of a search match.

12.4 Target III: the linear Brownian-excursion value is still unresolved

Let $C = C_{3,1}$ be the established Brownian-excursion stopping boundary and put $z = C^2/2$. The preceding boundary theorem proves that C and z are transcendental, but it does not classify $\mathcal{V}_{3,1}$. Its exact reduction is

$$M(1/2; 3/2; z) = \int_0^1 e^{zt^2}\, dt, \qquad 2M(1/2; 3/2; z) = e^z, \qquad \mathcal{V}_{3,1} = 2Ce^{-C^2/2}. \tag{12.21}$$

Indeed differentiation of the integral gives $zf'(z) = (e^z - f(z))/2$, while smooth fit gives $zf' = f/2$. This proves the middle identity and then the objective formula. It agrees with the Brownian-excursion value formula in [34].

If the value were an algebraic number $v > 0$, its boundary variable would have to satisfy the two exact equations

$$2M(1/2; 3/2; z) - e^z = 0, \qquad 8z - v^2 e^{2z} = 0. \tag{12.22}$$

The specialization theorem for E-functions only controls their values at algebraic arguments; here z is already known to be transcendental. Nor does Lindemann–Weierstrass apply to this transcendental exponent. We have not proved a common-zero theorem that excludes (12.22). Thus neither irrationality nor transcendence of $\mathcal{V}_{3,1}$ is established here. No global historical open-status claim is made solely from our literature search. This target is explicitly recorded as unresolved by the present work, rather than being inferred from transcendence of the boundary.

Scope and accounting. There are two proved objective-value target families: the even-payoff transcendence theorem, whose main new step is polynomial elimination at a transcendental boundary, and the exact joint-relation theorem on algebraic-boundary values, with its infinite independent family. The latter uses classical Lindemann–Weierstrass and is not presented as a new transcendence theorem. The linear excursion value is a third investigated target without an unconditional arithmetic conclusion. General irrational payoff exponents, generic rational non-even payoffs outside the algebraic-boundary family, and ordinary exponential discounting are not resolved by the arguments above.

13 Transcendental extraction rates and an obstruction to holonomic boundary tables

The preceding arithmetic paper proved that the endpoint extraction constant $A(d)$ is irrational for every rational $d > 1/2$. It did not prove that these numbers are transcendental. We first examine $A(1)$ and $A(3/2)$ separately and make that limitation precise: under the remaining algebraic-irrational hypothesis, the relevant Kummer function is provably not an E-function. We then prove transcendence for a different, exactly specified constant of the same factorization problem: the critical intermediate rate for extracting an exponential factor. Finally, the large-order boundary asymptotic implies that the boundary table on every positive rational arithmetic progression is not polynomial-recursive over the algebraic numbers. These are transfers within the developed model; no named historical constant conjecture is asserted to be closed.

13.1 The two endpoint targets and a provable obstruction

We recall only the exact inputs needed from the earlier manuscript [30]. For $d > 1/2$ there is a unique

$$A(d) \in \big(\max(0, 1-d), 1/(4d)\big)$$

for which, with $a = A(d)$ and $\eta = a + d - 1$,

$$y(z) - \frac{z}{2(1-a)} y'(z) = 0, \qquad y(z) = M(a-1; \eta; z), \qquad z = 2(d-a). \tag{13.1}$$

Here $M = {}_1F_1$. The contact coordinate $x = d - a$ is a double zero of the boundary remainder density. The previous arithmetic argument excludes rational a by applying Beukers' theorem to (y, y'). That theorem cannot simply be extended by replacing "rational" with "algebraic" in the Kummer parameters.

Proposition 13.1 (The remaining algebraic hypothesis is outside the E class)**.** *Let $d > 0$ be rational and let a be algebraic irrational. Then $M(a-1; a+d-1; z)$ is not an E-function.*

Proof. Put $\alpha = a - 1$ and $\beta = a + d - 1$. Neither is an integer, and the series is nonpolynomial. If

$$M(\alpha; \beta; z) = \sum_{n \geq 0} \frac{(\alpha)_n}{(\beta)_n} \frac{z^n}{n!}$$

were an E-function, its formal Borel transform

$$\sum_{n \geq 0} \frac{(\alpha)_n}{(\beta)_n} z^n = {}_2F_1(\alpha, 1; \beta; z)$$

would be a G-function. Indeed the coefficient height and common-denominator bounds are the same, the latter series has positive radius of convergence, and its hypergeometric differential equation supplies holonomicity.

Galochkin's characterization, in the form reproved by Rivoal [36, Theorem 1], says that every irrational numerator parameter of a nonpolynomial hypergeometric G-function must pair with an irrational denominator parameter so that numerator minus denominator is a positive integer, after equal parameters have been cancelled. Here the only irrational numerator is α, the only irrational denominator is β, and $\alpha - \beta = -d < 0$. They do not cancel and cannot form such a pair. This contradicts the characterization. □

Thus the obstruction is an actual failure of the arithmetic-function hypothesis, not merely an omitted denominator estimate. Differential Galois calculations concern functional relations; they do not by themselves provide the missing arithmetic specialization statement.

Target $A(1)$: an incomplete-gamma reduction. Write $a = A(1)$, $p = 1-a$, so $0 < a < 1/4$ and $z = 2p$. The elementary contiguous identity

$$z y'(z) - p y(z) = -p e^z, \qquad y(z) = M(-p; 1-p; z),$$

follows termwise from its Taylor series. Equation (13.1) at $z = 2p$ is therefore equivalent to $y(z) = -e^z$. Using $y' = -(p/a) M(a; a+1; z)$ gives the exact positive-integral equation

$$(1-a) \int_0^1 t^{a-1} e^{2(1-a)t} \, dt = e^{2(1-a)}. \tag{13.2}$$

The integral is $M(a; a+1; 2(1-a))/a$, or the corresponding branch-consistent incomplete-gamma expression; see [39, 13.6.5]. If a were algebraic irrational, the order of that incomplete

gamma function would still be algebraic irrational. Murty and Saha's special-value theorem [37, Corollary 1] assumes rational order. It does not settle (13.2). Transcendence of the exponential on the right does not contradict equality with an unknown integral on the left. We obtain no transcendence proof for $A(1)$.

Target $A(3/2)$: a positive Euler integral. An alternative formulation makes the other obstruction explicit. For positive a, d with $a + d > 1$, put $U(z) = M(a; a + d; z)$. The inverse of $u^{-a}(u + 2)^{-d}$ is

$$k(x) = \frac{e^{-2x}x^{a+d-1}}{\Gamma(a + d)}U(2x),$$

and the endpoint remainder density is $k' + k$. At $x = d - a$, its vanishing is equivalent to

$$\int_0^1 \{2a - 1 + 2(d - a)t\}e^{2(d-a)t}t^{a-1}(1 - t)^{d-1}\,dt = 0. \tag{13.3}$$

This follows by differentiating the displayed k and applying Euler's integral for U, whose positive beta normalizer cancels. In particular, for $d = 3/2$ it becomes

$$\int_0^1 \{2a - 1 + (3 - 2a)t\}e^{(3-2a)t}t^{a-1}\sqrt{1 - t}\,dt = 0. \tag{13.4}$$

If a is algebraic irrational, t^{a-1} is not an algebraic function: continuation about zero multiplies it by $e^{2\pi i(a-1)}$, which is not a root of unity. Thus this is not an exponential integral with an algebraic integrand to which an algebraic-period assertion could automatically be applied. The same Kummer non-E obstruction in Proposition 13.1 applies. Differential transcendence of the incomplete gamma function as a function of its order [38] is not a theorem about its individual values. No irrational-order special-value theorem closing (13.4) has been established in this argument. The already proved irrationality of $A(3/2)$ remains the strongest individual arithmetic conclusion here.

13.2 A different boundary coordinate that is transcendental

Keep the first excess-power McKay transform

$$L(s) = \frac{b + s}{b}\left(\frac{b^2 - 1}{(b + s)^2 - 1}\right)^{\kappa}, \qquad b > 1, \quad \kappa > 1. \tag{13.5}$$

For a gamma factor of shape one and rate $\rho = b - c$, with $0 \le c \le 1$, the candidate remainder transform, after a positive normalization and a shift of the Laplace variable, is

$$F_{c,1}(w) = \frac{w(w + c)}{((w + c)^2 - 1)^{\kappa}}.$$

Define

$$\nu = \kappa - \frac{3}{2}, \qquad t = 2\kappa - 2, \qquad \ell_{\kappa}(x) = \frac{t}{x} + \frac{I_{\nu+1}(x)}{I_{\nu}(x)}, \quad x > 0. \tag{13.6}$$

The parameter t in this subsection is a positive constant, not the integration variable of (13.4).

We recall the classical arithmetic fact needed for this coordinate. For rational $\nu > -1$ and nonzero algebraic x, the ratio $I_{\nu+1}(x)/I_{\nu}(x)$, when interpreted through the normalized entire functions, is transcendental. Here is a verification tailored to our use. Put

$$H_{\nu}(z) = \sum_{n\ge 0} \frac{(z^2/4)^n}{n!(\nu + 1)_n}.$$

Its even exponential Taylor coefficients are $(1/2)_n/(\nu+1)_n$, and its odd coefficients vanish. Rational Pochhammer quotients have exponential coefficient and common-denominator bounds: at primes not dividing the fixed parameter denominators, the two arithmetic progressions have divisibility counts differing by at most one at each prime power, giving a factor dividing $\operatorname{lcm}(1,\dots,Cn)$; the finitely many remaining primes contribute $\exp(O(n))$. Thus H_ν and H'_ν are E-functions. They satisfy

$$H''_\nu + \frac{2\nu+1}{z} H'_\nu - H_\nu = 0.$$

Their logarithmic derivative cannot be rational. Otherwise an entire function of exponential type with finitely many zeros has form $e^{\lambda z}P(z)$; evenness forces $\lambda = 0$, whereas H_ν has infinitely many nonzero Taylor coefficients and is not a polynomial. Beukers' ordinary-point linear specialization theorem [35, Corollary 1.4] now applies to (H_ν, H'_ν) at every nonzero algebraic point. Since $H'_\nu/H_\nu = I_{\nu+1}/I_\nu$, it proves the asserted ratio transcendence. This is a classical input, already used in the preceding arithmetic paper.

Theorem 13.2 (A transcendental intermediate extraction rate)**.** *Suppose $\kappa > 1$ and ℓ_κ takes a value below one. Then it has a unique global minimum $c_\kappa \in (0,1)$ at a point $x_\kappa > 0$. Both satisfy the exact contact relation*

$$(1-c_\kappa^2)x_\kappa^2 + tc_\kappa x_\kappa - t = 0. \tag{13.7}$$

In the model (13.5), extraction of a shape-one gamma factor at rate $b-c$, $0 \le c \le 1$, is possible exactly when $c \le c_\kappa$. At the boundary its remainder density has exactly one positive zero, of order two, and the unrestricted maximal shape is exactly $H_{\mathrm{P}}(c_\kappa) = 1$.

If κ is rational, then c_κ and x_κ are both transcendental. The hypothesis that ℓ_κ takes a value below one holds for every real $1 < \kappa \le 7/6$. Consequently this supplies an explicit interval of rational input parameters producing transcendental critical rates. When $b > 1$ is algebraic, the physical threshold $b - c_\kappa$ is transcendental.

Proof. Let

$$f(x) = x^{\kappa-1/2} I_{\kappa-3/2}(x).$$

Positive constants do not matter in this proof. The Bessel series gives $f(x) = Cx^t H_\nu(x)$ with $C > 0$, so $f > 0$ and $f' > 0$, and $f'/f = \ell_\kappa$. At the two ends,

$$\ell_\kappa(x) \longrightarrow \infty \quad (x \downarrow 0), \qquad \ell_\kappa(x) = 1 + \frac{\kappa-1}{x} + O(x^{-2}) \quad (x \to \infty).$$

The latter is the standard Bessel expansion [39, 10.40.1]. Therefore a value below one implies an attained positive global minimum below one.

The Bessel equation transforms into

$$f'' - \frac{t}{x} f' + \left(\frac{t}{x^2} - 1\right) f = 0, \qquad \ell' = 1 + \frac{t\ell}{x} - \frac{t}{x^2} - \ell^2.$$

At a minimizing point this gives (13.7). For a fixed $0 < c < 1$, the polynomial $(1-c^2)x^2 + tcx - t$ is strictly increasing on the positive axis, starts negative, and tends to infinity. It has exactly one positive root. All global minimizers have the same value c_κ, so there is exactly one. Differentiating the Riccati identity at this stationary point gives

$$\ell''(x_\kappa) = \frac{t(2 - c_\kappa x_\kappa)}{x_\kappa^3} > 0, \tag{13.8}$$

since (13.7) implies $c_\kappa x_\kappa < 1$.

The inverse of $(w+c)/((w+c)^2-1)^\kappa$ is a positive constant times $e^{-cx}f(x)$. Since $t>0$, this function tends to zero at zero. Multiplication of its transform by w thus corresponds to differentiation, with no atom or boundary term:

$$g_{c,1}(x) = Ce^{-cx}f(x)(\ell_\kappa(x)-c), \qquad C>0.$$

After the prescribed positive exponential tilt it is an integrable signed density with total mass one. Uniqueness of Laplace transforms makes a probability remainder possible exactly when this density is nonnegative, that is, when $c \le c_\kappa$. At equality uniqueness of the minimizer and (13.8) give exactly one double zero.

If some larger shape $h>1$ were extractable at this rate, the shape-one remainder would be the convolution of its probability remainder μ with a gamma law of shape $h-1$. Put $M=2\kappa-1$. The quotient Laplace transform of μ is asymptotic to $Cs^{-(M-h)}$ at positive infinity, for $C>0$. Necessarily $h \le M$, since a probability Laplace transform is bounded. For $h<M$ this polynomial decay, and for $h=M$ its positive limit, show that μ has positive mass arbitrarily near zero: support bounded below by $\epsilon>0$ would instead force exponential decay. Convolving with a positive gamma density therefore gives a continuous strictly positive lower bound near every positive point, by restricting μ to $[0,x/2]$. This contradicts the continuous density's zero at x_κ. No shape larger than one is feasible, whereas shape one is feasible, so $H_{\mathrm{P}}(c_\kappa)=1$ without requiring a separate attainment theorem.

For the arithmetic assertion, first suppose c_κ were algebraic and κ rational. Equation (13.7), whose leading coefficient is nonzero, would force x_κ to be algebraic. Then

$$\frac{I_{\nu+1}(x_\kappa)}{I_\nu(x_\kappa)} = c_\kappa - \frac{t}{x_\kappa}$$

would be algebraic, contradicting the classical ratio result. Conversely, an algebraic x_κ makes (13.7) a nonzero quadratic equation over the algebraic numbers for c_κ, giving the same contradiction. Thus both numbers are transcendental. Subtracting c_κ from an algebraic b preserves transcendence.

It remains to verify the promised explicit range. Coefficientwise comparison of $H_{\nu+1}$ and H_ν gives, for $x>0$,

$$0 < \frac{I_{\nu+1}(x)}{I_\nu(x)} = \frac{x}{2(\nu+1)}\frac{H_{\nu+1}(x)}{H_\nu(x)} < \frac{x}{2\kappa-1}.$$

Taking $x=\sqrt{t(2\kappa-1)}$ therefore yields

$$\ell_\kappa(x) < 2\sqrt{\frac{2\kappa-2}{2\kappa-1}} \le 1 \qquad (1<\kappa\le 7/6).$$

The first inequality is strict also at $\kappa=7/6$, so the hypothesis holds throughout the stated interval. □

For example, $\kappa=9/8$ gives $x_\kappa \simeq 0.647079685$ and $c_\kappa \simeq 0.856687361$. These approximations merely identify the constants. Their transcendence follows from the exact argument above. This theorem changes the boundary coordinate: its unknown is the rate at fixed shape one, rather than the endpoint shape $\kappa - A(\kappa)$. It does not prove transcendence of $A(1)$, $A(3/2)$, or an arbitrary intermediate-rate shape.

13.3 The endpoint boundary table cannot obey an algebraic-coefficient recurrence

The older analytic manuscript established the precise asymptotic

$$A(d) \sim \frac{1}{2\sqrt{2\pi d}} \left(\frac{2}{e}\right)^d \qquad (d \to +\infty). \tag{13.9}$$

We use this as an inherited analytic input, not as a new estimate. It has a further exact arithmetic consequence that does not require knowing whether any individual $A(d)$ is transcendental.

Theorem 13.3 (No polynomial recurrence on a rational mesh)**.** *Let $u \in \mathbb{Q}$, $v \in \mathbb{Q}_{>0}$, and take all sufficiently large integers n so that $u+vn > 1/2$. There are no polynomials $p_0, \dots, p_r \in \overline{\mathbb{Q}}[X]$, not all zero, for which*

$$\sum_{j=0}^{r} p_j(n) A(u + v(n+j)) = 0 \tag{13.10}$$

holds for every sufficiently large n. Consequently the ordinary generating series of any such tail is not D-finite over $\overline{\mathbb{Q}}(z)$. In particular, $\sum_{n\geq 1} A(n) z^n$ has transcendental radius of convergence $e/2$ and satisfies no nonzero linear differential equation with algebraic polynomial coefficients. The function $A(d)$ itself is not algebraic over $\mathbb{C}(d)$.

Proof. Set $q = (2/e)^v$. This number is transcendental. Indeed if $v = r/s > 0$ in lowest terms and q were algebraic, then $q^s = 2^r e^{-r}$ would be algebraic, forcing e^r and then e algebraic, contrary to Hermite's theorem. From (13.9), for each fixed j,

$$\frac{A(u + v(n+j))}{A(u+vn)} \longrightarrow q^j.$$

Suppose (13.10) held. Let D be the maximum of the degrees of its nonzero coefficient polynomials, and let c_j be the coefficient of X^D in p_j. Divide by $n^D A(u+vn)$ and take the limit. The result is $\sum_j c_j q^j = 0$, a nonzero polynomial equation with algebraic coefficients, contradiction.

For completeness, equating Taylor coefficients in a polynomial-coefficient differential equation gives a polynomial recurrence for the series coefficients: derivatives contribute falling factorials and multiplication by a monomial shifts the index. A nonzero differential operator gives a nonzero such recurrence for all sufficiently large indices. Thus the excluded recurrence also excludes D-finiteness. The radius follows directly from $A(n)^{1/n} \to 2/e$.

Finally suppose A satisfied a nonzero polynomial relation over $\mathbb{C}(d)$. Clearing denominators gives $\sum_{j=0}^{m} P_j(d) A(d)^j = 0$, with $P_j \in \mathbb{C}[d]$. Let j_0 be the smallest index for which P_{j_0} is nonzero. After division by $A(d)^{j_0}$, every remaining term with $j > j_0$ tends to zero exponentially by (13.9), even after multiplication by its polynomial. This would force the nonzero polynomial $P_{j_0}(d)$ to tend to zero on the positive real axis, an impossibility. □

This is an obstruction to exact linear differential or recurrence compression of the boundary family over algebraic coefficients. It is not a proof that an individual entry is transcendental. The conclusion is stated over $\overline{\mathbb{Q}}$: the proof does not exclude polynomial recurrences with arbitrary complex coefficients. Nor does it establish differential transcendence of $A(d)$, an irrationality measure for its individual values, or the arithmetic of every noninteger-shape intermediate boundary.

Result accounting. There are two proved target families: the transcendental critical rate for an exponential factor, and the recurrence obstruction for the endpoint boundary table. The separate investigations of $A(1)$ and $A(3/2)$ remain unsuccessful at proving transcendence. Their failure is supported by an exact non-E theorem and explicit integral reductions. Examples at different rational κ, rational meshes, or algebraic base rates are not counted separately.

14 Positive arithmetic kernels and the two missing boundary values

This section makes three further attempts at individual-constant problems: Euler's constant γ, the Euler–Gompertz constant δ, and Catalan's constant G; the last argument also treats the same polynomial approach to $\zeta(5)$ and all integer zeta values. The new results are exact obstructions for larger kernel classes and an explicit limitation of arithmetic inference from finite differences. None identifies γ, δ, G, or $\zeta(5)$ as irrational.

We use

$$\gamma = -\int_0^\infty e^{-t}\log t\,dt, \qquad \delta_x = \int_0^\infty \frac{e^{-t}}{t+x}\,dt, \qquad \delta = \delta_1.$$

The preceding manuscript [31] solved the square-kernel problem for δ and a beta-kernel denominator calculation for G. Here the former is enlarged to *every* integer polynomial nonnegative on the half-line, and the latter to *every* integer polynomial nonnegative on the unit interval. These are restricted approximation problems, not assertions about all integer linear forms.

14.1 The full positive polynomial cone for the Gompertz constant

Lemma 14.1 (Two weighted leading-coefficient bounds)**.** *For $\alpha \in \{0,1\}$ and a real polynomial A of degree n with leading coefficient a, put*

$$J_\alpha(A) = \int_0^\infty \frac{t^\alpha e^{-t}A(t)^2}{1+t}\,dt.$$

Then

$$J_\alpha(A) \geq \frac{a^2 n!(n+\alpha)!}{2n+\alpha+2}. \tag{14.1}$$

Proof. Let $\pi_n^{(\alpha)}$ be the monic Laguerre polynomial for $t^\alpha e^{-t}dt$. Rodrigues' formula

$$t^\alpha e^{-t}\pi_n^{(\alpha)}(t) = (-1)^n \frac{d^n}{dt^n}(e^{-t}t^{n+\alpha})$$

shows, by integration by parts, that it is orthogonal to every polynomial of degree less than n, and that its squared norm is $n!(n+\alpha)!$. The coefficient of t^{n-1} is $-n(n+\alpha)$. Comparison with $\pi_{n+1}^{(\alpha)}$ and orthogonality therefore give

$$\int_0^\infty t^{\alpha+1}e^{-t}\big(\pi_n^{(\alpha)}(t)\big)^2 dt = (2n+\alpha+1)n!(n+\alpha)!.$$

The inner product of A with $\pi_n^{(\alpha)}$ is $an!(n+\alpha)!$. Cauchy–Schwarz, with factors $A/\sqrt{1+t}$ and $\sqrt{1+t}\,\pi_n^{(\alpha)}$, proves (14.1). □

Theorem 14.2 (Exact minimum over the full half-line cone)**.** *Let $\mathcal{P}$ be the set of nonzero $P \in \mathbb{Z}[t]$ such that $P(t) \geq 0$ for $t \geq 0$. Then*

$$\min_{P \in \mathcal{P}} \int_0^\infty \frac{e^{-t}P(t)}{1+t}\,dt = 4\delta - 2 > 0. \tag{14.2}$$

The unique minimizing polynomial is $P(t) = (t-1)^2$. Every integral in (14.2) belongs to $\mathbb{Z}\delta + \mathbb{Z}$.

Proof. We first record rational bounds with certificates. Direct polynomial division and the moments $\int_0^\infty t^k e^{-t}dt = k!$ give

$$\int_0^\infty \frac{e^{-t}(t-1)^2}{1+t}dt = 4\delta - 2 > 0.$$

With $Q(t) = t^4 - 20t^3 + 120t^2 - 240t + 120$, the same calculation gives

$$\int_0^\infty \frac{e^{-t}tQ(t)^2}{1+t}dt = 150300 - 251001\delta > 0.$$

These strictly positive integrals imply

$$\frac{1}{2} < \delta < \frac{100}{167} < \frac{3}{5}, \qquad 0 < 4\delta - 2 < \frac{2}{5}. \tag{14.3}$$

Every real polynomial nonnegative on $[0,\infty)$ has a representation

$$P(t) = A(t)^2 + tB(t)^2 \tag{14.4}$$

with real polynomials A, B. Here is an elementary factorization proof. Positive real roots have even multiplicity. A negative real root gives a factor $t + c = (\sqrt{c})^2 + t$, $c > 0$, and the factor t has the same form with zero first square. An irreducible real quadratic can be written

$$t^2 - 2at + a^2 + b^2 = (t-r)^2 + t\,2(r-a), \qquad r = \sqrt{a^2+b^2}, \quad b \neq 0.$$

All these factors have the required form, as does a positive constant. Finally the identity

$$(A^2 + tB^2)(C^2 + tD^2) = (AC - tBD)^2 + t(AD + BC)^2$$

proves (14.4). The positive highest terms of the two summands cannot cancel. Thus if $\deg P = 2n$ with leading coefficient p, then $\deg A = n$ and its leading coefficient squared is p; if $\deg P = 2n+1$, the corresponding statement holds for B. The integer p is at least one.

For even degree $2n \geq 4$, Lemma 14.1 gives a lower bound $(n!)^2/(2n+2) \geq 2/3$. For odd degree $2n+1 \geq 3$, it gives $n!(n+1)!/(2n+3) \geq 2/5$. Both sequences increase in their stated ranges. By (14.3) these degrees cannot minimize. A constant polynomial has integral at least δ. A polynomial $at + c$ nonnegative on the half-line has nonnegative integer coefficients, so its nonzero integral is at least $\min(\delta, 1-\delta) > 2/5$.

For degree two with leading coefficient $p \geq 2$, the even-degree bound is $p/4 \geq 1/2$. It remains to treat $P = t^2 + ut + v$ with $u, v \in \mathbb{Z}$. If $u \geq 0$, then $v \geq 0$ and the integral is at least δ. If $u = -k < 0$, nonnegativity implies $v \geq \lceil k^2/4 \rceil$, and the integral is

$$-k + (1 + k + v)\delta.$$

For $k \geq 4$, this exceeds $(k-2)^2/8 \geq 1/2$, using $\delta > 1/2$. For $k = 1$ the least value is $3\delta - 1 > 4\delta - 2$; for $k = 3$ it is $7\delta - 3 > 4\delta - 2$. For $k = 2$, the least value is $4\delta - 2$ and equality requires $v = 1$. This proves the minimum and uniqueness. Finally, division $P(t) = P(-1) + (t+1)R(t)$ with $R \in \mathbb{Z}[t]$ proves the integer linear-form assertion. □

What this closes, and what it leaves. The theorem solves a self-posed lattice optimization over all degrees; no priority assertion is made. It strictly includes the preceding square class because the factors in (14.4) need not have integer coefficients. Rational normalization cannot help within this construction: if $P = Q/d$, $Q \in \mathcal{P}$, clearing that specified denominator gives the integral of Q, bounded below as above. Signed kernels remain a different problem. Indeed $P(t) = q - p(1+t)$ realizes the arbitrary integer form $q\delta - p$, so dropping positivity would turn the proposed lower bound into an unproved arithmetic assertion about δ. The multiple-orthogonal approximants of Wolfs and Van Assche [40] are not excluded by Theorem 14.2.

14.2 A full compact-interval obstruction for Catalan and zeta kernels

Lemma 14.3 (A universal positive integer-polynomial bound)**.** *Suppose $w(t) \geq c(1-t)^r$ on $(0,1)$, with $c > 0$ and $r \geq 0$. For every nonzero $P \in \mathbb{Z}[t]$ nonnegative on $[0,1]$, of degree $d \geq 1$,*

$$\int_0^1 w(t)P(t)dt \geq \frac{c}{r+1}\, 4^{-d}(8d^3)^{-(r+1)}. \tag{14.5}$$

An infinite integral is allowed on the left.

Proof. Write $M = \max_{[0,1]} P > 0$. The monic Chebyshev extremal inequality implies $M \geq 2^{1-2d}$. For completeness, the shifted polynomial $2^{1-2d}T_d(2t-1)$ is monic and alternates between $\pm 2^{1-2d}$ at $d+1$ ordered points. A monic polynomial of strictly smaller supremum would differ from it by a degree-at-most-$d-1$ polynomial with at least d sign changes, which is impossible. Divide the leading coefficient of P, whose absolute value is at least one, to apply this argument.

Expand $P(t) = \sum_{j=0}^{d} c_j T_j(2t-1)$. The cosine integral formulas give $|c_j| \leq 2M$ for $j \geq 1$. Since $|T_j'(u)| \leq j^2$ on $[-1,1]$, we have

$$\|P'\|_\infty \leq 4M \sum_{j=1}^{d} j^2 \leq 4Md^3.$$

The derivative bound for T_j follows from $T_j(\cos\theta) = \cos(j\theta)$ and $|\sin(j\theta)| \leq j|\sin\theta|$, including endpoints by continuity. At a maximizing point, therefore, $P \geq M/2$ throughout an interval in $[0,1]$ of length at least $h = (8d^3)^{-1}$. Among intervals of length h, the integral of $(1-t)^r$ is smallest at the right endpoint and equals $h^{r+1}/(r+1)$. Thus the integral is at least $cMh^{r+1}/(2(r+1))$. The lower bound on M proves the lemma. □

Theorem 14.4 (Standard denominator clearance fails on both full cones)**.** *Let $P \in \mathbb{Z}[t]\setminus\{0\}$ be nonnegative on $[0,1]$, of degree $d \geq 1$. Put $D_d = \operatorname{lcm}(1,\dots,d)$ and $O_d = \operatorname{lcm}(1,3,\dots,2d-1)$. Define*

$$C(P) = \int_0^1 \frac{-\log x}{1+x^2} P(x^2)dx, \qquad Z_s(P) = \frac{1}{(s-1)!}\int_0^1 \frac{(-\log x)^{s-1}P(x)}{1-x}dx \quad (s \geq 2).$$

Then

$$C(P) \geq \frac{4^{-d}}{1024d^6}, \tag{14.6}$$

$$Z_s(P) \geq \frac{4^{-d}}{(s-1)!(s-1)(8d^3)^{s-1}}. \tag{14.7}$$

Moreover $O_d^2 C(P) \in \mathbb{Z}G + \mathbb{Z}$ and $D_d^s Z_s(P) \in \mathbb{Z}\zeta(s) + \mathbb{Z}$. Uniformly over these polynomials as $d \to \infty$,

$$\log(O_d^2 C(P)) \geq (4 - \log 4)d + o(d), \tag{14.8}$$

$$\log(D_d^s Z_s(P)) \geq (s - \log 4)d + o(d). \tag{14.9}$$

Thus no sequence in either cone, using these specified clearances, gives nonzero integer forms tending to zero.

Proof. After $t = x^2$, the Catalan weight is $(-\log t)/(4\sqrt{t}(1+t)) \geq (1-t)/8$. Use Lemma 14.3 with $c = 1/8, r = 1$. For zeta, $-\log t \geq 1 - t$ gives the weight lower bound $(1-t)^{s-2}/(s-1)!$, so use $c = 1/(s-1)!, r = s-2$.

Division of $P(t)$ by $1 + t$ gives

$$C(P) = P(-1)G + \sum_{k=0}^{d-1} \frac{q_k}{(2k+1)^2}, \qquad q_k \in \mathbb{Z}.$$

For $P(t) = \sum_{k=0}^{d} a_k t^k$, geometric-series integration gives

$$Z_s(P) = P(1)\zeta(s) - \sum_{k=0}^{d} a_k H_k^{(s)}, \qquad H_k^{(s)} = \sum_{j=1}^{k} j^{-s}.$$

These prove the integrality claims. The prime number theorem gives $\log D_d = d + o(d)$ and $\log O_d = 2d + o(d)$; exclusion of the power of 2 changes the latter logarithm by only $O(\log d)$. Combining these with (14.6)–(14.7) proves the logarithmic lower bounds, whose leading constants are positive for every integer $s \geq 2$. If degrees remain bounded, Lemma 14.3 gives a positive lower bound depending only on that bound on the degree. Constant polynomials also have a positive lower bound. This proves the final assertion for arbitrary sequences of degrees. □

This includes G, $\zeta(5)$, and $\zeta(7)$ as one obstruction family, not as separately solved conjectures. It improves the beta-only calculation by allowing arbitrary positive integer-polynomial shapes. It concerns the stated least-common-multiple multipliers, not the minimal reduced denominators: a proven arithmetic cancellation could still change the conclusion for a different clearance. Also the target coefficients $P(-1)$ and $P(1)$ must remain nonzero. The theorem is compatible with the much richer constructions of Brown–Zudilin [22].

14.3 Euler's constant: an indefinite lattice criterion and its circular input

The positive cone behaves differently for the logarithmic weight because $\log t$ changes sign. This gives a precise connection to a powerful homogeneous-dynamics theorem, but also exposes the missing hypothesis.

Proposition 14.5 (A ternary criterion for γ)**.** *For $v = (a, b, c) \in \mathbb{Z}^3$, put*

$$Q_\gamma(v) = \int_0^\infty e^{-t} \log t \, (a + bt + ct^2)^2 dt.$$

Its symmetric matrix is

$$M_\gamma = \begin{pmatrix} -\gamma & 1-\gamma & 3-2\gamma \\ 1-\gamma & 3-2\gamma & 11-6\gamma \\ 3-2\gamma & 11-6\gamma & 50-24\gamma \end{pmatrix}. \tag{14.10}$$

This form is nondegenerate of signature $(2, 1)$. The following are equivalent:

1. *γ is irrational;*

2. *Q_γ is not a real scalar multiple of a rational form;*

3. *$Q_\gamma(\mathbb{Z}^3)$ is dense in $\mathbb{R}$;*

4. *for every $\varepsilon > 0$, there is $v \in \mathbb{Z}^3$ with $0 < |Q_\gamma(v)| < \varepsilon$.*

Proof. The moments are $\int t^k e^{-t} \log t \, dt = k!(H_k - \gamma)$, which proves (14.10). Strict Jensen's inequality for an exponential random variable gives $\gamma > 0$; restricting its negative logarithmic integral to $(0, 1)$ gives $\gamma < \int_0^1 (-\log t) dt = 1$. The determinant is

$$\det M_\gamma = -11 + 3\gamma + 10\gamma^2 - 4\gamma^3 < -2,$$

since the polynomial increases from -11 to -2 on $[0, 1]$. The first diagonal entry is negative and the last positive. The negative determinant therefore forces exactly one negative eigenvalue.

If γ is rational, the matrix is rational. Conversely, if the matrix is a scalar multiple of a rational matrix, the ratio of its $(1, 2)$ and $(1, 1)$ entries is rational. That ratio is $1 - 1/\gamma$, so γ is rational. This proves the equivalence of the first two conditions. Margulis' theorem on the Oppenheim conjecture [41] applies to a nondegenerate indefinite irrational ternary form and gives density, proving the implication from the second to the third. Density implies the fourth. Finally, expansion gives

$$Q_\gamma(v) = A(v) - B(v)\gamma, \qquad A(v), B(v) \in \mathbb{Z}, \quad B(v) = \int_0^\infty e^{-t}(a + bt + ct^2)^2 dt > 0$$

for $v \neq 0$. If $\gamma = p/q$ were rational, every nonzero form would have absolute value at least $1/q$, contradicting the fourth condition. □

This is a self-posed equivalent criterion, not a proof of any of its conditions. Applying Oppenheim by asserting that the displayed matrix is irrational would assume exactly the desired irrationality of γ. The point of the calculation is that polynomial degree two already gives a genuine indefinite lattice problem; positivity of the polynomial square does not imply positivity of this logarithmic integral. A new effective search theorem for this specific form would still have to establish its required Diophantine hypothesis independently.

14.4 Curvature recovery and the two arithmetic boundary values

Let $\mathcal{F}_{\exp} = \overline{\mathbb{Q}}(e^a : a \in \overline{\mathbb{Q}})$. The preceding manuscript proves that the values $\mathrm{Ein}(x)$ at distinct nonzero algebraic x are algebraically independent over $\mathcal{F}_{\exp}$ [31]. This uses classical Kolchin–Ostrowski and Siegel–Shidlovskii theory; we do not count its following linear consequences as a new independence theorem. For an integer $b \geq 2$, put

$$F_j = \mathrm{Ein}(b^j), \quad Y_j = e^{-b^j} \delta_{b^j}, \quad C_j = Y_j - 2Y_{j+1} + Y_{j+2}, \qquad j \geq 0.$$

The classical split identity gives $Y_j = F_j - \gamma - j \log b$.

Proposition 14.6 (Exact recovery from independent curvatures)**.** *The sequence $(C_j)_{j \geq 0}$ is algebraically independent over $\mathcal{F}_{\exp}$, every C_j is positive, and*

$$Y_j = \sum_{r=j}^{\infty} (r - j + 1) C_r. \tag{14.11}$$

For $N \geq j$, the error after $r = N$ lies strictly between zero and

$$(N - j + 2)\frac{e^{-b^{N+1}}}{b^{N+1}}.$$

Moreover

$$\mathcal{F}_{\exp}(Y_j : j \geq 0) = \mathcal{F}_{\exp}(C_j : j \geq 0)(Y_0, Y_1).$$

Proof. The common affine terms cancel, so $C_j = F_j - 2F_{j+1} + F_{j+2}$. Any finite collection of these linear forms has independent coefficient rows: in a nonzero row combination, the row with largest index has a nonzero coefficient in its last column. Independent linear forms in algebraically independent variables are algebraically independent, proving the first assertion.

Write $Y(x) = \int_1^\infty e^{-xt}dt/t$. Then $Y(bx) \leq e^{-(b-1)x}Y(x)$, so $Y_{j+1} < Y_j/2$ for $b \geq 2$ and $j \geq 0$. Thus $C_j > 0$. Finite summation gives

$$\sum_{r=j}^{N}(r - j + 1)C_r = Y_j - (N - j + 2)Y_{N+1} + (N - j + 1)Y_{N+2}.$$

Since $0 < Y_j \leq e^{-b^j}/b^j$, passage to the limit proves the recovery formula and the stated positive error bound. The recurrence determines every Y_j from Y_0, Y_1 and finitely many C_r; conversely each C_r is a linear combination of three Y's. This proves the field identity. □

The recovery is analytically complete. Its arithmetic limitation is not removed by the positivity or the very rapid error bound. The following countermodel makes that limitation exact.

Theorem 14.7 (Independent curvatures can have two prescribed rational boundaries)**.** *Let $K \subset \mathbb{R}$ be a countable field, and let rational numbers $q_0 > 2q_1 > 0$ be given. Fix $b > 1$. There exists a positive sequence $(c_j)_{j\geq 0}$, algebraically independent over K, such that $c_j = O(e^{-b^j})$ as $j \to \infty$ and*

$$y_j = \sum_{r=j}^{\infty}(r - j + 1)c_r$$

satisfies $y_0 = q_0$, $y_1 = q_1$, and $y_j - 2y_{j+1} + y_{j+2} = c_j$.

Proof. Choose positive numbers $\varepsilon_j \leq e^{-b^j}$ for $j \geq 2$ such that $\sum_{j\geq 2} 2j\varepsilon_j < q_1/2$. Let the tail coordinates vary in the compact metrizable product $X = \prod_{j\geq 2}[\varepsilon_j, 2\varepsilon_j]$. For every point of X, define

$$c_1 = q_1 - \sum_{r\geq 2} rc_r, \qquad c_0 = q_0 - 2q_1 + \sum_{r\geq 2}(r - 1)c_r.$$

These are continuous functions on X, are positive, and give the two prescribed sums. All series converge uniformly on X.

For any nonzero polynomial over K in a finite selection of the coordinates c_j, its zero set in X is closed and has empty interior. To see the latter assertion, take a basic open cylinder. Besides the finitely many selected tail coordinates, vary two further coordinates c_k, c_l, with distinct $k, l \geq 2$, outside all indices involved in the polynomial or cylinder. Their contributions to (c_0, c_1) have matrix

$$\begin{pmatrix} k - 1 & l - 1 \\ -k & -l \end{pmatrix}, \qquad \det = l - k \neq 0.$$

Together with the selected tail coordinates this gives a linear map of full row rank onto those coordinates and (c_0, c_1). Varying them in open subintervals of their boxes produces an open

set in the finite coordinate space. A nonzero polynomial cannot vanish on that open set. This remains valid if one or both of c_0, c_1 do not occur in the polynomial, by projecting away unused coordinates.

There are only countably many such polynomials because K is countable. The Baire category theorem on the nonempty complete space X supplies a point outside all their zero sets. Its full sequence is algebraically independent over K. The bounds on the boxes give the stated tail decay, and absolute convergence permits termwise subtraction in the definition of y_j, giving its second difference and the required boundary values. □

Apply the countermodel with $K = \mathcal{F}_{\exp} \cap \mathbb{R}$, which is countable. Since $\mathcal{F}_{\exp}$ is closed under complex conjugation, independence of real numbers over this real subfield is equivalent to independence over $\mathcal{F}_{\exp}$: take real and imaginary parts of a putative polynomial relation. The countermodel does not supply a counterexample about the actual Gompertz integrals or the specific values F_j. It proves that algebraic independence of all curvatures, positivity, rapid tail decay, and exact boundary reconstruction alone do not classify even the first two boundary values: in the countermodel both are rational. Therefore the preceding defect-two family statement cannot be turned into an individual classification by those properties alone. One must use additional arithmetic information specific to γ, $\log b$, or the exponential-integral connection. The mixed arithmetic-Gevrey problems of Fischler–Rivoal [42] locate related missing separation statements; they are conjectural inputs, not hypotheses silently used here.

15 Prime block complexity, polynomial carries, and modular neighbours

We examine three specific transcendence targets,

$$P_2 = \sum_{p\ mprime} 2^{-p}, \qquad L_{\mathbb{P}} = \sum_{p\ mprime} \frac{1}{2^p - 1}, \qquad E = \sum_{n\geq 1} \frac{1}{2^n - 1}.$$

None is proved transcendental here. The first result is a quantitative transfer from Maynard's prime-cluster theorem to the block complexity of the prime indicator. It excludes a tempting transcendence method for P_2. The second result makes the additional carry information needed for $L_{\mathbb{P}}$ explicit. The third identifies three algebraically independent linear combinations of reciprocal-Mersenne moments using Nesterenko's theorem. This last transfer is a consequence of established modular transcendence theory, not a new solution of the problem for E.

15.1 Prime binary digits: superpolynomial complexity with zero entropy

Write $u_j = 1$ if j is prime and $u_j = 0$ otherwise, for $j \geq 1$. Let $p_u(n)$ be the number of distinct contiguous length-n words in u. There are infinitely many zero digits, so u is the canonical binary expansion of P_2. Let $p_u^{\mathrm{rec}}(n)$ count only words that occur infinitely often.

The analytic input is the following consequence of Maynard's Theorem 3.4 in [43]. There is an absolute constant C_0 such that for every positive integer k there is an integer $M = M(k)$, with $k \leq M \leq \exp(C_0 k)$, having this property: every fixed admissible family of M distinct positive integer linear forms has infinitely many arguments at which at least k forms are prime. Here admissible means that their product has no fixed prime divisor. The stated consequence follows by taking a fixed interval parameter in that theorem and then letting its size tend to infinity; all coefficients in our application are fixed during that limit.

Theorem 15.1 (Prime-word complexity transfer)**.** *Fix $k \geq 1$, choose $M = M(k)$ as above, and put $W = \prod_{p \leq M} p$ and $K = \lfloor n/W \rfloor$. For $K \geq M$,*

$$p_u^{\mathrm{rec}}(n) \geq \frac{\binom{K}{k}}{\binom{M}{k}}. \tag{15.1}$$

Consequently there are absolute positive constants c, C such that, for all sufficiently large n,

$$\exp\{c \log n \log\log n\} \leq p_u^{\mathrm{rec}}(n) \leq p_u(n) \leq \exp\{Cn/\log\log n\}. \tag{15.2}$$

In particular, the prime word has zero topological entropy, while its block complexity grows faster than every fixed power of n.

Proof. Let $G = \{W, 2W, \ldots, KW\} \subseteq \{1, \ldots, n\}$. For every M-element subset $H \subseteq G$, assign distinct primes $q_i > n$ to the positions $i \in \{1, \ldots, n\} \setminus H$. The Chinese remainder theorem supplies r modulo

$$A = W \prod_{i \notin H} q_i$$

such that $r \equiv 1 \pmod{W}$ and $r \equiv -i \pmod{q_i}$. Choose a positive representative r. Consider the M forms $At + r + h$, $h \in H$.

For a prime dividing W, their constant terms are all 1 modulo that prime. For $q_i \mid A/W$, the difference $h - i$ is nonzero and has absolute value less than q_i, so no constant term vanishes modulo q_i. For a prime $p \nmid A$, necessarily $p > M$; the M forms exclude at most M residues of t modulo p. The family is therefore admissible. Maynard's theorem supplies infinitely many t with at least k prime values among these forms. At every sufficiently large such t, each position outside H is composite because $At + r + i$ is a multiple of q_i exceeding q_i.

Among the finitely many resulting words, at least one recurs infinitely often. Its support S satisfies $S \subseteq H$ and $|S| \geq k$. Choose one such word for each H. A fixed output support S, of size $s \geq k$, can be chosen by at most $\binom{K-s}{M-s} \leq \binom{K-k}{M-k}$ different sets H. Counting the $\binom{K}{M}$ inputs proves

$$p_u^{\mathrm{rec}}(n) \geq \frac{\binom{K}{M}}{\binom{K-k}{M-k}} = \frac{\binom{K}{k}}{\binom{M}{k}}.$$

For the uniform lower bound, enlarge C_0 if necessary and take $k = \lfloor (\log\log n)/(4C_0) \rfloor$. Then $M \leq (\log n)^{1/4}$ and

$$\log W \leq \log(M!) \leq M \log M = o(\log n).$$

Thus $K/M \geq n^{1/2}$ for large n. Since $(K - j)/(M - j) \geq K/M$ for $0 \leq j < k$, the right side of (15.1) is at least $(K/M)^k$. This gives the lower bound in (15.2). There is no interchange of a prime-counting limit with k: for each fixed block length n, all the finitely many forms have already been fixed before Maynard's theorem is applied.

For the upper bound, let $y \geq 3$ and $Q = \prod_{p \leq y} p$. A block starting after position y can have ones only in residue classes coprime to Q. For each starting residue modulo Q, at most $n\varphi(Q)/Q + Q$ positions are allowed. There are at most Q starting residues and at most $\lceil y \rceil$ exceptional early blocks. Therefore

$$p_u(n) \leq \lceil y \rceil + Q\, 2^{n\varphi(Q)/Q + Q}. \tag{15.3}$$

The finite Euler product satisfies

$$\frac{Q}{\varphi(Q)} = \prod_{p \leq y} (1 - p^{-1})^{-1} \geq \sum_{m \leq y} \frac{1}{m} \geq \log y,$$

because every integer $m \le y$ has all its prime factors at most y. Choose $y = \lfloor \log n/(2\log\log n)\rfloor$. For large n, $Q \le y! \le y^y \le n^{1/2}$ and $\log y \asymp \log\log n$. Substitution in (15.3) proves the upper bound. □

Adamczewski–Bugeaud's criterion states that an irrational algebraic number must have $p(n)/n \to \infty$ in each integer base [44, Theorem 1]. Theorem 15.1 shows that this necessary condition is satisfied by P_2 with a large margin. In particular, obtaining zero entropy or a subexponential complexity upper bound cannot establish its transcendence by that criterion. This is a method exclusion, not evidence that P_2 is algebraic. The argument is a direct consequence of Maynard's theorem; we make no global priority claim for this consequence.

15.2 Prime Lambert series: finite certificates for polynomial carries

Absolute convergence gives

$$L_{\mathbb{P}} = \sum_{n\ge 1} \frac{\omega(n)}{2^n},$$

where $\omega(n)$ counts distinct prime factors. Its irrationality is already a theorem of Tao and Teräväinen [45, Theorem 1.3]. To seek transcendence one must exclude every nonconstant polynomial relation, not only linear ones. The following elementary certificate identifies exactly which coefficient combinations a carry argument would have to control.

Theorem 15.2 (Finite polynomial-carry certificate)**.** *Let $F(z) = \sum_{n\ge1} a_n z^n$, where a_n are integers with $|a_n| \le n$, and set $\xi = F(1/2)$. For a nonconstant polynomial $A(X) = \sum_{j=0}^{D} c_j X^j \in \mathbb{Z}[X]$, let $H = \max_j |c_j|$ and write*

$$A(F(z)) = \sum_{n\ge 0} b_n z^n.$$

For $N \ge 0$ and $J \ge 1$, define the rational number and explicit bound

$$\begin{aligned} S_{N,J} &= \sum_{m=1}^{J} \frac{b_{N+m}}{2^m}, \\ B_{N,J} &= HD(2D-1)!\, 2^{2D-1-J}(N+J+D+1)^{2D-1}. \end{aligned} \tag{15.4}$$

If $\operatorname{dist}(S_{N,J},\mathbb{Z}) > B_{N,J}$, then $A(\xi) \ne 0$. Furthermore, ξ is transcendental if and only if every nonconstant $A \in \mathbb{Z}[X]$ has such a certificate for some N, J. If ξ is transcendental, one may fix any N in advance and find a sufficiently large J separately for each A.

Proof. Put $s = 2D-1$. Coefficientwise absolute values in $F(z)^j$ are bounded by those in $(z/(1-z)^2)^j$. For $n \ge 1$ this gives

$$|b_n| \le H \sum_{j=1}^{D} \binom{n+j-1}{2j-1} \le HD(n+D)^s,$$

where a binomial coefficient is zero when $n < j$. All series below converge absolutely. Set $T_N = \sum_{m\ge1} b_{N+m} 2^{-m}$. Writing $m = J+1+t$ and using $(t+1)^s \le s!\binom{t+s}{s}$ gives

$$\begin{aligned} |T_N - S_{N,J}| &\le HD\, 2^{-J-1}(N+J+D+1)^s \sum_{t\ge0} (t+1)^s 2^{-t} \\ &\le HD\, 2^{-J-1}(N+J+D+1)^s s! 2^{s+1} = B_{N,J}. \end{aligned}$$

On the other hand,

$$2^N A(\xi) = \sum_{n=0}^{N} b_n 2^{N-n} + T_N. \tag{15.5}$$

The finite sum is an integer. Thus $A(\xi) = 0$ would force T_N to be an integer for every N, which is incompatible with the stated strict distance test.

If ξ is algebraic, take a nonconstant integer polynomial vanishing at ξ; no test for that polynomial can succeed. If ξ is transcendental, then $A(\xi)$ is transcendental for every nonconstant integer polynomial A: otherwise an algebraic equation for $A(\xi)$ would compose with A to give an equation for ξ. Consequently $T_N \notin \mathbb{Z}$ for every fixed N. Since $S_{N,J} \to T_N$ and $B_{N,J} \to 0$, the strict test eventually holds. This proves both directions. □

For $a_n = \omega(n)$, the coefficients in this test are explicitly

$$b_n = \sum_{j=1}^{D} c_j \sum_{\substack{n_1+\cdots+n_j=n \\ n_i \geq 1}} \omega(n_1)\cdots\omega(n_j) \qquad (n \geq 1). \tag{15.6}$$

For fixed D, H, the remainder in (15.4) can be made smaller than any fixed positive tolerance using $J = O_{D,H}(\log(N+2))$ (with an additional constant depending on the tolerance). But this does not prove the strict separation from integers. Nor is this a local algorithm using only $\omega(N+1), \ldots, \omega(N+J)$: the convolution coefficients require information throughout the prefix up to $N+J$. The missing arithmetic step is to produce strict certificates uniformly over all nonconstant polynomials. The known irrationality theorem supplies no such conclusion for (15.6) in degrees $D \geq 2$. Thus the criterion is an explicit reduction and a finite verification tool; its equivalence must not be counted as a transcendence proof.

15.3 Erdős–Borwein: modular combinations and the digit route

There is a useful positive comparison with modular transcendence. For $0 < q < 1$, put

$$A_r(q) = \sum_{n \geq 1} \left(\frac{q^n}{1-q^n}\right)^r, \qquad r \geq 1.$$

In particular $A_1(1/2) = E$.

Proposition 15.3 (Three modular combinations of six moments)**.** *For every real algebraic q with $0 < q < 1$, the following three numbers are algebraically independent over $\mathbb{Q}$:*

$$\begin{aligned} X_1 &= A_1(q) + A_2(q), \\ X_3 &= A_1(q) + 7A_2(q) + 12A_3(q) + 6A_4(q), \\ X_5 &= A_1(q) + 31A_2(q) + 180A_3(q) + 390A_4(q) + 360A_5(q) + 120A_6(q). \end{aligned}$$

Hence $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(A_1(q), \ldots, A_6(q)) \geq 3$.

Proof. For $|x| < 1$, differentiate the geometric series, or expand powers in falling factorials, to obtain

$$\sum_{m \geq 1} m^j x^m = \sum_{r=1}^{j+1} (r-1)! \left\{ {j+1 \atop r} \right\} \left(\frac{x}{1-x}\right)^r.$$

Here the braces are Stirling numbers of the second kind. For $j = 1, 3, 5$, summing this identity at $x = q^n$ gives the three displayed combinations and

$$X_j = \sum_{n,m \geq 1} m^j q^{nm} = \sum_{t \geq 1} \sigma_j(t) q^t.$$

Absolute convergence justifies the rearrangements. The three Ramanujan series are $1 - 24X_1$, $1 + 240X_3$, and $1 - 504X_5$. Nesterenko's theorem [46] says that these three values, together with q, generate a field of transcendence degree at least three. Since q is algebraic, the three values are algebraically independent. Invertible rational affine changes of the three coordinates preserve independence, proving the claim. □

This gives genuine transcendental quantities in the same denominator family as E, but it does not determine $A_1(1/2)$ alone. Even the first identity only proves that $E + A_2(1/2)$ is transcendental. Extracting one coordinate from an independent collection of linear combinations requires additional arithmetic relations; none are proved here. Proposition 15.3 is explicitly credited as a consequence of Nesterenko's established theorem.

The alternative digit route must also account for recent prior work. Campbell's 2026 preprint proves infinitely many occurrences of the binary word 11 [47]. A public September 2026 document package [48] claims every finite binary word, together with a quantitative occurrence bound; its accompanying Lean verification is explicitly conditional on prime-distribution inputs. We read the released proof, including its midpoint encoding, CRT modulus, divisor average and tail comparison. We found no obvious gap in those steps, but this is not a claim of specialist peer review or unconditional machine verification. We neither claim that result as new nor use it as a dependency of any theorem above. If its stated disjunctivity conclusion is taken as established, then $p_E(n) = 2^n$, so the low-complexity criterion cannot prove transcendence of E either. Disjunctivity asserts occurrence, not normality or prescribed limiting digit frequencies, and does not by itself imply transcendence.

16 Target register and the next missing arithmetic inputs

The register contains eighteen specified targets or target families. Rows 10–13 share one general obstruction and are not four independent solutions. The ten Pell examples belong together in row 3. A proved method obstruction is not a proved arithmetic classification of its target constant. This distinction is essential to the accounting.

No.	Target	Result here	Exact qualification
1	Mixed-sign rank-two threshold	Complete classification	One algebraic exceptional setting; all other algebraic settings give transcendental values.
2	$\mathcal{V}_{d,2m}$, rational $d > 0$	Transcendental	Every positive integer m; polynomial elimination also treats transcendental boundaries.
3	Pell-selected optimal values	Algebraically independent	Ten displayed values, and an infinite family; one selected family, not ten historical conjectures.
4	$c_{9/8}$ and the rational κ family	Transcendental	Fixed-shape critical extraction rate; a different coordinate from the endpoint shape.
5	$A(1)$	Transcendence unresolved	Earlier irrationality survives; an algebraic-irrational parameter would put the Kummer function outside the E class.
6	$A(3/2)$	Transcendence unresolved	Explicit positive-integral reduction; no suitable specialization theorem proved.

No.	Target	Result here	Exact qualification
7	$\mathcal{V}_{3,1}$	Arithmetic unresolved	Two explicit common-zero equations; boundary transcendence cannot decide the objective.
8	Euler–Gompertz δ	Positive-cone optimum proved	Unique minimizer $(t-1)^2$ over all nonnegative integer polynomials; no individual irrationality proof.
9	Euler γ	Equivalent lattice criterion	The needed irrational-form hypothesis is equivalent to the target; Oppenheim cannot bypass it.
10	Catalan G	Full polynomial-cone obstruction	Uses the stated standard denominator clearance; smaller reduced denominators are not ruled out.
11	$\zeta(5)$	Same obstruction	All positive integer-polynomial shapes covered; no irrationality proof.
12	$\zeta(7)$	Same obstruction	An instance of the uniform $s \geq 2$ theorem.
13	$\zeta(9)$	Same obstruction	An instance of that same theorem, not a separate mechanism.
14	Prime-binary P_2	Complexity route audited	The block-complexity theorem below does not establish transcendence of the real number.
15	Prime Lambert $L_{\mathbb{P}}(2)$	Transcendence unresolved	Known irrationality is credited to its source; carries prevent transferring prime digit conclusions directly.
16	Erdős–Borwein E	Transcendence unresolved	Existing irrationality and recent digit work are separated from an arithmetic proof.
17	Rank-three spectral example	Conditional transcendence	Follows from the displayed exponential-line bound, in particular Schanuel's conjecture.
18	Rational-mesh table $A(u+vn)$	No polynomial recurrence	Structural consequence of the inherited asymptotic; not an individual-value classification.

Where an additional breakthrough would have to enter. For the spectral theorem, a bound of transcendence degree at least $d-1$ on the specific exponential line would meet the proved geometric upper bound $d-2$. This is a concrete higher-rank target. The argument does not require the full force of Schanuel if that weaker line bound can be obtained directly for a chosen spectrum.

For an individual endpoint shape $A(d)$, rational-parameter E-function theory has already done all it can in the present proof: the remaining algebraic-irrational parameters produce a function outside that class. A theorem about those parameter-dependent special values, or a different arithmetic representation of the contact, is needed. The nonrecurrence theorem does not replace such an input.

For the Gompertz family, exact analytic reconstruction from all second differences still permits rational first values in the countermodel. An arithmetic connection statement specific

to the actual exponential integral, rather than another limiting or convexity argument alone, is necessary. For the positive Catalan and zeta kernels, a successful replacement must change the denominator analysis, the kernel class, or the arithmetic framework; merely optimizing a positive polynomial within the proved class is insufficient.

The present work therefore supplies definite new deductions relative to the preceding manuscript and several exact failure diagnoses. It does not supply a basis for announcing a resolution of a famous individual constant conjecture, ten historical closures, or an Annals-level breakthrough. Historical priority and journal significance require further comparison and external mathematical assessment.

17 Verification and source dependencies

This is a single-file LaTeX manuscript. The bibliography, tables, and proof text are embedded; no external input, bibliography database, or image is needed to compile it. Classical theorems remain cited mathematical inputs rather than being reproved in full.

The spectral factorization argument was checked independently for multiple components, repeated factors, a possibly complex factorization, and the real logarithmic Hessian. The stopping-value calculation was independently checked in both directions, including the degenerate Wronskian constant, the exceptional algebraic boundaries, and the relation-lattice proof. All ten Pell pairs satisfy their norm equations by exact integer arithmetic. The critical-rate proof was checked for uniqueness of the global minimum, the double-zero argument, and both directions of its algebraic finite-fiber implication. The positive-cone proof was independently checked in every degree and in its equality case; the compact-interval constants and specified denominator clearances were also checked. These are internal cross-reviews, not proof-assistant certificates or external refereeing.

The boundary asymptotic and the exponential-integral independence result are inherited from the identified project manuscripts. The former is the input to Theorem 13.3; the latter is the input to Proposition 14.6. Their consequences are distinguished from new foundational transcendence theorems. The full even-payoff and spectral proofs do not depend on the conjectural arithmetic of Euler's constant, Catalan's constant, or an individual odd zeta value.

Numerical approximations, where displayed, identify constants and do not prove their irrationality. Absence of a literature search match does not establish novelty. Unreviewed contemporary claims are not silently promoted to theorems. The explicitly conditional spectral corollary is not used in any statement marked unconditional.

Part III
Independence and Entropy Optima

Editorial integration note. The conditional arbitrary-rank spectral framework and the unconditional cubic branch have different scopes. The later planar stopping theorem is in Part IV; the field-closure development continues in Part V.

Original part framing.

Status. Working manuscript. Proofs are supplied for unconditional statements, external theorems are identified, and the squarefree-factorization implication is explicitly conditional. Independent internal cross-reading and symbolic checks are not external peer review or a formal proof certificate.

18 The research question and the actual increase in scope

This is a sequel to *Arithmetic of Optimal Values* [52], not a reprint of its results. The earlier 223-page foundation [49], 447-page sequel compilation [50], and first arithmetic continuation [51] supply context and explicitly identified analytic inputs. Page counts identify documents; they do not measure mathematical strength. The present question is whether those mechanisms can classify additional individual constants or prove relations substantially stronger than separate irrationality.

The strongest changes are shown below. Applications within one uniform theorem are not counted as separate historical conjectures.

Direction	Previous boundary	Result proved here
Spectral values	Higher-rank classification conditional on an exponential-line bound	Unconditional classification throughout real cubic fields, and an identified independent value/contact pair
Stopping values	Infinite independent family obtained by varying dimension and payoff	Infinite independent family in every fixed even dimension; all prime payoff indices in dimension two
Gamma extraction	Shape-one interior rate arithmetic	Every positive integer shape; exact double contacts and an explicit finite bound on their number
Information objectives	Selected arithmetic values and optimizer locations	All rational finite signed-entropy critical values, including nonconvex objectives and boundary supports
Classical constants	Bounds using standard denominator clearances	Actual reduced-denominator bounds, together with a proof that primitive pair reduction is a different obstruction
Modular transfer	Modular combinations of moment functions	Three-function algebraic independence over finite dilated modular differential fields; no value-specialization conclusion

18.1 Twenty-two targets and their outcomes

The register distinguishes arithmetic of a *number* from arithmetic of a family, and functional independence from independence after specialization. “Unresolved here” describes this

manuscript's result, not a blanket claim about every announcement in the literature. In particular, the September 24, 2026 OpenAI Math Release preprint [53] asserts the irrationality of Catalan's constant. Its full proof is neither audited nor used here. Its signed mixed-moment determinants fall outside our restricted positive-kernel theorem.

No.	Target	Outcome and exact limitation
1	Euler's constant γ: irrationality	Unresolved here. Finite differential cancellation eliminates its coefficient rather than isolates it (Proposition 23.3).
2	Euler–Gompertz δ: irrationality	Unresolved here. The new E-value division result does not classify this connection coefficient.
3	Catalan G: individual arithmetic	No new proof. Actual reduced-denominator obstruction for a specified class (Theorem 23.6); the external claim above is kept separate.
4	$\zeta(5)$: irrationality	Unresolved here. Reduced-denominator cancellation is controlled for an entire support-restricted kernel class (Theorem 23.5).
5	Erdős–Borwein $\sum(2^n-1)^{-1}$: transcendence	Unresolved as a number. Its generating function participates in Theorem 24.6.
6	Prime binary constant $\sum_p 2^{-p}$: transcendence	Unresolved as a number. All-order prime asymptotics yield functional independence, not specialization.
7	Prime Lambert constant $\sum_p(2^p-1)^{-1}$: transcendence	Same distinction; the triple functional theorem is one result, not three solved arithmetic conjectures.
8	Endpoint shape $A(1)$: transcendence	Unresolved. Irrationality was proved previously; integer differential operators do not address the remaining parameter.
9	Endpoint shape $A(3/2)$: transcendence	Unresolved for the same precise reason (Section 21).
10	Linear excursion optimum $\mathcal{V}_{3,1}$: transcendence	Conditional on the published squarefree-factorization conjecture; an explicit finite-multiple-zero reduction is proved.
11	All positive rational (d,q) bridge optima	The same conditional family theorem (Theorem 20.8); not an additional independent closure.
12	Algebraic values for cubic-field spectra	Complete unconditional classification (Theorem 19.4).
13	Joint arithmetic of a cubic two-atom value and contact	Algebraic independence, stronger than separate transcendence (Theorem 19.5).
14	Biquadratic optimized value Q in (19.25)	Value unresolved; its contact and elasticity are now individually transcendental.
15	Infinitely many independent optima for one fixed bridge dimension	Proved in every positive even dimension (Theorems 20.3 and 20.4), one uniform family.
16	Fixed integer-shape gamma critical rates	All interior critical rates and contacts classified as transcendental; exact contact multiplicity and count bound also proved.
17	Rational finite-channel capacity	Algebraic optimizer; algebraic exponential of capacity; effective rational/transcendental test in bits (Corollary 22.4).
18	Rational finite-source rate–distortion	Same arithmetic classification (Corollary 22.5), an application of the common entropy theorem.
19	Rational information-bottleneck optimum	Same conclusion despite nonconvexity, including all finite encoder alphabets without extra constraints (Corollary 22.6).

No.	Target	Outcome and exact limitation
20	$1/\operatorname{Ein}(1)$ in the ring of E-values	Nonmembership proved using a new external morphism theorem; an exact rational-function intersection is described (Theorem 23.2).
21	Entropy of Poisson(1): irrationality	Unresolved. Infinite logarithmic coefficients are outside the finite entropy proof.
22	The earlier optimized exponential-integral payoff	Unresolved. Algebraic-point specialization does not apply at its transcendental optimizer.

The register is an account of attempts. It does not turn related corollaries into twenty-two independent discoveries. In particular, the three information-theoretic classifications share one proof, the three prime/divisor functions share another, and the general rational stopping claim remains conditional. The additional reciprocal-moment targets $A_r(1/2)$ are covered by the same functional analysis and are not used to inflate the count.

18.2 Proof architecture and source dependence

The cubic argument is an intersection of two inequalities: tangency and indefinite curvature force transcendence degree at most one, whereas Brownawell's classical theorem forces at least two away from the unit contact. The fixed-dimension stopping argument instead makes algebraic boundary degrees carry new prime factors. These factors prevent rational linear relations between the boundary points; Lindemann–Weierstrass then excludes polynomial relations between their exponential optimal values. The entropy argument uses no infinite special-function expansion: rational logarithmic stationarity becomes a finite semialgebraic system, and a rational anchor on each face determines the arithmetic of its critical values.

These mechanisms are genuinely different. Their coexistence gives a broader research program, not evidence by itself for publication at a particular level. The precise priority and significance of the transfer theorems require a specialist literature review beyond the bounded primary-source comparisons recorded here.

19 Cubic spectra: unconditional classification and independent observables

For the optimized spectral boundary

$$S(x) = \sum_{j=1}^{m} w_j x^{r_j}, \qquad T(C) = \min_{x>0} \frac{1 + Cx}{S(x)}, \qquad 0 < r_j < 1, \tag{19.1}$$

the previous manuscript classified algebraic values when the rational span of $1, r_1, \dots, r_m$ has dimension two. The higher-dimensional extension was conditional on an exponential-line transcendence-degree bound. Here we remove that condition for every spectrum contained in a real cubic number field. The decisive classical input is a theorem of Brownawell that applies even when the base of the powers is transcendental. Its use is essential: the contact in (19.1) is not given to be algebraic.

There are two further conclusions. For a two-atom spectrum spanning a cubic field, the optimized value and its contact are algebraically independent. For any spectrum in an affine irrational lattice, including lattices outside cubic fields, the contact and the elasticity each have an exact algebraic-versus-transcendental classification. These are unconditional classifications of the present variational constants. The classical transcendence theorems are credited explicitly; no previously named arithmetic conjecture is claimed to be resolved.

19.1 Contact geometry and the precise cubic input

Throughout this section, weights and settings are positive real algebraic numbers, all exponents are real algebraic, and real powers use the real logarithm. Write

$$W = \sum_j w_j, \qquad M = \sum_j w_j r_j, \qquad C_* = \frac{M}{W - M}. \tag{19.2}$$

Partial derivatives with respect to C or a weight are taken in the open positive parameter domain, holding the exponents fixed.

Lemma 19.1 (Contact identities)**.** *The minimum in (19.1) has a unique contact $x_C > 0$. The contact and value depend real analytically on the positive parameters. With $x = x_C$, $\tau = T(C)$ and $H = S - xS'$, one has*

$$C = \frac{S'(x)}{H(x)}, \qquad S(x) = \frac{1 + Cx}{\tau}, \qquad S'(x) = \frac{C}{\tau}, \tag{19.3}$$

$$T_C = \frac{x}{S(x)} = \frac{xT}{1 + Cx}, \qquad B := T - CT_C = \frac{1}{S(x)} > 0, \qquad T_{w_j} = -\frac{Tx^{r_j}}{S(x)}. \tag{19.4}$$

Moreover $T_{CC} < 0$, and

$$x_C = 1 \quad \Longleftrightarrow \quad C = C_*, \qquad T(C_*) = \frac{1}{W - M}. \tag{19.5}$$

Proof. The objective tends to infinity at both ends of $(0, \infty)$, so a minimum exists. Each summand of H is $w_j(1 - r_j)x^{r_j} > 0$, while $S''(x) < 0$. Direct differentiation gives

$$\left(\frac{S'}{H}\right)' = \frac{S''S}{H^2} < 0.$$

The critical equation is $C = S'/H$, so there is at most one critical point and hence exactly one minimum. Its derivative with respect to C is the reciprocal of the displayed negative derivative. The analytic implicit-function theorem gives the asserted parameter dependence. The two value identities follow from stationarity. Differentiating the objective at its contact gives the envelope formulas, and $T_{CC} = H(x)x'_C/S(x)^2 < 0$. Finally $S(1) = W$ and $S'(1) = M$ yield (19.5). □

Lemma 19.2 (Classical cubic exponential-line bound)**.** *If $a > 0$, $a \neq 1$, and β is real algebraic of degree three, then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(a, a^{\beta}, a^{\beta^2}) \geq 2. \tag{19.6}$$

There is no algebraicity assumption on a.

Proof. Brownawell's Theorem 2 [54], with the real logarithm of a, asserts in particular that relatively prime nonzero integer polynomials $P(X, Y)$ and $Q(X, Y, Z)$ cannot both vanish at $(a, a^{\beta}, a^{\beta^2})$. We spell out the transcendence-degree consequence to retain the crucial arbitrary-base hypothesis.

Suppose the three numbers generate a field of transcendence degree at most one over $\mathbb{Q}$. If a is algebraic, take for $P(X)$ its minimal polynomial. The other two numbers are algebraically dependent, so a nonzero $Q(Y, Z) \in \mathbb{Z}[Y, Z]$ vanishes at them. These two polynomials are relatively prime, a contradiction. If a is transcendental, both other numbers are algebraic over $\mathbb{Q}(a)$. Their minimal polynomials, after clearing denominators and taking primitive parts, give irreducible polynomials $P(X, Y)$ and $Q(X, Z)$ over $\mathbb{Q}$, of positive degree in Y and Z, respectively. They remain irreducible upon adjoining the unused variable and cannot be

associates, so they are relatively prime. Again Brownawell's theorem gives a contradiction. Extending the base from $\mathbb{Q}$ to its algebraic closure does not change transcendence degree, proving (19.6). □

The input in Lemma 19.2 is entirely classical. It is stronger for the present purpose than the algebraic-base cubic power theorem alone. We use only its qualitative nonvanishing consequence, not its quantitative measure.

19.2 The algebraic tangency obstruction

We recall the curvature argument from the preceding manuscript, with its proof, because it identifies exactly how the cubic input enters. For a Laurent polynomial F and a positive point z, its *logarithmic Hessian* means the Hessian at $y = 0$ of $F(z_1 e^{y_1}, \dots, z_d e^{y_d})$.

Lemma 19.3 (Curvature forces codimension two)**.** *Let $d \geq 2$, let $b_1, \dots, b_d$ be rationally independent real algebraic numbers, and put*

$$D = \sum_{i=1}^{d} b_i Z_i \frac{\partial}{\partial Z_i}.$$

Suppose a Laurent polynomial F with real algebraic coefficients and a positive point z satisfy $F(z) = DF(z) = 0$. If the logarithmic Hessian of F at z is indefinite, then

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(z_1, \dots, z_d) \leq d - 2. \tag{19.7}$$

Proof. Work in the Laurent unique-factorization domain $R = \overline{\mathbb{Q}}[Z_1^{\pm 1}, \dots, Z_d^{\pm 1}]$. For every nonunit irreducible $H \in R$, the polynomials H, DH are relatively prime. Indeed, $DH = 0$ would force H to be constant: the weights $b \cdot k$, $k \in \mathbb{Z}^d$, are distinct, and only the constant monomial has weight zero. If $DH = AH$ with $A \neq 0$, Newton polytopes give

$$\operatorname{Newt}(A) + \operatorname{Newt}(H) = \operatorname{Newt}(DH) \subseteq \operatorname{Newt}(H).$$

Taking support functions in every direction forces $\operatorname{Newt}(A) \subseteq \{0\}$, so A is constant. The equation $DH = AH$ then makes H a single monomial, again a unit.

We also record why two relatively prime Laurent polynomials vanishing at z imply (19.7). Degree d is excluded by either equation. If the degree were $d - 1$, choose $d - 1$ coordinates forming a transcendence basis from among the coordinates of z. The remaining coordinate has a minimal polynomial over their rational function field. Clearing denominators and taking its primitive part gives a nonunit Laurent polynomial dividing every Laurent polynomial that vanishes at z, by Gauss's lemma. The two given polynomials would have a common factor, a contradiction.

Factor F in R. If two distinct irreducible factors vanish at z, the preceding paragraph applies. Otherwise there is one vanishing irreducible factor H, and write $F = H^e R_0$, with $R_0(z) \neq 0$. For $e = 1$, the equation $DF(z) = 0$ gives $DH(z) = 0$, so the first two paragraphs apply. It remains to exclude $e \geq 2$. In logarithmic coordinates write $h(y) = H(ze^y)$ and $r(y) = R_0(ze^y)$. Since $h(0) = 0$, the Hessian of $h^e r$ is zero if $e \geq 3$, and for $e = 2$ it is

$$2r(0)\, \nabla h(0) \nabla h(0)^{\mathsf{T}}.$$

This matrix has complex rank at most one. It is also real symmetric because F has real coefficients. Its real rank is therefore at most one, so it cannot be indefinite. This contradiction completes all cases, including nonreal irreducible factors and singular levels. □

Theorem 19.4 (Unconditional cubic-field classification)**.** *Let $K \subset \mathbb{R}$ be a number field of degree three. Suppose the exponents in (19.1) belong to K, and at least one of them is irrational. Then, for every positive algebraic setting C and positive algebraic weights,*

$$T(C) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad x_C = 1 \quad \Longleftrightarrow \quad C = C_*. \tag{19.8}$$

At the exceptional setting the value is $1/(W - M)$. There is no restriction on the signs of the coordinates of the exponents in a rational basis of K.

Proof. Choose a real cubic generator β of K. For a common positive integer q, write

$$r_j = \frac{m_{j0} + m_{j1}\beta + m_{j2}\beta^2}{q}, \qquad m_{ji} \in \mathbb{Z}.$$

At a contact $x = x_C$, put $u = x^{1/q}$, $v = u^{\beta}$, $z = u^{\beta^2}$. Suppose $\tau = T(C)$ is algebraic and form

$$P(U,V,Z) = \sum_j w_j U^{m_{j0}} V^{m_{j1}} Z^{m_{j2}},$$

$$F(U,V,Z) = P(U,V,Z) - \frac{1 + CU^q}{\tau},$$

$$D = U\partial_U + \beta V \partial_V + \beta^2 Z \partial_Z.$$

The contact equations give $F(u,v,z) = DF(u,v,z) = 0$. Along the exponential line, D acts as $qx\, d/dx$. Consequently

$$D^2 F(u,v,z) = q^2 x^2 S''(x) < 0. \tag{19.9}$$

The additional first-derivative term cancels by $S'(x) = C/\tau$. Some m_{ji} with $i \in \{1,2\}$ is nonzero, since one exponent is irrational. The corresponding coordinate field E_i is $V\partial_V$ or $Z\partial_Z$ and satisfies

$$E_i^2 F(u,v,z) = \sum_j w_j m_{ji}^2 x^{r_j} > 0.$$

Thus the logarithmic Hessian is indefinite. The rational independence of $1, \beta, \beta^2$ and Lemma 19.3 give $\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(u,v,z) \le 1$. If $x \neq 1$, then $u \neq 1$ and Lemma 19.2 gives the contradictory lower bound two. Therefore $x = 1$. The converse and the exceptional value follow from Lemma 19.1. □

For spectra of rational rank three, Theorem 19.4 removes a genuine conditional hypothesis from the preceding result. For spectra already contained in a two-dimensional rational space it is consistent with, and does not recount, the previous classification.

19.3 Two atoms identify an independent pair

Theorem 19.5 (The value and contact are algebraically independent)**.** *Suppose $m = 2$ and*

$$\operatorname{span}_{\mathbb{Q}}\{1, r_1, r_2\} = K, \qquad [K : \mathbb{Q}] = 3. \tag{19.10}$$

At every positive algebraic setting $C \neq C_*$ and for positive algebraic weights, the pair $(T(C), x_C)$ is algebraically independent over $\overline{\mathbb{Q}}$. Each of the pairs

$$(T, T_C), \qquad (x_C, T_C), \qquad (T, \eta)\,, \quad \eta = \frac{CT_C}{T},$$

is also algebraically independent. The triple (T, x_C, T_C) has transcendence degree exactly two and satisfies

$$(1 + Cx_C)T_C = x_C T. \tag{19.11}$$

At $C = C_*$ all these numbers are algebraic.

Proof. Use the same β, q, u, v, z as in the preceding proof, and let $L = \overline{\mathbb{Q}}(u, v, z)$. Since $x_C \neq 1$, Brownawell's bound gives $\operatorname{trdeg} L \geq 2$. Stationarity gives the Laurent relation

$$\sum_{j=1}^{2} w_j[r_j + (r_j - 1)CU^q]U^{m_{j0}}V^{m_{j1}}Z^{m_{j2}} = 0 \quad \text{at } (u, v, z). \tag{19.12}$$

This Laurent polynomial is nonzero. Its first group of monomials has D-weights $qr_j \in (0, q)$, and its second group has weights $q(r_j + 1) \in (q, 2q)$; the groups cannot cancel. Within either group the two exponents are distinct and the coefficients are nonzero. Thus $\operatorname{trdeg} L \leq 2$, and equality holds.

Write $A_j = w_j x^{r_j}$ and, after interchanging the labels if necessary, assume $r_1 < r_2$. The equations $A_1 + A_2 = (1 + Cx)/T$ and $r_1A_1 + r_2A_2 = Cx/T$ yield

$$A_1 = \frac{r_2(1 + Cx) - Cx}{T(r_2 - r_1)}, \qquad A_2 = \frac{Cx - r_1(1 + Cx)}{T(r_2 - r_1)}. \tag{19.13}$$

Hence

$$\overline{\mathbb{Q}}(T, x) = \overline{\mathbb{Q}}(x, x^{r_1}, x^{r_2}).$$

The inclusion from left to right uses $T = (1 + Cx)/S(x)$, and the reverse inclusion uses (19.13) and nonzero weights. By (19.10), each of $1/q, \beta/q, \beta^2/q$ is a rational linear combination of $1, r_1, r_2$. Clearing denominators shows that u, v, z are algebraic over this field: a positive integer power of each is a Laurent monomial in x, x^{r_1}, x^{r_2}. Conversely those three powers already belong to L. Therefore L is algebraic over $\overline{\mathbb{Q}}(T, x)$, and the latter field has transcendence degree two.

The envelope identities and their inverses give

$$T_C = \frac{xT}{1 + Cx}, \qquad x = \frac{T_C}{T - CT_C}, \qquad T = \frac{(1 + Cx)T_C}{x}, \qquad \eta = \frac{Cx}{1 + Cx}, \qquad x = \frac{\eta}{C(1 - \eta)}.$$

Every denominator is nonzero. These rational field identities prove all the pair assertions and (19.11). At the exceptional setting $x = 1$ and the same formulas give algebraic numbers. □

Corollary 19.6 (The field of first-order observations)**.** *Suppose an arbitrary finite spectrum satisfies* $\operatorname{span}_{\mathbb{Q}}\{1, r_1, \ldots, r_m\} = K$*, where K is a real cubic field. At a nonexceptional positive algebraic parameter point,*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(T, T_C, T_{w_1}, \ldots, T_{w_m}) = 2. \tag{19.14}$$

In particular at least one of the first partial derivatives is algebraically independent of T.

Proof. The envelope formulas give the exact field identity

$$\overline{\mathbb{Q}}(T, T_C, T_{w_1}, \ldots, T_{w_m}) = \overline{\mathbb{Q}}(x, x^{r_1}, \ldots, x^{r_m}), \tag{19.15}$$

because $B = T - CT_C = 1/S(x)$, $x = T_C/B$ and $x^{r_j} = -T_{w_j}/(TB)$. The full-span assumption makes the cubic power field L algebraic over the right side, as above. Brownawell supplies degree at least two. The nonzero stationary Laurent relation (19.12), with the sum now over all j, supplies degree at most two. Repeated exponents may be combined; positive coefficients prevent cancellation within each of the two disjoint weight intervals. This proves (19.14). Theorem 19.4 makes T transcendental. If every derivative were algebraic over $\overline{\mathbb{Q}}(T)$, their finite joint extension would have transcendence degree one, a contradiction. □

19.4 An individual observable beyond cubic fields

Theorem 19.7 (Affine irrational lattices)**.** *Suppose that, after combining repeated exponents, the spectrum has at least two elements and*

$$r_j = \alpha + \frac{k_j\theta}{q}, \qquad \alpha \in \overline{\mathbb{Q}} \cap \mathbb{R}, \quad \theta \in (\overline{\mathbb{Q}} \cap \mathbb{R}) \setminus \mathbb{Q}, \quad q \in \mathbb{Z}_{>0}, \quad k_j \in \mathbb{Z}. \tag{19.16}$$

The integers k_j are distinct, and all r_j lie in $(0,1)$. There is no requirement that $\alpha \in \mathbb{Q} + \mathbb{Q}\theta$. Then

$$x_C \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \eta(C) := \frac{CT_C}{T} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad C = C_*. \tag{19.17}$$

Thus both the contact and elasticity are individually transcendental at every nonexceptional positive algebraic parameter point.

Proof. Suppose $x = x_C$ is algebraic and $x \neq 1$. By the classical Gelfond–Schneider theorem [55], $y = x^{\theta/q}$ is transcendental. On factoring x^{α} out of the stationary equation, however, one obtains

$$\sum_j w_j\,[(1 + Cx)r_j - Cx]\,y^{k_j} = 0. \tag{19.18}$$

Its coefficients are algebraic and its exponents are distinct integers. It is not the zero Laurent polynomial: otherwise every r_j would equal $Cx/(1 + Cx)$, contrary to distinctness. Multiplication by a suitable power of y now makes y algebraic, a contradiction. Consequently an algebraic contact must equal one. Conversely the exceptional contact is one by Lemma 19.1. Finally $\eta = Cx/(1 + Cx)$ is a nonconstant fractional linear function of x with algebraic coefficients and inverse $x = \eta/[C(1 - \eta)]$. This proves the other equivalence. □

Every pair of algebraic exponents with irrational difference satisfies (19.16): take $\alpha = r_1$, $\theta = r_2 - r_1$, $q = 1$ and $k_1 = 0, k_2 = 1$. The theorem therefore classifies an individual observed number even when neither the cubic-field theorem nor the earlier rank-two theorem classifies the optimized value. The assumption of at least two distinct exponents matters. A single atom has $x_C = r_1/[C(1 - r_1)]$, an algebraic contact for all the parameters under consideration.

19.5 A finite jet for an unrestricted finite spectrum

The next consequence requires more observations but no field-degree restriction. It is stated separately from the cubic value theorem: algebraicity of an entire jet is a substantially stronger hypothesis than algebraicity of T alone.

Proposition 19.8 (Finite-jet obstruction)**.** *Let the $m \geq 2$ exponents be distinct, with at least one irrational. At a positive algebraic parameter point, the m numbers*

$$T, T_C, \ldots, \partial_C^{m-1} T \tag{19.19}$$

are all algebraic if and only if $C = C_*$. At $C = C_*$ all derivatives of T of every finite order with respect to C are algebraic.

Proof. Assume first that (19.19) is algebraic. The identities $B = T - CT_C > 0$, $x = T_C/B$, $S(x) = 1/B$ and $S'(x) = C/T$ give algebraic $x, S(x), S'(x)$. If $m \geq 3$, differentiation gives

$$x'_C = \frac{TT_{CC}}{B^2} \neq 0, \qquad S''(x) = \frac{B^3}{T^3 T_{CC}}, \qquad S^{(\ell+1)}(x) = \frac{\partial_C[S^{(\ell)}(x_C)]}{x'_C}. \tag{19.20}$$

Induction shows that $S^{(\ell)}(x)$ is a rational expression in $C, T, \dots, \partial_C^{\ell} T$ for every $2 \le \ell \le m-1$; its denominators are nonzero by $T > 0$, $B > 0$ and $T_{CC} < 0$. Thus all $S^{(\ell)}(x)$ for $0 \le \ell \le m-1$ are algebraic. The identities

$$x^{\ell} S^{(\ell)}(x) = \sum_{j=1}^{m} w_j r_j (r_j - 1) \cdots (r_j - \ell + 1) x^{r_j}, \qquad 0 \le \ell \le m-1,$$

form an invertible linear system for the $w_j x^{r_j}$. Its matrix is the falling-factorial Vandermonde matrix; its determinant is the ordinary Vandermonde product in the distinct r_j, since the falling-factorial polynomials are monic of successive degrees. Hence every x^{r_j} is algebraic. An irrational r_j and Gelfond–Schneider force the positive algebraic x to equal one.

Conversely, at $x = 1$, every derivative of S is algebraic. The analytic function $\phi(x) = S'(x)/(S(x) - xS'(x))$ has algebraic Taylor coefficients there and $\phi'(1) \neq 0$. Its local inverse x_C therefore has algebraic Taylor coefficients at the algebraic point C_*: recursive comparison of coefficients divides only by the nonzero algebraic number $\phi'(1)$. Composing with $(1 + Cx_C)/S(x_C)$ proves the final assertion and the converse. □

19.6 Explicit constants and the remaining boundary

For an integer $n \ge 2$ that is not a perfect cube, put $\beta = \sqrt[3]{n}$ and choose an integer $q > n^{2/3}$. Define

$$A_{n,q}(C) = \min_{x>0} \frac{2(1 + Cx)}{x^{\beta/q} + x^{\beta^2/q}}. \tag{19.21}$$

Here $1, \beta/q, \beta^2/q$ span a cubic field. The equal weights are $1/2$, and their mean $M = (\beta + \beta^2)/(2q)$ is irrational. Hence $C_* = M/(1 - M)$ is irrational. Every positive rational C is nonexceptional. Theorems 19.4 and 19.5 prove that $A_{n,q}(C)$ and its contact are algebraically independent, as are its value and elasticity. In particular this applies to the three concrete constants

$$A_{2,2}(1) = \min_{x>0} \frac{2(1 + x)}{x^{\sqrt[3]{2}/2} + x^{\sqrt[3]{4}/2}}, \tag{19.22}$$

$$A_{3,3}(1) = \min_{x>0} \frac{2(1 + x)}{x^{\sqrt[3]{3}/3} + x^{\sqrt[3]{9}/3}}, \tag{19.23}$$

$$A_{5,3}(1) = \min_{x>0} \frac{2(1 + x)}{x^{\sqrt[3]{5}/3} + x^{\sqrt[3]{25}/3}}. \tag{19.24}$$

These are three applications of one theorem, not three independent resolutions of historical conjectures.

For comparison, retain the earlier higher-rank example

$$Q = \min_{x>0} \frac{2(1 + x)}{x^{\sqrt{2}/2} + x^{\sqrt{3}/3}}. \tag{19.25}$$

The exponents belong to a biquadratic field and have irrational difference. Their sum is not one, so the setting $C = 1$ is nonexceptional. Theorem 19.7 now proves that its contact and elasticity are each transcendental. Proposition 19.8 also excludes simultaneous algebraicity of its value and first C-derivative. The present arguments do *not* prove the transcendence of Q itself. Brownawell's cubic bound does not apply merely because the rational span of a spectrum has dimension three: its embedding into a cubic field is an essential hypothesis.

The new unconditional gain is therefore precise. Cubic-field thresholds are classified completely, two full-rank cubic atoms yield an identified independent pair, and affine-lattice contacts and elasticities are classified individually. Spectra outside these arithmetic structures, and infinite prime-frequency boundaries, retain their previously stated barriers.

20 Arithmetic of a single bridge and a finite-multiple-zero reduction

Three questions are investigated here: the arithmetic of the linear Brownian-excursion optimum, the classification of optimal values for general positive rational payoff powers, and joint independence of different payoff optima for one fixed dimension. The third question has an unconditional answer in every positive even integral dimension. In dimension two, all prime-indexed even-payoff optima, together with e, are algebraically independent. This conclusion uses a new Eisenstein calculation for the boundary polynomials; the earlier construction varied the dimension and used Pell equations.

For the first two questions we obtain a different kind of progress. We prove that an explicit E-function associated with a hypothetical algebraic optimum has only finitely many multiple zeros, including a known transcendental double zero. The squarefree-factorization conjecture of Fischler–Rivoal would exclude this configuration. Consequently that published conjecture implies transcendence of *every* rational-parameter bridge optimum, including the linear excursion value. The factorization conjecture is not proved here, and these individual conclusions remain conditional.

20.1 Notation and the already established inputs

For a Bessel bridge of dimension $d > 0$ from zero to zero in unit time, write $\mathcal{V}_{d,q}$ for the optimal expected payoff R_τ^q, where $q > 0$. Set $a = q/2$, $b = d/2$, and

$$f(z) = M(a; b; z) = \sum_{n\geq 0} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}.$$

The established stopping formula of Hobson–Liu [34] is

$$z_* f'(z_*) = a f(z_*), \qquad C_{d,q} = \sqrt{2z_*}, \qquad \mathcal{V}_{d,q} = \frac{(2z_*)^a}{f(z_*)}, \tag{20.1}$$

where $z_* > 0$ is unique. For integer d, this is the radial problem for an ordinary d-dimensional Brownian bridge; in particular the fixed dimensions used below do not require a fractional-dimensional interpretation.

Here is a short verification of the uniqueness and nondegeneracy facts needed later. With $c_n = (a)_n/((b)_n n!) > 0$, give the nonnegative integers probabilities $c_n z^n/f(z)$ for real $z > 0$. Then $u(z) = zf'(z)/f(z)$ is their mean and

$$zu'(z) = \mathrm{Var}_z(n) > 0. \tag{20.2}$$

The mean starts at zero and tends to infinity. To see the latter without differentiating an asymptotic expansion, for each fixed K the total weight on $n \leq K$ tends to zero, by comparison with the single term of index $K+1$. Thus $u(z) = a$ has one solution. Moreover $h(z) = f(z)/z^a$ has a strict, nondegenerate minimum at z_*, since

$$h''(z_*) = h(z_*)\frac{u'(z_*)}{z_*} > 0. \tag{20.3}$$

We recall, rather than recount as new, the earlier exact boundary classification [51, 52]:

$$a, b \in \mathbb{Q}_{>0}: \quad z_* \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad a - b \in \mathbb{Z}_{\geq 0}. \tag{20.4}$$

In the exceptional case $a = b + N$, Kummer's identity gives

$$f(z) = e^z P_{N,b}(z), \qquad P_{N,b}(z) = M(-N; b; -z) = \sum_{k=0}^{N} \binom{N}{k} \frac{z^k}{(b)_k}. \tag{20.5}$$

Its coefficients are positive. Then z_* is algebraic and $\mathcal{V}_{d,q}$ is a nonzero algebraic multiple of e^{-z_*}, so is transcendental by Lindemann–Weierstrass.

For clarity, the nonexceptional input in (20.4) uses Beukers' specialization theorem [56, Corollary 1.4]. The rational-parameter functions f, f' are E-functions with a rational differential system singular only at zero. They are linearly independent over $\overline{\mathbb{Q}}(z)$ unless $f = e^z P$ as in (20.5). Indeed a rational logarithmic derivative of an entire exponential-type solution gives $f = e^{\lambda z} P$; substitution in Kummer's equation gives either $a = -\deg P$ or $\lambda = 1$, $a - b = \deg P$. The first alternative is impossible for $a > 0$. Beukers' theorem then excludes the boundary relation at a nonzero algebraic point. We also use the classical Lindemann–Weierstrass theorem in the form that exponentials of distinct algebraic numbers are linearly independent over $\overline{\mathbb{Q}}$.

20.2 Target I: independent optimal values in one fixed dimension

The new ingredient is an irreducibility statement for the actual stopping-boundary polynomial, not an irreducibility assertion for a Laguerre polynomial substituted without checking its derivative term.

Lemma 20.1 (Prime-parameter boundary polynomials)**.** *Let $b \geq 1$ be an integer and let $p \geq b$ be prime. Set $N = p - b$, $D = N + 1$, and*

$$B_{p,b}(Z) = (b)_N \big[Z P'_{N,b}(Z) + (Z - p) P_{N,b}(Z) \big]. \tag{20.6}$$

Then $B_{p,b}$ is a monic integral polynomial of degree D, Eisenstein at p. In particular

$$[\mathbb{Q}(z_{2b,2p}) : \mathbb{Q}] = p - b + 1. \tag{20.7}$$

In dimension two it has the explicit form

$$B_{p,1}(Z) = \sum_{k=0}^{p} (2k - p) \frac{p!(p-1)!}{(k!)^2 (p-k)!} Z^k. \tag{20.8}$$

Proof. The scaled polynomial $(b)_N P_{N,b}$ is monic and integral, because its coefficient of Z^k is $\binom{N}{k}(b+k)_{N-k}$. Thus (20.6) is monic and integral. Its constant term is

$$-p(b)_N = -p \frac{(p-1)!}{(b-1)!},$$

which has p-adic valuation exactly one. For $1 \leq k \leq N$, the coefficient of Z^k equals

$$(b+k)_{N-k} \left[(k-p) \binom{N}{k} + (b+k-1) \binom{N}{k-1} \right] \tag{20.9}$$

$$= p \, \frac{2k - N - 1}{N - k + 1} \binom{N}{k} (b+k)_{N-k}. \tag{20.10}$$

The identity follows by using $\binom{N}{k-1}/\binom{N}{k} = k/(N-k+1)$ and $p = b + N$. The denominator $N - k + 1$ is a positive integer below p. Consequently each integral coefficient in (20.10) is divisible by p. All nonleading coefficients are divisible by p and the constant is not divisible by p^2; Eisenstein's criterion proves irreducibility. The case $N = 0$ is included: then $B_{p,p}(Z) = Z - p$. The stopping equation (20.1) is precisely $B_{p,b}(z_*) = 0$, which proves (20.7). Substituting $b = 1$, $N = p - 1$ in the coefficient formula, including the leading and constant terms, gives (20.8). □

The first planar polynomials, in increasing degree, are

$$\begin{aligned} B_{2,1}(Z) &= Z^2 - 2, \\ B_{3,1}(Z) &= Z^3 + 3Z^2 - 6Z - 6, \\ B_{5,1}(Z) &= Z^5 + 15Z^4 + 40Z^3 - 120Z^2 - 360Z - 120. \end{aligned}$$

Only their unique positive roots are stopping boundaries. Their other roots nevertheless matter in the field argument below.

Lemma 20.2 (Separation from earlier splitting fields)**.** *Let $\alpha_1, \dots, \alpha_n$ be algebraic numbers of degrees $D_1, \dots, D_n$, respectively. Suppose that for each j there is a prime $\ell_j \mid D_j$ exceeding every preceding degree D_i with $i < j$, and $D_1 > 1$. Then $1, \alpha_1, \dots, \alpha_n$ are linearly independent over $\mathbb{Q}$.*

Proof. The compositum K of the splitting fields of the minimal polynomials of $\alpha_1, \dots, \alpha_{j-1}$ is Galois over $\mathbb{Q}$. Its Galois group injects into the product of the corresponding symmetric groups. Thus $[K : \mathbb{Q}]$ divides $\prod_{i<j} D_i!$, and is not divisible by ℓ_j. If α_j belonged to K, its degree D_j would divide $[K : \mathbb{Q}]$, a contradiction. In a proposed rational linear relation, take the largest index with a nonzero coefficient. It would put that α_j in the preceding compositum. For $j = 1$ it would make α_1 rational, also impossible. □

Theorem 20.3 (One planar bridge supplies infinitely many independent optima)**.** *The collection*

$$\{e\} \ \cup \ \{\mathcal{V}_{2,2p} : p \text{ prime}\} \tag{20.11}$$

is algebraically independent over $\overline{\mathbb{Q}}$. Equivalently one can replace e by $\mathcal{V}_{2,2} = 2/e$. Thus all the constants come from different payoff powers for a single two-dimensional Brownian bridge.

Proof. Write $z_p = z_{2,2p}$. Its degree is p by Lemma 20.1. In increasing prime order these degrees meet Lemma 20.2, so every finite subset of $1, z_2, z_3, z_5, \dots$ is rationally linearly independent. The stopping formula gives

$$\mathcal{V}_{2,2p} = a_p e^{-z_p}, \qquad a_p = \frac{(2z_p)^p}{P_{p-1,1}(z_p)} \in \overline{\mathbb{Q}}_{>0}.$$

Distinct monomials in $e, e^{-z_{p_1}}, \dots, e^{-z_{p_n}}$ have distinct algebraic exponents by the stated linear independence. Lindemann–Weierstrass excludes every nonzero polynomial relation between them. Multiplication by the nonzero algebraic a_p preserves independence. Inverting and scaling the first variable preserves it as well, proving the equivalent formulation. □

The fixed dimension can be any positive even integer, at the cost of selecting a subsequence of primes. No assertion about prime pairs at a prescribed difference is needed.

Theorem 20.4 (Independent optima in every fixed even dimension)**.** *For each integer $b \geq 1$ there is a strictly increasing sequence of primes $p_j > b$ such that*

$$\{e\} \cup \{\mathcal{V}_{2b,2p_j} : j \geq 1\} \tag{20.12}$$

is algebraically independent over $\overline{\mathbb{Q}}$. For $b = 1$ all primes can be used. For $b > 1$, a recursive selection is: after choosing the earlier p_i, choose a prime $\ell_j > \max(b, p_1, \dots, p_{j-1})$ and then a larger prime

$$p_j \equiv b - 1 \pmod{\ell_j}. \tag{20.13}$$

Proof. For $b > 1$, the residue $b-1$ is nonzero modulo ℓ_j. Dirichlet's theorem on primes in arithmetic progressions [59] supplies arbitrarily large primes in (20.13). The degree $D_j = p_j - b + 1$ is divisible by ℓ_j, and ℓ_j exceeds all earlier D_i. Lemmas 20.1 and 20.2 show that $1, z_{2b,2p_1}, z_{2b,2p_2}, \dots$ are rationally linearly independent. Each objective is again a nonzero algebraic multiple of the exponential of its negative boundary. The same Lindemann–Weierstrass argument proves the claim. The case $b = 1$ is Theorem 20.3. □

The two theorems are counted as one fixed-dimension family. They do not prove independence of all even-payoff optima in every dimension, nor of the nonexceptional optima at transcendental boundaries. The distinction from the prior Pell construction is that dimension is held fixed, polynomial degrees grow, and the field separation is obtained from their prime factors.

20.3 Target II: general rational payoffs and finite multiple zeros

The following analytic observation is unconditional. It concerns complex critical points, although the stopping boundary itself is positive real. Controlling all complex critical points is needed to justify the arithmetic reduction; positivity on the real axis alone would not suffice.

Lemma 20.5 (Escape of the complex critical values)**.** *Let $a, b > 0$, $f = M(a; b; z)$, and suppose $b - a \notin \{0, -1, -2, \dots\}$. At zeros z of $D(z) = zf'(z) - af(z)$ tending to infinity,*

$$|f(z)| = \frac{\Gamma(b)}{|\Gamma(b-a)|}|z|^{-a}(1 + O(|z|^{-1})). \tag{20.14}$$

In particular the magnitude of any branch of the corresponding critical value $(2z)^a/f(z)$ satisfies

$$\left|\frac{(2z)^a}{f(z)}\right| = \frac{2^a|\Gamma(b-a)|}{\Gamma(b)}|z|^{2a}(1 + O(|z|^{-1})) \longrightarrow \infty. \tag{20.15}$$

Proof. We use the standard two-term Kummer expansion with separate remainders, as in [57, 13.7.2 and §13.7(ii)]. It is uniform on either closed half-plane: on the upper half-plane take $0 \le \arg z \le \pi$ and the plus sign, and on the lower one take $-\pi \le \arg z \le 0$ and the minus sign. These are closed subsectors of the respective validity sectors. Put

$$A = \frac{\Gamma(b)}{\Gamma(a)}, \qquad B_\pm = \frac{\Gamma(b)e^{\pm\pi i a}}{\Gamma(b-a)}.$$

Both are nonzero, and the expansion reads

$$f(z) = Ae^z z^{a-b}(1 + O(|z|^{-1})) + B_\pm z^{-a}(1 + O(|z|^{-1})). \tag{20.16}$$

The two error terms are bounded relative to their own displayed terms; this statement can also be obtained from the exact connection formula with U and its two standard remainder estimates. It is valid when one term is exponentially smaller than the other and does not divide by their possibly cancelling sum.

To avoid differentiating an unqualified asymptotic remainder, apply the same expansion to the exact identity $f' = (a/b)M(a+1; b+1; z)$. The exponential leading coefficient remains A, while the algebraic one is $-aB_\pm$. Therefore

$$D(z) = Ae^z z^{a-b+1}(1 + O(|z|^{-1})) - 2aB_\pm z^{-a}(1 + O(|z|^{-1})). \tag{20.17}$$

At a sufficiently large zero of D, the factors $1 + O(|z|^{-1})$ are nonzero. Equation (20.17) then gives

$$Ae^z z^{a-b} = 2aB_\pm z^{-a-1}(1 + O(|z|^{-1})).$$

Substitution into (20.16) yields $f(z) = B_\pm z^{-a}(1 + O(|z|^{-1}))$. Since a is real, $|z^{-a}| = |z|^{-a}$ for either chosen branch, and $|B_\pm| = \Gamma(b)/|\Gamma(b-a)|$. This proves (20.14) and (20.15). □

Proposition 20.6 (A finite-multiple-zero family)**.** *Let $a = r/s > 0$ be rational in lowest terms, with r, s positive integers, let $b > 0$, and let $c \in \mathbb{C}^*$. Then*

$$G_c(z) = M(a; b; z)^s - cz^r \tag{20.18}$$

has only finitely many multiple zeros in the complex plane. If $b \in \mathbb{Q}_{>0}$ and $c \in \overline{\mathbb{Q}}^*$, then G_c is an E-function.

Proof. The function is not identically zero since $G_c(0) = 1$. At a multiple zero $z \neq 0$, $f(z)^s = cz^r \neq 0$ and $sf(z)^{s-1}f'(z) = crz^{r-1}$. Dividing these nonzero quantities proves $zf'(z)/f(z) = r/s = a$. If $b - a$ is not a nonpositive integer, Lemma 20.5 gives at large such points

$$\left|\frac{f(z)^s}{z^r}\right| = \left(\frac{\Gamma(b)}{|\Gamma(b-a)|}\right)^s |z|^{-2r}(1 + O(|z|^{-1})) \longrightarrow 0.$$

This cannot equal the fixed positive number $|c|$. Consequently all multiple zeros lie in a bounded set. They are finite because the nonzero entire function G_c has no finite accumulation of zeros.

In the exceptional case, $f = e^z P$ with P polynomial as in (20.5). Then $D = e^z[zP' + (z - a)P]$, and the bracket is a nonzero polynomial of degree $\deg P + 1$. Every multiple zero of G_c is one of its finitely many zeros. Finally rational-parameter Kummer functions are E-functions, and this class is closed under products and addition of algebraic-coefficient polynomials. This proves the last assertion when $b \in \mathbb{Q}_{>0}$ and $c \in \overline{\mathbb{Q}}^*$. □

The arithmetic step needed to exploit this analytic result is exactly identifiable. Fischler and Rivoal's published Conjecture 1.4 [58] asserts a squarefree-factorization property for E-functions: a nonzero E-function is a finite product $\prod_j g_j^j$, where each g_j is an E-function with simple zeros and different g_j have disjoint zero sets. We call this statement (SF). It is an external conjecture, not an assumption silently included in a theorem claimed to be unconditional.

Lemma 20.7 (What (SF) implies for finitely many multiple zeros)**.** *Assume* (SF)*. If a nonzero E-function has only finitely many multiple zeros, all those multiple zeros are algebraic.*

Proof. First, any E-function g with finitely many zeros is a polynomial over $\overline{\mathbb{Q}}$ times $e^{\lambda z}$, with $\lambda \in \overline{\mathbb{Q}}$. Indeed the growth bound for an E-function and Hadamard factorization give $g = Pe^{\lambda z}$, initially with complex coefficients and complex λ, absorbing a constant factor into P. Take a nonzero differential equation $\sum_j p_j(z)g^{(j)}(z) = 0$ with algebraic polynomial p_j. If m is the greatest degree of the p_j and L is the leading coefficient of P, the coefficient of $z^{m+\deg P}$ after division by $e^{\lambda z}$ is

$$L \sum_{\deg p_j = m} \mathrm{lc}(p_j)\lambda^j.$$

The polynomial in λ is nonzero, so λ is algebraic. All Taylor coefficients of $ge^{-\lambda z} = P$ are then algebraic, proving the assertion about P and its zeros.

Now factor the original function according to (SF). For $j \geq 2$, each zero of g_j is a multiple zero of the original function, so g_j has finitely many zeros. The preceding argument makes all its zeros algebraic. Every multiple zero belongs to one of these factors, proving the lemma. □

Theorem 20.8 (Conditional classification of every rational-parameter optimum)**.** *Assume Fischler–Rivoal's squarefree-factorization conjecture* (SF)*. Then*

$$\mathcal{V}_{d,q} \text{ is transcendental} \qquad (d, q \in \mathbb{Q}_{>0}). \tag{20.19}$$

More precisely, outside $q - d \in 2\mathbb{Z}_{\geq 0}$, an algebraic optimum v would give an explicit counterexample to (SF)*:*

$$M(q/2; d/2; z)^s - \frac{2^r}{v^s} z^r, \qquad q/2 = r/s \text{ in lowest terms.} \tag{20.20}$$

This E-function would have finitely many multiple zeros and a transcendental zero of multiplicity exactly two at z_*.

Proof. The exceptional family already has an unconditional transcendence proof, recalled after (20.5). In all other cases (20.4) gives a transcendental z_*. If $v = \mathcal{V}_{d,q}$ were algebraic, raising (20.1) to the sth power shows that (20.20) vanishes at z_*. Differentiating and using $z_* f'(z_*) = a f(z_*)$ shows that its derivative also vanishes there.

This zero has order exactly two. On the positive axis put $h = f/z^a$. Locally the function in (20.20) is $z^r[h(z)^s - h(z_*)^s]$; its second derivative at z_* is $s z_*^r h(z_*)^{s-1} h''(z_*) > 0$ by (20.3). Proposition 20.6 gives only finitely many multiple zeros. Lemma 20.7 would make each algebraic under (SF), contradicting the transcendence of z_*. This proves both the implication and the explicit counterexample statement. □

The analytic finite-zero statement is unconditional; the use of (SF) is not. The known partial factorization theorem in [58, Theorem 1.5] assumes that *all* zeros have the same multiplicity (and additional hypotheses). Our functions are only known to have finitely many multiple zeros, so that theorem cannot replace the conjecture. Nor does the exponential-polynomial result under Schanuel's conjecture in that paper automatically apply to general Kummer functions.

20.4 Target III: the linear Brownian-excursion optimum

This target is singled out because it is a concrete established stopping constant, not merely an arbitrarily selected parameter. For $d = 3$, $q = 1$, set

$$f(z) = M(1/2; 3/2; z) = \int_0^1 e^{zt^2}\, dt.$$

Integration of $(te^{zt^2})'$ gives $z f'(z) = (e^z - f(z))/2$. Thus its unique positive stopping boundary satisfies

$$2f(z_*) = e^{z_*}, \qquad C = \sqrt{2z_*}, \qquad V := \mathcal{V}_{3,1} = 2Ce^{-C^2/2}. \tag{20.21}$$

Its boundary is already known to be transcendental. Theorem 20.8 proves the following precise new conditional arithmetic statement:

$$\text{(SF)} \quad \Longrightarrow \quad V \text{ is transcendental.} \tag{20.22}$$

Indeed if $V = v$ were algebraic, then

$$M(1/2; 3/2; z)^2 - \frac{2}{v^2} z \tag{20.23}$$

would be an E-function with finitely many multiple zeros, one of them the transcendental double zero z_*. The critical value escape in this particular case is

$$\left|\frac{\sqrt{2z}}{M(1/2;3/2;z)}\right| \sim \frac{2\sqrt{2}}{\sqrt{\pi}}|z| \quad \text{at unbounded complex contact points.}$$

The general proof therefore supplies the missing global control that a real-axis tangency calculation alone could not provide.

Unconditionally, the arithmetic of V is still undecided by this work. The two equations $2M(1/2;3/2;z_*) = e^{z_*}$ and $8z_* = v^2e^{2z_*}$ do not contradict Lindemann–Weierstrass, because z_* is transcendental. The same issue prevents using the algebraic-point specialization theorem for E-functions. The finite-multiple-zero construction locates a sufficient published conjecture rather than treating those algebraic-argument theorems as statements at arbitrary transcendental points.

Scope and result accounting. One unconditional arithmetic family is proved: infinitely many jointly independent actual stopping optima in every fixed even dimension, including a complete prime-indexed planar family. One unconditional analytic reduction is proved: the associated E-functions have only finitely many multiple zeros. This gives one conditional arithmetic family covering all positive rational dimensions and payoff exponents. The linear excursion optimum is an instance of that conditional family and is not counted again as an unconditional closure. The three targeted questions are therefore distinguished from the number of proved families. None is asserted to close a historically named arithmetic conjecture, and the published squarefree-factorization conjecture itself remains open here.

21 All integer-shape extraction rates and the multiplicity of their contacts

The preceding paper proved transcendence of an intermediate critical rate for extracting a gamma factor of shape one. We now treat every positive integer shape. The obstacle is not merely a larger polynomial calculation: after repeated differentiation, a rational coefficient row can vanish, so a naive determinant argument can lose the very special-function relation needed for transcendence. A third-order intertwining equation excludes simultaneous loss of both contact rows. It also proves that every interior zero of a boundary remainder in this integer-shape family is exactly double. No generic-parameter qualification or omitted half-integer Bessel exception is needed.

The endpoint constants $A(1)$ and $A(3/2)$ remain unresolved at the level of transcendence. The argument below uses finite differential operators and keeps the Bessel order rational; it does not remove the nonrational-parameter obstruction recorded in the preceding paper.

21.1 Notation and the inherited probability diagram

Fix $b > 1$ and $\kappa > 1/2$. We use the first excess-power McKay law with Laplace transform

$$L(s) = \frac{b+s}{b}\left(\frac{b^2-1}{(b+s)^2-1}\right)^{\kappa}. \tag{21.1}$$

Write $M = 2\kappa - 1$. A gamma factor has shape h and rate $\rho = b - c$, with $0 \le c \le 1$. The previously proved maximal shape for an arbitrary probability remainder is $H_\kappa(c)$. We use the following established analytic facts [50]:

1. H_κ is continuous and strictly decreasing on $[0,1]$, with $H_\kappa(0) = M$ and $H_\kappa(1) = \kappa - A(\kappa)$.
2. $0 < A(\kappa) < 1/(4\kappa)$, with the stronger lower bound $A(\kappa) > \max(0, 1-\kappa)$.
3. At an interior maximal shape the remainder has an analytic density on $(0,\infty)$, strictly positive near zero and infinity, and has a nonempty finite set of interior zeros, each of even order.

The earlier arithmetic result additionally says that $A(\kappa)$ is irrational for rational $\kappa > 1/2$ [51, 52]. These are inputs, not new conclusions of this section.

Let $D = d/dx$ and set

$$t = 2\kappa - 2, \quad j(x) = x^{\kappa-1/2} I_{\kappa-1/2}(x), \quad f(x) = j'(x) = x^{\kappa-1/2} I_{\kappa-3/2}(x). \tag{21.2}$$

The derivative identity follows from the standard Bessel recurrence [60, 10.29.2]. Direct substitution gives

$$xj'' - tj' - xj = 0, \tag{21.3}$$

$$f'' - \frac{t}{x} f' + \left(\frac{t}{x^2} - 1\right) f = 0. \tag{21.4}$$

For an integer $m \geq 1$ define

$$v_m = (D-c)^m j, \qquad g_m = (D-c)^m f = v_m'. \tag{21.5}$$

When $m < M$, multiplication of the shifted Laplace transform by w^m is ordinary differentiation without an initial-value term: $f(x)$ has order x^{M-1} at zero and its first $m-1$ derivatives vanish there. Up to a positive normalizer, the actual shape-m remainder density is

$$e^{-bx} g_m(x). \tag{21.6}$$

Indeed the intermediate inverse is $D^m(e^{-cx} f) = e^{-cx} g_m$, and its probability tilt is $e^{-(b-c)x}$. In particular, nonnegativity is exactly the condition $g_m \geq 0$.

21.2 The third-order equation removes every interior degeneracy

Lemma 21.1 (Intertwining and nonvanishing of the two contact rows)**.** *Let $m \geq 1$, let t, c be arbitrary complex constants with $c^2 \neq 1$, and work locally away from $x = 0$. For every solution j of (21.3), the function $v = (D-c)^m j$ satisfies*

$$xv''' + (m + 1 - t - cx)v'' + (ct - x)v' + (cx - m - 1)v = 0. \tag{21.7}$$

The map $j \mapsto v$ is injective on the two-dimensional solution space. Consequently it is impossible that both transformed basis functions $g_i = ((D-c)^m j_i)'$ satisfy

$$g_i(x_0) = g_i'(x_0) = 0 \qquad (i = 1, 2) \tag{21.8}$$

at the same nonzero point x_0.

Proof. Put $L = x(D^2 - 1) - tD$ and $T = D - c$. The commutator is $[T, L] = D^2 - 1$, and $D^2 - 1$ commutes with T. Hence

$$T^{m+1} L = L T^{m+1} + (m+1)(D^2 - 1) T^m.$$

Applying this to j yields $LTv + (m+1)(D^2-1)v = 0$, which expands to (21.7).

If $T^m j = 0$, then $j = e^{cx}P(x)$ for a polynomial P of degree at most $m-1$. Substitution in $Lj = 0$ gives

$$xP'' + (2cx - t)P' + \{(c^2-1)x - tc\}P = 0.$$

For a nonzero polynomial P, its highest-degree term multiplied by $(c^2-1)x$ cannot cancel, because $c^2 \neq 1$. Thus $P = 0$, proving injectivity.

Equation (21.7) is an ordinary third-order equation at $x_0 \neq 0$. Its solutions with $v'(x_0) = v''(x_0) = 0$ form a one-dimensional space, determined by the remaining initial value $v(x_0)$. Two independent images v_1, v_2 cannot both belong to that space. This proves the last assertion, since $g_i = v_i'$. □

To turn the preceding argument into a finite algebraic contact relation, define rational functions U_m, V_m by

$$(D-c)^m f = U_m f + V_m f'$$

for every solution of (21.4). They are obtained without any special-function evaluation from $U_0 = 1$, $V_0 = 0$ and

$$U_{m+1} = U_m' - cU_m + (1 - t/x^2)V_m, \tag{21.9}$$
$$V_{m+1} = U_m + V_m' + (t/x - c)V_m. \tag{21.10}$$

Put

$$\Delta_m = \det \begin{pmatrix} U_m & V_m \\ U_m' + (1-t/x^2)V_m & U_m + V_m' + tV_m/x \end{pmatrix}, \qquad P_m(t;x,c) = x^{2m}\Delta_m(t;x,c). \tag{21.11}$$

Lemma 21.2 (The exact contact polynomial and its fibers)**.** *For every $m \geq 1$, P_m belongs to $\mathbb{Q}[t,x,c]$. It has degree at most $2m$ in each of x, c, with leading coefficients*

$$[x^{2m}]P_m = (c^2-1)^m, \qquad [c^{2m}]P_m = x^{2m}. \tag{21.12}$$

For fixed t, c with $c^2 \neq 1$, its polynomial in x is nonzero. For fixed $t, x \neq 0$, its polynomial in c is nonzero. At every nonzero x and every $c^2 \neq 1$, the matrix in (21.11) is not the zero matrix.

Proof. Induction in (21.9)–(21.10) gives

$$U_m = x^{-m}u_m(t;x,c), \qquad V_m = x^{1-m}w_m(t;x,c),$$

where u_m, w_m are polynomials of degree at most $m, m-1$ in x. The determinant therefore has denominator dividing x^{2m}, with numerator of degree at most $2m$ in x. The binomial expansion of $(D-c)^m$ shows that the first row has leading c-term $(-c)^m(1,0)$ and the second row has leading term $(-c)^m(0,1)$. The determinant is consequently monic of degree $2m$ in c. This proves the second identity in (21.12) and the degree bound in c.

For the first identity, take the constant term at infinity in the rational coefficient matrix. Equation (21.4) there reduces to $f'' = f$. The transformation $D - c$ on the state vector (f, f') is then

$$K = \begin{pmatrix} -c & 1 \\ 1 & -c \end{pmatrix}.$$

The constant term of the transformed state matrix is K^m, with determinant $(c^2-1)^m$. This is the coefficient of x^{2m} after multiplication by x^{2m}.

Finally differentiation identifies the solution spaces in (21.3) and (21.4). An explicit inverse is $j = f' - tf/x$, for which $j' = f$ by (21.4). If the displayed matrix vanished at a nonzero point, every basis pair would satisfy (21.8), contradicting Lemma 21.1. □

The last assertion is stronger than saying that the determinant is not identically zero. For example, when $m = 2$, the first row can vanish at particular positive parameter values. It is the simultaneous vanishing of both rows that would invalidate the arithmetic argument, and the third-order equation rules it out for all orders. No search through finitely many small m proves that assertion; the intertwining identity does.

21.3 Arithmetic of the entire multiple-contact locus

We use the classical rational-order Bessel consequence of Beukers' linear-specialization theorem [56, Corollary 1.4]: for rational $\nu > -1$ and positive algebraic x,

$$I_{\nu+1}(x)/I_\nu(x) \quad \text{is transcendental.} \tag{21.13}$$

For completeness, the normalized even entire function

$$H_\nu(z) = \sum_{n\geq 0} \frac{(z^2/4)^n}{n!(\nu+1)_n}$$

has even exponential Taylor coefficients $(1/2)_n/(\nu+1)_n$ and odd coefficients zero. These coefficients and their common denominators have exponential bounds for rational ν; the bound follows by counting the two rational arithmetic progressions modulo prime powers, with the finitely many parameter-denominator primes treated separately. Thus H_ν, H'_ν are E-functions. They solve a system whose only finite possible singularity is zero. Since H_ν is entire of exponential type, a rational logarithmic derivative would make $H_\nu = e^{az}Q(z)$ for a polynomial Q; evenness forces $a = 0$, contrary to its infinitely many nonzero Taylor coefficients. The pair is therefore linearly independent over $\overline{\mathbb{Q}}(z)$. Beukers' theorem at a nonzero algebraic point implies that H'_ν/H_ν is transcendental, and this quotient is (21.13). This input is classical and was already used at shape one.

Theorem 21.3 (Every multiple contact has transcendental coordinates)**.** *Let $\kappa > 1/2$ be rational, $m \geq 1$ an integer, and $0 < c < 1$. If the function g_m of (21.5) has a multiple zero at $x > 0$, then both c and x are transcendental. They obey the explicit polynomial relation*

$$P_m(2\kappa - 2; x, c) = 0. \tag{21.14}$$

Each coordinate is algebraic of degree at most $2m$ over the field generated over $\overline{\mathbb{Q}}$ by the other. In particular their joint transcendence degree is exactly one. At an algebraic rate coordinate $c \in (0, 1)$ all positive zeros of g_m are simple.

Proof. The two equations $g_m(x) = g'_m(x) = 0$ are the matrix equation in (21.11) applied to $(f(x), f'(x))$. Since $f(x) > 0$ on the positive axis, its determinant vanishes. This proves (21.14).

Suppose first that c were algebraic. The first identity in (21.12) makes (21.14) a nonzero polynomial equation with algebraic coefficients for x. Thus x is algebraic. All entries of the contact matrix are then algebraic. By Lemma 21.2 at least one row is nonzero. Applied to $(f(x), f'(x))$, this row would force $f'(x)/f(x)$ algebraic; a row with zero second coefficient would instead contradict $f(x) > 0$ immediately. But

$$\frac{f'(x)}{f(x)} = \frac{2\kappa - 2}{x} + \frac{I_{\kappa-1/2}(x)}{I_{\kappa-3/2}(x)}$$

is transcendental by (21.13). This is a contradiction. If x were algebraic to begin with, the second identity in (21.12) would first force c algebraic, producing the same contradiction. Thus both coordinates are transcendental.

The two nonzero leading coefficients also give the two degree bounds over the corresponding one-coordinate fields. Since both coordinates are transcendental and are algebraically dependent, their joint transcendence degree is exactly one. The assertion at an algebraic c is the contrapositive of the first conclusion. □

21.4 All boundary contacts are exactly double

Theorem 21.4 (Multiplicity and contact bound at an integer-shape boundary)**.** *Let $\kappa > (m+1)/2$, $m \geq 1$, and $0 < c < 1$. Suppose the shape-m remainder density is nonnegative and has a positive zero. Then every positive zero of that density is exactly double and lies in*

$$0 < x < \frac{m+1}{c}. \tag{21.15}$$

There are at most $2m$ distinct positive zeros. These conclusions do not require rational κ.

Proof. Nonnegativity of the density is $g_m = v_m' \geq 0$. Near zero, the Bessel series gives

$$j(x) \sim Cx^{2\kappa-1}, \qquad v_m(x) \sim C(2\kappa-1)(2\kappa-2)\cdots(2\kappa-m)x^{2\kappa-1-m}, \qquad C > 0.$$

All factors and the exponent are positive under the stated strict inequality. Hence $v_m(0+) = 0$ and $v_m > 0$ near zero. Since $v_m' = g_m \geq 0$, it follows that $v_m(x) > 0$ for all $x > 0$.

At a positive zero of g_m, nonnegativity gives $g_m' = 0$. Equation (21.7) becomes

$$xg_m'' = (m+1-cx)v_m. \tag{21.16}$$

At a local minimum the left side is nonnegative, so $cx \leq m+1$. If equality held, then $g_m'' = 0$. Differentiate (21.7); using $v_m' = v_m'' = v_m''' = 0$ at that point gives

$$xg_m''' = -cv_m < 0.$$

The zero would have exact odd order three, contradicting nonnegativity on both sides. Thus $cx < m+1$, and (21.16) gives $g_m'' > 0$. This proves exact multiplicity two and (21.15).

Each zero satisfies the contact polynomial equation by the same matrix argument as before; no arithmetic assumption is needed for this step. For fixed t, c, this is a nonzero polynomial of degree $2m$ in x, so there can be at most $2m$ distinct positive zeros. □

The multiplicity statement strengthens the inherited assertion "finitely many zeros of even order." It does not assert that there is only one contact for every integer shape. Multiple contacts, if present, all lie in the same algebraic extension of the field generated by their common rate coordinate.

21.5 Complete arithmetic classification of the critical integer rate

Theorem 21.5 (The integer-shape phase diagram)**.** *Fix rational $\kappa > 1/2$, an integer $m \geq 1$, and restrict c to $[0,1]$. Exactly one of the following cases occurs.*

1. *If $m > 2\kappa - 1$, no rate in this interval permits a shape-m factor.*
2. *If $m = 2\kappa - 1$, the only allowable rate coordinate is $c = 0$. This is an endpoint with an atom in the remainder.*
3. *If $m < 2\kappa - 1$ and $m < \kappa - A(\kappa)$, every $c \in [0,1]$ permits the factor. The largest rate coordinate within the prescribed interval is the algebraic endpoint 1.*

4. *If $m < 2\kappa - 1$ and $m > \kappa - A(\kappa)$, there is one critical coordinate $c_{\kappa,m} \in (0,1)$, and extraction is possible exactly for $0 \le c \le c_{\kappa,m}$. This critical coordinate is transcendental. Its remainder has between one and $2m$ positive zeros; every one is exactly double, transcendental, and smaller than $(m+1)/c_{\kappa,m}$. The rate and all its contacts together generate a field of transcendence degree exactly one over $\overline{\mathbb{Q}}$.*

There is no equality case $m = \kappa - A(\kappa)$ at rational κ. In particular, for each integer $m \ge 2$, every rational

$$\frac{m+1}{2} < \kappa \le m \tag{21.17}$$

belongs to case 4. This includes integer Bessel parameters $\kappa = m$ without an exceptional reduction. For algebraic $b > 1$, the physical critical rate $b - c_{\kappa,m}$ is also transcendental.

Proof. The cases follow from the inherited strict monotonicity and endpoint values of H_κ. The equality $m = \kappa - A(\kappa)$ would make $A(\kappa) = \kappa - m$ rational, contradicting its established irrationality. In case 4, continuity and strict monotonicity give a unique interior solution $H_\kappa(c_{\kappa,m}) = m$ and the asserted closed feasible interval. Its analytic remainder has at least one positive zero by the inherited boundary theorem. Since $m < M$, the derivative representation (21.6) applies. Theorems 21.3 and 21.4 therefore give all the claimed arithmetic, multiplicity, and counting conclusions. Every contact is algebraic over the field of the common rate, so finitely many contacts together do not increase its transcendence degree.

At $m = M$ and $c = 0$, the shifted transform is $(1 - w^{-2})^{-\kappa}$. Its expansion has a positive constant term and nonnegative remaining inverse terms, giving the central endpoint remainder and its atom. Every positive c has smaller capacity. This also explains why the strict inequality $m < M$ was needed to use ordinary differentiation without an origin term.

Finally, (21.17) ensures $M > m$ and $\kappa - A(\kappa) < \kappa \le m$, strictly. Hence case 4 holds throughout that explicit interval. Translation by algebraic b proves the final assertion. □

The two integer values $c = \pm 1$ are the spectral endpoints in the intertwining argument and are deliberately excluded from its local rank lemma. This is not an unclassified interior exception. For example, at integer $\kappa = k \ge 1$, equation (21.3) has an exponential-polynomial solution $e^x Q_{k-1}(x)$; at $c = 1$ the operator $(D-1)^m$ kills it when $m \ge k$. For $0 < c < 1$, the highest-degree argument in Lemma 21.1 excludes such a kernel even when both Bessel solutions reduce to exponentials times polynomials. The phase theorem classifies the physical endpoints separately.

21.6 The endpoint transcendence targets still do not close

We also reexamined the two specific endpoint targets $A(1)$ and $A(3/2)$. Their previous irrationality and the exact integral reductions remain unchanged. The new operator identity does not apply to a hypothetical algebraic-irrational endpoint shape $h = \kappa - A(\kappa)$: $(D-c)^h$ is then not a finite polynomial in D, and the displayed commutator induction no longer supplies a third-order equation. Conversely, forcing an integer shape is precisely what moves the new unknown to an interior rate.

The previous non-E obstruction is exact: for algebraic irrational a and positive rational d, the Borel transform ${}_2F_1(a-1, 1; a+d-1; z)$ violates Galochkin's irrational-parameter pairing criterion, since its only possible irrational difference is $-d < 0$ [36, Theorem 1]. Thus the old Kummer function is not an E-function under the remaining algebraic hypothesis. We have not proved a replacement arithmetic specialization theorem. Transcendence of $A(1)$ and $A(3/2)$, and the general noninteger fixed-shape rate problem, remain unresolved here.

Accounting. The arithmetic theorem is one uniform extension from shape one to all positive integer shapes, not one new theorem per integer. The exact double-contact and finite-contact bound resolve a separate analytic multiplicity question within that same family. The endpoint investigations are recorded as unsuccessful. None of these internally posed questions is claimed to be a historical named conjecture or an Annals-level result on the basis of its label.

22 Arithmetic of finite entropy optimization

The previous exponential-integral arguments classified selected values at algebraic parameters and certain optimizer locations. Here the object is the *optimized information quantity itself*. Three targets are finite-channel capacity, finite-source rate–distortion, and the nonconvex information-bottleneck objective. A single theorem handles all three: for rational data, every face-stationary value is the logarithm of a positive algebraic number, and an algebraic optimizer exists. The theorem includes boundary supports and positive-dimensional critical sets; it assumes neither a unique optimizer nor convexity.

The mechanism is related to established algebraic statistics. Dukkipati already expressed maximum-entropy and minimum-divergence estimation through polynomial equations [63]. Sturmfels, Telen, Vialard, and von Renesse study the toric geometry of entropy regularization [64]. We do not claim that exponentiating rational logarithmic stationarity equations is new. Blakaj and Wolf already prove fixed-law entropy arithmetic dichotomies: algebraic probabilities give zero or transcendental entropy in nats, while rational probabilities give rational or transcendental entropy in an algebraic base [71, Theorem 2]. Our formulation records the consequences for all signed entropy combinations, all support faces, and the arithmetic of the attained values. Its historical priority has not been established.

All logarithms in this section are natural, and $0 \log 0 = 0$. An algebraic vector means a real vector with algebraic coordinates. A rational polytope is a bounded polyhedron given by rational linear equalities and inequalities.

22.1 All critical values, including nonisolated critical sets

Let $P \subset \mathbb{R}^d$ be a nonempty rational polytope. For $1 \le a \le A$, let

$$L_a(x) = (\ell_{a1}(x), \dots, \ell_{an_a}(x))$$

be an affine map with rational coefficients that takes P into a probability simplex. Thus each coordinate is nonnegative on P and $\sum_j \ell_{aj}(x) = 1$ there. Given $c_a \in \mathbb{Q}$, set

$$F(x) = \sum_{a=1}^{A} c_a H(L_a(x)), \qquad H(p) = -\sum_j p_j \log p_j. \tag{22.1}$$

For a nonempty face Q of P, a *face-stationary point* is a point in relint Q at which the derivative of F vanishes in every direction parallel to Q. This includes every vertex, where there are no tangent conditions.

Theorem 22.1 (Finite arithmetic description of entropy optima)**.** *For the rational data in* (22.1)*:*

1. *The set of face-stationary values over all faces is finite. Each such value is* $\log \alpha$ *for a positive algebraic number* α*, and is attained at an algebraic face-stationary point.*
2. *Both global extrema of* F *on* P *are attained at algebraic points, and their exponentials are algebraic.*

3. *From a finite exact rational description of the data, a terminating algorithm produces a finite list of algebraic points and algebraic positive numbers that represents every face-stationary value. It determines the global extrema by comparing those algebraic numbers.*

In particular every nonzero extremal value in nats is transcendental. No polynomial-time complexity bound is asserted.

Proof. Fix a nonempty face Q. An affine function that is nonnegative on Q and zero at a relative interior point is identically zero on Q: otherwise a sufficiently short line segment through that point in the direction of a point with positive value would have a negative value at its opposite end. Consequently each ℓ_{aj} is either identically zero on Q or strictly positive on $\operatorname{relint} Q$. Delete the identically zero coordinates. The remaining expression for F is smooth on that relative interior.

Choose a rational basis $v_1, \dots, v_k$ of the linear space parallel to Q. Since each probability vector has total mass one,

$$\sum_j \partial_{v_i} \ell_{aj} = 0.$$

The constant terms in the derivative of $t \log t$ therefore cancel, and stationarity is exactly

$$\sum_{a,j} c_a (\partial_{v_i} \ell_{aj}) \log \ell_{aj}(x) = 0, \qquad 1 \le i \le k. \tag{22.2}$$

All coefficients of these logarithms are rational. Clear their denominators, obtaining integers b_{iaj}. Because the arguments are positive, (22.2) is equivalent to the polynomial equation

$$\prod_{b_{iaj}>0} \ell_{aj}(x)^{b_{iaj}} = \prod_{b_{iaj}<0} \ell_{aj}(x)^{-b_{iaj}}. \tag{22.3}$$

An empty product is one. Thus the set S_Q of face-stationary points is a semialgebraic set defined over $\mathbb{Q}$.

We use the standard real-algebraic decomposition theorem: a semialgebraic set defined over the real algebraic numbers has a finite effective cell decomposition with algebraic sample points; each cell is connected by piecewise C^1 semialgebraic paths. These properties follow from cylindrical algebraic decomposition and semialgebraic path decomposition; see [61]. Along any such path in S_Q, its tangent vector is parallel to Q, and (22.2) makes the derivative of F zero on each smooth piece. The mean-value theorem and continuity at the finitely many joins make F constant on the path. Hence F is constant on each cell of a finite decomposition of S_Q. In particular it has finitely many values there, and each is attained at an algebraic sample point x_*. This step does not assume that the critical equations have finitely many solutions.

Choose a rational point $x_0 \in \operatorname{relint} Q$, which exists because Q is a nonempty rational polytope. Pair the derivative at x_* with $x_* - x_0$. Affineness and stationarity give

$$0 = -\sum_{a,j} c_a \big(\ell_{aj}(x_*) - \ell_{aj}(x_0)\big) \log \ell_{aj}(x_*).$$

Therefore

$$F(x_*) = -\sum_{a,j} c_a \ell_{aj}(x_0) \log \ell_{aj}(x_*). \tag{22.4}$$

The coefficients on the right are rational, while all logarithm arguments are positive algebraic numbers. Taking the exponential expresses $e^{F(x_*)}$ as a finite product of rational powers of positive algebraic numbers, so it is algebraic.

There are finitely many faces. Every global extremizer belongs to the relative interior of its unique minimal face and is stationary there. Continuity of F and compactness of P guarantee existence. The first two conclusions follow. For effectiveness, enumerate the faces, their rational tangent bases, the equations (22.3), and algebraic sample points of a cell decomposition of each S_Q. Formula (22.4) computes each positive algebraic exponential exactly, with its positive real branches. Algebraic-number order comparison selects the smallest and largest. Finally, a nonzero algebraic value of F would have a transcendental exponential by Hermite–Lindemann, contradicting what was proved [62]. □

Remark 22.2 (What the algorithm outputs). The output for a value is the symbolic expression $\log \alpha$, with α specified by a polynomial and an isolating interval; it is not a claim that a transcendental real has a finite decimal expansion. The input consists of exact finite rational data. This does not supply a continuous or computable selector from arbitrary real channels given by approximation oracles. In particular it does not contradict the optimizer noncomputability results for computable-channel representations in [66]. Likewise, only the rational-face *stationary sets* are being algebraized. No semialgebraic description of arbitrary entropy *level sets* is asserted; the latter can be nowhere semialgebraic, as proved in [71].

22.2 An exact test for rationality in bits

Theorem 22.3 (Norm test for rational versus transcendental values). *Let $\alpha > 0$ be algebraic, let $d = [\mathbb{Q}(\alpha) : \mathbb{Q}]$, and put $N = |N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)| \in \mathbb{Q}_{>0}$. Then $\log_2 \alpha$ is rational if and only if there exists an integer k such that*

$$N = 2^k, \qquad \alpha^d = 2^k. \tag{22.5}$$

In that case $\log_2 \alpha = k/d$. Otherwise it is transcendental. Both conditions are decidable by exact rational and algebraic arithmetic. Thus every extremum in Theorem 22.1 admits an exact rational-or-transcendental classification in bits.

Proof. Suppose $\log_2 \alpha = a/b$ with $a \in \mathbb{Z}$, $b \geq 1$. Then $\alpha^b = 2^a$, and taking absolute norms yields $N^b = 2^{ad}$. A positive rational number with an integral power equal to a power of two must be 2^k for an integer k, by unique prime factorization in $\mathbb{Q}$. Hence $kb = ad$, so $a/b = k/d$ and $\alpha^d = 2^k$. The converse follows by taking real logarithms. If the real number $\log_2 \alpha$ were algebraic irrational, Gelfond–Schneider would make $2^{\log_2 \alpha} = \alpha$ transcendental, a contradiction [62]. Factor the numerator and denominator of N to test its first condition, then test the second in the number field. □

22.3 Three information-theoretic targets

The optimizations below are established information-theory problems, not newly named constants. Channel capacity and rate–distortion are the two classical settings of Blahut's algorithms [65]; the third is the information bottleneck of Tishby, Pereira, and Bialek [68].

Corollary 22.4 (Rational finite-channel capacities, with rational constraints). *Let $W(y \mid x)$ be a finite stochastic matrix with rational entries, and let P be a nonempty rational polytope of input distributions. The maximum*

$$C_P(W) = \max_{p \in P} I_p(X;Y)$$

is attained by an algebraic input distribution, and $e^{C_P(W)}$ is algebraic. A positive value in nats is transcendental; in bits the value is rational or transcendental, with an exact decision procedure. This includes finitely many rational input-cost constraints.

Proof. The input law p, joint law $(p_x W(y \mid x))_{x,y}$, and output law $(\sum_x p_x W(y \mid x))_y$ are rational affine probability maps. Apply Theorem 22.1 to $I(X;Y) = H(X) + H(Y) - H(X,Y)$, then Theorem 22.3. □

The binary-symmetric rational-crossover special case is already explicitly discussed and proved in Appendix B of [66], and is not counted as a new result here. Liang studies algebraic solvability of general capacity formulas [67]. Polynomial descriptions for each fixed rational instance are not formulas in radicals, nor formulas algebraic in arbitrary real channel parameters. These distinctions prevent an incorrect claim to resolve that separate closed-form problem.

Corollary 22.5 (Rational finite-source rate–distortion values)**.** *Let p_X be a rational distribution on a finite alphabet, let the reproduction alphabet be finite, let the distortion entries $d(x,y) \geq 0$ and the threshold $D \geq 0$ be rational, and suppose the distortion constraint is feasible. Then*

$$R(D) = \min \left\{ I(X;Y) : P_X = p_X,\ \sum_{x,y} P_{XY}(x,y) d(x,y) \leq D \right\}$$

has an algebraic optimizing joint law, and $e^{R(D)}$ is algebraic. Every positive $R(D)$ in nats is transcendental, and the value in bits is rational or transcendental with an exact decision procedure.

Proof. The feasible joint laws form a compact rational polytope. Their joint and two marginal distributions are rational affine maps. Apply the two theorems to the same three-entropy expression for mutual information. Zero source probabilities simply give zero rows; no positive-support hypothesis is needed. For positive source probabilities, dividing the algebraic joint law by the rational source probabilities also produces an algebraic test channel. □

Corollary 22.6 (Finite information-bottleneck optima)**.** *Fix a rational joint law p_{XY} on finite alphabets, a positive integer m, and a rational $\beta \geq 0$. Over stochastic encoders $q(t \mid x)$ with $1 \leq t \leq m$, set*

$$B_{\beta,m} = \min_{q(t|x)} \{ I(X;T) - \beta I(Y;T) \}, \qquad T - X - Y.$$

There is an algebraic optimizing encoder and $e^{B_{\beta,m}}$ is algebraic. Every nonzero optimum in nats is transcendental, and its value in bits is rational or transcendental, decidably. The same assertions hold after imposing rational linear constraints on the encoder, provided the feasible set is nonempty. Convexity of this objective is not required. Without additional encoder constraints, the same assertions hold for the infimum over all finite encoder alphabets*: an optimum is attained with at most $|\operatorname{supp} p_X|$ symbols.*

Proof. The encoder space, with any stated rational linear constraints, is a rational polytope. The joint probabilities $p_{XY}(x,y)q(t \mid x)$ and all their marginals are rational affine functions of the encoder. Expand each mutual information as three entropies and apply the two theorems. Here m is fixed; no limiting argument over growing alphabets is needed.

For completeness, the unrestricted cardinality assertion follows by a direct support reduction. Delete zero-probability source letters and write $n = |\operatorname{supp} p_X|$. Any finite encoder has output weights $w_t > 0$ and posterior vectors $r_t = P_{X|T=t} \in \mathbb{R}^n$, with $\sum_t w_t r_t = p_X$. If V denotes the fixed channel $X \to Y$, the objective is

$$H(X) - \beta H(Y) + \sum_t w_t \{ \beta H(r_t V) - H(r_t) \}.$$

When there are more than n positive weights, the vectors r_t are linearly dependent. Choose nonzero real a_t with $\sum_t a_t r_t = 0$; summing coordinates gives $\sum_t a_t = 0$. The feasible interval of s for the weights $w_t + s a_t \geq 0$ has endpoints of both signs, and at each endpoint some weight vanishes. The displayed objective is affine in s. At one endpoint it is no greater than at $s = 0$. Keep that endpoint and repeat, preserving the mean posterior and reducing the output support until at most n symbols remain. The new posteriors and weights define an encoder by $q(t \mid x) = w_t r_t(x)/p_X(x)$. Consequently the infimum over all finite alphabets equals the minimum on the compact n-symbol encoder polytope. The preceding arithmetic conclusions apply to that polytope. □

Ordinary constrained maximum entropy is another immediate special case. Its polynomial stationarity mechanism is already present in the cited toric-statistics literature and is not counted as a fourth independent target. The signed nonconvex objective in the bottleneck corollary explains why a proof based solely on strict concavity would not have sufficed for the general theorem.

22.4 A globally optimized bottleneck value in closed form

To exhibit an actual nonzero optimized constant, let X be a fair bit and $Y = X \oplus Z$, with Z independent and $\mathbb{P}(Z = 1) = 1/4$. Consider all finite-valued encoders $T - X - Y$.

Proposition 22.7 (An explicit nonconvex optimized information constant)**.** *For this source,*

$$\inf_{T-X-Y} \{I(X;T) - 8I(Y;T)\} = \log \frac{11 + 5\sqrt{5}}{32}. \tag{22.6}$$

The infimum is attained by the binary symmetric encoder $T = X \oplus E$, with independent E and

$$\mathbb{P}(E = 1) = \frac{1 - 2\sqrt{\sqrt{5} - 2}}{2}.$$

The optimum is transcendental both in nats and in bits. Numerically it is approximately -0.3665295969 nats, or -0.5287904318 bits. The underlying binary bottleneck reduction is classical [69, 70]; the claim here is its exact evaluation and arithmetic verification at the stated parameters, without a historical priority claim.

Proof. Write $h(p) = -p \log p - (1-p)\log(1-p)$. We first recall and prove the one-bit convexity underlying Mrs. Gerber's lemma. For $0 < c < 1$, the function

$$u = h((1-s)/2) \longmapsto h((1-cs)/2), \qquad 0 \leq s \leq 1,$$

is convex. For $0 < s < 1$ its derivative with respect to u is $c \operatorname{arctanh}(cs)/\operatorname{arctanh}(s)$. The ratio $\operatorname{arctanh}(cs)/\operatorname{arctanh}(s)$ decreases with s: expand both series in s^2 after cancelling s, and observe that the ratios of their positive coefficients are c^{2n+1}, strictly decreasing in n. Explicitly, the numerator of the derivative of the ratio of these series is a sum over $r > n$ of terms proportional to $(r-n)(c^{2r+1} - c^{2n+1})(s^2)^{r+n-1} \leq 0$. Since u decreases with s, the asserted convexity follows; continuity gives the endpoints.

Apply this with $c = 1/2$ to each posterior law of X given $T = t$. Replacing its parameter by its complement if necessary does not change either binary entropy. Jensen's inequality yields

$$H(Y \mid T) \geq h(1/4 + \varepsilon/2), \qquad H(X \mid T) = h(\varepsilon), \quad 0 \leq \varepsilon \leq 1/2.$$

For every such ε, equality is attained by the binary symmetric encoder with crossover ε. As both X and Y are fair, the optimization over all finite T is therefore exactly the minimum over $0 \le \varepsilon \le 1/2$ of

$$8h(1/4 + \varepsilon/2) - h(\varepsilon) - 7\log 2.$$

Set $s = 1 - 2\varepsilon$. Its derivative with respect to s is

$$g(s) = \operatorname{arctanh} s - 4\operatorname{arctanh}(s/2).$$

For $0 < s < 1$, the equation $g(s) = 0$, after applying the strictly increasing hyperbolic tangent and using its quadruple-angle identity, is equivalent to

$$s^4 + 16s^2 - 16 = 0.$$

There is exactly one positive solution in $(0, 1)$, $s_* = 2\sqrt{\sqrt{5} - 2}$. Moreover $g'(0) = -1$ and $g(s) \to +\infty$ as $s \uparrow 1$. Thus the derivative is negative before s_* and positive after it, proving that this point is the global minimum, including the two endpoints.

Put $\varepsilon_* = (1 - s_*)/2$ and $r_* = 1/4 + \varepsilon_*/2$. Stationarity, or direct cancellation of the log terms proportional to $\varepsilon_* - 1/2$, rewrites the minimum as

$$-4\log(r_*(1 - r_*)) + \tfrac{1}{2}\log(\varepsilon_*(1 - \varepsilon_*)) - 7\log 2.$$

Since

$$\sqrt{\varepsilon_*(1 - \varepsilon_*)} = \frac{\sqrt{5} - 2}{2}, \qquad r_*(1 - r_*) = \frac{3 - \sqrt{5}}{4},$$

its exponential equals

$$\frac{\sqrt{5} - 2}{(3 - \sqrt{5})^4} = \frac{11 + 5\sqrt{5}}{32} =: \alpha.$$

Here $1/2 < \alpha < 1$, so the natural logarithm is nonzero and transcendental. The degree is two and the absolute norm is $1/256 = 2^{-8}$. A rational $\log_2 \alpha$ would therefore have to be -4 by Theorem 22.3, which contradicts $1/2 < \alpha < 1$. The value in bits is consequently transcendental. □

22.5 Scope, sharp hypotheses, and unsuccessful extensions

The three targets above concern finite distributions with exact rational data. Their common proof is an arithmetic consequence of semialgebraic critical equations, not a new general transcendence theorem and not a claim to solve a named historical conjecture. The following distinctions are essential.

1. *An algebraic optimizer alone is insufficient.* For $a = 1/\sqrt{2}$, the exponential of the entropy of the fixed law $(a, 1 - a)$ is

   $$e^{h(a)} = (1 - a)^{-1} \left(\frac{1 - a}{a}\right)^a,$$

   which is transcendental by Gelfond–Schneider. In the theorem, the rational point x_0 on the same stationary face changes algebraic log coefficients into rational ones. A general algebraic singleton polytope does not have this property.

2. *Rational inputs cannot be replaced indiscriminately by algebraic inputs.* A binary erasure channel with algebraic irrational erasure probability $\epsilon \in (0, 1)$ has capacity $1 - \epsilon$ bits, which is algebraic irrational. Indeed the erasure flag is independent of the input and an unerased symbol reveals it, so $I(X; Y) = (1 - \epsilon)H(X)$; the fair input is optimal. This is an explicit failure of the rational-or-transcendental classification outside the stated hypotheses.

3. *No arbitrary added rational linear cost is allowed.* Even adding the constant one to an identically zero entropy objective makes its optimized exponential equal to e. Rational costs in the capacity and rate–distortion statements are constraints defining the feasible polytope, not additive terms in (22.1).

4. *Finite alphabets matter.* For a Poisson law with algebraic positive mean λ, its Shannon entropy contains $e^{-\lambda}\sum_{n\geq 0}\lambda^n \log(n!)/n!$. The logarithmic coefficients do not meet the algebraic-coefficient E-function framework used for Ein. The finite critical-cell proof gives no arithmetic classification of this infinite series. No irrationality of the entropy of Poisson(1) is claimed.

5. *The previous optimized exponential-integral payoff remains unresolved here.* At the earlier transcendental critical point of $\mathrm{Ein}(x)/x^{\theta}$, the algebraic-point Siegel–Shidlovskii theorem cannot be invoked. The present finite entropy theorem does not repair that missing specialization.

The primary-source comparison is bounded. Maximum-entropy algebraization, toric stationarity, the binary entropy transcendence example, and the binary bottleneck tradeoff are credited above. The checked sources do not establish historical novelty of the full signed-entropy critical-value formulation or the norm-based decision corollary. These are proved research statements with uncertified priority, rather than announced resolutions of old open problems.

23 Exponential-integral division and genuinely reduced denominators

The individual targets in this section are Euler's constant γ, the Euler–Gompertz constant δ, Catalan's constant G, and $\zeta(5)$. None is classified here as irrational by a new proof. There are two substantive advances over the preceding polynomial-kernel analysis: an explicit application of the new arithmetic morphism of Fischler–Rivoal to rational functions of an exponential-integral value, and denominator obstructions using the *actual reduced rational remainder*, not a convenient common multiple. A constructive saturation theorem then explains exactly why further common-factor division is a different issue. These are self-posed results and dependent applications; no priority assertion is made for them.

We write

$$\delta_x = \int_0^{\infty} \frac{e^{-t}}{t+x}\,dt \quad (x>0), \qquad \delta = \delta_1, \qquad \mathrm{Ein}(z) = \int_0^{z} \frac{1-e^{-t}}{t}\,dt.$$

The familiar connection formula is

$$\mathrm{Ein}(x) = \gamma + \log x + e^{-x}\delta_x \quad (x>0). \tag{23.1}$$

It follows by differentiation and the defining limit of γ. We use its analytic continuation with the principal logarithm on the plane cut along the nonpositive real axis.

23.1 A new division obstruction for the exponential integral

Let $\mathcal{E}$ be the ring of values of E-functions at algebraic points. The distinction between this ring and its fraction field is essential. We use the following external theorem of Fischler–Rivoal [72, Proposition 3.1]: there is an injective ring morphism

$$\Phi_0 : \mathcal{E} \longrightarrow \mathcal{R} := \left\{ \sum_{\rho\in\overline{\mathbb{Q}}} c_\rho e^{\rho z} : c_\rho \in \mathbb{C}, \text{finite support} \right\}.$$

It fixes algebraic numbers. If $f(z) = \sum_\rho e^{\rho z} f_\rho(z)$ is the canonical large-sector asymptotic decomposition of an E-function, then $\Phi_0(f(1)) = \sum_\rho f_\rho(1) e^{\rho z}$, with the prescribed one-sided convention at an exceptional direction. This external result is not reproved or counted as our discovery.

Proposition 23.1 (The two coefficients of an exponential-integral value)**.** *Put* $u = \mathrm{Ein}(1)$. *Then*

$$\Phi_0(u)(z) = \gamma + \delta e^{-z}. \tag{23.2}$$

In particular u *is not a unit of* $\mathcal{E}$*, so that* $1/\mathrm{Ein}(1) \notin \mathcal{E}$*. More generally, for every positive algebraic* $a \geq 1$*,* $1/\mathrm{Ein}(a) \notin \mathcal{E}$*.*

Proof. The coefficients of $\mathrm{Ein}(z) = \sum_{n\geq 1} (-1)^{n+1} z^n/(n\, n!)$ have the required exponential denominator growth, and the function satisfies a linear differential equation over $\mathbb{Q}(z)$; thus it is an E-function. For complex z off the negative real axis,

$$\delta_z = \int_0^\infty \frac{e^{-t}}{t+z}\, dt \sim \sum_{k\geq 0} \frac{(-1)^k k!}{z^{k+1}}.$$

For completeness, on each closed sector $|\arg z| \leq \pi - \varepsilon$ one has $|t+z| \geq c_\varepsilon(t+|z|)$. Expanding $(t+z)^{-1}$ through N terms and integrating the remainder gives an error at most $c_\varepsilon^{-1} N! |z|^{-N-1}$. This is the requisite factorial asymptotic expansion on sectors of opening greater than π. Formula (23.1) therefore gives the canonical decomposition with phases $0, -1$ and coefficient functions $\gamma + \log z, \delta_z$. The external morphism theorem gives (23.2).

Both γ and δ are positive. Hence $\gamma + \delta e^{-z}$ has a complex zero. If u had an inverse in $\mathcal{E}$, its image would be an entire exponential polynomial whose product with this function were 1, which is impossible at that zero. Replacing z by az gives

$$\Phi_0(\mathrm{Ein}(a))(z) = \gamma + \log a + \delta_a e^{-az}.$$

For $a \geq 1$ both coefficients are positive, and the same argument applies. □

There is an exact, more informative rational-function consequence. The preceding proposition also shows that u is transcendental: every nonzero algebraic number and its inverse belong to $\mathcal{E}$. Consequently evaluation identifies $\overline{\mathbb{Q}}(T)$ with $\overline{\mathbb{Q}}(u)$.

Theorem 23.2 (The only possible exceptional denominator)**.** *Let* $R \in \overline{\mathbb{Q}}(T)$ *be nonpolynomial. If* $R(u) \in \mathcal{E}$*, then* γ *is algebraic, every finite pole of* R *is* γ*, and* δ *is a unit of* $\mathcal{E}$*. In fact*

$$\mathcal{E} \cap \overline{\mathbb{Q}}(u) = \begin{cases} \overline{\mathbb{Q}}[u, (u-\gamma)^{-1}], & \gamma \in \overline{\mathbb{Q}} \text{ and } \delta \in \mathcal{E}^\times, \\ \overline{\mathbb{Q}}[u], & \text{otherwise.} \end{cases} \tag{23.3}$$

Thus any rational function having a finite pole at an algebraic number outside $(0,1)$ *takes at* u *a value outside* $\mathcal{E}$*. In particular* $1/(u-k) \notin \mathcal{E}$ *for every integer* k*.*

Proof. Write $R = A/B$ in lowest terms. If $v = R(u) \in \mathcal{E}$, apply Φ_0 to $B(u)v = A(u)$. On the open set where its denominator is nonzero this yields

$$\Phi_0(v)(z) = R(\gamma + \delta e^{-z}).$$

The left side is entire. If $\alpha \neq \gamma$ were a finite pole of R, the equation $e^{-z} = (\alpha - \gamma)/\delta$ would have a complex solution. The derivative of $\gamma + \delta e^{-z}$ at that solution is $-(\alpha - \gamma) \neq 0$, so the pole of R would remain a pole of the composite, a contradiction. All finite poles are therefore

γ. A nonpolynomial rational function has a finite pole, and its poles are algebraic, proving $\gamma \in \overline{\mathbb{Q}}$.

Now $B(T)$ is a nonzero constant times $(T-\gamma)^m$ for some $m \geq 1$. Bezout's identity for the coprime polynomials A, B gives $SA + T_0 B = 1$ with $S, T_0 \in \overline{\mathbb{Q}}[T]$. Evaluation shows $1/B(u) = S(u)v + T_0(u) \in \mathcal{E}$. Multiplication by a suitable power of $u - \gamma$ gives $(u-\gamma)^{-1} \in \mathcal{E}$. Since $\delta = e(u-\gamma)$ and $e, e^{-1} \in \mathcal{E}$, the number δ is a unit. Conversely, if γ is algebraic and δ is a unit, the localization displayed in the first line of (23.3) lies in $\mathcal{E}$. The necessary pole condition proves that it contains the whole intersection. If either hypothesis fails, the first part of the theorem excludes every nonpolynomial rational function and gives the second line. Finally the elementary bounds $0 < \gamma < 1$ exclude the stated poles. □

The theorem isolates the precise unresolved exception instead of assuming that every transcendental E-value has an inverse in the ring. It does not decide whether γ is algebraic, nor whether δ belongs to $\mathcal{E}$. The mixed-Gevrey separation statements in [73] remain conjectural inputs; the new ring theorem does not supply them. The explicit nonmembership above is a dependent application of [72], rather than a new proof of individual transcendence for either connection coefficient.

23.2 Why a common deformation does not furnish a second equation

Proposition 23.3 (Finite differential cancellation removes the Euler term)**.** *Let $a_1, \dots, a_m$ be positive algebraic numbers, and let $L_j = \sum_{k=0}^{N_j} r_{jk}(z)(d/dz)^k$ with $r_{jk} \in \overline{\mathbb{Q}}(z)$. Put*

$$Y_j(z) = e^{-a_j z}\delta_{a_j z}, \qquad U(z) = \sum_{j=1}^{m} L_j Y_j(z).$$

If U is single valued on a punctured neighborhood of 0, then $\sum_j r_{j0}(z) = 0$. In the connection-formula expansion of U, the coefficient of γ consequently vanishes identically. Adding any single-valued meromorphic expression formed from E-functions and algebraic rational functions does not change this conclusion. If all the r_{jk} are regular at 1, then under the single-valued hypothesis

$$U(1) \in \mathcal{E} + \sum_{j=1}^{m} \overline{\mathbb{Q}} \log a_j.$$

Proof. One has $Y_j'(z) = -e^{-a_j z}/z$, so every positive derivative of Y_j is an exponential times an algebraic rational function. The only multivalued terms come from its zeroth derivative:

$$Y_j(z) = \mathrm{Ein}(a_j z) - \gamma - \log a_j - \log z.$$

The monodromy of U around 0 is therefore $-2\pi i \sum_j r_{j0}(z)$. Single valuedness forces this rational function to vanish. The same sum is the negative coefficient of γ. The assertion at 1 now follows directly from the remaining terms and closure of $\mathcal{E}$ under algebraic scalar multiplication and addition. □

For the individual Gompertz constant, a common-parameter derivative has no independent arithmetic content either. Differentiation under the integral and integration by parts give

$$\delta_x' = \delta_x - x^{-1}, \qquad \delta_x^{(k)} = \delta_x - \sum_{\ell=0}^{k-1} (-1)^\ell \ell! x^{-\ell-1} \quad (k \geq 1).$$

The second identity follows by induction. At $x = 1$ all these derivatives lie in $\delta + \mathbb{Z}$. Differentiating (23.1) cancels these occurrences of δ and yields the already known derivatives of Ein. This is an exact failure of this finite differential route, not a general impossibility theorem about analytic identities involving γ or δ.

23.3 Actual reduced denominators for positive zeta kernels

For an integer $s \geq 2$, define

$$\mathcal{Z}_s(P) = \frac{1}{(s-1)!} \int_0^1 \frac{(-\log x)^{s-1}}{1-x} P(x)\, dx.$$

If $P(x) = \sum_{k=0}^{N} a_k x^k \in \mathbb{Z}[x]$, integration term by term gives the *formal* coefficient identity

$$\mathcal{Z}_s(P) = P(1)\zeta(s) - R_s(P), \qquad R_s(P) = \sum_{k=0}^{N} a_k H_k^{(s)} \in \mathbb{Q}. \tag{23.4}$$

Here $H_0^{(s)} = 0$ and $H_k^{(s)} = \sum_{j=1}^{k} j^{-s}$. No irrationality hypothesis is used in defining these coefficients. Write $b_s(P)$ for the denominator of $R_s(P)$ in lowest terms, with denominator 1 for zero. This is the smallest positive integer clearing that rational remainder.

Lemma 23.4 (Endpoint amplitude controls a positive polynomial integral)**.** *If P is a nonnegative real polynomial on $[0,1]$, of degree at most $N \geq 1$, and $P(1) = q > 0$, then*

$$\mathcal{Z}_s(P) \geq \frac{q}{2(s-1)!(s-1)(8N^3)^{s-1}}.$$

Proof. Set $M = \max_{[0,1]} P \geq q$. Its shifted Chebyshev expansion has coefficients of absolute value at most $2M$ in positive degree, by the cosine integral formulas. Since $|T_j'| \leq j^2$ on $[-1,1]$, $\|P'\|_\infty \leq 4M \sum_{j=1}^{N} j^2 \leq 4MN^3$. On an interval of length $h = (8N^3)^{-1}$ adjacent to a maximum, and contained in $[0,1]$, one therefore has $P \geq M/2$. The weight satisfies $(-\log x)^{s-1}/(1-x) \geq (1-x)^{s-2}$. Among intervals of length h its last weight has least integral at the right endpoint, namely $h^{s-1}/(s-1)$. Substitution proves the claimed inequality. □

Theorem 23.5 (Prime cancellation is compensated by the target coefficient)**.** *Let $n \geq 2$, $0 \leq r < n$, and let $Q \in \mathbb{Z}[x]$ have degree at most r, be nonnegative on $[0,1]$, and satisfy $q = Q(1) > 0$. Put $P = x^n Q$, $N = n + r$, and*

$$\Pi_{n,r} = \prod_{(n+r)/2 < p \leq n} p,$$

the product being over primes. For every prime in this product,

$$v_p(b_s(P)) = \max\{0, s - v_p(q)\}. \tag{23.5}$$

Consequently $\Pi_{n,r}^s$ divides $b_s(P)q$, and

$$b_s(P)\mathcal{Z}_s(P) \geq \frac{\Pi_{n,r}^s}{2(s-1)!(s-1)(8N^3)^{s-1}}. \tag{23.6}$$

For every fixed $\eta < 1$, uniformly for $r \leq \eta n$ and for all such integer polynomials Q without a coefficient-height restriction,

$$\log(b_s(P)\mathcal{Z}_s(P)) \geq \frac{s(1-\eta)}{2} n + o(n) \longrightarrow +\infty.$$

In particular the reduced-denominator loophole does not rescue these kernels for $\zeta(5)$, or for any fixed integer $s \geq 2$.

Proof. Write $Q = \sum_{j=0}^{r} c_j x^j$. For a prime $(n+r)/2 < p \le n$, the only multiple of p among $1, \dots, n+r$ is p itself, and every harmonic sum $H_{n+j}^{(s)}$ contains its term p^{-s}. Thus

$$R_s(P) = \frac{\sum_j c_j}{p^s} + U_p = \frac{q}{p^s} + U_p, \qquad U_p \in \mathbb{Z}_{(p)}.$$

If $v_p(q) < s$, the first summand has negative p-adic valuation and the second has nonnegative valuation. Its denominator has exactly $p^{s-v_p(q)}$. If $v_p(q) \ge s$, both summands are p-integral. This proves (23.5), including the possibility that $R_s(P) = 0$. Multiplying the divisibilities over the distinct primes gives $b_s(P)q \ge \Pi_{n,r}^s$. Lemma 23.4 applied to P proves (23.6).

The prime number theorem, $\vartheta(X) = X + o(X)$, gives

$$\log \Pi_{n,r} = \vartheta(n) - \vartheta((n+r)/2) = (n-r)/2 + o(n).$$

The error is uniform for $0 \le r \le \eta n$, since both arguments lie in fixed proportional intervals. All losses in (23.6) are logarithmic in n for fixed s. This proves the last assertion. Only this classical weak form of the prime number theorem is needed; stronger unconditional estimates are available in [74]. □

This theorem replaces the standard $\operatorname{lcm}(1, \dots, N)^s$ by the actual reduced denominator. Its support hypothesis is material: the order of vanishing n at 0 exceeds the remaining degree by a fixed proportion, and $Q(1) \ne 0$. It does not cover every nonnegative polynomial or multi-integral construction. A canceled prime is allowed; the point is that its loss from the denominator is paid for by its appearance in q, which the positive integral also detects.

23.4 A reduced-denominator obstruction for Catalan kernels

Set

$$\mathcal{C}(P) = \int_0^1 \frac{-\log x}{1+x^2} P(x^2)\, dx = \int_0^1 w(t) P(t)\, dt, \qquad w(t) = \frac{-\log t}{4\sqrt{t}(1+t)} \ge \frac{1-t}{8}.$$

For $m \ge 0$, polynomial division gives

$$\mathcal{C}(t^m) = (-1)^m G + \sum_{j=0}^{m-1} \frac{(-1)^{m-1-j}}{(2j+1)^2}. \tag{23.7}$$

An empty sum is zero. Define the rational remainder $R_C(P)$ by $\mathcal{C}(P) = P(-1)G + R_C(P)$ using this identity, and let $b_C(P)$ be its actual reduced denominator.

Theorem 23.6 (Exterior amplitude and surviving odd primes)**.** *Let $n \ge 2$ and $0 \le r \le n/3$. Let $Q \in \mathbb{Z}[t]$ have degree at most r, be nonnegative on $[0,1]$, and satisfy $q = Q(-1) \ne 0$. Put $P = t^n Q$, $R = \max(1, r)$, and*

$$\Omega_{n,r} = \prod_{\substack{(2(n+r)-1)/3 < p \le 2n-1 \\ p \text{ odd prime}}} p.$$

Then $\Omega_{n,r}^2$ divides $b_C(P)|q|$, and

$$b_C(P)\mathcal{C}(P) \ge \frac{2^{-n} 14^{-r}}{8192 R^6}\, \Omega_{n,r}^2. \tag{23.8}$$

Uniformly in all such Q,

$$\log\bigl(b_C(P)\mathcal{C}(P)\bigr) \ge \left(\frac{20}{9} - \log 2 - \frac{\log 14}{3}\right) n + o(n) \longrightarrow +\infty.$$

The coefficient in parentheses is positive.

Proof. For a prime in the stated interval, the only odd multiple of p among $1, 3, \dots, 2(n+r)-1$ is p. Moreover $p \le 2n-1$, so its square denominator occurs in every moment of t^{n+j}, $0 \le j \le r$. Writing $Q = \sum_j c_j t^j$ and $k = (p-1)/2$, the coefficient of p^{-2} in (23.7) summed over these moments is

$$\sum_{j=0}^{r} c_j (-1)^{n+j-1-k} = (-1)^{n-1-k} Q(-1).$$

All remaining terms are p-integral. Exactly as in (23.5), $v_p(b_C(P)) = \max\{0, 2 - v_p(q)\}$, proving the divisibility.

Let $M = \max_{[1/2,1]} Q$. The exterior Chebyshev inequality on this interval yields

$$|Q(-1)| \le T_r(7) M \le 14^r M. \tag{23.9}$$

Here the affine coordinate is $u = 4t - 3$, for which $t = -1$ corresponds to $u = -7$. To recall the elementary exterior inequality, if a degree at most r polynomial bounded by 1 on $[-1, 1]$ exceeded $|T_r(u_0)|$ at a real $|u_0| > 1$, subtract the corresponding multiple of T_r. At its $r+1$ alternating extrema this difference has r interior sign changes; the additional root at u_0 is impossible. Finally $T_r(7) \le (7 + \sqrt{48})^r < 14^r$ for $r \ge 1$; degree zero is immediate.

For $r \ge 1$ the same cosine-coefficient derivative estimate used earlier, now on an interval of length $1/2$, gives $\|Q'\|_{[1/2,1]} \le 8Mr^3$. There is an interval of length $h = (16R^3)^{-1}$ in $[1/2, 1]$ on which $Q \ge M/2$. For $r = 0$ this assertion holds trivially. On this interval $t^n \ge 2^{-n}$; the least integral of $(1-t)$ over an interval of length h is $h^2/2$. Consequently

$$\mathcal{C}(t^n Q) \ge 2^{-n} \frac{M}{2} \frac{1}{8} \frac{h^2}{2} = \frac{2^{-n} M}{8192 R^6} \ge \frac{2^{-n} 14^{-r} |q|}{8192 R^6}.$$

Combining with $b_C(P)|q| \ge \Omega_{n,r}^2$ proves (23.8).

The prime number theorem gives, uniformly for $r \le n/3$,

$$\log \Omega_{n,r} = (4n - 2r)/3 + o(n).$$

The resulting logarithmic lower bound is $(8n - 4r)/3 - n \log 2 - r \log 14 - O(\log n) + o(n)$. Its minimum over this range of r is the asserted expression. For example $\log 2 < 0.7$ and $\log 14 < 2.64$ already show that its coefficient is greater than 0.64. □

This result covers genuine coefficient cancellation in the rational remainder. The restriction on the remaining degree and the requirement $Q(-1) \ne 0$ are part of the theorem, not conclusions about every approximating construction for G. An OpenAI Math Release preprint dated 24 September 2026 asserts the irrationality of G [53]. Its full proof is not audited or used here. Its signed mixed-moment determinant construction is outside our single positive polynomial kernel and support hypotheses; the present obstruction neither verifies nor refutes that claim.

23.5 Complete coefficient reduction: positivity has no remaining force

The smallest denominator of the rational remainder and the primitive normalization of the whole coefficient pair must not be conflated. If an integer kernel gives $A\xi + B$, $A \in \mathbb{Z}$, $B \in \mathbb{Q}$, and b is the denominator of B, the coefficients of $b(A\xi + B)$ may still share a common divisor. The next theorem describes the much larger class obtained by dividing that common divisor as well.

Theorem 23.7 (Constructive saturation of three positive kernel cones)**.** *Let ξ be any of δ, G, or $\zeta(s)$ for a fixed integer $s \geq 2$. For every relatively prime pair $p \in \mathbb{Z}$, $q \in \mathbb{Z}_{>0}$ with $q\xi - p > 0$, there is a nonzero integer polynomial P, nonnegative on the corresponding integration interval, whose integral has formal coefficients $Mq, -Mp$ for some positive integer M:*

$$\int_0^\infty \frac{e^{-t}P(t)}{1+t}\,dt = M(q\delta - p), \quad \text{or} \quad \mathcal{C}(P) = M(qG - p), \quad \text{or} \quad \mathcal{Z}_s(P) = M(q\zeta(s) - p).$$

Thus its rational remainder is already integral, and primitive reduction of its complete coefficient pair gives exactly $q\xi - p$.

Proof. In each case we construct a rational polynomial $K_n \geq 0$ with target coefficient 1 and integral $\xi - r_n > 0$, where $r_n \in \mathbb{Q}$ and $r_n \to \xi$. We also give a positive polynomial S with target coefficient 0 and integral 1.

For $\xi = \delta$, let

$$L_n(t) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{t^k}{k!}, \qquad K_n(t) = \frac{L_n(t)^2}{L_n(-1)^2}, \qquad S(t) = 1 + t.$$

The Laguerre orthogonality normalization is $\int_0^\infty e^{-t} L_n(t)^2 dt = 1$. It follows directly from Rodrigues' formula $L_n(t) = e^t (n!)^{-1} (d/dt)^n (e^{-t} t^n)$ by n integrations by parts. Also $L_n(-1) \geq n + 1$, so

$$0 < \int_0^\infty \frac{e^{-t}K_n(t)}{1+t}\,dt \leq \frac{1}{(n+1)^2} \longrightarrow 0.$$

Its target coefficient is $K_n(-1) = 1$. Division by $1+t$ and the factorial moments give a rational remainder. The assertions about S are immediate.

For $\xi = G$, take

$$K_n(t) = 2^{-n}(1-t)^n, \qquad S(t) = 1 + t.$$

Here $K_n(-1) = 1$ and $0 < \mathcal{C}(K_n) \leq 2^{-n} G \to 0$. Formula (23.7) gives its rational remainder, while $\mathcal{C}(1+t) = \int_0^1 -\log x\,dx = 1$ and $S(-1) = 0$.

For $\xi = \zeta(s)$, take $K_n(x) = x^n$ and $S(x) = 1 - x$. Then $r_n = H_n^{(s)} \to \zeta(s)$, while $S(1) = 0$ and

$$\mathcal{Z}_s(S) = \frac{1}{(s-1)!} \int_0^1 (-\log x)^{s-1} dx = 1.$$

Given the specified p, q, choose n with $r_n > p/q$. The rational polynomial

$$W = qK_n + (qr_n - p)S$$

is nonnegative, has target coefficient q, and has integral $q\xi - p$. Choose a positive integer M clearing all coefficients of W, and put $P = MW$. The formal coefficient pair is exactly $(Mq, -Mp)$, whose greatest common divisor is M because p, q are coprime. This proves every assertion, without presupposing irrationality of any of the three target constants. □

The theorem is a constructive statement about the image of a cone under its arithmetic moment map. In particular, a uniform positive lower bound for *all primitive-normalized* positive kernels cannot be deduced merely from positivity: that would be a bound for every positive primitive rational-approximation form $q\xi - p$ itself. The zeta and Catalan reduced-denominator theorems above concern the displayed integer kernels before this extra common-factor division. They do not close that logically different loophole. The saturation construction also does not control M; it gives no previously unknown small integer form for γ, δ, G, or $\zeta(5)$.

24 Three arithmetic generating functions beyond the modular cusp field

The three number-theoretic targets in this section are the Erdős–Borwein constant, the prime binary constant, and the prime Lambert constant,

$$E = \sum_{n\geq 1} \frac{1}{2^n - 1}, \qquad P_2 = \sum_{p \text{ prime}} 2^{-p}, \qquad L_{\mathbb{P}} = \sum_{p \text{ prime}} \frac{1}{2^p - 1}.$$

Their transcendence is not proved here. Instead, we prove that their generating functions are algebraically independent over a field containing every finite collection of dilated Eisenstein series and their derivatives. This answers a precise functional question behind an attempted modular transfer: no finite algebraic elimination within that field can isolate any of these three functions from the other two. The result extends to each individual reciprocal-Mersenne moment. It does not assert joint independence of all the moments, which would be false over this field.

For $|q| < 1$ and an integer $r \geq 1$, write

$$A_r(q) = \sum_{n\geq 1} \left(\frac{q^n}{1-q^n}\right)^r, \qquad P(q) = \sum_{p \text{ prime}} q^p, \qquad B(q) = \sum_{p \text{ prime}} \frac{q^p}{1-q^p}. \tag{24.1}$$

All series converge locally uniformly in the open unit disk. In particular, $E = A_1(1/2)$, $P_2 = P(1/2)$, and $L_{\mathbb{P}} = B(1/2)$. Functional independence of q-zeta series has a substantial prior literature; see, for example, Pupyrev [77] and Bézivin [78]. We make no priority claim for the general idea of using boundary asymptotics to establish functional independence. The statement proved below concerns the specified prime-supported functions and the explicit enlarged coefficient field.

24.1 A coefficient field that includes finite modular constructions

Throughout this section t is positive, $t \downarrow 0$, $q = e^{-t}$, and $L = \log(1/t)$. For $\lambda > 0$, define

$$\mathscr{K}_\lambda = \left\{ F(t, e^{-\lambda/t}) : F \in \operatorname{Frac} \mathbb{C}\{t, z\} \right\}. \tag{24.2}$$

Here $\mathbb{C}\{t, z\}$ denotes convergent power-series germs at $(0, 0)$, and the expressions in (24.2) are germs on the positive real axis. The next lemma verifies, in particular, that the evaluation is injective and that nonzero denominators cause no ambiguity near $t = 0$.

Lemma 24.1 (Leading order in the cusp field)**.** *For every nonzero $h \in \mathscr{K}_\lambda$ there exist integers a, m and $c \in \mathbb{C} \setminus \{0\}$ such that*

$$h(t) = ct^a e^{-m\lambda/t}(1 + O(t)). \tag{24.3}$$

The field contains t and e^{-t} and is closed under differentiation with respect to t.

Proof. For a nonzero convergent germ $G(t, z)$, choose the least $m_0 \geq 0$ for which the coefficient $g_{m_0}(t)$ of z^{m_0} is not identically zero. There are an integer $a_0 \geq 0$ and a nonzero c_0 such that $g_{m_0}(t) = c_0 t^{a_0}(1 + O(t))$. Convergence in a polydisk gives

$$G(t, z) = z^{m_0}\big(g_{m_0}(t) + zH(t, z)\big)$$

with H analytic and bounded in a smaller polydisk. Substitution of $z = e^{-\lambda/t}$ shows that the second term inside the parentheses is $o(t^N)$ for every N, and proves (24.3) for G. Thus

the evaluated germ is nonzero and has no zeros for all sufficiently small positive t. Taking quotients proves the assertion for the field and proves injectivity of evaluation. Finally, the chain rule gives

$$\frac{d}{dt}F(t,e^{-\lambda/t}) = \left(F_t + \lambda z t^{-2} F_z\right)(t,e^{-\lambda/t}),$$

and the meromorphic germ on the right belongs to the same fraction field. □

Let $\mathcal{E}_2, \mathcal{E}_4, \mathcal{E}_6$ denote the normalized Eisenstein series, whose constant terms are 1.

Proposition 24.2 (Finite modular fields lie in a cusp field)**.** *Let $\mathcal{D}$ be a finite nonempty set of positive integers and $D = \operatorname{lcm}\mathcal{D}$. The differential field over $\mathbb{C}$ generated by q and*

$$\mathcal{E}_2(q^d), \quad \mathcal{E}_4(q^d), \quad \mathcal{E}_6(q^d) \qquad (d \in \mathcal{D}),$$

where the derivation is $q\,d/dq$, is contained, after $q = e^{-t}$, in $\mathscr{K}_{4\pi^2/D}$.

Proof. The modular transformation and its weight-two correction, in the normalization of [76, Section 1], give

$$\begin{aligned}
\mathcal{E}_2(e^{-t}) &= -\frac{4\pi^2}{t^2}\mathcal{E}_2(e^{-4\pi^2/t}) + \frac{12}{t},\\
\mathcal{E}_4(e^{-t}) &= \frac{16\pi^4}{t^4}\mathcal{E}_4(e^{-4\pi^2/t}),\\
\mathcal{E}_6(e^{-t}) &= -\frac{64\pi^6}{t^6}\mathcal{E}_6(e^{-4\pi^2/t}).
\end{aligned}$$

Replace t by dt and set $z = e^{-4\pi^2/(Dt)}$. The transformed argument is $z^{D/d}$, an integer power of z, and each Eisenstein series is analytic at argument zero. Thus all the displayed generators lie in the indicated field. The derivation is $-d/dt$, so Lemma 24.1 completes the proof. □

24.2 The logarithmic coordinate of every reciprocal-Mersenne moment

The following estimate retains a power-saving remainder after removal of the polar terms. That remainder, rather than only a leading equivalent, is useful for the subsequent independence argument.

Lemma 24.3 (A flat logarithmic coordinate)**.** *Define the real coefficients $\kappa_{r,j}$ by the Laurent expansion*

$$(e^x - 1)^{-r} = \sum_{j=1}^{r} \kappa_{r,j} x^{-j} + O(1) \qquad (x \longrightarrow 0).$$

Then $\kappa_{r,1} = (-1)^{r-1}$. There is a real constant C_r such that

$$A_r(e^{-t}) = \sum_{j=2}^{r} \kappa_{r,j}\zeta(j)t^{-j} + \kappa_{r,1}\frac{L}{t} + \frac{C_r}{t} + O(1). \tag{24.4}$$

Consequently the function

$$X_r(t) = \frac{1}{\kappa_{r,1}}\left(tA_r(e^{-t}) - \sum_{j=2}^{r}\kappa_{r,j}\zeta(j)t^{1-j} - C_r\right) \tag{24.5}$$

satisfies $X_r(t) = L + O(t)$.

Proof. The coefficient of x^{-1} is the residue at zero. Under $z = e^x - 1$,

$$(e^x - 1)^{-r}\,dx = \frac{z^{-r}}{1+z}\,dz,$$

so that residue is $(-1)^{r-1}$. Put $\kappa = \kappa_{r,1}$ and

$$g_r(x) = (e^x - 1)^{-r} - \sum_{j=2}^{r} \kappa_{r,j} x^{-j} - \kappa \frac{e^{-x}}{x}.$$

The singularity of g_r at zero is removable. The function extends to a continuously differentiable function on $[0, \infty)$, and both g_r and g_r' are integrable there: at infinity the subtracted powers have order at most x^{-2}, and all the other terms decay exponentially. Let $C_r = \int_0^\infty g_r(x)\,dx$.

For each interval $[(n-1)t, nt]$, comparison with its right endpoint gives

$$\left| t g_r(nt) - \int_{(n-1)t}^{nt} g_r(x)\,dx \right| \le t \int_{(n-1)t}^{nt} |g_r'(x)|\,dx.$$

Summation yields $\sum_{n\ge1} g_r(nt) = C_r/t + O(1)$. The removed terms sum exactly:

$$\sum_{n\ge1} (nt)^{-j} = t^{-j}\zeta(j), \qquad \sum_{n\ge1} \frac{e^{-nt}}{nt} = -\frac{1}{t}\log(1-e^{-t}) = \frac{L}{t} + O(1).$$

This proves (24.4) and (24.5). No unproved estimate for the divisor problem is required. □

24.3 The prime coordinate has a divergent inverse-logarithmic series

Lemma 24.4 (All inverse-logarithmic orders)**.** *For every fixed positive integer N,*

$$Y(t) := tP(e^{-t}) = \sum_{k=0}^{N-1} \frac{a_k}{L^{k+1}} + O_N(L^{-N-1}), \qquad a_k = (-1)^k \Gamma^{(k)}(1). \tag{24.6}$$

The formal power series $\widehat{S}(z) = \sum_{k\ge0} a_k z^{k+1}$ is transcendental over $\mathbb{C}(z)$.

Proof. Write $\pi(x)$ for the prime-counting function and $\mathrm{Li}_2(x) = \int_2^x du/\log u$ for $x \ge 2$. The exponential error estimate in Trudgian's Theorem 2 [75] implies, for every fixed $A > 0$,

$$\pi(x) - \mathrm{Li}_2(x) = O_A(x/\log^A x).$$

The normalization of the logarithmic integral changes its value only by a constant. Stieltjes integration and then integration by parts give

$$tP(e^{-t}) - t\int_2^\infty \frac{e^{-tx}}{\log x}\,dx = t^2 \int_2^\infty e^{-tx}(\pi(x) - \mathrm{Li}_2(x))\,dx.$$

On $2 \le x \le t^{-1/2}$ the integral on the right is $O(t)$, using $\pi(x) + |\mathrm{Li}_2(x)| = O(x)$. On the rest of the interval, $\log x \ge L/2$, so it is $O_A(L^{-A})$. Changing variables $u = tx$ therefore gives

$$Y(t) = \int_{2t}^\infty \frac{e^{-u}}{L + \log u}\,du + O_A(L^{-A}). \tag{24.7}$$

The portion with $2t \le u \le t^{1/2}$ is $O(t^{1/2})$, since the denominator is at least $\log 2$. The portion with $u \ge t^{-1/2}$ is smaller than every power of t. On the middle interval, $|\log u| \le L/2$, and

$$\frac{1}{L+\log u} = \sum_{k=0}^{N-1} \frac{(-\log u)^k}{L^{k+1}} + \frac{(-\log u)^N}{L^N(L+\log u)}.$$

The last term, after integration against e^{-u}, is $O_N(L^{-N-1})$, because $\int_0^\infty e^{-u}|\log u|^N du < \infty$. Extending the coefficient integrals to $(0,\infty)$ introduces errors smaller than every inverse power of L. Taking $A = N+1$ in (24.7) proves (24.6).

The function $\Gamma(1+w) - (1+w)^{-1}$ is analytic in $|w| < 2$. For every fixed $1 < \rho < 2$, Cauchy's coefficient estimate gives

$$a_k = k!(1 + O_\rho(\rho^{-k})). \tag{24.8}$$

Thus $\widehat{S}$ has radius of convergence zero. We recall why this excludes algebraicity, rather than assuming any summability assertion. Every formal complex power series algebraic over $\mathbb{C}(z)$ is convergent. Indeed, let an irreducible polynomial $Q(z,w)$ annihilate a formal series $S(z)$. In characteristic zero, $Q_w(z, S(z)) \neq 0$; let its order be m. For a sufficiently long truncation S_N, with $N+1 > m$, write $S = S_N + z^{N+1}U$. The equation

$$z^{-(N+1+m)}Q(z, S_N + z^{N+1}U) = 0$$

is analytic in (z,U): its constant-in-U term has no negative powers, its coefficient of U has a nonzero constant term, and its higher coefficients vanish at $z = 0$. Its derivative in U at the formal constant term $U(0)$ is nonzero. The analytic implicit function theorem supplies a convergent solution, whose formal series is unique. Hence S converges. Applying this fact to (24.8) proves the asserted formal transcendence. □

Lemma 24.5 (The prime Lambert coordinate)**.** *As $t \downarrow 0$,*

$$Z(t) := tB(e^{-t}) \sim \log L. \tag{24.9}$$

Proof. Put $X = 1/t$. Uniformly for $p \leq X$, $(e^{p/X} - 1)^{-1} = X/p + O(1)$. For $p > X$ the summand is at most a constant times $e^{-p/X}$, and comparison with all integers bounds the tail by $O(X)$. Therefore

$$B(e^{-1/X}) = X \sum_{p \leq X} \frac{1}{p} + O(X).$$

Partial summation from $\pi(x) \sim x/\log x$ gives $\sum_{p \leq X} 1/p \sim \log\log X$. Dividing by X proves the claim. □

24.4 The independence theorem and its exact scope

Theorem 24.6 (Three-function independence over the cusp field)**.** *For every $\lambda > 0$ and every fixed integer $r \geq 1$, the three germs*

$$A_r(e^{-t}), \qquad P(e^{-t}), \qquad B(e^{-t})$$

are algebraically independent over $\mathscr{K}_\lambda$. Consequently, for every finite set $\mathcal{D}$ of positive integer dilations, the three analytic functions $A_r(q), P(q), B(q)$ are algebraically independent over the differential field generated by $\mathbb{C}(q)$ and the functions $\mathcal{E}_2(q^d), \mathcal{E}_4(q^d), \mathcal{E}_6(q^d)$, $d \in \mathcal{D}$.

Proof. The affine changes (24.5), $Y = tP(e^{-t})$, and $Z = tB(e^{-t})$ have coefficients in $\mathscr{K}_\lambda$ and nonzero slopes. It is enough to prove algebraic independence of $X = X_r, Y, Z$.

First consider a nonzero polynomial $R(U,V) \in \mathbb{C}[U,V]$. Because $\widehat{S}$ is transcendental over $\mathbb{C}(z)$,

$$R(z^{-1}, \widehat{S}(z)) \neq 0$$

as a formal Laurent series. It has a first nonzero term cz^b, where $b \in \mathbb{Z}$ and $c \neq 0$. By taking sufficiently many terms in (24.6) and using $X = L + O(t)$, we obtain

$$R(X(t), Y(t)) \sim cL^{-b}. \tag{24.10}$$

This step only uses a finite truncation for each specified polynomial; no convergence of $\widehat{S}$ is asserted.

Now let $Q(U, V, W) \in \mathbb{C}[U, V, W]$ be nonzero, and write $Q = \sum_j R_j(U, V)W^j$ with its nonzero coefficients R_j. By (24.10), $R_j(X, Y) \sim c_j L^{b_j}$ for suitable integers b_j and nonzero constants c_j. Choose the largest b_j, and among the indices attaining that largest value choose the largest j. Since $Z \sim \log L$, the corresponding term uniquely dominates:

$$Q(X, Y, Z) \sim c_j L^{b_j} (\log L)^j. \tag{24.11}$$

In particular, no such polynomial can vanish, or be smaller than every inverse power of L.

Suppose, finally, that a nonzero polynomial with coefficients in $\mathscr{K}_\lambda$ vanishes at (X, Y, Z). Write each nonzero coefficient in the form (24.3). Among this finite set choose the least exponential order m, and then the least power a among coefficients having that exponential order. Divide the polynomial identity by $t^a e^{-m\lambda/t}$. The leading constants of the coefficients with these two minimal orders give a nonzero polynomial $Q \in \mathbb{C}[U, V, W]$. All other terms, including the errors in those selected coefficients, are bounded by $O(t)$ times a fixed power of $L + \log L + 1$, plus terms exponentially small in $1/t$ times a fixed power of t^{-1} and $L + \log L + 1$. Here we used $X = O(L)$, $Y = O(L^{-1})$, and $Z = O(\log L)$. The identity would therefore force $Q(X, Y, Z)$ to be smaller than every inverse power of L. This contradicts (24.11). The functional conclusion follows from Proposition 24.2; an analytic identity in the disk would also hold for the corresponding germs. □

Corollary 24.7 (Algebraic extensions do not remove the obstruction)**.** *The independence in Theorem 24.6 persists after replacing $\mathscr{K}_\lambda$ by any algebraic extension contained in a common field of germs. Thus adjoining finitely many algebraic branches of finite modular constructions does not supply a functional algebraic relation among the three functions.*

Proof. If $K'/\mathscr{K}_\lambda$ is algebraic, then $K'(X, Y, Z)/\mathscr{K}_\lambda(X, Y, Z)$ is algebraic. Transcendence degree is unchanged by algebraic extension, so

$$\operatorname{trdeg}_{K'} K'(X, Y, Z) = \operatorname{trdeg}_{\mathscr{K}_\lambda} \mathscr{K}_\lambda(X, Y, Z) = 3.$$

□

There is a useful exact check on the proposed $q \mapsto -q$ route. Directly from the coefficients $d(n)$ of $A_1(q)$,

$$A_1(q) + A_1(-q) = 4A_1(q^2) - 2A_1(q^4). \tag{24.12}$$

Indeed $d(2n) = 2d(n) - \mathbf{1}_{2|n} d(n/2)$, as is seen by writing $n = 2^a m$ with m odd. Comparing the even coefficients proves (24.12). It introduces further unknown Lambert values and is not a modular expression for A_1. For completeness, the analogous divisor-sum identity gives

$$\mathcal{E}_{2k}(-q) = 2(1 + 2^{2k-1})\mathcal{E}_{2k}(q^2) - 2^{2k}\mathcal{E}_{2k}(q^4) - \mathcal{E}_{2k}(q), \qquad k = 1, 2, 3.$$

Thus adding these negative-argument Eisenstein series still stays inside a finite dilation field covered by the theorem.

The result leaves a precise arithmetic gap. Functional algebraic independence does not imply algebraic independence, or even individual transcendence, at $q = 1/2$. For example,

multiplying any functionally independent family over $\mathbb{C}(q)$ by $q - 1/2$ preserves functional independence but makes all its specialized values zero. A successful proof for E, P_2, or $L_{\mathbb{P}}$ therefore still needs a specialization theorem with verified hypotheses, or a direct arithmetic approximation argument. Neither is supplied here. The theorem also does not exclude differential equations involving derivatives of the target functions themselves; the derivatives allowed in its coefficient field are derivatives of the modular generators. Its contribution is a proved, explicit boundary on one proposed transfer method, not a resolution of the three transcendence problems.

25 What has been resolved, and what would constitute the next step

The unconditional gains concern structured variational constants. They include a full cubic-field classification, an identified independent pair of observables, infinite independence in a fixed stochastic model, all integer extraction shapes, and finite arithmetic descriptions of signed entropy optima. Their proofs use classical transcendence inputs in explicitly checked settings. They do not establish a new general transcendence principle for arbitrary special-function values.

The conditional stopping result identifies a narrower target than a blanket appeal to Schanuel's conjecture. For rational $a = r/s > 0$ and $b > 0$, the function $M(a; b; z)^s - cz^r$ has only finitely many multiple zeros. It would therefore suffice to prove that multiple zeros of this specific E-function family, for algebraic $c \neq 0$, are algebraic. That restricted statement would already classify every rational bridge payoff. The full squarefree-factorization conjecture is sufficient, but it is not proved and is not known here to be necessary.

For the spectral program, the first concrete gap is the biquadratic value in (19.25). Its optimizing location and elasticity are transcendental, but these facts do not determine its value. The cubic proof explains the missing requirement: a lower bound of two for the relevant exponential-line transcendence degree. Rational dimension three alone does not provide the cubic-field theorem's hypothesis.

For fixed even-dimensional stopping, an additional question is whether all even-payoff optima, beyond the prime-indexed or selected subsequence, are algebraically independent, or which explicit relations their boundary polynomials permit. The splitting-field argument used here makes no such claim when the degree factor is already present in the earlier fields. An answer would require more arithmetic of those specific polynomials.

For classical constants, the reduced-denominator and saturation theorems show why normalization must be specified before an approximation is called small. Positivity controls actual denominator-cleared forms in the displayed support classes. After primitive reduction of the entire coefficient pair, however, these positive cones realize every positive primitive rational-approximation form. That universality supplies no new small form and cannot be counted as an irrationality proof.

The prime/divisor functional theorem has a similarly precise boundary: even algebraic independence over a large modular coefficient field does not force transcendence at $q = 1/2$. A valid specialization theorem or a direct arithmetic argument is still needed. Replacing that missing step by numerical evidence or a functional identity would change the result into an unsupported claim.

Verification and priority. The main sections were independently cross-read within this project. The checks covered the arbitrary-base hypothesis in Brownawell's theorem, the

curvature factorization argument, the Eisenstein coefficients and splitting-field separation, the complex Kummer remainder sectors, the third-order intertwiner and contact multiplicity, the entropy face and cardinality arguments, the exact reduced-denominator prime windows, and the cusp-field asymptotics. Exact small-parameter polynomial calculations were used to check formulas, not to replace uniform proofs. The cited external conjecture remains conditional. Historical priority is not certified, and no newly solved named individual-constant conjecture is reported.

Part IV
Rigidity of Optimization

Editorial integration note. The all-planar independence theorem strengthens earlier subfamilies. The differential classifications and exponential-value closure results are further developed in Part V. Statements about functions do not automatically classify their values at a specified point.

Original part framing.

Status. Research working manuscript. Proofs and external inputs are separated, conditional implications are identified, and internal cross-reading is not external peer review or formal verification. Historical priority for the particular transfers has not been established by an exhaustive literature review.

26 The question, the gain, and the remaining arithmetic gap

This paper continues *Arithmetic Independence of Variational Constants* [79], the preceding 45-page manuscript. The 223-page observation-and-recovery foundation and its collected sequels are earlier inputs, rather than results reprinted or recounted here. The research objective is to turn their analytic and identification methods into proofs about the arithmetic of specific constants. One must distinguish the arithmetic of an optimizer from that of its value, and independence of functions from independence of values after specialization. Both distinctions determine the successful transfers and the remaining failures below.

26.1 The most substantial increase in scope

The leading result is Theorem 27.1. If

$$V_n = \sup_{0\le\tau\le1} \mathbb{E}\|B_\tau\|^{2n}$$

for one standard planar Brownian bridge, then

$$\boxed{V_1, V_2, V_3, \dots \text{ are algebraically independent.}}$$

There is no nonzero polynomial with algebraic coefficients relating any finite selection of these values. This is stronger than saying that each is transcendental. The preceding manuscript proved the corresponding statement only for e and prime-indexed V_p; the present theorem includes every composite index as well. Because $V_1 = 2/e$, adjoining e separately to the full sequence would create a relation, and is deliberately not asserted.

The mechanism is also reusable. The boundary polynomial has degree n and a doubly transitive Galois action. Its centered roots generate an irreducible rational representation of dimension $n-1$. Different degrees therefore supply different representations even when splitting fields intersect. This removes the need to prove disjointness of those fields. Lindemann–Weierstrass then converts linear independence of algebraic boundary parameters into algebraic independence of optimal values.

In information theory, Theorem 28.6 gives an exact arithmetic transition. For rational $0 < c < 1$ and algebraic $\beta > c^{-2}$, the unique positive optimal bias satisfies

$$\operatorname{arctanh} s_\beta = \beta c \operatorname{arctanh}(c s_\beta), \qquad s_\beta \in \overline{\mathbb{Q}} \iff \beta \in \mathbb{Q}.$$

For algebraic irrational β, every finite optimal encoder has a transcendental entry. This is an exact classification in a familiar optimization model; it is not a proof of transcendence of the optimized objective itself. Its proof uses Gelfond–Schneider, and its analytic binary bottleneck boundary is credited to the established literature.

The other changes are summarized without counting individual examples as independent conjectures.

Direction	Earlier result or limitation	Present result
Spectral optimization	Arithmetic of selected values; a biquadratic value unresolved	D-finite exactly for rational exponent gap; exact nonlinear order and a complete derivative-field barrier
Gamma extraction	Integer-shape contact arithmetic and double contact	Double contacts for all real shapes, piecewise analytic frontier and slope comparison; exact rank three at noninteger shape
Contact observables	Contact and critical rate individually transcendental	At integer Bessel parameter, the contact exponential is also transcendental; the three quantities remain algebraically dependent
Entropy values	Upper inclusion in logarithms of positive algebraic numbers	Exact converse realization by four signed entropy terms in a pure-saddle game
E-value arithmetic	Rational-function intersection and nonunits	Exact rational-power membership; a finite-cover obstruction and an unconditional all-but-one relative-closure theorem
Prime generating series	Functional algebraic independence	Joint independence of every derivative in the stated linear sense, over an enlarged coefficient field

26.2 Twenty-one targets, with different kinds of outcome

The register is an account of attempts, not a count of famous open problems solved. "Unresolved here" reports this paper's conclusion. Known irrationality results and results proved in the earlier manuscripts are not added to the present success count.

No.	Target	Outcome and precise scope
1	Euler's constant γ: irrationality	Unresolved. A nontrivial algebraic extension inside the specified E-value field would force γ algebraic, but no such extension is constructed.
2	Euler–Gompertz δ: irrationality	Unresolved. The new morphism and divisor argument does not determine this connection coefficient.
3	Erdős–Borwein $\sum_{n\geq 1}(2^n-1)^{-1}$: transcendence	Unresolved as a number. A functional coefficient-field enlargement is proved; known linear-form estimates do not settle arbitrary algebraic degree.
4	Prime binary $\sum_p 2^{-p}$: transcendence	Unresolved as a number. The full derivative family is linearly independent over the stated cusp field with added moment and Lambert coefficients.
5	Prime Lambert $\sum_p(2^p-1)^{-1}$: transcendence	Unresolved as a number. Joint derivative independence and a prime-forced shift equation are established; external irrationality is credited.

No.	Target	Outcome and precise scope
6	Endpoint shape $A(1)$: transcendence	Unresolved. The rational noninteger-shape theorem does not permit an algebraic irrational varying Kummer parameter.
7	Endpoint shape $A(3/2)$: transcendence	Unresolved for the same parameter obstruction; the previous irrationality result is not recounted.
8	Linear excursion optimum $\mathcal{V}_{3,1}$	Unconditional arithmetic unresolved. Mixed zero multiplicities and unbounded exponential denominators exclude a specified available partial factorization theorem.
9	Biquadratic spectral optimum Q in (30.1)	Value unresolved. Completing an exponential table adds an uncontrolled fourth power; every setting derivative remains in the first-derivative field.
10	$H(\mathrm{Poisson}(1))$: irrationality	Unresolved. Infinite logarithmic coefficients remain outside finite entropy and E-function arguments.
11	$\max_{x>0} \mathrm{Ein}(x)/x^{\theta}$, algebraic $0 < \theta < 1$	Value unresolved. The maximizing location is transcendental, so algebraic-argument specialization cannot be applied there.
12	All planar even-payoff optimal values jointly	Algebraic independence of the entire sequence, Theorem 27.1. One infinite-family theorem.
13	Algebraic-weight binary bottleneck optimizer	Exact rational/algebraic-irrational classification above threshold, Theorem 28.6. Includes every finite output alphabet.
14	Range of rational pure-saddle entropy values	Exactly $\log \overline{\mathbb{Q}}_{>0}$, with four signed entropy terms, Theorem 28.2.
15	Rational monomials in $\mathrm{Ein}(a)$ inside the E-value field	Necessary and sufficient integral-exponent criterion for the stated separated family, Theorem 29.2.
16	Algebraic extensions over $\overline{\mathbb{Q}}(\mathrm{Ein}(a))$	Relative algebraic closure for all but at most one algebraic $a \geq 1$, Corollary 29.5; the exception is not identified.
17	Arithmetic at rational noninteger extraction shapes	Normalized values and logarithmic derivatives at nonzero algebraic points are transcendental; zeros and physical stationary locations are transcendental under the stated rate hypotheses.
18	Fractional-shape critical rate itself	Unresolved. Exact rank three explains why two contact equations do not reproduce the integer-shape elimination.
19	e^{2x} at integer extraction contacts	Transcendental and in $\overline{\mathbb{Q}}(x, c)$; the triple has transcendence degree one, Proposition 31.8.
20	Two-atom optimal-value function and contact map	Complete D-finite classification and exact minimal nonlinear differential order, Theorems 30.2–30.4.
21	All-real-shape extraction frontier	Exact double contact, locally finite analytic branches, signed slope formula and strict comparison law, Theorems 31.2 and 31.4.

26.3 Conventions and external inputs

Throughout, algebraic and transcendental mean over $\mathbb{Q}$; joint algebraic independence is equivalently stated over $\overline{\mathbb{Q}}$. All real powers use the real logarithm unless an analytic continuation is explicitly specified. The notation D-finite refers to a homogeneous linear differential equation with polynomial coefficients; it is different from satisfying a nonlinear algebraic differential equation.

The stopping formula, Schur irreducibility, Jordan–Marggraff, Lindemann–Weierstrass, Gelfond–Schneider, Beukers specialization, and Lind's Perron realization are external inputs.

The arithmetic-ring section additionally uses a recent Fischler–Rivoal morphism theorem, identified there as an external preprint result. The analytic existence, continuity and uniform endpoint positivity of the gamma-extraction capacity are inherited from the earlier project manuscript and stated explicitly where used. All new deductions are given with proofs; these dependency statements do not certify the earlier manuscripts or external papers beyond the specific inputs checked.

For the two endpoint targets in the register, the earlier exact definition is recalled here to remove a notation dependency. For $d > 1/2$, $A(d)$ is the unique value

$$a \in (\max(0, 1-d), 1/(4d))$$

for which, with $y(z) = M(a-1; a+d-1; z)$ and $z = 2(d-a)$,

$$y(z) - \frac{z}{2(1-a)} y'(z) = 0.$$

Its existence, uniqueness and irrationality at rational d are earlier results, not proved anew here. The present question for $A(1), A(3/2)$ is their remaining transcendence classification.

27 All planar even-payoff optima are algebraically independent

This section answers the self-posed arithmetic question for the entire sequence of even-power stopping optima of a single planar Brownian bridge. The preceding paper treated prime-indexed powers. Here every positive integer index is included: the resulting optimal values are jointly algebraically independent. The argument combines Schur's irreducibility theorem, a Newton-polygon calculation, the classical Jordan–Marggraff theorem, and a representation-theoretic separation of centered roots. In particular, it requires no disjointness assertion about the splitting fields.

A second target is the unconditional arithmetic of the linear Brownian-excursion optimum. That question remains unresolved. We prove two concrete obstructions to applying an available partial E-function factorization theorem: the relevant functions have mixed zero multiplicities and fail its global-boundedness hypothesis. These are limitations of a specified route, not a proof of impossibility of the desired arithmetic conclusion.

27.1 The stopping constants and the main result

Let R_t be the norm of a standard two-dimensional Brownian bridge from zero to zero on $[0, 1]$. For $n \geq 1$ set

$$V_n = \sup_{0 \leq \tau \leq 1} \mathbb{E}[R_\tau^{2n}], \tag{27.1}$$

where the supremum is over stopping times for its natural filtration. The established stopping formula of Hobson–Liu [34] gives

$$f_n(z) = M(n; 1; z), \qquad z_n f_n'(z_n) = n f_n(z_n), \qquad V_n = \frac{(2z_n)^n}{f_n(z_n)}. \tag{27.2}$$

Here $z_n > 0$ is the unique solution and the stopping boundary for the radial process is $\sqrt{2z_n(1-t)}$.

For completeness, uniqueness of the positive solution follows directly from the positive Taylor coefficients $c_j = (n)_j/(j!)^2$. The probabilities $c_j z^j / f_n(z)$ have mean $u(z) = z f_n'(z)/f_n(z)$ and satisfy

$$z u'(z) = \mathrm{Var}_z(j) > 0.$$

Their mean starts at zero and tends to infinity: for every fixed K, the total weight on $j \le K$ tends to zero by comparison with the term of index $K+1$. Thus $u(z) = n$ has exactly one positive solution. The probabilistic optimality assertion itself is the cited stopping theorem, rather than a new verification theorem here.

Theorem 27.1 (The full planar sequence)**.** *The numbers*

$$V_1, V_2, V_3, \dots$$

are algebraically independent over $\overline{\mathbb{Q}}$. Equivalently, $e, V_2, V_3, \dots$ are algebraically independent. The associated squared boundary parameters satisfy

$$1, z_2, z_3, \dots \quad \text{are linearly independent over } \mathbb{Q}. \tag{27.3}$$

Every assertion about an infinite family means the corresponding assertion for every finite subfamily.

Since $V_1 = 2/e$, the first formulation does not append e to a family already containing V_1. In particular the ten consecutive optimal values for payoff powers $2, 4, 6, \dots, 20$ are algebraically independent. This is one uniform theorem, not ten separate closures.

27.2 Irreducibility of every boundary polynomial

Kummer's transformation gives

$$f_n(z) = e^z P_{n-1}(z), \qquad P_{n-1}(z) = M(1-n; 1; -z) = \sum_{k=0}^{n-1} \binom{n-1}{k} \frac{z^k}{k!}. \tag{27.4}$$

In particular $P_{n-1}(z) > 0$ for $z > 0$. Define

$$B_n(z) = (n-1)!\{z P'_{n-1}(z) + (z-n) P_{n-1}(z)\}. \tag{27.5}$$

Its unique positive zero is z_n.

Lemma 27.2 (Schur form of the stopping polynomial)**.** *For every $n \ge 1$, B_n is a monic irreducible polynomial in $\mathbb{Z}[z]$ of degree n. More explicitly,*

$$B_n(z) = n! \sum_{k=0}^{n} a_{n,k} \frac{z^k}{k!}, \qquad a_{n,k} = \binom{n-1}{k-1} - \binom{n-1}{k}, \tag{27.6}$$
$$a_{n,0} = -1, \qquad a_{n,n} = 1.$$

Out-of-range binomial coefficients are zero. The sum of all roots of B_n is $n(2-n)$.

Proof. Taking coefficients in (27.5) gives

$$\frac{(n-1)!}{k!}\left[k\binom{n-1}{k-1} + (k-n)\binom{n-1}{k}\right].$$

The expression in brackets is $n[\binom{n-1}{k-1} - \binom{n-1}{k}]$, proving (27.6). Its integrality and monicity follow immediately. Schur's classical theorem says that $\sum_{k=0}^{n} a_k z^k/k!$ is irreducible over $\mathbb{Q}$ when $a_k \in \mathbb{Z}$ and $|a_0| = |a_n| = 1$. This exact formulation is stated and generalized in Jindal–Khanduja [80, Theorem 1.1 and introduction]. It applies to the coefficients above; multiplying by $n!$ does not affect irreducibility. Finally the coefficient of z^{n-1} is $n(n-2)$, giving the stated trace by Vieta's formula. □

For comparison with classical orthogonal polynomials,

$$B_n(z) = n!\{L_n(-z) - 2L_{n-1}(-z)\}.$$

This identity follows from the same coefficient calculation. Irreducibility of an individual Laguerre polynomial would not by itself justify irreducibility of this particular linear combination; the Schur form above is the needed input.

27.3 Double transitivity from one local valuation

Lemma 27.3 (A long prime cycle)**.** *Let $n \geq 3$ and let p be a prime with $n/2 < p < n$. The Galois group of B_n contains a p-cycle.*

Proof. Write $n = p + r$ with $1 \leq r < p$. Normalize the p-adic valuation by $v_p(p) = 1$. The constant coefficient of B_n is $-n!$, of valuation one. Every coefficient of degree $k < p$ has valuation at least one, since $n!/k!$ is divisible by p.

For the coefficient of degree p, observe that, modulo p,

$$(1 + X)^{p+r-1} \equiv (1 + X^p)(1 + X)^{r-1}.$$

Consequently $\binom{n-1}{p-1} \equiv 0$ and $\binom{n-1}{p} \equiv 1 \pmod{p}$, and hence $a_{n,p} \equiv -1 \pmod{p}$. Since $n!/p!$ is a p-adic unit, the coefficient of degree p has valuation zero. All other coefficients are integral and the leading coefficient is one.

The lower Newton polygon therefore has a segment from $(0, 1)$ to $(p, 0)$ and then a horizontal segment to $(n, 0)$. The Newton-polygon theorem gives roots of valuation $1/p$. Thus the ramification index of a splitting field over $\mathbb{Q}_p$ is divisible by p. A completion of the global splitting field has that splitting property, so its local degree, and therefore the order of the global Galois group, is divisible by p. Cauchy's theorem gives an element of order p. Because $n < 2p$, its permutation on the n roots is a single p-cycle. □

Proposition 27.4 (Double transitivity in every degree)**.** *For every $n \geq 2$, the Galois group G_n of B_n acts doubly transitively on its roots.*

Proof. For $n = 2$, $B_2(z) = z^2 - 2$, so the assertion is immediate. For $n = 3$ choose $p = 2$. For $n \geq 4$, Bertrand's theorem applied to $m = \lfloor n/2 \rfloor \geq 2$ gives a prime $m < p < 2m$, and this prime satisfies $n/2 < p < n$. Lemma 27.3 supplies a p-cycle in G_n. The group is transitive by Lemma 27.2.

It is also primitive. Indeed, a nontrivial invariant block system would have $s \geq 2$ blocks of common size $t \geq 2$, with $st = n$. Both s and t are at most $n/2 < p$. The element of order p cannot act nontrivially on the s blocks. Its nontrivial orbit of size p would therefore have to lie within one block of size $t < p$, also impossible.

The classical Jordan–Marggraff theorem states that a primitive permutation group containing a cycle with k fixed points is $(k + 1)$-transitive. An elementary proof is given by Levingston–Taylor [81, Theorem A]; Jones [82, Section 3] discusses its original attribution to Jordan. Here $k = n - p \geq 1$, so G_n is at least doubly transitive. Only this classical theorem is used, not a classification of finite simple groups. □

27.4 Separating roots without separating splitting fields

The following elementary representation-theoretic observation is useful because the degrees of the polynomials need not be prime, coprime, or ordered by ramification.

Lemma 27.5 (Distinct degrees and centered root modules)**.** *Let $F_1, \dots, F_s \in \mathbb{Q}[z]$ be separable irreducible polynomials of pairwise distinct degrees $d_i \geq 2$, whose Galois groups are doubly transitive. Choose one root α_i of each. Then $1, \alpha_1, \dots, \alpha_s$ are linearly independent over $\mathbb{Q}$.*

Proof. Let K be the compositum of the splitting fields and $G = \mathrm{Gal}(K/\mathbb{Q})$. Its action on the d_i roots of F_i is doubly transitive, since restriction surjects onto the Galois group of the corresponding splitting field.

Consider the rational permutation module $\mathbb{Q}^{d_i}$ and its augmentation submodule

$$W_i = \{(v_1, \dots, v_{d_i}) : v_1 + \cdots + v_{d_i} = 0\}.$$

This is an irreducible $\mathbb{Q}G$-module. Here is a direct verification. Double transitivity means that the commuting matrices on $\mathbb{Q}^{d_i}$ are precisely the matrices constant on the diagonal and constant off the diagonal, a two-dimensional space. The decomposition $\mathbb{Q}^{d_i} = \mathbb{Q}\mathbf{1} \oplus W_i$ has no nonzero homomorphisms between its two summands, because the only invariant vectors and invariant linear forms in the permutation module are constant. Hence $\dim_{\mathbb{Q}} \mathrm{End}_G(W_i) = 1$. Maschke's theorem then makes W_i irreducible: a decomposition into two nonzero submodules would provide two independent commuting projections.

List the roots as $\alpha_{i,1}, \dots, \alpha_{i,d_i}$ and put $\mu_i = d_i^{-1} \sum_j \alpha_{i,j} \in \mathbb{Q}$. The map

$$(v_j)_j \longmapsto \sum_j v_j \alpha_{i,j} \quad \text{from } W_i \text{ into } K$$

is G-equivariant and nonzero, since the roots are distinct. It is therefore injective. Its image U_i is an irreducible submodule of dimension $d_i - 1$, and it contains $\alpha_i - \mu_i$, the image of a suitable vector $e_j - d_i^{-1}\mathbf{1}$.

The modules U_i have different dimensions and are thus pairwise nonisomorphic. None is the trivial module: W_i has no nonzero invariant vector. This also deals with degree two, when $\dim U_i = 1$. Semisimplicity now gives the direct sum

$$\mathbb{Q} \cdot 1 \ \oplus \ U_1 \ \oplus \cdots \oplus \ U_s \ \subseteq K.$$

For clarity, directness follows by induction: an irreducible submodule intersecting a direct sum of nonisomorphic irreducible submodules would embed into that sum, and some coordinate projection would make it isomorphic to a summand, a contradiction.

Each $\alpha_i - \mu_i$ is nonzero, since an irreducible polynomial of degree at least two has no rational root. A rational linear relation among $1, \alpha_i$ can be rewritten using these centered roots. Directness forces every coefficient to vanish. □

Proof of Theorem 27.1. Apply Lemma 27.5 to any finite selection of the polynomials B_n, $n \geq 2$, using Proposition 27.4. This proves (27.3). In this application the centered root is particularly explicit:

$$z_n - \mu_n = z_n + n - 2.$$

For $n = 1$, $z_1 = 1$. Hence all the z_n, $n \geq 1$, are linearly independent over $\mathbb{Q}$.

Equations (27.2)–(27.4) give

$$V_n = A_n e^{-z_n}, \qquad A_n = \frac{(2z_n)^n}{P_{n-1}(z_n)} \in \overline{\mathbb{Q}}^{\times}. \tag{27.7}$$

A nonzero polynomial relation over $\overline{\mathbb{Q}}$ among finitely many V_n would therefore give a nonzero algebraic linear combination of exponentials

$$\exp\left(- \sum_n k_n z_n \right).$$

Distinct monomials have distinct algebraic exponents, by the linear independence just proved. Lindemann–Weierstrass rules out this linear combination. Finally $P_0 = 1$ and $z_1 = 1$ give $V_1 = 2/e$, proving the equivalent formulation. □

The first boundary polynomials are

$$B_1(z) = z - 1, \qquad B_2(z) = z^2 - 2,$$
$$B_3(z) = z^3 + 3z^2 - 6z - 6,$$
$$B_4(z) = z^4 + 8z^3 - 48z - 24.$$

The new statement includes $V_4, V_6, V_8, V_9, V_{10}$ and all other composite-indexed optima together with the preceding prime family. Merely observing that every B_n is irreducible would not prove this conclusion; the centered-module step controls possible relations across their splitting fields.

27.5 Target II: a precise obstruction for the excursion value

The linear Brownian-excursion optimum is the established constant

$$V = \mathcal{V}_{3,1} = 2\sqrt{2z_*}\, e^{-z_*}, \qquad 2f(z_*) = e^{z_*}, \qquad f(z) = M(1/2; 3/2; z) = \int_0^1 e^{zt^2}\, dt. \tag{27.8}$$

The preceding arithmetic analysis proves that z_* is transcendental and that a published squarefree-factorization conjecture would imply transcendence of V [79, 83]. We do not count those earlier results again. We investigate whether the particularly simple first-order identity

$$2zf'(z) + f(z) = e^z \tag{27.9}$$

allows the known unconditional factorization theorem to apply.

Proposition 27.6 (Two failed hypotheses in the excursion family)**.** *For $c \in \mathbb{C}^\times$, put*

$$G_c(z) = f(z)^2 - cz. \tag{27.10}$$

Every multiple zero of G_c has multiplicity exactly two. There are only finitely many such zeros and infinitely many simple zeros. If $c \in \overline{\mathbb{Q}}^\times$, then G_c is an E-function but is not globally bounded in the exponential-coefficient sense used by Fischler–Rivoal.

Proof. Since $G_c(0) = 1$, a multiple zero w is nonzero. Its two defining equations give

$$f(w)^2 = cw, \qquad 2wf'(w) = f(w), \qquad e^w = 2f(w).$$

In particular $f(w) \neq 0$. The Kummer equation

$$zf'' + (3/2 - z)f' - \tfrac{1}{2}f = 0$$

then gives the exact identity at this point

$$G_c''(w) = \frac{c(2w-1)}{w}. \tag{27.11}$$

The only candidate for a zero of multiplicity at least three is $w = 1/2$. But $f(1/2) > 1$ whereas $e^{1/2} < 2$, so $2f(1/2) = e^{1/2}$ is impossible. Every multiple zero is therefore exactly double.

Finiteness of multiple zeros is the special case already proved in the preceding paper; the short calculation here records its global analytic content. On the closed upper and lower

half-planes separately, the two-term Kummer expansion [57, §13.7], with the corresponding branch of $z^{-1/2}$, is

$$f(z) = \frac{e^z}{2z}\{1 + O(|z|^{-1})\} + B_\pm z^{-1/2}\{1 + O(|z|^{-1})\}, \qquad B_\pm = \pm\frac{i\sqrt{\pi}}{2}.$$

The error bounds apply separately to the two displayed terms. At an unbounded sequence of contacts $e^z = 2f(z)$ this implies

$$e^z = 2B_\pm z^{-1/2}\{1 + O(|z|^{-1})\}, \qquad \frac{f(z)^2}{z} = -\frac{\pi}{4z^2}\{1 + O(|z|^{-1})\} \longrightarrow 0.$$

The additional condition $f(z)^2/z = c \neq 0$ is thus impossible outside a fixed disk. Zeros of the nonzero entire function G_c have no finite accumulation point, proving finiteness.

The integral defining f gives $|f(z)| \le e^{|z|}$, so G_c has order at most one. Laplace's endpoint estimate on the positive real axis gives

$$f(x) = \frac{e^x}{2x}\{1 + O(x^{-1})\}, \qquad G_c(x) = \frac{e^{2x}}{4x^2}\{1 + O(x^{-1})\}.$$

If G_c had only finitely many zeros, Hadamard factorization would give $G_c = P(z)e^{az+b}$ with a nonzero polynomial P. Comparison of logarithmic absolute values along $x > 0$ first forces $\Re a = 2$ and then $\deg P = -2$, a contradiction. There are infinitely many zeros, and the finite-multiple-zero conclusion makes infinitely many of them simple.

For the arithmetic assertion, write

$$f(z) = \sum_{n\ge0} \frac{1}{2n+1}\frac{z^n}{n!}, \qquad f(z)^2 = \sum_{n\ge0} b_n \frac{z^n}{n!}, \qquad b_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{(2k+1)(2n-2k+1)}.$$

This rational-parameter Kummer function is an E-function; the closure of E-functions under products and algebraic polynomials gives the same assertion for G_c. In the terminology of [83], global boundedness would require an integer $D \ge 1$ for which D^{n+1} times every exponential Taylor coefficient is an algebraic integer.

Take an arbitrary prime $p \ge 5$ and $n = (p-1)/2 \ge 2$. The endpoint terms $k = 0, n$ in b_n have sum $2/p$. All remaining denominators are prime to p, because both their positive odd factors lie strictly between 1 and p. Thus

$$v_p(b_{(p-1)/2}) = -1. \tag{27.12}$$

Subtracting cz changes only the coefficient of index one. For any proposed D, choose a prime $p \ge 5$ not dividing D. The coefficient in (27.12) remains nonintegral after multiplication by D^{n+1}. A rational number is an algebraic integer exactly when it is an integer, so this contradicts global boundedness, even when c is algebraic but not rational. □

If $V = v$ were algebraic, then $c = 2/v^2$ would be algebraic and z_* would be a double zero of G_c. Proposition 27.6 shows that this function has both simple and double zeros, and that it is not globally bounded. Fischler–Rivoal's unconditional Theorem 1.5 assumes global boundedness and that all zeros have the same multiplicity $m \ge 2$, in addition to an order restriction on a differential equation. Hence that theorem fails two independently verified hypotheses here. The first-order identity (27.9) does not repair either of them.

The remaining arithmetic target can be stated without referring to any factorization terminology. Transcendence of V is equivalent to the following assertion: for every $c \in \overline{\mathbb{Q}}^\times$, the two entire functions

$$e^z - 2f(z), \qquad e^{2z} - 4cz$$

have no common positive real zero. The unique positive zero of the first function is z_*; at this point the second equation is exactly $V^2 = 2/c$. Algebraic-point specialization theorems for E-functions do not settle this common-zero question, because z_* is already transcendental.

Scope and provenance. The main new result relative to the preceding manuscript is the full, consecutive planar family in Theorem 27.1, including every composite payoff index. The root-module lemma and its application are proved explicitly; Schur irreducibility, Bertrand's theorem, the Newton-polygon theorem, Jordan–Marggraff, Maschke, and Lindemann–Weierstrass remain classical inputs. A search of primary literature on Laguerre Galois groups, Schur-type irreducibility, and bridge stopping did not verify a prior arithmetic statement for this exact optimal-value sequence. This search does not establish a historical priority claim.

The excursion proposition is an unconditional diagnosis of an unsuccessful route, not an unconditional irrationality or transcendence theorem for V. General rational nonexceptional payoff optima likewise remain outside the unconditional value classification obtained here. No named historical conjecture is claimed to have been closed by this section.

28 The range of entropy-game values and an arithmetic bottleneck transition

Two questions go beyond classifying extrema of rational entropy programs. First, which logarithms of algebraic numbers actually occur as finite entropy-game values? Second, does an algebraic optimizer still exist when a rational information-bottleneck weight is replaced by an algebraic irrational one? We answer the first question exactly for games with a pure saddle, and the second exactly for the binary symmetric model above its nontriviality threshold. The constructions use established results of Lind and Gelfond–Schneider. They do not resolve the arithmetic of Poisson entropy or the optimized exponential-integral payoff discussed at the end.

The number-theoretic representation used below is already known in a stronger form: Dubickas proves that every positive algebraic number is a quotient of two Pisot numbers in its own number field [84, Theorem 3]. Lind's realization of Perron numbers as spectral radii is also classical [85, Theorem 1]. Our exact entropy-game formulation combines these facts with the previous rational-entropy stationary-value theorem. We make no historical priority claim for this formulation. The binary information-bottleneck boundary is likewise established [70, Section IV-A]; the arithmetic distinction between rational and algebraic irrational weights is the additional consequence proved here.

All logarithms are natural, and $0 \log 0 = 0$. Write $\overline{\mathbb{Q}}_{>0}$ for the positive real algebraic numbers.

28.1 An exact converse for finite entropy games

A *rational entropy functional* on a nonempty compact rational polytope P is

$$F(z) = \sum_{a=1}^{A} c_a H(L_a(z)), \qquad H(p) = -\sum_j p_j \log p_j, \tag{28.1}$$

where $c_a \in \mathbb{Q}$ and every L_a is a rational affine map from P to a finite probability simplex. A point is *face-stationary* when it is in the relative interior of a face and the derivative vanishes in every direction parallel to that face. This definition includes vertices.

We recall the stationary-value fact from the preceding work, with its proof so that the converse below has explicit hypotheses.

Lemma 28.1 (Previously established upper inclusion)**.** *Every face-stationary value of* (28.1) *belongs to* $\log \overline{\mathbb{Q}}_{>0}$. *For fixed rational data there are only finitely many such values, each attained at an algebraic face-stationary point.*

Proof. On the relative interior of a face Q, each affine probability coordinate is either identically zero or positive. Delete the zero coordinates and choose a rational tangent basis v_i. Since each L_a has total mass one, stationarity is exactly

$$\sum_{a,j} c_a(\partial_{v_i}\ell_{aj})\log \ell_{aj}(z) = 0.$$

Clearing denominators and exponentiating turns these conditions into polynomial equations, with positive arguments, over $\mathbb{Q}$. The stationary set is therefore semialgebraic over $\mathbb{Q}$. A finite semialgebraic cell decomposition supplies algebraic sample points and piecewise C^1 paths within each cell [61]. The derivative of F along every such path is zero, so F is constant on each cell. Every stationary value is thus attained at an algebraic point z_*.

Choose a rational point z_0 in the relative interior of Q. Pair stationarity at z_* with $z_* - z_0$. Affineness yields

$$F(z_*) = -\sum_{a,j} c_a \ell_{aj}(z_0)\log \ell_{aj}(z_*).$$

The coefficients are rational and the arguments positive algebraic. Exponentiation gives a positive algebraic number. There are finitely many cells and finitely many faces, proving the assertion. □

A *rational entropy game* has strategy polytopes P, Q of the same rational kind and payoff $F(x, y)$ of the form (28.1) on $P \times Q$. A pure saddle is a pair (x_*, y_*) such that

$$F(x, y_*) \le F(x_*, y_*) \le F(x_*, y) \quad (x \in P,\ y \in Q). \tag{28.2}$$

The maximizing player chooses x and the minimizing player chooses y. We do not assume that every rational entropy game has such a pair.

Theorem 28.2 (Exact range of pure-saddle entropy values)**.** *The following three sets of real numbers are equal:*

1. $\log \overline{\mathbb{Q}}_{>0}$*;*
2. *the values at pure saddles of all rational entropy games;*
3. *the face-stationary values of all rational entropy functionals.*

Moreover, every number in this common set is realized by a separable concave–convex game with exactly four signed entropy terms,

$$F(x, y) = H(x) - H(\pi(x)) - H(y) + H(\pi(y)), \tag{28.3}$$

on two rational directed-graph flow polytopes. The constructed game has an algebraic pure saddle in the relative interior. The sizes of the graphs are allowed to depend on the algebraic number.

We give the representation argument in two steps. The first is weaker than Dubickas's Pisot quotient theorem, and is included only as a short direct proof of the precise fact required here.

Lemma 28.3 (Perron quotient representation; known consequence)**.** *For every $\alpha \in \overline{\mathbb{Q}}_{>0}$ there are Perron numbers $\beta, \gamma > 1$ in $\mathbb{Q}(\alpha)$ with $\alpha = \gamma/\beta$. Here a Perron number is a positive algebraic integer strictly larger in absolute value than all its other conjugates.*

Proof. Put $K = \mathbb{Q}(\alpha)$ and choose a positive integer D such that $D\alpha$ is an algebraic integer. In the Minkowski space $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ of K, the image of the ideal $D\mathcal{O}_K$ is a full lattice. Take the first real coordinate to be the given embedding $K \subset \mathbb{R}$. A bounded fundamental parallelepiped implies that for every real $T > 0$ there is $\beta_T \in D\mathcal{O}_K$ with

$$\beta_T = T + O(1), \qquad |\sigma(\beta_T)| = O(1) \quad (\sigma \neq \mathrm{id}),$$

where the constants are independent of T. Indeed, express $(T, 0, \dots, 0)$ in a real lattice basis and round each coefficient to an integer. The resulting error lies in a fixed bounded parallelepiped; both real coordinates of each complex embedding are bounded.

Both β_T and $\gamma_T = \alpha\beta_T$ are algebraic integers: writing $\beta_T = Db_T$ gives $\gamma_T = (D\alpha)b_T$ with $b_T \in \mathcal{O}_K$. At the distinguished embedding they tend to $+\infty$, while every other embedding of either number remains bounded. For sufficiently large T, both distinguished values exceed one and strictly dominate all other embedding values in modulus. Every conjugate of an element of K is obtained by an embedding of K, so these two numbers are Perron. Their quotient is α. This proves the weaker form needed here; the stronger Pisot conclusion is [84, Theorem 3]. □

Let A be a primitive nonnegative integral matrix and regard A_{ij} as the number of directed edges from i to j, retaining parallel edges as distinct elements of the edge set E_A. Its stationary flow polytope is

$$P_A = \left\{ x \geq 0 : \sum_{e \in E_A} x_e = 1, \quad \sum_{s(e)=i} x_e = \sum_{t(e)=i} x_e \text{ for every } i \right\}. \tag{28.4}$$

Write $\pi_i(x) = \sum_{s(e)=i} x_e$ and $h_A(x) = H(x) - H(\pi(x))$. These are rational affine probability maps and a rational entropy functional. The following is the finite-state entropy variational principle, traditionally associated with Shannon and Parry; see the historical account in [85, Section 1]. We supply its elementary form.

Lemma 28.4 (Graph entropy maximum)**.** *For $\lambda = \rho(A)$,*

$$\max_{x \in P_A} h_A(x) = \log \lambda.$$

The maximum has a positive algebraic maximizing flow. The functional h_A is concave.

Proof. Choose positive right and left Perron eigenvectors r, l, so $Ar = \lambda r$ and $l^{\mathsf{T}}A = \lambda l^{\mathsf{T}}$. They can be chosen algebraic: the simple eigenvalue λ is algebraic and its eigenvectors are solutions of linear equations over the real algebraic numbers. Give each edge $e : i \to j$ the transition probability

$$p_e = \frac{r_j}{\lambda r_i}.$$

The probabilities leaving each vertex sum to one. For any $x \in P_A$ and every i with $\pi_i(x) > 0$, put $q_e = x_e/\pi_i(x)$ for outgoing edges. Nonnegativity of relative entropy, followed by flow conservation, gives the exact identity

$$\begin{aligned} h_A(x) &= -\sum_e x_e \log p_e - \sum_{i:\pi_i(x)>0} \pi_i(x) D(q_i \| p_i) \\ &= \log \lambda - \sum_{i:\pi_i(x)>0} \pi_i(x) D(q_i \| p_i) \leq \log \lambda. \end{aligned}$$

In the second equality, the incoming and outgoing averages of $\log r_i$ cancel. The positive algebraic flow

$$x^*_e = \frac{l_i r_j}{\lambda \sum_k l_k r_k} \quad (e : i \to j)$$

has both incoming and outgoing masses $\pi_i^* = l_i r_i / \sum_k l_k r_k$, has total mass one, and has $q_i = p_i$. It attains equality. Finally, conditional entropy is concave in the joint edge law: each vertex contribution $-\sum_{s(e)=i} x_e \log(x_e/\pi_i)$ is the perspective of the concave entropy function. Summing proves concavity. □

Proof of Theorem 28.2. A pure saddle is face-stationary on the product of the minimal faces containing its two strategies: each of the two partial derivatives vanishes along that face by (28.2). Their tangent spaces span the product tangent space. Thus (2) is contained in (3), and Lemma 28.1 gives (3) contained in (1).

Conversely fix $\alpha > 0$ algebraic. Use Lemma 28.3 to write $\alpha = \gamma/\beta$ with $\gamma, \beta > 1$ Perron. Lind's realization theorem gives primitive nonnegative integral matrices A, B with spectral radii γ, β, respectively [85, Theorem 1]. On $P_A \times P_B$ set $F(x, y) = h_A(x) - h_B(y)$. If x_*, y_* are the maximizing flows of Lemma 28.4, then

$$F(x, y_*) \le \log \gamma - \log \beta \le F(x_*, y).$$

This is a pure saddle of value $\log \alpha$. Its coordinates are algebraic and positive, so it is in the relative interior of the product flow polytope. Formula (28.3) and concavity of the two conditional entropies prove all the remaining claims. □

Remark 28.5 (What the converse does and does not say). The result is an exact range statement, with no additive constants inserted into the objective. In particular, every logarithm of a positive algebraic number occurs using only four entropy terms with coefficients $1, -1, -1, 1$ and rational linear constraints. Every nonzero value in this class is transcendental by Hermite–Lindemann. This is a consequence of established arithmetic and entropy theory, not a newly solved classical transcendence conjecture. We have not shown that every such logarithm is the global maximum of a *single* rational entropy functional; the second player supplies a minimum. No conclusion is asserted for arbitrary min–max games without a pure saddle.

28.2 An exact arithmetic transition for binary information bottleneck

Let X be fair binary and let $Y = X \oplus Z$, where Z is independent Bernoulli with rational parameter $0 < \delta < 1/2$. Set $c = 1 - 2\delta \in (0, 1) \cap \mathbb{Q}$. For $\beta > 0$, consider the usual information-bottleneck objective

$$V_c(\beta) = \inf_{T - X - Y} \{I(X;T) - \beta I(Y;T)\}, \tag{28.5}$$

where T ranges over arbitrary finite alphabets and arbitrary stochastic encoders. The model and its binary symmetric optimum are classical [68, 70]. The next result classifies the arithmetic of its optimizers, including encoders with more than two output symbols.

Theorem 28.6 (Algebraic weights and transcendental optimal encoders)**.** *Let $\beta > 0$ be algebraic.*

1. *If $\beta \le c^{-2}$, then $V_c(\beta) = 0$ and a rational encoder attains it. Every optimal encoder makes X and T independent.*
2. *If $\beta > c^{-2}$, there is a unique $s_\beta \in (0, 1)$ satisfying*

$$\operatorname{arctanh} s_\beta = \beta c \operatorname{arctanh}(c s_\beta). \tag{28.6}$$

A binary symmetric encoder with crossover $\varepsilon_\beta = (1 - s_\beta)/2$ attains the infimum. In every finite optimal encoder, each positive-probability output has posterior probability

$$\mathbb{P}(X = 1 \mid T = t) \in \{\varepsilon_\beta, 1 - \varepsilon_\beta\}.$$

The total output probability assigned to each of these two posterior values is $1/2$.

3. *In the second case, s_β and ε_β are algebraic if and only if β is rational. Consequently an optimal encoder with all entries algebraic exists if and only if β is rational. If β is algebraic irrational, every finite optimal encoder has at least one transcendental entry.*

Proof. Write $h(p) = -p\log p-(1-p)\log(1-p)$. For an arbitrary finite encoder let $w_t = \mathbb{P}(T=t)$ and $p_t = \mathbb{P}(X=1 \mid T=t)$, omitting outputs of zero probability. The two marginal laws X, Y are fair and $\mathbb{P}(Y=1 \mid T=t) = \delta + cp_t$. Hence its objective is

$$(1-\beta)\log 2 + \sum_t w_t\{\beta h(\delta + cp_t) - h(p_t)\}. \tag{28.7}$$

The expression in braces is symmetric under $p_t \mapsto 1-p_t$. Its minimum over $[0,1]$ is therefore attainable under the prior constraint $\sum_t w_t p_t = 1/2$: take its two symmetric minimizing arguments with weights $1/2$. This is exactly a binary symmetric encoder (or an independent encoder if the minimum is at $p = 1/2$). Conversely, an average cannot lie below its pointwise minimum. It follows that (28.5) is attained and equals the minimum over $0 \le s \le 1$ of

$$f_\beta(s) = \beta h((1-cs)/2) - h((1-s)/2) - (\beta-1)\log 2. \tag{28.8}$$

Furthermore, equality in (28.7) is possible only when every p_t is a pointwise minimizer.

For $0 < s < 1$,

$$f'_\beta(s) = \operatorname{arctanh} s - \beta c \operatorname{arctanh}(cs).$$

The function

$$b_c(s) = \frac{\operatorname{arctanh} s}{c \operatorname{arctanh}(cs)}$$

is strictly increasing from c^{-2} to $+\infty$. Here is a direct proof of strictness. Put $u = s^2$ and $a_n = 1/(2n+1)$. Then

$$\frac{\operatorname{arctanh}(cs)}{\operatorname{arctanh} s} = \frac{\sum_{n\ge 0} a_n c^{2n+1} u^n}{\sum_{n\ge 0} a_n u^n}.$$

For $0 < u < 1$, differentiate this quotient. After multiplying by the square of its denominator, pair the terms with indices $m > n$. Their sum is

$$a_m a_n (m-n)(c^{2m+1} - c^{2n+1}) u^{m+n-1} < 0.$$

The series and their derivatives converge absolutely on compact subintervals, so this computation is valid. The quotient is strictly decreasing, proving the claim about b_c. Its endpoint limits follow from $\operatorname{arctanh} s \sim s$ at zero and its divergence at one.

If $\beta \le c^{-2}$, then $f'_\beta(s) > 0$ for every $s > 0$, so the unique minimizing bias is $s = 0$ and $f_\beta(0) = 0$. All optimal posteriors therefore equal $1/2$, which is independence. If $\beta > c^{-2}$, the equation $b_c(s) = \beta$ has exactly one solution $s_\beta \in (0,1)$. The derivative of f_β is negative below that point and positive above it. Thus the only minimizing posterior values are $\varepsilon_\beta, 1-\varepsilon_\beta$. The condition that their weighted average is $1/2$, and their distinctness, force total weights $1/2$ and $1/2$.

Exponentiating (28.6) gives

$$\frac{1+s_\beta}{1-s_\beta} = \left(\frac{1+cs_\beta}{1-cs_\beta}\right)^{\beta c}, \tag{28.9}$$

where the power has its positive real meaning. If β is rational, write $\beta c = a/b$ with positive integers a, b. Then s_β is a root of

$$(1+s)^b(1-cs)^a - (1-s)^b(1+cs)^a = 0.$$

This is a nonzero rational polynomial, since at $s = 1$ its value is $2^b(1-c)^a > 0$. Thus s_β and the optimal binary encoder are algebraic.

If β is algebraic irrational and s_β were algebraic, both ratios in (28.9) would be algebraic numbers greater than one, while βc would be algebraic irrational. The Gelfond–Schneider theorem makes the right side transcendental, a contradiction [62]. Consequently s_β and ε_β are transcendental. In any finite encoder with all entries algebraic, both w_t and the positive-probability posteriors are algebraic, because the input law is rational and Bayes' formula uses only field operations. The posterior classification already proved therefore rules out every such optimal encoder. This proves the theorem. □

For a concrete instance, take $\delta = 1/4$ and $\beta = 4\sqrt{2}$. The unique positive optimal bias solves

$$\operatorname{arctanh} s = 2\sqrt{2}\operatorname{arctanh}(s/2).$$

It is transcendental, as is the optimal binary crossover $(1-s)/2$, and no enlargement of the finite output alphabet permits an entirely algebraic optimal encoder. The example is an instance of a classified family, rather than a separately counted constant.

Remark 28.7 (Optimizer arithmetic is not value arithmetic). At the nontrivial stationary point, the elementary identity

$$h((1-s)/2) = \log 2 - \tfrac{1}{2}\log(1-s^2) - s\operatorname{arctanh} s$$

and (28.6) give

$$V_c(\beta) = \tfrac{1}{2}\log(1-s_\beta^2) - \tfrac{\beta}{2}\log(1-c^2 s_\beta^2). \tag{28.10}$$

For rational β, the algebraic optimizer makes the exponential of this value algebraic. For algebraic irrational β the arguments in (28.10) are transcendental; the proof above does not classify the arithmetic of the value itself. In particular, transcendence of an optimizer is not a proof of transcendence of its optimum value.

28.3 Two unresolved transfers

The Shannon entropy of a Poisson law with intensity one is

$$H(\text{Poisson}(1)) = 1 + e^{-1}\sum_{n=0}^{\infty}\frac{\log(n!)}{n!}.$$

The sum converges, but its coefficients $\log(n!)$ are not algebraic coefficients of an E-function. Neither the finite entropy stationary-value theorem nor the exponential-integral independence theorem applies to this series. Finite truncation followed by convergence supplies no arithmetic closure principle: limits of algebraic or transcendental values can have either arithmetic type. The bounds of Adell, Lekuona, and Yu [86] concern analytic approximation and do not supply the missing transcendence argument. We obtain no irrationality or transcendence result for this individual entropy.

Likewise, for $0 < \theta < 1$ algebraic, the previously studied maximum of $\mathrm{Ein}(x)/x^\theta$ has a transcendental maximizing location. Siegel–Shidlovskii controls suitable E-function values at algebraic arguments; it cannot simply be evaluated at that transcendental optimizer. No arithmetic classification of this optimized payoff follows here. These two failed transfers are recorded as remaining questions, not as solved targets.

29 Radical saturation and the only possible finite branch point

This section distinguishes a ring-membership problem from a transcendence problem. Put $k = \overline{\mathbb{Q}}$, let $\mathcal{E}$ be the ring of values of E-functions at algebraic points, and put $K = \mathrm{Frac}(\mathcal{E})$. We prove two structural results: an exact classification of rational monomials in a separated family of exponential-integral values, and a restriction on every finite algebraic extension of $k(\mathrm{Ein}(1))$ occurring inside K. These are applications of the external morphism theorem of Fischler–Rivoal [72, Proposition 3.1]. The paper's nonunit criterion and discussion of factorization are the relevant antecedents; the statements here are not attributed to that paper. No independent priority claim is made for these applications.

The logarithmic transcendence target is deliberately not counted again. The earlier theorem on algebraic independence of distinct Ein-values over the field generated by algebraic exponentials already implies the transcendence of every nonzero rational linear combination of their logarithms. Neither that conclusion nor its one-value special case is a new result in this section.

29.1 Input morphism and its divisor interpretation

We use the notation

$$\delta_a = \int_0^\infty \frac{e^{-t}}{t+a}\,dt, \qquad u_a = \mathrm{Ein}(a) = \int_0^a \frac{1-e^{-t}}{t}\,dt \quad (a>0).$$

The connection formula and the morphism computation established in the preceding division analysis are

$$u_a = \gamma + \log a + e^{-a}\delta_a, \qquad \Phi_0(u_a)(z) = A_a + B_a e^{-az}, \qquad A_a = \gamma + \log a, \quad B_a = \delta_a. \tag{29.1}$$

Here a is algebraic, and

$$\Phi_0 : \mathcal{E} \hookrightarrow \mathcal{R} = \left\{ \sum_{\rho\in k} c_\rho e^{\rho z} : c_\rho \in \mathbb{C},\ \text{finite support} \right\}$$

fixes k. It extends injectively to $K \to \mathrm{Frac}(\mathcal{R})$. The external theorem constructs Φ_0 from the canonical large-sector asymptotic decomposition. In the present case this decomposition is $\mathrm{Ein}(az) = \gamma + \log a + \log z + e^{-az}\delta_{az}$; indeed $\delta_{az} \sim \sum_{n\ge 0}(-1)^n n!/(az)^{n+1}$ on the required sectors. Evaluating the coefficient functions at $z = 1$ gives (29.1). This recalls the input rather than counting its proof as another result.

For $a \ge 1$, $A_a, B_a > 0$. All zeros of $\Phi_0(u_a)$ are simple, and have real part

$$c(a) = \frac{1}{a}\log\frac{B_a}{A_a}. \tag{29.2}$$

Their imaginary parts are odd integer multiples of π/a. The entire-versus-meromorphic distinction is useful: images of elements of $\mathcal{E}$ have nonnegative integral orders at finite points, whereas images of elements of K have arbitrary integral orders.

Lemma 29.1 (A uniform separation range)**.** *The function $c(a)$ is strictly increasing for real $a \ge 3$. Consequently, the zero sets of $\Phi_0(u_a)$ and $\Phi_0(u_b)$ are disjoint whenever $a, b \ge 3$ are distinct. Independently of this range, for $a, b \ge 1$ an irrational ratio a/b also ensures disjoint zero sets.*

Proof. Let X have density e^{-x} on $[0,\infty)$. The exact identity

$$\frac{1}{a+x} = \frac{1}{a+1} - \frac{x-1}{(a+1)^2} + \frac{(x-1)^2}{(a+1)^2(a+x)}$$

and Cauchy–Schwarz give

$$\delta_a = \frac{1}{a+1} + \frac{1}{(a+1)^2}\mathbb{E}\frac{(X-1)^2}{a+X} \geq \frac{1}{a+1} + \frac{1}{(a+1)^2(a+3)} = \frac{(a+2)^2}{(a+1)^2(a+3)}.$$

Here $\mathbb{E}(X-1)^2 = 1$ and $\mathbb{E}[(a+X)(X-1)^2] = a+3$; the inequality is the weighted Cauchy–Schwarz inequality applied to $X-1$. Also $\delta_a < 1/a$ and differentiation followed by integration by parts gives $\delta_a' = \delta_a - 1/a$. Write $A = \gamma + \log a$ and $F = \log(A/\delta_a)$, so that $c = -F/a$. Then

$$aF' - F = \frac{1}{A} + \frac{1}{\delta_a} - a - \log(A/\delta_a) < H(a) := \frac{1}{A} + 1 - \frac{a+1}{(a+2)^2} - \log(aA).$$

A direct differentiation yields

$$H'(a) = \frac{a}{(a+2)^3} - \frac{1}{a}\left(1 + \frac{1}{A} + \frac{1}{A^2}\right) < 0.$$

The elementary bounds $\gamma > 1/2$ and $\log 3 > 13/12$ imply $A|_{a=3} > 19/12$. Since $\log(19/4) > 3/2$,

$$H(3) < \frac{12}{19} + \frac{21}{25} - \frac{3}{2} < 0.$$

Thus $c'(a) = (F - aF')/a^2 > 0$ for $a \geq 3$. For explicit rational certificates for the small numerical bounds, $e < 11/4$ and $e^{1/12} < 12/11$ give $e^{13/12} < 3$, while $(11/4)^{3/2} < 19/4$. Furthermore $\gamma > H_6 - \log 7$ and $\sum_{j=0}^{6}(39/20)^j/j! > 7$ give $\gamma > 1/2$. Finally, a common zero for $a, b \geq 1$ would have imaginary parts $(2j+1)\pi/a = (2\ell+1)\pi/b$, forcing a/b rational. □

Theorem 29.2 (Exact rational-power saturation)**.** *Let $a_1, \ldots, a_m \geq 1$ be positive algebraic numbers such that the zero sets of $\Phi_0(u_{a_j})$ are pairwise disjoint. For arbitrary $r_1, \ldots, r_m \in \mathbb{Q}$, take the positive real powers and put $v = \prod_{j=1}^{m} u_{a_j}^{r_j}$. Then*

$$v \in K \quad \Longleftrightarrow \quad r_j \in \mathbb{Z} \quad (1 \leq j \leq m), \tag{29.3}$$

$$v \in \mathcal{E} \quad \Longleftrightarrow \quad r_j \in \mathbb{Z}_{\geq 0} \quad (1 \leq j \leq m). \tag{29.4}$$

In particular, both equivalences hold for arbitrary distinct algebraic $a_j \geq 3$, with no rational-independence assumption on the parameters. For one parameter they hold for every algebraic $a \geq 1$.

Proof. Choose $q \geq 1$ with $m_j = qr_j \in \mathbb{Z}$. If $v \in K$, the identity $v^q = \prod_j u_{a_j}^{m_j}$ can be passed through the extended morphism. At a zero z_j of $\Phi_0(u_{a_j})$, no other factor vanishes. Taking orders of meromorphic functions gives

$$q \operatorname{ord}_{z_j} \Phi_0(v) = m_j.$$

The order on the left is an integer multiple of q, hence r_j is an integer. If $v \in \mathcal{E}$, its image is entire and this integer is nonnegative. Conversely, integral powers belong to the fraction field, and nonnegative integral powers belong to the ring. The separation lemma proves the stated parameter range. For $m = 1$ there is no pairwise separation condition. □

Thus, for example, $u_3^{1/2}$ is outside even K, while u_3^{-1} is in K but outside $\mathcal{E}$. Mixed values such as $u_3^{2/3}u_4^{-5/7}$ are classified by exactly the same criterion. This is an arithmetic membership theorem for a self-posed family, not a new irrationality proof for γ or δ. The range $a \geq 3$ is a proved sufficient condition, not a claimed optimal boundary or a classification of every possible collision in $1 \leq a < 3$.

29.2 Rational exponential functions and finite algebraic covers

The next argument is not limited to radicals. It concerns every algebraic element of K over $k(u)$, where $u = u_1$.

Lemma 29.3 (One-frequency algebraic descent)**.** *Let $W \in \mathrm{Frac}(\mathcal{R})$ be algebraic over $\mathbb{C}(e^{-z})$. There exists an integer $q \geq 1$ such that $W \in \mathbb{C}(e^{-z/q})$.*

Proof. Choose a $\mathbb{Q}$-basis $1, \beta_2, \ldots, \beta_s$ of the $\mathbb{Q}$-span of 1 and the finitely many phases occurring in a numerator and denominator of W. After clearing their rational coordinates with a positive integer q, all these exponential functions are Laurent monomials in

$$X = e^{-z/q}, \qquad Y_j = e^{\beta_j z/q} \quad (2 \leq j \leq s).$$

These functions are algebraically independent over $\mathbb{C}$: a polynomial relation would be a linear relation among finitely many exponentials with distinct exponents, contradicted by differentiating at 0 and the Vandermonde determinant. Hence W is a rational function in the independent variables $X, Y_2, \ldots, Y_s$. It is algebraic over $\mathbb{C}(X)$ because it is algebraic over the smaller field $\mathbb{C}(X^q)$.

In characteristic zero a field F is relatively algebraically closed in $F(Y_2, \ldots, Y_s)$. Here is a short proof in the needed form. Differentiate a separable minimal polynomial for an algebraic element h over F with respect to each Y_j; its nonvanishing derivative forces $\partial h/\partial Y_j = 0$. The only rational functions in a variable with zero derivative are independent of that variable, and iteration gives $h \in F$. Apply this with $F = \mathbb{C}(X)$. □

Theorem 29.4 (The only possible finite branch point)**.** *Let $v \in K$ be algebraic over $k(u)$, and let $d = [k(u, v) : k(u)]$. If $d > 1$, then $\gamma \in k$. More precisely, after extension of constants to $\mathbb{C}$, the corresponding cover of the U-line is cyclic of degree d, is totally ramified at $U = \gamma$ and at infinity, and is unramified elsewhere. Consequently, under the explicit hypothesis $\gamma \notin k$,*

$$\{v \in K : v \text{ is algebraic over } k(u)\} = k(u), \tag{29.5}$$

$$\{v \in \mathcal{E} : v \text{ is algebraic over } k(u)\} = k[u]. \tag{29.6}$$

The hypothesis in these last two equalities is not established here.

Proof. The image $U = \Phi_0(u) = \gamma + \delta e^{-z}$ is nonconstant; since Φ_0 fixes k and is injective, u is transcendental. Choose a primitive irreducible polynomial $P(U, V) \in k[U, V]$ for v over $k(u)$. Because k is algebraically closed, P remains irreducible over $\mathbb{C}(U)$. One elementary justification is Gauss's lemma followed by descent of a hypothetical polynomial factorization: the normalized coefficients of factors satisfy finitely many polynomial equations over k; a solution over $\mathbb{C}$ gives a solution over k by the weak Nullstellensatz. Nonzero normalizing coefficients are encoded by adjoining their inverses.

Apply Φ_0 to the relation. With $W = \Phi_0(v)$, it gives $P(\gamma + \delta e^{-z}, W) = 0$. The descent lemma puts W in $\mathbb{C}(X)$ for $X = e^{-z/q}$ and some $q \geq 1$, with $U = \gamma + \delta X^q$. Irreducibility shows that

$$[\mathbb{C}(U, W) : \mathbb{C}(U)] = d.$$

The extension $\mathbb{C}(X)/\mathbb{C}(U)$ is cyclic Galois, with $X \mapsto \zeta X$ for $\zeta^q = 1$. Every subgroup is cyclic, and the fixed field of its subgroup of order h is $\mathbb{C}(X^h)$. Thus

$$\mathbb{C}(U, W) = \mathbb{C}(X^h), \qquad h \mid q, \qquad d = q/h.$$

Writing $Y = X^h$, the cover is $U = \gamma + \delta Y^d$. If $d > 1$, it is totally ramified at γ and infinity and unramified elsewhere, as follows directly by differentiating or taking local orders.

All finite branch points of the cover defined by P are algebraic. Indeed, outside the zeros of the leading coefficient of P in V and its nonzero V-discriminant, its roots are locally distinct holomorphic functions by the implicit function theorem. These two polynomials have coefficients in k. Since γ is a branch point when $d > 1$, it must be algebraic. This proves the unconditional implication and (29.5).

For (29.6), assume $\gamma \notin k$. We have just shown that every relevant $v \in \mathcal{E}$ is $R(u)$ for $R \in k(U)$. If R had a finite pole $\alpha \in k$, then $\alpha \neq \gamma$ and $R(\gamma + \delta e^{-z})$ would have a pole at each solution of $e^{-z} = (\alpha - \gamma)/\delta$. The derivative of the inner function there is $-(\alpha - \gamma) \neq 0$. This contradicts the entire image $\Phi_0(v)$. Hence R is polynomial. The reverse inclusion is immediate. □

The implication in Theorem 29.4 is useful even without knowing whether γ is algebraic: an algebraic curve with two distinct finite branch values, or with a finite branch value outside $(0, 1)$, cannot acquire a nonrational solution at $U = u$ inside K. There is no assertion that an arbitrary complex algebraic extension of $k(u)$ embeds into this fraction field.

Corollary 29.5 (An unconditional all-but-one family)**.** *Among all positive algebraic parameters $a \geq 1$, at most one can fail one of the following equalities:*

$$\{v \in K : v \text{ is algebraic over } k(u_a)\} = k(u_a),$$
$$\{v \in \mathcal{E} : v \text{ is algebraic over } k(u_a)\} = k[u_a].$$

In particular, for any prescribed m distinct such parameters, both equalities hold for at least $m - 1$ of them. This does not identify a possible exceptional parameter, nor assert its existence.

Proof. Repeat the proof of Theorem 29.4 with e^{-az} in place of e^{-z} and $A_a + B_a e^{-az}$ in place of $\gamma + \delta e^{-z}$. In the descent lemma start the rational basis with a, giving $X = e^{-az/q}$. Every step is unchanged. Hence either equality can fail only if $A_a = \gamma + \log a$ is algebraic: for the second equality, the degree-one pole argument also requires a pole at A_a. If distinct a, b both had algebraic centers, then $\log(a/b) = A_a - A_b$ would be a nonzero algebraic number whose exponential is algebraic. This contradicts the Hermite–Lindemann theorem [87]. Thus there is at most one possible exception in the entire parameter family. □

Corollary 29.6 (Explicit radical exclusions inside the fraction field)**.** *Let $R \in k(U)^\times$, let $p \in \mathbb{Z}$, $q \geq 1$ be coprime, and suppose that R has a zero or pole at some $\alpha \in k$ with $\alpha \neq \gamma$ and order $m \neq 0$. If a number v satisfies $v^q = R(u)^p$ and belongs to K, then q divides m. In particular, for every $n \geq 1$ and every nonintegral $r \in \mathbb{Q}$,*

$$(u^n + 1)^r \notin K. \tag{29.7}$$

All choices of the complex root have the same nonmembership conclusion.

Proof. At a solution z_0 of $\gamma + \delta e^{-z_0} = \alpha$, the inner function is unramified. The order of $R(\gamma + \delta e^{-z})$ there is exactly m. Applying the extended morphism to the power identity and taking orders gives $q \operatorname{ord}_{z_0} \Phi_0(v) = pm$. Coprimality forces $q \mid m$. For $R(U) = U^n + 1$, all roots are simple and none equals $\gamma \in (0, 1)$, so $q = 1$ would be necessary. Multiplication by a root of unity does not change membership because all roots of unity belong to $k \subset \mathcal{E}$. □

29.3 What these results settle and what remains

The two new structural targets are the exact saturation Theorem 29.2 and the finite-cover obstruction Theorem 29.4; Corollaries 29.5 and 29.6 are dependent applications, the former

removing the hypothesis for all but at most one parameter in the entire family. They address division and algebraic extraction in the arithmetic ring and its fraction field, which algebraic independence of a preassigned finite family does not settle. The explicit examples $\sqrt{u^n+1}$, $1 \le n \le 10$, are ten instances of a single theorem, not ten independent famous conjectures. Their transcendence already follows from that of u; the new property proved here is nonmembership in K.

The attempted individual targets γ and δ remain separated from these claims. A nontrivial algebraic extension in K would force γ algebraic, but no such extension is constructed, and no contradiction from γ algebraic is proved. The positive coefficient δ controls the exponential coordinate in the cover; its presence does not determine whether it is rational or algebraic. Finally, the old logarithmic target $\log(\gamma + \delta/e)$ is already settled by the earlier algebraic-independence argument, so it provides no extra advance in this round.

30 The differential complexity of a spectral optimum

The preceding continuation classified cubic-field critical values and left the biquadratic value

$$Q = \min_{x>0} \frac{2(1+x)}{x^{1/\sqrt{3}} + x^{1/\sqrt{2}}} \tag{30.1}$$

unresolved. We investigated that value again. The conclusion remains open here; we identify precisely why completing a biquadratic exponential table does not supply the needed contradiction.

The successful direction concerns the entire optimal-value function. For every two-atom positive spectrum, we classify exactly when that function satisfies a linear differential equation with polynomial coefficients. We also obtain an explicit nonlinear differential equation, prove its converse, and determine its least possible algebraic differential order. These conclusions include (30.1) as a specialization without making an unwarranted inference about its individual arithmetic value.

30.1 The parametric curve of values and contacts

In the functional results below, the exponents and weights may be arbitrary real numbers satisfying

$$0 < a < b < 1, \qquad w_1, w_2 > 0, \qquad S(x) = w_1 x^a + w_2 x^b, \qquad T(C) = \min_{x>0} \frac{1 + Cx}{S(x)} \quad (C > 0). \tag{30.2}$$

Write $d = b - a$, let $x(C)$ be the unique contact, and put $H(x) = S(x) - xS'(x)$. Positive real powers use the real logarithm. The objective tends to infinity at zero and infinity, and

$$\left(\frac{S'}{H}\right)' = \frac{S''S}{H^2} < 0.$$

These facts give a unique minimum and an analytic contact map with $x'(C) < 0$. Its value curve has the parametrization

$$C = \Phi(x) := \frac{S'(x)}{H(x)}, \qquad T = \Psi(x) := \frac{1}{H(x)}. \tag{30.3}$$

In particular

$$T' = \frac{x}{S(x)} > 0, \qquad B := T - CT' = \frac{1}{S(x)} > 0, \qquad x = \frac{T'}{B}, \tag{30.4}$$

$$T'' < 0, \qquad x' = \frac{TT''}{B^2}, \qquad S''(x) = \frac{B^3}{T^3 T''}. \tag{30.5}$$

Here and below primes on T denote differentiation in C and primes on S denote differentiation in x.

We use the standard term *D-finite* for an analytic germ that satisfies a nonzero linear differential equation with polynomial coefficients, as in [88]. All function classifications are over $\mathbb{C}(C)$. They therefore exclude such equations even when arbitrary complex constants are allowed as coefficients. Algebraic differential order means the smallest $k \geq 0$ for which a nonzero polynomial over $\mathbb{C}(C)$ vanishes at $(T, T', \ldots, T^{(k)})$.

Lemma 30.1 (Algebraicity of an optimum)**.** *For a finite sum $S(x) = \sum_j w_j x^{r_j}$ with distinct real $0 < r_j < 1$ and positive weights, the analytic germ of its optimized value is algebraic over $\mathbb{C}(C)$ if and only if every r_j is rational.*

Proof. If T is algebraic, so is T', and the envelope formula makes $x(C) = T'/(T - CT')$ algebraic. This function is nonconstant. Its local inverse $C(x)$ is consequently algebraic, and the identity $S(x) = 1/[T(C(x)) - C(x)T'(C(x))]$ makes S algebraic over $\mathbb{C}(x)$.

An algebraic germ has finitely many branches. Continuing the displayed finite power sum around zero a suitable positive number N of times must therefore return to the original branch. Thus

$$\sum_j w_j (e^{2\pi i N r_j} - 1) x^{r_j} = 0.$$

Distinct real powers are linearly independent on an interval: after writing $x = e^t$, the Vandermonde matrix obtained by successive derivatives proves this assertion. Hence $N r_j \in \mathbb{Z}$ for every j, proving necessity. Conversely, rational exponents make S, H, Φ, Ψ algebraic in x. The nonconstant local inverse of Φ and its composition with Ψ are algebraic. This proves sufficiency. □

30.2 A complete linear-differential classification

Theorem 30.2 (Rational gaps are exactly the D-finite case)**.** *For (30.2), the following statements are equivalent:*

1. *the gap $d = b - a$ is rational;*
2. *the contact germ $x(C)$ is algebraic over $\mathbb{C}(C)$;*
3. *the contact germ $x(C)$ is D-finite;*
4. *the optimized-value germ $T(C)$ is D-finite.*

In addition, $T(C)$ is algebraic if and only if both a and b are rational. If d is irrational, analytic continuations of the real germs of both T and x have infinitely many distinct finite branch points. At the explicit branch points constructed below, the contact has exponent $1/2$ and the value has exponent $1/2$ when $a + b \neq 1$; for $a + b = 1$ these exponents are respectively $1/3$ and $2/3$.

Proof. First suppose $d = p/q$ for positive integers p, q. Set $z = x^{1/q}$. Stationarity becomes the polynomial equation

$$Cz^q\big((1-a)w_1 + (1-b)w_2 z^p\big) = aw_1 + bw_2 z^p. \tag{30.6}$$

It is nonzero as a polynomial in C, z, so $z(C)$ and $x(C)$ are algebraic. The value has the form

$$T = \frac{z^{-qa}}{(1-a)w_1 + (1-b)w_2 z^p}.$$

Its logarithmic derivative belongs to the finite algebraic extension $L = \mathbb{C}(C, z)$, which is closed under differentiation. Consequently $T^{(k)} = R_k T$ with $R_k \in L$ for all $k \geq 0$. If $[L : \mathbb{C}(C)] = e$, then $R_0, \ldots, R_e$ are linearly dependent over $\mathbb{C}(C)$. Clearing denominators gives a nonzero linear differential equation with polynomial coefficients for T. The same finite-dimensional argument proves that every algebraic function, in particular $x(C)$, is D-finite.

Now let d be irrational. Work on the logarithmic parameter plane $x = e^t$, on which all powers in (30.3) are single valued and $\Phi(e^t), \Psi(e^t)$ are meromorphic. The zeros of $S(e^t)$ are

$$t_n = \frac{\log(w_1/w_2) + (2n+1)\pi i}{d}, \qquad x_n = e^{t_n}, \qquad n \in \mathbb{Z}. \tag{30.7}$$

At such a point, put $A_n = w_1 x_n^a \neq 0$, with the branch determined by t_n. Then

$$S(x_n) = 0, \qquad S'(x_n) = -\frac{dA_n}{x_n}, \qquad H(x_n) = dA_n \neq 0, \tag{30.8}$$

$$\Phi(x_n) = c_n = -x_n^{-1}, \qquad S''(x_n) = \frac{d(1-a-b)A_n}{x_n^2}. \tag{30.9}$$

The values c_n are distinct: equality for $n \neq m$ would require $(n-m)/d$ to be an integer and thus make d rational.

The parameter derivatives are

$$\Phi'(x) = \frac{S''(x)S(x)}{H(x)^2}, \qquad \Psi'(x) = \frac{xS''(x)}{H(x)^2}. \tag{30.10}$$

If $a + b \neq 1$, the zero of S is simple and $S''(x_n) \neq 0$. Writing $h = x - x_n$ in its local branch gives

$$\Phi(x) - c_n = \alpha_n h^2 + O(h^3), \qquad \Psi(x) - \Psi(x_n) = \beta_n h + O(h^2), \quad \alpha_n \beta_n \neq 0. \tag{30.11}$$

If $a + b = 1$, then instead

$$S''(x_n) = 0, \qquad S'''(x_n) = \frac{ab\,dA_n}{x_n^3} \neq 0.$$

Thus (30.10) gives

$$\Phi(x) - c_n = \widetilde{\alpha}_n h^3 + O(h^4), \qquad \Psi(x) - \Psi(x_n) = \widetilde{\beta}_n h^2 + O(h^3), \quad \widetilde{\alpha}_n \widetilde{\beta}_n \neq 0. \tag{30.12}$$

These are genuine branch points of the locally inverted curve. For example, if the value were holomorphic in C at c_n, its nonconstant composition with Φ would have vanishing order a multiple of two in (30.11), or a multiple of three in (30.12); the displayed orders are respectively one and two. The contact has order one in h in either case, and is likewise ramified.

It remains to show that these branches are reachable from the positive real germ; a list of critical values alone would not suffice. Suppose either germ satisfied a linear equation, and let

Σ be its finite set of possible finite singularities, the zeros of the leading polynomial coefficient. Choose $c_n \notin \Sigma$ and a positive initial setting outside Σ. In the t-plane remove the poles of $\Phi(e^t)$ and $\Psi(e^t)$, the zeros of $d\Phi(e^t)/dt$, and the inverse image of $\Sigma \cup \{c_n\}$. This is a discrete set; all functions involved are meromorphic and Φ is nonconstant. Its complement is path connected. A path from the real initial parameter can therefore approach t_n, with only its endpoint in the removed set. Along the path, the local inverse of Φ provides analytic continuation of the original contact and value germs. The image path avoids Σ. Near its endpoint the continuations are precisely the ramified branches in (30.11) or (30.12). But solutions of a linear differential equation are holomorphic at an ordinary point $c_n \notin \Sigma$, including every branch reached by continuation away from Σ. This contradiction proves that neither germ is D-finite.

We have proved all four equivalences, since an algebraic germ is D-finite. The last algebraicity assertion follows from Lemma 30.1. □

The result distinguishes two different restrictions. A common irrational shift of two rationally spaced exponents prevents algebraicity of the value, but still permits a linear differential equation. An irrational gap prevents every such linear equation. This statement concerns functions and is independent of the arithmetic nature of a value at one specified setting.

30.3 An exact nonlinear equation and its least order

Theorem 30.3 (A differential characterization of two atoms)**.** *Every optimum in* (30.2) *satisfies*

$$T^2\big[abT + (1-a-b)CT'\big]T'' + (T - CT')^2(T')^2 = 0. \tag{30.13}$$

Conversely, let a real analytic function T on an interval of positive settings satisfy (30.13) *and*

$$T > 0, \quad T' > 0, \quad B := T - CT' > 0, \quad T'' < 0, \qquad a < \frac{CT'}{T} < b. \tag{30.14}$$

Then, on that interval, T is the optimized value for unique positive constants w_1, w_2 in (30.2)*.*

Proof. The two powers satisfy the Euler equation

$$x^2S'' + (1-a-b)xS' + abS = 0. \tag{30.15}$$

Insert $x = T'/B$, $S = 1/B$, $S' = C/T$ and the last identity in (30.5). Multiplication by BT^3T'' gives exactly (30.13). No differentiation of order three is needed, and all its coefficients are independent of the weights.

For the converse, define $x = T'/B$. Direct differentiation gives $x' = TT''/B^2 < 0$, so it has a local real analytic inverse. Set $S(x(C)) = 1/B(C)$. The chain rule gives $S' = C/T$ and the formula for S'' in (30.5). Thus (30.13) implies (30.15). Since $a \neq b$, the general solution is $S(x) = w_1x^a + w_2x^b$ with constants w_1, w_2. Writing $\eta = CT'/T$ and using $S = 1/B$ gives

$$w_1x^a = \frac{b-\eta}{dB}, \qquad w_2x^b = \frac{\eta - a}{dB}. \tag{30.16}$$

Both are positive by (30.14). Moreover $S - xS' = 1/T$, so $C = S'/(S - xS')$ and $T = 1/(S - xS')$. The unique-contact argument preceding (30.3) now identifies T with the global minimum for these positive weights. The Euler representation and (30.16) give uniqueness. Local representations agree on overlaps, so the same two constants work on the entire interval. □

Theorem 30.4 (Exact algebraic differential order)**.** *Let*

$$r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{1, a, b\}.$$

The least algebraic differential order of $T(C)$ is exactly $r-1$. Thus it is zero, one or two. In particular, if $1, a, b$ are rationally independent, then T and T' are algebraically independent over $\mathbb{C}(C)$, and (30.13) has the least possible order.

Proof. All the following fields are fields of meromorphic germs, not fields of values at a fixed number. Parametrize by $x = e^t$. The contact and envelope formulas give the exact identity

$$\mathbb{C}(C, T, T') = \mathbb{C}(x, x^a, x^b). \tag{30.17}$$

Indeed, the inclusion from left to right follows from (30.3) and $T' = x/S$. For the reverse inclusion, $x = T'/(T - CT')$, while solving the two contact equations gives

$$w_1 x^a = \frac{b(1+Cx) - Cx}{dT}, \qquad w_2 x^b = \frac{Cx - a(1+Cx)}{dT}. \tag{30.18}$$

The transcendence degree over $\mathbb{C}$ of the right side of (30.17) is exactly r. To verify this directly, choose a rational basis $\lambda_1, \dots, \lambda_r$ from the exponent span. The functions $e^{\lambda_i t}$ are algebraically independent: any polynomial relation would be a linear relation among finitely many exponentials with distinct exponents, contradicted by a Vandermonde matrix. Clearing denominators in the rational changes of basis makes the two power fields algebraic over one another, which preserves transcendence degree. Since C is a nonconstant analytic function, it is transcendental over the constant field. Hence

$$\operatorname{trdeg}_{\mathbb{C}(C)} \mathbb{C}(C, T, T') = r - 1. \tag{30.19}$$

If $r = 3$, the two functions T, T' are algebraically independent over $\mathbb{C}(C)$, excluding every equation of order at most one. Equation (30.13) supplies order two. If $r = 2$, the same field degree supplies an equation of order at most one. At least one exponent is irrational, so Lemma 30.1 excludes order zero. If $r = 1$, both exponents are rational and that lemma gives order zero. □

Thus a failure of linear differential equations is not differential transcendence. Irrational-gap optima are differential algebraic, and their minimal nonlinear order is known exactly. For example, a rationally independent cubic pair and the biquadratic pair in (30.1) both have order two and are not D-finite, even though the present arithmetic information about their individual values differs.

Corollary 30.5 (Higher setting derivatives add no arithmetic field)**.** *Suppose a, b and the positive setting C_0 are algebraic. At that setting the entire derivative jet generates exactly the field*

$$\overline{\mathbb{Q}}(T(C_0), T'(C_0), T''(C_0), \dots) = \overline{\mathbb{Q}}(T(C_0), T'(C_0)). \tag{30.20}$$

This assertion does not require the weights to be algebraic.

Proof. The coefficient $A = abT + (1-a-b)CT' = T[ab + (1-a-b)\eta]$ in (30.13) is positive. Indeed $\eta = CT'/T$ belongs to (a, b), and the affine expression in brackets takes the positive endpoint values $a(1-a)$ and $b(1-b)$. Define the rational function

$$R(C, Y, Z) = -\frac{(Y - CZ)^2 Z^2}{Y^2[abY + (1-a-b)CZ]}.$$

Then $T'' = R(C, T, T')$, and successively differentiating shows that every higher derivative is a rational function of C, T, T' over $\mathbb{Q}(a, b)$. For instance $T''' = R_C + T'R_Y + RR_Z$ evaluated at (C, T, T'). Induction introduces only powers of the displayed nonzero denominators. Evaluation at the algebraic setting proves one field inclusion in (30.20); the other is immediate. □

30.4 The biquadratic value: an exact reduction and a failed transfer

For this subsection take

$$a = \frac{1}{\sqrt{3}}, \qquad b = \frac{1}{\sqrt{2}}, \qquad w_1 = w_2 = \frac{1}{2}, \qquad C = 1,$$

and write $\xi = x(1)$ and $\tau = T(1) = Q$. The previous continuation proved that ξ and the elasticity are individually transcendental, using the irrational gap; those results are not recounted as new here [79]. The weighted-mean contact identity gives $a < \xi/(1+\xi) < b$, so

$$\frac{a}{1-a} < \xi < \frac{b}{1-b}. \tag{30.21}$$

In particular $\xi > 1$. The value is less than two because the objective equals two at one and has negative derivative there.

Proposition 30.6 (What an algebraic value would imply)**.** *If τ is algebraic, then*

$$\xi^a = \frac{2[b-(1-b)\xi]}{(b-a)\tau}, \qquad \xi^b = \frac{2[(1-a)\xi - a]}{(b-a)\tau}, \tag{30.22}$$

and

$$\overline{\mathbb{Q}}(\xi, \xi^a, \xi^b) = \overline{\mathbb{Q}}(\xi), \qquad \operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\xi, \xi^a, \xi^b) = 1. \tag{30.23}$$

Under this same hypothesis, the missing fourth power $\xi^{ab} = \xi^{1/\sqrt{6}}$ is transcendental over $\overline{\mathbb{Q}}(\xi)$.

Proof. The two affine formulas are (30.18) with $C = 1$ and weights $1/2$. They give the field identity. The contact ξ is transcendental by the preceding result; alternatively, assuming it algebraic, the stationary equation would make ξ^{b-a} algebraic, contrary to Gelfond–Schneider since $\xi > 1$ and $b - a$ is algebraic irrational. This proves (30.23).

For clarity we specify the classical input governing the fourth power. Corollary 1.2(a), case $t = 1$, of Roy–Waldschmidt [89] implies the following qualitative fact: for two rationally independent families $u_1, \dots, u_p$ and $v_1, \dots, v_q$, the field generated by the pq numbers $e^{u_i v_j}$ has transcendence degree at least two if $pq \geq 2(p+q)$.

Apply it with $p = q = 4$, the first family $(1, a, b, ab)$, and the second family $(\log\xi, a\log\xi, b\log\xi, ab\log\xi)$. Both are rationally independent, since $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has basis $1, a, b, ab$ and $\log\xi \neq 0$. The equality $16 = 2(4+4)$ meets the hypothesis. As $a^2 = 1/3$ and $b^2 = 1/2$, all the products $u_i v_j$ are rational multiples of $\log\xi, a\log\xi, b\log\xi, ab\log\xi$. Consequently the exponential-table field is algebraic over $\mathbb{Q}(\xi, \xi^a, \xi^b, \xi^{ab})$, and it contains that field. The latter therefore has transcendence degree at least two. Extending the base to $\overline{\mathbb{Q}}$ does not change this bound. In view of (30.23), ξ^{ab} must be transcendental over $\overline{\mathbb{Q}}(\xi)$. □

The proposition explains the precise limitation of this attempted quartic-field transfer. The full multiplication table does provide a second transcendence degree, but it may be contributed entirely by the fourth coordinate, which the two-atom contact equations do not constrain. There is no contradiction. The ordinary conclusion of a four- or six-exponential argument, that some entry is transcendental, is already compatible with (30.23). Nor can an algebraic-base theorem simply be applied at the contact: that contact is itself transcendental. Corollary 30.5 makes the derivative limitation equally exact: if τ were algebraic, all setting derivatives would lie in $\overline{\mathbb{Q}}(\xi)$, whereas the fourth power would be transcendental over that field. Taking more derivatives with respect to C would not supply the missing coordinate.

There is also a simple obstruction to applying the particular exponential-table criterion used above while keeping only the three controlled powers. Put $E = \operatorname{span}_{\mathbb{Q}}\{1, a, b\}$. If rationally

independent families $(u_i)_{i=1}^{p}$ and $(v_j)_{j=1}^{q}$ have every product $u_i v_j \in E$, then fixing one nonzero member in either family gives $p, q \le 3$. It follows that $pq < 2(p+q)$, so that criterion cannot apply. This is a limitation of this specified rectangle construction, not a claim that every transcendence method must fail.

A sufficient missing input for the previous curvature strategy would be

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(x, x^{1/\sqrt{3}}, x^{1/\sqrt{2}}) \ge 2 \qquad (x > 0,\ x \ne 1), \tag{30.24}$$

or a weaker assertion at the particular contact (30.21). We have not proved (30.24), and the full-table argument above does not prove it. Thus the irrationality and transcendence of the individual number Q remain unresolved here.

Accounting. The positive results classify two new functional questions: the linear-differential class of the optimum/contact map, and the exact minimal algebraic differential order together with a nonlinear characterization. The individual biquadratic value was investigated but not closed. Functional algebraic independence in (30.19) must not be confused with algebraic independence after specializing C to an algebraic number. No historical named conjecture is counted as solved.

31 Fractional extraction: exact contact geometry and a sharp arithmetic rank barrier

The integer-shape argument of the preceding continuation used a two-dimensional Bessel system. Fractional extraction changes that dimension, but it does not destroy all of the useful differential structure. We prove two results. First, every interior boundary remainder, at every real shape, has only double zeros. The entire capacity curve is locally a finite minimum of analytic contact curves, with an exact integral slope formula and a global comparison inequality. Second, for a noninteger shape the normalized inverse has differential order exactly three. At rational parameters it is an E-function, and its normalized values and logarithmic derivatives at algebraic points are transcendental. The rank calculation explains precisely why the preceding two-row elimination does not settle fractional critical-rate transcendence.

The individual endpoint targets $A(1)$ and $A(3/2)$ are revisited at the end. Their transcendence is not proved here.

31.1 An inverse and a third-order equation for every real shape

Fix $\kappa > 1/2$, $b > 1$, and write

$$M = 2\kappa - 1, \qquad t = 2\kappa - 2.$$

The original law has Laplace transform

$$L(s) = \frac{b+s}{b}\left(\frac{b^2-1}{(b+s)^2-1}\right)^{\kappa}.$$

For a gamma shape $0 < h < M$ and rate $b - c$, where $0 < c < 1$, put $\eta = M - h > 0$. Define coefficients and functions by

$$(1 - cz)^h (1 - z^2)^{-\kappa} = \sum_{n \geq 0} b_n z^n, \tag{31.1}$$

$$v(x) = \sum_{n \geq 0} \frac{b_n x^{\eta + n}}{\Gamma(\eta + n + 1)}, \qquad g(x) = v'(x), \tag{31.2}$$

$$H(x) = \Gamma(\eta + 1) x^{-\eta} v(x), \qquad G(x) = \eta H(x) + x H'(x). \tag{31.3}$$

Here H and G are entire, whereas v and g use the positive real branch when η is not an integer. The series converge locally uniformly, together with all derivatives away from zero; their parameter dependence is real analytic in the strict domain. The actual candidate remainder density is a positive constant times $e^{-bx} g(x)$. In particular it is a probability density precisely when $g \geq 0$. The transform identities are

$$V(w) := \int_0^\infty e^{-wx} v(x)\, dx = \frac{(w - c)^h}{(w^2 - 1)^\kappa}, \qquad \mathcal{L}g(w) = wV(w), \qquad \Re w > 1. \tag{31.4}$$

They follow first for sufficiently large real w by termwise integration and then by analytic continuation. The factor $v(0+) = 0$ justifies the second identity.

We use the previously established capacity theorem as an input [91]: the maximal permissible shape $\mathcal{H}(c) = H_\kappa(c)$ is continuous and strictly decreasing on $[0, 1]$, lies strictly between 0 and M in the interior, and is attained. An interior boundary remainder is analytic on $(0, \infty)$, positive near zero and infinity, and has finitely many nonempty interior contacts. The endpoint positivity estimates are uniform in compact parameter neighborhoods with $h < M$. These earlier analytic facts are not counted again.

Lemma 31.1 (Fractional-shape differential identity)**.** *For every real $0 < h < M$ and $0 < c < 1$, the inverse v satisfies*

$$xv''' + (h + 1 - t - cx)v'' + (ct - x)v' + (cx - h - 1)v = 0 \qquad (x > 0). \tag{31.5}$$

Proof. The logarithmic derivative of (31.4) gives

$$(w - c)(w^2 - 1)V' = \{h(w^2 - 1) - 2\kappa w(w - c)\}V.$$

Under Laplace inversion, V' corresponds to $-xv$ and multiplication by w to differentiation on the positive axis. Thus, with $D = d/dx$, the equation becomes

$$-(D - c)(D^2 - 1)(xv) = \{h(D^2 - 1) - 2\kappa D(D - c)\}v.$$

Expansion gives (31.5). This manipulation can be verified without assuming that v'' or v''' is integrable at zero: insert the convergent series (31.2), compare coefficients in the preceding first-order equation for V, and then differentiate on a compact positive interval. The resulting coefficient identities are exactly (31.5). In particular no rule for commuting a fractional differential power with a variable coefficient has been assumed. □

Theorem 31.2 (Every real-shape contact is exactly double)**.** *Suppose $0 < h < M$, $0 < c < 1$, and the candidate remainder is nonnegative. Every positive zero x of its density is exactly double and satisfies*

$$0 < x < \frac{h + 1}{c}. \tag{31.6}$$

This includes every interior probability boundary and every noninteger shape; no rationality assumption is required.

Proof. Since $g = v' \geq 0$ and $v(x) \sim x^\eta/\Gamma(\eta+1)$ at zero, one has $v(x) > 0$ for every $x > 0$. At a zero of the nonnegative analytic function g, we have $g' = 0$. Equation (31.5) then gives

$$xg'' = (h + 1 - cx)v. \tag{31.7}$$

The left side is nonnegative, so $cx \leq h + 1$. In the equality case $g'' = 0$. Differentiating (31.5) at that point, where $v' = v'' = v''' = 0$, would give $xg''' = -cv < 0$. This would be an odd zero of exact order three, contrary to nonnegativity on both sides. Therefore $cx < h + 1$, and (31.7) gives $g'' > 0$. □

31.2 The full frontier has analytic contact branches and a signed response

Partial derivatives below hold κ fixed. At a boundary contact x, define the positive finite integrals

$$A_x = \int_0^x e^{cy} g(x-y)\, dy, \qquad B_x = \int_0^x \frac{e^{cy} g(x-y)}{y}\, dy. \tag{31.8}$$

Finiteness near $y = x$ follows from $\eta > 0$, and near $y = 0$ from the double zero just proved.

Lemma 31.3 (Parameter derivatives at a contact)**.** *At every boundary contact,*

$$g_c(x) = -hA_x < 0, \qquad g_h(x) = -B_x < 0. \tag{31.9}$$

Proof. Differentiation of wV with respect to c multiplies its transform by $-h/(w-c)$. Convolution therefore gives the first identity for every positive x.

For the second derivative, the multiplier is $\log(w-c)$. The regularized identity

$$\log(w-c) = \int_0^\infty \frac{e^{-y} - e^{-(w-c)y}}{y}\, dy$$

gives, on the positive axis,

$$g_h(x) = -(\gamma + \log x)g(x) + \int_0^x \frac{g(x) - e^{cy} g(x-y)}{y}\, dy, \tag{31.10}$$

where γ is Euler's constant. To justify it directly, truncate the preceding transform integral at $y = \epsilon$, invert the resulting convolution, and use $\int_\epsilon^\infty e^{-y} dy/y = -\gamma - \log\epsilon + o(1)$. The subtracted integrand converges locally at $y = 0$ because g is differentiable at $x > 0$, and is integrable at $y = x$ because $g(u) = O(u^{\eta-1})$. Thus the limit is (31.10); it also agrees with the differentiated locally convergent inverse series. At a contact $g(x) = 0$, this is precisely $-B_x$. Both integrals are strictly positive, since g is nonnegative and positive near the origin. □

Theorem 31.4 (Regularity and a universal slope inequality)**.** *On every compact subinterval of $(0,1)$, the capacity $\mathcal{H}(c)$ is piecewise real analytic with only finitely many possible switching points. Locally it is the minimum of finitely many real analytic functions. It is locally Lipschitz; its one-sided derivatives exist everywhere in $(0,1)$, and*

$$-\frac{\mathcal{H}(c)(\mathcal{H}(c)+1)}{c} < \mathcal{H}'_\pm(c) < 0. \tag{31.11}$$

At a switching point the derivative can jump only downward: $\mathcal{H}'_-(c) \geq \mathcal{H}'_+(c)$.

More precisely, a contact branch through (c, h, x) has slope

$$h'(c) = -h\frac{A_x}{B_x}, \qquad -hx < h'(c) < 0 \tag{31.12}$$

whenever that branch realizes the probability boundary. The normalized frontier

$$c \longmapsto \frac{c\mathcal{H}(c)}{1+\mathcal{H}(c)} \tag{31.13}$$

is strictly increasing. In particular, for $0 < c_1 < c_2 < 1$,

$$\frac{c_1\mathcal{H}(c_1)}{c_2 + (c_2 - c_1)\mathcal{H}(c_1)} < \mathcal{H}(c_2) < \mathcal{H}(c_1). \tag{31.14}$$

Proof. Fix a boundary point (c_0, h_0) and list its finitely many contacts $x_1, \dots, x_N$. The equations $g = 0$, $g_x = 0$ have, as equations in (h, x), Jacobian determinant $g_h g_{xx} \neq 0$ at each contact, by Theorem 31.2 and Lemma 31.3. The analytic implicit-function theorem therefore gives contact curves $(h_j(c), x_j(c))$.

Here these curves describe the full local probability boundary, not only necessary conditions for it. Choose disjoint small intervals around the contacts. Their strict minima persist, since $g_{xx} > 0$, and their minimum values have negative h-derivative. On the intervening compact set the original density has a positive minimum. Uniform endpoint positivity, an inherited input stated above, also persists in a parameter neighborhood. Consequently, in that neighborhood,

$$g \geq 0 \quad \Longleftrightarrow \quad h \leq \min_{1 \leq j \leq N} h_j(c).$$

Continuity of the already established capacity ensures that its graph stays in this neighborhood. Thus $\mathcal{H}(c) = \min_j h_j(c)$ there.

Two analytic branches either agree identically locally or have isolated intersections. A finite family of such branches has only finitely many switching points in a sufficiently small neighborhood of a fixed interior point. Compactness then gives the asserted local piecewise analyticity and local Lipschitz property. At a multiple active branch, the left derivative is the largest active slope and the right derivative is the smallest, which proves the direction of a possible jump.

Differentiating $g(c, h(c), x(c)) = 0$ and using $g_x = 0$ gives $h' = -g_c/g_h = -hA_x/B_x$. The quotient A_x/B_x is the mean of y under the positive measure with density $e^{cy}g(x-y)/y$ on $(0, x)$. It lies strictly between 0 and x. This proves (31.12); combining it with (31.6) proves (31.11), including both one-sided derivatives at a switch.

On each analytic piece, differentiation gives

$$\left(\frac{c\mathcal{H}}{1+\mathcal{H}}\right)' = \frac{\mathcal{H}}{1+\mathcal{H}} + \frac{c\mathcal{H}'}{(1+\mathcal{H})^2} > 0.$$

The function is continuous at the finitely many switches on a compact interval, so it is strictly increasing throughout that interval. Rearranging this comparison at c_1, c_2 yields the first inequality of (31.14); the second is the inherited strict decrease of the capacity. □

The theorem allows corners and more than one contact. It does not assert global analyticity or a unique contact at every rate. What it excludes is a flat contact, a locally infinite accumulation of switches in the interior, and an unbounded local slope.

31.3 Noninteger shapes have exact differential order three

The following rank result needs no arithmetic assumptions. It is the obstruction that the integer-shape calculation did not encounter.

Theorem 31.5 (Three singularities force rank three)**.** *Let $\kappa > 1/2$, $0 < c < 1$, and $0 < h < M$ with $h \notin \mathbb{Z}$. Then*

$$1, H, H', H'' \tag{31.15}$$

are linearly independent over $\mathbb{C}(x)$. In particular the minimal homogeneous differential equation of H has order exactly three. The same minimal-order conclusion holds for G.

Proof. Suppose a rational relation in (31.15) existed. The substitution $H = \Gamma(\eta+1)x^{-\eta}v$ changes the three derivative coefficients by an invertible triangular rational matrix. Clearing denominators and multiplying by x^2 would therefore give

$$P_0(x)v + P_1(x)v' + P_2(x)v'' = Q(x)x^{\eta}, \tag{31.16}$$

where the P_i, Q are polynomials, all divisible by x^2, and at least one P_i is nonzero. If all P_i were zero, the original relation would already force $Q = 0$.

Write $P_i(x) = \sum_n p_{in}x^n$ and let N be the largest index for which some $p_{iN} \neq 0$. Thus $N \geq 2$ and

$$R_N(w) = \sum_{i=0}^{2} p_{iN}w^i$$

is a nonzero polynomial of degree at most two. Laplace transformation of (31.16) is legitimate after the multiplication by x^2: every integrand is integrable at zero, and integration by parts has no boundary term. The transform of the left side is

$$\sum_{i=0}^{2}\sum_{n\geq 2} p_{in}(-D_w)^n\{w^iV(w)\}. \tag{31.17}$$

The right side is a finite linear combination of powers $w^{-\eta-n-1}$. It is therefore holomorphic locally at each of the three nonzero points $c, 1, -1$, with compatible branches chosen by analytic continuation.

At these three points λ, the transform has the form

$$V(w) = (w-\lambda)^{a_\lambda}U_\lambda(w), \qquad U_\lambda(\lambda) \neq 0, \qquad a_c = h, \quad a_1 = a_{-1} = -\kappa.$$

If $R_N(\lambda) \neq 0$, the leading local term of (31.17) is a nonzero multiple of

$$(a_\lambda)_{\underline{N}}R_N(\lambda)U_\lambda(\lambda)(w-\lambda)^{a_\lambda - N}.$$

No lower derivative order has this exponent. At $\lambda = c$, the exponent is nonintegral and the falling factorial is nonzero because h is not an integer. At $\lambda = \pm 1$, the exponent is negative and the falling factorial is nonzero because $-\kappa < 0$. In all cases this term prevents a holomorphic right side: it is either a pole or the first nonintegral power of the local branch. Hence $R_N(c) = R_N(1) = R_N(-1) = 0$. A nonzero polynomial of degree at most two cannot have these three distinct roots, contradiction.

Equation (31.5), after the rational gauge $v = x^{\eta}H/\Gamma(\eta+1)$, supplies an order-three equation for H, so its minimal order is exactly three. Finally the rational state transformation from (v, v', v'') to (g, g', g'') has determinant $(h+1-cx)/x$, by (31.5). This determinant is not identically zero. Thus differentiation preserves the functional rank three, and the further gauge $g = x^{\eta-1}G/\Gamma(\eta+1)$ proves the assertion for G. □

The pole case at $\lambda = \pm 1$ is included, so integer κ is not an exception. In contrast, when h is an integer the factor $(w-c)^h$ is locally a polynomial and the third singularity disappears. That is precisely the reduction which allowed the preceding two-row contact determinant.

31.4 Arithmetic consequences for every rational noninteger shape

Lemma 31.6 (The normalized inverse is an E-function)**.** *If κ, h are rational in the preceding strict domain and c is algebraic, then H, H', H'' are E-functions. They satisfy a first-order system over $\overline{\mathbb{Q}}(x)$ whose only finite possible singularity is zero.*

Proof. The exponential Taylor coefficients of H are

$$a_n = \frac{n! b_n}{(\eta+1)_n}.$$

The coefficients of each rational-power binomial in (31.1) have denominators dividing C^n for some integer C. Indeed, for a rational parameter A/B in lowest terms, primes not dividing B cancel the factorial denominator by counting a complete residue class modulo every prime power. At primes dividing B, its denominator valuation is at most $n\{v_p(B) + 1/(p-1)\}$. Clearing a fixed algebraic denominator of c therefore makes $C^n b_n$ algebraic integral in a fixed number field. Cauchy's estimate applied to (31.1) under each embedding bounds all conjugates of b_n geometrically.

It remains to account for $n!/(\eta+1)_n$. Write $\eta + 1 = A/B > 0$ in lowest terms. For a prime not dividing B, counting the terms $A + Bj$, $0 \le j < n$, divisible by p^k differs from $\lfloor n/p^k \rfloor$ by at most one. No such term exceeds Cn, for a fixed integer C. Thus the denominator valuation is at most $\lfloor \log(Cn)/\log p \rfloor$; primes dividing B add no denominator. Consequently the common denominator for these ratios through index N divides $\operatorname{lcm}(1, \dots, CN)$, which has exponential growth. The ratios themselves have polynomial growth, by the gamma-ratio estimate, so the coefficients a_n and their common denominators satisfy the exponential bounds in the definition of an E-function.

The rational gauge in (31.5) gives a third-order equation for H with algebraic coefficients and possible poles only at zero. This proves the required differential condition. Derivative closure gives the same properties for H' and H''. □

Theorem 31.7 (Normalized values and logarithmic derivatives)**.** *Assume $\kappa, h \in \mathbb{Q}$, $0 < h < 2\kappa - 1$, $h \notin \mathbb{Z}$, and $c \in \overline{\mathbb{Q}} \cap (0,1)$. For every nonzero algebraic ξ, the four numbers*

$$1, H(\xi), H'(\xi), H''(\xi) \tag{31.18}$$

are linearly independent over $\overline{\mathbb{Q}}$. In particular

$$G(\xi) \quad \text{and} \quad \frac{G'(\xi)}{G(\xi)} \quad \text{are transcendental.} \tag{31.19}$$

Every positive zero of the candidate remainder density is transcendental. If $b > 1$ is also algebraic, every positive stationary point of that density is transcendental as well.

Proof. Adjoin the constant 1 to the first-order E-function system in Lemma 31.6. Its only finite possible singularity is zero, and its four entries are linearly independent by Theorem 31.5. Beukers' linear specialization theorem [35, Corollary 1.4] proves (31.18) at every nonzero algebraic point.

The expression $G(\xi) = \eta H(\xi) + \xi H'(\xi)$ is a nonzero algebraic linear form in the last three entries. It cannot be algebraic, since that would give a relation with the first entry 1. If $G'(\xi)/G(\xi) = a$ were algebraic, then

$$-a\eta H(\xi) + (\eta + 1 - a\xi) H'(\xi) + \xi H''(\xi) = 0,$$

another nonzero relation because $\xi \neq 0$. This proves (31.19).

On the positive axis the candidate physical density is a nonzero constant times $e^{-bx}x^{\eta-1}G(x)$. Its positive zeros are exactly the zeros of G, so no such zero is algebraic. Where it is nonzero, its logarithmic derivative is

$$-b+\frac{\eta-1}{x}+\frac{G'(x)}{G(x)}.$$

At positive algebraic x, this is transcendental, and in particular nonzero, if b is algebraic. Thus a stationary point cannot be algebraic either. □

The normalization in (31.19) matters. We have not asserted transcendence of the physical density value after multiplication by its gamma and exponential normalizers. The logarithmic derivative and the zero or stationary locations do not have that ambiguity.

31.5 What the new rank theorem does not prove

For a fixed rational noninteger h, a boundary contact still imposes only $g(x)=g'(x)=0$. At an algebraic candidate rate, Theorem 31.7 forces the contact x to be transcendental, but does not contradict its existence. The rank-three system has a one-dimensional space of initial states satisfying those two equations. Hence the two-row determinant from the integer argument has no rank-three counterpart of the same form. This is a proved structural obstruction to that specific elimination method, not a proof that fractional-rate transcendence is impossible by other methods.

The two endpoint questions $A(1)$ and $A(3/2)$ remain separate. If one of these constants were algebraic irrational, its corresponding maximal shape would be algebraic irrational, and the normalized Kummer function in the endpoint representation would fail the E-function criterion. The earlier exact application of Galochkin's classification, as formulated by Rivoal [36, Theorem 1], continues to apply: the only possible irrational numerator–denominator parameter pair has difference $-d<0$. The present rational-parameter argument therefore supplies no arithmetic specialization at that hypothetical endpoint. Nor does linear independence of functions at a fixed order imply transcendence of a varying algebraic-irrational order.

31.6 An exponential observable at the integer contacts

There is nevertheless a further arithmetic question that can be settled when the original Bessel parameter is an integer. It concerns the exponential of the contact, rather than an exponential of the critical rate.

Proposition 31.8 (The contact exponential is also transcendental)**.** *Let $\kappa=k\geq 2$ and $1\leq m<2k-1$ be integers. Suppose the shape-m candidate has a positive multiple contact x at $0<c<1$. Then e^{2x} is transcendental. In fact*

$$e^{2x}\in\overline{\mathbb{Q}}(x,c),\qquad \operatorname{trdeg}_{\overline{\mathbb{Q}}}\overline{\mathbb{Q}}(x,c,e^{2x})=1. \tag{31.20}$$

Thus the contact, the rate coordinate, and the contact exponential are individually transcendental but are algebraically dependent. The exponential e^{rx} is transcendental for every nonzero rational r as well.

Proof. Up to a nonzero constant factor, the half-integer Bessel expression $x^{k-1/2}I_{k-3/2}(x)$ has the form

$$f(x)=e^{x}P(x)+e^{-x}Q(x),$$

where $P, Q \in \overline{\mathbb{Q}}[x]$ both have degree $d = k-1$ and nonzero leading coefficients p, q. The removed constant has no effect on contacts. This representation follows from the elementary formulas for $I_{-1/2}, I_{1/2}$ and the usual Bessel recurrence [90, 10.39 and 10.29]. The two exponential terms separately solve the base Bessel equation and form an independent solution basis. Put

$$A = (D + 1 - c)^m P, \qquad B = (D - 1 - c)^m Q.$$

At a contact, writing $E = e^{2x}$ gives the polynomial equations

$$EA(x, c) + B(x, c) = 0, \qquad E(A_x + A)(x, c) + (B_x - B)(x, c) = 0. \tag{31.21}$$

The nonzero-contact-row lemma of the preceding integer-shape paper [79] says that these two coefficient rows cannot both vanish when $x \neq 0$ and $c^2 \neq 1$. Consequently one equation has a nonzero coefficient of E; otherwise its constant coefficient would also vanish at the actual contact. Solving that equation proves the field inclusion in (31.20).

We now show that for every fixed complex $E \neq 0$, the two equations in (31.21) permit only finitely many x-coordinates. First their common zeros cannot have $|x| \to \infty$. Along such a hypothetical sequence, if c has a bounded subsequence with limit c_0, division by x^d gives the two limiting equations

$$Ep(1 - c_0)^m + q(-1 - c_0)^m = 0, \qquad Ep(1 - c_0)^m - q(-1 - c_0)^m = 0.$$

They would require simultaneously $c_0 = 1$ and $c_0 = -1$. If instead $|c| \to \infty$, divide by $x^d c^m$. The terms involving derivatives of P, Q are smaller by a factor tending to zero, and the two limits are $(-1)^m(Ep + q) = 0$ and $(-1)^m(Ep - q) = 0$, also impossible.

For completeness, the required algebraic elimination follows from this boundedness without a generic-fiber assumption. Regard the two polynomials as polynomials in c over $\mathbb{C}(x)$. If their greatest common divisor had positive c-degree, clearing denominators in a common factor would produce a common zero in c for every sufficiently large x outside a finite exceptional set. This contradicts the boundedness just proved. Therefore their greatest common divisor over $\mathbb{C}(x)[c]$ is one. A Bezout identity over $\mathbb{C}(x)$, with denominators cleared, yields a nonzero polynomial in x belonging to their ideal; all common x-coordinates are among its finitely many roots. If E is algebraic, the same computation takes place over $\overline{\mathbb{Q}}(x)$ and gives a polynomial with algebraic coefficients.

Suppose now that the actual $E = e^{2x}$ were algebraic. The preceding elimination would force x algebraic. Since $x > 0$, the Hermite–Lindemann theorem would then make e^{2x} transcendental, contradiction. Thus E is transcendental. The preceding integer-shape theorem already proves that x, c are individually transcendental with joint transcendence degree one; the field inclusion gives the second assertion of (31.20). Finally, algebraicity of e^{rx} for nonzero rational $r = a/b$ would make $e^{2ax} = E^a$ algebraic, and hence E algebraic, another contradiction. □

This proves dependence rather than independence for the contact and its own exponential. It gives no conclusion about e^c, and no application of Lindemann–Weierstrass at a transcendental argument has been assumed.

Accounting. The first new family is the all-real-shape contact and frontier regularity theorem, including its exact response and comparison law. The second is the exact noninteger differential-rank theorem and its rational-parameter arithmetic specialization. The contact-exponential proposition closes an additional observable question inside the preceding integer family, and is recorded as a related consequence rather than a new family. Individual examples of a rational shape or algebraic sample point are not separate discoveries. Transcendence of the fractional critical rate itself and of $A(1), A(3/2)$ remains unresolved. No historical named constant conjecture is claimed to have been closed.

32 A linear differential obstruction for the two prime series

We revisit the three fixed-value targets

$$E = \sum_{n\ge 1} (2^n - 1)^{-1}, \qquad P_2 = \sum_{p \text{ prime}} 2^{-p}, \qquad L_{\mathbb{P}} = \sum_{p \text{ prime}} (2^p - 1)^{-1}.$$

No transcendence assertion for these numbers is proved here. The new result concerns a more permissive functional transfer than algebraic elimination: we allow arbitrarily many derivatives of both prime series, and coefficients rational in a reciprocal-Mersenne moment and finite modular constructions. Even a nontrivial linear differential relation is impossible. The proof separates two meromorphic Borel transforms by their different sets of poles. This does not establish nonlinear differential algebraic independence.

Write, for $|q| < 1$,

$$A_r(q) = \sum_{n\ge 1} \left(\frac{q^n}{1-q^n}\right)^r, \qquad P(q) = \sum_p q^p, \qquad B(q) = \sum_p \frac{q^p}{1-q^p}.$$

For the boundary analysis put $q = e^{-t}$, $L = \log(1/t)$ and $\partial = -t\, d/dt$, so $\partial L = 1$. For $\lambda > 0$ let

$$\mathscr{K}_\lambda = \{F(t, e^{-\lambda/t}) : F \in \operatorname{Frac} \mathbb{C}\{t, z\}\}.$$

These are germs along $t \downarrow 0$. A nonzero element has the form $ct^a e^{-m\lambda/t}(1 + O(t))$, with $a, m \in \mathbb{Z}$ and $c \neq 0$: expand a nonzero convergent numerator and denominator in z, take their first nonzero coefficients, and then take the first nonzero powers of t in those coefficients. The remaining z terms are smaller than every power of t. This also proves that evaluation is injective. The field is closed under d/dt, by the chain rule.

32.1 Two formal transforms with different infinite pole sets

Let

$$F(w) = \sum_{k\ge 0} a_k w^{-k-1}, \quad a_k = (-1)^k \Gamma^{(k)}(1), \qquad G(w) = \log w + M + \sum_{k\ge 1} \frac{(-1)^{k+1} m_k}{k w^k}, \tag{32.1}$$

where M is any complex constant and

$$H(s) = s\Gamma(s+1)\zeta(s+1), \qquad m_k = H^{(k)}(0). \tag{32.2}$$

The singularity of H at zero is removable and $H(0) = 1$. The following statement is about formal series; it assumes no convergence of (32.1).

Lemma 32.1 (Formal independence of both derivative families)**.** *The family*

$$1, \quad F, F', F'', \ldots, \quad G, G', G'', \ldots$$

is linearly independent over $\mathbb{C}(w)$, with derivatives taken formally with respect to w.

Proof. Clear denominators in a putative finite relation so that all its coefficients are polynomials in w. Only the undifferentiated G contains $\log w$, so its coefficient must vanish. The remaining terms are formal Laurent series with finitely many nonnegative powers. Define their formal Borel transform by

$$\mathcal{B}\left(\sum_{k\ge 0} c_k w^{-k-1} + \text{a polynomial in } w\right)(s) = \sum_{k\ge 0} \frac{c_k}{k!} s^k.$$

Direct coefficient comparison gives

$$\mathcal{B}(wU) = \frac{d}{ds}\mathcal{B}(U), \qquad \mathcal{B}(U') = -s\mathcal{B}(U).$$

In particular,

$$\mathcal{B}(F) = \Gamma(1-s), \qquad \mathcal{B}(G') = H(-s) = -s\Gamma(1-s)\zeta(1-s). \tag{32.3}$$

Both right-hand sides converge near zero and have meromorphic continuations to the complex plane. Thus the transformed relation is

$$U_1\Gamma(1-s) + U_2H(-s) = 0, \tag{32.4}$$

where U_1, U_2 are polynomial-coefficient differential operators in s. Substitution $w \mapsto d/ds$, $d/dw \mapsto -s$ is an invertible map of Weyl algebras; hence a nonzero original differential operator gives a nonzero U_i. Equivalently, one may verify this by writing operators uniquely as finite sums of powers of d/ds with polynomial coefficients.

The function $\Gamma(1-s)$ has a simple pole at every positive integer. The function $H(-s)$ is holomorphic at every odd integer $s \geq 3$, because the trivial zero of $\zeta(1-s)$ cancels the gamma pole. It has a simple pole at every positive even integer, since $\zeta(1-2n) \neq 0$ for $n \geq 1$. These facts also follow directly from the functional equation of ζ.

If $U_1 = \sum_{j=0}^{d} c_j(s)(d/ds)^j$ is nonzero, take d maximal. At all but finitely many positive odd integers, c_d is nonzero and $U_1\Gamma(1-s)$ has a pole of exact order $d+1$: the derivatives of lower order cannot cancel its leading polar term. In contrast, $U_2H(-s)$ is holomorphic there. This contradicts (32.4), so $U_1 = 0$. The same argument at the positive even integers forces $U_2 = 0$. Consequently all coefficients of $F^{(j)}$ and $G^{(j)}$, $j \geq 1$, in the original relation vanish. The remaining polynomial must also vanish, proving the lemma. □

32.2 All-order asymptotics, including every fixed derivative

Let M now denote the Meissel–Mertens constant, determined by

$$M(x) := \sum_{p \leq x} \frac{1}{p} = \log\log x + M + o(1).$$

We use a precise convention for the next lemma: an asymptotic formal expansion means that truncation at any prescribed power of L^{-1} has an error of the next order. The expansion is asserted separately for each fixed derivative order, not uniformly in that order.

Lemma 32.2 (Differentiated prime expansions)**.** *For $Y(t) = tP(e^{-t})$ and $Z(t) = tB(e^{-t})$, one has, to every inverse-logarithmic order and for every fixed $j \geq 0$,*

$$\partial^j Y(t) \sim F^{(j)}(L), \qquad \partial^j Z(t) \sim G^{(j)}(L), \tag{32.5}$$

where F, G are (32.1) with this value of M.

Proof. The prime-counting estimate in Trudgian's Theorem 2 [92] implies, for every $A > 0$,

$$\pi(x) - \int_2^x \frac{du}{\log u} = O_A(x/\log^A x).$$

Partial summation also gives

$$M(x) = \log\log x + M + O_A((\log x)^{-A}) \tag{32.6}$$

for every fixed A: the tail integral of the prime-counting error is bounded by a constant times $\int_x^\infty du/(u\log^{A+1}u)$, after increasing its requested exponent if necessary.

For clarity we justify differentiation of the expansions by integral formulas, rather than by differentiating an unqualified remainder. Define polynomials $Q_0(u)=1$ and $Q_{j+1}(u)=(1-u)Q_j(u)+uQ_j'(u)$. Then

$$(t\,d/dt)^jY(t)=t\int_{2^-}^\infty Q_j(tx)e^{-tx}\,d\pi(x).$$

Integration by parts against the prime-counting error, splitting at $x=t^{-1/2}$, shows that the right-hand side equals

$$\int_{2t}^\infty \frac{Q_j(u)e^{-u}}{L+\log u}\,du+O_{A,j}(L^{-A}). \tag{32.7}$$

On the first part of the split use $\pi(x)=O(x)$ and the boundedness of the fixed polynomial kernel and its derivative near zero; its contribution to the error is $O_j(t)$. On the second part use $\log x\ge L/2$ and the integrability of polynomial multiples of e^{-u}.

In (32.7), discard $u<t^{1/2}$ and $u>t^{-1/2}$ at an error smaller than every inverse power of L. On the remaining interval expand $(L+\log u)^{-1}$ to any fixed order. All logarithmic moments are integrable. Finally,

$$\int_0^\infty Q_j(u)e^{-u}u^s\,du=(-s)^j\Gamma(1+s).$$

This identity follows by induction and integration by parts, first for $\Re s>0$ and then near zero by analyticity. It identifies the resulting coefficients as those of $(-d/dL)^jF(L)$ for $(t\,d/dt)^jY$, proving the first part of (32.5).

For the second part put $h(u)=u/(e^u-1)$ and

$$h_j(u)=(u\,d/du)^jh(u),\qquad w_j(u)=-h_j'(u).$$

Each w_j is bounded near zero and decays exponentially times a polynomial at infinity. Since $Z(t)=\int h(tx)\,dM(x)$, integration by parts gives

$$(t\,d/dt)^jZ(t)=\int_0^\infty w_j(u)M(u/t)\,du, \tag{32.8}$$

where $M(x)=0$ for $x<2$. The boundary terms vanish. The part $u<t^{1/2}$ is $O_j(t^{1/2}(1+\log L))$ and the part $u>t^{-1/2}$ is smaller than every inverse power of L. In between, (32.6) replaces $M(u/t)$ by $\log(L+\log u)+M$ with an error $O_A(L^{-A})$. Taylor expansion of this logarithm is valid uniformly because $|\log u|\le L/2$; its integrated remainder has the requested inverse-logarithmic order.

The moment-generating identity is

$$\int_0^\infty w_j(u)u^s\,du=(-s)^jH(s). \tag{32.9}$$

For $j=0$, integration by parts for $\Re s>0$ gives $s\int_0^\infty u^s/(e^u-1)\,du=s\Gamma(s+1)\zeta(s+1)$; geometric expansion of the denominator justifies the latter identity by absolute convergence. Repeated integration by parts gives the factor $(-s)^j$. Both sides are analytic near zero, so (32.9) holds there too. For $j=0$ it yields precisely the series G in (32.1); for general j it yields its jth derivative with the sign corresponding to $t\,d/dt=-\partial$. This proves the second part without differentiating an asymptotic error term. □

32.3 No linear differential transfer even with a moment coefficient

We record the coordinate estimate needed to include A_r in the coefficient field. If $(e^x-1)^{-r}=\sum_{k=1}^{r}\kappa_{r,k}x^{-k}+O(1)$, then $\kappa_{r,1}=(-1)^{r-1}$, by substituting $v=e^x-1$ in its residue. Subtract from $(e^x-1)^{-r}$ the powers with $k\ge 2$ and the function $\kappa_{r,1}e^{-x}/x$. The residual g_r extends smoothly to zero, and g_r, g_r' are integrable on $[0,\infty)$. Comparing the integral with the right-endpoint Riemann sum gives $\sum_{n\ge1} g_r(nt)=t^{-1}\int g_r+O(1)$. Thus, with $C_r=\int_0^\infty g_r(x)\,dx$,

$$X_r(t):=\kappa_{r,1}^{-1}\left(tA_r(e^{-t})-\sum_{k=2}^{r}\kappa_{r,k}\zeta(k)t^{1-k}-C_r\right)=L+O(t). \tag{32.10}$$

This supporting coordinate estimate is the same one used for the preceding functional algebraic-independence result; it is not counted as a new result here.

Theorem 32.3 (Independence of the prime derivative families)**.** *Fix $\lambda>0$ and an integer $r\ge1$. Over the coefficient field $\mathscr{K}_\lambda(A_r(e^{-t}))$, the following germs are linearly independent:*

$$1,\quad Y,\partial Y,\partial^2 Y,\dots,\quad Z,\partial Z,\partial^2 Z,\dots. \tag{32.11}$$

Equivalently, replace Y,Z by $P(e^{-t}),B(e^{-t})$ and ∂ by d/dt in this assertion. In particular there is no nontrivial finite inhomogeneous linear differential relation between the two prime series with coefficients rational in A_r and elements of this cusp field.

Proof. The fields obtained by adjoining $A_r(e^{-t})$ and $X_r(t)$ are the same, by (32.10). Clear the finitely many denominators of a putative relation. Its coefficients can then be written as polynomials in $X=X_r$ with coefficients in $\mathscr{K}_\lambda$, not all zero. Among all the nonzero cusp-field coefficients choose the least exponential order m, then the least power of t, say a, at that order. Divide by $t^a e^{-m\lambda/t}$ and take the leading constants of those coefficients. This produces a nontrivial relation expression

$$C(X)+\sum_{j=0}^{J}U_j(X)\partial^j Y+\sum_{j=0}^{J}V_j(X)\partial^j Z,\qquad C,U_j,V_j\in\mathbb{C}[X], \tag{32.12}$$

which the original exact relation would force to be smaller than every inverse power of L. Indeed the discarded coefficients are $O(t)$ after normalization, or exponentially small in $1/t$ times fixed powers of t^{-1}; all the functions multiplying them have at most a fixed power of $L+\log L+1$ as growth.

By (32.10), replacing X by L changes (32.12) by less than every inverse power of L. Lemma 32.2 gives its formal expansion

$$C(L)+\sum_j U_j(L)F^{(j)}(L)+\sum_j V_j(L)G^{(j)}(L).$$

Lemma 32.1 says this is nonzero. It therefore has a first nonzero power of L, or a first nonzero power times $\log L$; any such leading term contradicts the required smallness. This proves (32.11).

Multiplication of each target by t, and replacement of d/dt by $-t\,d/dt$, induce invertible triangular changes between any finite initial derivative families, with coefficients in $\mathscr{K}_\lambda$. This proves the stated equivalence. □

Corollary 32.4 (Rational Lambert coefficients)**.** *The family $1,Y,\partial Y,\partial^2 Y,\dots$ is linearly independent over the field*

$$\mathscr{K}_\lambda(A_r(e^{-t}),B(e^{-t})).$$

Proof. Clear denominators, use $X = X_r$ and $Z = tB(e^{-t})$, and select the least cusp orders exactly as in Theorem 32.3. A putative relation would force a nonzero polynomial-in-Z expression

$$\sum_{h=0}^{H} Z^h \left(C_h(X) + \sum_{j=0}^{J} U_{hj}(X)\partial^j Y \right), \qquad C_h, U_{hj} \in \mathbb{C}[X],$$

to be smaller than every inverse power of L. For each nonzero coefficient expression, the formal independence of $1, F, F', \dots$ in Lemma 32.1 gives a nonzero first Laurent term $c_h L^{b_h}$. By (32.10) and Lemma 32.2, the actual coefficient has this same leading equivalent. Choose the largest b_h and then the largest h among ties. Since $Z \sim \log L$, the corresponding term $c_h L^{b_h} (\log L)^h$ uniquely dominates. It cannot be smaller than every inverse power of L. □

For a finite collection of integer dilations d, take $D = \operatorname{lcm} d$ and $\lambda = 4\pi^2/D$. The usual Eisenstein transformation laws imply that all $\mathcal{E}_2(q^d), \mathcal{E}_4(q^d), \mathcal{E}_6(q^d)$ belong to $\mathscr{K}_\lambda$ after $q = e^{-t}$: their transformed arguments are integer powers of $e^{-4\pi^2/(Dt)}$, their prefactors are powers of t, and the weight-two correction is a constant multiple of t^{-1}. Their derivatives also belong to this field. The normalizations and differential closure are given in [93, Section 1]. Consequently Theorem 32.3 excludes the same linear differential relation when its coefficients are rational expressions in a single A_r, q, these finite modular generators, and their derivatives. Derivatives of A_r are not included in that coefficient field by this statement.

32.4 The arithmetic routes checked, and the steps still missing

For the Erdős–Borwein target, the primary q-logarithm approximation literature already proves considerably more than irrationality. For example, Zudilin's Theorems 1–2 and their corollary [94] give quantitative rational linear independence for appropriate triples involving $A_1(1/b)$, $A_1(1/b^2)$, and the q-dilogarithm, for integer $b \geq 2$. Those theorems are not new consequences of our work. They estimate linear forms; they do not give the estimates for polynomials of unbounded degree in E that would prove its transcendence. Finite irrationality measure alone does not distinguish an algebraic irrational from a transcendental number.

There is also an exact distinction between the functional equation used by the ordinary q-logarithm and its prime-restricted version. For a fixed integer $b \geq 2$, define

$$F_b(x) = \sum_{n\geq 1} \frac{x^n}{b^n - 1}, \qquad F_{b,\mathbb{P}}(x) = \sum_{p} \frac{x^p}{b^p - 1}.$$

Both are analytic for $|x| < b$, and for $|x| < 1$ direct subtraction gives

$$F_b(bx) - F_b(x) = \frac{x}{1-x}, \qquad F_{b,\mathbb{P}}(bx) - F_{b,\mathbb{P}}(x) = P(x). \tag{32.13}$$

Thus the rational forcing term in the ordinary shift equation is replaced by the prime generating function. The identity itself is elementary, but it identifies exactly why one cannot simply insert prime support into the ordinary q-logarithm proof. The new theorem shows that this forcing function cannot be eliminated by the stated finite linear differential constructions, even when A_r and the prime Lambert function are available. It does not rule out other q-difference or arithmetic approximation methods.

For the prime Lambert target, the irrationality theorem of Tao and Teräväinen [95] is already an external result. Its connection to this target is exact:

$$B(1/2) = \sum_{n\geq 1} \omega(n) 2^{-n},$$

obtained by interchanging the absolutely convergent divisor sums. We do not re-count this settled irrationality statement as a new closure. For P_2, our previous prime-word complexity analysis and the present differential obstruction likewise do not prove transcendence at $q = 1/2$.

Finally, the Borel argument above must not be upgraded silently to nonlinear differential transcendence. It uses the exact linear rules $w \mapsto d/ds$ and $d/dw \mapsto -s$. Products of formal series produce convolution under this transform, not pointwise products. The distinct pole sets therefore prove the asserted linear independence, but do not by themselves exclude nonlinear differential polynomials. Both that stronger functional question and the three fixed-value transcendence questions remain unresolved by this section.

33 What would constitute the next arithmetic step?

The results above close several precisely stated family questions. They do not establish a new proof of irrationality for Euler's constant, Euler–Gompertz, or Poisson entropy, nor a new proof of transcendence for the three prime and reciprocal-Mersenne targets. The following remaining steps are substantive mathematical requirements, rather than requests for additional numerical precision.

For the linear excursion optimum, a squarefree-factorization result beyond the currently applicable partial theorem would be needed for the route investigated here. The associated function fails global boundedness and contains infinitely many simple zeros, alongside its multiple zeros. Neither failure can be removed by checking a larger finite collection of Taylor coefficients or zeros.

For the biquadratic optimum, the field-degree argument must control the three powers in (30.24), without relying on a fourth power that is unconstrained by the contact. The full-jet identity (30.20) proves that repeatedly differentiating with respect to the setting cannot create this missing arithmetic coordinate. The cubic-field theorem from the preceding paper therefore does not silently become a theorem for every three-dimensional exponent span.

For fractional extraction rates, a third independent arithmetic constraint or a different specialization argument is needed. The rank-three theorem does yield normalized-value and location arithmetic, but two contact equations leave a one-dimensional state space. The all-real-shape frontier theorem supplies geometric control, not the missing transcendence constraint.

For the prime constants, the derivative-family obstruction is a functional theorem. A new arithmetic specialization or approximation argument must control values at $q = 1/2$. Conversely, the obstruction does not exclude nonlinear differential relations or all possible q-difference constructions. It excludes only the exactly stated coefficient fields and linear differential relations.

The most complete positive result in this continuation is the full planar independence theorem: it removes an infinite set of missing composite indices and proves a genuinely joint assertion. The binary bottleneck arithmetic transition is a second exact classification, with a particularly short transcendence mechanism once its global optimization structure is established. These are concrete candidates for specialist review and a more focused publication project. Their journal level and historical novelty remain questions for that review; they cannot be inferred from the number of targets or pages.

Verification record. Each of the six subject sections was cross-read independently within this research session. Particular checks covered the Newton polygon and representation argument, arbitrary finite encoder alphabets, branch accessibility in the non-D-finite proof,

normalization in E-function specialization, fixed-exponential elimination, cyclic-cover ramification, and leading-term extraction with flat coefficients. Compilation and page rendering check the artifact, not the truth of its theorems. No external referee decision, machine-checked proof, or exhaustive novelty certification is claimed.

Part V
Structural Rigidity and Adaptive Recovery

Editorial integration note. The exact, adaptive, approximate, and noisy observation models used here are distinct. The finite-entropy counterexample explains why mixing does not preserve every earlier pure-saddle classification. The following part tests transfers to logic without treating recovery alone as a decision procedure.

Original part framing.

Status. Research working manuscript with complete arguments for the stated deductions and explicit external inputs. Internal proof cross-reading is not external peer review or formal verification. The targeted literature comparison does not certify global priority. Equivalent formulations and routine consequences are not counted as independent discoveries.

34 What changes when arithmetic is no longer the only objective

The 223-page foundation [96], its 447-page collected sequels [97], and the subsequent arithmetic manuscripts supply several different kinds of observations: replicated random variables, optimized value curves, adaptive scalar measurements, and morphisms of special-value rings. The preceding manuscript [98] concentrated on arithmetic rigidity. Here we ask whether those observations determine hidden structure exactly, how much information they require, and which obstructions survive changes of model.

The strongest transfers remove hypotheses or give necessary and sufficient conditions for entire families. They are not obtained by listing many special numerical constants. The six technical sections are independent enough to permit separate scrutiny; their common theme is to find an observation coordinate in which an otherwise nonlinear ambiguity becomes a rigid support, divisor, or fiber constraint.

34.1 Two principal classification results

Let (Q, R) have an arbitrary probability law on $\mathbb{R} \times (0, \infty)$, and conditionally on it let

$$Y_j = Q + R \log E_j, \qquad j = 1, 2,$$

where E_1, E_2 are independent unit exponential variables. Theorem 35.1 proves

$$\boxed{\mathcal{L}(Y_1, Y_2) \text{ uniquely determines } \mathcal{L}(Q, R).}$$

The theorem actually proves injectivity for finite complex mixing measures. There is no compact support, moment, or scale restriction. In the equivalent transform $\exp\{-B(u^\alpha + v^\alpha)\}$, it removes the earlier lower bound on α while also permitting arbitrary positive Weibull shapes. The improvement is substantive: the old triangular argument needed a first nonzero scale, whereas the new weighted contraction starts even when the support accumulates at zero. A nonzero Fourier frequency supplies the exponentially small gamma factor that makes the contraction possible.

This is population identifiability from a joint law, not estimation from one realized pair of numbers. One replicate fails by stable subordination. Nor does injectivity imply a uniformly

well-conditioned inverse: Proposition 35.3 gives latent laws a fixed distance apart whose observed laws become arbitrarily close for every fixed group size.

For the second result, let

$$S(x) = \sum_{j=1}^{N} w_j x^{r_j}, \qquad T(C) = \min_{x>0} \frac{1 + Cx}{S(x)}, \qquad w_j > 0, \quad 0 < r_1 < \cdots < r_N < 1.$$

Theorems 36.3 and 36.4 give

$$T \text{ is } D\text{-finite} \iff r_i - r_j \in \mathbb{Q} \quad \text{for all } i, j,$$
$$\operatorname{ord}_{\text{alg.diff}}(T) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{1, r_1, \ldots, r_N\} - 1.$$

Thus linear differential finiteness depends on exponent differences, whereas minimal nonlinear differential order depends on a rational rank that also includes 1. The previous two-atom theorem did not by itself handle collisions of complex zeros or repeated zeros for arbitrary N. The present proof does: Skolem–Mahler–Lech controls zero progressions, and every projected zero produces an accessible pole or branch point of the optimized value.

34.2 Exact gains and their limits

Direction	Previous restriction or question	Present conclusion
Latent mixtures	Two-replicate inversion with index bounded away from zero	Arbitrary joint mixing without cutoffs or moments; exact threshold two in the stated Weibull and centered stable models.
Finite spectra	Two-atom differential classification	Necessary and sufficient D-finite criterion and exact nonlinear order for every finite number of atoms.
Observation design	Classical moment certificates; nonlinear optimized observations	Exact jet budget $2N$, or $2N-1$ with known mass away from contact $x = 1$; sharp local square-root stability.
Entropy programs	Rational-weight elimination and special binary arithmetic transitions	A terminating existence test for algebraic optimizers in the promised smooth concave class with algebraic weights.
Mixed entropy games	Pure-saddle values lie in logarithms of positive algebraic numbers	Rational Shannon games encode finite logarithmic matrices; the pure-saddle range fails under mixing.
Special-value fields	Rational-expression intersection and one-coordinate closure	Exact Kummer algebraic closure of exponential fields, plus a joint multivariate exceptional-coordinate theorem.
Gamma extraction	Locally finite contact branches for general real order	No switches for $1/2 < \kappa \leq 1$; finitely many overall for each order; a two-order mixture has an atomic plateau and a genuine corner.
Adaptive topology	Arbitrary-accuracy recovery in Euclidean dimension	Exact recovery lies between d and $d+1$ on nonempty compact metric spaces of dimension d; closed d-manifolds require $d+1$ for $d \geq 1$.

Direction	Previous restriction or question	Present conclusion
Sparse recovery	Fixed linear measurements and stable/noisy models	Exactly $s+1$ adaptive linear queries for all s-sparse vectors in $\mathbb{R}^m$, $s<m$, with a finite final-query menu; s queries in the stated class have error at least $R/2$ on the radius-R set.

The last two conclusions concern exact real-valued observations and allow discontinuous dependence of the next query on the preceding transcript. The compact-space construction nevertheless has a Borel selection rule and decoder; the sparse construction has a semialgebraic rule. Neither result asserts numerical stability, finite precision, or a fast implementation. In fact exact and arbitrarily accurate recovery differ: on S^3, three queries have error infimum zero but cannot attain exact recovery, which requires four.

34.3 Eighteen target questions

The first twelve rows are resolved within the precise models proved below. They are not twelve externally documented famous conjectures. The last six record limits encountered during the same investigation. Multiple corollaries of one mechanism are grouped in one row.

No.	Question	Outcome
1	Can two replicates identify arbitrary joint index–scale mixing down to index zero?	Yes, Theorem 35.1 and Corollary 35.2; no atom at index zero is allowed, but accumulation there is unrestricted.
2	Which finite positive spectra produce D-finite optimized values?	Exactly the rational-gap spectra, Theorem 36.3.
3	What is the least nonlinear algebraic differential order?	Exactly rational rank minus one, Theorem 36.4.
4	How many value derivatives certify finitely many atoms against every positive continuous competitor?	Sharp normalization-dependent orders and local stability, Theorems 36.5 and 36.6.
5	Does the pure-saddle arithmetic range survive arbitrary mixed strategies?	No. A universal finite-logarithmic-game embedding gives two explicit counterexample candidates with an unconditional at-least-one conclusion, Theorems 37.1 and 37.2.
6	Can algebraic-optimum existence be decided with algebraic tradeoff weights?	Yes in the specified smooth concave rational entropy class, Theorem 37.5; no complexity bound is claimed.
7	What is the algebraic closure of an exponential field inside $\mathrm{Frac}(\mathcal{E})$?	Exactly its rational-exponent Kummer closure, with the ring intersection and integral closure, Theorem 38.3.
8	Can an algebraic extension depend jointly on Ein coordinates at $\mathbb{Q}$-linearly independent algebraic parameters $a_i \geq 1$?	Every possible nontrivial finite extension is a cyclic radical in at most one exceptional coordinate, Theorem 38.5.
9	Can low-order extraction contacts switch, and can all-order switches accumulate at endpoints?	No switches for $1/2 < \kappa \leq 1$; only finitely many for each fixed general order, Theorem 39.2 and Proposition 39.3.

No.	Question	Outcome
10	Can a richer positive mixture create a genuine extraction phase corner?	Yes. An atomic plateau ends at one corner, with an explicit rational-parameter example, Theorem 39.4 and Corollary 39.5.
11	What changes for exact adaptive recovery on compacta and closed manifolds?	$d \le Q(X) \le d+1$ for nonempty compact metric X of finite dimension d; $Q(M) = d+1$ for closed d-manifolds with $d \ge 1$, Theorems 40.1 and 40.2.
12	What is the exact adaptive linear query count for sparse vectors?	$s+1$ for $1 \le s < m$, Theorem 40.4; with $s-1$ initial linear queries and one continuous final query, radius-R worst-case error is at least $R/2$, Proposition 40.5.
13	Is two-query arbitrary-accuracy recovery possible on all of $\mathbb{R}^3$?	Unresolved. The exact cohomology obstruction does not yield a positive error lower bound; (40.5) shows why.
14	Do the full E-value unit and factorization conjectures follow?	No. A hypothetical unit can remain transcendental over the exponential field; the closure theorem supplies no missing algebraicity hypothesis.
15	Do all-index stopping-value independence results extend to every higher even dimension?	Unresolved. The unshifted Schur criterion used for the planar case does not apply to the shifted-factorial coefficients arising in the proposed extension; no replacement irreducibility theorem is established here.
16	Which of the two explicit mixed-game values lies outside the logarithmic algebraic span?	Unresolved individually. The at-least-one conclusion uses the established strong six-exponentials theorem and does not prove four exponentials.
17	Is the optimal Chebyshev–Bernstein exponent $\log 2$?	Unresolved. An apparently stronger contour shortcut is disproved exactly in Section 41; the older constant-one bound is not counted anew.
18	What are the minimal group sizes with unknown stable location or skewness?	Unresolved. Five and six remain sufficient after removing the index cutoff; minimality is not proved.

34.4 Dependencies, terminology, and priority

"Determines" refers to injectivity on the stated class, not automatically to a statistically stable or computable inverse. An L-jet includes the value and all derivatives through order L. D-finiteness means a linear differential equation with polynomial coefficients; algebraic differential order permits nonlinear equations and is a different invariant.

The principal external inputs are explicitly credited in their sections: Skolem–Mahler–Lech and Hadamard factorization; classical moment certificates and the separated-reference moment comparison of Wu–Yang; Baker, Gelfond–Schneider and Roy; real algebraic geometry; the Fischler–Rivoal morphism for E-values; and the Hurewicz, sheaf-cohomology, and van Kampen–Flores theorems. The gamma-extraction section also states the exact analytic inputs inherited from the earlier project manuscripts. No external input becomes a new theorem of this paper merely by being used in a different model.

In particular, the finite-mixture grouped-sample literature differs from the unrestricted mixing problem in Section 35, and the fixed-scale Gumbel example in Powers–Xu differs from the arbitrary joint location–scale pair theorem. Those distinctions motivate further literature review; they are not themselves proof of priority.

35 Two observations identify unrestricted Weibull and stable mixtures

The preceding identification theorem required the stable index to remain bounded away from zero. Here that restriction is removed. In fact the proof identifies an arbitrary joint location–scale mixing law for the log-exponential kernel from two conditionally independent observations. Its scale may approach both zero and infinity, and no latent moment is assumed. The equivalent Weibull formulation allows all positive shapes and rates; the stable formulation allows the entire index range $(0, 2]$.

The new mechanism is a weighted uniqueness principle for integer dilations. Ordinary recursion needs a first nonempty support interval. A nonzero Fourier frequency supplies exponential flatness at the missing endpoint, and a contraction recovers that initial interval. The Gamma estimate establishing this flatness also justifies the infinite-series exchanges at arbitrarily small shape. All identification statements concern the population *joint law* of observations sharing a latent draw, not recovery of a distribution from a single realized pair.

35.1 The unrestricted theorem

Let E_1, E_2 be independent mean-one exponential variables. For a finite complex Borel measure ρ on $\mathbb{R}\times(0,\infty)$ define the induced finite complex observation measure by mixing the conditional pair

$$Y_j = Q + R\log E_j, \qquad j = 1, 2. \tag{35.1}$$

For probability mixing laws, the latent pair (Q, R) is independent of (E_1, E_2).

Theorem 35.1 (Two observations without support or moment restrictions)**.** *The map from ρ to the joint observation measure in (35.1) is injective on all finite complex measures on $\mathbb{R}\times(0,\infty)$. Equivalently, the map*

$$\mu \longmapsto \left\{ \int e^{-B(u^\alpha+v^\alpha)}\, d\mu(\alpha, B) : u, v > 0 \right\} \tag{35.2}$$

is injective on all finite complex measures on $(0,\infty)^2$. No support cutoff, moment, finite-component assumption, or latent independence is required.

The change of coordinates between the two formulations is

$$R = \alpha^{-1}, \qquad Q = -\alpha^{-1}\log B. \tag{35.3}$$

Indeed the upper-orthant probabilities of (35.1) are $\exp\{-B(e^{\alpha x} + e^{\alpha y})\}$. Upper orthants determine finite complex measures by inclusion and exclusion and finite-variation limits at unbounded endpoints. Thus the equivalence is valid for complex measures as well as probability laws.

Corollary 35.2 (Exact replication threshold for Weibull and stable mixing)**.** *For Weibull kernels with survival function $\mathbb{P}(W > x \mid \alpha, B) = e^{-Bx^\alpha}$, $x > 0$, two conditionally independent observations identify every Borel mixing law on $(\alpha, B) \in (0,\infty)^2$. For symmetric stable kernels with characteristic function $\exp\{-B|u|^\alpha\}$, the same conclusion holds on the entire range $(\alpha, B) \in (0, 2]\times(0,\infty)$. One observation does not identify all probability mixing laws in either class. Consequently two is the exact minimal number.*

The two formulations in this corollary are consequences of the same transform theorem and are counted as one identification family. The unrestricted log-exponential statement also applies, after reflection, to ordinary Gumbel location–scale mixtures.

Proposition 35.3 (No uniform inverse stability on the unrestricted class)**.** *For any fixed group size $k \geq 1$, there are two sequences of probability mixing laws separated by a fixed bounded-Lipschitz distance, whose induced k-observation laws converge to one another in total variation. Thus exact identification does not provide a uniform inverse modulus on the unrestricted parameter space.*

Proof. Use the mixing point masses $\rho_R = \delta_{(0,R)}$ and $\widetilde{\rho}_R = \delta_{(1,R)}$, with $R \to \infty$. Equip the parameter space with the bounded metric

$$d((Q,R),(Q',R')) = \min\{1, |Q-Q'| + |\log R - \log R'|\}.$$

The function $h(Q,R) = \max\{0, \min\{1, Q\}\}$ is bounded by one and is one-Lipschitz for this metric. Its integrals under the two mixing laws differ by one.

The density of $\log E$ is $g(x) = e^{x-e^x}$ and

$$\int_{\mathbb{R}} |g'(x)|\,dx = \int_0^\infty |1-u|e^{-u}\,du = \frac{2}{e}.$$

The one-observation laws under the two mixing point masses are translations by one at scale R, so their total-variation distance is at most $1/(eR)$. The telescoping product bound gives distance at most $k/(eR)$ for the joint k-observation laws. This tends to zero while the latent bounded-Lipschitz separation remains at least one. □

35.2 A weighted principle for dilation inversion

Lemma 35.4 (Starting a dilation recursion at an accumulation point)**.** *Let η be a locally finite complex measure on $(0,\infty)$, with finite variation on $(0,L]$ for every $L < \infty$. Let c_m be Borel functions, $c_1 = 1$, and set $D_m(\alpha) = m\alpha$. Suppose the series*

$$M = \sum_{m\geq 1} (D_m)_*(c_m\eta) \tag{35.4}$$

converges in total variation after multiplication by e^{-dx} for every $d > 0$, and $\int e^{-dx}\,dM(x) = 0$ for every $d > 0$. Suppose also that there exist $\varepsilon > 0$, a positive Borel weight w, and $\kappa < 1$ such that

$$\int_{(0,\varepsilon]} \frac{d|\eta|(\alpha)}{w(\alpha)} < \infty, \tag{35.5}$$

$$\sum_{m\geq 2} \sup_{0<\alpha\leq\varepsilon/m} |c_m(\alpha)|\frac{w(\alpha)}{w(m\alpha)} \leq \kappa. \tag{35.6}$$

Then $\eta = 0$.

Proof. First $M = 0$ by uniqueness of the Laplace transform. To see this without a finiteness assumption on the unweighted measure, fix $d_0 > 0$, multiply by $e^{-d_0 x}$, and push forward by $y = e^{-x}$. This is a finite complex measure on $[0,1]$ with all nonnegative integer moments zero. Polynomial density gives the zero measure. The positive tilt is invertible on bounded intervals, proving $M = 0$.

On $(0,\varepsilon]$, divide (35.4) by w. Writing $\sigma = \eta/w$ there, the total variation of the sum of terms with $m \geq 2$ is at most $\kappa\|\sigma\|$ by (35.6). That same bound justifies the weighted series exchange. Since the term with $m = 1$ is σ, the equation $M = 0$ implies $\|\sigma\| \leq \kappa\|\sigma\|$, and hence $\eta = 0$ on $(0,\varepsilon]$.

If $\eta = 0$ on $(0,L]$, all terms with $m \geq 2$ contributing to $(0,2L]$ originate in $(0,L]$ and vanish. Thus $M = 0$ implies $\eta = 0$ on $(0,2L]$. Iteration covers $(0,\infty)$. □

The endpoint hypothesis is material. The lemma does not assert that an arbitrary integer-dilation equation is uniquely solvable without control on mass accumulating at zero.

35.3 The Gamma estimate that supplies the endpoint control

Lemma 35.5 (A globally integrable series envelope)**.** *For every real $t \neq 0$ and $d > 0$, the function*

$$S_{t,d}(\alpha) = \alpha \sum_{m \geq 1} \frac{|\Gamma(m + 1 + it/\alpha)|}{(m-1)!} e^{-m\alpha d} \tag{35.7}$$

is bounded on the entire interval $(0, \infty)$. Moreover, for fixed $0 < \theta_0 < \pi/2$,

$$|\Gamma(2 + it/\alpha)| \leq (\cos\theta_0)^{-2} e^{-|t|\theta_0/\alpha}. \tag{35.8}$$

Proof. Rotation of Euler's Gamma integral into a ray of argument $\operatorname{sgn}(T)\theta$ gives, for an integer $k \geq 1$,

$$|\Gamma(k + iT)| \leq \Gamma(k)(\cos\theta)^{-k} e^{-|T|\theta}, \qquad 0 < \theta < \pi/2. \tag{35.9}$$

Indeed the rotated integrand has the factor $e^{i\theta \operatorname{sgn}(T)(k+iT)}$; its modulus contributes $e^{-|T|\theta}$. The radial integral is bounded by $\int_0^\infty e^{-x\cos\theta} x^{k-1}\, dx$. Contour rotation is justified by exponential decay on the large arc and the vanishing small-arc contribution for $k > 0$. Taking $k = 2$ proves (35.8).

If $\alpha d \leq 1$, choose $\theta = \sqrt{\alpha d}/2 \leq 1/2$. The elementary inequality $-\log\cos\theta \leq \theta^2$ on this interval gives

$$\frac{e^{-\alpha d}}{\cos\theta} \leq e^{-3\alpha d/4}.$$

Using (35.9) with $k = m + 1$ and $\Gamma(m + 1)/(m - 1)! = m$, we obtain

$$\begin{aligned} S_{t,d}(\alpha) &\leq C\alpha e^{-|t|\sqrt{d}/(2\sqrt{\alpha})} \sum_{m \geq 1} m e^{-3m\alpha d/4} \\ &\leq C_d \alpha^{-1} e^{-|t|\sqrt{d}/(2\sqrt{\alpha})}. \end{aligned}$$

This is bounded down to $\alpha = 0$. If $\alpha d \geq 1$, Euler's integral without rotation gives $|\Gamma(m + 1 + it/\alpha)| \leq m!$, and hence

$$S_{t,d}(\alpha) \leq \frac{\alpha e^{-\alpha d}}{(1 - e^{-\alpha d})^2}.$$

This is bounded over the remaining range and tends to zero as $\alpha \to \infty$. □

The nonzero frequency in this lemma is essential to the displayed bound at zero. The proof below recovers the missing zero Fourier frequency only at the end, by continuity.

35.4 Proof of unrestricted two-observation identification

Proof of Theorem 35.1. Let ρ be a finite complex mixing measure inducing the zero joint observation measure in (35.1). Fix $t \in \mathbb{R} \setminus \{0\}$ and define the finite complex measure

$$\nu_t(dr) = \int e^{itq}\, \rho(dq, dr).$$

Fourier-weight the first observed coordinate by e^{itY_1} and project to $D = Y_2 - Y_1$. The resulting zero measure has density

$$T_t(d) = \int \frac{\Gamma(2 + itr)}{r} \frac{e^{d/r}}{(1 + e^{d/r})^{2+itr}}\, d\nu_t(r). \tag{35.10}$$

To derive it, write $E_1 = z$, $E_2 = ze^{d/r}$ in the two exponential densities. The Jacobian is $ze^{d/r}/r$, and the Fourier weight contributes $e^{itq}z^{itr}$. Integration in $z > 0$ gives (35.10).

Its absolute integrand is bounded by the unweighted conditional difference density, because $|\Gamma(2 + itr)| \le 1$. This also justifies the density calculation for finite complex mixing measures by Fubini. On a compact subinterval of $d > 0$, the same integrand is bounded uniformly over all $r > 0$ by $\alpha e^{-\alpha d}$, where $\alpha = 1/r$. Thus T_t is continuous for positive d and, being a zero density, vanishes at every such d.

Push ν_t forward by $\alpha = 1/r$, calling the resulting finite complex measure λ_t. The binomial expansion at $e^{-\alpha d}$ gives

$$e^{itd}T_t(d) = \sum_{m\ge1} \int \alpha \frac{(-1)^{m-1}\Gamma(m + 1 + it/\alpha)}{(m-1)!} e^{-m\alpha d}\, d\lambda_t(\alpha), \qquad d > 0. \tag{35.11}$$

The factor e^{-itd} cancels exactly in this expansion. Lemma 35.5 bounds the sum of absolute integrands uniformly in α, so the exchange of series and integration is valid without any shape or scale moment.

Define

$$d\eta_t(\alpha) = \alpha\Gamma(2 + it/\alpha)\, d\lambda_t(\alpha), \tag{35.12}$$

$$c_m(\alpha) = \frac{(-1)^{m-1}(2 + it/\alpha)_{m-1}}{(m-1)!}. \tag{35.13}$$

The measure η_t is finite on every $(0, L]$ by (35.8). Its dilation series (35.4) has finite exponentially tilted total variation by the same envelope used for (35.11), and its Laplace transform vanishes.

To verify the contraction at zero, choose $0 < c < |t|\theta_0$ for some fixed $0 < \theta_0 < \pi/2$, and put $w(\alpha) = e^{-c/\alpha}$. Equation (35.8) gives

$$\frac{\alpha|\Gamma(2 + it/\alpha)|}{w(\alpha)} \le (\cos\theta_0)^{-2}\alpha e^{-(|t|\theta_0 - c)/\alpha}.$$

This is bounded on $(0, \varepsilon]$, so (35.5) holds for every fixed $\varepsilon > 0$.

For $m \ge 2$, put $\beta = m\alpha$. Each factor in the Pochhammer symbol in (35.13) has modulus at most $m(1 + |t|/\beta)$. Using $m^{m-1}/(m-1)! \le e^m$ gives

$$|c_m(\alpha)|\frac{w(\alpha)}{w(m\alpha)} \le e\left[e(1 + |t|/\beta)e^{-c/\beta}\right]^{m-1}. \tag{35.14}$$

The quantity

$$r_\varepsilon = \sup_{0<\beta\le\varepsilon} e(1 + |t|/\beta)e^{-c/\beta}$$

tends to zero as $\varepsilon \downarrow 0$. Choose ε so small that $er_\varepsilon/(1 - r_\varepsilon) < 1$. Summing (35.14) proves (35.6). Lemma 35.4 now gives $\eta_t = 0$ on $(0, \infty)$.

The multiplier in (35.12) never vanishes. Its inverse is bounded on compact subintervals of $(0, \infty)$, so $\lambda_t = 0$ and then $\nu_t = 0$. This holds for every nonzero t. For each Borel set A of scale coordinates, dominated convergence shows that $t \mapsto \int_{\mathbb{R}\times A} e^{itq}\, d\rho(q, r)$ is continuous. It is therefore zero also at $t = 0$. Fourier uniqueness gives $\rho(C \times A) = 0$ for all Borel $C \subseteq \mathbb{R}$, $A \subseteq (0, \infty)$, proving $\rho = 0$. The coordinate change (35.3) proves the equivalent formulation. □

Proof of Corollary 35.2. The Weibull pair has joint survival $\int e^{-B(u^\alpha + v^\alpha)}\, d\mu$. The symmetric stable pair has that same expression as its joint characteristic function when $u, v > 0$. Theorem 35.1 proves both positive assertions.

For minimality choose $0 < a < b$ and a positive stable random variable S of index a/b, whose Laplace transform is $\mathbb{E}e^{-sS} = e^{-s^{a/b}}$. This classical law exists with Lévy measure

$$\frac{a/b}{\Gamma(1-a/b)} x^{-1-a/b}\, dx;$$

see also [101]. It is strictly positive because its Laplace transform tends to zero at infinity. The identity

$$e^{-u^a} = \mathbb{E}e^{-Su^b}, \qquad u > 0,$$

identifies the one-observation laws of the distinct mixing measures concentrated respectively at $(a, 1)$ and at (b, S). It proves failure for Weibull mixing; choosing $a < b \leq 2$ proves failure for stable mixing. □

35.5 Unrestricted stable location and skewness: inherited bounds

We record the consequence for correlated latent coordinates, while keeping its observation count separate from the exact threshold in Corollary 35.2. Use the stable parametrization

$$\widehat{K}_{A,\alpha,B,\beta}(t) = \exp\{itA - B|t|^\alpha + iB\beta\psi_\alpha(t)\}, \tag{35.15}$$

where $A \in \mathbb{R}$, $B > 0$, $0 < \alpha \leq 2$, $|\beta| \leq 1$, and

$$\psi_\alpha(t) = \begin{cases} \tan(\pi\alpha/2)\,\operatorname{sgn}(t)|t|^\alpha, & \alpha \notin \{1, 2\}, \\ -(2/\pi)t\log|t|, & \alpha = 1, \\ 0, & \alpha = 2. \end{cases}$$

Set $\psi_\alpha(0) = 0$ and impose $\beta = 0$ at $\alpha = 2$ to remove the unobservable Gaussian skewness coordinate.

Corollary 35.6 (Finite replication with unrestricted stable parameters)**.** *For arbitrary Borel joint mixing of (A, α, B) in the symmetric case $\beta = 0$, the joint law of five conditionally independent observations identifies the mixing law. For arbitrary Borel joint mixing of all four parameters in (35.15), six observations suffice. No moments, index cutoff, or latent independence is assumed. These are sufficient bounds, not assertions of minimality.*

Proof. The contrast-and-anchor construction is the one in the preceding paper [103]. Its only former index restriction came from the two-observation theorem, now replaced by Theorem 35.1. We give the steps to check that no new endpoint assumption enters.

In the symmetric case retain X_1 as an anchor and use $D_1 = X_2 - X_3$, $D_2 = X_4 - X_5$. Conditionally on the latent variables, these contrasts are independent symmetric stable observations with exponent coefficient $2B$, independent of the anchor. For a difference ρ of two candidate mixing laws and fixed real t, the anchor weight $e^{itA-B|t|^\alpha}$ defines a finite complex measure on (α, B). The joint transform with (D_1, D_2) places it in the kernel of (35.2); hence it is zero. The real multiplier $e^{-B|t|^\alpha}$ can be removed on bounded B intervals and then by exhaustion. Fourier uniqueness in A, after restriction to each Borel set of (α, B), proves the five-observation assertion.

For six observations retain X_1, X_2 and use $D_1 = X_3 - X_4$, $D_2 = X_5 - X_6$. Oddness of ψ_α cancels both location and skewness in the contrasts. Applying Theorem 35.1 for each pair of anchor frequencies (p, q) and removing the positive real multiplier gives, after disintegration with respect to (α, B),

$$\int e^{i(p+q)A + iB\beta(\psi_\alpha(p)+\psi_\alpha(q))}\, d\rho_{\alpha,B}(A, \beta) = 0. \tag{35.16}$$

The identities hold simultaneously outside one null set: first use countably many rational pairs (p, q), then continuity in frequencies to obtain all real pairs.

Fix such an (α, B) with $\alpha < 2$ and a nonzero sum $u = p + q$. It suffices to inspect $u > 0$; reflection of both frequencies gives the same assertion for $u < 0$. As p varies between 0 and u, the value $B[\psi_\alpha(p) + \psi_\alpha(u - p)]$ ranges over a nondegenerate interval. For $\alpha \neq 1$ this follows from the nonvanishing second derivative of $p^\alpha + (u - p)^\alpha$ on the appropriate same-sign interval. For $\alpha = 1$ it follows from the strict convexity of $p \log p + (u - p) \log(u - p)$. For each fixed u, the Fourier transform in β of $e^{iuA}\rho_{\alpha,B}$ is entire, since $|\beta| \leq 1$. By (35.16) it vanishes on that interval and therefore everywhere. Continuity supplies $u = 0$ and Fourier uniqueness in (A, β) gives $\rho_{\alpha,B} = 0$. At $\alpha = 2$ the imposed condition $\beta = 0$ reduces the argument directly to Fourier uniqueness in A. □

Relation to prior results and remaining questions. The compact-index two-observation theorem and the contrast-and-anchor architecture were proved in the preceding project manuscript [103]. The added result here is the removal of both shape endpoints for log-exponential/Weibull mixing, including the formerly inaccessible accumulation at stable index zero. Lemma 35.4 isolates the reusable mechanism. The one-observation subordination identity is classical.

Grouped finite-mixture identification, such as [99, 100], and continuous-mixture transform transfers, such as [102], provide relevant prior frameworks. The present class permits arbitrary continuous mixing of all kernel coordinates and is not inferred from a bound on the number of mixture components. A primary-source search did not verify an earlier theorem of this exact unrestricted scope; it does not establish historical priority or the closure of a named external conjecture.

Minimal replication with unknown stable location or skewness remains unresolved here. So do quantitative stability and estimation from finitely many noisy observations. Separately investigated higher-dimensional stopping-polynomial irreducibility also remains unresolved: its shifted-factorial coefficients do not satisfy the planar Schur theorem's hypotheses. These questions are recorded in the research ledger rather than counted as additional theorems.

36 Finite spectra: differential complexity and robust certification

Let μ be a nonzero finite positive Borel measure supported on a fixed interval $[a, b] \subset (0, 1)$, and set

$$S_\mu(x) = \int_a^b x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}, \qquad C > 0. \tag{36.1}$$

For arbitrary finite spectra, we classify linear differential finiteness and determine the least nonlinear algebraic differential order exactly. Against arbitrary positive continuous mixtures, we give finite-jet certification and a sharp local stability bound.

The arithmetic-progression input below is the classical Skolem–Mahler–Lech theorem [104, 105]. Moment certificates and polynomial interpolation are classical; Wu and Yang [106, Lemma 4 and Proposition 5] give moment identifiability and, at separated reference atoms, a square-root comparison against unrestricted measures. Our assertions concern the nonlinear optimal-value observation, including its differential classification and the conversion of its jets into moment certificates. They do not assert a new general theorem on moment problems.

36.1 Observation derivatives as spectral moments

Write $x = x(C)$ for the contact, $H = S - xS'$, and $B = T - CT'$. Differentiation under the integral gives $S' > 0$, $S'' < 0$, and $H > 0$. The objective diverges at both ends of the positive axis, and $(S'/H)' = S''S/H^2 < 0$. Thus the contact is unique and analytic. The identities needed throughout are

$$C = \frac{S'}{H}, \qquad T = \frac{1}{H}, \qquad T' = \frac{x}{S}, \qquad B = \frac{1}{S}, \qquad x = \frac{T'}{B}, \tag{36.2}$$

$$x' = \frac{TT''}{B^2} < 0, \qquad S' = \frac{C}{T}, \qquad S'' = \frac{B^3}{T^3 T''}. \tag{36.3}$$

Primes on S and T refer to their respective variables. Define an operator acting on observed functions by

$$\mathcal{L}_T = \frac{T'B}{TT''}\frac{d}{dC}, \qquad m_k = \mathcal{L}_T^k(B^{-1}). \tag{36.4}$$

It is exactly $x\,d/dx$ in the contact coordinate, so

$$m_k(C) = \int r^k x(C)^r \, d\mu(r). \tag{36.5}$$

Lemma 36.1 (A finite-jet change of coordinates)**.** *For $L \geq 2$, the L-jet of T at C_0 determines $x_0 = x(C_0)$ and the moments $m_0, \dots, m_L$ by rational formulas. The first jet already determines x_0, m_0, m_1. Conversely, x_0 and $m_0, \dots, m_L$, with the stationary relation at C_0, determine the same L-jet, locally analytically. All these maps are locally Lipschitz near every positive spectrum.*

Proof. The apparent extra derivative in $\mathcal{L}_T$ cancels on its first application:

$$m_0 = B^{-1}, \qquad m_1 = \frac{CT'}{TB}.$$

Applying $\mathcal{L}_T$ once more uses only the second jet, and each further application increases the required order by one. The denominators T, B, T'' are nonzero.

For the converse, the first two moments give S, S' at x_0. Ordinary derivatives of S through order L are invertible triangular linear combinations of the Euler derivatives $m_0, \dots, m_L$. The formulas above recover T, T', T'', with $T'' = B^3/(T^3 S'')$. Successively differentiating gives the rest: $d/dC = (H^2/(S''S))d/dx$, and after the cancellation in T'', each application increases the required S-order by one. These are rational expressions with nonzero denominators. This also proves local Lipschitz continuity. □

36.2 A classification for every finite positive spectrum

An exponential sum below has constant coefficients, not polynomial coefficients. A germ is called D-finite if it satisfies a nonzero linear differential equation with polynomial coefficients.

Lemma 36.2 (Finitely many projected zeros)**.** *Let $E(t) = \sum_{j=1}^{N} c_j e^{r_j t}$, with distinct real r_j and nonzero complex c_j. If $\{e^t : E(t) = 0\}$ is finite, then $r_j - r_1 \in \mathbb{Q}$ for every j.*

Proof. For $N = 1$ the assertion is immediate. No zero for $N \geq 2$ has multiplicity exceeding $N - 1$: the first N vanishing derivatives would give a nonsingular Vandermonde system for $c_j e^{r_j t}$. Choose logarithms τ_h of the finitely many projected zeros. All zeros lie among $\tau_h + 2\pi i n$, $n \in \mathbb{Z}$. For each h and $0 \leq k \leq N - 2$, the sequence $E^{(k)}(\tau_h + 2\pi i n)$ is a linear recurrence over $\mathbb{C}$. Skolem–Mahler–Lech makes its zero set a finite union of complete arithmetic progressions,

augmented by a finite set. The two-sided version follows from [105, Theorem 1.6], or by applying the ordinary theorem in both directions: a zero progression on a ray extends to the whole progression by Vandermonde independence of its distinct characteristic roots.

Take a common period Q for these finitely many progressions. On each $\tau_h + 2\pi i(\ell + Q\mathbb{Z})$ the zero multiplicity has a constant baseline $d_{h\ell} \in \{0, \dots, N-1\}$, with only finitely many exceptions, all increases. The baseline counts the successive initial derivatives that vanish on the entire progression.

Division of this divisor requires no factorization conjecture. If a real exponential sum F vanishes on $t_0 + 2\pi i Q\mathbb{Z}$, group its exponents modulo $(1/Q)\mathbb{Z}$:

$$F(t) = \sum_{\gamma} e^{\gamma t} P_\gamma(e^{t/Q}),$$

where P_γ are Laurent polynomials. On the progression, $e^{t/Q} = y_0 = e^{t_0/Q}$ is constant and the characteristic numbers $e^{2\pi i Q\gamma}$ are distinct. Vandermonde independence implies $P_\gamma(y_0) = 0$ for every γ. Division by $e^{t/Q} - y_0$ therefore leaves a finite real exponential sum. Repeat this operation for all baseline multiplicities. The roots $e^{(\tau_h + 2\pi i\ell)/Q}$, $0 \le \ell < Q$, are distinct for distinct (h, ℓ). We obtain

$$E(t) = A(e^{t/Q})R(t),$$

where A is a polynomial and R a finite real exponential sum with finitely many zeros. Hadamard factorization gives $R(t) = e^{\alpha t+\beta}P(t)$ with a polynomial P, since R has order at most one. If ρ is its largest real exponent, real-axis asymptotics give $R'/R \to \rho$, hence $\alpha = \rho$. Also $e^{-\rho t}R(t)$ has a finite nonzero limit as $t \to +\infty$, forcing P constant. Thus $E(t) = e^{\rho t+\beta}A(e^{t/Q})$. Independence of distinct exponentials puts all original exponents in $\rho + (1/Q)\mathbb{Z}$, proving the claim. □

Theorem 36.3 (Exact linear differential classification)**.** *Suppose*

$$S(x) = \sum_{j=1}^{N} w_j x^{r_j}, \qquad 0 < r_1 < \cdots < r_N < 1, \qquad w_j > 0.$$

The optimized-value germ $T(C)$ is D-finite over $\mathbb{C}(C)$ if and only if every gap $r_j - r_1$ is rational. If some gap is irrational, analytic continuations of the positive real value germ have infinitely many distinct finite singularities. In the rational-gap case the contact $x(C)$ is algebraic and the logarithmic derivative T'/T is algebraic.

Proof. First write $r_j = r_1 + k_j/q$ in the rational-gap case, with nonnegative integers k_j and $q \ge 1$. Put

$$z = x^{1/q}, \quad P(z) = \sum_j w_j z^{k_j}, \quad R(z) = r_1 P(z) + q^{-1} z P'(z), \quad Q(z) = P(z) - R(z).$$

Stationarity and the value formula become

$$C z^q Q(z) - R(z) = 0, \qquad T = \frac{z^{-qr_1}}{Q(z)}. \tag{36.6}$$

The polynomial equation is nonzero, so $z(C)$ and $x(C)$ are algebraic. Logarithmic differentiation puts T'/T in the finite algebraic field $\mathbb{C}(C, z)$, which is closed under differentiation. Writing $T^{(k)} = R_k T$ in that field and using its finite dimension gives a linear differential equation over $\mathbb{C}(C)$. Clearing denominators proves D-finiteness.

Suppose now that some gap is irrational. Lemma 36.2 gives infinitely many distinct projected zeros $x_0 = e^{t_0}$ of $S(e^t)$. Near any such t_0, $x = e^t$ is a valid coordinate and the

corresponding branch of S is analytic. Write its zero multiplicity as m and let $\Phi = S'/H$, $\Psi = 1/H$. For $m \geq 2$, putting $h = x - x_0$ gives

$$\Phi(x) = -\frac{1}{x_0} + \frac{1 - 1/m}{x_0^2} h + O(h^2).$$

Its derivative is nonzero, while H vanishes to order $m-1$. Thus the value has a pole of order $m-1$ at $c_0 = -1/x_0$. For $m = 1$, $H(x_0) \neq 0$. If S'' vanishes to order $k \geq 0$ there, the identities

$$\Phi' = \frac{S''S}{H^2}, \qquad \Psi' = \frac{xS''}{H^2}$$

show that $\Phi - c_0$ and $\Psi - \Psi(x_0)$ have orders $k+2$ and $k+1$. The order k is finite since S'' is not identically zero. A holomorphic function of Φ would have order divisible by $k+2$, so the value is ramified at c_0. Distinct projected zeros give distinct c_0.

These branches are accessible from the real germ. If T satisfied a linear equation, let Σ be the finite zero set of its leading polynomial coefficient. Choose $c_0 \notin \Sigma$ and a positive initial setting outside Σ. In the t-plane remove the poles of $\Phi(e^t), \Psi(e^t)$, the zeros of $d\Phi(e^t)/dt$, and the inverse image of $\Sigma \cup \{c_0\}$. This is a locally finite discrete set. Its complement is path connected, so a path from the real contact can approach t_0 with only its endpoint removed. Local inversion along the path continues the real value to the pole or ramified branch just found. Its image avoids Σ. A solution of a linear differential equation is holomorphic at every ordinary point, including all continuations to c_0. This contradiction proves the theorem. □

Repeated zeros are essential here: they give poles rather than the branch points used in the two-atom calculation. We do not classify D-finiteness of the contact in the irrational-gap case, because the contact is locally regular at these repeated zeros.

36.3 The exact nonlinear differential order

Algebraic differential order is the least $k \geq 0$ for which a nonzero polynomial over $\mathbb{C}(C)$ vanishes at $T, T', \dots, T^{(k)}$.

Theorem 36.4 (Rational rank equals differential complexity)**.** *For the finite spectrum of Theorem 36.3, put*

$$R = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{1, r_1, \dots, r_N\}.$$

The least algebraic differential order of T is $R-1$, and

$$\mathbb{C}(C, T, T', T'', \dots) = \mathbb{C}(x, x^{r_1}, \dots, x^{r_N}). \tag{36.7}$$

An explicit equation, of order at most N, is

$$\prod_{j=1}^{N} (\mathcal{L}_T - r_j)(B^{-1}) = 0. \tag{36.8}$$

Conversely, an analytic T on a connected positive interval satisfying $T, T', B > 0$, $T'' < 0$, and (36.8) comes from these exponents with positive fixed weights if and only if the system

$$\sum_{j=1}^{N} A_j r_j^k = m_k, \qquad 0 \leq k < N, \tag{36.9}$$

has strictly positive entries A_j at one setting.

Proof. Let $K = \mathbb{C}(x, x^{r_1}, \dots, x^{r_N})$, a field of meromorphic germs in $t = \log x$. It is stable under $D = d/dt$. Both C, T belong to K, and $d/dC = (DC)^{-1}D$ preserves K. Thus the left field of (36.7) lies in K. Conversely, that field contains $x = T'/B$, $S = B^{-1}$, and all $D^k S = \mathcal{L}_T^k(B^{-1})$. The Vandermonde system (36.9) recovers every $w_j x^{r_j}$ and proves equality.

The transcendence degree of K over $\mathbb{C}$ is R. Indeed, exponentials with rationally independent exponents are algebraically independent as functions: a polynomial relation would be a linear relation among exponentials with distinct exponents, contradicted by a Vandermonde matrix. Clearing denominators in a rational exponent basis changes the field only by algebraic extensions. Since C is nonconstant, the relative transcendence degree over $\mathbb{C}(C)$ is $R - 1$.

Let k be the first order at which the jet is algebraically dependent; such an order exists by finite transcendence degree. Then $T, \dots, T^{(k-1)}$ are algebraically independent and $T^{(k)}$ is algebraic over their field. Its minimal polynomial has nonzero separant in characteristic zero. Differentiating that polynomial expresses $T^{(k+1)}$ rationally in $\mathbb{C}(C, T, \dots, T^{(k)})$. Repetition puts every later derivative in the same field, whose transcendence degree is k. Hence $k = R - 1$. This argument also covers $k = 0$.

The Euler equation $\prod_j (D - r_j)S = 0$ gives (36.8). Lemma 36.1 bounds its order by N after denominators are cleared. It is not a formal zero: a single power with exponent different from all r_j produces an admissible curve on which its left side is nonzero.

For the converse, define $x = T'/B$ and $S(x(C)) = B^{-1}$. The inequalities make x locally invertible and give (36.3). The equation becomes $\prod_j (d/dt - r_j)S(e^t) = 0$. Thus $S = \sum_j w_j x^{r_j}$ with real constants w_j. The system (36.9) gives $A_j = w_j x^{r_j}$, so positivity at one point is precisely $w_j > 0$ for every j. The contact identities then give the unique global minimum. The constants agree on overlapping intervals. □

Adding atoms need not increase differential order: the controlling quantity is the rational rank together with 1. This statement concerns functions, not algebraic independence of their values at one specified setting.

36.4 Exact atomic certification against continuous mixtures

The competitors in this subsection are all finite positive measures on $[a, b]$; they need not have a finite support. An L-jet includes the value and derivatives through order L.

Theorem 36.5 (Exact jet budget, including normalization)**.** *Fix $\mu_0 = \sum_{j=1}^N w_j \delta_{r_j}$ with $a < r_1 < \cdots < r_N < b$, $w_j > 0$, a setting $C_0 > 0$, and contact x_0. Among all positive measures in (36.1), the $2N$-jet at C_0 uniquely determines μ_0, including its atom count. No smaller jet certifies it if total mass is unrestricted.*

If total mass is specified in advance, the exact minimum order for this certification is

$$\begin{cases} 2N, & x_0 = 1, \\ 2N - 1, & x_0 \neq 1. \end{cases} \tag{36.10}$$

These are pointwise assertions at every reference spectrum satisfying the stated interior-node conditions.

Proof. For a supplied $2N$-jet, Lemma 36.1 recovers x_0 and the moments of the tilted finite measure $\rho(dr) = x_0^r \, d\mu(r)$ through order $2N$. Put $q(r) = \prod_j (r - r_j)^2$. At μ_0 its integral is zero; any competing measure with the same moments also has $\int q \, d\rho = 0$. Positivity forces support on the same N nodes. The first N moments give the weights by Vandermonde inversion. Multiplication by x_0^{-r} recovers μ_0 itself.

This is constructive even when the nodes are initially unknown. Let $M_s = (m_{i+j})_{0\le i,j\le s}$. The conditions $M_{N-1} > 0$ and $\det M_N = 0$ certify exactly N atoms for an actually positive representing measure. Indeed, solve for the unique monic degree-N polynomial p orthogonal to $1, r, \dots, r^{N-1}$ using M_{N-1}. The zero Schur complement gives $\int p^2\, d\rho = 0$. Thus its roots contain the support, while $M_{N-1} > 0$ forces at least N support points. These roots and Vandermonde weights recover the spectrum.

To prove sharpness without known mass, write the reference tilted weights as $A_j = w_j x_0^{r_j}$. The map

$$(A_1, \dots, A_N, r_1, \dots, r_N) \longmapsto \left(\sum_j A_j r_j^k \right)_{k=0}^{2N-1}$$

has invertible Jacobian. A left null vector would be a polynomial of degree at most $2N-1$ vanishing with its derivative at all N distinct nodes, hence the zero polynomial. Choose a new node $z \in (a,b)$, different from all r_j, and give it tilted weight $\varepsilon > 0$. The implicit function theorem adjusts the old nodes and weights by $O(\varepsilon)$ so that all moments through order $2N-1$ stay unchanged. They remain positive and interior. Untilt by x_0^{-r}. The stationary relation at C_0 is unchanged, so x_0 is still the contact, and Lemma 36.1 gives identical $(2N-1)$-jets. The competitor has $N+1$ distinct atoms. When $x_0 = 1$, the zeroth moment is total mass, so this construction already proves sharpness with mass fixed.

Now assume mass is fixed and $t = \log x_0 \neq 0$. Equality of $(2N-1)$-jets supplies equal tilted moments through degree $2N-1$. Let P be the degree-at-most-$2N-1$ Hermite interpolant to $f(r) = e^{-tr}$, matching both value and derivative at the N reference nodes. The real interpolation remainder gives

$$f(r) - P(r) = \frac{t^{2N} e^{-t\xi_r}}{(2N)!} \prod_j (r - r_j)^2 \tag{36.11}$$

for some $\xi_r \in [a,b]$ when r is not a node. It is strictly positive off the nodes. Equal mass is exactly equality of $\int f\, d\rho$; the polynomial integrals also agree. The reference remainder integral is zero, so positivity forces the competitor onto the same nodes.

For sharpness at $2N-2$ when $N \ge 2$, apply the added-atom construction to the $2N$ functions

$$1,\ e^{tr},\ re^{tr}, \dots, r^{2N-2} e^{tr}$$

using the original, untilted weights. Its square Jacobian is invertible. Otherwise a nonzero function $c + e^{tr} P_{2N-2}(r)$ would have N double zeros. After multiplication by e^{-tr}, repeated Rolle's theorem would force its derivative of order $2N-1$ to have a zero. That derivative is $c(-t)^{2N-1} e^{-tr}$, impossible if $c \neq 0$; if $c = 0$, the polynomial degree gives the contradiction. The new measure has the same mass and the same tilted moments through $2N-2$, and hence the same jet of that order.

For $N = 1$, only the failure of the zeroth jet needs separate verification. Split its atom into equal weights at $r + s - h$ and $r + s + h$, preserving the prescribed mass. The optimum is analytic in s, h near zero. Its derivative in s at zero equals $-T(C_0) \log x_0 \neq 0$ by the envelope formula. The implicit function theorem gives $s(h) = O(h^2)$ preserving the single observed value. For $h > 0$ small this is a distinct positive two-atom competitor. □

The reduction in (36.10) is an effect of observation geometry: away from $x_0 = 1$, a known total mass supplies a nonpolynomial tilted moment. At $x_0 = 1$ it merely repeats m_0.

36.5 A sharp stability certificate with no atom bound on competitors

For a finite nonzero measure, let $\bar{\mu} = \mu/\mu([a,b])$. For a fixed C_0 and integer L, define

$$d_L(\mu,\nu) = \max_{0\le k\le L} |T_\mu^{(k)}(C_0) - T_\nu^{(k)}(C_0)|.$$

These are derivative observations; bounded errors in function values alone do not supply derivative errors without a further argument.

Theorem 36.6 (Sharp local square-root stability)**.** *Fix the reference spectrum and setting of Theorem 36.5. There are $K, \delta_0 > 0$, depending on that reference and $[a,b]$, such that every finite positive competing measure ν with $\delta = d_{2N}(\mu_0,\nu) \le \delta_0$ satisfies*

$$W_1(\bar{\mu}_0,\bar{\nu}) \le K\sqrt{\delta}. \tag{36.12}$$

No support-cardinality bound or regularity condition on ν is used. The exponent $1/2$ cannot be improved, even for competitors having the same total mass as μ_0.

More precisely, let $\pi_\nu(dr) = x_\nu^r\, d\nu(r)/S_\nu(x_\nu)$ be its normalized tilted measure at C_0, and let $R_0 = \{r_1,\dots,r_N\}$. Partition $[a,b]$ into nearest-node cells V_j, resolving ties arbitrarily. Then

$$\int \operatorname{dist}(r,R_0)^2\, d\pi_\nu(r) \le K\delta, \tag{36.13}$$

$$|\pi_\nu(V_j) - \pi_{\mu_0}(\{r_j\})| \le K\delta. \tag{36.14}$$

Proof. The rational formulas of Lemma 36.1, on a compact neighborhood of the reference jet, give $|x_\nu - x_0| \le K\delta$ and

$$\left|\int r^k\, d(\pi_\nu - \pi_{\mu_0})\right| \le K\delta, \qquad 0 \le k \le 2N.$$

The constants here are uniform over all competitors whose jets belong to that neighborhood. For $q(r) = \prod_j (r-r_j)^2$, it follows that $0 \le \int q\, d\pi_\nu \le K\delta$. Distinct fixed nodes on a compact interval give $q(r) \ge c\,\operatorname{dist}(r,R_0)^2$: relative to a nearest node, every other factor has distance at least half the minimum node separation. For $N = 1$ the inequality is an equality. This proves (36.13).

Let h_j be the degree-at-most-$2N-1$ Hermite polynomial with $h_j(r_i) = \mathbf{1}_{\{i=j\}}$ and $h_j'(r_i) = 0$. Taylor's theorem on each cell gives

$$|h_j(r) - \mathbf{1}_{V_j}(r)| \le K\operatorname{dist}(r,R_0)^2.$$

Combine this with the moment bound and (36.13) to obtain (36.14). Project π_ν to its nearest nodes. The transport cost is at most $K\sqrt{\delta}$ by Cauchy–Schwarz, and the projected weights differ from those of π_{μ_0} by $O(\delta)$. Hence $W_1(\pi_\nu,\pi_{\mu_0}) \le K\sqrt{\delta}$.

Finally the untilting operation

$$U_x(\pi)(dr) = \frac{x^{-r}\, d\pi(r)}{\int x^{-u}\, d\pi(u)}$$

is jointly Lipschitz in W_1 and in x on a compact positive x-interval. Indeed, the weights, their derivatives in r, x, and reciprocal denominators are uniformly bounded; apply the dual Lipschitz formula for W_1 to the weighted integrals. Since $U_{x_\nu}(\pi_\nu) = \bar{\nu}$, this proves (36.12).

For sharpness, replace one reference atom $w_j\delta_{r_j}$ by $\frac{1}{2}w_j(\delta_{r_j-h} + \delta_{r_j+h})$. For small h all nodes remain interior and the mass is unchanged. Taylor's theorem gives an $O(h^2)$ change in the derivatives of S of any prescribed finite order on a contact neighborhood. Analytic stationary inversion therefore gives $d_{2N}(\mu_0,\nu_h) \le K_0h^2$. The one-dimensional transport cost of this split is exactly $W_1(\bar{\mu}_0,\bar{\nu}_h) = w_jh/\mu_0([a,b])$ for small h. Taking $h = \sqrt{\delta/K_0}$ proves the matching lower modulus. □

If observed jets have coordinate errors at most ε, any positive candidate within ε of the data is within 2ε of the true jet. The theorem is therefore a deterministic certificate for every such candidate near the fixed reference. It is not a claim that arbitrary noisy jets are feasible, nor a uniformly conditioned inverse when reference nodes collide.

36.6 Relation to existing open problems

Two relevant existing problems were checked rather than silently identified with the present classification. Shapiro's common-zero conjecture for exponential sums is stated in [107, Section 2]. It concerns arbitrary pairs with infinitely many common zeros. Lemma 36.2 assumes that all zeros project to finitely many points and divides explicit periodic progressions; it does not settle that conjecture. The effective Skolem problem asks for a decision procedure for a zero in a specified recurrence. Recent work still identifies effectivity as the missing ingredient [108]. Our use of Skolem–Mahler–Lech is non-effective and does not solve that decision problem.

The completed targets are the finite-spectrum functional classification, exact differential-order law, and local certification theory in the stated observation model. The contact's own D-finite classification for arbitrary spectra with irrational gaps, quantitative locations of the complex singularities, and uniform collision-sensitive jet conditioning remain outside these results.

37 Arithmetic changes under mixing and algebraic tradeoffs

All logarithms are natural, and $H(p) = -\sum_i p_i \log p_i$, with $0 \log 0 = 0$. Two structural questions motivate this section. First, does the logarithmic algebraicity of rational entropy programs with a pure saddle survive passage to mixed strategies? Second, when rational entropy objectives are combined with algebraic weights, can the existence of an algebraic optimizer still be decided? We give a negative answer to the first question and a positive answer to the second in a specified smooth concave class. The transcendence inputs are classical theorems of Baker, Gelfond–Schneider, and the six-exponentials theory; we do not claim a new transcendence theorem or a solution of the four-exponentials conjecture.

37.1 Finite logarithmic games inside rational entropy payoffs

Write $\Delta_r = \{x \in \mathbb{R}^r_{\geq 0} : \sum_i x_i = 1\}$ and

$$\mathcal{L}_{\mathrm{rat}} = \mathrm{span}_{\mathbb{Q}}\{\log n : n \in \mathbb{Z},\ n \geq 2\}.$$

A *rational Shannon payoff* on a product of rational polytopes is a finite rational signed linear combination of functions $H(L(x, y))$, where each L is a rational affine map taking that product into a finite probability simplex. Constant probability maps are permitted. This definition does not permit arbitrary real coefficients or an arbitrary real additive constant. A mixed strategy is any Borel probability measure on the player's pure-strategy polytope; the players randomize independently. The first player maximizes and the second minimizes the expected payoff.

Theorem 37.1 (An entropy embedding of finite logarithmic games)**.** *Given a finite $r \times s$ real matrix M with entries in $\mathcal{L}_{\mathrm{rat}}$, one can effectively construct a rational Shannon payoff F on $\Delta_r \times \Delta_s$ such that:*

(i) $F(e_i, e_j) = M_{ij}$ at every pair of vertices;

(ii) *$F(\cdot, y)$ is strictly convex for each y, and $F(x, \cdot)$ is strictly concave for each x, whenever the corresponding simplex has positive dimension;*

(iii) *the mixed equilibria of F are exactly the mixed equilibria of M, interpreted as measures supported on simplex vertices.*

In particular, the full continuous-strategy game has the same mixed value as M. Statement (iii) holds for arbitrary Borel mixed strategies, not just for finitely supported strategies chosen in advance.

Proof. The cases of a one-point simplex follow by deleting that variable, so suppose $r, s \geq 2$. Put $h(t) = H(t, 1-t)$ and, for $n \geq 2$, set

$$P_n(u, v) = h((u+v)/n), \qquad 0 \leq u, v \leq 1.$$

For a function K on this square write $\delta K = K(1,1) - K(1,0) - K(0,1) + K(0,0)$. Direct calculation gives $\delta P_2 = -2\log 2$ and, for $n \geq 3$,

$$\delta P_n = -\log n - \frac{2}{n}\log 2 - \frac{n-2}{n}\log(n-2) + \frac{2(n-1)}{n}\log(n-1).$$

Here the term involving $\log 1$ is zero. Define $K_1 = 0$, $K_2 = -P_2/2$, and recursively

$$K_n = -P_n - \frac{2}{n}K_2 - \frac{n-2}{n}K_{n-2} + \frac{2(n-1)}{n}K_{n-1} \quad (n \geq 3).$$

Induction gives $\delta K_n = \log n$. Each K_n is a rational linear combination of $P_2, \ldots, P_n$, is symmetric, and vanishes at $(0,0)$. Let $k_n = K_n(1,0) = K_n(0,1)$, and set

$$C_n(u, v) = K_n(u, v) - (u+v)k_n.$$

Its four corner values are zero except $C_n(1,1) = \log n$.

The linear correction is itself an allowed rational Shannon expression. Indeed, k_n belongs to $\mathcal{L}_{\mathrm{rat}}$ because the entropy of a rational probability vector is a rational combination of logarithms of positive integers. For $z \in [0,1]$ and an integer $m \geq 2$, the identity

$$z\log m = H(1-z, \underbrace{z/m, \ldots, z/m}_{m \text{ entries}}) - h(z) \tag{37.1}$$

represents every required linear logarithmic term. In particular no coefficient k_n is inserted as a new, nonrational entropy coefficient.

Choose a finite expression $M_{ij} = \sum_n q_{ijn}\log n$ with $q_{ijn} \in \mathbb{Q}$, and put

$$F_0(x, y) = \sum_{i,j,n} q_{ijn} C_n(x_i, y_j).$$

This is an allowed payoff and agrees with M at the vertices. We next make its separate curvature have the desired signs without changing those vertex values.

Write $K_n = \sum_{m=2}^{n} a_{nm} P_m$, with $a_{nm} \in \mathbb{Q}$, and set

$$B_n = \sum_m |a_{nm}|, \qquad A = \sum_{i,j,n} |q_{ijn}| B_n.$$

For $0 < u < 1$ and $0 \leq v \leq 1$,

$$|\partial_u^2 P_m(u, v)| = \frac{1}{(u+v)(m-u-v)} \leq \frac{1}{u(1-u)}. \tag{37.2}$$

To check the inequality, minimize the concave quadratic $z(m-z)$ on $z \in [u, u+1]$: both endpoint products are at least $u(1-u)$ for $m \geq 2$. The same bound holds after interchanging u and v. For x in the relative interior of Δ_r and a tangent vector a with $\sum_i a_i = 0$, it follows that

$$|D_x^2 F_0(x,y)[a,a]| \leq A \sum_i \frac{a_i^2}{x_i(1-x_i)} \leq rA \sum_i \frac{a_i^2}{x_i}.$$

For the last inequality, use $1/(x_i(1-x_i)) = 1/x_i + 1/(1-x_i)$ and Cauchy–Schwarz in the form

$$\frac{a_i^2}{1-x_i} = \frac{(\sum_{k\neq i} a_k)^2}{\sum_{k\neq i} x_k} \leq \sum_{k\neq i} \frac{a_k^2}{x_k}.$$

Choose the rational number $K = 1 + \max(r,s)A$ and define

$$F(x,y) = F_0(x,y) - KH(x) + KH(y). \tag{37.3}$$

Since $D^2H(x)[a,a] = -\sum_i a_i^2/x_i$, the claimed strict convexity and strict concavity follow in relative interiors. They also hold on every nonconstant segment in the closed simplices: restrict to the smallest face containing the segment, apply the same estimates to its active coordinates on the open segment, and then use continuity at the endpoints. The face size only decreases the curvature bound. No derivative at a zero coordinate is used. The added entropies vanish at vertices, proving (i) and (ii).

For completeness, the reduction of arbitrary mixed strategies follows from the two pointwise inequalities

$$F(x,y) \leq \sum_i x_i F(e_i, y), \qquad F(x,y) \geq \sum_j y_j F(x, e_j). \tag{37.4}$$

The first is strict for a nonvertex x, and the second is strict for a nonvertex y. Replacing a measure μ on Δ_r by the vertex measure with masses $\int x_i \, d\mu(x)$ weakly increases its payoff against every opponent measure. Replacing the second player's measure by the corresponding vertex measure weakly decreases the payoff against every first-player measure. All integrations are justified by bounded continuity on a compact product. The finite matrix minimax theorem therefore supplies an equilibrium of the full game with the same value.

If an equilibrium measure assigned positive mass to nonvertices, its replacement would strictly improve that player's expected payoff: the appropriate strict gap in (37.4) is a positive measurable function at every nonvertex and against every opponent pure strategy. Its integral is strictly positive whenever the measure of those nonvertices is positive. Thus every equilibrium is supported on vertices, and its masses must form a matrix-game equilibrium. This proves (iii). □

37.2 An unconditional failure of logarithmic algebraicity

Theorem 37.2 (Two rational entropy games beyond the pure-saddle range)**.** *There are two explicitly constructible rational Shannon games, each on $\Delta_2 \times \Delta_2$, with respective mixed values*

$$V_1 = \frac{(\log 2)^2}{\log 7}, \qquad V_2 = \frac{\log 2 \log 5}{\log 7}, \tag{37.5}$$

such that each has a unique mixed equilibrium, all its nonzero equilibrium masses are transcendental, and neither has a pure saddle. At least one of e^{V_1}, e^{V_2} is transcendental.

More strongly, at least one of V_1, V_2 lies outside

$$\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \operatorname{span}_{\overline{\mathbb{Q}}}\{z \in \mathbb{C} : e^z \in \overline{\mathbb{Q}}^{\times}\}.$$

Consequently, for at least one of the two games, both its value and the exponential of its value are transcendental. These assertions do not identify which of the two games has this property.

Proof. Let $a = \log 2$, $b = \log 7$, and $d = \log 5$. For $0 < u, v < b$, consider

$$M(u, v; b) = \begin{pmatrix} u & 0 \\ u + v - b & v \end{pmatrix}.$$

The unique equilibrium assigns first-row probability $1 - u/b$ and first-column probability v/b, and its value is uv/b. Indeed, the difference between the two column payoffs is $u - b + pb$ when the first-row probability is p, while the difference between the two row payoffs is $qb - v$ when the first-column probability is q. The strict inequalities $0 < u, v < b$ put both intersection probabilities in $(0, 1)$ and give uniqueness and the absence of a pure saddle. Apply Theorem 37.1 to $M(a, a; b)$ and $M(a, d; b)$. Their entries are rational combinations of logarithms of integers, and the construction is fully explicit through C_2, C_5, C_7 and the rational curvature bound in (37.3).

The ratios a/b and d/b are irrational by unique prime factorization. If either ratio were algebraic irrational, the Gelfond–Schneider theorem applied to the base 7 would contradict $7^{a/b} = 2$ or $7^{d/b} = 5$. Both ratios, and their complements, are therefore transcendental. The embedding theorem transfers the unique matrix equilibria, with exactly these masses, to the full games.

For the exponential assertion apply the classical six-exponentials theorem to the rationally independent pair $(1, a/b)$ and triple (b, a, d); see [111, Theorem 3]. The six products are

$$\begin{pmatrix} b & a & d \\ a & a^2/b & ad/b \end{pmatrix}.$$

All entries except the last two already exponentiate to algebraic numbers. Thus at least one of e^{V_1}, e^{V_2} is transcendental.

For the stronger assertion use Roy's strong six-exponentials theorem, in the formulation recorded in [110, Theorem 3] and credited there to [109, Corollary 2]: if two complex numbers are linearly independent over $\overline{\mathbb{Q}}$ and three others are likewise independent, one of their six products lies outside $\widetilde{\mathcal{L}}$. Here $(1, a/b)$ is independent by the preceding Gelfond–Schneider argument, and (b, a, d) is independent by Baker's theorem applied to their rational independence. Four products are already in $\widetilde{\mathcal{L}}$, leaving the claimed alternative. This space contains both all algebraic numbers and all logarithms of nonzero algebraic numbers, which proves the last conclusion. □

Remark 37.3 (What the counterexample does and does not settle). The conclusion is existential but unconditional: it disproves the universal assertion that every mixed rational Shannon game has logarithmic algebraic value. It does not prove either individual exponential in (37.5) transcendental. The usual four-exponentials conjecture would imply both individual assertions, by applying it to $(1, a/b)$ and respectively (b, a) or (b, d). These quotient-of-products questions belong to the classical problem family explicitly discussed in [110, Section 1.2]. The new structural step here is their realization, with exact equilibrium control, by rational Shannon payoffs; the number-theoretic alternative is a direct consequence of classical theory. The word "entropy game" here refers to a one-shot Shannon payoff game, not to the combinatorial growth-rate games studied in [112]; no complexity result for the latter is being claimed.

37.3 Baker splitting of algebraic tradeoffs

The next result handles a different obstruction: an algebraic irrational tradeoff weight cannot in general be absorbed by clearing denominators in stationarity equations.

Lemma 37.4 (Splitting and exact signs for algebraic logarithmic forms)**.** *Let $\theta_1, \dots, \theta_d$ be real algebraic numbers linearly independent over $\mathbb{Q}$. If each L_k is a rational linear combination of real logarithms of positive algebraic numbers, then*

$$\sum_{k=1}^{d} \theta_k L_k = 0 \quad \Longleftrightarrow \quad L_1 = \cdots = L_d = 0.$$

Given finitely many real algebraic coefficients β_i and positive algebraic numbers α_i, the sign of $\sum_i \beta_i \log \alpha_i$ is decidable from exact algebraic-number representations. No time bound is asserted.

Proof. Choose a rational basis $\lambda_1, \dots, \lambda_m$ for the vector space generated by the finitely many logarithms that occur. Baker's linear-independence theorem states that these rationally independent logarithms are also independent over $\overline{\mathbb{Q}}$ [62]. Write $L_k = \sum_j r_{kj} \lambda_j$ with rational r_{kj}. A vanishing sum in the assertion implies $\sum_k \theta_k r_{kj} = 0$ for every j, hence all $r_{kj} = 0$. The converse is immediate.

For the algorithm one need not compute a basis of the logarithms. Compute instead a rational basis θ_k of the span of the algebraic coefficients β_i, using exact number-field linear algebra, and write $\beta_i = \sum_k r_{ki} \theta_k$. By the first assertion, the form is zero if and only if, for every k, $\sum_i r_{ki} \log \alpha_i = 0$. Clearing rational denominators makes each of these conditions an exact identity $\prod_i \alpha_i^{n_{ki}} = 1$ with integer exponents. Such identities are decidable by algebraic-number arithmetic. If the zero test fails, successively refined certified intervals for the algebraic numbers and their real logarithms eventually exclude zero and determine the sign. Termination follows from nonvanishing, not from an assumed separation bound. □

Fix a nonempty compact rational polytope P. Call a function on P a *smooth rational logarithmic entropy expression* if it has the form

$$G(z) = \sum_a c_a H(L_a(z)) + \sum_b \ell_b(z) \log r_b, \tag{37.6}$$

where $c_a \in \mathbb{Q}$, $r_b \in \mathbb{Q}_{>0}$, ℓ_b are rational affine functions, and L_a are rational affine probability maps all of whose coordinates are strictly positive throughout P. Identically zero coordinates may be deleted before applying this definition. In particular these functions are smooth on a neighborhood of P in its affine hull. Their derivatives along rational vectors are rational linear combinations of logarithms of positive rational affine functions and positive rational constants: the nonlogarithmic terms cancel because each L_a has total mass one.

Theorem 37.5 (Common stationarity and decidable algebraic optimizers)**.** *Let $G_1, \dots, G_m$ be expressions of the form (37.6), let $\lambda_1, \dots, \lambda_m$ be specified real algebraic numbers, and suppose $F = \sum_i \lambda_i G_i$ is concave on P. Choose a rational basis $\theta_1, \dots, \theta_d$ of the span of the λ_i, write $\lambda_i = \sum_k r_{ki} \theta_k$, and put $A_k = \sum_i r_{ki} G_i$.*

(i) At an algebraic point in the relative interior of any face Q of P, stationarity of F along Q is equivalent to simultaneous stationarity of all A_k along Q.

(ii) There is an effective finite list of algebraic points of P which contains a global maximizer whenever any algebraic global maximizer exists.

(iii) It is decidable whether F has an algebraic global maximizer. If one exists the procedure returns one; otherwise it certifies that every global maximizer has at least one transcendental coordinate.

The hypotheses specify a promised concave input class; the procedure does not purport to decide concavity of arbitrary entropy expressions. It has no polynomial-time guarantee.

Proof. For (i), choose a rational tangent basis of the face. At an algebraic point each derivative of A_k along such a vector is a rational linear combination of real logarithms of positive algebraic numbers. Apply Lemma 37.4 separately to every tangent derivative.

For each face Q, let S_Q be the subset of its relative interior where all A_k are stationary along Q. Each equation defining S_Q is of the form

$$\sum_j t_j \log u_j(z) = 0, \qquad t_j \in \mathbb{Q}, \quad u_j(z) > 0,$$

where the u_j are rational affine functions or positive rational constants. Exponentiating, clearing rational exponents, and moving negative integer powers to the other side gives a polynomial equation over $\mathbb{Q}$. Positivity makes this conversion reversible. Together with the relative-interior conditions for Q, these give an effective semialgebraic description over $\mathbb{Q}$.

Semialgebraic decomposition algorithms supply finitely many connected cells covering S_Q, each with an algebraic sample point [61]. On each cell every A_k is constant: connect two of its points by a piecewise C^1 semialgebraic path inside the cell and differentiate along that path; its tangent vectors lie in the tangent space of Q, on which the derivatives vanish. Thus F is constant on that cell as well. Include all sample points from all faces in the candidate list, including zero-dimensional faces, for which stationarity is vacuous. If an algebraic maximizer z exists, it lies in the relative interior of its minimal face and is stationary there. Part (i) puts it in S_Q. The sample point of its cell has the same values of all A_k, hence the same maximal value of F. This proves (ii).

For each candidate z, concavity and differentiability give the exact global optimality criterion

$$dF_z(v - z) \le 0 \quad \text{for every vertex } v \text{ of } P. \tag{37.7}$$

Necessity follows by differentiating a feasible segment from a maximizer; sufficiency follows from the concave tangent-plane bound and the fact that vertices generate P. Each left-hand side is an algebraic linear combination of logarithms of positive algebraic numbers. Although $v - z$ need not be rational, it is algebraic, which is all that the exact-sign part of Lemma 37.4 needs. The mass-one cancellation in (37.6) still applies in these directions. Test the finitely many inequalities exactly for each candidate. Part (ii) proves that rejecting every candidate is equivalent to absence of an algebraic maximizer. A real maximizer exists by compactness and continuity, proving (iii). □

Corollary 37.6 (Weighted information objectives with exact input data)**.** *Let $W_1, \dots, W_m$ be finite rational channels with a common input alphabet and strictly positive transition probabilities. Let P be a nonempty compact rational polytope of input distributions, and let $\lambda_i > 0$ be specified real algebraic weights. It is decidable whether*

$$\max_{p \in P} \sum_i \lambda_i I(p; W_i)$$

admits an input distribution all of whose coordinates are algebraic. If it does, one such distribution can be returned exactly.

Proof. Write

$$I(p; W_i) = H(W_i p) - \sum_x p_x H(W_i(\cdot \mid x)).$$

The first term has strictly positive rational affine probability coordinates, and the second is an affine rational combination of logarithms of positive rationals. Mutual information is concave in the input distribution, so the positive weighted sum satisfies all hypotheses of Theorem 37.5. □

Example 37.7 (A positive two-channel family with no algebraic optimizer). Let W be the binary-input, binary-output channel with $W(1 \mid 0) = 1/4$ and $W(1 \mid 1) = 1/2$, and let W^{rev} be obtained by interchanging the two input symbols. If p is the probability of input 1, put

$$J(p) = I(p; W) = h(1/4 + p/4) - (1 - p)h(1/4) - ph(1/2).$$

For every positive algebraic β, the strictly concave objective $J(p) + \beta J(1 - p)$ has a unique maximizer $p_\beta \in (0, 1)$. Moreover,

$$p_\beta \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \beta \in \mathbb{Q}.$$

To prove this, $J''(p) < 0$, $J(0) = J(1) = 0$, and $J(p) > 0$ on $(0, 1)$, so the weighted maximum is interior and unique. Let p_0 maximize J. The derivative identity

$$J'(1/2) = \frac{1}{4} \log(80/81) < 0$$

shows $p_0 < 1/2$, while the maximizer of $J(1 - p)$ is $1 - p_0 > 1/2$. For algebraic irrational β, an algebraic stationary point of $J(p) + \beta J(1 - p)$ would, by Baker splitting, be stationary for both summands separately, which is impossible. For rational β, the stationarity equation exponentiates after clearing denominators to a polynomial equation over $\mathbb{Q}$. This polynomial is not identically zero, because the strictly decreasing derivative has only one zero. Hence the unique optimizer is algebraic. This example uses ordinary smooth concave information objectives and strictly positive channels; it does not rely on a singular boundary channel or a nonconcave information-bottleneck problem.

Scope and comparison with prior work. The algebraic stationary splitting is an application of Baker's classical linear-independence theorem, and the finite sampling step uses standard real algebraic geometry. The decision statement concerns exact rational and algebraic input descriptions with a concavity promise. It does not contradict impossibility results for selecting capacity-achieving distributions uniformly from computable-real channel descriptions, a different input model studied in [113]. Nor does it settle the arithmetic of a transcendental optimizer's objective value. The mixed-game embedding and the algebraic-optimizer test are the reusable structural contributions established here; historical priority for these formulations has not been certified by the targeted literature check.

38 Algebraic closure inside the field of E-values

Let $k = \overline{\mathbb{Q}}$, let $\mathcal{E}$ denote the ring of values of E-functions at algebraic points, and put $K = \mathrm{Frac}(\mathcal{E})$. This section studies finite algebraic extensions inside K, rather than another list of individual rational powers. Its two targets are an exact algebraic-closure theorem for exponential values and a joint finite-cover theorem for several exponential-integral values. Both depend on the injective arithmetic morphism of Fischler–Rivoal [72, Proposition 3.1]. Their Corollary 3.2 already classifies the intersection of $\mathcal{E}$ with the field of rational expressions in algebraic exponentials; that rational-expression statement is an antecedent and is not counted again. The results below extend the question to arbitrary algebraic elements. They are proved

applications, with no independently established priority claim and no claim to settle the full unit or factorization questions posed in [72].

We use the external morphism and its extension to fraction fields:

$$\Phi : \mathcal{E} \hookrightarrow \mathcal{R}, \qquad K \hookrightarrow \operatorname{Frac}(\mathcal{R}), \qquad \mathcal{R} = \left\{ \sum_{\rho \in k} c_\rho e^{\rho z} : c_\rho \in \mathbb{C},\ \text{finite support} \right\}. \tag{38.1}$$

It fixes k, and $\Phi(e^{\alpha}) = e^{\alpha z}$ for $\alpha \in k$. For the exponential integral, the preceding morphism computation gives

$$u_a = \operatorname{Ein}(a), \qquad \Phi(u_a) = A_a + B_a e^{-az}, \qquad A_a = \gamma + \log a, \quad B_a = \delta_a = \int_0^\infty \frac{e^{-t}}{t+a}\, dt \tag{38.2}$$

for positive algebraic $a \geq 1$. The constants A_a, B_a are positive. The choices of real logarithm in this formula are fixed throughout.

38.1 Two descent lemmas

Lemma 38.1 (Finite phase descent)**.** *Let $\alpha_1, \dots, \alpha_m \in k$ be linearly independent over $\mathbb{Q}$. If $W \in \operatorname{Frac}(\mathcal{R})$ is algebraic over $\mathbb{C}(e^{\alpha_1 z}, \dots, e^{\alpha_m z})$, then, for some positive integer q,*

$$W \in \mathbb{C}(e^{\alpha_1 z/q}, \dots, e^{\alpha_m z/q}).$$

If $W \in \mathcal{R}$, it belongs to the Laurent polynomial ring in these same functions, after increasing q if necessary.

Proof. Extend the given α_i to a rational basis of the span of these numbers and the finitely many phases in a numerator and denominator of W. Clear all rational coordinates with one positive integer q. Writing $X_i = e^{\alpha_i z/q}$ and $Y_j = e^{\beta_j z/q}$ for the extra basis vectors gives $W \in \mathbb{C}(X_1, \dots, X_m, Y_1, \dots, Y_s)$. These variables are algebraically independent over $\mathbb{C}$: a polynomial relation would be a linear relation among finitely many exponentials with distinct exponents, and successive derivatives at zero give an invertible Vandermonde matrix.

Put $F = \mathbb{C}(X_1, \dots, X_m)$. An element of $F(Y_1, \dots, Y_s)$ algebraic over F belongs to F. Indeed, apply each derivation $\partial/\partial Y_j$ to its separable minimal polynomial over F. The derivative of the polynomial at the element is nonzero, so every such derivative of the element vanishes. A rational function in characteristic zero with zero derivative in a variable is independent of that variable. This proves the field assertion. In the ring case W began as a Laurent polynomial in X, Y. Independence of the Y_j then removes all its nonzero Y-monomials, leaving a Laurent polynomial in the X_i. □

Lemma 38.2 (Descent of constant coefficients)**.** *For independent indeterminates $T_1, \dots, T_m$, the field $k(T_1, \dots, T_m)$ is relatively algebraically closed in $\mathbb{C}(T_1, \dots, T_m)$. Further, a finite function-field extension over $k(T_1, \dots, T_m)$ retains its degree, normality, and Galois group, when applicable, after extension of constants to $\mathbb{C}$.*

Proof. For the first assertion let $h \in \mathbb{C}(T)$ satisfy a nonzero polynomial over $k(T)$. Every automorphism of $\mathbb{C}/k$, acting on coefficients and fixing T, sends h to a root of the same polynomial, so h has a finite orbit. Express $h = A/B$ with relatively prime polynomials and normalize one specified nonzero coefficient of B to one. This representation is unique, hence every coefficient of A and B has a finite orbit under these automorphisms. A complex number transcendental over k has an infinite orbit: include it in a transcendence basis, send it to itself

plus an arbitrary integer while fixing the other basis elements, and extend these isomorphisms to the algebraic closure $\mathbb{C}$. All the coefficients therefore lie in k.

For the second assertion, take a primitive element of a finite extension and clear denominators in its irreducible polynomial. A polynomial irreducible over an algebraically closed coefficient field stays irreducible after extension of that field. In this setting one can see this directly: a putative factorization has finitely many normalized coefficients satisfying polynomial equations over k; a solution over $\mathbb{C}$ gives a solution over k by the weak Nullstellensatz. Nonzero normalizing coefficients are handled by adjoining their inverses. Gauss's lemma transfers the assertion between polynomial and rational-function coefficients. Thus the primitive-element polynomial has the same degree after the constant extension. Apply the same argument to the normal closure: its original automorphisms extend and already account for the preserved degree. Its Galois group, and the subgroups corresponding to the original intermediate fields, are unchanged. Normality and the stated Galois-group assertion follow. □

38.2 Exact algebraic and integral closure for exponential values

Fix rationally independent $\alpha_1, \dots, \alpha_m \in k$ and write

$$F_\alpha = k(e^{\alpha_1}, \dots, e^{\alpha_m}), \qquad R_{\alpha,q} = k[e^{\alpha_1/q}, e^{-\alpha_1/q}, \dots, e^{\alpha_m/q}, e^{-\alpha_m/q}], \qquad F_{\alpha,q} = \mathrm{Frac}(R_{\alpha,q}).$$

The exponential values are algebraically independent over k; this follows either from Lindemann–Weierstrass or by applying Φ and the exponential Vandermonde argument above. Thus these really are Laurent polynomial rings and rational function fields in m variables.

Theorem 38.3 (Exact exponential algebraic closure)**.** *One has*

$$\{v \in K : v \text{ is algebraic over } F_\alpha\} = \bigcup_{q \geq 1} F_{\alpha,q}, \tag{38.3}$$

$$\{v \in \mathcal{E} : v \text{ is algebraic over } F_\alpha\} = \bigcup_{q \geq 1} R_{\alpha,q}. \tag{38.4}$$

Moreover, $\bigcup_{q\geq 1} R_{\alpha,q}$ is exactly the integral closure of $R_{\alpha,1}$ inside K. The field on the right of (38.3) is a Galois extension of F_α with Galois group $\widehat{\mathbb{Z}}^m$.

Proof. Let $v \in K$ be algebraic over F_α. Its image $W = \Phi(v)$ is algebraic over $k(e^{\alpha_1 z}, \dots, e^{\alpha_m z})$. Lemma 38.1 gives variables $X_i = e^{\alpha_i z/q}$ for some q, with $W \in \mathbb{C}(X_1, \dots, X_m)$. Since W is algebraic over $k(X_1, \dots, X_m)$, the constant-descent lemma gives

$$W \in k(X_1, \dots, X_m).$$

Write $W = S(X)$ with S rational over k. The element $S(e^{\alpha_1/q}, \dots, e^{\alpha_m/q})$ belongs to K and has image W. Its denominator is nonzero by algebraic independence. Injectivity of Φ therefore identifies it with v. This proves one inclusion in (38.3); the reverse inclusion follows from $(e^{\alpha_i/q})^q = e^{\alpha_i}$.

If $v \in \mathcal{E}$, the ring assertion of the phase-descent lemma puts W in $\mathbb{C}[X_1^{\pm 1}, \dots, X_m^{\pm 1}]$. Constant descent and uniqueness of Laurent coefficients now give

$$W \in k[X_1^{\pm 1}, \dots, X_m^{\pm 1}].$$

Injectivity gives $v \in R_{\alpha,q}$. Conversely, all generators of these Laurent polynomial rings are E-values, proving (38.4).

Every $R_{\alpha,q}$ is integral over $R_{\alpha,1}$, including its negative-power generators, because each satisfies a monic power polynomial with coefficient in $R_{\alpha,1}$. Conversely, if $v \in K$ is integral

over $R_{\alpha,1}$, the field assertion puts v in $F_{\alpha,q}$ for some q. It is then also integral over $R_{\alpha,q}$. This Laurent polynomial ring is a unique factorization domain, hence integrally closed in its fraction field; therefore $v \in R_{\alpha,q}$.

For each q, the extension $F_{\alpha,q}/F_\alpha$ has group $(\mathbb{Z}/q\mathbb{Z})^m$, acting independently by multiplication of the qth-root generators by qth roots of unity. Independence of the generators proves that its degree is q^m. The directed union uses divisibility of q, and the restriction maps are the usual reduction maps. Its inverse-limit Galois group is consequently $\widehat{\mathbb{Z}}^m$. □

Corollary 38.4 (Exponential-field dichotomy)**.** *Put*

$$F_{\exp} = k(e^\alpha : \alpha \in k), \qquad R_{\exp} = \left\{ \sum_{j=1}^{n} c_j e^{\alpha_j} : c_j, \alpha_j \in k \right\}.$$

Then $F_{\exp}$ is relatively algebraically closed in K, and $R_{\exp}$ is integrally closed inside K. In particular, an E-value is either a finite algebraic-coefficient exponential sum or is transcendental over $F_{\exp}$. Any counterexample to the unit conjecture $\mathcal{E}^\times = k^\times \exp(k)$ in [72] would have to be transcendental over $F_{\exp}$.

Proof. A polynomial relation over $F_{\exp}$ uses only finitely many exponential phases in its coefficients. Choose a rational basis of their span and clear denominators. Theorem 38.3 then puts every element of K algebraic over $F_{\exp}$ back in $F_{\exp}$; for elements of $\mathcal{E}$ it puts them in $R_{\exp}$. The same finite-support reduction applied to a monic relation proves the relative integral-closure assertion.

If a unit $v \in \mathcal{E}^\times$ were algebraic over $F_{\exp}$, both v and v^{-1} would belong to $R_{\exp}$. Choose one finite rational basis and denominator for all their phases. Their images are mutually inverse Laurent polynomials. A unit of a Laurent polynomial ring is a nonzero constant times a Laurent monomial: for example, the Newton polytope of a product is the Minkowski sum of those of its factors, so a product equal to one forces both supports to consist of one point. Hence $v = ce^\alpha$ with $c \in k^\times$, $\alpha \in k$. □

An earlier restricted result of Rivoal–Roques [114, Theorem 5] treats reciprocal values of two E-functions that form a common two-dimensional differential system. Our restriction instead concerns algebraicity over $F_{\exp}$ and does not assume such a common system. These are distinct hypotheses.

The exact closure theorem is not a proof of the full unit conjecture. It excludes algebraic counterexamples over $F_{\exp}$ and leaves open possible units transcendental over that field. Similarly, a factor of an exponential sum need not be algebraic over the exponential field merely because its product with a second factor lies there. That missing implication cannot be inserted into a factorization argument.

38.3 Joint exponential-integral fields have at most one ramified coordinate

Let $a_1, \dots, a_m \geq 1$ be positive algebraic numbers linearly independent over $\mathbb{Q}$. Write

$$u_i = u_{a_i}, \qquad A_i = \gamma + \log a_i, \qquad B_i = \delta_{a_i}, \qquad F = k(u_1, \dots, u_m).$$

The u_i are algebraically independent: their images are the affine coordinates $A_i + B_i X_i$ in the algebraically independent exponential variables $X_i = e^{-a_i z}$. This simple argument needs the stated rational independence of the parameters; no claim is made here for arbitrary rationally dependent frequency families.

Theorem 38.5 (A joint cyclic-coordinate obstruction)**.** *Let $v \in K$ be algebraic over F, and set $d = [F(v) : F]$. If $d > 1$, there exists an index i such that $A_i \in k$, and*

$$F(v) = F\big((u_i - A_i)^{1/d}\big). \tag{38.5}$$

In particular this extension is cyclic. Among the entire parameter family there is at most one possible such index, independent of v. If none of the A_i is algebraic, then F is relatively algebraically closed in K. Unconditionally, one can remove at most one of the parameters so that the field jointly generated by all the remaining u_i is relatively algebraically closed in K.

Proof. Apply Φ to a minimal polynomial for v over F. Set $U_i = A_i + B_i e^{-a_i z}$ and $W = \Phi(v)$. The phase-descent lemma, with phases $-a_i$, gives an integer $q \geq 1$ and

$$W \in \mathbb{C}(X_1, \dots, X_m), \qquad X_i = e^{-a_i z/q}, \qquad U_i = A_i + B_i X_i^q.$$

The degree of the extension is still d over $\mathbb{C}(U_1, \dots, U_m)$, by Lemma 38.2. The extension

$$\mathbb{C}(X_1, \dots, X_m)/\mathbb{C}(U_1, \dots, U_m)$$

is Galois with group $G = (\mathbb{Z}/q\mathbb{Z})^m$. Let H be the subgroup whose fixed field is $\mathbb{C}(U, W)$. As G is abelian, this intermediate extension is Galois with group G/H.

For each i, let G_i be the coordinate subgroup acting on X_i alone. At the generic point of the hyperplane $D_i = \{U_i = A_i\}$ its inertia group in the full extension is exactly G_i. To check this explicitly, $U_i - A_i = B_i X_i^q$ has order q along $X_i = 0$, while the other X_j are nonzero units there. The transformations of X_j with $j \neq i$ act nontrivially on the residue field, whereas those of X_i act trivially on it. Thus the inertia in the intermediate extension is the image of G_i in G/H.

If this image is nontrivial, D_i is a branch divisor of the cover defined over k by the minimal polynomial of v. More concretely, clear denominators to obtain a polynomial in $k[U_1, \dots, U_m, V]$. Its leading coefficient in V times its nonzero V-discriminant vanishes on every branch divisor. This follows from the implicit function theorem off the zero set of that product, where its roots are distinct local holomorphic functions. Consequently a nonzero polynomial $D \in k[U_1, \dots, U_m]$ vanishes identically when $U_i = A_i$. Expand D in the other variables. Any nonzero coefficient polynomial in U_i then has A_i as a root, proving $A_i \in k$.

At most one A_i is algebraic. Otherwise, for distinct i, j, $A_i - A_j = \log(a_i/a_j)$ would be a nonzero algebraic number with algebraic exponential, contrary to Hermite–Lindemann [87]. If none is algebraic, every G_i lies in H; since the coordinate subgroups generate G, this gives $H = G$ and $d = 1$. If an algebraic center exists, denote its index by i. All other G_j lie in H. The quotient G/H is therefore a cyclic quotient of G_i, of order d, and

$$\mathbb{C}(U, W) = \mathbb{C}(U_1, \dots, U_m, (U_i - A_i)^{1/d}). \tag{38.6}$$

Indeed the fixed field of the subgroup of order q/d of the ith coordinate group is generated by $X_i^{q/d}$; its nonzero constant factor $B_i^{1/d}$ is available over $\mathbb{C}$.

It remains to descend the exact field description, not just its geometric degree. Lemma 38.2 shows that $F(v)/F$ itself is cyclic of degree d. As k contains the dth roots of unity, elementary Kummer theory gives an element y with $F(v) = F(y)$ and $y^d = f(u_1, \dots, u_m)$ for some $f \in k(U)^\times$. For completeness, if σ generates the cyclic group, the operator $\sum_{j=0}^{d-1} \zeta^{-j} \sigma^j$ is nonzero by linear independence of field automorphisms. Its nonzero value y satisfies $\sigma(y) = \zeta y$ for a primitive dth root ζ; therefore $y^d \in F$ and y has degree d.

By (38.6), over $\mathbb{C}(U)$ this eigenvector is a rational multiple of $(U_i - A_i)^{b/d}$ for some b coprime to d. Hence the order of f at every irreducible polynomial divisor other than $U_i - A_i$ is

divisible by d, and its order n at $U_i - A_i$ is coprime to d. Unique factorization in $k[U_1, \dots, U_m]$ therefore gives

$$f(U) = c\,(U_i - A_i)^n R(U)^d, \qquad c \in k^\times, \quad R \in k(U)^\times.$$

Divide y by $c^{1/d}R(u)$. Bezout's identity for n, d then produces an element $t \in F(v)$ with $t^d = u_i - A_i$. This element has degree d, by the valuation at $U_i - A_i$ (or the irreducibility of $T^d - (U_i - A_i)$). Consequently $F(v) = F(t)$, which proves (38.5). All choices of the root generate the same field, since their ratios are roots of unity in k.

Finally remove the unique index with algebraic center if it exists. For the remaining family the already proved no-center case establishes joint relative algebraic closure. If there was no such index, no removal is necessary. □

This conclusion is stronger than separate relative-algebraic-closure statements for each single $k(u_i)$. Separate statements do not rule out a new algebraic extension depending simultaneously on several coordinates. The joint theorem rules out every noncyclic extension and every cyclic extension ramified along a genuinely mixed divisor. For example, for rationally independent $a, b \geq 1$, no root of

$$T^3 - u_a T - u_b = 0 \tag{38.7}$$

belongs to K. This is a consequence of the joint theorem, not an additional structural target.

Verification of the cubic example. Over $k(U, V)$ the polynomial $T^3 - UT - V$ is irreducible. Indeed it defines the degree-three rational-function extension $k(U, T)/k(U, T^3 - UT)$: over $k(U)$ the polynomial map $T \mapsto T^3 - UT$ has degree three, so the function-field degree is three. Its discriminant is $4U^3 - 27V^2$, which is not a square in $k(U, V)$; it is an irreducible polynomial, since U^3 is not a square in $k(U)$ and the polynomial is quadratic in V. An irreducible cubic has transitive Galois group A_3 or S_3, and the discriminant is a square precisely in the first case. Thus its group is S_3 and its degree-three root extension is not Galois. Algebraic independence of u_a, u_b identifies their field with $k(U, V)$, preserving these assertions. Theorem 38.5 would make any such root extension in K cyclic, a contradiction. □

38.4 Limits of the structural transfer

The first theorem gives an exact unconditional closure computation and a corresponding restriction on the unit conjecture. The second gives an unconditional description of every possible joint finite extension, with a single exceptional connection coefficient left explicitly unclassified. It does not assert that any exceptional extension exists. The full ring of E-values may contain units transcendental over the exponential field; the present argument neither constructs nor rules out such units. It also does not identify all factorizations of a nonexponential E-value. These are concrete remaining gaps in the open ring questions, not consequences suppressed by the notation.

39 A convexity mechanism for gamma extraction and an atomic corner in mixtures

The preceding continuation left open whether different interior contacts can exchange control of a gamma-extraction boundary. We settle the question throughout the order interval $1/2 < \kappa \leq 1$: every candidate inverse, including an inadmissible one, has a unique global minimum. Consequently its probability boundary has exactly one contact and is analytic throughout the intermediate-rate interval. The argument is a signed Cauchy-transform calculation, followed by a second-order strict differential inequality.

A second investigation concerns positive mixtures of two different orders. The same convexity survives, but the geometry changes: the maximal extractable shape can remain constant on an initial interval, with an atom in the remainder, and then leave that interval through a genuine corner. We give a complete implicit phase diagram and prove that this phenomenon occurs. These are extensions of the manuscript's factorization problems, not claimed resolutions of a named conjecture in the external Bessel-distribution literature [115].

39.1 Notation and the inherited boundary facts

Fix $b > 1$. For $\kappa > 1/2$, let X_κ have transform

$$L_\kappa(s) = \frac{b+s}{b}\left(\frac{b^2-1}{(b+s)^2-1}\right)^\kappa.$$

Write $M = 2\kappa - 1$. Extraction of a gamma variable with shape h and rate $b - c$, $0 < c < 1$, leads, after a positive normalization and the tilt e^{-bx}, to the inverse

$$F_{\kappa,c,h}(w) = \frac{w(w-c)^h}{(w-1)^\kappa(w+1)^\kappa}, \tag{39.1}$$

$$g_{\kappa,c,h}(x) = \sum_{n\ge 0} \frac{a_n(c,h)x^{\eta+n-1}}{\Gamma(\eta+n)}, \qquad \eta = M - h, \tag{39.2}$$

$$(1-cz)^h(1-z^2)^{-\kappa} = \sum_{n\ge 0} a_n(c,h)z^n. \tag{39.3}$$

For $0 < h < M$ the series and its derivatives converge locally uniformly on $x > 0$. They depend real analytically on all strict parameters. The candidate is a probability remainder exactly when $g \ge 0$.

We explicitly use the earlier capacity and contact theorems [116, 118]. The maximal shape $H_\kappa(c)$ is attained, belongs to $(0, M)$ for $0 < c < 1$, and is continuous and strictly decreasing on $[0, 1]$. A boundary inverse has a nonempty finite contact set; every contact is double and satisfies $x < (h+1)/c$. At a contact of a nonnegative inverse,

$$g_c = -h\int_0^x e^{cy}g(x-y)\,dy < 0, \qquad g_h = -\int_0^x \frac{e^{cy}g(x-y)}{y}\,dy < 0. \tag{39.4}$$

The finite-contact boundary is locally the minimum of finitely many analytic contact branches. Positivity estimates at zero and infinity are uniform in compact strict parameter neighborhoods. These are inputs, rather than results counted again below.

39.2 A signed cut representation becomes strict convexity

Theorem 39.1 (Strict convexity and a unique minimum)**.** *Suppose* $1/2 < \kappa \le 1$, $0 < c < 1$, *and* $0 < h < 2\kappa - 1$. *Then*

$$g''_{\kappa,c,h}(x) + g'_{\kappa,c,h}(x) > 0 \qquad (x > 0). \tag{39.5}$$

The function $r \mapsto g_{\kappa,c,h}(-\log r)$ *is strictly convex on* $(0, 1)$ *and tends to* $+\infty$ *at both ends. It has exactly one minimum, which is nondegenerate. In particular, the inverse is nonnegative on the whole positive axis if and only if its value at this single minimum is nonnegative.*

Proof. First assume $\kappa < 1$. Then

$$0 < h < M < \kappa < 1, \qquad 0 < \eta = M - h < 1.$$

Take principal powers in (39.1), with cut $(-\infty, 1]$, and set $W(s) = -\pi^{-1} \operatorname{Im} F(s + i0)$. Direct boundary values give

$$W(s) = \begin{cases} \dfrac{-s\sin(\pi\eta)}{\pi} \dfrac{(c-s)^h}{(1-s)^\kappa(-1-s)^\kappa}, & s < -1, \\ \dfrac{-s\sin(\pi(h-\kappa))}{\pi} \dfrac{(c-s)^h}{(1-s)^\kappa(1+s)^\kappa}, & -1 < s < c, \\ \dfrac{s\sin(\pi\kappa)}{\pi} \dfrac{(s-c)^h}{(1-s)^\kappa(1+s)^\kappa}, & c < s < 1. \end{cases} \tag{39.6}$$

Thus W is positive on $(-\infty, -1)$ and $(0, 1)$, and negative on $(-1, 0)$. The integrable singularities at ± 1 have exponent $-\kappa > -1$; the point c causes no singularity. At negative infinity $W(s) = O(|s|^{-\eta})$.

For completeness, this signed cut really inverts the transform. Apply Cauchy's formula to $F(z)/(z - w)$ on a cut disk, with $w > 1$ fixed, and then let its radius tend to infinity. The outer-circle contribution is $O(R^{-\eta})$, because $F(z) = O(R^{-\eta})$ there. The small circles around ± 1 contribute $O(\epsilon^{1-\kappa})$, and that around c tends to zero. The two sides of the cut consequently give

$$F(w) = \int_{-\infty}^{1} \frac{W(s)}{w - s}\, ds.$$

The integral is absolutely convergent. Fubini's theorem, justified by the same absolute integral, shows that

$$g(x) = \int_{-\infty}^{1} e^{sx} W(s)\, ds \qquad (x > 0). \tag{39.7}$$

Indeed the right side has Laplace transform $F(w)$, and uniqueness agrees with (39.2). Every number of x derivatives can be passed under this integral on a compact positive interval.

The quadratic $s(s+1)$ has exactly the same sign as $W(s)$, apart from endpoints. Therefore

$$g''(x) + g'(x) = \int_{-\infty}^{1} s(s+1) e^{sx} W(s)\, ds > 0.$$

For $r = e^{-x}$, the operator $D_x^2 + D_x$ is $r^2 D_r^2$, which proves the asserted convexity. The endpoint estimates are

$$g(x) \sim \frac{x^{\eta-1}}{\Gamma(\eta)} \quad (x \downarrow 0), \qquad g(x) \sim \frac{(1-c)^h}{2^\kappa \Gamma(\kappa)} e^x x^{\kappa-1} \quad (x \longrightarrow \infty). \tag{39.8}$$

The first follows from the series, and the second follows from (39.7) at its right endpoint (or from the inherited uniform tail estimate). Both limits are positive infinity. Strict convexity therefore gives a unique interior minimum. At that minimum $g' = 0$, and (39.5) gives $g'' > 0$.

Finally fix $\kappa = 1$, $0 < h < 1$, and let $\kappa_j \uparrow 1$ with $h < 2\kappa_j - 1$. The series gives local convergence of g and its first two derivatives, so the limit in (39.5) is nonnegative. It is strictly positive: choose any compact interval $I \subset (0, c)$. On I the positive integrand in the preceding formula converges uniformly to a strictly positive function, since $-\sin \pi(h - \kappa_j) \to \sin \pi h > 0$. All the remaining integrals are nonnegative. The integral over I supplies a fixed positive lower bound. The endpoint estimates (39.8) still hold, with $\eta = 1 - h \in (0, 1)$. This proves all assertions also at $\kappa = 1$ without an unjustified integrable-cut formula at its pole endpoints. □

Theorem 39.2 (No contact switches in the full low-order interval)**.** *For every $1/2 < \kappa \le 1$ and $0 < c < 1$, the maximal probability remainder has exactly one positive zero $x_\kappa(c)$. This zero is double. Both $H_\kappa(c)$ and $x_\kappa(c)$ are real analytic throughout $(0, 1)$. Thus no exchange between distinct controlling contacts occurs in this order interval. The exact slope is*

$$H'_\kappa(c) = -H_\kappa(c) \frac{\int_0^{x_\kappa(c)} e^{cy} g(x_\kappa(c) - y)\, dy}{\int_0^{x_\kappa(c)} e^{cy} g(x_\kappa(c) - y)\, dy/y}.$$

Proof. The attained boundary has at least one zero. By Theorem 39.1 it must be the unique minimum, and its second derivative is positive. At this contact $g_h < 0$ by (39.4). The Jacobian of (g, g_x) with respect to (h, x) is consequently $g_h g_{xx} \neq 0$. The analytic implicit-function theorem gives analytic local functions $h(c), x(c)$. Uniqueness identifies these local functions on overlaps with the actual capacity and its contact. The first equation differentiated in c gives the displayed slope formula. □

39.3 Endpoint continuation and finitely many switches for every order

Here the order is no longer restricted to $\kappa \leq 1$. The earlier endpoint theorem supplies the analytic scalar $A(\kappa)$ and the unique double contact

$$h_1 = H_\kappa(1) = \kappa - A(\kappa), \qquad x_1 = h_1, \qquad M - h_1 > 0$$

[117]. The earlier central-rate bifurcation theorem also proves, for every fixed $\kappa > 1/2$, uniqueness of the boundary contact for all sufficiently small positive c, with $M - H_\kappa(c)$ and $x_\kappa(c)/c$ analytic in c^2 [116]. In particular that theorem establishes global uniqueness near zero, not merely a local formal expansion.

Proposition 39.3 (Endpoint control and global finiteness)**.** *For every fixed $\kappa > 1/2$, the boundary has a unique contact for all $c < 1$ sufficiently close to 1. The capacity and contact extend as analytic contact curves through $c = 1$, and*

$$-h_1^2 < H_\kappa'(1-) < 0.$$

There are only finitely many points in $(0, 1)$ at which the capacity changes its controlling analytic contact branch. This does not assert absence of switches when $\kappa > 1$.

Proof. Let $c_j \uparrow 1$ and let x_j be any boundary contacts. Continuity gives $H_\kappa(c_j) \to h_1$, while the inherited bound $x_j < (H_\kappa(c_j) + 1)/c_j$ bounds them above. The local series at zero, uniformly for η near $M - h_1 > 0$, bounds them away from zero. Every subsequential limit must therefore be a zero of the endpoint inverse. There is only one, namely x_1.

At that point $g_{xx} > 0$ and $g_h < 0$. The inverse series is analytic in parameters near this positive point, also for c slightly larger than 1. The response formula remains valid at $c = 1$, since its finite integrals require only $\eta > 0$. The implicit-function theorem gives a unique local contact curve. All boundary contacts for c close to 1 lie in this neighborhood, so they coincide with it. In the slope formula the ratio of the two integrals is a strict weighted mean of $y \in (0, x_1)$. It lies in $(0, x_1)$, giving the stated derivative bounds.

There are no switches in a small interval at either endpoint, by this argument and the inherited central-rate theorem. The remaining compact interval is covered by finitely many neighborhoods on which the capacity is a finite minimum of analytic functions. Differences of distinct analytic germs have isolated zeros; a finite collection has only finitely many crossings on a sufficiently small closed neighborhood. A finite subcover proves the claim. The analytic continuation past $c = 1$ is only a continuation of the contact equations; it is not an assertion of probabilistic feasibility there. □

39.4 Two orders: an atomic plateau and a genuine corner

Choose

$$\frac{1}{2} < \kappa_0 < \kappa_1 \leq 1, \qquad M_i = 2\kappa_i - 1, \qquad \delta = M_1 - M_0 \in (0, 1), \qquad \epsilon > 0. \tag{39.9}$$

Consider the probability law with transform

$$L(s) = \frac{\Phi(b+s)}{\Phi(b)}, \qquad \Phi(w) = \frac{w}{(w^2-1)^{\kappa_0}} + \epsilon \frac{w}{(w^2-1)^{\kappa_1}}. \tag{39.10}$$

It is a positive mixture of the two laws already considered. Let $K(c)$ be the greatest gamma shape extractable at rate $b-c$. For $0 < h < M_0$ put

$$q_{c,h}(x) = g_{\kappa_0,c,h}(x) + \epsilon g_{\kappa_1,c,h}(x).$$

At the endpoint $h = M_0$, the first summand has an atom of unnormalized mass one. Its continuous part is

$$k_c(x) = \sum_{n\geq 1} \frac{a_n(c, M_0) x^{n-1}}{\Gamma(n)}, \qquad k_c(0+) = -M_0 c, \tag{39.11}$$

where the coefficients use $\kappa = \kappa_0$. Define

$$Q_c(x) = k_c(x) + \epsilon g_{\kappa_1,c,M_0}(x). \tag{39.12}$$

Thus the full-shape candidate, before normalization and the tilt, is $\delta_0 + Q_c(x)\,dx$. More precisely the actual probability candidate is this tilted measure divided by $(b-c)^{M_0}\Phi(b)$.

Theorem 39.4 (Complete implicit diagram for a two-order mixture)**.** *Under* (39.9)*, the following assertions hold.*

1. *For $0 < c < 1$ and $0 < h < M_0$, $q_{c,h}$ has a unique nondegenerate global minimum and tends to $+\infty$ at both ends. The same assertions hold for Q_c. Positivity in either case is therefore an exact one-point minimum test.*

2. *There is a number $c_* \in (0,1]$ such that $K(c) = M_0$ for $0 \leq c \leq c_*$ within $[0,1)$, and $0 < K(c) < M_0$ for $c_* < c < 1$. For $0 < c < c_*$ the full-shape continuous density Q_c is strictly positive. For $c_* < c < 1$ the capacity has a unique double contact and is real analytic and strictly decreasing.*

3. *If $c_* < 1$, then Q_{c_*} has exactly one zero x_*, which is double. A unique analytic contact branch $h(c)$ passes through (c_*, M_0, x_*), and locally*

$$K(c) = \min\{M_0, h(c)\}. \tag{39.13}$$

Its right derivative is strictly negative whereas its left derivative is zero. Writing $Q = Q_{c_*}$, its exact slope is

$$K'(c_*+) = -M_0 \frac{e^{c_* x_*} + \int_0^{x_*} e^{c_* y} Q(x_* - y)\,dy}{e^{c_* x_*}/x_* + \int_0^{x_*} e^{c_* y} Q(x_* - y)\,dy/y} < 0. \tag{39.14}$$

Consequently the mixture has at most one corner, and it is a transition from an atomic shape ceiling to a moving contact, rather than a competition between two interior contacts.

Proof. For $h < M_0$, both component inverses satisfy Theorem 39.1. Their positive combination satisfies $q'' + q' > 0$ and has positive infinite limits at both ends. At $h = M_0$, the series gives local convergence $g_{\kappa_0,c,h} \to k_c$ together with derivatives away from zero. Hence $k_c'' + k_c' \geq 0$. The second summand has strictly positive $g'' + g'$ since its remaining exponent is $\delta \in (0,1)$. Thus $Q_c'' + Q_c' > 0$. At zero,

$$Q_c(x) = -M_0 c + O(x) + \frac{\epsilon x^{\delta-1}}{\Gamma(\delta)}(1 + O(x)), \tag{39.15}$$

and at infinity the component of order κ_1 dominates with the positive leading term in (39.8). This proves the first assertion, including nondegeneracy.

No shape larger than M_0 can be feasible: its normalized candidate Laplace transform grows as a positive constant times s^{h-M_0} as $s \to \infty$, whereas a probability Laplace transform is bounded by one. At $h = M_0$ the transform is the one of $\delta_0 + Q_c\,dx$, because subtracting its limit one at infinity leaves exactly the series (39.11) and the second component. The exponential tilt makes its total variation finite. The full shape is therefore feasible exactly when $Q_c \geq 0$.

At $c = 0$, all nonzero coefficients of the two inverse series are positive: in the first summand they are the even coefficients of $(1 - z^2)^{-\kappa_0}$, and the second has remaining exponent $\delta > 0$. Thus $Q_0(x) > 0$ for every $x > 0$. This strict positivity persists for small positive c. Indeed (39.15) is uniform for c in a small compact interval, so the positive divergent term controls a common neighborhood of zero. The positive large-x leading term is uniform there as well; its coefficient is bounded away from zero since c stays below one. The intervening compact interval is handled by continuity and its positive minimum. This proves an initial plateau of positive length.

For fixed shape, feasibility is preserved on decreasing c. To see this without a density comparison, if $c_1 < c_2$, the ratio of the normalized candidate transforms is the transform of a compound Poisson variable whose finite Levy measure is

$$h\frac{e^{-(b-c_2)x} - e^{-(b-c_1)x}}{x}\,dx > 0.$$

Its density away from its atom is strictly positive on the whole positive axis. Thus the feasible full-shape parameters form an initial interval. When a full-shape remainder at c_2 is feasible, it has a positive atom at zero, so its convolution with this compound Poisson variable has strictly positive continuous density at every $x > 0$. It follows that $Q_{c_1} > 0$. Pointwise continuity shows that the feasible interval is closed at any endpoint below one. Its endpoint is the asserted $c_* > 0$, with $c_* = 1$ if it fills $[0, 1)$.

For each strict c, some positive shape is feasible: choose h smaller than both individual capacities, and combine the two nonnegative inverses. Feasibility in shape is an interval, because decreasing a shape convolves the remainder with a gamma variable. It is closed: for a convergent sequence of feasible shapes the normalized transforms converge pointwise to a function continuous at $s = 0$ with value one; the continuity theorem for Laplace transforms gives a probability limit. The explicit inverse identifies this limit, including the possible atom at $h = M_0$. Thus $K(c)$ is attained.

If $K(c) < M_0$, its inverse must have a zero. Otherwise the uniform positive endpoint estimates and a positive minimum on the intervening compact set would permit a slightly larger shape. Its zero is the unique minimum and is double, by the first assertion. The response identities (39.4) hold for q as well, because both summands have the same transform multipliers under c and h differentiation. They give $q_c < 0$ and $q_h < 0$ at the contact. The implicit-function theorem, uniform endpoint estimates, and the strict unique minimum identify a locally analytic capacity, with derivative $-q_c/q_h < 0$. Uniqueness patches these branches throughout $(c_*, 1)$.

Suppose now $c_* < 1$. If Q_{c_*} were everywhere positive, the same compact-set and uniform-endpoint argument would preserve full-shape feasibility beyond c_*. It therefore has its unique zero x_*, with $Q''_{c_*}(x_*) > 0$. For fixed $x > 0$, the series (39.2) extends analytically in h through M_0: $1/\Gamma(\eta)$ is entire and its zero at $\eta = 0$ removes the $n = 0$ term pointwise. Thus its pointwise value there is exactly $Q_c(x)$.

The parameter responses at this atomic contact are

$$q_c(x_*) = -M_0 \left(e^{c_* x_*} + \int_0^{x_*} e^{c_* y} Q(x_* - y)\, dy \right) < 0, \tag{39.16}$$

$$q_h(x_*) = -\frac{e^{c_* x_*}}{x_*} - \int_0^{x_*} \frac{e^{c_* y} Q(x_* - y)}{y}\, dy < 0. \tag{39.17}$$

For the first identity the multiplier $-h/(w-c)$ acts on the measure $\delta_0 + Q\,dx$, and its atom gives the first displayed term. For the second, use the regularized identity

$$\log(w-c) = \int_0^\infty \frac{e^{-y} - e^{-(w-c)y}}{y}\, dy.$$

On the positive axis the atom contributes $-e^{cx}/x$. On the continuous part its inverse is

$$-(\gamma + \log x)Q(x) + \int_0^x \frac{Q(x) - e^{cy} Q(x-y)}{y}\, dy.$$

This follows by cutting the integral off at $y = \rho$, then letting $\rho \downarrow 0$. At the contact $Q(x_*) = 0$ it reduces to (39.17). The integrals converge at $y = 0$ by the double zero, and at $y = x_*$ because $Q(u) = O(u^{\delta-1}) + O(1)$ with $\delta > 0$. This calculation also agrees with the differentiated inverse series, and does not discard the atom in the logarithmic response.

The contact Jacobian is $q_h Q' \neq 0$, so there is a unique analytic contact curve $h(c), x(c)$ through this point. The strict negative slope $h' = -q_c/q_h$ gives (39.14). Locally on its right, $h(c) < M_0$; the unique minimum and uniform endpoint positivity identify it with the capacity. To justify uniformity while $h \uparrow M_0$, put $\eta = M_0 - h \geq 0$. The first component is a nonnegative term $x^{\eta-1}/\Gamma(\eta)$ plus a uniformly bounded remainder for small x; the second has the uniformly positive leading term $\epsilon x^{\delta+\eta-1}/\Gamma(\delta+\eta)$. Choose η small enough that $\delta + \eta < 1$. This term dominates the bounded remainder uniformly. The tail estimate is uniform because c_* is strictly below one. On the left the shape ceiling forces $K = M_0$. This proves (39.13) and completes the diagram. □

Corollary 39.5 (An explicit mixture with a nonzero corner)**.** *The source* (39.10) *with*

$$\kappa_0 = \frac{3}{4}, \qquad \kappa_1 = 1, \qquad \epsilon = \frac{1}{100}$$

has $0 < c_* < 1$*, and hence has the genuine corner in* (39.13)*. This holds for every* $b > 1$*.*

Proof. Here $M_0 = \delta = 1/2$. We show that its full-shape inverse is negative at $c = 1$ and $x = 1/100$, which by continuity is also true for some $c < 1$. For either $\kappa = 3/4$ or 1, the coefficients of $(1-z)^{1/2}(1-z^2)^{-\kappa}$ obey $|a_n| \leq 2^{n+1}$ by Cauchy's estimate on $|z| = 1/2$: the modulus there is at most $\sqrt{3/2}\,(4/3) < 2$. Since $a_1 = -1/2$, the first continuous component satisfies

$$k_1(x) \leq -\frac{1}{2} + 4(e^{2x} - 1) \leq -\frac{1}{2} + \frac{4}{49} \quad (x = 1/100).$$

For the second component, $(1/2)_n \geq n!/2^n$ gives

$$|g_{1,1,1/2}(x)| \leq \frac{2x^{-1/2}}{\sqrt{\pi}} e^{4x} < 20\frac{25}{24} = \frac{125}{6}.$$

We used $e^u \leq (1-u)^{-1}$ for $0 \leq u < 1$. Consequently

$$Q_1(1/100) < -\frac{1}{2} + \frac{4}{49} + \frac{5}{24} = -\frac{247}{1176} < 0.$$

The initial plateau has positive length by Theorem 39.4, and this strict inequality forces its endpoint to lie below one. The value of b changes only the positive tilt and normalization, not these signs. □

40 Exact adaptive recovery: dimension, topology, and sparse models

This section changes the recovery criterion in a deliberate way. The quantity $N(m)$ in the earlier observation papers concerns an algorithm for every prescribed positive error; the algorithm may change with the error. Here an *exact* algorithm must recover every input without error. We obtain a dimension theorem for this model, a cohomological obstruction sharp on closed manifolds, and an exact measurement theorem for sparse vectors with adaptive linear observations. These are structural transfers of classical topology and finite-candidate separation. They do not settle the unrestricted two-query approximation problem in $\mathbb{R}^3$, and their global priority is not asserted.

40.1 The exact model and its dimension interval

For a nonempty compact metrizable space X, let $Q(X)$ be the least number of deterministic adaptive continuous scalar queries that recover every $x \in X$ exactly. At a transcript $(y_1, \dots, y_{j-1})$, the algorithm selects a continuous function $f_{y_1,\dots,y_{j-1}} : X \to \mathbb{R}$ and receives its exact value. The selection rule and decoder are initially unrestricted. Algorithms with fewer than the allowed number of queries can be padded by constant queries. Set $Q(X) = \infty$ if no finite number suffices. Thus $Q(X) = 0$ precisely for a singleton. The feasible set at each realized transcript is compact. Queries defined only on such a feasible set may equivalently be extended to X by the Tietze extension theorem.

We use covering dimension. Two classical dimension theorems are the inputs: for a continuous map of compact metric spaces,

$$\dim X \le \dim f(X) + \sup_y \dim f^{-1}(y), \tag{40.1}$$

and a compact metric space of dimension at most d admits a continuous map to $[0,1]^d$ whose fibers have dimension zero. The first is the Hurewicz dimension-lowering theorem, recalled in [119]; the latter light-map characterization is stated in [120, Theorem 2.3]. No manifold or local connectedness hypothesis is imposed on the fibers.

Theorem 40.1 (Exact dimension interval)**.** *If X is a nonempty compact metrizable space of finite dimension d, then*

$$d \le Q(X) \le d + 1. \tag{40.2}$$

The upper bound can be realized with the first d queries fixed in advance, a Borel measurable rule selecting the last query in $C(X)$ with its uniform norm, and a Borel decoder. If $\dim X = \infty$, then $Q(X) = \infty$.

Proof. We first prove the lower bound by induction on the query budget n. For $n = 0$ the space is a singleton. For $n \ge 1$, let f be the first query. Its restriction to each compact fiber leaves an exact algorithm with $n - 1$ queries. The induction hypothesis and (40.1) give

$$\dim X \le \dim f(X) + \sup_y \dim f^{-1}(y) \le 1 + (n-1) = n.$$

This also proves the infinite-dimensional assertion.

For the upper bound choose a light map $g : X \to [0,1]^d$ and query its coordinates. Each fiber $F_y = g^{-1}(y)$ is a zero-dimensional compact metric space, so it embeds into the real line. An embedding of F_y extends continuously to X. Querying that extension separates all

remaining candidates and proves the upper bound with unrestricted selection. For $d = 0$ this construction is just a scalar embedding of the zero-dimensional compactum.

Here are details ensuring that no nonmeasurable choice is necessary. Write $Y = g(X)$ and $E = C(X)$ with its uniform norm, a separable Banach space. Fix a compatible metric ρ on X. For each $j \geq 1$ let

$$U_j = \{(y, h) \in Y \times E : h(x) \neq h(x') \text{ whenever } x, x' \in F_y,\ \rho(x, x') \geq 1/j\}.$$

The set U_j is open. Indeed, its complement is the projection of the closed conditions $g(x) = g(x') = y$, $\rho(x, x') \geq 1/j$, and $h(x) = h(x')$; compactness of $X \times X$ makes this projection closed. Every section $(U_j)_y$ is dense in E. To approximate a given h, partition the zero-dimensional compactum F_y into finitely many clopen sets of diameter less than $1/j$, on each of which the oscillation of h is small. Replace its values there by distinct nearby constants. The resulting small continuous perturbation on F_y extends with the same uniform bound to X, and belongs to $(U_j)_y$.

For completeness, a Borel selector from $\bigcap_j (U_j)_y$ can be constructed directly. Choose countable bases of Y and of open balls in E. Recursively, after a ball $B_{j-1}(y)$ has been chosen, take the first pair of basic sets V, B_j satisfying

$$y \in V, \qquad \overline{B_j} \subset B_{j-1}(y), \qquad \operatorname{diam} B_j \leq 2^{-j}, \qquad V \times \overline{B_j} \subset U_j.$$

At the first step omit the inclusion in the previous ball. Openness of U_j and density of its sections guarantee such a pair. For each fixed previous-ball index, the eligible pairs form a fixed countable list, and choosing the first one with $y \in V$ is Borel. The centers of these balls are Borel maps of y, and their nested closures have a unique limit $h_y \in E$. This limit is Borel and belongs to every $(U_j)_y$, hence is injective on F_y.

The transcript map $x \mapsto (g(x), h_{g(x)}(x))$ is therefore a Borel injection into $\mathbb{R}^{d+1}$. The Lusin–Souslin theorem gives a Borel image and a Borel inverse on that image [121, Corollary 15.2]. Extend the decoder arbitrarily at unrealized transcripts. This proves the stated measurable version. □

40.2 A cohomological obstruction sharp on closed manifolds

All cohomology below is Čech cohomology, with coefficients in an abelian group G. For compact metrizable spaces it agrees with cohomology of the constant sheaf. This choice matters because the feasible fibers may be wild compacta.

Theorem 40.2 (Top cohomology obstructs exact acquisition)**.** *If X has an exact adaptive continuous algorithm with $n \geq 1$ queries, then*

$$\check{H}^q(A; G) = 0 \qquad (q \geq n) \tag{40.3}$$

for every compact subset $A \subset X$ and every abelian group G. Consequently, if $\dim X = d \geq 1$ and $\check{H}^d(X; G) \neq 0$ for some G, then $Q(X) = d + 1$. In particular, for every nonempty closed d-manifold, $d \geq 1$,

$$Q(M) = d + 1. \tag{40.4}$$

Proof. We prove the assertion for X by induction on n; restricting the algorithm to a compact subset then proves the hereditary formulation. If $n = 1$, the query is a continuous injection into $\mathbb{R}$ and is a homeomorphism onto its compact image. Compact subsets of the real line have zero positive-degree Čech cohomology: their arbitrarily small neighborhoods can be taken to be finite unions of intervals.

Suppose $n \geq 2$ and write $f : X \to Y = f(X) \subset \mathbb{R}$ for the first query. The map is proper. Each fiber F_y is exactly recoverable with $n-1$ queries and therefore has $\check{H}^q(F_y; G) = 0$ for $q \geq n-1$. Proper base change gives

$$(R^q f_* \underline{G})_y \cong H^q(F_y; \underline{G}|_{F_y}) = \check{H}^q(F_y; G),$$

so $R^q f_* \underline{G} = 0$ for $q \geq n-1$. We use here the topological proper-base-change theorem, which applies to compact Hausdorff spaces without local contractibility assumptions [122, Tag 09V6].

The compact metric space $Y \subset \mathbb{R}$ has covering dimension at most one. Cohomology of every sheaf of abelian groups on Y vanishes above degree one. This is the standard covering-dimension bound for sheaf cohomology; for precise modern statements see [123, Theorem 7.2.3.6 and Corollary 7.2.2.30]. The Leray spectral sequence

$$E_2^{p,q} = H^p(Y, R^q f_* \underline{G}) \implies H^{p+q}(X; \underline{G})$$

thus has possible nonzero entries only for $p \leq 1$ and $q \leq n-2$. It follows that $H^k(X; \underline{G}) = 0$ for $k \geq n$, completing the induction. The dimension interval now proves the first consequence. A nonempty closed d-manifold has nonzero top cohomology with $\mathbb{F}_2$ coefficients, so the same argument proves (40.4). □

The interval in Theorem 40.1 is sharp at both ends: $Q([0,1]^d) = d$ by coordinate observations, whereas closed manifolds require $d+1$. Vanishing of top cohomology is not a converse to the obstruction. A tripod has dimension one and trivial positive-degree cohomology, but cannot embed into the real line, so its exact query count is two.

There is also a precise reason why this theorem does not sharpen the earlier metric-width bound. On the unit sphere $S^3 \subset \mathbb{R}^4$,

$$Q(S^3) = 4, \qquad \inf_{\text{three-query algorithms}} \sup_{x \in S^3} \|x - \widehat{x}\|_2 = 0. \tag{40.5}$$

The second statement follows by restricting the three-query arbitrary-precision algorithm of Krieg–Novak–Ullrich on $\mathbb{R}^4$ [124, Theorem 1]; if outputs are required on the sphere, radial normalization changes an error $\varepsilon < 1$ to at most 2ε. Thus the infimum is genuinely unattained. No universal positive-error conclusion can follow solely from the cohomological exact-recovery obstruction.

40.3 Finite polyhedra: adaptive linear observations beat embedding dimension

Proposition 40.3 (A finite last-query menu)**.** *Let $X \subset \mathbb{R}^M$ lie in the union of L affine subspaces, each of dimension at most d. There are d fixed linear queries followed by one adaptively chosen linear query that recover X exactly. The last query can be selected from a fixed finite menu of at most*

$$(M-1)\binom{L}{2} + 1 \tag{40.6}$$

linear functionals. For a finite union of affine subspaces the rule and decoder can be chosen semialgebraic; for a finite polyhedron the same statement holds after restricting candidates to its simplices.

Proof. Choose a linear map $A : \mathbb{R}^M \to \mathbb{R}^d$ whose restriction to each of the finitely many direction spaces is injective. Such maps form a nonempty open dense set, as witnessed by the

nonvanishing of the relevant rank minors. Each affine subspace then contributes at most one candidate to a fiber $A^{-1}(y)$, so there are at most L distinct candidates.

For distinct real numbers $t_1, \dots, t_T$, consider the functionals

$$\ell_t(x) = x_1 + t x_2 + \cdots + t^{M-1} x_M, \qquad T = (M-1)\binom{L}{2} + 1.$$

For each pair of distinct candidates, the polynomial $\ell_t(x - x')$ is nonzero and has degree at most $M-1$. Hence at most $(M-1)\binom{L}{2}$ menu entries fail to separate some pair. Select the first entry separating all candidates, and use its exact value to identify the input. A singleton fiber requires no separation and may use any menu entry.

For the measurability claim, the candidate in a particular affine subspace, when it exists, is an affine function of y on an affine set. Candidate validity and the finitely many tests of equality or inequality of scalar values are semialgebraic. Choosing the first successful entry and the first consistent candidate is therefore semialgebraic. Membership in a fixed simplex adds only finitely many linear inequalities. For a general subset X in the stated union, one may use all candidates from the containing affine subspaces; this already recovers every point of X. □

Let K_d be the d-skeleton of a $(2d+2)$-simplex, $d \geq 1$, in its usual Euclidean realization. Let $Q_{\mathrm{na}}(K_d)$ denote the minimum number of fixed continuous scalar queries for exact recovery. Then

$$Q(K_d) = d + 1, \qquad Q_{\mathrm{na}}(K_d) = 2d + 1. \tag{40.7}$$

Indeed, K_d has nonzero dth cohomology: the boundary of any $(d+1)$-simplex is a nonzero simplicial d-cycle, and there are no $(d+1)$-chains in K_d. The cohomology theorem proves the adaptive lower bound, and Proposition 40.3 gives an adaptive *linear* algorithm attaining it. Fixed continuous exact queries are equivalent to an embedding in their Euclidean transcript space. The van Kampen–Flores theorem rules out an embedding of K_d in $\mathbb{R}^{2d}$; see the primary formulation in [125]. A generic placement of its vertices in $\mathbb{R}^{2d+1}$ gives a linear embedding: any two simplices involve at most $2d+2$ vertices, which are affinely independent, so their convex hulls intersect only in their common face. This proves (40.7). The separation concerns exact deterministic acquisition, rather than a stable compressed-sensing guarantee.

40.4 An exact adaptive linear threshold for sparse vectors

Write

$$\Sigma_s^m = \{x \in \mathbb{R}^m : |\operatorname{supp} x| \leq s\}, \qquad 1 \leq s < m.$$

All measurement values in the next theorem are exact real numbers. Adaptive choices need not vary continuously with previous values.

Theorem 40.4 (Sparse exact linear query count)**.** *The minimum number of deterministic adaptive linear scalar queries recovering every $x \in \Sigma_s^m$ exactly is $s+1$. This remains the minimum for $\Sigma_s^m \cap \overline{B}_2(0, R)$ for every $R > 0$. An optimal algorithm uses s fixed queries followed by one query selected semialgebraically from a finite menu. For the lower bound, the last query may even be an arbitrary continuous function, provided the preceding $s-1$ queries are linear.*

Proof. For the upper bound apply Proposition 40.3 to the $\binom{m}{s}$ coordinate subspaces of dimension s. Explicitly, take an $s \times m$ matrix A all of whose $s \times s$ column minors are nonzero, for example a Vandermonde matrix with distinct nodes. Every support set S of size s gives

one candidate $x^S(y)$, obtained by solving $A_S x_S = y$ and setting the other coordinates to zero. Every input with smaller support is included by extending that support to size s. Duplicate candidates are harmless. The finite menu in (40.6) separates the distinct candidates after s fixed observations. The same algorithm recovers a bounded sparse class without modification.

Suppose, conversely, that s queries suffice with the first $s-1$ linear and the last continuous. Follow the transcript generated by $x = 0$ for the first $s-1$ queries. The selected linear functionals are then fixed. Restrict their common kernel to a coordinate $(s+1)$-plane $E \subset \mathbb{R}^m$; the resulting linear space $W \subset E$ has dimension at least two. We claim that $W \cap \Sigma_s^m$ contains two distinct line segments crossing at zero.

If W lies in one coordinate hyperplane of E, every vector of W is s-sparse and any two independent directions work. Otherwise, all $s+1$ coordinate functionals restrict nontrivially to W. If their kernels all coincided, these functionals would be proportional, and their common kernel would have codimension one in W. But the common kernel of all coordinates of E is zero, which would force $\dim W = 1$, a contradiction. Thus two of their kernels are distinct. Each has positive dimension; choose nonzero, nonproportional vectors from two such kernels. Each vector, and its entire line, has a zero coordinate in E and is s-sparse. Scaling the segments puts their cross inside the ball of any prescribed radius $R > 0$.

All points of this cross produce the same first $s-1$ observations as zero, by induction along that transcript. The final selected continuous scalar query would therefore have to be injective on the cross. This is impossible. A continuous injection of a compact cross into $\mathbb{R}$ would be a homeomorphism onto a compact interval, whereas removing its crossing point gives four connected components and removing a point from an interval gives at most two. This contradiction proves the lower bound. □

Proposition 40.5 (A uniform sparse approximation obstruction)**.** *Every s-query algorithm on $\Sigma_s^m \cap \overline{B}_2(0,R)$, $1 \le s < m$, whose first $s-1$ queries are linear and whose final query is continuous has worst-case Euclidean error at least $R/2$. The decoder and the selection of queries are unrestricted.*

Proof. Follow the zero transcript for the first $s-1$ observations, and form W inside a coordinate $(s+1)$-plane as in the preceding proof. If W is contained in a coordinate hyperplane, it contains a Euclidean two-dimensional disk of radius R consisting entirely of sparse vectors. The final scalar query identifies two antipodal boundary points by the Borsuk–Ulam theorem, giving the stronger error bound R.

Otherwise choose a two-dimensional subspace $V \subset W$ that is contained in none of the coordinate hyperplanes. Such a choice exists: take a vector of W outside their finite union and extend it to a two-dimensional subspace. Identify V isometrically with $\mathbb{R}^2$. Its $s+1$ coordinate functionals are nonzero row vectors r_i satisfying $\sum_i r_i r_i^{\mathsf{T}} = I_2$. If u_i is a unit direction perpendicular to r_i, and $w_i = \|r_i\|_2^2 > 0$, then

$$\sum_i w_i u_i u_i^{\mathsf{T}} = I_2, \qquad \sum_i w_i = 2. \tag{40.8}$$

The line segments $[-Ru_i, Ru_i]$ belong to the sparse feasible set.

Write f for the final selected query. If an endpoint has value $f(0)$, it and zero already form a same-value pair at distance R. If the two endpoints of any segment have values strictly on the same side of $f(0)$, take the endpoint whose value is closer to $f(0)$ and use the intermediate value theorem on the opposite half-segment. This again gives a same-value pair at distance at least R.

In the remaining case orient each u_i so that $f(Ru_i) > f(0)$; orientation does not alter (40.8). At least two of these unit vectors have nonpositive inner product. To see this, suppose

all pairwise inner products were positive. The vector $v = \sum_i w_i u_i$ has positive inner product with every u_i, so their angular representatives lie in a single open semicircle. Pairwise positivity then places all of them in an arc of angular length strictly less than $\pi/2$. For its bisector a we have $(a \cdot u_i)^2 > 1/2$ for every i, contradicting $1 = \sum_i w_i (a \cdot u_i)^2 > \frac{1}{2} \sum_i w_i = 1$.

Choose u_i, u_j with $u_i \cdot u_j \le 0$, and suppose without loss that $f(Ru_i) \le f(Ru_j)$. By the intermediate value theorem there is $t \in [0, R]$ with $f(tu_j) = f(Ru_i)$. Their distance satisfies

$$\|Ru_i - tu_j\|_2^2 = R^2 + t^2 - 2Rt(u_i \cdot u_j) \ge R^2.$$

They have the same entire transcript, so any common reconstruction errs by at least half their distance on one of them. □

For comparison, the exact nonadaptive linear count on Σ_s^m is $\min(2s, m)$. Indeed, injectivity is equivalent to the kernel containing no nonzero vector supported on at most $2s$ coordinates: every such vector is a difference of two s-sparse vectors. Fewer than $\min(2s, m)$ rows always have such a kernel vector by a column-dependence argument, while a full-spark matrix with $\min(2s, m)$ rows attains the bound. Thus the adaptive result improves the exact deterministic count from $2s$ to $s + 1$ when $m \ge 2s$. This comparison should not be confused with the much stronger noise-stable and approximately sparse recovery requirements studied, for example, in [126]. Our finite-candidate algorithm has no uniform noise guarantee, and its candidate enumeration can be exponentially expensive in m.

40.5 The non-dyadic approximation target remains open

The primary version of [124], dated September 2026, explicitly asks whether two adaptive continuous queries approximate all of $\mathbb{R}^3$ uniformly to arbitrary prescribed accuracy. The earlier width theorem in this project gives $\lceil \log_2(m+1) \rceil \le N(m) \le \lceil \log_2 m \rceil + 1$, and hence $2 \le N(3) \le 3$. The present dimension and cohomology arguments do not remove that gap. They start from injective final fibers, whereas an approximate algorithm only forces its final fibers to have small diameter. Replacing one condition by the other would invalidate both inductions. Equation (40.5) is an explicit counterexample to that replacement as a general principle. Accordingly, the non-dyadic approximation problem is recorded as an attempted but unresolved external target, not as a consequence of the exact theorems.

41 Failed shortcuts and the next proof obligations

The successful transfers do not remove the distinction between an exact classification and a route toward an external open problem. This section records specific failures that affect how the results can be used.

41.1 A contour bound cannot simply lose its prefactor

In the earlier collected manuscript, shifted Chebyshev polynomials have Bernstein coefficients defined by

$$T_m(2x - 1) = \sum_{j=0}^{t} C(t, m, j) \binom{t}{j} x^j (1 - x)^{t-j}.$$

The constant-one exponential estimate was already proved there. The remaining best-constant question is whether $|C(t, m, j)| \le 2^{m^2/t}$ for all $m \le t$, with the inherited bounds $\log 2 \le c_* \le 1$ for the optimal exponent constant. It is not a new closure in this paper.

One possible strengthening of the contour method would be

$$|C(t,m,j)| \overset{?}{\le} \exp\{t\mathsf{H}(m/t)\}, \qquad \mathsf{H}(a) = \tfrac{1}{2}[(1+a)\log(1+a) + (1-a)\log(1-a)].$$

This strengthening is false, even though the same expression with a prefactor occurs in a valid earlier estimate. Indeed, at $(t,m,j) = (6,4,1)$,

$$C(6,4,1) = 1 - \frac{32}{6} = -\frac{13}{3}, \qquad \exp\{6\mathsf{H}(2/3)\} = \frac{3125}{729}, \qquad \frac{13}{3} = \frac{3159}{729} > \frac{3125}{729}.$$

The coefficient identity follows from $T_4(2x-1)$ having constant coefficient 1 and derivative -32 at zero, and the identity $p'(0) = t(b_1 - b_0)$ for degree-t Bernstein coefficients. This is a counterexample to the proposed shortcut, not to the weaker $2^{m^2/t}$ conjecture.

41.2 Four boundaries that remain after the new theorems

Identification versus rates. The unrestricted two-replicate theorem leaves a quantitative inverse problem. Its proof uses a nonzero Fourier frequency and a weight depending on that frequency; it does not provide a uniform finite-data estimator. The explicit large-scale instability forbids uniform continuity in the stated unrestricted latent metric for any fixed group size. A useful next theorem must therefore specify tail constraints, a weaker metric, or a local parameter class before claiming a rate.

Differential classification versus special values. The finite-spectrum classification concerns functions and their analytic continuations. It does not decide whether an optimized value at one algebraic setting is algebraic or transcendental. The exact jet field identifies the available functional information; it is not a specialization theorem. Similarly, the moment-based square-root rate is a credited classical comparison after a new observation-coordinate conversion.

Exact acquisition versus approximation. The manifold obstruction applies to a single algorithm attaining zero error. It does not exclude a sequence of algorithms with errors tending to zero. The sphere example proves that such an inference would be false. The problem $N(3)$ therefore needs a genuinely quantitative obstruction or a new approximation construction; exact top cohomology alone cannot settle it. The sparse theorem likewise uses exact scalar values and a discontinuous adaptive choice; it makes no claim against noisy recovery lower bounds.

Relative algebraic closure versus all units. The arithmetic closure theorem deals exactly with elements already algebraic over a specified exponential field. It excludes an algebraic route to an exotic unit, but does not prove that every unit must take that route. Nor does a product lying in that field make either factor algebraic over it. The joint Ein theorem similarly classifies all possible finite extensions without asserting that its one possible exception actually occurs.

41.3 Research judgment

The unrestricted pair-identification theorem and the finite-spectrum differential classification are the strongest candidates for focused follow-up papers: each resolves a complete family and has a proof mechanism not restricted to a single numerical instance. The algebraic optimizer decision procedure is another reusable result, but its current input model and smoothness

promise must be kept explicit. The extraction corner provides a concrete analytic transition, and the exact-query theorems give sharp results in a different observation model.

Whether any of these formulations has publication-level novelty requires broader specialist comparison and external checking. The present work does not justify claiming an Annals-level breakthrough, a Fields-level result, or a newly solved historical named conjecture. It does provide explicit statements, proofs, counterexamples, and a finite list of next obligations on which such an assessment can be based.

Part VI
Logic and Effective Transfer

Editorial integration note. The fifty-seven-entry register is preserved in full, including version distinctions and qualified external announcements. Its status columns are dated source assessments, not a fresh literature survey performed for this compilation. Solved restricted cases are not counted as solutions of the unrestricted listed problems.

Original part framing.

Reading convention. "Proved here" describes the argument supplied in this document; it does not certify novelty. "Recent preprint" identifies the authors' stated result and its checked scope, not external peer review or formal validation. A proof audit of selected steps is not a certification of an entire paper. All bibliographic status statements refer to sources checked on 16–17 October 2026. The original A/B/C labels are retained as identifiers, not as a ranking endorsed by this document.

42 Scope, evidence, and the strongest transfer

The task is to try every item, not to assume that a list of celebrated conjectures must contain one that the available methods can settle. We therefore keep the user's numbering fixed. Two entries explicitly contain two questions: Skolem and positivity (37), and BMTO and MTWO (43). Both components are addressed. The two Shelah categoricity labels (7 and 29) are retained separately and their potentially different meanings are explained. Broad labels such as "main gap", "universality spectrum", and "cardinal transfer for omitted types" require a specified class, language, or cardinal range before they have a single truth value.

Earlier observation and optimization manuscripts motivate three routes: arithmetic information in logarithmic identities, spectral information in finite observations, and Borel coding of objects. The present document states the hypotheses needed for each transfer explicitly. It does not infer a theorem about arbitrary models, arbitrary Borel equivalence relations, or arbitrary proof systems merely from a result about a finite or identifiable observation model.

An exact boundary rather than an analogy. Let $\mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}$. Consider polynomial predicates over $\mathbb{A}$ together with predicates

$$P_i(\mathbf{x}) > 0 \ (1 \le i \le m), \qquad \sum_{i=1}^{m} \alpha_i \log P_i(\mathbf{x}) = 0, \quad \alpha_i \in \mathbb{A} \text{ fixed.}$$

Arbitrary first-order formulas formed from these predicates are effectively equivalent, on $\mathbb{A}$, to formulas in the language of real closed fields. In contrast, adding just

$$R(x,y) \quad \Longleftrightarrow \quad y > 0 \ \text{ and } \ x \log 2 = \log y$$

defines $\mathbb{Q}$ by $\exists y\, R(x,y)$ and makes the full first-order theory undecidable. Theorem 43.1 supplies the proof, including the algebraic-domain restriction and the external transcendence and definability inputs. This does not decide the theory of the real exponential field, and it does not assert undecidability of an existential fragment. Its significance is a precise failure point for transferring fixed-coefficient arithmetic tests to variable-coefficient logic.

What a successful closure would require. For an original named problem, the target statement, hypotheses, and conclusion must match those in its source; the new argument must discharge the missing step. A finite subcase, a conditional reduction, a recent result of another author, or a reformulation of the obstruction is useful research information but is not such a closure. No original closure of a previously open named problem is established here. In particular, the number of separate entries or proved auxiliary statements is not evidence of Annals-level significance.

How to use the dossier. The original-order index below locates all fifty-seven targets. The six technical sections group related proof attempts so that shared assumptions and failed transfers can be checked together. Each entry records a concrete attempt and its remaining obstruction. The final synthesis identifies which missing conclusions would actually advance the original targets.

42.1 Original-order index: all fifty-seven entries

The short status column is a navigation aid; qualifications, versions, sources, attempts, and verdicts appear at the linked pages. "Unresolved" does not certify the absence of every unpublished claim.

ID	Requested target	Status or necessary distinction	Page
1 (A)	Partition Principle implies Choice in ZF	Unresolved; countermodel claim withdrawn	210
2 (A)	Superpolynomial lower bounds for Frege proofs	Unrestricted bound unresolved	233
3 (B)	Continuum problem after Cohen	CH independent of ZFC; new-axiom question differs	211
4 (B)	Vaught's conjecture	General first-order case unresolved	207
5 (B)	Martin's conjecture on Turing degrees	General case unresolved; order-preserving case known	219
6 (B)	Tarski's exponential-function problem	Unresolved over all reals; restricted boundary proved here	207
7 (B)	Shelah's categoricity conjecture	Exact infinitary threshold distinguished from eventual transfer	207
8 (B)	Shelah's PCF conjecture	Two variants; recent relative counterexample claim	212
9 (B)	Woodin's Omega conjecture	Completeness direction unresolved	213
10 (B)	HOD conjecture	Conditional negative results; consistency premise essential	214
11 (B)	Consistency of Reinhardt cardinals	Choice-free case unresolved; 2026 claim withdrawn	215
12 (B)	Automorphisms of the Turing degrees	Global rigidity unresolved	225
13 (B)	NIP field conjecture	General case unresolved; finite-rank results differ	208
14 (B)	Weiss's hyperfiniteness conjecture	General Borel case unresolved	219
15 (B)	Fraïssé in ATR_0	Laver theorem known; ATR implication unresolved	234
16 (B)	Kolmogorov-Loveland versus Martin-Lof randomness	Equality unresolved; encoded result does not imply it	226
17 (B)	Π^1_1 conservation for RT^2_2	Partial theorem; new full claim not verified	235
18 (B)	Kreisel's conjecture	Kleene-presentation version unresolved	226
19 (B)	Sacks's embedding conjecture	General embedding problem unresolved	227
20 (B)	Borel Ruziewicz problem	Borel version distinguished from completed-measure theorem	220
21 (B)	Roitman: $\mathfrak{d} = \aleph_1 \Rightarrow \mathfrak{a} = \aleph_1$	Recent positive proof claim; core construction checked	215
22 (B)	Consistency strength of PFA	General equiconsistency unresolved	216
23 (B)	Topological Vaught conjecture	General Polish-action case unresolved	220
24 (B)	Reverse-mathematical strength of Hindman's theorem	Exact strength unresolved; new avoidance result partial	235
25 (B)	Ordinal analysis of Π^1_2-CA_0	Exact axiom system needs matching ordinal calibration	237
26 (B)	Friedman's grand conjecture concerning EFA	Publication corpus and formalization must be specified	237
27 (B)	Zilber's quasiminimality conjecture	Full complex exponential case unresolved	208
28 (C)	Cherlin-Zilber conjecture	Finite-Morley-rank conjecture unresolved	209
29 (C)	Shelah's eventual categoricity conjecture	Full AEC proof claims; long proofs not certified here	207

ID	Requested target	Status or necessary distinction	Page
30 (C)	Main gap conjecture	Canonical countable first-order case already a theorem	209
31 (C)	Kueker's conjecture	General case unresolved; stable case known	209
32 (C)	Stable field conjecture	General case unresolved; large-field case known	208
33 (C)	Stable forking conjecture	Recent external counterexample preprint	210
34 (C)	Universality of C-free graphs	General classification not verified solved; tree case known	241
35 (C)	Universality spectrum problem	Precise spectrum convention selected; reduction proved	241
36 (C)	Generalized star-height problem	Recent height-three bound claim; height-one question remains	242
37 (C)	Skolem problem and LRS Positivity	Both general problems unresolved; promised class decided here	228
38 (C)	Finite-model property of Henson graphs	Pseudofiniteness, not a finite model of the full theory	243
39 (C)	Finiteness of reducts of homogeneous structures	Finite relational version unresolved; arity reformulation proved	221
40 (C)	O-minimal theories with a superexponential function	Transexponential target; competing indexed claim unverified	244
41 (C)	Categoricity transfer in atomic classes	Underspecified; precise high-cardinal benchmark examined	222
42 (C)	Algebraic closure of characteristic-zero minimal fields	Characteristic-zero case unresolved	245
43 (C)	Decidability of BMTO and MTWO	BMTO decidable; full MTWO not verified resolved	229
44 (C)	Decidability of Laurent series fields	Positive-characteristic pure-field target unresolved	230
45 (C)	Compact logics with Beth and Delta interpolation	Literal existence known; interpolation separation differs	217
46 (C)	Cardinality transfer for models omitting types	Sharp Hanf bound known; specific transfers need parameters	222
47 (C)	Barendregt-Geuvers-Klop conjecture	Whole-PTS implication unresolved	238
48 (C)	Decidability of multiplicative-exponential linear logic	General case unresolved; effective bound missing	246
49 (C)	Borel boundedness of countable Borel equivalence relations	General boundedness and converse questions unresolved	223
50 (C)	Are all Polish groups graphic?	Many graphic classes known; general case unresolved	223
51 (C)	Scott sets as standard systems of PA models	At most aleph-one known; arbitrary size unresolved	231
52 (C)	Increasing unions of hyperfinite countable Borel equivalence relations	General increasing-union problem unresolved	224
53 (C)	Word problem for the S combinator	Normalization decidable; word problem unresolved	246
54 (C)	Expansion postponement in pure type systems	Normalizing-types case known; unrestricted version unresolved	239
55 (C)	Proof-theoretic strength of the graph minor theorem	GMT true; exact proof strength not calibrated here	239
56 (C)	Decidability of three-pair Post correspondence	Ordinary three-pair problem not verified resolved	247
57 (C)	The $\forall\exists$ theory of c.e. degrees	Pure-order two-block theory not verified resolved	232

43 Model theory: an effective logarithmic boundary and eleven distinct problems

The eleven labels considered here do not denote eleven interchangeable conjectures. Some have several inequivalent formulations, the canonical countable first-order main gap is already a theorem, and two labels have very recent primary manuscripts announcing resolutions. We distinguish those external claims from results proved in this section. The status review is dated 17 October 2026. Our positive mathematical contribution is the following precisely delimited transfer of the logarithmic splitting method; its priority as a formulation is not certified.

43.1 Fixed versus variable logarithmic coefficients

Let $A = \overline{\mathbb{Q}} \cap \mathbb{R}$, with its usual ordered-field structure. Algebraic inputs are represented by integer minimal polynomials and rational isolating intervals. All logarithms below are ordinary real logarithms, evaluated externally in $\mathbb{R}$. For a finite list of specified coefficients $\lambda_i \in A$ and polynomials $P_i \in A[X_1, \dots, X_n]$, introduce the relation

$$Z_{\lambda,P}(x) \quad \Longleftrightarrow \quad \bigwedge_i P_i(x) > 0 \quad \text{and} \quad \sum_i \lambda_i \log P_i(x) = 0, \qquad x \in A^n. \tag{43.1}$$

In particular, the relation is false outside its positivity guard. The subscripts are effective finite syntactic data, not quantified variables. Let $\mathcal{A}_{\mathrm{fix}}$ have all these effectively indexed relation symbols and names for algebraic constants.

Theorem 43.1 (A decidability and independence boundary)**.** *The following assertions hold.*

1. *Every relation (43.1) has an effectively computable semialgebraic definition over A. Consequently the complete first-order theory of $\mathcal{A}_{\mathrm{fix}}$ is decidable, with arbitrary quantifier alternation over A. This is a definitional expansion of a real closed field; its theory is o-minimal and NIP.*

2. *Add instead just the binary relation*

$$P(x, y) \quad \Longleftrightarrow \quad y > 0 \text{ and } x \log 2 = \log y, \qquad x, y \in A. \tag{43.2}$$

   In $(A; 0, 1, +, -, \cdot, <, P)$ the sets $\mathbb{Q}$ and $\mathbb{Z}$ are definable without parameters. Its full first-order theory is undecidable and has the independence property; the structure is not o-minimal.

These conclusions do not settle decidability of the real exponential field. No existential undecidability conclusion is asserted.

Proof. We use Baker's theorem that logarithms of nonzero algebraic numbers which are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$ [62]. In the present proof all logarithms are real, so branch choices do not arise.

Compute a $\mathbb{Q}$-basis $\theta_1, \dots, \theta_d$ of the span of the finitely many λ_i, and write $\lambda_i = \sum_k r_{ki}\theta_k$, $r_{ki} \in \mathbb{Q}$. This is effective: embed the input numbers in a finite number field using exact algebraic arithmetic, express them in a rational basis of that field, and perform rational row reduction. Zero coefficients and the case $d = 0$ cause no difficulty.

Fix an algebraic tuple x satisfying the positivity guard. Choose a $\mathbb{Q}$-basis $\rho_1, \dots, \rho_s$ of the span of the real numbers $\log P_i(x)$. Each ρ_j can be chosen from those logarithms. Write

$\log P_i(x) = \sum_j q_{ij}\rho_j$ with rational q_{ij}. Baker's theorem and then the independence of the θ_k give

$$\begin{aligned}\sum_i \lambda_i \log P_i(x) = 0 &\iff \sum_k \theta_k \sum_i r_{ki} q_{ij} = 0 \quad \text{for every } j \\ &\iff \sum_i r_{ki} q_{ij} = 0 \quad \text{for every } k, j \\ &\iff \sum_i r_{ki} \log P_i(x) = 0 \quad \text{for every } k.\end{aligned}$$

If all logarithms vanish, this is the same argument with an empty basis. Choose positive integers D_k clearing denominators and put $n_{ki} = D_k r_{ki} \in \mathbb{Z}$. By positivity, the last conditions are equivalent to the polynomial equations

$$\prod_{i:n_{ki}>0} P_i(x)^{n_{ki}} = \prod_{i:n_{ki}<0} P_i(x)^{-n_{ki}} \qquad (1 \le k \le d). \tag{43.3}$$

An empty product is one. Importantly, the algorithm computes the r_{ki}, but never needs to find a basis of the logarithms at an input point. That latter basis was used only to prove correctness.

Replace each occurrence of (43.1) in a formula by its guard and (43.3). Structural induction on formulas preserves truth, including negation and every alternation of quantifiers, because the pointwise equivalence holds at every tuple in A. The field A is real closed. Effective real-closed-field quantifier elimination with exact algebraic coefficients therefore decides every resulting sentence [127]. Algebraic constant names can also be removed using their polynomial and isolating-interval definitions. The new relations introduce no new definable sets; o-minimality and NIP follow from the corresponding properties of real closed fields. This proves the first assertion.

For the second assertion set $D(x) := \exists y\, P(x, y)$. If $x = p/q$ is rational, with $q > 0$, the positive real number $2^{p/q}$ is algebraic, so $D(x)$ holds. If $x \in A \setminus \mathbb{Q}$, the Gelfond–Schneider theorem (also treated in [62]) makes 2^x transcendental, so $D(x)$ fails. Thus $D(A) = \mathbb{Q}$. Julia Robinson's definition of the integers in the rational field, with every quantifier relativized to D, now defines $\mathbb{Z}$ in our structure [128]. The same relativization interprets the full first-order theory of the rational field. Its undecidability, proved by Robinson, gives the claimed undecidability.

For completeness, let Z denote the resulting integer predicate and consider the single formula

$$\delta(p, n) := Z(p) \wedge Z(n) \wedge \exists k\, (Z(k) \wedge n = pk).$$

For any r choose distinct positive rational primes $p_1, \dots, p_r$. For each subset $S \subseteq \{1, \dots, r\}$ take $n_S = \prod_{i \in S} p_i$, with $n_\varnothing = 1$. Unique factorization yields $\delta(p_i, n_S)$ if and only if $i \in S$. This is the independence property. Finally, the definable set $D(A) = \mathbb{Q}$ is dense and codense in the ordered field A, so it is not a finite union of points and intervals. Hence the expansion is not o-minimal. □

Remark 43.2 (Why this is not a solution of Tarski's problem). The equivalence (43.3) is proved on algebraic tuples, not on arbitrary real tuples. In addition, A is not a definable domain in the o-minimal real exponential field. Restricting an external relation to A and quantifying over A changes the structure: the sentence $\forall x \exists y\, P(x, y)$ is true over $\mathbb{R}$ and false over A, for instance at $x = \sqrt{2}$. Thus the negative half of the theorem cannot be transferred to $\mathbb{R}_{\exp}$ by merely reusing the displayed formula. The undecidability proof uses full first-order arithmetic of $\mathbb{Q}$; it does not settle Hilbert's tenth problem over $\mathbb{Q}$.

43.2 ID 4: the first-order Vaught conjecture

The precise statement is that a complete theory T in a countable first-order language, with infinite models, has either at most countably many or exactly $2^{\aleph_0}$ countable models up to isomorphism. Pillay–Tanović's August 2026 revision still treats this as open [129]. Results or announcements for a single $L_{\omega_1,\omega}$ sentence are not automatically results for this first-order formulation.

The attempted transfer is to construct complete measurable observational invariants of countable models. Its precise sufficient condition is classical: if isomorphism on the standard Borel space $\mathrm{Mod}_\omega(T)$ is coanalytic, Silver's dichotomy gives countably many classes or a perfect set of pairwise nonisomorphic models [130]. In the latter case there are $2^{\aleph_0}$ classes; in the former there are at most countably many. The model space is Borel because satisfaction of each first-order sentence is Borel and T is countable. This proves the conditional transfer. In general isomorphism is analytic, and no coanalyticity argument follows from finite-dimensional recovery. Producing the required complete invariant or complexity bound is the missing step. Verdict: no resolution of the general first-order conjecture.

43.3 ID 6: Tarski's real exponentiation decision problem

The target is an algorithm deciding every sentence in $\mathrm{Th}(\mathbb{R}; 0, 1, +, -, \cdot, <, \exp)$, with quantifiers over all real numbers. The unconditional problem remains open in the reviewed primary literature. Macintyre–Wilkie's conditional decidability theorem under the appropriate real Schanuel conjecture is reviewed in Krapp's 2024 paper [131]. Theorem 43.1 supplies an unconditional algorithm for an explicit logarithmic equality language over A, and locates an undecidable extension of that language. The exact failures of a transfer to Tarski's problem are the algebraicity hypothesis in Baker's theorem and the nondefinability of the quantified domain A, as explained in Remark 43.2. Neither arbitrary real exponential zeros nor inequalities with variable logarithmic coefficients have been eliminated here. Verdict: a proved restricted decision boundary, not a closure of ID 6.

43.4 IDs 7 and 29: two categoricity thresholds

For ID 7 we select the infinitary formulation explicitly posed in Shelah's Problem 6.9 [132]: a countable-language $\psi \in L_{\omega_1,\omega}$ categorical in one $\lambda \geq \beth_{\omega_1}$ should be categorical in every $\lambda' \geq \beth_{\omega_1}$. If ordinary countable first-order categoricity is intended instead, its uncountable transfer is Morley's theorem, not an open conjecture. Vasey's established universal-sentence theorem proves an eventual transfer at the larger bound $\beth_{\beth_{\omega_1}}$ [133]; it does not give the displayed threshold for arbitrary infinitary sentences.

For ID 29 the general abstract-elementary-class formulation is: for every infinite cardinal λ there is $\mu(\lambda)$ such that every AEC K with $\mathrm{LS}(K) \leq \lambda$ which is categorical in one $\kappa \geq \mu(\lambda)$ is categorical in all $\kappa' \geq \mu(\lambda)$. Here categoricity includes existence, and amalgamation, tameness, and absence of maximal models are not added as assumptions.

There are recent full proof claims that a current status audit must record. OpenAI's 24 September 2026 primary manuscript states precisely this uniform assertion as Theorem 1.2 [134]. Espíndola's 2019 preprint, revised in 2022, also announces a general proof [136]. This section checked the statements and their scopes; it does not independently certify their long proofs. Accordingly the status recorded here is "recent full proof claims, not independently verified in this report", rather than either an unqualified open or an independently proved closed status. An accompanying OpenAI manuscript claims a CH obstruction to a prescribed AEC threshold [135]. A counterexample to one specified bound is compatible with qualitative

eventual categoricity.

The attempted transfer between these two IDs is rigorous but limited. Choose a countable fragment F of $L_{\omega_1,\omega}$ containing ψ and closed under subformulas and the usual fragment operations. Then $(\operatorname{Mod}\psi, \preccurlyeq_F)$ is an AEC with Löwenheim–Skolem number $\aleph_0$. Coherence is inherited from agreement on formulas; the union-of-chains property follows by induction on formulas in F, with the existential step being the Tarski–Vaught argument. A countable closure under chosen witnesses for the countably many formulas gives downward Löwenheim–Skolem, and the same construction over a set B has size at most $|B| + \aleph_0$. Isomorphism invariance is immediate. Therefore a uniform AEC eventual theorem implies eventual categoricity for infinitary sentences with some common threshold. It supplies no inequality $\mu(\aleph_0) \leq \beth_{\omega_1}$ and hence does not by itself settle the specified ID 7. Our observational recovery arguments do not supply the missing amalgamation, type locality, or cardinal bound. Neither external proof claim is counted as our result.

43.5 IDs 13 and 32: NIP fields and stable fields

The field formulation of Shelah's NIP conjecture used here is: every NIP field is finite, real closed, separably closed, or admits a nontrivial henselian valuation. Johnson's 2026 ICM article states it as Conjecture 4.2 [137]. His dp-finite theorem establishes the relevant classification under finite dp-rank [138]. NIP provides finite VC dimension separately for each formula; it does not assert a uniform finite dp-rank for the field. Thus interpreting finite-observation bounds as a finite-rank bound would add an unproved hypothesis. The arbitrary NIP case is not closed here.

The stable fields conjecture says that every infinite field with stable pure field theory is separably closed. Separably closed cannot be replaced by algebraically closed in positive characteristic. Johnson–Tran–Walsberg–Ye prove the assertion for large stable fields [139]; Kaplan–Matthews–Walsberg's July 2026 work gives a further proof and applications in the large-field setting [140]. The general case is not resolved by either hypothesis-restricted statement.

Proposition 43.3 (A classical conditional reduction)**.** *The NIP field conjecture in the displayed formulation implies the stable fields conjecture.*

Proof. Let K be an infinite stable field. Stability implies NIP. Under the NIP field conjecture, K is real closed, separably closed, or nontrivially henselian. The real-closed case is impossible: the order is definable by nonzero squares and has the order property. A field with a nontrivial henselian valuation is large. The established theorem that large stable fields are separably closed then handles the remaining case [139]. □

This reduction makes the attempted transfer and its obstruction explicit: it remains to obtain henselianity or largeness without assuming finite dp-rank or largeness at the outset. It is not a new proof of either field conjecture.

43.6 ID 27: quasiminimality of the full complex exponential field

The selected conjecture says that every unary definable subset, with parameters, of $(\mathbb{C}; 0, 1, +, -, \cdot, \exp)$ is countable or cocountable. Bays–Kirby prove it conditional on exponential algebraic closedness [141]. Gallinaro–Kirby prove the corresponding complex-powers reduct quasiminimal in 2024 [142]; that theorem is not the full exponential conjecture.

There is a direct restricted transfer from analytic zero sets. Every one-variable term in the ring-exponential language is entire. Consequently an atomic equality is either identically

true or has a discrete, hence countable, set of solutions: intersecting a discrete zero set with successively larger compact disks proves countability. Finite Boolean combinations of countable or cocountable sets are again countable or cocountable. Thus the quantifier-free unary fragment satisfies the desired conclusion. This elementary argument does not survive arbitrary projections of several-variable zero sets. The missing step is a quantifier-level theorem such as the required exponential algebraic closedness, not independence of a finite list of special values. Verdict: the full conjecture is not closed by our analytic or transcendence arguments.

43.7 ID 28: the finite-Morley-rank Cherlin–Zilber conjecture

The formulation used is that every infinite simple group of finite Morley rank is an algebraic group over an algebraically closed field. It is distinct from broader conjectures about arbitrary ω-stable groups. Tent's June 2026 primary exposition still describes the general problem as open and explains its relation to nonsplit sharply 2-transitive groups [143].

We tested the tempting counterexample transfer: use constructions of nonsplit sharply 2-transitive groups to leave the algebraic category. The missing hypothesis is finite Morley rank. Their existence as abstract groups gives no bound on the ranks of all definable sets in their theories; finite-dimensional numerical observations are not such a bound. Tent explicitly identifies a suitable finite-rank construction as a potential route, not a completed construction. Thus neither an abstract-group example nor an analytic rank analogy is a counterexample to the stated conjecture. Verdict: no closure.

43.8 ID 30: which main gap?

For a complete countable first-order theory T, the canonical numerical main-gap statement is the following established theorem: either $I(T, \kappa) = 2^{\kappa}$ for every uncountable cardinal κ, or

$$I(T, \aleph_\alpha) < \beth_{\omega_1}(|\alpha| + \aleph_0) \qquad (\alpha \geq 1).$$

Here the beth expression denotes the beth iteration with the specified initial infinite cardinal. Shelah's classification theory and the complete spectrum analysis of Hart–Hrushovski–Laskowski are existing results [144]; the displayed formulation is also recalled in the 2026 survey [129].

Consequently this interpretation of ID 30 was already closed before the present work. A different, explicitly recorded research problem is Shelah's Problem 4.7 on extending the main gap to uncountable theories [132]. The bare name "main gap conjecture" does not specify an AEC or an uncountable-theory variant, and we do not silently assign one a fabricated uniform cardinal bound. Our attempted structural transfer would classify models by trees of observable invariants. Its unproved step is constructing primary/prime models and decomposition systems over nonforking configurations; finite-parameter inverse theorems supply neither. Verdict: the precise countable statement is credited as classical; no new uncountable or non-elementary main-gap theorem is asserted.

43.9 ID 31: Kueker's conjecture and a failed stability reduction

For a complete countable theory T, the conjecture is: if every uncountable model of T is $\aleph_0$-saturated, then T is categorical in some infinite cardinality. Saturation here means realizing all types over finite parameter sets; the conclusion is not restricted to uncountable categoricity. Hrushovski proves the stable case [145]; Tanović's later paper gives additional structural results [146]. No general resolution was located in the reviewed primary sources.

An attractive attempted reduction is to derive stability from the saturation premise and invoke Hrushovski. That reduction is false. Consider the theory of the random graph. Quantifier elimination shows that a type over a finite parameter set prescribes only equalities and finitely many adjacency choices for any fixed finite tuple. The extension axioms realize each consistent prescription, successively for its distinct coordinates, in every model. Thus every model is $\aleph_0$-saturated. On the other hand adjacency has the independence property: for any distinct vertices $a_1, \dots, a_r$ and any subset S, the extension axiom gives a vertex adjacent exactly to those a_i with $i \in S$. The theory is therefore unstable. It is $\aleph_0$-categorical, so it satisfies Kueker's conclusion and is not a counterexample to the conjecture. This explicit example rules out the proposed proof reduction while leaving the conjecture itself untouched.

43.10 ID 33: a 2026 counterexample announcement to stable forking

The stable forking conjecture says that, in a simple theory, every forking type $\mathrm{tp}(a/B)$ over $C \subseteq B$ contains an instance $\varphi(x, b)$ of a stable formula, with b from B, which itself forks over C. Freitag–Mutchnik's primary preprint posted 31 August 2026 states the negation as Theorem 1.3 [147]. Their construction uses a module over a division ring built from a quantum graph algebra; the paper proves supersimplicity and then excludes stable witnesses for the specified forking. This is a recent external counterexample claim, not a counterexample constructed or independently fully checked here.

The attempted positive transfer would replace nonforking by a linear-disjointness or rank criterion and then treat its witnesses as stable. The reported example identifies the failure of that inference: noncommutative incidence can encode a half-graph even when the independence calculus has the required simple-theory axioms. A rank criterion alone does not prove stability of a witnessing formula. The paper's Question 4.7 separately asks about the finite-SU-rank supersimple case. We therefore retire the unrestricted positive target in light of the counterexample announcement and do not conflate it with that remaining finite-rank question. Verification in this report covers the formulation and announced mechanism, not a replacement proof of the source's construction.

Accounting. Theorem 43.1 is one proved decision/definability boundary with a full proof from classical inputs. The conditional Silver transfer, field reduction, quantifier-free quasiminimality, and random-graph obstruction are explicitly credited elementary or classical deductions. None is counted as a new solution of a named historical conjecture. The eleven-ID ledger preserves the distinction between a theorem already known, an externally announced resolution, a restricted result proved here, and a target left unresolved.

44 Set-theoretic and abstract-logical targets

This section treats items 1, 3, 8–11, 21, 22 and 45 in the user's fixed numbering. The evidence cut-off is 17 October 2026, Korean time. "A proof is supplied below" and "a source announces a proof" are different statuses. None of the historical results or contemporary authors' claims is counted as a new solution by this investigation. In particular, old analytic results proved in ZFC cannot be imported as choice-free lemmas merely by changing the ambient theory to ZF.

44.1 Item 1: the Partition Principle versus choice

Exact question and current evidence. The Partition Principle is the ZF assertion

$$\mathrm{PP}: \quad (\exists f : X \twoheadrightarrow Y) \Longrightarrow (\exists i : Y \hookrightarrow X).$$

The question is whether ZF $\vdash$ PP $\Rightarrow$ AC. The injection need not be a section of the specified surjection. Gilson's current revision, dated 10 September 2026, explicitly withdraws its claimed countermodel; its abstract and correction notice identify the stronger splitting principle, a misuse of forcing-name symmetry and a failed limit truncation argument [148]. Thus the earlier search result claiming a solution is superseded. The ordinary implication is not resolved here.

Attempt: force local inverses, then preserve the choice failure. Here is a self-contained explanation of the splitting obstruction acknowledged in that correction. It does not require analysis of the iteration.

Proposition 44.1 (Localized splitting already well-orders the seed)**.** *In ZF let A be a set, $S = A^\omega$ and $T = \mathcal{P}(S)$. Suppose every surjection between two subsets of T has a right inverse. Then A is well-orderable.*

Proof. The empty case is immediate. For nonempty $B \subseteq A$ let

$$C(B) = \{(b, b, b, \ldots) : b \in B\} \subseteq S.$$

For $a \in B$ put

$$E(a, B) = \{(a, b, a, a, \ldots) : b \in B\} \subseteq S.$$

The nonempty set $E(a, B)$ determines a from the zeroth coordinate and B from the first coordinates. Consequently

$$X = \{C(B) : \varnothing \neq B \subseteq A\}, \qquad Y = \{E(a, B) : a \in B \subseteq A\}$$

are subsets of T, and $E(a, B) \mapsto C(B)$ is a well-defined surjection $Y \to X$. A section determines a function c on all nonempty subsets of A, with $c(B) \in B$. Transfinite recursion now repeatedly selects $c(A \setminus \{a_\xi : \xi < \eta\})$ whenever the remainder is nonempty. This process must terminate before the Hartogs ordinal of A, since otherwise it gives an injection of that ordinal into A. The resulting enumeration well-orders A. No choices are made in this recursion beyond the given c. □

For the genuine PP the same construction gives only an arbitrary injection $X \to Y$; its value at $C(B)$ can encode an element of an entirely different subset of A. Recovering a fiber-respecting map is exactly the unjustified step. The proposition rules out the strengthened seed-preserving construction, not a model of ordinary PP without choice. This is an audited obstruction, not a new proof of either answer to item 1.

44.2 Item 3: what remains of the continuum problem

Exact question and status. The sentence CH is $2^{\aleph_0} = \aleph_1$ in ZFC. Its independence, relative to the consistency of ZFC, is a theorem of Gödel and Cohen; Cohen's 1963 paper is [149]. A later machine-checked treatment is [150], its main independence theorem. The question of a preferred additional axiom deciding CH is not the question whether ZFC decides CH. Woodin's 2026 discussion still separates the proposed Ultimate-L axiom from the conjectures needed to justify its compatibility with large cardinals [151].

Attempt: identify the continuum through old observation theorems. The obstruction is not insufficient numerical precision. The standard forcing calculation shows why no ZFC theorem about the previous analytic constructions can settle CH.

Proposition 44.2 (The Cohen calculation, recalled)**.** *If $M \models \mathrm{ZFC}$ and M has an infinite cardinal κ with $\kappa^{\aleph_0} = \kappa$, forcing over M with $\mathrm{Fn}(\kappa \times \omega, 2, <\omega)$ preserves cardinals and forces $2^{\aleph_0} = \kappa$.*

Proof. An uncountable family of finite conditions has an uncountable delta-system of domains. Thinning to equal values on the root gives compatible conditions, so the forcing is ccc and preserves cardinals. For each $\alpha < \kappa$ the generic union gives a real c_α. For different α, β, conditions making their values differ at a fresh integer are dense; hence the reals are distinct. Conversely, a nice name for a real consists of countably many countable antichains, since the forcing is ccc. There are at most $\kappa^{\aleph_0} = \kappa$ such names. These two estimates give the result. The familiar ccc preservation and nice-name lemmas are used here as standard forcing lemmas, not reproved as part of this calculation. □

Starting from a model of GCH, take $\kappa = \aleph_2$; its extension satisfies ¬CH while the constructible model gives the positive relative consistency direction. Every ZFC theorem proved in the earlier papers remains a theorem in both settings. More formally, adding already provable sentences to a theory does not change its deductive consequences: replace each use of an added axiom by its proof. A new nonconservative set-theoretic axiom, with an independent justification, is the missing ingredient. There is no new resolution of the post-Cohen axiom-selection problem here.

44.3 Item 8: distinguish two PCF questions

Formalization and evidence. For a progressive set A of infinite regular cardinals ($|A| < \min A$), put

$$\mathrm{pcf}(A) = \{\mathrm{cf}(\textstyle\prod A/U) : U \text{ an ultrafilter on } A\}.$$

One strong PCF conjecture asserts $|\,\mathrm{pcf}(A)| = |A|$ for every such A. The numerical assertion

$$\aleph_\omega \text{ strong limit} \quad \Longrightarrow \quad 2^{\aleph_\omega} < \aleph_{\omega_1} \tag{PCF$_\omega$}$$

is a different target. Shelah's established bound has ω_4 in place of ω_1 [152, §6]. Gitik's author-hosted manuscript, revised 3 March 2026, reports a countable A with $\mathrm{otp}(\mathrm{pcf}(A)) = \omega_1 + 1$; the critical spectrum calculation is Lemma 7.12, with large-cardinal settings in §1 [153]. This is a reported relative counterexample to the unrestricted cardinality conjecture, not a proof of its negation in every model. Its §8, Question 3, expressly leaves moving the construction down to $\aleph_\omega$ unresolved. We have checked this scope, not independently certified its extender-forcing construction or established a publication status.

Attempt: reconstruct the spectrum from finite restrictions. The following elementary calculation exposes a failure of the finite-observation analogy. It does not improve Shelah's bound.

Proposition 44.3 (Finite decomposition does not extend to countable decomposition)**.** *In ZFC, if $A = A_0 \cup \cdots \cup A_{m-1}$, then*

$$\mathrm{pcf}(A) = \bigcup_{i<m} \mathrm{pcf}(A_i),$$

where empty pieces are omitted. In contrast, if $\langle \kappa_n : n < \omega \rangle$ is strictly increasing and consists of uncountable regular cardinals, and $\lambda = \sup_n \kappa_n$, every free ultrafilter U on ω satisfies

$$\mathrm{cf}(\textstyle\prod_n \kappa_n / U) > \lambda.$$

In particular the PCF spectrum is not the union of the spectra of its singleton pieces.

Proof. Refine a finite cover to a disjoint partition. An ultrafilter contains exactly one piece; restriction to that piece identifies the corresponding reduced product. Conversely an ultrafilter on a piece extends to an ultrafilter on A concentrated there. Monotonicity under inclusion then gives the stated formula for the original cover.

Fix a family $F \subseteq \prod_n \kappa_n$ of cardinality $\mu < \lambda$. For all sufficiently large n we have $\mu < \kappa_n$. Regularity gives

$$g(n) = \sup\{f(n) + 1 : f \in F\} < \kappa_n$$

on those coordinates; give g arbitrary legal values on the remaining ones. Then $[g]_U > [f]_U$ for every $f \in F$. Thus no family of size less than λ is cofinal in the reduced product. The quotient is a linear order without a largest element. The cofinality of such an order is a regular cardinal: a singular least cofinal size would, by taking upper bounds of successively larger proper subfamilies, give a smaller cofinal family. Since λ has cofinality ω, the quotient cofinality cannot equal λ, and is therefore larger. Finally the only ultrafilter on $\{\kappa_n\}$ is principal, giving spectrum $\{\kappa_n\}$. □

The missing step in the proposed reconstruction is control of free ultrafilters and true cofinalities, invisible to all finite coordinate restrictions. For (PCF_ω), a forcing counterexample would additionally need the difficult collapses at successors of singular cardinals while preserving the required scales. Neither step is supplied here.

44.4 Item 9: Woodin's Ω conjecture

Exact formulation. Assume a proper class of Woodin cardinals. For a countable theory T, $T \models_\Omega \varphi$ means that every rank segment satisfying T in every set-forcing extension satisfies φ. One standard proof-relation convention is Larson's Definition 6.1: $T \vdash_\Omega \varphi$ if some universally Baire set $A \subseteq \omega^\omega$ has the property that every countable transitive A-closed model M of ZFC containing T satisfies "$T \models_\Omega \varphi$". Concretely, A-closed means that for each poset $P \in M$ there is a countable family D of open dense subsets of P, including all such sets belonging to M, such that $A \cap M[G] \in M[G]$ whenever a filter G meets every member of D. The members of D need not belong to M. The conjecture is, for all sentences φ,

$$\varnothing \models_\Omega \varphi \quad \Longrightarrow \quad \varnothing \vdash_\Omega \varphi.$$

Soundness is already a theorem. The stronger definability conjectures sometimes also called an Ω conjecture are not being identified with this statement.

Source and actual reduction. Larson [154, Definition 6.1, Theorems 6.4 and 9.2] gives soundness and a conditional completeness route: suitable countable collapses of rank segments with universally Baire iteration strategies yield the required forcing countermodels to unprovability. No proof of the unrestricted completeness direction was verified in the current search. The unresolved step in this route is obtaining these strategies for the relevant collapses of V, rather than only for canonical iterable inner models. Finite-cover uniqueness from the preceding analytic work does not produce an iteration strategy or certify all its countable iterates.

Attempt: replace iteration strategies by compactness. This fails at an exact, elementary boundary. If a consistent theory T extends ZF, adjoin constants c_n and sentences saying that all c_n are ordinals and $c_{n+1} \in c_n$ for every $n < \omega$. Every finite subset has a model: interpret its finitely many constants as a sufficiently long descending list of finite ordinals in a model of T. First-order compactness therefore gives a model of the entire enlarged theory. Its membership relation is externally ill-founded, so it cannot be a transitive model. Thus the Henkin construction alone cannot provide the transitive A-closed objects required in Ω logic. This proves an obstruction to this attempted method, not a refutation of Ω completeness. The exact missing strategy hypothesis remains unsolved by our methods.

44.5 Item 10: HOD conjecture and the stationarity gap

Formalization and current status. HOD is the class of hereditarily ordinal-definable sets. A regular uncountable θ is ω-strongly measurable in HOD if some infinite cardinal $\delta < \theta$ satisfies $(2^\delta)^{\mathrm{HOD}} < \theta$ and HOD has no partition of S^θ_ω into δ sets stationary in V. The HOD Hypothesis asserts a proper class of regular cardinals not ω-strongly measurable in HOD. With an extendible cardinal, the HOD dichotomy connects it with correct singular-cardinal successors in HOD. The HOD *conjecture*, in the formulation of [155, §1.2, Theorems 1.1–1.2 and B], is a provability claim over ZFC plus an extendible cardinal, not merely the sentence $V = \mathrm{HOD}$. That paper proves a conditional negative result from the joint consistency of an exacting cardinal above an extendible cardinal. It does not establish this joint consistency from ZFC alone. Woodin's published 2026 lectures continue to formulate the associated program as conjectural [151]. Neither an unconditional proof nor an unconditional consistency refutation is obtained here.

Attempt: use HOD's choice to manufacture the required partitions. HOD satisfies AC, but the sets it regards as stationary need not be stationary in the outer universe. The following standard club-shooting argument shows that even agreement about ω_1 cannot repair this inference.

Proposition 44.4 (Stationarity is not upward absolute)**.** *Let $M \models \mathrm{ZFC}$ and let $S \subseteq \omega_1^M$ be stationary and costationary in M. There is an ω-distributive forcing extension $M[G]$ preserving ω_1 in which S contains a club. Hence the ground-model stationary set $\omega_1 \setminus S$ becomes nonstationary.*

Proof. Use the poset of countable closed bounded subsets of S, ordered by end-extension. It adds an unbounded closed union, since extending the maximum beyond any prescribed countable ordinal is dense. To prove ω-distributivity, fix a condition p and open dense sets D_n. Choose a countable elementary submodel $N \prec H_\chi$ containing these parameters with $\delta = N \cap \omega_1 \in S$; stationarity supplies N. Choose in the ambient universe an increasing sequence cofinal in δ. Recursively choose $p_n \in N$ below p and earlier conditions, in D_n, with maxima exceeding successive terms of that sequence. Elementarity and unboundedness of S make each choice possible inside N. The set

$$q = \bigcup_{n<\omega} p_n \ \cup \ \{\delta\}$$

is a countable closed subset of S: every limit point below δ already occurs in some end-extended closed condition, and the only new limit point is δ. It is a lower bound lying in every D_n. This proves ω-distributivity and hence that no countable sequence of ordinals is added; in particular ω_1 is preserved. The generic club is disjoint from the old complement of S. □

This classical construction is included to isolate the obstruction, not to claim that M in the proposition is HOD of its extension. Our attempted HOD argument needs a theorem transferring stationarity of its proposed partition pieces to V. HOD's internal well-order and uniqueness of a definition supply no such theorem. The HOD dichotomy and modern exacting cardinal results show why this missing ingredient is substantive.

44.6 Item 11: consistency of Reinhardt cardinals

Framework and source correction. The intended assertion is a nonidentity elementary class embedding $j : V \to V$, with critical point κ, in a choice-free class theory, or in the usual ZF(j) framework with the stated class-replacement instances. It is not the assertion that a set function has domain V, and it is not restricted to parameter-definable embeddings. These variants must not be interchanged. Kunen's inconsistency uses choice; see [156] for the standard stationary-set argument and precise variants. McCallum's 2026 inconsistency claim is withdrawn in revision 3 of 23 August 2026, with an identified problem concerning continuity of Skolem functions [157]. It therefore supplies no inconsistency theorem. The unrestricted choice-free consistency problem remains unresolved by the verified sources and by this investigation.

Attempt: localize Kunen's argument to exactly what choice supplied.

Proposition 44.5 (Conditional stationary-partition obstruction)**.** *Work in the class-embedding framework just specified. Suppose $j : V \to V$ has critical point κ. If $\theta > \kappa$ is a regular uncountable cardinal fixed by j, and S^θ_ω has a partition $\langle S_\alpha : \alpha < \kappa\rangle$ into stationary sets, then a contradiction follows.*

Proof. Write $\langle T_\alpha : \alpha < j(\kappa)\rangle = j(\langle S_\alpha : \alpha < \kappa\rangle)$. Since $j(\theta) = \theta$ and $j(\omega) = \omega$, this is a partition of the same S^θ_ω into stationary sets. The set

$$C = \{\beta < \theta : j\text{“}\beta \subseteq \beta\}$$

is club. It is closed by taking suprema; it is unbounded because starting above any $\eta < \theta$ we can repeatedly close under j and take the supremum of the resulting countable sequence below the regular θ. The required sequences are sets by the replacement assumptions on j. Pick $\beta \in C \cap T_\kappa$. It has cofinality ω. For an increasing cofinal sequence $\langle \beta_n : n < \omega\rangle$ in β, elementarity and $j\text{“}\beta \subseteq \beta$ give $j(\beta) = \sup_n j(\beta_n) \leq \beta$, while embeddings of ordinals never decrease them. Thus $j(\beta) = \beta$. There is $\alpha < \kappa$ with $\beta \in S_\alpha$, so $\beta = j(\beta) \in j(S_\alpha) = T_\alpha$, since $j(\alpha) = \alpha$. This contradicts the disjointness of T_α and T_κ. □

This is the classical obstruction with the hypotheses exposed. In ZFC one gets a fixed regular successor above the supremum of the critical sequence and the required stationary partition. In the choice-free setting these steps are unavailable in this generality. A construction of local rank embeddings, or the consistency results at $V_{\lambda+2}$, is not a construction of $j : V \to V$; conversely our conditional contradiction does not remove its partition hypothesis. The missing step is exactly such a choice-free global construction or an argument dispensing with that hypothesis.

44.7 Item 21: Roitman's problem and the September 2026 claim

Formalization and version-sensitive status. Here $\mathfrak{d}$ is the least size of an eventually dominating subset of ω^ω, and $\mathfrak{a}$ the least size of an infinite maximal almost-disjoint family in

$[\omega]^\omega$. The question is whether

$$\mathrm{ZFC} \vdash \mathfrak{d} = \aleph_1 \Longrightarrow \mathfrak{a} = \aleph_1.$$

The purported counterexample of Mildenberger, arXiv:1404.7343, is withdrawn; the arXiv record identifies its Lemma 3.2 as flawed [158]. Pelayo-Gómez's 30 September 2026 preprint now claims the positive theorem and preservation under outer models retaining domination [159, Theorems 1.1 and 1.4]. We read its full core construction, Lemmas 3.1–3.3, and maximality argument; two independent checks in this investigation found no local gap. Publication, priority and wider independent validation were not established. The correct evidence label is therefore *recent author proof claim, core checked here*, not an unqualified "still open", and not our own solution.

Our attempted reduction and its interface with that claim. Normalize a size-ω_1 dominating family by closing under finite modifications, obtaining pointwise domination. For a bounding function d and an enumeration $c = \langle c_k : k < \omega \rangle$ of earlier almost-disjoint sets, use the two observable objects

$$K_d = \{x \in [\omega]^\omega : e_x \leq d\}, \qquad G(c, g) = \omega \setminus \bigcup_k (c_k \setminus [0, g(k))).$$

If x is almost disjoint from every c_k, the trace $t_x(k) = 1 + \max(x \cap c_k)$, with value zero for the empty intersection, satisfies $t_x \leq g \Rightarrow x \subseteq G(c, g)$. Thus domination controls both enumeration and finite intersections, but it does not alone build a single new almost-disjoint set hitting every member of the corresponding compact class. This simultaneous capture is the precise step missing from the naive recursion.

The cited preprint's protected-interval fusion is designed to supply that step. Our audit checked that its successor stage leaves infinitely many reserved intervals for each active index, and that countable limit stages take infinite pseudointersections of the decreasing reservations. The maximality proof then uses the trace implication above. These checks do not replace the full proof by an analogy with inverse observation. Our outcome is a technical audit of an external claimed resolution, with no additional theorem or claimed closure count.

44.8 Item 22: consistency strength of PFA

Formalization and established bounds. PFA asserts that for every proper poset and every family of ω_1 dense subsets there is a filter meeting all of them. The target is the equiconsistency of ZFC+PFA with ZFC plus a supercompact cardinal, not the false literal claim that PFA makes ω_2 supercompact in V. The supercompact upper bound is classical. A verified lower bound is Trang's theorem giving a transitive inner model containing all reals and ordinals and satisfying $\mathrm{AD}_{\mathbb{R}}$ + "Θ is regular" [160, Theorems 0.2 and 0.4]. This is an established bound, not a claim that this search determined the latest optimal lower bound. Viale–Weiß prove that a *proper standard iteration* from an inaccessible κ forcing PFA with $\kappa = \omega_2$ requires that κ was supercompact [161, Corollary 6.11]. Their qualification cannot be deleted to obtain a general lower bound.

Attempt: descend branches to recover a large-cardinal ground. The core descent mechanism can be stated without any ambiguity about forcing presentations.

Lemma 44.6 (Approximation transfers an already supplied branch)**.** *Let $M \subseteq W$ be transitive models of ZFC with the same ordinals and the κ-approximation property: if $b \in W$ is a subset of a set of M and $b \cap x \in M$ for every $x \in M$ of M-size less than κ, then $b \in M$. Let $\lambda \in M$*

be an ordinal and $\mathcal{T} = \langle T_x : x \in (P_\kappa\lambda)^M \rangle \in M$*, where* $T_x \subseteq \mathcal{P}(x)$*. If* $b \subseteq \lambda$ *in* W *satisfies* $b \cap x \in T_x$ *for every such* x*, then* $b \in M$ *and is a branch for the same system in* M*.*

Proof. Every member of $T_x \in M$ belongs to M by transitivity. For any $y \in M$ of M-size less than κ, set $x = y \cap \lambda$. Then $b \cap y = b \cap x \in T_x \subseteq M$. Approximation yields $b \in M$. Membership of each restriction in T_x is absolute between the transitive models, proving the second assertion. □

This explains the attraction of guessing-model and approximation arguments, but does not construct an appropriate ground M from an arbitrary model of PFA. Nor does PFA automatically provide branches for every system whose levels are indexed only by the old small subsets: new small subsets in W require additional analysis. The missing recovery theorem is an inner model or ground with the required extender/approximation structure, not an estimate on finitely many measurements. The general equiconsistency question is not closed here.

44.9 Item 45: compactness, Beth and Δ interpolation

Disambiguation. Taken literally, existence of a compact logic having Beth definability and Δ interpolation has the elementary positive answer $L_{\omega\omega}$. The nontrivial problem is a regular abstract logic extending first-order logic that is compact and has ordinary Beth definability and Δ interpolation *but fails Craig interpolation.* Countable compactness, full compactness, weak Beth and projective Beth are distinct variants. Here $\mathrm{PC}(L)$ denotes classes of reducts of models of an L sentence; $\Delta(L)$ comprises classes for which both the class and its complement are $\mathrm{PC}(L)$. Δ interpolation means $\Delta(L) = L$, up to definability; Craig interpolation separates arbitrary disjoint projective classes arising from an implication. Ordinary Beth concerns an implicitly unique relation, whereas projective Beth permits additional hidden relations.

Verified neighboring results. Shelah's 1985 construction is compact and has Beth, but explicitly fails even Δ interpolation [162, Conclusion 6.11(4)]. Its Claim 6.12 concerns an enlarged closure; it does not assert failure of Craig interpolation for that enlarged logic. The current detailed account [163, Proposition 14 and the following discussion] also warns that the Δ closure of the cofinality logic need not have ordinary Beth. Thus neither of these frequently conflated constructions settles the conjunction requested here. No verified resolution of that precise conjunction was found.

Attempt: alternate Beth and Δ closure. Let φ and ψ have common vocabulary τ and satisfy $\varphi \models \psi$. Define

$$K = \mathrm{red}_\tau \,\mathrm{Mod}(\varphi), \qquad J = \mathrm{red}_\tau \,\mathrm{Mod}(\neg\psi).$$

These are disjoint $\mathrm{PC}(L)$ classes. If one could show that the complement of K were also $\mathrm{PC}(L)$, Δ interpolation would define K by a τ sentence θ. Then every model of φ satisfies θ, and no model of θ satisfies $\neg\psi$, giving Craig interpolation. This two-line reduction is valid; its premise is the missing assertion. Disjointness $K \cap J = \varnothing$ does not imply that $K \cup J$ is the class of all τ structures. Ordinary Beth does not turn existence of an arbitrary expansion into an implicitly unique expansion.

Alternating the two closures therefore has three unproved requirements: compactness at the closure limit, persistence of ordinary Beth after adding projective complementary definitions, and preservation of a concrete failure of Craig interpolation. The cited constructions show that each property must actually be checked, rather than inherited by name. Under the additional

countable downward Löwenheim–Skolem property and finite occurrence (each sentence uses a finite vocabulary), the classical Lindström theorem already forces a compact regular extension back to first-order logic; that restricted route cannot produce a counterexample. We obtain a precise failure diagnosis and the literal first-order answer, but no new solution of the intended separation problem.

Outcome of the nine-target attempt. The outputs are explicit proof barriers and reductions, a corrected distinction between PCF variants, and audits of recent or withdrawn claims. CH independence is historical; the current Roitman proof claim belongs to its named author. No unresolved external problem is recorded as newly solved by this section.

45 Descriptive and model-theoretic targets: ten precise tests

This section addresses targets 5, 14, 20, 23, 39, 41, 46, 49, 50, and 52. The literature audit is dated 17 October 2026. None of the general open conjectures in this group is claimed to be settled here. We give explicit reductions, counterexamples to proposed extensions, and proofs identifying the additional hypotheses that an observation argument would have to supply. The elementary propositions below are method audits; no priority claim is made for their mathematical content. In particular, items 41 and 46 are families of questions, not uniquely specified conjectures.

45.1 A common obstruction: recovering points does not classify classes

Our earlier observation constructions recover individual points by Borel adaptive choices. The following distinction is necessary before applying them to an equivalence relation. A complete invariant for an equivalence relation E on X is a map c such that

$$c(x) = c(y) \quad \Longleftrightarrow \quad xEy.$$

The existence of a Borel complete invariant in a standard Borel space is the definition of smoothness.

Proposition 45.1 (Point information and invariant information)**.** *Let X be compact metrizable. A finite adaptive scheme whose query selection maps into $C(X)$ are Borel, and whose decoder is Borel, has a Borel output map. Such a scheme can output a complete invariant for E only if E is smooth. Continuous injective point coding does not imply this condition, even for a hyperfinite countable Borel equivalence relation.*

Proof. Equip $C(X)$ with the uniform norm. Evaluation $(f, x) \mapsto f(x)$ is continuous. Induction on the number of queries therefore shows that every transcript coordinate is Borel; composing with the decoder proves the first assertion. The second is the definition just given.

For the last assertion take $X = 2^{\mathbb{N}}$ and eventual equality E_0. The map

$$x \longmapsto \sum_{j \geq 0} 2x(j)3^{-j-1}$$

is continuous and injective. Nevertheless E_0 has no Borel complete invariant. Here is a direct proof. If $c : X \to Y$ were such an invariant, choose a countable family (B_j) of Borel sets separating the points of the standard Borel space Y. Each $c^{-1}(B_j)$ is unchanged by changing finitely many bits. Under fair Bernoulli product measure it is thus a tail event and has measure zero or one. For completeness, invariance under all changes of the first n bits

makes it independent of the first n coordinates; passing to the sigma-algebra they generate makes it independent of itself, giving this zero–one assertion. Intersecting the measure-one choices of $c^{-1}(B_j)$ or its complement gives a measure-one set on which all the separating tests agree. On that set c is constant. Its fiber would be one countable E_0-class, which has measure zero, a contradiction. Finally, agreement outside the first n coordinates defines a finite Borel equivalence relation, and these relations increase to E_0. □

The nonsmoothness of E_0 is classical. Notice also that it does not contradict a perfect-set dichotomy: partition $\mathbb{N}$ into infinite sets I_j and repeat the jth bit on I_j. This continuous injection has compact, hence perfect, image and sends distinct sequences to different E_0-classes. Thus point recovery, smooth classification, and constructing a perfect antichain are three different tasks.

45.2 Target 5: Martin's conjecture and countable adaptive branching

Write $x \leq_T y$ for Turing reducibility and let a cone be $\{x : a \leq_T x\}$. One standard formulation assumes $\mathsf{ZF} + \mathsf{AD} + \mathsf{DC}_{\mathbb{R}}$ and considers all Turing-invariant functions $f : 2^{\mathbb{N}} \to 2^{\mathbb{N}}$. Part I asserts that either the degree of $f(x)$ is constant on a cone, or $x \leq_T f(x)$ on a cone. Part II asserts the prewellordering, modulo cones, of functions above the identity, with pointwise Turing jump increasing rank by one. The Borel versions over choice-based set theory are distinct formulations.

Lutz and Siskind prove Part I for order-preserving functions [164, Theorem 4.8], and for Borel order-preserving functions without the extra determinacy assumption [164, Theorem 4.13]. Yu's October 2026 preprint studies weaker regularity hypotheses and consistency strength for that order-preserving case [165, Theorems 1.2 and 1.4]. These results do not establish the conjecture for arbitrary Turing-invariant functions.

Proposition 45.2 (Countable cone patching)**.** *In ZFC, a countable Borel partition (A_n) of $2^{\mathbb{N}}$ into Turing-invariant sets has a member containing a cone. Consequently a Turing-invariant, Borel, countably valued branch selector is constant on a cone. Patching countably many functions on such a partition cannot create a cone behavior different from that of every branch.*

Proof. The Borel cone theorem, a consequence of Borel determinacy, says that each A_n or its complement contains a cone. If no A_n contains a cone, choose bases a_n for cones in their complements. The real $a = \bigoplus_n a_n$ computes every a_n, so the cone above a misses every A_n, contrary to the partition property. On the cone contained in A_n, a patched function equals its nth branch. If that branch has a specified property on another cone, intersect the two cones by taking the join of their bases. □

The cone theorem is an external input; see [164, Theorem 1.1 and Section 2]. The obstruction to the proposed transfer is exact: general real-valued transcripts are not countable partitions, and an arbitrary Borel selector need not be Turing-invariant. Neither Borel measurability nor exact numerical recovery supplies uniform indices for Turing reductions. The attempt therefore excludes a simple patching strategy and leaves both full parts unresolved.

45.3 Target 14: the Borel Weiss problem

The formulation used here asks whether every Borel action of a countable amenable group on a standard Borel space has hyperfinite orbit relation. Hyperfinite means an increasing union of finite Borel equivalence relations on the whole space. Removing a null set is not allowed. Naryshkin and Vaccaro's 2026 result proves the treeable case [166, Corollary 3]; their Theorem

2 uses the stronger property of hyper-u-amenability. Their introduction explicitly retains the general Weiss question.

The finite-subgroup mechanism can be checked completely. If $\Gamma = \bigcup_n \Gamma_n$ with finite subgroups $\Gamma_n \subseteq \Gamma_{n+1}$, set

$$xF_n y \quad \Longleftrightarrow \quad (\exists g \in \Gamma_n)\ g \cdot x = y.$$

Each F_n is Borel, is an equivalence relation, and has classes of size at most $|\Gamma_n|$. Their increasing union is the orbit relation. This is the familiar locally finite special case, not a new solution.

Replacing subgroups by Følner sets destroys the argument at a precise point: an orbit fragment indexed by a finite set need not be an equivalence class. On the translation action of $\mathbb{Z}$ on itself, the symmetric one-step edges $k \leftrightarrow k+1$ already generate one infinite class. Taking transitive closure can therefore destroy finiteness before any limiting argument begins. The missing input is a Borel construction of coherent finite partitions; selecting Følner sets separately at each point does not provide one. The treeable theorem is an applicable external partial result, but no arbitrary amenable action is shown here to satisfy its extra hypothesis.

45.4 Target 20: the Borel Ruziewicz problem

Fix $n \geq 2$ and normalized spherical measure σ on S^n. The Borel problem asks whether every finitely additive isometry-invariant probability on the Borel subsets of S^n equals σ. This differs from the classical uniqueness theorem on the completed Lebesgue sigma-algebra. Conley and Marks explicitly distinguish the two and prove a related exotic-measure example for a different group action [167, Introduction and Theorem 1.7]. That example is not a sphere counterexample.

Proposition 45.3 (The null-charge obstruction)**.** *Assume the classical completed-measure Ruziewicz theorem in dimension $n \geq 2$. If a finitely additive invariant Borel probability ν vanishes on every σ-null Borel set, then $\nu = \sigma$. Thus any counterexample to the Borel problem must charge a Borel null set.*

Proof. Every set A in the completion differs from a Borel set B by a subset of a Borel null set. Define $\widetilde{\nu}(A) = \nu(B)$. Two choices of B differ inside a Borel null set, so finite additivity and positivity of ν make this well defined. For disjoint completed sets, their Borel representatives have null intersection; removing that intersection verifies finite additivity of $\widetilde{\nu}$. Isometries preserve Borel sets and σ-null sets, so invariance also extends. The completed-measure uniqueness theorem now gives $\widetilde{\nu} = \sigma$, and restriction proves the claim. □

This is the standard reduction underlying the distinction, not a new case of the conjecture. It locates the failure of a proposed spectral transfer: a functional charging null sets does not descend to $L^\infty(\sigma)$ modulo almost-everywhere equality. A spectral-gap or L^2 observation inequality cannot be applied to it without first proving null-set annihilation. That step remains unproved here.

45.5 Target 23: the topological Vaught conjecture

For a continuous action of a Polish group G on a Polish space X, the chosen statement is: either there are countably many orbits, or there is a perfect set of pairwise orbit-inequivalent points. This is stronger in formulation than merely excluding intermediate quotient cardinalities. Becker and Kechris relate the topological and Borel-action versions [168]; Melleray's 2026

exposition explicitly records the general problem as open immediately after Silver's theorem [169, Theorem 15.10 and following discussion].

The direct reduction attempt is to make the orbit relation coanalytic and apply Silver's dichotomy. It works if the orbit relation is Borel: it is then coanalytic, and Silver yields exactly the required alternative. But the assertion that each individual orbit is Borel does not say that the relation is Borel in $X \times X$. General Polish orbit relations are analytic and can fail to be Borel. The needed global complexity bound is absent. Proposition 45.1 also prevents substituting point codes for orbit invariants; its explicit perfect E_0-antichain emphasizes that smoothness is not required by the actual Vaught conclusion. No reduction from the general case to the coanalytic case is obtained.

45.6 Target 39: Thomas's finite-reduct conjecture

We use the finite relational version: a countably infinite ultrahomogeneous structure in a finite relational language has finitely many parameter-free first-order reducts up to interdefinability. Cameron restates this as an open problem in 2026 [170, Question 8.1]. Dropping the relational hypothesis changes its status: Bodor, Cameron and Szabó construct homogeneous structures with infinitely many reducts, including the countable atomless Boolean algebra in its functional signature [171].

Finite relational homogeneity implies ω-categoricity: for each tuple length there are only finitely many atomic diagrams, and equal diagrams are carried to one another by homogeneity. The following reformulation isolates what finite observation would have to control.

Proposition 45.4 (Uniform arity is the missing finite budget)**.** *Let M be a countable ω-categorical structure in a countable language, and let o_k be the number of $\mathrm{Aut}(M)$-orbits on M^k. For fixed r, the number of interdefinability classes of reducts generated by relations of arity at most r is at most*

$$2^{N_r}, \qquad N_r = \sum_{k=1}^{r} 2^{o_k}.$$

Moreover, M has finitely many reducts if and only if some r works for every reduct, in the sense that every reduct is interdefinable with one generated by relations of arity at most r.

Proof. Every parameter-free definable k-ary relation is a union of $\mathrm{Aut}(M)$-orbits, so there are at most 2^{o_k} choices. A reduct generated in arities at most r therefore chooses a subset of a collection of size at most N_r. This proves the bound and the reverse implication in the equivalence.

For the forward implication assume there are only finitely many reduct classes. Enumerate the distinct relations of a reduct R as $R_1, R_2, \ldots$; there are only countably many parameter-free formulas in the countable language. The reducts generated by $R_1, \ldots, R_j$ form an increasing sequence in the partial order of definability closures. A monotone sequence in a finite partial order stabilizes. After stabilization every later R_j is definable from the same finite initial list. Thus R has a finite relational generating list. Choose such a list for a representative of each of the finitely many classes, and take the maximum of their arities. □

The fixed-r counting statement does not settle Thomas's conjecture: the required r must work simultaneously for all reducts, including ones initially specified using arbitrarily large tuple arities. Homogeneity of the original structure bounds neither that generating arity nor the number of definability closures by the argument above. This is a proved reformulation, not an established uniform bound.

45.7 Target 41: categoricity transfer for atomic classes

The supplied label does not state a cardinal threshold, set-theoretic hypothesis, or amalgamation assumption, so it has no single truth value. Fix a countable complete first-order theory T and write $\mathrm{At}(T)$ for its atomic models: every finite tuple realizes a type isolated over the empty set. As a precise benchmark, one may ask whether categoricity in some $\lambda \geq \beth_{\omega_1}$ implies categoricity in every $\mu \geq \beth_{\omega_1}$ for this class. This is a specialization of the standard $L_{\omega_1,\omega}$ categoricity conjecture stated in [172, Conjecture 1.2], rather than a uniquely determined interpretation of the original label. Known transfer theorems for short AECs impose structural hypotheses such as amalgamation over sets; these hypotheses cannot simply be deleted.

Here is an exact translation and a counterexample to a much stronger, naive transfer. For every n, list the T-consistent formulas $\varphi_{n,i}(\bar{x})$ that isolate complete n-types. Then

$$\Psi_T = \bigwedge T \ \wedge \bigwedge_{n\geq 1} \forall \bar{x} \bigvee_i \varphi_{n,i}(\bar{x})$$

is an $L_{\omega_1,\omega}$ sentence, and its models are exactly $\mathrm{At}(T)$. Indeed the displayed disjunction says precisely that each tuple has an isolated type; all conjunctions and disjunctions are countable. This encoding gives access to infinitary categoricity questions, but proves no excellence or amalgamation property.

Categoricity in one arbitrary infinite cardinal does not transfer for atomic classes. Let T be dense linear order without endpoints. Quantifier elimination makes the finite order-and-equality pattern of each tuple an isolating formula, so every model is atomic. The countable model is unique. Put $L = \sum_{\alpha<\omega_1} \mathbb{Q}$, with lexicographic order, and $M = \sum_{j<\omega} L$. Both are dense without endpoints and have size $\aleph_1$. The cofinalities are respectively ω_1 and ω, so they are not isomorphic. This elementary counterexample does not refute the stated high-cardinal benchmark.

The attempt from exact observations stops at the same distinction: identifying every finite tuple's type gives atomicity, but does not give compatible amalgams of independent diagrams or uniqueness of large models. The outcome for this underspecified target is a precise formalization and a rejected low-cardinal extrapolation, not a decision of all atomic categoricity-transfer problems.

45.8 Target 46: cardinal transfer while omitting types

Again a threshold and a class must be specified. We consider models of a countable first-order theory omitting a specified countable family of partial types in finitely many variables over the empty set. This is not the same question as a two-cardinal transfer for a named predicate. The classical Hanf bound for this countable omitting-types problem is $\beth_{\omega_1}$; below it no smaller universal bound works, even after restricting to dependent theories, by the examples of Chernikov and Shelah [173, Theorem 4.5]. The following calculations test the most direct extension strategy.

Proposition 45.5 (The obstruction is extension, not chain union)**.** *There is a complete countable theory and a nonprincipal partial type having a countable omitting model but no uncountable omitting model. On the other hand, the union of an elementary chain of models omitting fixed finite-tuple partial types also omits those types.*

Proof. Take the equality language with constants c_n for $n \in \mathbb{N}$, and let T be the complete theory of a countably infinite set in which the c_n name all elements without repetition. Set $p(x) = \{x \neq c_n : n \in \mathbb{N}\}$. The named model omits p, and any omitting model has no elements

beyond the interpretations of the constants, hence is countable. The type is nonprincipal: a consistent formula mentions only finitely many constants; in the pure equality theory with these finitely many names, any unnamed element has the same type as a constant not among those names. Consequently no such formula can imply all inequalities in p. Finite subsets of p are plainly satisfiable.

For the second assertion let (M_i) be an elementary chain and $M = \bigcup_i M_i$. Any finite tuple $\bar{a}$ of M lies in one M_i. If it realized a fixed partial type p in M, elementarity would make every formula of p true of $\bar{a}$ already in M_i, a contradiction. Apply this separately to each specified type. □

Thus ordinary upward Löwenheim–Skolem or compactness cannot supply the necessary omitting extensions. The chain-union step is valid; constructing proper extensions that continue to omit the types is the missing step. The general Hanf theorem and its sharpness are existing results, not new closures credited to the observation method. More specific cardinal-transfer conjectures require their omitted hypotheses before they can be evaluated.

45.9 Target 49: Borel boundedness and representative coherence

A countable Borel equivalence relation E on X is Borel bounded if for every Borel $\phi : X \to \mathbb{N}^{\mathbb{N}}$ there is Borel ψ such that

$$\phi(x) \leq^* \psi(x), \qquad xEy \implies \psi(x) =^* \psi(y).$$

The stars mean eventual domination and eventual equality, not exact equality. There are two distinct questions: whether every countable Borel equivalence relation is Borel bounded, and whether Borel boundedness implies hyperfiniteness. Both are explicitly listed as open by Day and Marks in 2026 [174, Section 3, before Question 3.6]. Coskey and Schneider give the definition and its connection with the union problem [175, Definitions 1.2 and 3.3].

The known implication from hyperfiniteness can be seen by an explicit observation budget. If $E = \bigcup_n F_n$, with F_n finite Borel equivalence relations increasing in n, put

$$\psi(x)(n) = \max\{\phi(y)(n) : yF_nx\}.$$

Finite-section Borel uniformization makes the maximum Borel. It dominates $\phi(x)(n)$, and for xEy the F_n-classes of x, y agree for every sufficiently large n. Hence the required eventual equality holds. This is the standard Boykin–Jackson observation, not a new implication.

For a general relation, enumerating each class and taking diagonal maxima supplies domination but need not supply coherence. For example, let E be the one-class relation on $\mathbb{Z}$, take $\phi(k)(n) = |k|$, and enumerate from x the offsets $0, 1, -1, 2, -2, \ldots$. The maximum over the first $2n+1$ entries is $n + |x|$. The outputs for $x = 0$ and $x = 1$ are different at every n. The relation itself is smooth and bounded; this is a counterexample to that proposed construction, not to boundedness. A Borel selector of adequate pointwise bounds from our earlier framework would still have to satisfy the eventual class-coherence condition. No such general selection theorem is proved here.

45.10 Target 50: are all Polish groups graphic?

A Polish group is graphic if the orbit relation of every Borel action on a standard Borel space X is the connectedness relation of a Borel graph on that same X. Graph degrees need not be countable. This does not ask whether the group is the automorphism group of a countable graph. Arant, Kechris and Lutz prove graphicness for connected groups and many other

classes, and closure under countable products; they retain the general question and the stronger closed-subgroup question [176, Section 4, Question 3]. They also prove that every Polish orbit relation is strongly Borel 3-hypergraphable [176, Theorem 65 in the arXiv v2 numbering].

The tempting observation transfer is to encode the third point and then project each hyperedge to its pairs. The following exact example shows why Borelness does not follow from that operation.

Proposition 45.6 (A Borel hypergraph can have a non-Borel pair shadow)**.** *There is a Borel* 3*-uniform hypergraph whose relation of pairs lying together in a hyperedge is not Borel.*

Proof. Take an analytic non-Borel set A in a Polish space X, represented as the projection of a closed set $C \subseteq X \times \mathbb{N}^{\mathbb{N}}$. Use the disjoint-union vertex space

$$Z = (X \times \{0\}) \sqcup (X \times \{1\}) \sqcup (X \times \mathbb{N}^{\mathbb{N}} \times \{2\}).$$

For $(x, y) \in C$ put one hyperedge $\{(x, 0), (x, 1), (x, y, 2)\}$. The corresponding symmetric relation of ordered triples is Borel, since membership is tested by tags, equality of the X-coordinates, and membership in C. The pair $((x, 0), (x, 1))$ lies in its shadow exactly when $x \in A$. The continuous map $x \mapsto ((x, 0), (x, 1))$ would therefore reduce A to a Borel set if the shadow were Borel, a contradiction. □

This example is not a counterexample to graphicness of a Polish group. It invalidates the specific witness-erasing projection step. Keeping the extra witness changes the vertex space, while the definition requires a graph on the original space. A different Borel encoding that also preserves the required orbit components is still needed.

45.11 Target 52: increasing unions of hyperfinite relations

The question is whether $D_0 \subseteq D_1 \subseteq \cdots$, each a hyperfinite countable Borel equivalence relation, implies that $E = \bigcup_k D_k$ is hyperfinite. Frisch, Shinko and Vidnyánszky retain this as Question 0.2 in their September 2026 revision; their Theorem 0.3 shows that a counterexample would make the collection of hyperfinite relations $\mathbf{\Sigma}^1_2$-complete [177]. It is not legitimate to interchange the two increasing limits without a coherence argument.

Write $D_k = \bigcup_m F_{k,m}$, with each row increasing in m and each $F_{k,m}$ finite Borel. The following form of the known Boykin–Jackson reduction identifies exactly the missing choice.

Proposition 45.7 (Coherent rates yield a finite exhaustion)**.** *Suppose a Borel $\psi : X \to \mathbb{N}^{\mathbb{N}}$ satisfies, for every xEy, both*

$$\psi(x) =^* \psi(y), \qquad xF_{k,\psi(x)(k)}y \quad \textit{for all sufficiently large } k.$$

Then E is hyperfinite. In particular, if E is Borel bounded, these rates exist and E is hyperfinite.

Proof. Define H_n by

$$xH_ny \iff \forall k \geq n \ \big(\psi(x)(k) = \psi(y)(k) \text{ and } xF_{k,\psi(x)(k)}y\big).$$

For each fixed k the displayed condition is an equivalence relation: on each level set of $\psi(\cdot)(k)$ it is the corresponding $F_{k,m}$. Thus H_n is a Borel equivalence relation, being a countable intersection of such relations. Its class at x is contained in the finite $F_{n,\psi(x)(n)}$-class, and $H_n \subseteq H_{n+1}$. Also $H_n \subseteq E$. The two hypotheses ensure that every pair in E belongs to some H_n, proving $E = \bigcup_n H_n$.

For the last assertion enumerate each E-class by Borel maps $f_i(x)$, allowing repetitions, using countable-section uniformization. Set $\phi(x)(k)$ to the maximum, over $i \le k$ for which $xD_k f_i(x)$, of the least m with $xF_{k,m}f_i(x)$; take maximum 0 for the empty set. These least values and finite maxima are Borel. Borel boundedness supplies ψ eventually dominating ϕ and eventually equal on E-classes. Fix xEy and choose i with $y = f_i(x)$. For all sufficiently large k, one has $i \le k$, $xD_k y$, and $\psi(x)(k) \ge \phi(x)(k)$, which gives the second hypothesis by monotonicity of the row $F_{k,m}$. □

The implication in the last sentence is the existing theorem of Boykin and Jackson, recorded as Theorem 1.3 by Coskey and Schneider [175]; the proof above makes its missing rate explicit. Joining finitely many finite equivalence relations is not an alternative: on $\mathbb{Z}$, the pair partitions $\{2j, 2j+1\}$ and $\{2j-1, 2j\}$ have an infinite join. Those two relations are not nested, so this is only a failure of a proposed construction, not a counterexample to the increasing-union conjecture. Neither the coherent rates nor Borel boundedness of the arbitrary union are established here. Targets 49 and 52 consequently remain linked open problems rather than two closures of one selection lemma.

46 Computability and decision problems: nine audited targets

This section concerns catalogue entries 12, 16, 18, 19, 37, 43, 44, 51, and 57, with a literature checkpoint of 17 October 2026. Entry 37 contains two different decision problems, and entry 43 contains two different monadic theories. We distinguish externally established results, proved restricted transfers, and questions left open here. No proof below is claimed to settle a previously open unrestricted problem. In particular, qualitative identification of an object, finite-dimensional representation, and a uniform decision algorithm are different conclusions.

46.1 12. Automorphisms of the Turing degrees

Let $\mathcal{D} = (\mathcal{P}(\mathbb{N})/\equiv_T, \le_T)$. The rigidity conjecture asks whether every order automorphism of $\mathcal{D}$ is the identity. This is a question about all Turing degrees, not just the computably enumerable degrees or degrees below a particular jump. Slaman and Woodin formulate rigidity as Conjecture 7.5.4 and prove its equivalence to their biinterpretability statement in Theorem 7.5.3 of [178]. The introduction of [179] records the finite-automorphism-base theorem while distinguishing the still unresolved global rigidity problem from its own results on Δ^0_2 degrees. No resolution was verified in the present source search.

Here is the precise limitation of a reconstruction argument. An *automorphism base* of a structure $\mathcal{A}$ is a subset B whose pointwise stabilizer in $\mathrm{Aut}(\mathcal{A})$ is trivial.

Proposition 46.1 (Anchoring a reconstruction base)**.** *If B is an automorphism base, restriction to B is an injective map from $\mathrm{Aut}(\mathcal{A})$ to the set of maps $B \to A$. If each element of B is individually definable without parameters, then $\mathcal{A}$ is rigid.*

Proof. If automorphisms f, g agree on B, then $g^{-1}f$ fixes B pointwise and hence is the identity. Every automorphism preserves a parameter-free formula with a unique solution. Under the second assumption all automorphisms therefore fix B pointwise. □

This elementary proposition isolates an actual missing step. A finite base is a collection of identifying coordinates; it need not be a collection of invariantly named coordinates. In the two-element structure with equality only, a singleton is an automorphism base but the transposition is an automorphism. Thus neither a finite-base theorem nor an analogue of the

earlier mixture-identification results proves rigidity. One would still have to establish suitable parameter-free anchors, or another argument forcing the automorphisms to fix a base. That task is not achieved here.

46.2 16. Kolmogorov–Loveland versus Martin–Löf randomness

Work on Cantor space with fair-coin measure. A nonmonotonic betting strategy chooses, computably from the bits already inspected, a fresh coordinate and a fair nonnegative wager on that bit. Success means unbounded capital along the resulting play. In the usual Kolmogorov–Loveland setting partial computable strategies are allowed; only completed legal steps contribute to a play. The equality question asks whether avoiding success of all such strategies is equivalent to Martin–Löf randomness. Requiring all strategies to be total is a convention that must be stated separately; see the definitions and comparison in [180].

Proposition 46.2 (The effective maximal-inequality direction)**.** *Every Martin–Löf random sequence defeats every computable nonmonotonic fair betting strategy, including a partial strategy on its completed legal play.*

Proof. Normalize the initial capital to one. A finite legal history of k queried bits specifies a cylinder on k distinct coordinates and has measure 2^{-k}. At any history the two successor histories query the same new coordinate and have equal conditional probability; fairness makes capital a nonnegative martingale on this binary history tree. Stopping at the first completed history whose capital exceeds 2^m and truncating the tree at depth N, conservation of expected capital gives measure at most 2^{-m} for the stopped histories. Histories at which a partial computation does not finish can be made terminal; this does not increase that measure. Letting N increase proves the same bound for all first-hit histories.

Completed histories are effectively enumerable. Their finite coordinate cylinders are effectively open in the usual product topology: each is a finite union of initial-segment cylinders. Thus the union U_m of histories with capital exceeding 2^m is uniformly effectively open and has measure at most 2^{-m}. A successful play belongs to every U_m, contrary to Martin–Löf randomness. □

The reverse direction is not proved. An especially recent result must not be misreported as doing so. Petrović's preprint of 9 October 2026, [181, Theorem 4.1], constructs an encoding E and two total computable nonmonotonic strategies which succeed on the encoded points associated with a non-Martin–Löf random input. The encoding in its Section 2.8 is given through clopen partitions; the theorem is not a success theorem on the unchanged input. Its Section 5 explicitly explains a failure of the analogous unencoded construction. Our attempted transfer would need to remove this encoding while preserving computable betting and success on the original sequence. Injectivity or measure preservation alone supplies no such conversion. The missing effective conversion, and the total/partial convention, remain essential.

46.3 18. Which Kreisel conjecture?

The primary interpretation here is the bounded-proof-steps conjecture for *PA with Kleene's specified presentation*. If $A(x)$ has only the displayed free variable and some fixed k bounds the number of inference steps in proofs of $A(\overline{n})$ for every standard n, must PA prove $\forall x A(x)$? The numeral convention, language, axioms, and proof calculus form part of the problem.

Piazza's 2026 paper [182, Sections 1 and 5] states this version as Friedman's Problem 34 and proves a finite algebraic *target* theorem, not the conjecture. Parikh's positive result for a relational presentation PA^* and Hrubeš's negative result for a conservative definitional

extension PA(−) are not contradictory: they concern different proof systems. See [183, Introduction, Theorem 3 and Corollary 1]. A bound by a total recursive function on *proof codes* is yet another formulation; the adapted formulation in [184, Theorem 2.1] is false.

Proposition 46.3 (Uniform proof search is weaker than universal proof)**.** *For any recursively axiomatized theory T, if $T \vdash A(\overline{n})$ for every standard n, a total computable function can select a T-proof of each instance. Even for consistent PA this conclusion does not imply* $\mathrm{PA} \vdash \forall x A(x)$.

Proof. On input n, enumerate T-proofs until a proof of $A(\overline{n})$ occurs. The assumption makes this algorithm total. For the second assertion let $A(n)$ say that there is no PA-proof of $0 = 1$ with code at most n, using the standard arithmetization of proofs. If PA is consistent, each individual assertion is a true finite primitive-recursive calculation and is provable in PA. Their universal closure is the usual consistency statement for PA, which PA cannot prove by the second incompleteness theorem. □

This does not refute Kreisel's conjecture: the proof-step lengths in this example have not been uniformly bounded. The attempted finite-state transfer is correspondingly precise. Finitely many proof skeletons at a fixed bound yield finitely many combinatorial states, but the states still require arithmetic realizations. Piazza's Section 5 separates external membership in these realization sets from internally proving their coverage and transporting that coverage to $A(x)$. Neither our earlier finite spectral descriptions nor Proposition 46.3 fills that arithmetic gap.

46.4 19. Sacks's conjecture on order embeddings

The conjecture is that every locally countable partial order of cardinality at most $2^{\aleph_0}$ embeds into $\mathcal{D}$ by an order embedding, preserving both comparisons and incomparisons. Local countability means that every principal lower cone is countable. These are necessary conditions: there are only countably many sets computable from any fixed oracle, and there are $2^{\aleph_0}$ Turing degrees. They are not presently known to be sufficient in ZFC.

Higuchi and Lutz prove the height-two case and a serious obstruction at height three in [185, Theorems 1.1 and 1.2]. Miller and Greenberg obtain further height-three cases in [186, Theorem 1.1 and Proposition 1.2]. Their restriction counts elements of *depth three*, with depth defined by chains beginning at an element; it is not a claim about arbitrarily many uppermost elements under a reversed convention.

Proposition 46.4 (Obstruction to a continuous bottom-layer transfer)**.** *A scheme which always embeds a Cantor-indexed bottom layer using a continuous injective map into $2^{\mathbb{N}}$ cannot prove the general Sacks conjecture by subsequently extending that map to the other layers.*

Proof. A continuous injection from the compact space $2^{\mathbb{N}}$ into $2^{\mathbb{N}}$ is a homeomorphism onto a compact subset without isolated points, hence has perfect image. Higuchi–Lutz [185, Theorem 1.2] construct a locally countable height-three partial order of size continuum such that an embedding of its bottom layer whose image contains a perfect set of representatives cannot extend to an embedding of the whole order. Apply the proposed scheme to that order. Its prescribed bottom-layer image has exactly the property forbidden by that theorem. □

This is an obstruction to a specific regular construction, not a counterexample to Sacks's conjecture in ZFC. To use analytic reconstruction here one must additionally construct the required Turing reductions and prevent all unwanted reductions across an uncountable family. A perfect-set coding of independent coordinates does not provide that extension theorem.

46.5 37. Skolem and Positivity: an effective spectral subcase

For a rational linear recurrence sequence $(u_n)_{n\geq 0}$ specified by its coefficients and initial values, the *Skolem problem* asks whether some $u_n = 0$. The *Positivity problem* asks whether every $u_n \geq 0$; it is distinct from eventual positivity. Integer and rational formulations are effectively interchangeable by an appropriate positive exponential rescaling. The 2026 survey [187, Theorem 25, Sections 5.3, 7 and 9] records decidability of Skolem up to order four and leaves order five open. For Positivity, general order six and simple order ten remain open; general order at most five and simple order at most nine are known decidable. These statements are literature results, not consequences of this manuscript.

The projected-zero argument in the preceding structural manuscript used Skolem–Mahler–Lech qualitatively. It gave arithmetic progressions plus a finite exceptional set; it did not compute a bound for that exceptional set. The following restricted transfer makes the extra effective hypothesis explicit. All algebraic inputs are given by exact defining polynomials and isolating data.

Theorem 46.5 (Decision with a common-power positive-spectrum promise)**.** *Let (u_n) be a real algebraic linear recurrence sequence of arbitrary finite order. Suppose an integer $M \geq 1$ is supplied such that λ^M is positive real for every nonzero characteristic root λ. There is an algorithm computing its zero set as a finite set together with finitely many arithmetic progressions. The same algorithm determines the eventual sign on every residue class modulo M and decides both Skolem and Positivity for this input class.*

Proof. Effective algebraic linear algebra computes the generalized eigenspace expansion of the recurrence. The zero characteristic root contributes only a computably bounded finite transient, which we first check directly. Choose a computable K_* such that $Mk + r$ is beyond the transient whenever $k \geq K_*$ and $0 \leq r < M$. Combining equal bases gives, on each residue,

$$u_{Mk+r} = \sum_{j=1}^{s_r} P_{r,j}(k)\rho_{r,j}^k, \qquad 0 < \rho_{r,1} < \cdots < \rho_{r,s_r}, \quad k \geq K_*. \tag{46.1}$$

The coefficients and bases are algebraic and effectively computable. They are real: conjugating the expansion leaves the real sequence unchanged, and independence of exponential-polynomial sequences with distinct positive bases forces each combined polynomial to equal its conjugate. Alternatively one can group conjugate terms directly. Exact algebraic arithmetic decides which polynomials vanish.

If all polynomials vanish, this residue contributes the entire progression $\{Mk + r : k \geq K_*\}$. Otherwise delete zero polynomials and suppress r in the notation. Let the largest remaining base be ρ_s, and write $P_s(k) = ak^d + \sum_{i<d} a_i k^i$, with $a \neq 0$. Choose rational $a_* > 0$ with $a_* < |a|$, and rational $B > \sum_{i<d} |a_i|$. For $k \geq 1$, the lower-degree sum is at most Bk^{d-1} when $d \geq 1$. Therefore, for an effectively chosen $K_0 \geq K_*$, the polynomial $P_s(k)$ has sign $\operatorname{sgn}(a)$ and

$$|P_s(k)| \geq ck^d \quad (k \geq K_0) \tag{46.2}$$

for a computable rational $c > 0$, for instance $c = a_*/2$. For $d = 0$ this holds directly.

If $s = 1$, the conclusion already follows. Otherwise algebraic comparison supplies rational $q \in (0, 1)$ exceeding every ρ_j/ρ_s for $j < s$. Polynomial coefficient bounds give a rational $C > 0$ and integer $D \geq 0$ such that, for all $k \geq 1$,

$$\left| \sum_{j<s} P_j(k)(\rho_j/\rho_s)^k \right| \leq Ck^D q^k.$$

Set $h = \max(D - d, 0)$. We now give an effective tail cutoff rather than merely using exponential decay. Choose rational q' with $q < q' < 1$. Search for $K_1 \geq \max(K_0, 1)$ such that $q(1 + 1/K_1)^h < q'$. The ratio

$$\frac{(k+1)^h q^{k+1}}{k^h q^k} = q(1 + 1/k)^h$$

then remains below q' for $k \geq K_1$. Search for $K \geq K_1$ with $CK^h q^K < c/2$. This search terminates by the same geometric ratio bound, and all its comparisons are rational. For every $k \geq K$ the lower bases contribute less than $(c/2)k^d$, whereas (46.2) contributes at least ck^d. Thus (46.1) is nonzero with the fixed sign of a on that tail.

Repeat over the finitely many residues. Check all remaining initial terms exactly by real algebraic arithmetic. The zero progressions and the finite zeros found in these checks are the full zero set. For Positivity, every fixed tail sign and every checked initial sign must be nonnegative. This is a terminating decision procedure. □

The promise is verifiable for the supplied M by algebraic arithmetic. The theorem covers arbitrary Jordan multiplicities and root-of-unity phases of bounded common order. It is an explicit version of a classical dominance argument for a promised subcase; we make no historical novelty claim. General rational recurrences need not satisfy the promise. Several dominant roots with non-root-of-unity relative phases produce oscillatory cancellation; the proof's strict positive-base separation then fails.

Proposition 46.6 (Dimension one does not replace exact arithmetic)**.** *If initial values are supplied as arbitrary uniformly computable real numbers, there is no uniform Skolem algorithm even for the fixed recurrence $u_{n+1} = u_n$. There is likewise no uniform Positivity algorithm for that input representation.*

Proof. Given a program e, let

$$c_e = \begin{cases} 2^{-T}, & \text{if } e \text{ first halts at step } T \geq 1, \\ 0, & \text{if } e \text{ never halts.} \end{cases}$$

This is a uniformly computable real. To approximate within 2^{-m}, run e for m steps, returning 2^{-T} if it has halted and zero otherwise. A later halt changes the value by less than 2^{-m}. The constant sequence with initial value c_e has a zero exactly when e does not halt. A decision algorithm would decide the halting problem. For Positivity use the constant sequence with initial value $-c_e$: all its terms are nonnegative exactly when e does not halt. □

Here every zero set is either empty or all of $\mathbb{N}$, and every spectrum consists of the single root 1. Thus even this strongest qualitative zero-set description does not supply an algorithm for an arbitrary computable-real representation. This proposition does *not* prove the classical rational-input Skolem or Positivity problem undecidable. It identifies the exact arithmetic assumption that an attempted transfer must retain.

46.6 43. BMTO has a new answer; MTWO is a different question

BMTO means the *Borel monadic theory of the real order*: first order variables range over $\mathbb{R}$, the relation is $\leq$, and monadic second-order variables range over Borel subsets of $\mathbb{R}$. It is decidable by Manthe [188, Theorem 1.1], whose second preprint version is dated 9 March 2026. This is an external resolution and must be removed from an unqualified list of unsolved problems. We do not claim to have proved Manthe's theorem.

MTWO means the *monadic theory of well-ordering*, with full subset quantification. We interpret the historical question as asking whether it is consistent with ZFC that the set of monadic sentences valid in all wellorders is recursive. Satisfiability in some wellorder is the equivalent complementary decision problem. It is not the theory of the single order $(\omega, <)$, and it is not BMTO. Shelah's proposed sufficient set-theoretic hypothesis appears as Conjecture 0.3 in the author-posted version of [189]. Neeman's primary account [190, Theorems 11–16 and final discussion] separates independence at ω_2, consistency of decidability at ω_3, and still further questions about all ordinals. No later resolution of the full historical MTWO question was verified in this audit; its status here is retained as unresolved, with that explicit verification limit.

Proposition 46.7 (The missing effective table)**.** *Fix a set-theoretic universe. Suppose an algorithm, given r, returns the realized members of a fixed effective finite list of rank-r monadic Hintikka sentence types, and this returned list is exactly the set of types realized by wellorders in that universe. Then MTWO in that universe is decidable.*

Proof. An effective finite Hintikka list, allowing unrealizable members, is obtained by induction on rank and number of free individual and set variables in this finite relational language. Given a sentence, one computes its rank and the types on which it holds by the same syntactic induction. It is valid in all wellorders if and only if it holds on every type returned by the hypothesized algorithm. This uses no decision procedure for consistency of the candidate types; computing which candidates are realized is exactly the hypothesis. □

The hypothesis is not established. Merely knowing that there are finitely many rank-r types does not compute which are realized. Neeman's full-limit automata make the obstacle concrete: their uncountable-limit transitions consult formulas in the almost-all theory of the run, involving club and stationary-set behavior. This is additional semantic data, not an ordinary finite transition table that can simply be executed on a finite input. The previous finite-state or finite-spectrum arguments do not compute it. Manthe's restriction to Borel subsets of the real line does not eliminate full-subset quantification over arbitrary uncountable ordinals, so it supplies no asserted solution of MTWO.

46.7 44. Laurent series: specify both field and language

The principal open target is the full first-order theory of $\mathbb{F}_q((t))$ in the pure ring language, for a fixed finite field $\mathbb{F}_q$ of positive characteristic p. The notation $\mathbb{Z}_p$ in informal lists must not silently identify the finite field $\mathbb{F}_p$ with the ring of p-adic integers, which is not itself a field. Characteristic-zero Laurent fields have a different Ax–Kochen–Ershov theory. Naming a uniformizer, adding valuation predicates, or adding a predicate for its integral powers also changes the question.

Anscombe and Fehm [191, Introduction and Theorems 1.1–1.2] explicitly distinguish the open full positive-characteristic problem from known existential decidability without a named uniformizer and from results conditional on resolution assumptions in stronger languages. In a substantially stronger expansion, Gitin [192, Theorem 9] proves existential undecidability when a positive-valuation element α and the cyclic subgroup $\alpha^{\mathbb{Z}}$ are named. That is not undecidability of the pure field.

Proposition 46.8 (An exact Frobenius fragment)**.** *For an explicitly given rational function $a \in \mathbb{F}_q(t)$, the following conditions are equivalent and decidable:*

$$(\exists x \in \mathbb{F}_q((t)))\ x^p = a, \qquad \frac{da}{dt} = 0, \qquad a \in \mathbb{F}_q(t^p).$$

When they hold, a root can already be computed in $\mathbb{F}_q(t)$.

Proof. In characteristic p, a Laurent series raised to the pth power has only exponents divisible by p, so its formal derivative vanishes. The rational-function derivative is the restriction of the Laurent-series derivative. Conversely, $1, t, \dots, t^{p-1}$ is a basis of $\mathbb{F}_q(t)$ over $\mathbb{F}_q(t^p)$. Writing $a = \sum_{i=0}^{p-1} b_i(t^p)t^i$ and differentiating shows $a' = \sum_{i=1}^{p-1} i b_i(t^p)t^{i-1}$. Basis independence forces $b_i = 0$ for $i > 0$ when $a' = 0$. Finally, the finite coefficient field is perfect. If $a = b(t^p)$, applying inverse Frobenius to the coefficients of a numerator and denominator of b gives a rational function whose pth power is a. Polynomial arithmetic computes the derivative and tests whether it is identically zero, and the same basis calculation computes the root. □

This is a restricted parameterized family of existential instances, not new quantifier elimination. It demonstrates where exact coefficient information makes a decision possible. It does not address arbitrary alternating quantifiers or supply the missing positive-characteristic model theory.

46.8 51. Scott sets and standard systems

A Scott set is a nonempty Turing ideal $S \subseteq \mathcal{P}(\mathbb{N})$ such that every infinite binary tree coded in S has a path in S. For a nonstandard model $M \models \mathrm{PA}$, its standard system $\mathrm{SSy}(M)$ consists of the traces on the standard natural numbers of sets coded in M. Scott's problem asks whether every Scott set is $\mathrm{SSy}(M)$ for some such M. It is known for countable Scott sets and for Scott sets of cardinality at most $\aleph_1$; consequently CH gives a positive answer for all Scott sets. The unrestricted ZFC question remains open in the primary accounts [193, Theorems 1.2–1.3, Question 1.4] and [194, Theorems 1.2–1.3, Corollary 1.4]. Wang's manuscript is dated 2020; a later crawl date is not a new publication or a new resolution.

Proposition 46.9 (Where the chain construction needs an extension theorem)**.** *Let S be a Scott set of uncountable cardinality κ. Assume a countable nonstandard $M_0 \models \mathrm{PA}$ has $\mathrm{SSy}(M_0) \subseteq S$. Suppose that whenever $M \models \mathrm{PA}$ has cardinality less than κ and $\mathrm{SSy}(M) \subseteq S$, each $A \in S$ can be placed in $\mathrm{SSy}(N)$ for an elementary extension $N \succeq M$ satisfying*

$$\mathrm{SSy}(N) \subseteq S, \qquad |N| \leq \max(|M|, \aleph_0).$$

Then S is the standard system of a model of PA.

Proof. Enumerate $S = \{A_\alpha : \alpha < \kappa\}$. At successor stage use the assumed extension property to put A_α into the standard system. At limit stages take the elementary union. Every parameter coding a subset of $\mathbb{N}$ in such a union already belongs to an earlier model. Elementarity preserves its membership tests at each standard integer. Hence the standard system of a chain union is the union of the earlier standard systems. By induction $|M_\alpha| \leq \max(\aleph_0, |\alpha|)$ for infinite stages $\alpha < \kappa$, so the hypothesis remains applicable even when κ is singular. The final union has standard system contained in S and contains every A_α, as required. □

This is the standard Ehrenfeucht extension strategy; compare [194, Theorem 1.7]. A countable initial model with standard system inside S is available by first closing a countable subfamily under the Scott-set operations and applying the countable case. For $\kappa = \aleph_1$ the required countable extension lemma is known; see [193, Theorem 2.1]. For larger κ, the uncountable extension property, with control of all new coded traces, is the unresolved input. Recovering a probability law or a finite family of invariants does not provide that elementary-extension construction.

46.9 57. The universal–existential theory of the c.e. degrees

Let $\mathcal{R}$ be the degrees containing computably enumerable subsets of $\mathbb{N}$, with the pure order relation $\leq_T$. The target asks whether the set of true sentences $\forall\bar{x}\exists\bar{y}\,\theta(\bar{x},\bar{y})$, with quantifier-free θ in this language, is decidable. The one-block theory is decidable; the three-block theory is undecidable by Lempp–Nies–Slaman [195, Introduction and main theorem]. Their two-block question is still recorded as open in the author's later account [196, slides 2–4]; no subsequent resolution was verified in this search. Results for weak truth-table degrees, all Turing degrees, or languages with additional join, meet, or jump operations must not be substituted.

Proposition 46.10 (Exact finite-menu reformulation)**.** *For any fixed nonempty partial order D, its pure-order $\forall\exists$ theory is decidable if and only if the following finite-menu extension problem is decidable: given finite labeled partial orders $P \subseteq Q_1, \ldots, Q_s$, does every embedding of P into D extend to an embedding of at least one Q_j? Finite conjunctions of such menu instances are allowed without changing decidability.*

Proof. For a sentence $\forall\bar{x}\exists\bar{y}\,\theta$, enumerate all complete atomic diagrams on the finitely many x variables, including equality and all comparisons and noncomparisons. Discard those incompatible with a partial order. Quotient each remaining diagram by its equalities to obtain a finite labeled partial order P. Enumerate all complete diagrams on $(\bar{x},\bar{y})$ which extend that diagram and satisfy θ. Quotienting equalities gives a finite menu of extensions Q_j of P, recording that some y variables may name old or repeated points. The sentence is true if and only if the corresponding menu condition holds for every initial diagram. An unrealizable initial diagram contributes a vacuously true condition.

Conversely let $\delta_P(\bar{x})$ and $\delta_{Q_j}(\bar{x},\bar{y}_j)$ be complete order diagrams, using distinct variables for distinct labeled points. The menu condition is

$$\forall\bar{x}\left(\delta_P(\bar{x}) \;\longrightarrow\; \bigvee_{j=1}^{s} \exists\bar{y}_j\,\delta_{Q_j}(\bar{x},\bar{y}_j)\right).$$

Move the finite existential blocks together. Variables in unused disjuncts can be filled arbitrarily because D is nonempty. This is a $\forall\exists$ sentence. An empty menu is handled by a false consequent. Finite conjunctions of $\forall\exists$ sentences remain in the same fragment after renaming and merging their quantifier blocks. □

This is a standard structural translation, explicitly used in [196, slides 3–4], and is included to expose the remaining computational task. A decision procedure for a *single* prescribed extension Q does not automatically decide a menu. In general

$$\forall e \bigvee_j \mathrm{Ext}(e,Q_j) \quad\not\Longleftrightarrow\quad \bigvee_j \forall e\ \mathrm{Ext}(e,Q_j).$$

For two possible embeddings, one extension may work only for the first and another only for the second. The left condition then holds and the right fails. One needs a uniform structural classification of the menu property in the actual c.e. degree structure. Neither the reduction nor the previous analytic identification results provide it.

46.10 Outcome of this branch

The source audit corrects an obsolete status for BMTO, an external solution. It also prevents three concrete false transfers: the recent encoded randomness theorem is not KL=ML; a finite algebraic target for proof skeletons is not Kreisel's conjecture; and Skolem–Mahler–Lech is not

a general zero-finding algorithm. The fully proved algorithm in Theorem 46.5 addresses an exact promised spectral class and both decision questions in entry 37. Proposition 46.6 shows that dropping exact arithmetic can destroy uniform decidability even in dimension one. The remaining propositions give explicit reductions or construction obstructions, with their missing hypotheses stated. None is counted as a new closure of an unrestricted historical conjecture.

47 Proof theory and reverse mathematics: nine precisely delimited targets

This section audits catalogue items 2, 15, 17, 24, 25, 26, 47, 54 and 55 as of 17 October 2026. None of the nine unrestricted targets is claimed to be solved here. The positive statements proved below are transfer criteria and explicit obstructions to proposed shortcuts; they are not counted as nine new solutions. In particular, an analytic inverse theorem, a uniquely specified optimizer, or an existence proof over the real numbers supplies neither a short propositional proof nor a derivation in a weak arithmetic theory without an additional, quantitative formalization theorem.

We use the usual second-order arithmetic systems RCA_0, WKL_0, ACA_0, ATR_0, and Π^1_n-CA_0 with set parameters in the indicated comprehension schemes. A statement of conservativity always specifies both its base theory and its syntactic class. Statements about models in the arguments below are metatheorems in ordinary classical mathematics, not implicit formalizations inside the weak theory under discussion.

47.1 Item 2: unrestricted Frege lower bounds

Exact target and verified status. Fix a standard finite Frege calculus F, with proofs measured by total symbol length, and let $s_F(\tau)$ be the length of a shortest proof of the tautology τ. The target is that F is not polynomially bounded:

$$\forall d\,\forall C\,\exists \tau \in \mathrm{TAUT} \qquad s_F(\tau) > C(1+|\tau|)^d.$$

An explicit uniformly constructible hard family is a stronger, usual research goal. Extended Frege, bounded-depth Frege, and a restriction on individual line size are different targets. Filmus–Pitassi–Santhanam's Theorem 2 gives a subexponential bounded-depth simulation of short Frege proofs [197]. A 2026 result of de Rezende–Engström–Ghannane–Risse proves superpolynomial length lower bounds for *tree-like semantic systems with bounded line size*; its hypotheses do not yield the displayed unrestricted statement [198]. Our current primary-source search found no verified unrestricted resolution.

Proposition 47.1 (The quantitative hypothesis required for a transfer)**.** *Suppose a map sends every F-proof π of τ to a certificate in a system Q for an instance $R(\tau)$, with length at most $C(|\pi|+|\tau|)^d$, where $C>0$ and $d\geq 1$ are fixed. If all Q-certificates for $R(\tau)$ have length at least $L(\tau)$, then*

$$s_F(\tau) \geq \left(L(\tau)/C\right)^{1/d} - |\tau|.$$

Consequently, a superpolynomial lower bound in $|\tau|$ for these certificates transfers to F.

Proof. Apply both bounds to the image of a shortest F-proof and solve the inequality $L(\tau) \leq C(s_F(\tau)+|\tau|)^d$. If the conclusion had a polynomial upper bound, this same inequality would give a polynomial upper bound on L. □

Attempt and obstruction. The attempted transfer encodes a tautology as a finite positivity or identification assertion and then uses the earlier analytic uniqueness results to certify it. Even granting such a sound encoding, we have neither a map from *every* short Frege proof to a bounded-size analytic certificate nor a lower bound on the encoded certificates. An algorithm that happens to produce large certificates does not lower-bound other proofs. The proposition locates the two missing statements exactly. It is an elementary conditional transfer, not a new Frege lower bound. Even a bounded-depth lower bound must have the strength required by the simulation in [197]; a lower-bound exponent that degenerates with depth cannot be silently made uniform.

47.2 Item 15: Fraïssé's theorem over ATR_0

Exact target and verified status. Let FRA state that every sequence $(L_i)_{i\in\mathbb{N}}$ of countable linear orders has $i < j$ and an order embedding $L_i \hookrightarrow L_j$. The question is $\mathsf{ATR}_0 \vdash \mathsf{FRA}$, equivalently the remaining implication in the proposed equivalence over RCA_0. This is not the truth of Fraïssé's conjecture, which Laver proved. Freund's 2024 paper, Theorem 1.2, gives the stronger-than-ATR_0 upper theory of pseudo Π^1_1-comprehension with the indicated Γ closure; its introduction records the lower implication to ATR_0 and the unresolved converse [199]. That upper bound is not a proof in ATR_0.

Lemma 47.2 (Bounded witnesses are the missing compactness datum)**.** *Over WKL_0, let $T \subseteq \mathbb{N}^{<\mathbb{N}}$ be a coded prefix-closed tree and let $b : \mathbb{N} \to \mathbb{N}$ be a function such that $\sigma(i) \le b(i)$ for every $\sigma \in T$ and $i < |\sigma|$. If T has nodes of every length, it has an infinite path.*

Proof. Encode the ith entry by a binary block of a fixed positive length large enough to encode $0, \dots, b(i)$. Form the binary tree of prefixes of the encoded members of T. Membership of a binary string is decidable relative to T, b: at a partial last block there are only boundedly many completions to test. This is an infinite binary tree. A path given by weak König's lemma decodes at block endpoints into a path through T. □

For orders with underlying set $\mathbb{N}$, suppose that for each n the induced order on $\{0, \dots, n-1\}$ in L embeds in M, with the image of i bounded by the *same* function $b(i)$ for all n. The tree of such finite embeddings satisfies the lemma; its path is an embedding $L \hookrightarrow M$. This is a valid transfer of finite consistency to a global object.

Proposition 47.3 (Finite ages do not provide that bound)**.** *Every finite suborder of ω^* embeds in ω, but ω^* does not embed in ω. In particular, equality of the finite ages of two countable linear orders does not imply that either embeds in the other.*

Proof. Every finite linear order embeds in both orders. An embedding of ω^* in ω would give an infinite strictly descending sequence of natural numbers. Starting with a_0, induction gives $a_n \le a_0 - n$ for $n \le a_0 + 1$, a contradiction. Similarly an embedding of ω in ω^* would give a descending sequence in the usual natural-number order. The tree of finite descending sequences in ω has a node at each level but no infinite path; it has no uniform bound on its first coordinate. □

Attempt and obstruction. Finite observation profiles of orders, or finite embeddings recovered from those profiles, do not produce the coherent bounds in Lemma 47.2. Nor does that lemma choose a good pair from an arbitrary sequence of orders. The explicit pair ω^*, ω refutes the first shortcut before any reverse-mathematical issue is reached. Obtaining a good pair and its embedding using only arithmetical transfinite recursion remains the missing step; neither is furnished by the earlier analytic identifiability results.

47.3 Item 17: the exact conservation problem for Ramsey's theorem

Exact target and current evidence. Write $B = \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2$. The target is

$$B + \mathsf{RT}^2_2 \vdash \varphi \quad \Longrightarrow \quad B \vdash \varphi \qquad (\varphi \in \Pi^1_1),$$

where RT^2_2 concerns two-colourings of unordered pairs and Π^1_1 permits a universal set quantifier followed by an arbitrary arithmetical formula. Le Houérou–Levy Patey–Yokoyama, corrected version 3 of 6 May 2026, Theorem 1.4, prove $\forall\Pi^0_4$ conservativity, even with WKL_0 added. Their Theorem 1.2 reports the exact equivalence of the full target to $\forall\Pi^0_5$ conservativity; Question 1.1 states the full problem [200]. The correction in version 3 is material.

An author-posted manuscript by Hank Yu-Shiang Hwang, dated 3 October 2026, claims the full result [201]. We located its full text and its binary-partition/model-extension mechanism, but have not independently verified the complete proof. Thus the status recorded here is *a verified partial theorem plus a recent unverified full-resolution claim*, not an assertion that no such claim exists, and not a claimed solution by this manuscript.

Proposition 47.4 (A sufficient model-extension criterion)**.** *Let $B \subseteq T$ be countable theories of second-order arithmetic interpreted as two-sorted first-order theories. Suppose every countable model $(M, \mathcal{S}) \models B$ has an extension $(M, \mathcal{S}') \models T$ with $\mathcal{S} \subseteq \mathcal{S}'$ and with the same first-order structure and the same membership relation on old sets. Then T is Π^1_1 conservative over B.*

Proof. Suppose $B \nvdash \forall X\, \theta(X)$, where θ is arithmetical. By completeness and the downward Löwenheim–Skolem theorem there is a countable $(M, \mathcal{S}) \models B$ and $A \in \mathcal{S}$ satisfying $\neg\theta(A)$. Extend this model by the hypothesis. Induction on the arithmetical formula shows that its truth with parameter A is unchanged: all number quantifiers have the same domain and all atomic relations are the same. Hence the extension is a model of $T + \neg\theta(A)$, so $T \nvdash \forall X \theta(X)$. Several universal set variables can be treated in the same way, or coded into one set. □

Attempt and obstruction. Adding homogeneous sets in the standard natural numbers is not the hypothesis of Proposition 47.4: the relevant countable models may have nonstandard first-order parts. Their induction and bounding schemes must survive the extension. This is a genuine distinction. If a consistent recursively axiomatized B is strong enough for the second incompleteness theorem and has an ω-model, every ω-model satisfies the true arithmetical sentence $\mathrm{Con}(B)$, while B does not prove that sentence. Consequently, a proof valid in all standard-domain models is not a syntactic conservation proof. Our analytic constructions supply no nonstandard-model preservation theorem, and no $\forall\Pi^0_5$ conservation argument. Auditing the new claim would specifically require checking the coded-history, jump-regularity, and bounding-preservation steps rather than replacing them by standard compactness.

47.4 Item 24: Hindman's theorem and the iteration obstruction

Exact target and verified updates. For every k and $c : \mathbb{N} \to k$, full HT asks for an infinite H such that all nonempty sums of *distinct* members of H have one colour. The reverse-mathematical question is its exact strength over RCA_0, between ACA_0 and ACA^+_0, where the latter asserts existence of each ωth Turing jump. Exactly-two-sum, at-most-two-sum, and iterated versions must not be interchanged. For example, the two-colour exactly-two-sum version follows from RT^2_2, whereas full HT implies ACA_0 [202].

The 2026 preprint of Liu–Patey, Main Theorem 1.5, proves that any arithmetic instance has a solution avoiding a prescribed non-arithmetic Turing cone. Its introduction explicitly explains why this one-application result does not separate HT from ACA^+_0 [203]. A new negative result

of Liao gives a computable colouring with no infinite Π^0_3 solution [204]. Neither result supplies the exact reverse-mathematical classification.

Proposition 47.5 (A bounded finite-sums transfer)**.** *Over* WKL_0*, fix* $c : \mathbb{N} \to k$ *and* $b : \mathbb{N} \to \mathbb{N}$*. Suppose that for each* n *there are* $x_0 < \cdots < x_{n-1}$ *with* $x_i \le b(i)$ *whose nonempty finite sums all have the same colour. Then* c *has an infinite Hindman solution.*

Proof. The strictly increasing tuples with monochromatic finite sums form a prefix-closed tree. Checking its membership is a finite computation relative to c. The bounds $b(i)$ and Lemma 47.2 give an infinite path (x_i). Every two nonempty finite sums occur in one sufficiently long prefix of this path, so have the same colour. Its range is the required infinite set, which exists in RCA_0 for a strictly increasing sequence. □

The missing bound is a single b valid simultaneously for arbitrarily long solutions; bounds that grow with the target length do not suffice. Thus finite-sums compactness does not resolve the infinite theorem in a weak base. The recent cone-avoidance theorem suggests a sharper attempted route. Write $C \le_{\mathrm{ar}} Z$ if $C \le_T Z^{(n)}$ for some $n < \omega$.

Proposition 47.6 (The iteration hypothesis that would separate the upper bound)**.** *Let* P *be a second-order combinatorial principle with sets coding its instances and solutions and with arithmetical instance and solution predicates, as for Hindman's theorem. These predicates are absolute to the* ω*-models used below. Assume the following relativized arithmetic-cone avoidance property:*

$$\begin{aligned} &C \not\le_{\mathrm{ar}} Z, \quad I \le_T Z, \quad I \text{ a } P\text{-instance} \\ \Longrightarrow \quad &\exists H\,[H \text{ solves } I \ \wedge\ C \not\le_{\mathrm{ar}} Z \oplus H]. \end{aligned} \tag{AC}$$

Then for every non-arithmetic C *there is an* ω*-model of* $\mathsf{ACA}_0 + P$ *not containing* C*. In particular, if (AC) holds for* $P = \mathsf{HT}$*, then* $\mathsf{ACA}_0 + \mathsf{HT}$ *does not imply* ACA_0^+*.*

Proof. Start with $Z_0 = \varnothing$. Build an increasing Turing sequence (Z_s), maintaining $C \not\le_{\mathrm{ar}} Z_s$. Schedule all pairs (n, e), with $n \le s$, so every pair is eventually treated after Z_n has been defined. If the eth computation relative to Z_n is a code for a P-instance I, it is computable from Z_s. By (AC) choose a solution H with $C \not\le_{\mathrm{ar}} Z_s \oplus H$, and put $Z_{s+1} = (Z_s \oplus H)'$. Otherwise put $Z_{s+1} = Z_s'$. Taking a jump preserves the invariant, since being arithmetic in a finite jump of an oracle is equivalent to being arithmetic in that oracle. This construction takes place in the metatheory; deciding whether a computation codes a valid instance is not asserted to be effective.

Let $\mathcal{I} = \{X : (\exists s) X \le_T Z_s\}$. It is a Turing ideal, is closed under finite joins and under the Turing jump, and hence $(\mathbb{N}, \mathcal{I}) \models \mathsf{ACA}_0$. Every instance in $\mathcal{I}$ is eventually scheduled and has its chosen solution in $\mathcal{I}$, giving P. The invariant excludes C from $\mathcal{I}$. Finally choose $C = \varnothing^{(\omega)}$, which is not arithmetic. Any ω-model of ACA_0^+ contains this set, so the constructed model separates the theories as asserted. □

Attempt and obstruction. This is the usual ideal-construction argument with its necessary invariant written explicitly; no novelty is claimed for that method. Liu–Patey's conclusion is Turing nonreducibility, not (AC). From $C \not\le_T H$ one cannot infer $C \not\le_{\mathrm{ar}} H$: take any H and $C = H'$. Thus closing the constructed ideal under jumps can immediately destroy the weaker invariant. Proving (AC) for full Hindman solutions, or finding another preserved invariant strong enough for iteration, is the exact missing step in this attempted separation. The earlier analytic inverse results do not supply it.

47.5 Item 25: what "ordinal analysis of Π^1_2-comprehension" means

Formalization and a necessary status correction. The exact theory is $T = \Pi^1_2\text{-}\mathsf{CA}_0$, with arbitrary set parameters and its specified induction axioms. Its proof-theoretic ordinal may be defined as the supremum of order types of recursive orders whose well-foundedness T proves. An ordinal *analysis* normally demands an explicit effective notation system, proof reduction/cut elimination giving an upper bound, and matching well-ordering or reflection results giving the lower bound. Merely naming the supremum is not such an analysis.

Rathjen's 1995 paper announced analyses around Π^1_2-comprehension [205]; his 2005 complete paper analyzes a *parameter-free* Π^1_2 scheme over Δ^1_2-comprehension plus bar induction, not automatically T [206]. More recently, Arai's preprint, version 4 of 12 August 2025, Theorems 1.1–1.3, gives explicit ordinal analyses for Π_N-Collection and associated systems including Σ^1_{N+2}-dependent choice plus bar induction. Theorem 1.4 treats Z_2 in the author's convention, which includes dependent choice [207]. Accordingly, the broad assertion that analysis has not reached beyond Π^1_2 is not an accurate description of the current literature. These stronger-theory results do not, just by inclusion, identify the exact ordinal of the weaker, precisely specified T. We do not certify that exact calibration here and do not count this status correction as our solution.

Proposition 47.7 (Two limits on an ordinal shortcut)**.** *If $T \subseteq S$, then $|T| \leq |S|$ for the preceding definition of ordinal. An upper analysis of S alone supplies no equality. Furthermore, let T be a consistent recursively axiomatized arithmetic theory to which the second incompleteness theorem applies. If T proves $\mathrm{WF}(O) \to \mathrm{Con}(T)$ for a coded order O, it cannot also prove $\mathrm{WF}(O)$.*

Proof. Every recursive order whose well-foundedness is proved by T is also proved well-founded by S, yielding the inclusion of the two sets of order types and hence the supremum inequality. It contains no reverse inclusion. For the second assertion, the two supposed proofs would, by modus ponens, give a T-proof of $\mathrm{Con}(T)$, contradicting the second incompleteness theorem. □

Attempt and obstruction. A transfinite rank on the earlier inverse or extraction constructions ranks those constructions, not arbitrary proofs with Π^1_2 comprehension. An ordinal analysis would require a translation of the latter proofs and verification that each reduction decreases the assigned notation, followed by a cofinal lower-bound argument. None has been obtained. Arai's results provide relevant existing upper technology, but a claim of equality for T requires matching the exact axiom system and the lower-bound argument.

47.6 Item 26: Friedman's publication-scoped EFA conjecture

Which conjecture? Friedman's FOM posting of 16 April 1999 states two "grand conjectures". The EFA one concerns arithmetical *statements* of theorems published in the *Annals of Mathematics*; the other concerns eliminating categorical methods. Here EFA is $I\Delta_0 + \exp$. The former is not the claim that all true arithmetic is provable in EFA [208]. In his own 2010 clarification Friedman emphasizes ordinary core mathematics and the need for an actual arithmetical theorem, rather than a strong intermediate lemma in one proof [209].

For a precise corpus version one must fix a publication cutoff, a list $\mathcal{A}$ of included theorem statements, and their chosen arithmetizations $\widehat{\theta}$. The resulting metamathematical assertion is

$$\forall \theta \in \mathcal{A} \qquad \mathsf{EFA} \vdash \widehat{\theta}.$$

No cutoff or exclusions are specified in the catalogue label. We use this publication-scoped interpretation and record its unresolved/underspecified status. An overbroad replacement by

"every true arithmetical sentence" is false, assuming EFA is consistent, by its own consistency sentence and the second incompleteness theorem. That observation does not refute Friedman's actual corpus claim.

Proposition 47.8 (What an elementary witness bound really permits)**.** *Let $R(x, y)$ be bounded in the language of EFA and let $t(x)$ be an elementary term (allowing a definitional extension by elementary functions). If*

$$\mathsf{EFA} \vdash \forall x\, \exists y \le t(x)\ R(x, y),$$

then bounded search gives an elementary function f and EFA proves $R(x, f(x))$ for every x.

Proof. Evaluate the bounded predicate successively at $y = 0, \dots, t(x)$ and return the first successful y, using a default value if none exists. Bounded quantification and bounded search over elementary bounds preserve elementary computability: the bounded truth value can be computed by finite products and sums of 0–1 values. Bounded induction proves that the recorded first success is at most $t(x)$ and satisfies R, whenever there is a success. The assumed EFA theorem eliminates the default case. □

Attempt and obstruction. Replacing analytic existence by an explicit search in the earlier work would help only after both the bound and its correctness have been proved in EFA. An externally true estimate does not give that EFA proof. For a Π^0_1 statement such as Fermat's Last Theorem, bounding individual finite checks is likewise not a proof of their universal closure. No identified Annals theorem has been reduced to Proposition 47.8 in this work, so no new case of the grand conjecture is claimed.

47.7 Item 47: weak versus strong normalization of pure type systems

Exact target and verified status. A pure type system is specified by sorts, axioms between sorts, and product formation rules. Weak normalization means that *every typable term* has some finite β-reduction to normal form; strong normalization means that no typable term has an infinite reduction sequence. The Barendregt–Geuvers–Klop conjecture is that weak normalization of a whole PTS implies strong normalization of that PTS. It is not the false assertion that each individually weakly normalizing untyped term is strongly normalizing. The problem-originator's TLCA entry records the conjecture and the non-dependent partial result of Barthe–Hatcliff–Sørensen [210, 211]. No unrestricted resolution was verified in the current search.

Proposition 47.9 (Confluence and unique normal forms do not close the gap)**.** *There is a finitely branching, acyclic, confluent reduction relation in which every object is weakly normalizing and has the same unique normal form, but which is not strongly normalizing.*

Proof. Take objects $z, a_0, a_1, \dots$ and rules

$$a_n \longrightarrow z, \qquad a_n \longrightarrow a_{n+1} \quad (n \ge 0).$$

The only normal form is z, and every a_n reaches it in one step. Every two reducts of a common object reduce to z, which proves confluence. Each object has at most two immediate reducts, and no path returns to an earlier object. Nevertheless $a_0 \to a_1 \to a_2 \to \cdots$ is infinite. □

Attempt and obstruction. A unique analytic terminal state or a terminating chosen reduction strategy cannot establish termination of all reductions: the proposition is an exact counterexample to that abstract argument. It is *not* a counterexample PTS. To use the non-dependent positive results for an arbitrary PTS one would need a translation eliminating dependency, preserving the global weak normalization hypothesis, and reflecting strong normalization. The substitution and dependent-product steps of such a translation are the unproved requirements. A well-founded rank decreasing along every typed reduction would also suffice, but the ranks in the earlier analytic work do not rank these reductions.

47.8 Item 54: expansion postponement and the solved normalizing case

Exact statement and verified correction. For a PTS, write $\vdash_r$ for the typing relation in which conversion is restricted to type reduction, with type expansion removed. Expansion postponement asks whether

$$\Gamma \vdash a : A \quad \Longrightarrow \quad \exists A' \, [\Gamma \vdash_r a : A' \ \wedge \ A \longrightarrow_\beta^* A']. \tag{EP}$$

The final restoration of the type A can then be done by expansion. Poll's 1998 paper proves this for every PTS whose typable types have normal forms, which is weaker than requiring strong normalization of all terms. Precisely, Theorem 12 gives $\Gamma \vdash_{nf} a : \mathrm{nf}(A)$ and Theorem 8 embeds $\vdash_{nf}$ in $\vdash_r$; Corollary 13 is expansion postponement [212]. Thus the normalizing-types case is already solved and cannot be counted as a new result. The unrestricted all-PTS version is the remaining target; no verified unrestricted resolution was found here.

Attempted extension and exact failed inference. A natural route takes a conversion $A =_\beta B$, uses confluence to choose a common reduct C, and tries to replace each conversion in a typing derivation by reductions to C. The required derivation is not supplied by the term rewrite diagram: an abstraction premise has a dependent product type, and substitution changes the codomain. Reconstructing a reduction-only typing derivation requires a subject-reduction or substitution statement for the modified typing system, not just confluence of β-reduction. Poll explicitly distinguishes weak subject reduction for $\vdash_r$ from the ordinary subject-reduction statement, which does not hold for that relation in general. His normal-form intermediate relation solves this reconstruction problem using existence of normal forms. If types need not normalize, $\mathrm{nf}(A)$ is undefined and Theorem 12 cannot be applied. This identifies the failed proof step, without asserting an unproved counterexample to (EP).

Relation to item 47. Even a global proof of weak normalization for a PTS would already put its types in Poll's solved class; no appeal to strong normalization is needed for that conclusion. Conversely, eliminating expansion in typing judgments says nothing by itself about whether an arbitrary reduction path terminates. The two catalogue items are distinct and are not two consequences of one abstract uniqueness theorem.

47.9 Item 55: full graph minors, bounded treewidth, and finite size

Exact target and verified bounds. The graph minor theorem (GMT) says that every infinite sequence of finite graphs has $i < j$ with G_i a minor of G_j. The open target is its exact proof-theoretic strength, not its truth. Krombholz–Rathjen's 2019 preprint (published 2020), Section 1 and Sections 2–3, gives the upper theory

$$\Pi^1_1\text{-}\mathsf{CA}_0 + \Pi^1_3\text{-}\mathsf{IND} + \Pi^1_2\text{-}\mathsf{BI}.$$

Their Theorem 5 records equivalences for the *uniform bounded-treewidth* version, including extended Kruskal, planar GMT, and well-ordering of the indicated collapsing-function notation; Theorem 12 gives an explicit ordinal lower bound for extended Kruskal [213]. Each fixed treewidth instance and the universal statement over all bounds must be distinguished. These upper and lower results do not match into an exact full-GMT calibration; no verified later exact calibration was found in our search.

Proposition 47.10 (Finite vertex bounds have an elementary good-pair bound)**.** *For finite simple graphs with at most N vertices, every sequence of length*

$$K_N + 1, \qquad K_N = \sum_{r=0}^{N} 2^{\binom{r}{2}},$$

contains $i < j$ such that G_i is isomorphic to G_j, hence is a minor of G_j. The finite statement and its proof can be formalized in EFA with the usual finite-graph coding.

Proof. Relabel each graph on r vertices by $\{0, \dots, r-1\}$, choosing, for example, the least adjacency-matrix code among all relabellings. There are at most $2^{\binom{r}{2}}$ codes for this r, hence at most K_N in all. Two of the $K_N + 1$ codes coincide by the finite pigeonhole principle. Equal codes give an isomorphism, and every graph is a minor of an isomorphic graph. All searches, relabellings, and finite sums here have elementary bounds; for example $K_N \le (N+1)2^{N^2}$. Bounded induction proves the finite pigeonhole principle by induction on the size of the finite codomain, deleting one value and, if necessary, its unique preimage. This gives an EFA proof for coded finite sequences. □

Attempt and obstruction. Finite observational recovery of graphs can be combined with this proposition only when it bounds their number of vertices. Bounded treewidth does not do so: the paths have treewidth at most one and arbitrarily many vertices. Thus a finite list of recovered graph types is not a rank for a general bad minor sequence. The strong uniform bounded-treewidth theorem and the full GMT require well-quasi-order arguments beyond finite pigeonhole. No reduction of those arguments to the earlier finite identification bounds has been constructed. The elementary finite-size proposition is a classical special case with an explicit bound, not a new solution of the proof-strength problem.

47.10 Outcome of the nine-target attack

The strongest concrete transfer developed in this section is the exact arithmetic-cone invariant in Proposition 47.6; the missing hypothesis is exposed rather than assumed. The finite-age and abstract reduction counterexamples decisively invalidate two proposed shortcuts. The other positive statements isolate the quantitative or formalization hypotheses needed to use analytic results in logic. These are useful proof checks, but they provide no new unrestricted resolution of the nine named problems. In particular, neither recent external claims nor already solved restricted cases are added to a count of newly closed conjectures.

48 Finite structures, universality, and effective witnesses

This section investigates items 34, 35, 36, 38, 40, 42, 48, 53, and 56. The status cutoff is 17 October 2026. None of the nine unrestricted problems is claimed solved by this section. The propositions below are proved reductions, elementary certificates, or obstructions to proposed transfers; they are not nine new resolutions. Classical constructions and recent

external advances are explicitly separated from our deductions. In particular, an external 2026 star-height preprint changes the status of one boundedness variant, but does not settle the height-one question.

48.1 34: universality with forbidden ordinary subgraphs

Fix a finite family $\mathcal{C}$ of finite simple graphs. A graph is $\mathcal{C}$-free when it contains no member of $\mathcal{C}$ as an ordinary subgraph, not necessarily an induced subgraph. The question is whether there exists a countable $\mathcal{C}$-free graph into which every countable $\mathcal{C}$-free graph embeds as an *induced* subgraph. This is strong universality. Replacing the forbidden relation by induced subgraphs or minors changes the problem.

Cherlin–Shelah–Shi develop the algebraic-closure approach for finite families of connected constraints [214]. A substantial solved case is Cherlin–Shelah's Tree Theorem: for one finite tree T, weak and strong countable universality are equivalent, and hold precisely when T is a path or a near-path (a path with one extra pendant edge) [215, Theorem 1]. This is not a classification of arbitrary finite families. No such full classification was verified in our search.

The tempting transfer is to build a universal object by unrestricted free amalgamation. The following exact obstruction shows its scope.

Proposition 48.1 (Exact scope of free amalgamation)**.** *Remove redundant members of $\mathcal{C}$ so that it is an antichain under ordinary-subgraph containment. Assume its members have at least two vertices. The class of finite $\mathcal{C}$-free graphs is closed under free amalgamation over induced subgraphs if and only if every member of $\mathcal{C}$ is complete. The empty family also has this property.*

Proof. A free amalgam of B and D over A has no edges between $B \setminus A$ and $D \setminus A$. A complete forbidden graph cannot meet both these sets, so if all constraints are complete the amalgam remains free of them. Conversely, let $F \in \mathcal{C}$ have nonadjacent vertices u, v. Set $A = F \setminus \{u, v\}$, $B = F \setminus \{v\}$, and $D = F \setminus \{u\}$. Both B, D are $\mathcal{C}$-free: any constraint occurring in either would be a subgraph of F, contrary to minimality and the smaller number of vertices. Their free amalgam is exactly F. □

For complete constraints, the usual iterative extension construction now gives a countable universal graph. Start with the empty graph and successively realize every finite allowed one-point extension of every finite subgraph already constructed, repeatedly scheduling all requests. Free amalgamation preserves the constraint. A countable allowed graph embeds by enumerating its vertices and extending its embedding one vertex at a time. This proves the claimed restricted construction without assuming the desired universal object. It does not prove nonuniversality when free amalgamation fails: the near-path cases in the Tree Theorem are explicit counterexamples to that inference. The missing ingredient is control of the nontrivial algebraic closure generated by amalgamation.

48.2 35: which universality spectrum problem?

For a complete theory T in a countable language with an infinite model, put

$$\mathrm{Univ}_{\preceq}(T) = \{\kappa \geq \aleph_0 : \text{some } M \models T, |M| = \kappa, \text{ elementarily contains every } N \models T \text{ of size } \kappa\}.$$

We interpret "minimum spectrum" as a *least set under inclusion* among these spectra, in a fixed set-theoretic universe. Eventual inclusion modulo initial segments and comparison in forcing extensions are separate variants. Allowing complete theories with only finite models

would give an uninteresting empty minimum. Shelah formulates the classification problem in Definition 2.1 and Problem 2.2, distinguishes the smallest-spectrum and eventual variants in Problem 2.3, and warns that cardinal arithmetic is essential [216]. We have not verified a solution of the least-spectrum variant just specified.

Proposition 48.2 (Countable intersection reduction)**.** *For any sequence $(T_i)_{i<\omega}$ of complete countable theories with infinite models there is a complete countable theory $T_\oplus$, with infinite models, such that for every infinite cardinal κ,*

$$\mathrm{Univ}_{\preceq}(T_\oplus) = \bigcap_{i<\omega} \mathrm{Univ}_{\preceq}(T_i).$$

Consequently, if a countable family of these spectra is coinitial under inclusion among all such spectra, a least spectrum exists.

Proof. First use the disjoint many-sorted sum, with one infinite sort for each T_i and no symbols connecting sorts. Its sentences reduce to Boolean combinations of sentences in finitely many sorts, so it is complete. If U is universal of total size κ, testing it against a model whose ith sort has size κ shows that $|U_i| = \kappa$. Testing arbitrary κ-sized models of T_i in that sort shows that U_i is universal. Conversely, choose universal U_i of size κ for all i. Any sort of any model of total size κ has size at most κ and has an elementary extension of size κ, by upward Löwenheim–Skolem. It therefore embeds elementarily into U_i. The union of these maps is elementary in the disjoint sum, whose total size is $\aleph_0\kappa = \kappa$.

For completeness, this does not require enlarging the usual one-sorted class of theories. Relationalize the T_i, use disjoint unary predicates P_i, relativize T_i to P_i, and require each relation belonging to T_i to be false off P_i. No infinite disjunction asserting that the P_i cover the universe is imposed. Take the complete theory of such a sum with an additional infinite set of points outside all P_i. Completeness and elementarity follow from the following direct formula decomposition: a formula mentions only finitely many P_i and component languages; partition its quantified variables among those sorts and their complement. Relations with mixed sorts are false, equalities between different sorts are false, and on the complement only equality remains. The complement is infinite, because unused infinite sorts remain. Induction on formulas reduces quantification to formulas in the individual components and equality in an infinite set. This also shows that the points outside every P_i can be mapped by any injection into κ additional such points. Thus the same equivalence holds for the one-sorted complete theory.

If (T_i) is coinitial, its intersection is contained in every spectrum and is itself a spectrum by the construction. It is therefore least. □

This reduction identifies a concrete missing statement: a countable coinitial family, or another countable-language coding of the full intersection. There are continuum many countable theories, and taking one named sort for every one of them does not preserve a countable language. The proposition supplies no such compression and does not remove forcing dependence.

48.3 36: generalized star height and the 2026 boundedness claim

A generalized regular expression uses letters, $\varnothing$, $\{\varepsilon\}$, union, concatenation, complement in the same free monoid Σ^*, and Kleene star. Height counts nested stars; complement does not increase it. The classical unresolved question is whether some regular language has generalized height at least two, equivalently whether height one always suffices. A uniform finite upper bound, an algorithm computing the exact height, and this height-one question are different

assertions. Every individual regular language having *some* finite height is already immediate from ordinary regular expressions.

OpenAI's *Generalized Star Height at Most Three*, dated 25 September 2026, states the uniform bound $h_\Sigma(L) \leq 3$ in Theorem 1.1. Its Section 1.1 explicitly leaves the height-one question unanswered [217]. The repository also supplies a Lean scope document asserting formalization of the same top-line bound [218]. We checked the statement, the distinction of variants, and the periodic-template step below; we did not rebuild the Lean development or independently audit its entire 38-page construction. Accordingly this is recorded as an attributed 2026 boundedness result with a specified verification scope, not as our theorem or as a verified resolution of the classical height-one question.

Proposition 48.3 (A fully checked template transfer)**.** *Let L be recognized by a deterministic automaton with n states. For fixed words u, v, w, with $v \neq \varepsilon$, the language $L \cap uv^*w$ has an effectively constructible ordinary regular expression of star height at most one. The same holds with a finite union of such templates in place of uv^*w.*

Proof. From the state reached after u, repeatedly apply the transition associated to v. A state repeats within $n+1$ terms. Thus there are integers $0 \leq \mu < n$ and $1 \leq p \leq n$ such that the state after uv^j, and hence acceptance after the additional suffix w, is periodic with period p for $j \geq \mu$. The accepted exponents are a finite subset below μ, together with selected progressions $r + p\mathbb{N}$ with $\mu \leq r < \mu + p$. Express a progression by $uv^r(v^p)^*w$ and add the finitely many exceptional words. Finite unions preserve the bound. □

This verifies the ultimate-periodicity mechanism behind [217, Lemma 10.1] once its template hypotheses hold. The nontrivial additional obligation in that construction is to prove the independently guessed split tests are sound and their entire component languages satisfy the template hypotheses (in particular Proposition 7.1 and Proposition 10.2). We have not replaced that obligation by the much weaker assertion that the tests are finite-state. Nor does the proposition force height one for arbitrary languages: even $\{a, b\}^*$ cannot be covered by finitely many uv^*w templates, since such a union has only boundedly many words of each fixed length. This last observation is an obstruction to that cover, not a lower bound on generalized height, because complement is available.

48.4 38: Henson graphs and finite extension certificates

For $n \geq 3$, let H_n be the countable homogeneous universal K_n-free graph. The relevant problem is whether H_n is pseudofinite: every first-order sentence true in H_n has a finite model. It is *not* whether its full theory has a finite model; the latter contains sentences asserting arbitrarily many vertices. The triangle-free case remains explicitly open in Macpherson–Tent (2025) [219, Introduction]. Letzter–Sahasrabudhe identify the already difficult finite 4-extension case and give quantitative nonexistence bounds depending on graph order [220].

Call a finite K_n-free graph k-extendible if, for disjoint U, V with $|U| + |V| \leq k$ and U containing no K_{n-1}, there is a vertex outside $U \cup V$ adjacent to every vertex of U and to no vertex of V.

Proposition 48.4 (Exact finite-fragment reduction)**.** *H_n is pseudofinite if and only if a finite k-extendible K_n-free graph exists for every k. In the triangle-free case, a 2-extendible example exists on ten vertices. Every finite k-extendible triangle-free graph has at least $k + 2^k$ vertices.*

Proof. The graph axioms, the absence of K_n, and all one-point extension axioms axiomatize $\mathrm{Th}(H_n)$. Indeed they construct a back-and-forth isomorphism between any two countable

models, and they force infinitude; downward Löwenheim–Skolem then gives completeness. For fixed k the relevant extension assertions form a finite conjunction. Pseudofiniteness supplies a finite model of this conjunction together with the graph and forbidden-clique axioms. Conversely, every finite subset of the axiom system is implied by the axioms through some depth k. If all these have finite models, any sentence entailed by the complete axiom system has a finite model, by compactness applied to its proof from a finite subset.

For the ten-vertex example take the two-element subsets of a five-element set, adjacent precisely when disjoint (the Petersen graph). Three pairwise disjoint two-element sets do not exist, so it is triangle-free. Every vertex has degree three, and two distinct nonadjacent vertices have exactly one common neighbor: the complement of their three-element union. This handles $|U| = 2$. If $|U| = |V| = 1$, adjacent u, v have no common neighbor and u has two neighbors other than v; nonadjacent u, v have just one common neighbor, leaving two usable neighbors of u. For $U = \varnothing$, the union of the closed neighborhoods of two vertices has size at most eight, so a suitable vertex exists. Smaller requests follow from these cases or directly from degree three.

Finally build an independent set A of size k by repeatedly applying extension with $U = \varnothing$. For every subset $B \subseteq A$, the request $(U, V) = (B, A \setminus B)$ has a witness outside A. Two different subsets require different witnesses because their adjacency patterns on A differ. Thus there are at least 2^k vertices outside A. □

The exponential lower bound is an elementary check, weaker than the published quantitative bound in [220]; it is not a new best bound. The exact obstruction to a closure is producing finite witnesses at every depth, already beyond the checked depth two. An infinite Fraïssé construction realizes the requests in new vertices forever and does not supply finite witnesses or pseudofiniteness.

48.5 40: transexponential growth and a failed spectral transfer

Here "superexponential" must mean eventually larger than every fixed iterate $\exp^{\circ m}$, rather than merely larger than exp. The latter is already witnessed by $\exp \circ \exp$ in the o-minimal real exponential field. The intended question asks for an o-minimal expansion of an ordered exponential field which is not exponentially bounded. Krapp–Kuhlmann state it as Question 2.3 and prove that their particular axiomatized transexponential fields are *not* o-minimal (Theorem 3.6) [221]. Their existence theorem for transexponential fields therefore does not answer this question. A contemporaneous Palomar index advertises a result titled *A transexponential o-minimal structure* [222]. We could retrieve the indexed description but not its underlying proof project or exact checked statement. This competing claim is recorded as unverified here; the older open-status references alone do not settle its validity or establish the unrestricted problem's October 2026 status.

Our preceding power-mixture optimization suggests a precise transfer to test, including continuous spectra. It fails for a quantitative reason.

Proposition 48.5 (Compact positive spectra cannot supply the growth)**.** *Let μ be a nonzero finite positive measure supported on $[a, b] \subset (0, 1)$, let $M = \mu([a, b])$, and put*

$$S(x) = \int x^r \, d\mu(r), \qquad T(C) = \min_{x>0} \frac{1 + Cx}{S(x)} \quad (C > 0).$$

The minimizer x_C is unique. With $A = a/(1-a)$ and $B = b/(1-b)$,

$$A \le Cx_C \le B, \qquad \frac{C^a}{MB^a} \le T(C) \le \frac{1+B}{MA^b} C^b \quad (C \ge B).$$

For finite support, T is definable, with parameters, in $\mathbb{R}_{\exp}$.

Proof. Differentiation under the integral is valid locally on $x > 0$, and $S' > 0$, $S'' < 0$. The objective tends to infinity at both endpoints, so it has a minimum. Its derivative has the sign of $N(x) = CS(x) - (1 + Cx)S'(x)$, and $N'(x) = -(1 + Cx)S''(x) > 0$. Hence there is exactly one minimizer. Writing $m = xS'(x)/S(x)$, a weighted average of the exponents, stationarity gives $Cx = m/(1 - m)$ and the first bounds follow. For $C \geq B$ one has $x_C \leq 1$, so $Mx_C^b \leq S(x_C) \leq Mx_C^a$. Substituting $A \leq Cx_C \leq B$ gives both displayed inequalities. For finitely many atoms, $x^r = \exp(r \log x)$ is definable in $\mathbb{R}_{\exp}$, as are finite sums and the quantified condition that x minimizes the objective. Uniqueness makes this a definable function, and hence makes T definable. □

Thus neither non-D-finiteness nor differential-algebraic complexity of these optimum maps implies transexponential growth. Even arbitrary continuous mixtures in this compact positive regime give polynomially bounded values. Definability of an arbitrary continuous mixture is not asserted. A successful transfer would have to leave this regime and still prove o-minimality, the missing structural requirement.

48.6 42: characteristic-zero minimal fields

Podewski's conjecture says that every infinite minimal field is algebraically closed. Minimality means that every parameter-definable subset of the field itself is finite or cofinite; it is weaker than strong minimality of the theory. The positive-characteristic case is known. Krupiński–Tanović–Wagner reduce the remaining case to definable partial orders, and rule out the almost-linear case [223, Theorems 1 and 3]. The March 2026 version of Johnson–Tran–Walsberg–Ye still states the conjecture and explains a finite-closed-topology obstruction [224, Question 11.1 and Conjecture 11.3].

The elementary power-map attack stops exactly where arbitrary polynomial images cease to be subgroups.

Proposition 48.6 (Two exact algebraic/topological reductions)**.** *Let K be an infinite minimal field. For every integer $m \geq 1$, the power map $K^\times \to K^\times$, $x \mapsto x^m$, is surjective. Suppose in addition that K admits a nondiscrete T_1 topology in which translations are homeomorphisms and every nonconstant polynomial image $f(K)$ is closed. Then K is algebraically closed.*

Proof. The power-map image is definable and infinite (each fiber has at most m points), hence cofinite. It is a subgroup of $K^\times$. If it were proper, any other coset would be infinite and disjoint from it, contradicting cofiniteness. This first assertion is the classical argument of [223, Lemma 2.1].

For a nonconstant f, the same finite-fiber argument shows that $f(K)$ is infinite and therefore cofinite. Under the additional topological hypothesis, its complement is finite and open. If nonempty, a T_1 space has an open singleton there: remove the other finitely many closed singletons from that open set. Translations would make every singleton open, contradicting nondiscreteness. Thus every nonconstant polynomial is surjective, and in particular has a root at value zero. □

The second part makes the topological step explicit; it does not produce the required topology. In the finite-closed topology polynomial images are closed, but excluding discreteness in the perfect characteristic-zero case is exactly the outstanding Question 11.1 of [224]. Replacing this unknown nondiscreteness by ordinary continuity of polynomials would be invalid: continuity does not make their images closed. Likewise power-map surjectivity alone has not supplied roots of arbitrary polynomials in the present argument.

48.7 48: MELL and the effective branching bound

The question is provability of finite sequents in propositional multiplicative-exponential linear logic, with the standard single pair of modalities $!, ?$ and no additives. Straßburger diagnoses a fatal gap in an earlier decidability proof; his Theorem 4.4 proves decidability of *RMELL*, with unrestricted contraction, not MELL [225]. The equivalence with branching vector-addition-system reachability is the relevant transfer, not the already decidable plain VASS problem.

Recent work gives a sharper positive boundary. Bizière and coauthors prove that a uniform Strahler bound exists for every BVASS (Corollary 4.4), decide reachability with a supplied Strahler bound (Section 6), and compute semilinear reachability sets for two-dimensional BVASS and five-dimensional *one-state* BVAS (Theorem 7.5). Their Remark 5.7 explains the missing effective stabilization test; unrestricted reachability remains open [226]. The two-dimensional advance also appears in [227].

For a finite rooted tree, its Strahler number is zero at a leaf, is unchanged at a unary node, and at a branching node is the largest child number, increased by one if that maximum is attained at least twice.

Proposition 48.7 (Exact effectivity gap)**.** *Using the bounded-reachability algorithm of [226, Section 6], general BVASS reachability is decidable if and only if there is a total computable function $b(\mathcal{B}, c)$ such that every reachable configuration c has a witnessing run of Strahler number at most $b(\mathcal{B}, c)$. A computable uniform bound depending only on $\mathcal{B}$ would suffice, but is not asserted by this equivalence.*

Proof. Given the bound, apply bounded reachability at that bound. Conversely, first decide whether c is reachable. If not, output zero. If yes, enumerate finite labeled run trees until a valid run with target c appears; validity is decidable by checking all local transitions and nonnegative counters. Output its computable Strahler number. The search terminates precisely because the decision was positive. □

The distinction between an instance-dependent computable bound and an existing system-wide finite bound is essential. No quantitative estimate from our earlier finite spectral reconstruction controls the finite basis of BVASS runs in the cited well-quasi-order argument. Replacing that basis by a computed approximation without a stopping certificate would simply reintroduce the open problem.

48.8 53: the word problem for the single combinator S

Terms here are ground binary application trees with the sole constant S, and the only rewrite rule is

$$(((Sx)y)z) \longrightarrow (xz)(yz).$$

The word problem asks whether two terms are convertible by this rule and its inverse. It is not the word problem for SK, and it is not merely normalization. Waldmann proves decidability of normalization of ground S-terms [228]; Manzonetto's 2023 Problem 4 explicitly retains the pure-S word problem as open [229, Section 6.4]. We found no primary resolution by the cutoff.

Proposition 48.8 (A decidable region and the missing certificate)**.** *Convertibility is decidable on pairs for which at least one term has a normal form. More generally, it is semidecidable, and becomes decidable if a computable bound on the leaf size of a common reduct is supplied for every convertible input pair.*

Proof. The rule is left-linear and has no nonvariable overlap with itself, so the standard orthogonal-rewriting confluence theorem applies. Thus convertible terms have a common reduct, and normal forms, when they exist, are unique. Use Waldmann's decision procedure on both inputs. If exactly one normalizes they cannot be convertible, since a common reduct with a normal form forces both inputs to normalize. If both normalize, enumerate their reductions until their normal forms are found and compare them.

For arbitrary inputs enumerate pairs of finite reduction paths and accept when their endpoints coincide. To check a supplied bound B, observe that leaf size never decreases: a redex with subterm leaf sizes $|x|, |y|, |z|$ changes from $1 + |x| + |y| + |z|$ to $|x| + |y| + 2|z|$, a difference $|z| - 1 \geq 0$. Every path to a reduct of size at most B stays among the finitely many ground binary trees of that size. Search that finite graph from both inputs and test whether their reachable sets meet. If a computable valid B is available for all convertible pairs, this decides convertibility, returning no when the finite search finds no meeting. □

Conversely a word-problem decision procedure would compute such an input-dependent bound by accepting negative cases immediately and enumerating a common reduct on positive cases. Thus the genuine remaining region consists of pairs of nonnormalizing terms without a computable common-reduct bound. Decidable normalization or strong normalization of a different typed calculus does not supply that bound.

48.9 56: the three-pair Post correspondence problem

We use the ordinary finite-word PCP: three indexed pairs (u_i, v_i) over a finite alphabet, with repetitions of indices allowed, ask whether some nonempty word $i_1 \cdots i_m$ satisfies $u_{i_1} \cdots u_{i_m} = v_{i_1} \cdots v_{i_m}$. Empty tile words may be allowed; the witness sequence must be nonempty. The nonerasing variant is a restriction, not silently identified with it. An injective fixed-length binary encoding of the alphabet preserves the number of pairs and all equalities, so alphabet size is not the issue.

Neary's published Theorem 11 proves undecidability for five pairs and explicitly leaves three and four pairs open [230]. An early four-pair title found in secondary indexes is not the published theorem. Recent results for free-group, reverse, or bi-infinite correspondence problems concern different variants. No primary proof deciding the ordinary three-pair problem was verified here.

Proposition 48.9 (Finite residual certificate)**.** *For a PCP instance over an alphabet of size $q \geq 1$ and $B \geq 0$, it is decidable whether a solution exists whose unmatched suffix after each tile has length at most B. If such a solution exists, one exists using at most*

$$N_B = 1 + 2 \sum_{j=1}^{B} q^j$$

tiles. General three-pair PCP is decidable if and only if a total computable bound $B(I)$ of this kind exists for every positive three-pair instance I.

Proof. After deleting the longest common prefix of two partial concatenations, either they disagree at their next letters, in which case no extension can repair them, or one residual is empty. Record the latter case as 0, or $(+, r)$, or $(-, r)$, according as both are empty, only the upper residual is r, or only the lower residual is r. Keep only nonempty r of length at most B. These are exactly N_B states. From $(+, r)$ and tile (u_i, v_i) cancel the common prefix of (ru_i, v_i); from $(-, r)$ cancel that of (u_i, rv_i); from 0 cancel that of (u_i, v_i). Discard a letter mismatch or a residual longer than B. This is a finite directed labeled graph.

Solutions with the bound are precisely nonempty walks from 0 back to 0. A shortest such walk has no repeated internal vertex and no intermediate visit to 0, because removing the corresponding cycle would shorten it. Its length is at most the number of vertices. Thus a finite graph search decides the bounded problem and constructs the stated witness.

A computable bound reduces unrestricted three-pair PCP to this finite search. Conversely, given a decision procedure, output zero on a negative instance. On a positive instance enumerate index words until a solution is found, and output the largest residual length along it. This terminates and supplies a total computable bound. □

This is an exact bounded-delay algorithm, valid even with empty tile words, not an algorithm for unrestricted PCP. Every individual solution has a finite delay, but that observation gives no computable stopping bound on negative instances. Group cancellation, approximate equality, or replacing words by commutative letter counts destroys the prefix information used in the proof and cannot justify such a bound.

48.10 Outcome of the nine attempts

The strongest verified new work in this section is the explicit countable-intersection reduction and the quantified obstruction to using positive spectral mixtures for transexponential growth; we do not claim priority for these deductions. The remaining propositions make precise which finite certificates would be needed and reproduce restricted cases with complete proofs. The external 2026 star-height and BVASS advances are attributed to their authors. No unrestricted problem in this nine-item group is declared closed by our work.

49 Synthesis: what the attempts establish and what would change the targets

All fifty-seven requested entries have been treated. This includes separate answers to both components of entries 37 and 43, and explicit distinctions between the two categoricity entries. The mathematical output consists of proved restricted algorithms, definability boundaries, conditional transfers, and counterexamples to proposed generalizations. The status audit separately records established results, recent external claims, withdrawn arguments, independence, and ambiguous formulations. There is no newly proved solution of a previously open named conjecture to add to an original-closure count.

49.1 Four transfers that can be checked without accepting a conjecture

Logarithmic arithmetic becomes a logical boundary. Theorem 43.1 is the central positive transfer. Its fixed-coefficient half handles arbitrary first-order quantifier alternation, not merely the evaluation of a single numerical expression. Its negative half needs only one additional binary relation to interpret the rational field. The effective coefficient-basis computation is uniform in the input syntax; no algorithm for discovering rational relations among the logarithms at each quantified point is assumed. This precisely states what the earlier arithmetic tests can contribute to a decision problem. The mathematical ingredients are classical, and independent novelty of this combined formulation has not been established.

Spectral representation becomes a decision algorithm only with a cutoff. Theorem 46.5 computes a finite cutoff after which each residue class has a fixed sign or vanishes identically. The proof gives rational searches that provably terminate. The computable-real

example in Proposition 46.6 shows why exact input representation is essential even in dimension one. These two statements prevent the qualitative Skolem–Mahler–Lech conclusion from being mistaken for a general Skolem algorithm.

Borel bounds must agree on equivalence classes. Proposition 45.7 constructs an actual increasing finite exhaustion from an eventual class-coherent rate. Its construction uses intersections with a finite anchor, so it avoids the infinite classes that can appear when finite partitions are joined. This is a fully proved form of a known conditional reduction. It does not construct the missing rates for an arbitrary increasing union, and so leaves targets 49 and 52 open at precisely that step.

Universality spectra admit a genuine countable intersection. Proposition 48.2 includes the one-sorted coding needed to keep the output in the original class of countable theories. The result reduces existence of a least spectrum to a countable coinitial-family hypothesis. Naming a sort for every countable theory would violate the language restriction; no countable compression of that entire family is asserted. Thus the result is an exact reduction, with an identifiable remaining set-theoretic and model-theoretic problem.

49.2 The common obstructions are different mathematical requirements

The following comparisons are consequences of the explicit attempts, not a claim that all these problems share one hidden solution.

Targets	What is available	What a closure would additionally require
1, 11	An injection or a conditional stationary argument	A fiber-respecting section, or a choice-free global replacement for the missing partition hypothesis.
2, 26	Semantic truth or an external algorithm	A quantitative proof translation, or a derivation of the needed bound inside the specified weak theory.
4, 23, 50	Point codes, analytic orbit relations, or hypergraph witnesses	The required global Borel/coanalytic complexity bound while preserving the original objects.
7, 29, 41	Categoricity of one class at one scale	The exact cardinal threshold and structural transfer, with no unproved amalgamation or locality assumption.
15, 24, 55	Finite approximations or finite-size bounds	Uniform coherent witnesses, a preserved iteration invariant, or a rank for the unbounded problem.
17, 25	Standard-model arguments or stronger-theory upper bounds	Preservation over nonstandard models, or matching analysis for the exact axiom system.
37, 48, 53, 56	Finite witnesses or qualitative finite structure	An effective bound or stopping certificate for negative instances in the original input class.
39, 49, 52	Finite arity counts or pointwise bounds	One bound uniform across all reducts, or eventual coherence on every equivalence class.

Targets	What is available	What a closure would additionally require
46, 51	A valid chain-union operation	Proper elementary extensions preserving all omitted types or all permitted standard traces.

These requirements explain why repeatedly transferring a finite inverse theorem to another field does not, by itself, close the corresponding unrestricted conjecture. In each row the missing conclusion has different quantifiers, invariance, effectivity, or foundational assumptions from the available result.

49.3 Research priorities suggested by the proofs

Three directions merit a more focused investigation after this audit. First, classify useful intermediate logarithmic languages between the fixed-coefficient equality fragment and the single relation interpreting $\mathbb{Q}$. Any such project must specify the domain, allowed coefficients, inequalities, and quantifier alternation; the theorem here already supplies two exact endpoints, but does not classify everything between them. Second, seek a genuinely new effective tail bound for an oscillatory algebraic recurrence class beyond the supplied positive-spectrum promise. The proof above identifies the dominant-term estimate that must be replaced. Third, investigate conditions producing the coherent rates in Proposition 45.7 that are not simply restatements of hyperfiniteness or Borel boundedness. A new sufficient condition would need an independently verified class of examples outside known cases.

These are proposed next investigations, not three additional open problems claimed to be new to the literature. The present dossier provides complete arguments for its stated deductions and a source-linked account of the remaining work. Internal cross-reading does not certify global priority, formal correctness, or the validity of every external recent preprint.

Part VII
Arithmetic Beyond Convergence

Editorial integration note. The logarithmic-root and quantile classifications lead to the inverse constructions of Part VIII. The bound $N \le 8r^2$ in the tilted-arcsine analysis bounds a cyclotomic search in that family; it is not a universal algebraicity test for arbitrary constants. Later determinant constructions address limitations of the low-rank families considered here.

Original part framing.

Scope and attribution. A proof supplied here establishes its stated conclusion; it does not by itself establish historical novelty. Baker, Gelfond–Schneider, Hermite–Lindemann, and Mahler value theorems are external inputs, credited where used. The new manuscript's results are not formally verified in Lean. Independent checks described below concern the written proofs and exact computations, not a machine certification. Bibliographic status was checked on 16–17 October 2026; an external preprint or repository is identified with its actual verification scope. Full-text overlap with the closely related 2025 incomplete-beta paper remains unverified, so priority for the quantile formulation is not asserted.

50 From failed transfers to exact arithmetic questions

The preceding logic manuscript examined failures of transferring finite descriptions to effective or infinite conclusions. Three parts survive as useful mathematics here: exact logarithmic zero tests at algebraic inputs, the need for a computable bound before finite search becomes an algorithm, and the distinction between recovering an analytic object and determining its arithmetic type. We restate every hypothesis used; none of the preceding manuscripts' unverified assertions is used as an axiom.

The strongest transfer is positive. In an arcsine-type primitive, Baker's theorem forces an algebraic correction to vanish at an algebraic-probability algebraic quantile. Gelfond–Schneider forces a rational angle. Every conjugate of the resulting cyclotomic coordinate remains in $(0, 1)$, so the number of available real roots bounds the cyclotomic degree. The bound $\varphi(N) \le 2r$ then turns an infinite search into $N \le 8r^2$. This is the precise arithmetic information that a numerical inverse calculation alone does not provide.

Concrete complete classifications. Theorem 53.1 treats every rational polynomial tilt of the arcsine law. Theorem 52.3 sharpens the result for all positive half-integral beta shapes. Theorems 53.2 and 53.3 give the corresponding Student and Fisher conclusions. Theorem 51.1 decides the arithmetic type and multiplicity of all isolated roots in its rational-derivative logarithmic class, and Theorem 51.6 covers binary entropy at every algebraic input. Theorem 54.2 and Corollary 54.3 give exact algebraic exceptions in two related infinite-series families. These are uniform results, rather than a count inflated by listing many instances.

What a named-constant closure still requires. For irrationality via integer linear forms one must exhibit $A_n, B_n \in \mathbb{Z}$ such that $0 < |A_n + B_n C| \to 0$. Analytic smallness before denominator clearance is insufficient. For transcendence, excluding rational values alone is insufficient; a proof needs a criterion excluding all algebraic values, with its hypotheses checked. An applicable classical value theorem can supply this last step for a new family, while failing entirely for a superficially similar limit.

The fifteen named targets are retained separately in the following index. Finding a recent external solution corrects the target list but is not an original solution by this project. No

famous constant closure has been obtained from the present attempts. The complete family theorems and the quantified obstructions are the actual mathematical deliverables; their publication level requires an assessment beyond this investigation.

50.1 Fifteen targets, explicit attempts, and outcomes

The target column distinguishes irrationality from transcendence. The outcome column reports this investigation's result; it is not a guarantee that no unindexed manuscript exists elsewhere.

ID	Constant and target	Attempt	Outcome
1	Euler γ: irrationality	Nonlinear common-log elimination; published approximants after integer clearance	Not closed. Finite covers remove the common log; factorial denominators remain. Section 55.
2	Euler–Gompertz δ: irrationality	Positive rational Laplace kernels and twisted residues	Not closed. Fixed-denominator gap proved; growing pole order and actual gcd savings remain distinct loopholes. Section 55.
3	Catalan G: irrationality and transcendence	Positive tail Hankel forms; exact normalization	Recent external irrationality result and formalization reported, not ours; transcendence not obtained. Section 56.
4	$\zeta(5)$: irrationality and transcendence	Determinant smallness, rank, and denominator savings	Recent external irrationality result reported; no new proof here and no transcendence conclusion. Section 56.
5	$\zeta(3)$: transcendence	Raw tail moments and determinant growth	Irrationality is known; the present rank-one family does not supply a transcendence proof. Section 56.
6	$\Gamma(1/5)$: irrationality or transcendence	Reflection, multiplication, and known transcendence degree	Not closed; explicit constraint models do not distinguish this entry. Section 52.1.
7	$\Gamma(2/5)$: irrationality or transcendence	The separate second gamma entry under the same relations	Not closed; the rank statement does not identify it. Section 52.1.
8	Glaisher–Kinkelin A: irrationality or transcendence	Period representation and hyperfactorial approximants	Not closed. A differs from $\log A$; transcendence of every finite approximant does not pass to the limit. Section 52.2.
9	Erdős–Borwein E: transcendence	Finite Mahler certificates and full Lambert singularities	Not closed. Sparse families classified; the full Lambert function has no finite linear Mahler equation. Section 54.
10	$\sum_{n\geq 1} F_n^{-1}$: transcendence	Periodic sparse weights, Binet, additive functional equations	Not closed. Periodic doubling subseries classified; full spectrum does not satisfy the proposed additive equation. Section 54.
11	Prime Mertens B_1: irrationality	Finite prime logarithms, signs, and explicit tails	Not closed. Finite noncancellation proved; infinite cancellation with γ remains. Section 54.6.
12	$e+\pi$: irrationality or transcendence	Algebraic-input logarithmic elimination	Not closed; transformed logarithmic bases are not known algebraic. Section 51.
13	$e\pi$: irrationality or transcendence	Separate product equation and algebraic constraints	Not closed. The elementary disjunction does not identify this number. Section 51.
14	Khinchin K: irrationality	Ergodic recovery, unbounded averages, tail control	Not closed. Exact integrability condition and a normal Liouville counterexample isolate the missing input. Section 57.
15	Feigenbaum δ_F: irrationality	Certified fixed points and isolated unstable eigenvalues	Not closed. Uniform contraction, a spectral gap, and rational coefficients do not decide arithmetic type. Section 57.

51 Effective arithmetic classification of logarithmic roots

The useful residue of the earlier logical obstruction is an exact zero test on algebraic inputs. In one variable, a rational derivative turns this local test into a global classification of every real root. We give the algorithm and its proof, then a separate algebraic-input classification for binary entropy. All logarithms are natural unless a base is displayed. The inputs from transcendence theory are classical theorems of Baker and Hermite–Lindemann; no new transcendence theorem of that generality is claimed.

51.1 A terminating classifier, including multiplicities

Put $A = \overline{\mathbb{Q}} \cap \mathbb{R}$. An algebraic number is specified by a defining polynomial and an isolating interval. A semialgebraic input is given by a finite Boolean combination of polynomial sign conditions with coefficients in A. Consider

$$F(x) = R(x) + \sum_{i=1}^{s} \lambda_i \log P_i(x), \qquad R \in A(x), \quad \lambda_i \in A, \quad P_i \in A[x]. \tag{51.1}$$

Let $D \subseteq \mathbb{R}$ be a semialgebraic set over A on which R has no poles and all P_i are strictly positive. A positive algebraic constant is allowed as a constant polynomial P_i.

Theorem 51.1 (Effective root classification)**.** *There is a terminating algorithm with the following outputs for* (51.1) *on* D*.*

1. *It determines every interval component on which* F *vanishes identically. All remaining zeros form a finite set, which it isolates in pairwise disjoint rational intervals and classifies individually as algebraic or transcendental. Algebraic zeros are returned by exact algebraic representations. Each isolating interval can be refined to any prescribed positive rational width.*

2. *If* $F' \not\equiv 0$ *as a rational function, write* $F' = N/Q$ *in reduced form. Every multiple zero is algebraic. At any zero* a *the analytic multiplicity is*

$$1 + \operatorname{ord}_a N.$$

   In particular every transcendental zero is simple, and the algorithm computes the multiplicities of all zeros.

3. *If* D *has* c *nondegenerate interval components and* s_0 *isolated points, and* $F' \not\equiv 0$*, the total number of zeros in* D *is at most* $\deg N + c + s_0$*.*

In the constant case, isolated points of D *are tested separately; analytic multiplicity is not assigned to an identically zero germ. No polynomial-time complexity estimate is asserted.*

Proof. We first record two exact arithmetic facts. Baker's linear independence theorem says that logarithms of algebraic numbers independent over $\mathbb{Q}$ remain independent over $\overline{\mathbb{Q}}$. His inhomogeneous consequence says that a nonzero algebraic-coefficient linear form in logarithms of algebraic numbers is transcendental [231, 232].

Compute a rational basis $\theta_1, \dots, \theta_d$ of the span of the fixed coefficients λ_i, and write $\lambda_i = \sum_k r_{ki}\theta_k$, $r_{ki} \in \mathbb{Q}$. Exact arithmetic in a common number field computes this basis. Choose positive integers clearing the denominators of each row and write the resulting integer coefficients as n_{ki}. Define

$$G_k(x) = \prod_{n_{ki}>0} P_i(x)^{n_{ki}} - \prod_{n_{ki}<0} P_i(x)^{-n_{ki}}.$$

Empty products are one. At every algebraic $a \in D$, Baker's theorem gives

$$F(a) = 0 \quad \Longleftrightarrow \quad R(a) = 0 \text{ and } G_1(a) = \cdots = G_d(a) = 0. \tag{51.2}$$

Indeed, the inhomogeneous theorem first separates the algebraic term $R(a)$ from the logarithmic sum. For that sum choose any rational basis ρ_j from the finitely many logarithms $\log P_i(a)$. Writing them in this basis and applying Baker gives zero algebraic coefficients of each ρ_j; independence of the θ_k then gives $\sum_i r_{ki} \log P_i(a) = 0$ for every k. Exponentiation on positive real bases gives precisely $G_k(a) = 0$. This reasoning does not require an algorithm to find the basis of logarithms.

Conversely, the equations on the right of (51.2) imply $F(x) = 0$ for *every real* $x \in D$ satisfying them. Thus they define a semialgebraic subset S of the zero set that contains exactly all algebraic zeros, together with any intervals it may contain. Clear the denominator of R in forming S; its denominator is already excluded by the domain guard.

The same argument decides equality to zero of every number $b + \sum_i \lambda_i \log a_i$, where $b \in A$ and $a_i \in A_{>0}$. If it is nonzero, its sign is computable by certified approximations until an interval excluding zero is obtained. Such approximations exist effectively, for example by the convergent series

$$\log a = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \qquad z = \frac{a-1}{a+1}, \qquad |z| < 1,$$

and its explicit geometric tail bound. Termination uses the exact zero test, not a claim that small numerical values are nonzero.

Now differentiate:

$$F' = R' + \sum_i \lambda_i \frac{P_i'}{P_i} \in A(x). \tag{51.3}$$

Polynomial arithmetic decides whether this rational function is zero. If it is zero, F is constant on every interval component of D; an algebraic sample point and the exact zero test decide whether that constant is zero. The finitely many isolated points and included endpoints of D can be tested in the same way. This completes this case.

Suppose instead that the reduced numerator N in (51.3) is nonzero. Isolate its real algebraic roots and partition each interval component of D at those roots. On every remaining open interval F' has constant nonzero sign, so F is strictly monotone. All partition points that belong to D are algebraic and can be tested exactly. The endpoints of these intervals are algebraic or infinite. We next show that the signs, including zero, of all one-sided limits are computable.

At a finite endpoint b, factor $P_i(x) = (x-b)^{m_i} U_i(x)$ with $U_i(b) \neq 0$. Along a permitted side of the domain,

$$\log P_i(x) = m_i \log |x - b| + \log |U_i(b)| + o(1).$$

If R has a pole at b, its leading Laurent term dominates all logarithms and determines the infinite limit and its sign. If R has no pole, the coefficient $\sum_i \lambda_i m_i$ determines any logarithmic divergence. If that coefficient is zero, the finite limit is $R(b) + \sum_i \lambda_i \log |U_i(b)|$, to which the exact sign test applies. At either infinity use polynomial division for R and

$$\log P_i(x) = \deg(P_i) \log |x| + \log |\operatorname{lc}(P_i)| + o(1).$$

A nonconstant polynomial part of R dominates; otherwise the coefficient of $\log |x|$ determines divergence, or the constant limit has the same decidable algebraic-plus-logarithmic form. All orders, leading coefficients, and cancellations are exact algebraic computations.

A strictly monotone open interval has one zero precisely when zero lies strictly between its two endpoint limits. A zero limit at an excluded endpoint is not an interior root. When a root

exists, enumerate rational points in that interval until points of opposite sign are found, and refine by the sign test and bisection. This terminates by continuity and strict monotonicity. Roots at partition points have already been tested separately. Consequently the full zero set is finite and effectively isolated.

The set S above is now finite. Its points are effectively computable algebraic numbers: equivalently one may take the common real roots of the nonzero polynomials defining it and filter by the domain conditions. At least one such polynomial is nonzero unless S is empty or D has only isolated points; an interval of solutions would contradict $N \neq 0$. Standard univariate real algebraic algorithms handle all cases [127]. Exact order comparisons place each point of S in its monotonicity interval or at its partition point. An interval containing a zero but no point of S therefore contains a transcendental zero. Refining finitely many isolators separates all roots and gives any requested accuracy.

Since F is analytic near every point of D, its derivative has no pole there. Hence $Q(a) \neq 0$ at a zero, and integration of the first nonzero Taylor term of F' proves the multiplicity formula. A multiple zero must be a root of N, and is therefore algebraic. Finally Rolle's theorem bounds the number of zeros in an interval component by one plus the number of critical points there. Summing and adding the isolated domain points gives the stated bound. □

General root-isolation procedures for elementary functions predate this argument, including Strzeboński's exp–log–arctan algorithm [233]. Our statement concerns the exact arithmetic classification and the explicit terminating zero test in the much smaller rational-derivative class. It does not assert priority over that algorithmic literature. In particular, it uses neither a general oracle for elementary constants nor Schanuel's conjecture.

51.2 Three natural inverse families

Proposition 51.2 (Erlang quantiles)**.** *For any integer $n \geq 1$ and algebraic $q \in (0,1)$, the q-quantile of the unit-scale gamma distribution with shape n is transcendental. It is the unique positive zero of*

$$x - \log\left(\sum_{k=0}^{n-1} \frac{x^k}{k!}\right) + \log(1-q).$$

The zero is simple and effectively isolatable.

Proof. Integrating the density $x^{n-1}e^{-x}/(n-1)!$ gives survival function $e^{-x}S_n(x)$, where $S_n(x) = \sum_{k=0}^{n-1} x^k/k!$. The displayed function has value $\log(1-q) < 0$ at zero, tends to infinity, and has derivative $x^{n-1}/((n-1)!S_n(x)) > 0$ for $x > 0$. It has exactly one simple positive zero. At an algebraic positive zero the nonzero rational term x would have to vanish by (51.2), a contradiction. Equivalently, this last conclusion follows directly from Hermite–Lindemann applied to $e^{-x}S_n(x) = 1 - q$. □

This is a classical-transcendence consequence for an entire quantile family, not a claim that individual gamma quantiles were a named open problem. Algebraic positive changes of scale preserve the conclusion.

Proposition 51.3 (The inverse logarithmic mean and its exceptional root)**.** *Extend $L(x) = (x-1)/\log x$ continuously by $L(1) = 1$ on $(0,\infty)$. For algebraic $a > 0$, $L(x) = a$ has one solution. It is 1 when $a = 1$, and is transcendental when $a \neq 1$. For the un-divided equation $x - 1 - a\log x = 0$, there are two distinct simple roots when $a \neq 1$: the algebraic root 1 and the genuine inverse-mean solution. At $a = 1$ they coalesce to a double root at 1.*

Proof. For $x \neq 1$, the sign of $L'(x)$ is the sign of $\log x - 1 + 1/x$, which is positive: this expression has its unique minimum zero at $x = 1$. The expansion at 1 gives $L'(1) = 1/2$. Also $L(x) \to 0$ at zero and $L(x) \to \infty$ at infinity. Thus there is a unique inverse-mean solution. If it is algebraic and different from 1, the equation says that $\log x = (x-1)/a$ is a nonzero algebraic number, contradicting Hermite–Lindemann. For the un-divided equation, $F' = 1 - a/x$ and $F'' = a/x^2 > 0$. Its minimum is $a - 1 - a \log a < 0$ unless $a = 1$, and its two endpoint limits are positive infinity. These facts prove the two-root assertion and simplicity. When $a = 1$, $F(1) = F'(1) = 0$ and $F''(1) = 1$. □

Proposition 51.4 (A normalized rational-density quantile)**.** *On* $[0, 1]$ *normalize the positive rational weight* $w(x) = 1+(1+x)^{-1}$ *to a probability density. Every interior algebraic-probability quantile is transcendental.*

Proof. The cumulative distribution is $(x + \log(1 + x))/(1 + \log 2)$. For algebraic $q \in (0, 1)$ its quantile is the unique zero in $(0, 1)$ of

$$F_q(x) = x - q + \log(1 + x) - q \log 2.$$

Uniqueness follows from $F_q' > 0$ and the opposite endpoint signs. If the root were algebraic, (51.2) would force $x = q$ and $\log(1+q) = q \log 2$. Strict concavity of log gives $\log(1+q) > q \log 2$ for $0 < q < 1$, a contradiction. □

These examples illustrate three distinct tasks: an entire distribution family, an inverse mean with a removable-root trap, and quantiles whose normalizing integral is itself transcendental. None is obtained by declaring an arbitrary numerical approximation irrational.

51.3 Binary entropy: classification for every algebraic input

Write

$$h_2(p) = \frac{-p \log p - (1 - p) \log(1 - p)}{\log 2}, \qquad 0 < p < 1.$$

Boche–Schaefer–Poor prove that $h_2(p)$ is transcendental for rational $p \notin \{0, 1/2, 1\}$ [234, Appendix B, Theorem 4]. Blakaj–Wolf establish the general zero-or-transcendental dichotomy for natural-log entropy at algebraic spectra and the rational-or-transcendental dichotomy for algebraic-base entropy at rational spectra [236, Theorem 2]. The next proof covers all real algebraic binary inputs in base 2. The logarithmic coefficients now depend on p, so this is a separate argument rather than an application of Theorem 51.1.

Lemma 51.5. *If* $u, v \in \mathbb{Q}$ *and* $2^u + 2^v = 1$*, then* $u = v = -1$*.*

Proof. Choose a common positive denominator n and put $t = 2^{1/n}$. The polynomial $X^n - 2$ is Eisenstein at 2, so $1, t, \ldots, t^{n-1}$ are linearly independent over $\mathbb{Q}$. Writing each integer exponent of t as $kn + r$, $0 \leq r < n$, expresses each summand as a positive rational multiple of t^r. Independence and positivity force both remainders to be zero. Hence u, v are integers. Both are negative, and two negative integer powers of 2 sum to one only if both are $1/2$. □

Theorem 51.6 (Algebraic binary probabilities have no intermediate algebraic entropy)**.** *For* $p \in A \cap (0, 1)$*,*

$$h_2(p) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad p = \tfrac{1}{2}.$$

Consequently, for every algebraic $q \in (0, 1)$ *the two solutions of* $h_2(p) = q$ *are transcendental and effectively isolatable. A binary symmetric channel with algebraic crossover probability strictly between* 0 *and* $1/2$ *has transcendental capacity in bits.*

Proof. Suppose $q = h_2(p)$ is algebraic. The defining relation is

$$-\log(1-p) + p\log\frac{1-p}{p} - q\log 2 = 0. \tag{51.4}$$

First suppose p is algebraic irrational. If q does not belong to the rational span of $1, p$, Baker's coefficient splitting applied to the independent coefficients $1, p, q$ forces $\log 2 = 0$, which is impossible. Thus $q = u + vp$ with $u, v \in \mathbb{Q}$. Splitting (51.4) in the independent coefficients $1, p$ then gives

$$\log(1-p) = -u\log 2, \qquad \log\frac{1-p}{p} = v\log 2.$$

It follows that $1 - p = 2^{-u}$ and $p = 2^{-u-v}$. Lemma 51.5 forces $p = 1/2$, contradicting irrationality.

Now let $p = m/n$ in lowest terms, with $0 < m < n$. Baker splitting in (51.4) first implies $q \in \mathbb{Q}$, since otherwise the coefficients $1, q$ would again force $\log 2 = 0$. Exponentiating gives

$$\frac{n^n}{m^m(n-m)^{n-m}} = 2^{nq}.$$

After clearing the denominator of q, the prime valuation of the left-hand side at every odd prime is zero. The three positive integers $m, n-m, n$ are pairwise coprime. Therefore none has an odd prime factor: each is a power of 2. Since $n \geq 2$, coprimality forces $m = n - m = 1$, so $p = 1/2$ and $q = 1$. The converse is immediate.

The function h_2 is symmetric, strictly increasing from 0 to 1 on $(0, 1/2)$, and strictly decreasing on $(1/2, 1)$. Its derivative is $\log((1-p)/p)/\log 2$. Thus every $q \in (0, 1)$ has precisely two roots, and the classification proves their transcendence when q is algebraic. For effective isolation, use rational bisection on $(0, 1/2)$ and symmetry. At every rational interior sample point the difference from q is a nonzero computable real, so certified logarithmic approximation eventually decides its sign. Finally the binary symmetric capacity is $1 - h_2(p)$; subtracting a transcendental number from 1 preserves transcendence. □

The rational-input part is expressly credited to [234]. The algebraic-input extension is proved here from Baker's theorem; priority beyond the primary sources checked is not certified. This statement is special to the binary entropy measured in bits. For example, with $p = (\sqrt{5}-1)/2$ and $b = 1/p$, the identity $1 - p = p^2$ gives entropy to base b equal to $2 - p$, an algebraic irrational. Changing the base therefore changes the arithmetic conclusion.

Explicit isolators. The following decimal endpoints are exact rational numbers, not rounded assertions about a root. They give four concrete instances of the proved families:

Root	Rational isolating interval	Arithmetic type
Shape-2 Erlang median	$(1.678346, 1.678347)$	transcendental
$L(x) = 2$	$(3.512862, 3.512863)$	transcendental
Rational-density median	$(0.464832, 0.464833)$	transcendental
$h_2(p) = 1/2$, $p < 1/2$	$(0.110027, 0.110028)$	transcendental

For an entirely rational check of the signs, truncate the logarithm series used above after 240 terms. For $z = (a-1)/(a+1)$ the absolute remainder is bounded by $2|z|^{481}/(481(1-|z|^2))$. Rational interval arithmetic in the displayed defining equations gives a negative value at each left endpoint and a positive value at each right endpoint. The monotonicity proved above establishes isolation. These checks locate the roots; their transcendence follows from the proofs, not from the intervals.

51.4 Named targets that the classifier does not close

We tested $e + \pi$ and $e\pi$ separately. The irrationality of these individual constants is not established by the reviewed results; Waldschmidt lists the classical gap [235], and a 2026 study still treats $e + \pi$ as open [237]. The useful failure is a specific hypothesis failure, rather than an inconclusive decimal search.

For the sum, assuming $e + \pi = a \in A$ gives $\log(a - \pi) = 1$. The logarithm's argument is not known algebraic: it is e. Encoding π by a complex logarithm of -1 makes this a nested logarithmic equation, outside the fixed algebraic polynomial-base class. Baker's theorem cannot be applied to that unknown base. For the product, $e\pi = a \in A$ similarly gives $\log(a/\pi) = 1$ and again introduces a transcendental argument and a nested period relation. Treating π as a coefficient also violates the algebraic-coefficient input condition. Numerical isolation or a rational derivative in a different variable does not repair these arithmetic failures.

The elementary disjunctive conclusion remains valid: at least one of $e + \pi$ and $e\pi$ is transcendental. If both were algebraic, e would be a root of $X^2 - (e + \pi)X + e\pi \in \overline{\mathbb{Q}}[X]$, contradicting its transcendence. This does not identify either individual number and is not counted as a new result. Both named targets remain open in this branch's ledger.

52 Gamma values, a logarithmic obstruction, and arithmetic of beta quantiles

The three pre-existing named targets in this section are $\Gamma(1/5)$, $\Gamma(2/5)$, and the Glaisher–Kinkelin constant A. We do not prove any of these three numbers irrational or transcendental. We identify exactly why the available transfers do not do so, and prove a separate effective classification for a natural two-parameter family of probability quantiles. Related prior work on incomplete beta values by Dhillon–Saha must be taken into account [245]. This classification is an application of classical transcendence theorems with a conjugacy argument; priority for its particular formulation is not asserted. All status checks refer to 17 October 2026.

52.1 The two fifth-denominator gamma values

Write $G_1 = \Gamma(1/5)$ and $G_2 = \Gamma(2/5)$. The recent geometric transcendence theorem for the gamma function does not settle its values at these individual inputs. Sheth's 2026 paper explicitly leaves $\Gamma(1/5)$ open in the discussion following Conjecture 4.2 [238]. The classical result of Grinspan, together with the related work of Vasil'ev, gives

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(\pi, G_1, G_2) \geq 2; \tag{52.1}$$

see [239] and the precise account in [240, Section 3.3.2, p. 91]. This does imply that at least one of G_1, G_2 is transcendental even over $\mathbb{Q}(\pi)$. It does not identify which one. Neither individual irrationality question is resolved here, and no primary resolution of either individual transcendence question was found in the present search.

Reflection and the fivefold multiplication formula give

$$G_1\Gamma(4/5) = \frac{\pi}{\sin(\pi/5)}, \qquad G_2\Gamma(3/5) = \frac{\pi}{\sin(2\pi/5)}, \tag{52.2}$$

$$\prod_{j=1}^{4} \Gamma(j/5) = \frac{4\pi^2}{\sqrt{5}}. \tag{52.3}$$

The sine factors are algebraic. Since $\sin(\pi/5)\sin(2\pi/5) = \sqrt{5}/4$, the last displayed equation adds no independent relation to the first two.

Proposition 52.1 (A concrete limit of reflection and rank transfers)**.** *The relations* (52.2)–(52.3)*, positivity of their four entries, and the rank bound* (52.1) *do not, as a system of numerical constraints, force either specified entry G_1 or G_2 to be transcendental.*

Proof. Fix the actual number π and a positive real number T transcendental over $\overline{\mathbb{Q}}(\pi)$; such numbers exist because that field and its algebraic closure are countable. Set $g_1 = 2$, $g_2 = T$, and define

$$g_4 = \frac{\pi}{2\sin(\pi/5)}, \qquad g_3 = \frac{\pi}{T\sin(2\pi/5)}.$$

These four positive numbers satisfy all the displayed relations, and $\mathbb{Q}(\pi, g_1, g_2)$ has transcendence degree two, but g_1 is rational. Interchanging the roles of g_1 and g_2 and redefining their reflection partners gives the corresponding example with g_2 rational. □

These are auxiliary numerical tuples, not alternative gamma functions. The proposition proves only the insufficiency of the listed constraints. A genuine value theorem needs extra arithmetic information beyond those constraints. For example, complete beta integrals naturally involve ratios such as $B(1/5, 1/5) = G_1^2/G_2$, rather than isolating either G_j. The gamma function itself is differentially transcendental, so it cannot simply be inserted as an E-function into a Siegel–Shidlovskii value theorem. Likewise, a limiting value at infinity of an auxiliary function is not a value at a regular algebraic point. These are missing hypotheses, not cosmetic changes of representation.

52.2 Glaisher–Kinkelin: the constant is not its logarithm

Our normalization is

$$\log A = \frac{1}{12} - \zeta'(-1), \qquad A = \lim_{N\to\infty} \frac{\prod_{k=1}^{N} k^k}{N^{N^2/2+N/2+1/12} e^{-N^2/4}}. \tag{52.4}$$

See the primary formulas and discussion in [241, 242]. The first Stirling–Ramanujan constant in Muñoz–Pérez-Marco is $S_1 = \log A$, not A. Their transcendence conjecture concerns S_1. Furthermore, their Section 1 explicitly distinguishes their transalgebraic periods over $\mathbb{Q}(t, e^{-t})$ from the more restrictive exponential periods of Kontsevich–Zagier. Consequently the phrase "exponential period" in that paper is not a theorem that A is irrational, and it cannot be substituted into a period theorem having different hypotheses. No individual irrationality or transcendence result for A is claimed here.

There are two precise failed transfers. First, even a proof that $\log A$ is transcendental would not by itself prove A transcendental: $\log 2$ is transcendental whereas its exponential is algebraic. Baker's theorem applies to finite linear forms in logarithms of algebraic numbers. Writing an unknown constant as an integral or as an infinite logarithmic series does not supply that finite representation.

Second, the natural finite approximants already have a completely known arithmetic type, but this does not pass to their limit.

Proposition 52.2 (Transcendental hyperfactorial approximants)**.** *For each positive integer N, the finite expression*

$$A_N = \frac{\prod_{k=1}^{N} k^k}{N^{N^2/2+N/2+1/12}} e^{N^2/4}$$

is transcendental. This assertion, together with $A_N \to A$, does not imply that A is irrational.

Proof. The factor multiplying $e^{N^2/4}$ is a nonzero algebraic number, because all exponents of positive integers in that factor are rational. The exponential is transcendental by the Hermite–Lindemann theorem. Multiplication by a nonzero algebraic number preserves transcendence. As a general implication, transcendence is not closed under limits: the transcendental numbers $e^{1/N}$ converge to the rational number 1. □

Thus the logical failure identified earlier for noneffective finite descriptions has a direct arithmetic analogue. An exact finite formula and a convergence proof do not furnish a Diophantine lower bound for the limit. A new approximation argument would have to control nonzero integer linear forms in A itself, including their denominators and errors, or establish another applicable arithmetic value theorem. We have not done so.

52.3 A complete effective classification for half-integral beta shapes

For $a, b > 0$, define

$$I_x(a,b) = \frac{1}{B(a,b)} \int_0^x t^{a-1}(1-t)^{b-1}\,dt, \qquad 0 \le x \le 1.$$

This is the standard beta distribution function [243, Equations 8.17.1–8.17.4]. For each $0 < q < 1$ there is a unique $Q_{a,b}(q) \in (0,1)$ with $I_{Q_{a,b}(q)}(a,b) = q$: the density is strictly positive in the open interval, and the distribution function is continuous with endpoint values zero and one.

For nonnegative integers m, n, set $N = m + n$ and

$$K_{m,n} = \frac{\pi}{B(m+1/2, n+1/2)} = \frac{4^N m!n!N!}{(2m)!(2n)!} \in \mathbb{Q}_{>0}. \tag{52.5}$$

If $N > 0$, let $P_{m,n}(X) \in \mathbb{Q}[X]$ be the polynomial of degree at most $N-1$ satisfying

$$2X(1-X)P'_{m,n}(X) + (1-2X)P_{m,n}(X) = 2\big(K_{m,n}X^m(1-X)^n - 1\big). \tag{52.6}$$

Existence and uniqueness, as well as an explicit way to compute it, are proved below. Set $P_{0,0} = 0$.

Theorem 52.3 (Beta quantiles: algebraic and transcendental cases)**.** *Let $a, b \in \frac{1}{2}\mathbb{Z}_{>0}$, and let $q \in \overline{\mathbb{Q}} \cap (0,1)$.*

1. *If at least one of a, b is an integer, $Q_{a,b}(q)$ is algebraic.*
2. *For $a = b = 1/2$, the quantile is algebraic if and only if q is rational. In that case it is $\sin^2(\pi q/2)$.*
3. *Write $a = m + 1/2$, $b = n + 1/2$ with $m, n \ge 0$ and $m + n > 0$. If $mn = 0$, every such quantile is transcendental.*
4. *If $m, n > 0$, the quantile is algebraic if and only if one of the following exact rational tests succeeds:*

q	*test*	$Q_{m+1/2,n+1/2}(q)$
$1/3$	$P_{m,n}(1/4) = 0$	$1/4$
$1/2$	$m = n$	$1/2$
$2/3$	$P_{m,n}(3/4) = 0$	$3/4$.

For a fixed pair (m, n) at most one row can succeed. All other algebraic levels have transcendental quantiles.

Thus this is an effective decision of the arithmetic type for every exactly specified input in this family. It does not assert that every quantile is transcendental.

Lemma 52.4 (The correction polynomial and its real zero)**.** *For $m, n \geq 0$,*

$$I_x(m+1/2, n+1/2) = \frac{2\arcsin\sqrt{x} + \sqrt{x(1-x)}P_{m,n}(x)}{\pi}, \qquad 0 < x < 1. \tag{52.7}$$

For $m+n > 0$ the polynomial $P_{m,n}$ is nonzero. If exactly one of m, n is zero, it has no zero in $(0,1)$. If $m, n > 0$, it has exactly one distinct zero in $(0,1)$. That zero is $1/2$ if and only if $m = n$.

Proof. Put $x = \sin^2\theta$, $0 < \theta < \pi/2$. The change of variables turns the beta differential into $2\sin^{2m}\theta\cos^{2n}\theta\,d\theta$. The function $w(\theta) = \sin^{2m}\theta\cos^{2n}\theta$ is a rational-coefficient polynomial in $\cos(2\theta)$ and therefore a rational linear combination $c_0 + \sum_{j=1}^{N} c_j\cos(2j\theta)$. Its integral over $[0, \pi/2]$ is $c_0\pi/2$, so $c_0 = 1/K_{m,n} > 0$. Integrating and normalizing gives

$$I_{\sin^2\theta}(m+1/2, n+1/2) = \frac{2\theta}{\pi} + \frac{1}{\pi}\sum_{j=1}^{N}\frac{c_j}{jc_0}\sin(2j\theta).$$

Use $\sin(2j\theta) = \sin(2\theta)U_{j-1}(\cos(2\theta))$, where $U_{j-1} \in \mathbb{Z}[X]$ is a Chebyshev polynomial. Since $\sin(2\theta) = 2\sqrt{x(1-x)}$ and $\cos(2\theta) = 1 - 2x$, this proves (52.7) with $P_{m,n} \in \mathbb{Q}[X]$ of degree at most $N-1$. Differentiating yields (52.6). For uniqueness, the homogeneous equation implies that $\sqrt{x(1-x)}P(x)$ is constant on $(0,1)$. A polynomial solution has this product tending to zero at the endpoints, so the constant and the polynomial are zero. Nonzero N makes w nonconstant, proving that the correction is nonzero.

Let $H(\theta) = I_{\sin^2\theta}(m+1/2, n+1/2) - 2\theta/\pi$. It vanishes at both endpoints and has derivative

$$H'(\theta) = \frac{2}{\pi}(K_{m,n}w(\theta) - 1).$$

If $m = 0 < n$, w is strictly decreasing, and its mean lies strictly between its endpoint values. Consequently H' changes sign once, from positive to negative. Hence $H > 0$ in the interior. The case $n = 0 < m$ is the reflection, with $H < 0$.

If $m, n > 0$, w is strictly increasing up to its unique maximum and strictly decreasing thereafter, and vanishes at both endpoints. Its maximum exceeds its mean. Thus H' has exactly the sign pattern negative, positive, negative, with two transition points. The first local extremum of H is strictly negative, and the second is strictly positive because the final decreasing segment ends at zero. There is exactly one zero on the strictly increasing middle segment, and no other interior zero. By (52.7), zeros of H are precisely those of $P_{m,n}$ under $x = \sin^2\theta$.

For $m = n$, symmetry gives $I_{1/2} = 1/2$, so that zero is $1/2$. If $m > n$, then on $0 < x < 1/2$ the beta density is strictly smaller than its reflected density, since their ratio is $(x/(1-x))^{m-n} < 1$. Integration gives $I_{1/2} < 1/2$. For $m < n$ the reverse strict inequality holds. This proves the last assertion. □

Proof of Theorem 52.3. Suppose first that $a = k$ is a positive integer and b is a positive half-integer or integer. In the defining integral use $u = \sqrt{1-t}$. The differential becomes

$$-2(1-u^2)^{k-1}u^{2b-1}\,du,$$

which is a polynomial differential with rational coefficients. Its integral between zero and one is a positive rational number. Thus the equation $I_x(a,b) = q$ is a nonconstant polynomial

equation over $\overline{\mathbb{Q}}$ for $u = \sqrt{1-x}$. It forces u and hence x to be algebraic. Symmetry handles an integer b instead.

Now both parameters have the form $m+1/2, n+1/2$. Assume that a quantile x is algebraic, and set

$$\theta = \arcsin\sqrt{x}, \quad z = 1 - 2x + 2i\sqrt{x(1-x)}, \quad R = \sqrt{x(1-x)}P_{m,n}(x).$$

Here z is algebraic on the upper unit semicircle, $\log z = 2i\theta$ on the principal branch, and R is algebraic. Equation (52.7) gives

$$\log z - q\log(-1) = -iR, \qquad \log(-1) = i\pi. \tag{52.8}$$

Baker's theorem states that a nonzero linear form with algebraic coefficients in logarithms of nonzero algebraic numbers is transcendental [244]. Therefore (52.8) forces $R = 0$ and $2\theta = \pi q$. If q were algebraic irrational, the Gelfond–Schneider theorem would make $z = \exp(q\log(-1))$ transcendental. Hence q is rational. Conversely, when q is rational, $x = \sin^2(\pi q/2)$ is algebraic; it is the desired quantile exactly when $P_{m,n}(x) = 0$.

For $m = n = 0$ the correction is identically zero, giving part 2. The no-zero statement of Lemma 52.4 gives part 3. It remains to sharpen the rational-angle condition when $m, n > 0$. Since q is rational, z is a root of unity of order greater than two, and

$$x = \frac{2 - z - z^{-1}}{4}.$$

Every algebraic conjugate of x has the form $(2 - z^j - z^{-j})/4$ with $\gcd(j, \operatorname{ord}(z)) = 1$. Each lies strictly between zero and one. All are zeros of the rational polynomial $P_{m,n}$. Lemma 52.4 allows only one such real zero, so all conjugates are equal and x is rational. Now $z + z^{-1} = 2 - 4x$ is a rational algebraic integer strictly between -2 and 2. It is therefore $-1, 0$, or 1, giving exactly $x = 3/4, 1/2$, or $1/4$ and respectively $q = 2/3, 1/2$, or $1/3$. The midpoint test is equivalent to $m = n$ by the lemma. Uniqueness of the interior zero also proves that at most one row can hold. □

The tests in the theorem require only rational arithmetic. For explicit implementation write $K_{m,n}X^m(1-X)^n = \sum_{k=0}^{N} w_k X^k$ and $P_{m,n}(X) = \sum_{k=0}^{N-1} p_k X^k$. Coefficient comparison gives

$$p_0 = 2(w_0 - 1), \qquad p_k = \frac{2w_k + 2kp_{k-1}}{2k+1} \quad (1 \le k < N). \tag{52.9}$$

The final coefficient equation $-2Np_{N-1} = 2w_N$ follows from existence of the polynomial already proved. Exact comparison of an algebraic input q with the three rational candidates is also effective.

Corollary 52.5 (Median classification and a genuine exceptional quantile)**.** *For positive half-integral beta shapes: if at least one shape is integral, the median is algebraic; if both shapes are nonintegral half-integers, the median is algebraic exactly when the shapes are equal, in which case it is* $1/2$*. In every asymmetric nonintegral half-integer case the median is transcendental. Moreover,*

$$Q_{3/2,5/2}(1/3) = 1/4, \qquad Q_{5/2,3/2}(2/3) = 3/4.$$

All other algebraic levels for these two distributions have transcendental quantiles.

Proof. The median statements follow from the theorem at $q = 1/2$. For the explicit exception, (52.9) gives

$$P_{1,2}(X) = -2 + \frac{28}{3}X - \frac{16}{3}X^2 = \frac{2}{3}(4X-1)(3-2X).$$

Its only zero in $(0,1)$ is $1/4$. The theorem proves the first formula and excludes all other algebraic levels. Reflection of the beta distribution proves the second formula and its exclusion statement. □

Four individually transcendental examples are the beta medians with shape pairs $(1/2,3/2)$, $(1/2,5/2)$, $(3/2,5/2)$ and $(5/2,7/2)$. These are examples of one uniform family, not four independent historical conjectures. The asymmetric exception above is also a concrete reason not to replace the theorem by the false claim that symmetry supplies the only algebraic interior beta quantiles.

Prior-art boundary. Dhillon and Saha's 2025 paper studies the arithmetic nature of incomplete beta values [245]. Their author slides explicitly treat positive rational noninteger a with $a+b\in\mathbb{N}$ by logarithmic linear forms and Baker's theory [246, slide 4/4, compiled PDF p. 222]. This includes the nonintegral half-shape parameter range above, so neither that range nor the underlying logarithmic method is claimed as new. The slides do not state the normalized inverse classification or its three rational-angle tests. The paper's full theorem statements were not available in the legal open sources checked: the institutional record links back to the publisher DOI. Exact overlap with the present classification therefore remains unverified. We present a self-contained proved application without a claim of historical priority.

52.4 What was transferred and what remains open

The useful transfer is from the effective logarithmic splitting principle to a normalized inverse problem. Normalization introduces the period π, so ordinary algebraic-level inversion alone is insufficient. Baker's theorem removes the algebraic correction, Gelfond–Schneider restricts the level to a rational angle, and the one-real-zero property forces the entire Galois orbit to collapse. That last step reduces an apparently unbounded cyclotomic search to three exact rational tests.

This classification does not apply to beta shapes $1/5$ and $2/5$: the required polynomial correction of an arcsine primitive is absent, and the normalizing gamma ratios are not algebraically classified. It likewise gives no finite logarithmic representation for A. The three named targets therefore stay open in this investigation. The beta theorem is a fully proved connected-family result; it is not presented as an Annals-level assessment or as a resolution of those named constants.

53 Finite cyclotomic search and the Student and Fisher distributions

The beta argument has a wider consequence. Its arithmetic step does not require a beta density: it requires a rational polynomial correction to an arcsine primitive. Counting the real roots of that correction bounds the degree of an entire cyclotomic orbit. This gives an explicit finite decision procedure for polynomial tilts of the arcsine distribution.

53.1 All rational polynomial tilts of the arcsine law

Let $w(X)=\sum_{j=0}^{d} w_j X^j\in\mathbb{Q}[X]$ be nonzero and nonnegative on $[0,1]$. Put

$$c_w=\frac{1}{\pi}\int_0^1 \frac{w(t)}{\sqrt{t(1-t)}}dt=\sum_{j=0}^{d} w_j\frac{\binom{2j}{j}}{4^j}\in\mathbb{Q}_{>0},\qquad F_w(x)=\frac{1}{\pi c_w}\int_0^x \frac{w(t)}{\sqrt{t(1-t)}}dt.$$

The moment identity follows from $t = \sin^2\theta$ and the elementary recurrence for integrals of even powers of sine. A nonzero nonnegative polynomial is positive apart from finitely many points; thus F_w is strictly increasing from zero to one. Denote its unique q-quantile by $Q_w(q)$.

Theorem 53.1 (An exact finite test for polynomially tilted quantiles)**.** *Suppose w is not constant. There is a unique nonzero polynomial $P_w \in \mathbb{Q}[X]$ of degree at most $d-1$ such that*

$$F_w(x) = \frac{2\arcsin\sqrt{x} + \sqrt{x(1-x)}P_w(x)}{\pi}. \tag{53.1}$$

Let r be the number of its distinct real zeros in $(0,1)$. There are at most r algebraic numbers $q \in (0,1)$ whose quantiles are algebraic. All of them, with their quantiles, are obtained by this finite exact procedure:

1. *Enumerate $3 \le N \le 8r^2$ and integers $1 \le j < N/2$ with $\gcd(j,N) = 1$ and $\varphi(N) \le 2r$.*
2. *Set $q = 2j/N$ and $x = \sin^2(\pi j/N)$, an explicitly represented real algebraic number.*
3. *Retain (q,x) if and only if $P_w(x) = 0$.*

Every other algebraic $q \in (0,1)$ has a transcendental quantile. If $r = 0$ the list is empty. If w is constant, the separate arcsine rule applies: $Q_w(q)$ is algebraic exactly for rational q.

Proof. Expand $w(\sin^2\theta)$ as a rational linear combination of $1, \cos(2\theta), \dots, \cos(2d\theta)$. Its constant term is c_w. Integrating, normalizing, and using $\sin(2j\theta) = \sin(2\theta)U_{j-1}(\cos(2\theta))$ gives (53.1). Uniqueness and nonvanishing follow by differentiation; explicitly

$$2X(1-X)P_w'(X) + (1-2X)P_w(X) = 2\big(w(X)/c_w - 1\big).$$

The coefficient recurrence in (52.9), with w/c_w in place of $K_{m,n}X^m(1-X)^n$, computes P_w. Real algebraic root isolation computes r exactly.

For algebraic q, x with $F_w(x) = q$, put $z = 1 - 2x + 2i\sqrt{x(1-x)}$. As in (52.8), Baker's theorem applied to $\log z - q\log(-1) = -i\sqrt{x(1-x)}P_w(x)$ forces $P_w(x) = 0$ and $z = \exp(i\pi q)$. Gelfond–Schneider forces $q \in \mathbb{Q}$. Conversely a rational q gives an algebraic $x = \sin^2(\pi q/2)$, and it is the desired quantile exactly when $P_w(x) = 0$.

Let $N > 2$ be the order of the resulting root of unity z. Every conjugate of $x = (2 - z - z^{-1})/4$ lies in $(0,1)$ and is a zero of P_w. Moreover $[\mathbb{Q}(x) : \mathbb{Q}] = \varphi(N)/2$: z satisfies a quadratic equation over the real field $\mathbb{Q}(z + z^{-1})$, is nonreal, and has cyclotomic degree $\varphi(N)$. Consequently $\varphi(N) \le 2r$.

For completeness, the elementary bound $\varphi(N) \ge \sqrt{N/2}$ makes the search finite without an unspecified inverse-totient step. By prime factorization,

$$\frac{\varphi(N)}{\sqrt{N}} = \prod_{p^e \| N} p^{e/2-1}(p-1).$$

Every factor is at least one except possibly $p = 2, e = 1$, whose factor is $1/\sqrt{2}$. Thus $N \le 8r^2$. The enumeration includes every primitive upper-semicircle root of unity of each allowed order. Each retained pair satisfies the CDF equation by (53.1); there are at most r pairs because distinct levels give distinct roots in $(0,1)$. This proves both completeness and soundness of the procedure. □

This theorem also applies to any explicitly given rational convex mixture of the nonintegral half-integer beta laws: each density is a rational polynomial times the arcsine density. The number and locations of algebraic exceptions may increase after mixing; the finite search, rather than the three-candidate beta rule, is the applicable conclusion. The classical transcendence inputs are credited explicitly. We assert no historical priority for this formulation without a wider literature comparison.

53.2 A complete parity classification for Student quantiles

Let T_ν have the standard Student distribution with positive integer degrees of freedom ν, and let $t_\nu(q)$ be its unique q-quantile. Its conventional density is

$$f_\nu(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2};$$

see [278] for the normalization and beta-CDF relation.

Theorem 53.2 (Arithmetic parity of Student quantiles)**.** *For every algebraic $q \in (0,1)$,*

degrees of freedom	*arithmetic type of $t_\nu(q)$*
ν even	*algebraic for every such q,*
$\nu = 1$	*algebraic if and only if $q \in \mathbb{Q}$,*
$\nu \geq 3$ odd	*algebraic if and only if $q = 1/2$.*

In the last row the sole algebraic quantile is zero; every other algebraic-probability quantile is transcendental.

Proof. The strictly increasing change of variables

$$U = \frac{1}{2}\left(1 + \frac{T_\nu}{\sqrt{\nu + T_\nu^2}}\right), \qquad T_\nu = \frac{\sqrt{\nu}(2U-1)}{2\sqrt{U(1-U)}}$$

has range $(0,1)$ and transforms the Student density into the Beta$(\nu/2, \nu/2)$ density. This can also be checked directly: $1 + t^2/\nu = 1/(4u(1-u))$ and $dt/du = \sqrt{\nu}/(4[u(1-u)]^{3/2})$; the duplication formula for gamma gives the beta normalization. The two displayed transformations are algebraic with algebraic coefficients, so t is algebraic if and only if its image u is algebraic. They preserve the probability level. Apply Theorem 52.3. Even ν gives integer beta shapes; $\nu = 1$ gives the arcsine law; odd $\nu \geq 3$ gives equal nonintegral half-integer shapes with a single interior correction zero at $u = 1/2$. Its probability level is $1/2$, and its inverse image is $t = 0$. □

For example, at $q = 39/40$ the standard t_3 critical value is transcendental, while the standard t_4 critical value is algebraic. This distinction is exact; decimal accuracy cannot detect it. It is a classification of a familiar statistical family, not a solution of a previously named conjecture about Euler's constant or a gamma value.

53.3 Fisher quantiles and the asymmetric exceptional value

Write $f_{u,v}(q)$ for the standard Fisher F-distribution quantile with positive integer degrees of freedom u, v. If X has that distribution then $Y = uX/(v+uX)$ has the Beta$(u/2, v/2)$ distribution, and $X = (v/u)Y/(1-Y)$. These are mutually inverse rational maps over $\mathbb{Q}$.

Corollary 53.3 (Arithmetic decision for integer-degree Fisher laws)**.** *At every algebraic interior probability level, Fisher quantiles are algebraic if at least one degree of freedom is even. If both degrees are odd, all algebraic exceptions and all transcendental cases are exactly those prescribed by Theorem 52.3 for $(a,b) = (u/2, v/2)$, transferred by $x \mapsto (v/u)x/(1-x)$. In particular,*

$$f_{3,5}(1/3) = \frac{5}{9}, \qquad f_{5,3}(2/3) = \frac{9}{5},$$

and these are the only algebraic-probability algebraic quantiles for their respective distributions.

Proof. The beta transformation preserves both algebraicity and the probability level. For the displayed cases use $Q_{3/2,5/2}(1/3) = 1/4$ and its reflected counterpart in Corollary 52.5. Substitution gives $5/9$ and $9/5$. The exclusion statements are inherited from the same corollary. □

54 Lambert constants: finite certificates and the limit obstruction

This section investigates three specified targets:

$$E = \sum_{n\geq 1} \frac{1}{2^n - 1}, \qquad R_F = \sum_{n\geq 1} \frac{1}{F_n}, \qquad B_1 = \lim_{x\to\infty} \left(\sum_{p\leq x} \frac{1}{p} - \log\log x \right),$$

where $F_1 = F_2 = 1$. The questions are transcendence for the first two numbers and irrationality for the third. None of these three questions is resolved here. The positive conclusions concern explicitly specified lacunary families, and the negative conclusions identify exact failures of proposed transfers to the three original constants.

54.1 Status and scope of the arithmetic input

Duverney–Tachiya explicitly distinguish the established irrationality of E from its unknown transcendence [247, Remark 1.1]. Campbell's May 2026 preprint proves a different assertion, namely infinitely many occurrences of the binary block 11 in E; it does not claim transcendence [248]. Waldschmidt's survey lists the full reciprocal Fibonacci sum as irrational, while leaving its transcendence unresolved [249, slide 59]. This must not be confused with transcendence results for even reciprocal powers or odd-index subseries. The prime constant B_1 has effective analytic representations and high-precision computations [250, 251]; no verified irrationality proof for B_1 was located in the present search, completed on 17 October 2026. Those primary sources support the representations used below, not a claim that a computation proves irrationality.

The functional transcendence inputs used in this section are classical. For a rational function R over a number field, regular at zero with $R(0) = 0$, the solution of

$$f(z) - f(z^d) = R(z), \qquad f(0) = 0, \qquad d \geq 2,$$

is either rational or transcendental as a function; at a nonzero algebraic point of the disk whose forward orbit avoids the poles of R, a transcendental solution has a transcendental value. These are the rational–transcendental dichotomy and the degree-one case of Mahler's value theorem; see [252, Theorem 4.1 and its discussion]. General rational Mahler solving is already algorithmic [253]; complete discrete-residue obstructions to rational summability are due to Arreche–Zhang [254]. The elementary certificate below is a bounded-pole specialization of that theory, with all of its finite bounds proved explicitly.

54.2 An exact finite certificate for a cyclotomic kernel

Theorem 54.1 (Finite pole certificate)**.** *Let K be a number field, $d \geq 2$, and $R \in K(z)$ satisfy $R(0) = 0$. Suppose all finite poles of R are N-th roots of unity and have order at most m. If R is a polynomial, one may take $m = 0, N = 1$. Let $B \geq 0$ bound the degree of its polynomial part, counting a constant or zero polynomial part as degree zero. Define, for $|z| < 1$,*

$$f_R(z) = \sum_{k=0}^{\infty} R(z^{d^k}), \qquad D(z) = (1 - z^N)^m, \qquad M = mN + \lfloor B/d \rfloor.$$

For every algebraic α with $0 < |\alpha| < 1$, the following conditions are equivalent:

1. *$f_R(\alpha)$ is algebraic;*
2. *f_R is rational;*
3. *there is $A \in K[z]$, with $A(0) = 0, \deg A \le M$, such that*

$$A(z)D(z^d) - A(z^d)D(z) = R(z)D(z)D(z^d). \tag{54.1}$$

Equation (54.1) is a finite linear system over K; its right side is polynomial. Its normalized solution, when present, is unique and $f_R = A/D$. In particular, a failure certificate for this linear system proves transcendence of every such disk value.

Proof. Since $R(z) = O(z)$ at zero and has no poles in the open disk, the series converges locally uniformly and satisfies $f_R(z) - f_R(z^d) = R(z)$. This is the unique analytic solution with zero constant term: the first nonzero coefficient of a difference of two such solutions would contradict $h(z) = h(z^d)$. The classical inputs just stated prove the equivalence of (1) and (2); every forward iterate of α is in the punctured disk, so the required regularity holds.

Suppose $f_R = g$ is rational. A pole of g outside the set $\mu_N = \{\zeta : \zeta^N = 1\}$, other than zero, forces poles at all of its successive d-th roots. Indeed a d-th root of a point outside μ_N is still outside μ_N, R is regular there, and $g(z) - g(z^d) = R(z)$ forces cancellation by a pole of $g(z)$. There are arbitrarily many distinct such roots, which is impossible for a rational function. Zero is regular because $g = f_R$ there. Thus all finite poles of g belong to μ_N.

If g had a pole of maximal order $s > m$, the same backward argument would propagate poles of order s to every successive d-th root: the rational function R, whose pole orders are at most m, cannot cancel the order-s principal part. Again this would give infinitely many poles. Hence Dg is polynomial. If the polynomial part of g has positive degree b, then the polynomial part of $g(z) - g(z^d)$ has degree exactly db. Proper rational parts remain proper under substitution, so $db \le B$. It follows that $g = A/D$ with $\deg A \le M$. Substitution gives (54.1).

Although this argument initially permits complex coefficients for A, the finite linear system has coefficients in K. Its unique normalized solution therefore belongs to K, by Gaussian elimination. Conversely any solution of (54.1) yields a rational function analytic in the disk with zero constant term satisfying the required equation. Uniqueness identifies it with f_R. □

The explicit degree bound is useful because it converts an existence statement into a terminating exact procedure. It does not extend Mahler's theorem to an arbitrary infinite spectrum.

Theorem 54.2 (All algebraic linear relations in a residue family)**.** *Fix $d \ge 2$, an algebraic number $0 < |\alpha| < 1$, and*

$$U_j(\alpha) = \sum_{k=0}^{\infty} \frac{\alpha^{jd^k}}{1 - \alpha^{d^{k+1}}}, \qquad 1 \le j \le d-1.$$

For algebraic $c_1, \dots, c_{d-1}$,

$$\sum_{j=1}^{d-1} c_j U_j(\alpha) \text{ is algebraic} \quad \Longleftrightarrow \quad c_1 = \cdots = c_{d-1}.$$

If their common value is c, the sum is $c\alpha/(1-\alpha)$. Thus the vector space over $\overline{\mathbb{Q}}$ spanned by $1, U_1(\alpha), \dots, U_{d-1}(\alpha)$ has dimension exactly $d-1$.

Proof. Apply Theorem 54.1 to

$$R(z) = \frac{\sum_{j=1}^{d-1} c_j z^j}{1 - z^d}.$$

If its normalized sum g is rational, its poles are simple and belong to μ_d, and its polynomial part is constant. If $\zeta \in \mu_d \setminus \{1\}$ were a pole of g, any $\beta^d = \zeta$ would lie outside μ_d. Then $g(z^d)$ would have a pole at β, whereas g and R are both regular there, a contradiction. Therefore g has at most a simple pole at 1, is bounded at infinity, and vanishes at zero. Consequently $g(z) = cz/(1 - z)$. The identity

$$\frac{z}{1 - z} - \frac{z^d}{1 - z^d} = \frac{z + z^2 + \cdots + z^{d-1}}{1 - z^d}$$

proves both necessity and sufficiency of equal coefficients. For the dimension statement, any algebraic linear relation involving the U_j's and 1 has all U_j-coefficients equal; the displayed identity gives the unique relation up to a scalar. □

For an integer $b \geq 2$, the same theorem classifies exactly the algebraic numbers in the finite-parameter family

$$\sum_{k \geq 0} \sum_{j=1}^{d-1} \frac{c_j b^{(d-j)d^k}}{b^{d^{k+1}} - 1} :$$

the algebraic locus is the diagonal $c_1 = \cdots = c_{d-1}$, with value $c/(b - 1)$. This is an explicit application of classical Mahler theory, not a newly solved historical conjecture. For comparison, transcendence of the reciprocal Fermat sum $\sum_{k \geq 0} (2^{2^k} + 1)^{-1}$ was already known; Coons also proved its irrationality exponent is 2 [257]. It is not counted as an additional discovery here.

Corollary 54.3 (Periodic weights on doubling Fibonacci indices)**.** *Let $(a_k)_{k \geq 0}$ be a periodic sequence of algebraic numbers. Then*

$$V(a) = \sum_{k \geq 0} \frac{a_k}{F_{2^k}}$$

is algebraic if and only if (a_k) is constant. In the constant case $a_k = a$,

$$V(a) = a \, \frac{7 - \sqrt{5}}{2}.$$

Proof. For a period h, put $d = 2^h$ and

$$H_a(z) = \sum_{k \geq 0} a_k \frac{z^{2^k}}{1 - z^{2^{k+1}}}.$$

Grouping k modulo h, its kernel for dilation by d is

$$\sum_{r=0}^{h-1} \frac{a_r z^{2^r}}{1 - z^{2^{r+1}}} = \frac{\sum_{j=1}^{d-1} a_{v_2(j)} z^j}{1 - z^d}.$$

Indeed each $1 \leq j < 2^h$ has a unique representation $j = 2^r(2s + 1)$. Theorem 54.2 says that $H_a(\alpha)$ is algebraic precisely when $a_0 = \cdots = a_{h-1}$.

Let $\varphi = (1 + \sqrt{5})/2$ and $q = \varphi^{-1}$. Binet's formula, treating the odd index $2^0 = 1$ separately, gives

$$V(a) = a_0 + \sqrt{5} \sum_{k \geq 1} a_k \frac{q^{2^k}}{1 - q^{2^{k+1}}}.$$

The last sum is $H_b(q^2)$, where $b_k = a_{k+1}$. The shift of a periodic sequence is constant exactly when the original sequence is constant. In that case $H_b(q^2) = aq^2/(1-q^2) = aq$, proving the formula. □

The constant-weight identity and special nonconstant-weight transcendence results have a long history; Nguyen records, for example, Mignotte's result for $2+(-1)^k$ [255]. The corollary is a uniform proof for this restricted periodic family, not a priority claim. Nguyen's much stronger Theorem 1.2 proves transcendence for arbitrary Fibonacci/Lucas choices at indices with $n_{k+1}/n_k \geq c > 2$. The exact algebraic doubling identity shows why the strict inequality cannot simply be replaced by $c = 2$.

54.3 Why the complete Lambert spectrum does not fit the certificate

Let

$$L(z) = \sum_{n\geq 1} \frac{z^n}{1-z^n} = \sum_{n\geq 1} d(n) z^n, \qquad |z| < 1.$$

The next obstruction also follows from Bell–Smertnig's 2026 classification of multiplicative Mahler series [256]. We give a direct proof because it identifies the local quantities that fail to close under a finite dilation system.

Lemma 54.4 (Radial logarithmic coefficient)**.** *If ζ is a root of unity of order q, then, as $t \downarrow 0$,*

$$L(\zeta e^{-t}) = \frac{1}{qt}\log\frac{1}{t} + O_q(t^{-1}).$$

Proof. The terms with $q \mid n$ contribute $L(e^{-qt})$. For $0 < u \leq 1$, the function $(e^u-1)^{-1} - u^{-1}$ is bounded. Splitting at $n = \lfloor 1/t \rfloor$, and bounding the remaining terms by a constant times e^{-nt}, gives

$$L(e^{-t}) = \frac{1}{t}\sum_{n\leq 1/t}\frac{1}{n} + O(t^{-1}) = t^{-1}\log(1/t) + O(t^{-1}).$$

For $q \nmid n$, the denominators $|1-\zeta^n e^{-nt}|$ are bounded below by a positive constant depending only on q. Their total absolute contribution is therefore $O_q(\sum_{n\geq 1} e^{-nt}) = O_q(t^{-1})$. □

Theorem 54.5 (No finite linear Mahler equation for L)**.** *For every $d \geq 2$, every $r \geq 0$, and rational functions $a_0, \ldots, a_r, b \in \mathbb{C}(z)$, an identity*

$$\sum_{j=0}^{r} a_j(z) L(z^{d^j}) = b(z) \tag{54.2}$$

in the disk implies $a_0 = \cdots = a_r = b = 0$.

Proof. Fix $k \in \{0, \ldots, r\}$, and choose infinitely many distinct roots ζ of orders $d^k\ell$, where ℓ ranges over primes coprime to d, avoiding the finitely many poles of all the rational functions in (54.2). The order of ζ^{d^j} is $d^{\max(k-j,0)}\ell$. Apply Lemma 54.4 to $z = \zeta e^{-t}$. Since all coefficients and b are regular at ζ, the coefficient of $t^{-1}\log(1/t)$ must vanish. It is

$$\frac{1}{\ell}\left(d^{-k}\sum_{j=0}^{k} a_j(\zeta) + \sum_{j=k+1}^{r} d^{-j} a_j(\zeta)\right).$$

Terms arising from variation of the rational coefficients are only $O(\log(1/t))$; the other remainders are $O(t^{-1})$. Consequently the rational function

$$P_k(z) = d^{-k} \sum_{j=0}^{k} a_j(z) + \sum_{j=k+1}^{r} d^{-j} a_j(z)$$

has infinitely many zeros, and so vanishes identically. For $1 \le k \le r$,

$$P_{k-1} - P_k = (d^{-(k-1)} - d^{-k}) \sum_{j=0}^{k-1} a_j = 0.$$

Successively $a_0, \dots, a_{r-1}$ vanish, and $P_r = 0$ then gives $a_r = 0$. For $r = 0$, $P_0 = 0$ gives the same conclusion directly. Finally $b = 0$. □

In particular the full Lambert function cannot be a component of a finite linear Mahler system over $\mathbb{C}(z)$: successive dilates of any one component of such a system belong to a finite-dimensional vector space and would satisfy a nontrivial relation of the forbidden form. This is an obstruction to this specific function-theoretic approach, not a theorem that $E = L(1/2)$ cannot admit some unrelated Mahler representation at a single point.

For the Fibonacci target define

$$\mathcal{F}(z) = \sum_{n \ge 1} \frac{z^n}{1 - (-1)^n z^{2n}}, \qquad R_F = \sqrt{5}\, \mathcal{F}(\varphi^{-1}).$$

Its even-index part is $L(z^2) - L(z^4)$, while for real $0 < z < 1$ its odd-index part is bounded above by $\sum_{n \text{ odd}} z^n$. It follows rigorously that

$$\mathcal{F}(e^{-t}) = \frac{1}{4t} \log \frac{1}{t} + O(t^{-1}).$$

Thus, for every $d \ge 2$,

$$\mathcal{F}(z) - \mathcal{F}(z^d) \notin \mathbb{C}(z) :$$

along $z = e^{-t}$ the difference has leading term $(1 - d^{-1})(4t)^{-1} \log(1/t)$, incompatible with a rational function. This rules out the direct additive certificate, but does not purport to exclude every possible higher-order equation for $\mathcal{F}$.

54.4 Denominators and finite prime logarithms

For clarity, the familiar positive-tail argument has the following precise limitation. If $s_N = P_N/D_N$, $P_N \in \mathbb{Z}$, $D_N \in \mathbb{Z}_{>0}$, and $S - s_N = \varepsilon_N > 0$, then

$$D_N \varepsilon_N \longrightarrow 0 \quad \Longrightarrow \quad S \notin \mathbb{Q}.$$

Indeed, if $S = u/v$, the positive integer $vD_N(S - s_N) = uD_N - vP_N$ would tend to zero. For the full Erdős–Borwein sum, the product denominator $D_N = \prod_{n=1}^{N} (2^n - 1)$ gives

$$D_N \varepsilon_N > 2^{N(N-1)/2 - N},$$

because $2^n - 1 \ge 2^{n-1}$ and $\varepsilon_N > \sum_{n>N} 2^{-n} = 2^{-N}$. For the Fibonacci sum, $F_n \ge \varphi^{n-2}$ and $F_n \le \varphi^{n-1}$, so the product denominator gives

$$\left(\prod_{n=1}^{N} F_n \right) \sum_{n>N} \frac{1}{F_n} \ge \varphi^{N(N-3)/2 - N}.$$

Both grow without bound. These calculations reject the stated product-denominator certificate; they are not lower bounds for the reduced denominators and do not reject optimized approximants. Most importantly, even a successful irrationality certificate would still not prove either requested transcendence statement.

The prime target admits

$$B_1 - \gamma = \sum_p \Delta_p, \qquad \Delta_p = \log(1 - 1/p) + 1/p. \tag{54.3}$$

This follows by adding the second and logarithmic third Mertens formulas, with the convention for B_1 fixed above [250, 251].

Proposition 54.6 (No finite rational cancellation of prime corrections)**.** *Every nonzero finite rational linear combination of the numbers Δ_p is transcendental. Consequently $\{1\} \cup \{\Delta_p : p \text{ prime}\}$ is linearly independent over $\mathbb{Q}$. For every integer $X \geq 2$, however, the omitted tail of (54.3) satisfies only the explicit bounds*

$$\frac{1}{8X^2} < -\sum_{p>X} \Delta_p \leq \frac{1}{2X}.$$

Proof. After clearing rational coefficients, the logarithmic part is $\log A$, with $A = \prod_p ((p-1)/p)^{n_p} \in \mathbb{Q}_{>0}$. If not all integers n_p vanish, take the largest prime P with $n_P \neq 0$. No numerator $p - 1$, for $p \leq P$, is divisible by P; among the denominators only P is. Hence $v_P(A) = -n_P \neq 0$, and $A \neq 1$. Hermite–Lindemann gives transcendence of the real logarithm $\log A$. Adding the rational contribution $\sum n_p/p$ preserves transcendence, proving the first assertion and the linear independence.

The convergent expansion

$$-\Delta_p = \sum_{j \geq 2} \frac{1}{jp^j}$$

lies between $1/(2p^2)$ and $1/(2p(p-1))$. Bertrand's postulate supplies a prime $X < p < 2X$, giving the strict lower bound. Summing the upper bound over all integers $n > X$, rather than only primes, telescopes to $1/(2X)$. □

This proposition is a direct classical logarithmic-transcendence transfer, not new information about the irrationality of B_1. It makes the missing step exact: finite noncancellation is available, but neither an arithmetic statement about the infinite sum $\sum_p \Delta_p$ nor a noncancellation theorem between that sum and γ follows. Convergence, explicit tail signs, and finite linear independence do not supply either statement.

Outcome. The three named targets remain unclosed. The completed work gives an explicit finite certificate, a full algebraic-exception locus for a lacunary family, a periodic Fibonacci-weight classification, and direct obstruction proofs for transfers to the full spectra. These results reuse classical machinery and contain known special cases. They must not be reported as three new resolutions of the original constant problems.

55 Euler and Gompertz constants: finite algebraic extraction and fixed-kernel lattices

The two external targets are the irrationality or transcendence of Euler's constant γ and of the Euler–Gompertz constant

$$\delta = \int_0^\infty e^{-t} \log(1+t)\, dt = \int_0^\infty \frac{e^{-t}}{1+t}\, dt.$$

The equality follows by integration by parts. Neither individual target is resolved here. The published approximation results of Wolfs–Van Assche [258, Theorems 1.2–1.3], and the primary papers [259, 260], explicitly distinguish these targets from the known disjunctive transcendence statement. Our current-source check found no independently verified individual resolution. The progress below consists of two reusable obstruction theorems, with complete proofs: finite algebraic extraction cannot retain a common logarithmic coordinate while giving finite monodromy; and every fixed positive polynomial denominator gives an attained positive gap on the integer-polynomial numerator cone. These strengthen earlier linear cancellation and single-denominator arguments. Their priority as standalone formulations is not claimed.

Write

$$\delta_a = \int_0^\infty \frac{e^{-t}}{t+a}\,dt, \qquad \mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\,n!}, \qquad e^{-a}\delta_a = \mathrm{Ein}(a) - \gamma - \log a \quad (a>0). \tag{55.1}$$

The differential identity for the last expression, followed by the usual defining limit of γ, proves the connection formula. In particular $\gamma + \delta/e = \mathrm{Ein}(1)$; it is not $\gamma + \delta$. The earlier E-function arguments apply to this combination. The new ring morphism in [261, Proposition 3.1] has also been checked against its primary text: its coefficients need not be algebraic, so it does not separate γ and δ simply by displaying them as two coefficients. Neither obstruction theorem below assumes that ring morphism or any conjecture about its image.

55.1 Finite covers do not eliminate a logarithm by retaining its constant term

The failed logical passage from finite consistency to a suitably well-founded model has an arithmetic analogue here: a finite algebraic cover is a much stronger constraint than convergence to an analytic boundary constant. The following precise statement supplies the constraint; the analogy itself supplies no proof.

Theorem 55.1 (Finite-orbit algebraic elimination)**.** *Let F be a field of characteristic zero, let $T_1, \dots, T_r$ be algebraically independent over F, and let w be algebraic over $F(T_1, \dots, T_r)$. Suppose that, in a common algebraic extension, for each i there is an automorphism τ_i fixing F, sending T_i to $T_i + c_i$ for some $c_i \in F^\times$, and fixing the other T_j. If each orbit $\{\tau_i^n(w) : n \in \mathbb{Z}\}$ is finite, then w is algebraic over F. If additionally $w \in F(T_1, \dots, T_r)$, then $w \in F$.*

Proof. First eliminate T_1, putting $F_1 = F(T_2, \dots, T_r)$. Clear the denominators of an algebraic relation to obtain a nonzero $P(T_1, w) = 0$ with $P(X, V) \in F_1[X, V]$. Finiteness of the orbit provides an integer $N \ge 1$ with $\tau_1^N(w) = w$. Applying its integer powers gives

$$P(T_1 + nNc_1, w) = 0 \qquad (n \in \mathbb{Z}).$$

The polynomial $P(T_1 + X, w)$ over the extension field has infinitely many distinct roots $X = nNc_1$, hence is identically zero. Translation of the indeterminate is invertible, so $P(X, w)$ is identically zero. Writing $P(X, V) = \sum_j p_j(V) X^j$, at least one $p_j \in F_1[V]$ is nonzero and satisfies $p_j(w) = 0$. Thus w is algebraic over F_1. Apply the same argument to T_2, and continue. No commutativity of the automorphisms on the algebraic extension is needed.

Finally F is relatively algebraically closed in a pure rational function field over F. For completeness, if h is rational and algebraic over F, differentiating its separable minimal polynomial with respect to each T_i gives $\partial h/\partial T_i = 0$. In characteristic zero a rational function with zero derivative in one variable is independent of that variable. Repetition proves $h \in F$. □

Corollary 55.2 (Nonlinear common-logarithm obstruction)**.** *Let $a_1, \dots, a_m > 0$ be algebraic, let $k = \overline{\mathbb{Q}}(\log a_1, \dots, \log a_m)$ with real logarithms, and consider the single-valued meromorphic-germ field*

$$F = k(z, e^{-a_1 z}, \dots, e^{-a_m z}, \mathrm{Ein}(a_1 z), \dots, \mathrm{Ein}(a_m z)).$$

On a logarithmic covering of a punctured neighborhood of zero put

$$T = \gamma + \log z, \qquad Y_j = e^{-a_j z}\delta_{a_j z} = \mathrm{Ein}(a_j z) - \log a_j - T.$$

Suppose that a germ w algebraic over $F(Y_1, \dots, Y_m)$ admits continuation through repeated turns around zero, and has a finite orbit under that continuation. Then w is algebraic over F. If w is rational in the Y_j over F, then $w \in F$. If in addition w is a constant germ, then respectively w is algebraic over k, or $w \in k$ in the rational case. All finite derivatives of the displayed data can be included without changing these assertions.

Proof. Every element of F is single valued and has a Laurent expansion at zero with coefficients in k. A turn fixes F and sends T to $T + 2\pi i$. The element T is transcendental over F: an alleged polynomial relation, continued around the origin, would give infinitely many roots of a nonzero polynomial at any point where its leading coefficient is nonzero. Although $2\pi i$ need not belong to k, we may first enlarge F by this constant for this transcendence argument. For the elimination itself one does not need $2\pi i \in F$: the polynomial-in-X proof of Theorem 55.1 takes place in an extension field containing the nonzero shift, and its coefficient polynomials still lie in F. Thus it applies without changing the coefficient field in the conclusion.

The displayed identities give $F(Y_1, \dots, Y_m) = F(T)$. Analytic continuation gives the required translation and its action on the algebraic germs. More directly, applying the proof just given to an algebraic relation $P(T, w) = 0$ and a turn that fixes w yields w algebraic over F. Relative algebraic closedness gives the rational assertion.

For a constant w algebraic over F, clear an algebraic relation into $\sum_j f_j(z) w^j = 0$, where not all $f_j \in F$ vanish. Choose the lowest Laurent power with a nonzero coefficient in some f_j. Its coefficient yields a nonzero polynomial over k vanishing at w. If $w \in F$ is constant, its constant Laurent coefficient lies in k. Finally F is a differential field, and $Y_j' = -e^{-a_j z}/z \in F$. All higher derivatives of Y_j therefore already belong to F. □

For $a_1 = 1$, the coefficient field is simply $\overline{\mathbb{Q}}$. The statement covers arbitrary finite polynomial equations and rational operations, rather than just linear combinations or differentiation. It does not assert that the numerical value of every single-valued expression at $z = 1$ is algebraic. Such values can be transcendental E-values. Nor does it show that γ is algebraic: a formula extracting the constant germ γ by the stipulated operations has not been constructed. At the positive boundary,

$$\gamma = \lim_{x \to \infty} (\mathrm{Ein}(x) - \log x), \qquad 0 < \mathrm{Ein}(x) - \log x - \gamma = e^{-x}\delta_x < e^{-x}/x.$$

This limit is outside the finite algebraic operation hypothesis. Exponential convergence is analytic information, not a height estimate for rational linear forms in γ.

55.2 A positive gap for every fixed rational Laplace kernel

The next result turns positivity into an effective finite-dimensional reduction. It is not an inference from compactness alone: the factorial coercivity below supplies the missing uniform degree bound.

Lemma 55.3 (Weighted Laguerre lower bound)**.** *Let $D(t) = \sum_{j=0}^{m} d_j t^j \in \mathbb{R}[t]$ be positive on $[0,\infty)$, put $C_D = \sum_j |d_j|$, and let Q have degree n and leading coefficient q_n. For $\alpha \in \{0,1\}$,*

$$\int_0^\infty \frac{t^\alpha e^{-t} Q(t)^2}{D(t)}\,dt \geq \frac{q_n^2 n!(n+\alpha)!}{C_D[4(n+m+1)+2]^m}. \tag{55.2}$$

Proof. Let $L_n^{(\alpha)}$ denote the usual Laguerre polynomial, with leading coefficient $(-1)^n/n!$. Normalize it by $\ell_n = L_n^{(\alpha)}/\sqrt{h_n}$, where $h_n = (n+\alpha)!/n!$. Rodrigues' formula and integration by parts show that the ℓ_n are orthonormal for $t^\alpha e^{-t}dt$ and

$$\int_0^\infty t^\alpha e^{-t} Q(t)\ell_n(t)dt = (-1)^n q_n \sqrt{n!(n+\alpha)!}.$$

Weighted Cauchy–Schwarz bounds the square of this expression by the left side of (55.2) times $\int t^\alpha e^{-t}\ell_n(t)^2 D(t)dt$.

Multiplication by t has a three-term recurrence. Its diagonal is $2j+\alpha+1$; its off-diagonal magnitudes are $\sqrt{(j+1)(j+\alpha+1)}$. In its truncation to indices $0,\dots,K$ every absolute row sum is at most $4(K+1)+2\alpha$. The symmetric matrix therefore has operator norm at most that number. Take $K = n+m$. For $0 \leq j \leq m$, the matrix element calculating $\int t^{j+\alpha}e^{-t}\ell_n^2 dt$ is unchanged by this truncation: a length-j path starting at index n cannot reach an index above $n+m$. Hence that integral is at most $[4(n+m+1)+2]^j$. Summing with the coefficients $|d_j|$ bounds the remaining factor by $C_D[4(n+m+1)+2]^m$, proving the claim. □

Theorem 55.4 (Fixed-denominator lattice gap and finite candidate reduction)**.** *Let $D \in \mathbb{Q}[t]$ be fixed and strictly positive on $[0,\infty)$. For nonzero $P \in \mathbb{Z}[t]$ nonnegative on that half-line put*

$$J_D(P) = \int_0^\infty e^{-t}\frac{P(t)}{D(t)}dt.$$

Then

$$\inf_P J_D(P) > 0,$$

and this infimum is attained. A finite list of integer polynomials containing every minimizer can be computed from D. This means an exact finite candidate reduction, not a claim to decide equality of arbitrary transcendental constants. More precisely, if $\deg P = 2n$ then

$$J_D(P) \geq \frac{(n!)^2}{C_D[4(n+m+1)+2]^m},$$

and if $\deg P = 2n+1$ the numerator can be replaced by $n!(n+1)!$.

Proof. A real polynomial nonnegative on $[0,\infty)$ can be written $P = A^2 + tB^2$ with real polynomials A, B. Here is a factorization proof. A positive real root has even multiplicity and contributes a square. The factors t and $t+a$ $(a>0)$ have the required form. For an irreducible quadratic $t^2 - 2ut + u^2 + v^2$, write

$$(t - \sqrt{u^2+v^2})^2 + 2(\sqrt{u^2+v^2} - u)t.$$

Positive constants also have the required form, and multiplication preserves it by

$$(A^2+tB^2)(C^2+tE^2) = (AC - tBE)^2 + t(AE+BC)^2.$$

This proves the representation. The nonnegative leading terms show that for degree $2n$, A has degree n, B has degree at most $n-1$, and the square of the leading coefficient of A is the

leading coefficient of P. For degree $2n+1$ the analogous leading term is that of tB^2. The leading coefficient of a nonzero such integer polynomial is at least one. Applying Lemma 55.3 to this leading square proves both displayed bounds.

They tend to infinity with n. Their even-degree lower bound is increasing once $n \geq \max(m, 1)$: the ratio of consecutive terms is

$$(n+1)^2 \left(1 - \frac{1}{n+m+5/2}\right)^m \geq (n+1)^2 \left(1 - \frac{m}{n+m+5/2}\right) > 1.$$

A rational $b_D > 0$ with $D(t) \geq b_D$ for all $t \geq 0$ is effectively obtainable by real algebraic root isolation and an elementary leading term bound outside a bounded interval. Thus $J_D(1) \leq B := 1/b_D$. The preceding monotonicity gives a computable degree bound $N \geq 1$ for every P with $J_D(P) \leq B$.

There is also a computable coefficient bound at those degrees. On $[0,1]$, $e^{-t}/D(t) \geq 1/(3C_D)$, so $\int_0^1 P \leq 3C_D B$. Set $M = \max_{[0,1]} P$. The shifted Chebyshev expansion, whose coefficients in positive degree have absolute value at most $2M$, gives $\|P'\|_\infty \leq 4MN^3$. On an interval of length $1/(8N^3)$ next to a maximizer, $P \geq M/2$. Consequently $M \leq 48N^3 C_D B$. The coefficient ℓ^1 norm of $T_j(2t-1)$ is at most 7^j, by its recurrence; hence every monomial coefficient of P has absolute value at most

$$144N^3 C_D B\, 7^N.$$

There are only finitely many integer polynomials satisfying these bounds. Their nonnegativity on the half-line is decidable by real root isolation. The resulting finite list contains 1 and every possible minimizer. Every nonzero member has strictly positive integral, so their minimum is positive and attained; any polynomial outside the list has integral greater than $B \geq J_D(1)$. □

The constants are deliberately coarse; the theorem covers an entire fixed rational kernel, not just a predetermined degree or a particular orthogonal-polynomial sequence. It does not apply uniformly when D varies. For example $J_{(1+t)^m}(1) \leq 1/(m-1)$ for $m > 1$, by omitting e^{-t}, so the unscaled integrals tend to zero as pole order grows. The arithmetic clearance of those small integrals is a separate matter, addressed next.

55.3 Exact arithmetic reduction and the remaining Gompertz loophole

Proposition 55.5 (Twisted residues classify the finite kernel quotient)**.** *Let k be a characteristic-zero subfield of $\mathbb{C}$, and let $R \in k(t)$ have poles only among $-a_1, \ldots, -a_s$, where the $a_i \in k$ are distinct positive real numbers. There are unique $Q \in k(t)$ and $b_i \in k$, with Q having poles only at these points, such that*

$$R = Q' - Q + \sum_i \frac{b_i}{t + a_i}. \tag{55.3}$$

If the principal part at $-a_i$ is $\sum_{j=1}^{m_i} c_{ij}(t+a_i)^{-j}$, then

$$b_i = \sum_{j=1}^{m_i} \frac{(-1)^{j-1} c_{ij}}{(j-1)!}, \qquad \int_0^\infty e^{-t} R(t)\,dt = -Q(0) + \sum_i b_i \delta_{a_i}.$$

Thus the quotient by rational twisted derivatives $Q' - Q$ has precisely one coordinate per allowed pole. This is a classification of rational functions modulo that differential operator; it is not a claim that the numerical constants $1, \delta_{a_1}, \ldots, \delta_{a_s}$ are linearly independent.

Proof. For a pole of order $j \geq 2$, the leading term of $(d/dt-1)(t+a_i)^{-j+1}$ is $-(j-1)(t+a_i)^{-j}$. Successively remove these terms; only simple poles remain. On polynomials the operator is invertible, its inverse at H being $-\sum_{j\geq 0} H^{(j)}$, a finite sum. This proves existence. If the difference of two decompositions were nonzero, a pole of Q of maximal order would give a pole of strictly larger order in $Q'-Q$, which simple poles cannot cancel. Thus Q would be polynomial; then its leading term shows $Q'-Q$ cannot be a nonzero sum of simple poles or zero. This proves uniqueness. Multiplication by e^{-t} makes $Q'-Q$ an exact derivative, so its residue at every finite pole is zero. Taking the residue of $e^{-t}R$ at $-a_i$ and dividing by e^{a_i} gives the formula for b_i. Integration of $(e^{-t}Q)'$ yields the stated boundary term, since Q is regular at zero and $e^{-t}Q(t) \to 0$ at positive infinity. □

For the individual target δ, take $D = (1+t)^m$, $m \geq 1$, and $P \in \mathbb{Z}[t]$. Ordinary division and partial fractions give

$$\frac{P(t)}{(1+t)^m} = H(t) + \sum_{j=1}^{m} \frac{c_j}{(1+t)^j}, \qquad H \in \mathbb{Z}[t], \quad c_j \in \mathbb{Z}.$$

Put $I_j = \int_0^\infty e^{-t}(1+t)^{-j}dt$. Integration by parts shows $(j-1)I_j = 1 - I_{j-1}$ for $j \geq 2$, with $I_1 = \delta$. Consequently there are explicitly computable integers $A_m(P), B_m(P)$ such that

$$(m-1)!J_{(1+t)^m}(P) = A_m(P) + B_m(P)\delta, \quad B_m(P) = (m-1)!\sum_{j=1}^{m} \frac{(-1)^{j-1}c_j}{(j-1)!}. \tag{55.4}$$

The coefficient identity is formal and makes no irrationality assumption. If $P \geq 0$ is nonzero, the form is strictly positive, giving the required nonvanishing certificate. For any fixed m, Theorem 55.4 proves that these positive integer forms cannot tend to zero, regardless of how numerator degrees or coefficients grow.

There are precise ways outside this theorem. If $g_m(P) = \gcd(A_m(P), B_m(P))$ is unbounded, division by this actual common factor replaces the numerator by the rational polynomial $P/g_m(P)$; the integer lattice hypothesis is lost. No bound on that gcd is proved here. Allowing m to grow or allowing sign-changing numerators also escapes the theorem. In the elementary growing-order example $P = 1$, however, (55.4) has $B_m = \pm 1$, so there is no gcd saving at all, and

$$(m-1)!I_m \geq (m-1)!\frac{e^{-1/m}}{m(1+1/m)^m} \geq \frac{(m-1)!}{9m} \longrightarrow \infty.$$

Here the integral was restricted to $0 \leq t \leq 1/m$ and the last bound uses $e < 3$. Thus this explicit exponentially weighted rational kernel attempt supplies nonzero forms and exact denominators, but its cleared errors grow instead of shrink. It does not resolve δ.

55.4 The separate approximation test for Euler's constant

For comparison, write $d_n = \operatorname{lcm}(1, \ldots, n)$. The verified published construction [258, Theorem 1.2] gives integer approximants $P_n = n!d_nF^{(I)}_{n;1}(1)$ and $Q_n = n!d_nF^{(I)}_{n;2}(1) > 0$ with

$$\left|\gamma - \frac{P_n}{Q_n}\right| \leq O\Big(e^{-4n^{3/4}+n^{1/2}+(3/8)n^{1/4}}\Big),$$

while

$$Q_n \asymp n!d_n\, n^{-9/8}e^{4n^{3/4}-(1/2)n^{1/2}-(3/8)n^{1/4}}.$$

These estimates yield only $|Q_n\gamma - P_n| \leq O(n!d_nn^{-9/8}e^{n^{1/2}/2})$. This is a growing upper bound, not a proof that the forms grow, and not a proof that they approach zero. Moreover a quoted

upper bound alone is not a nonvanishing certificate. In reduced terms, with $g_n = \gcd(P_n, Q_n)$, a successful use of this particular upper estimate would require g_n to dominate that factorial-scale bound along an infinite subsequence, together with nonvanishing there. Neither input is supplied by finite-cover elimination, by analytic convergence, or by the prior uniqueness/observation arguments. The primary paper itself states that its improvement does not prove irrationality.

Thus the individual γ attempt has an exact nonlinear algebraic barrier and a separate explicit denominator/error deficit. The individual δ attempt has an exact kernel quotient, nonvanishing positive integer forms, and a proved fixed-kernel gap. The two new obstruction theorems are structural progress within specified routes; they are not replacements for either missing individual arithmetic proof.

56 A rank obstruction for arithmetic moment determinants

The named targets of this section are Catalan's constant G, the irrationality and transcendence of $\zeta(5)$, and the transcendence of $\zeta(3)$. The principal result below concerns an actual attempted method, rather than claiming a new resolution of one of these targets. It proves that the superexponential smallness of certain positive Hankel determinants is lost under *every* rational normalization to an integer polynomial. The reason is the rank of the dependence on the desired constant. This gives a precise limitation on transferring the earlier moment-positivity and finite-witness arguments to arithmetic.

56.1 Current status and external results, checked on 17 October 2026

There are important changes after the usual older accounts of these constants. Fauzan's September 2026 preprint proves irrationality of the real number $\zeta(5)$ by normalized polynomial determinants [262]. Independent Lean developments have since been released. We inspected the actual target, definition and axiom files of the Romik repository: its target is irrationality of $\sum_{v\geq 0}(v+1)^{-5}$, and its declared additional axiom is precisely the prime number theorem in the form $\vartheta(x)/x \to 1$ [263]. We did not replay its complete Lean build in this investigation. Lai's October 2026 research paper explicitly records Fauzan's result as Theorem 1.5 [264]. Thus it would be inaccurate to present irrationality of $\zeta(5)$ as an untouched open target, or to count it as our own achievement.

For G, Sun's September claim [265] must be distinguished from the later OpenAI release. Wachs identifies an unaccounted quadratic 2-adic contribution in Sun's denominator calculation [266]; that criticism does not address the different OpenAI proof. The official OpenAI family 005 scope states that Catalan irrationality has been formalized, and supplies the paper and a comparator statement for the exact alternating series [267]. Lai also records this later result. A comparator is a specification, not by itself the proof implementation; no independent kernel replay is claimed here. These are externally announced/formalized irrationality results. We found no verified individual transcendence result for G, $\zeta(5)$ or $\zeta(3)$ in the primary material examined.

Brown's 2026 Mellin-period criterion and Fauzan's work demonstrate that positive determinant methods can have genuine arithmetic power [268, 262]. The obstruction below applies to a particular rank-constrained construction; it does not refute those richer constructions. The elementary determinant and Christoffel identities used in its proof are classical. Our contribution here is their explicit combination to diagnose this proposed transfer and to calculate its exact loss.

56.2 Integer coefficients versus the rank of a positive perturbation

Theorem 56.1 (Rank-sensitive integer-normalization bound)**.** *Let $\theta \in \mathbb{R}$, let M be a symmetric positive definite $n \times n$ real matrix, and let U be an $n \times r$ real matrix of rank $r \geq 1$. Suppose*

$$D(X) = \det(M + (X - \theta)UU^{\mathsf{T}}) \in \mathbb{Q}[X].$$

Put $K = U^{\mathsf{T}}M^{-1}U$. Then D has degree r, its leading coefficient is $\det(M)\det(K) > 0$, and every nonzero rational c for which $P = cD \in \mathbb{Z}[X]$ satisfies

$$|P(\theta)| \geq \frac{1}{\det K}. \tag{56.1}$$

In particular, if the columns u_j satisfy $u_j^{\mathsf{T}}M^{-1}u_j \leq L$, then $|P(\theta)| \geq L^{-r}$. For $r = 0$, a nonzero integral normalization is a nonzero constant integer, so its absolute value is at least one.

Proof. The matrix determinant identity gives

$$D(X) = \det(M)\det(I_r + (X - \theta)K).$$

Since M is positive definite and U has full column rank, K is positive definite. Its determinant is positive, proving both the degree and the leading-coefficient assertion. The leading coefficient of P is a nonzero integer. Its absolute value is therefore at least one. As $P(\theta) = c\det(M)$, division by $\det K$ gives (56.1). Hadamard's inequality applied to the positive definite matrix K yields $\det K \leq \prod_j K_{jj} \leq L^r$. □

For example, if $r = r_n$ and $L \leq CN_n^a$, then

$$-\log|P_n(\theta)| \leq r_n(\log C + a\log N_n).$$

Consequently $r_n \log(2N_n) = o(n^2)$ rules out a bound $0 < |P_n(\theta)| \leq e^{-\delta n^2}$ for any fixed $\delta > 0$. This is an obstruction independent of any conjecture about θ and independent of how accurately denominators are estimated.

Corollary 56.2 (Polynomial moment spaces and evaluation rank one)**.** *Let μ be a finite positive measure with infinite support, and let $V \subset \mathbb{R}[x]$ be a finite-dimensional space with basis $p_1, \ldots, p_n$. Define*

$$M_{ij} = \int p_i p_j \, d\mu, \qquad u_i = p_i(a), \qquad K_V(a,a) = \sup_{0 \neq p \in V} \frac{|p(a)|^2}{\int p^2 \, d\mu}.$$

If $u \neq 0$, then $K_V(a,a) = u^{\mathsf{T}}M^{-1}u$ and

$$\frac{\det M}{[X]\det(M + (X - \theta)uu^{\mathsf{T}})} = \frac{1}{K_V(a,a)}. \tag{56.2}$$

Here $[X]$ denotes the coefficient of X; the determinant is affine. If that determinant has rational coefficients, every nonzero rational normalization to an integer polynomial has value at θ at least $1/K_V(a,a)$ in absolute value. If $V \subseteq \mathbb{R}[x]_{\leq N}$, the same lower bound holds with $K_{\leq N}(a,a)$ in place of $K_V(a,a)$.

Proof. A polynomial $p = \sum c_i p_i$ has norm squared $c^{\mathsf{T}}Mc$ and value $c^{\mathsf{T}}u$. Cauchy–Schwarz after multiplication by $M^{1/2}$ shows that the supremum is $u^{\mathsf{T}}M^{-1}u$, attained by $c = M^{-1}u$. Theorem 56.1 proves the remaining statements, and inclusion of spaces can only increase the supremum. Positive definiteness of M follows because a nonzero polynomial cannot vanish on an infinite support. □

Both results are unchanged by a rational invertible change of basis, or by left and right rational invertible matrix transformations: their effect on the determinant is only a nonzero rational scalar. Thus a clever rational basis does not evade the conclusion for a fixed moment space.

56.3 The zeta tail attempt: a polynomial lower bound after every normalization

For an integer $s \geq 2$, put $\theta_s = \zeta(s)$ and

$$H_m^{(s)} = \sum_{k=1}^{m} k^{-s}, \qquad m_j^{(s)} = \theta_s - H_j^{(s)}.$$

The integral identity

$$m_j^{(s)} = \int_0^1 x^j w_s(x)\, dx, \quad w_s(x) = \frac{(-\log x)^{s-1}}{\Gamma(s)(1-x)} \tag{56.3}$$

follows by expanding $(1-x)^{-1}$ and integrating its nonnegative terms. The individual integrals are

$$\int_0^1 x^k (-\log x)^{s-1} dx = \frac{\Gamma(s)}{(k+1)^s}.$$

The measure is finite and strictly positive on $(0,1)$.

The proposed determinant is

$$D_{n,s}(X) = \det(X - H_{i+j}^{(s)})_{0 \leq i,j < n}.$$

At $X = \theta_s$ it is strictly positive. Its dependence on X is the rank-one matrix $\mathbf{1}\mathbf{1}^{\mathsf{T}}$, corresponding exactly to polynomial evaluation at $a = 1$. It therefore has degree one for every n. To quantify the resulting obstruction we calculate the endpoint kernel.

Lemma 56.3 (Endpoint kernel bound)**.** *For fixed integer $s \geq 2$, there are constants $0 < c_s < C_s < \infty$ such that for every $N \geq 0$,*

$$c_s (N+1)^{2s-2} \leq K_{\leq N}^{w_s}(1,1) \leq C_s (N+1)^{2s-2}.$$

Proof. Write $b = s - 2 \geq 0$. The inequality $-\log x \geq 1 - x$ gives

$$w_s(x) \geq \Gamma(s)^{-1} (1-x)^b.$$

Also $w_s(x) \leq C x^{-1/2} (1-x)^b$ for a finite constant C: the ratio is

$$\frac{x^{1/2}}{\Gamma(s)} \left(\frac{-\log x}{1-x} \right)^{s-1},$$

which extends continuously to $[0,1]$, with values zero at zero and $1/\Gamma(s)$ at one. It suffices to compute the kernel for the two Jacobi weights $x^a(1-x)^b$, $a = 0, -1/2$.

Here is the calculation, including the normalization. For the parameters $a = 0, -1/2$ and $b = s - 2 \geq 0$ let

$$J_k(x) = \frac{(-1)^k}{k!\, x^a (1-x)^b} \frac{d^k}{dx^k} \{ x^{k+a} (1-x)^{k+b} \}.$$

Leibniz's rule shows that this is a polynomial. Integration by parts k times proves its orthogonality to degrees below k, since the boundary terms vanish. The same calculation with J_k itself, followed by the beta integral, gives

$$J_k(1) = \frac{\Gamma(k+b+1)}{k!\Gamma(b+1)}, \qquad h_k = \int_0^1 J_k^2 x^a (1-x)^b dx = \frac{\Gamma(k+a+1)\Gamma(k+b+1)}{(2k+a+b+1)k!\Gamma(k+a+b+1)}.$$

For the norm formula one can use the leading coefficient $\Gamma(2k+a+b+1)/(k!\Gamma(k+a+b+1))$ and integrate $x^{k+a}(1-x)^{k+b}$. These normalizations also agree with the standard Jacobi table [269]. Thus

$$K_{\le N}^{a,b}(1,1) = \sum_{k=0}^{N} \frac{J_k(1)^2}{h_k}, \qquad \frac{J_k(1)^2}{h_k} \sim \frac{2k^{2b+1}}{\Gamma(b+1)^2}.$$

The last relation follows directly from Stirling's formula; summation gives $K_{\le N}^{a,b}(1,1) \asymp (N+1)^{2b+2}$. Finally, increasing a measure decreases its evaluation kernel. Applying this observation to the two weight comparisons proves the assertion. □

Theorem 56.4 (No exponential arithmetic gain for the entire polynomial tail family)**.** *Let $s \ge 2$ be a fixed integer and define on $\mathbb{Q}[x]$ the rational polynomial-valued functional*

$$\Lambda_X(x^j) = X - H_j^{(s)}.$$

Let $V_N \subseteq \mathbb{Q}[x]_{\le N}$ be any nonzero finite-dimensional space with any rational basis p_i. Put $D_{V_N}(X) = \det(\Lambda_X(p_i p_j))$. For every nonzero rational c with $P = cD_{V_N} \in \mathbb{Z}[X]$,

$$0 < |P(\zeta(s))| \quad \text{and} \quad |P(\zeta(s))| \ge c_s'(N+1)^{-(2s-2)}, \tag{56.4}$$

where $c_s' > 0$ depends only on s. In particular, if N is bounded by a fixed power of a construction index n, no such normalized determinant can be $e^{-\delta n}$, or $e^{-\delta n^2}$, for all sufficiently large n. For the full space $V = \mathbb{Q}[x]_{<n}$, writing $D_{n,s}(X) = A_{n,s}X + B_{n,s}$ gives

$$A_{n,s} > 0, \qquad \zeta(s) + \frac{B_{n,s}}{A_{n,s}} = \frac{1}{K_{<n}^{w_s}(1,1)} \asymp n^{-(2s-2)}. \tag{56.5}$$

Proof. For a polynomial p, the coefficient of X in $\Lambda_X(p)$ is $p(1)$. At $X = \zeta(s)$ the functional is integration against w_s. Consequently its Gram matrix is a positive definite matrix plus $(X - \zeta(s))uu^{\mathsf{T}}$, where $u_i = p_i(1)$. If $u = 0$, the determinant is a nonzero rational constant and any nonzero integral normalization has absolute value at least one. Otherwise Corollary 56.2 and Lemma 56.3 give (56.4). The exact ratio and the two-sided kernel bound prove (56.5). A fixed power of N cannot decay exponentially in n when N is polynomial in n, proving the last assertion. □

For $s = 3$ the exponent is 4; for $s = 5$ it is 8. These are statements about this precise family, not about all rational approximations to the constants. In particular, they do not contradict Apéry's or Fauzan's constructions. They show why replacing their arithmetic input by positivity of the simplest tail matrices cannot work as proposed.

56.4 Catalan tails: the exact exterior-evaluation exponential rate

Write

$$S_{j-1} = \sum_{k=0}^{j-1} \frac{(-1)^k}{(2k+1)^2}, \qquad t_j = (-1)^j(G - S_{j-1}), \qquad S_{-1} = 0.$$

A geometric-series integration and $y = x^2$ give

$$t_j = \int_0^1 \frac{x^{2j}(-\log x)}{1+x^2}dx = \int_0^1 y^j w_G(y)dy, \qquad w_G(y) = \frac{-\log y}{4\sqrt{y}(1+y)}. \tag{56.6}$$

The geometric expansion is justified either first on $[0, r]$, $r < 1$, and then by dominated convergence, or by its uniformly bounded partial sums. The coefficient of G is $(-1)^j$, so the associated polynomial functional has evaluation at $a = -1$ as its rank-one coefficient.

Lemma 56.5 (Exterior kernel)**.** *Let $\rho = 3 + 2\sqrt{2}$. For the weight w_G there are constants $c, C > 0$ such that, for $N \geq 0$,*

$$c\rho^{2N} \leq K_{\leq N}^{w_G}(-1,-1) \leq C(N+1)^2\rho^{2N}.$$

In particular $N^{-1}\log K_{\leq N}^{w_G}(-1,-1) \to 2\log\rho$.

Proof. Since $-\log y \geq 1 - y$ and $4\sqrt{y}(1+y) \leq 8$, we have $w_G(y) \geq (1-y)/8$. The shifted Jacobi polynomial for the latter weight is $J_k(y) = P_k^{(1,0)}(2y-1)$, with squared norm $1/(2k+2)$. The elementary identity

$$P_k^{(1,0)}(t) = \frac{P_k(t) - P_{k+1}(t)}{1-t}$$

follows, for example, by comparing orthogonality and leading coefficients; P_k is the Legendre polynomial. At $t = -3$ its magnitude is at most $(\rho^k + \rho^{k+1})/4$. Indeed, the elementary Laplace representation

$$P_k(3) = \frac{1}{\pi}\int_0^\pi (3 + \sqrt{8}\cos u)^k du$$

gives $|P_k(-3)| = P_k(3) \leq \rho^k$. Both polynomial identities can also be checked by their three-term recurrences and initial values. Summing $J_k(-1)^2/h_k$ and using the weight comparison proves the upper bound.

For the lower bound test the variational kernel with $p(y) = T_N(2y-1)$, where T_N is a Chebyshev polynomial. On $[0,1]$ its absolute value is at most one, so $\int p^2 w_G \leq G$. At $y = -1$, $|T_N(-3)| = (\rho^N + \rho^{-N})/2 \geq \rho^N/2$. Hence the ratio in the variational definition is at least $\rho^{2N}/(4G)$. The limiting statement follows by taking logarithms. □

Theorem 56.6 (Catalan integer-normalization obstruction)**.** *Define $\Lambda_X^G(y^j) = (-1)^j(X - S_{j-1})$. For any rational polynomial space $V \subseteq \mathbb{Q}[y]_{\leq N}$, any rational basis of V, and any nonzero rational normalization*

$$P(X) = c\det(\Lambda_X^G(p_i p_j)) \in \mathbb{Z}[X],$$

we have

$$0 < |P(G)| \quad \textit{and} \quad |P(G)| \geq c_0(N+1)^{-2}\rho^{-2N}$$

for an absolute $c_0 > 0$. Thus bounded linear degree $N = O(n)$ rules out quadratic exponential decay $e^{-\delta n^2}$ after any *integer normalization. For the full space of degrees below n, its affine polynomial $D_{n,G}(X) = A_{n,G}X + B_{n,G}$ satisfies*

$$\lim_{n\to\infty}\frac{1}{n}\log\left(G + \frac{B_{n,G}}{A_{n,G}}\right) = -2\log(3+2\sqrt{2}).$$

Proof. The coefficient functional is $p \mapsto p(-1)$, and evaluation at G is the positive measure (56.6). Apply Corollary 56.2 and Lemma 56.5. If all polynomials in V vanish at -1, the normalized determinant is a nonzero integer constant and the conclusion still holds after decreasing c_0. For the full polynomial space use the exact ratio (56.2) and the kernel limit. □

This argument does not rule out an ordinary exponentially small Catalan linear form with satisfactory arithmetic. It rules out the tempting inference that increasing the size of these raw tail matrices automatically produces a quadratic-exponential irrationality criterion. The needed arithmetic saving must come from a different functional or different family, not a rescaling of these same determinants.

56.5 The exact geometric decay and a complete denominator ledger

The obstruction is not caused by an incorrectly estimated analytic decay. The unnormalized determinants really do decay on a quadratic exponential scale. We give that scale and then account for the denominators explicitly.

Proposition 56.7 (Sharp geometric scale)**.** *Let μ be a finite positive measure on $[a,b]$, $a<b$, having a density that is continuous and strictly positive on every compact subinterval of (a,b). Set*

$$\Delta_n(\mu)=\det\left(\int x^{i+j}\,d\mu(x)\right)_{0\le i,j<n}.$$

Then

$$\lim_{n\to\infty}\frac{\log\Delta_n(\mu)}{n^2}=\log\frac{b-a}{4}.$$

In particular, both $D_{n,s}(\zeta(s))$ and $D_{n,G}(G)$ have limit $-\log 4$ after taking their logarithms and dividing by n^2.

Proof. An affine change of variable multiplies the determinant by $(b-a)^{n(n-1)}$, apart from a total-mass normalization contributing only $O(n)$ to its logarithm. It suffices to prove the result on $[0,1]$. Gram–Schmidt expresses Δ_n as the product of the squared norms of the monic orthogonal polynomials of degrees $0,\dots,n-1$. The monic shifted Chebyshev polynomial of degree $k\ge 1$ has supremum 2^{1-2k} on $[0,1]$. Since the orthogonal polynomial minimizes the norm among monic polynomials of its degree, if $M=\mu([0,1])$,

$$\Delta_n\le M^n 2^{-2(n-1)^2}.$$

This gives the upper limit $-\log 4$.

For $0<\varepsilon<1/2$, let $m_\varepsilon>0$ be a lower bound for the density on $[\varepsilon,1-\varepsilon]$. The Gram matrix dominates m_ε times the Lebesgue Gram matrix on this interval. Positivity and conjugation by the inverse square root of the latter matrix give the same inequality for determinants, with factor m_ε^n. The monic shifted Legendre norms give the exact identity

$$\Delta_n(dx|_{[u,v]})=(v-u)^{n^2}\prod_{k=0}^{n-1}\frac{1}{(2k+1)\binom{2k}{k}^2}.$$

Indeed the degree-k Legendre polynomial on $[0,1]$ has leading coefficient $\binom{2k}{k}$ and squared norm $1/(2k+1)$; rescaling gives the displayed formula. Stirling's formula now gives a lower limit $\log((1-2\varepsilon)/4)$. Letting $\varepsilon\downarrow 0$ proves the claim. The two weights used above satisfy all the hypotheses. □

Put $d_m=\operatorname{lcm}(1,\dots,m)$ and $o_m=\operatorname{lcm}(1,3,\dots,2m-1)$, with $d_0=o_0=1$. The ordinary prime number theorem implies

$$\log d_m=m+o(m),\qquad \log o_m=2m+o(m).$$

For the second identity, deleting the powers of 2 from the least common multiple through $2m-1$ changes its logarithm by only $O(\log m)$.

Proposition 56.8 (Certified denominator cost)**.** *The explicit integers*

$$C_{n,s} = \prod_{i=0}^{n-1} d_{n-1+i}^{s}, \qquad C_{n,G} = \prod_{i=0}^{n-1} o_{n-1+i}^{2}$$

satisfy $C_{n,s}D_{n,s}(X), C_{n,G}D_{n,G}(X) \in \mathbb{Z}[X]$ *and*

$$\lim_{n\to\infty} \frac{\log(C_{n,s}D_{n,s}(\zeta(s)))}{n^2} = \frac{3s}{2} - \log 4 > 0, \qquad \lim_{n\to\infty} \frac{\log(C_{n,G}D_{n,G}(G))}{n^2} = 6 - \log 4 > 0.$$

Thus these particular, completely justified integer normalizations grow on the quadratic exponential scale.

Proof. The denominator of $H_j^{(s)}$ divides d_j^s. Multiplication of row i by d_{n-1+i}^s therefore makes all its polynomial entries integral. The alternating partial sum S_{j-1} has denominator dividing o_j^2, which proves the Catalan assertion by the same row operation. Summing $\log d_m = m + o(m)$ over $m = n-1, \dots, 2n-2$ gives $\log C_{n,s} = (3s/2)n^2 + o(n^2)$; the Catalan sum gives $\log C_{n,G} = 6n^2 + o(n^2)$. Proposition 56.7 completes the calculation. The little-oh errors can be summed uniformly because their arguments lie between $n-1$ and $2n-2$. □

This also identifies precisely what a hypothetical content improvement would have to do. Let g_n be the greatest common divisor of the coefficients of one of the integer polynomials C_nD_n. If $\log g_n/n^2 \to \gamma$ and $\log C_n/n^2 \to \alpha$, then its primitive normalization has logarithmic value

$$\lim_{n\to\infty} n^{-2} \log |(C_n/g_n)D_n(\theta)| = \alpha - \gamma - \log 4.$$

A negative limit would require $\gamma > \alpha - \log 4$. Theorems 56.4 and 56.6 prove that this strict inequality is impossible for the present raw-tail families. The argument therefore rules out even a presently undiscovered content saving of a fixed positive quadratic margin. It does not merely exhibit one inefficient denominator estimate.

Proposition 56.9 (Rational dilation does not improve the quadratic balance)**.** *Let* $s \geq 2$ *and* $z = a/b \in (0,1]$*, with* a, b *coprime positive integers. Write*

$$L = \mathrm{Li}_s(z), \qquad D_{n,s,z}(X) = \det\left(X - \sum_{k=1}^{i+j} \frac{z^k}{k^s}\right)_{0\leq i,j<n}.$$

Then $D_{n,s,z}$ *is affine and positive at* L*, and*

$$b^{n(n-1)}C_{n,s}D_{n,s,z}(X) \in \mathbb{Z}[X], \qquad \lim_{n\to\infty} \frac{1}{n^2} \log\left(b^{n(n-1)}C_{n,s}D_{n,s,z}(L)\right) = \log a + \frac{3s}{2} - \log 4 > 0.$$

In particular, shrinking a rational argument with numerator one produces no improvement at all in this certified quadratic-scale balance.

Proof. Expansion of $(1-x)^{-1}$ and the substitution $x = zu$ give

$$L - \sum_{k=1}^{j} \frac{z^k}{k^s} = \int_0^z x^j \frac{\log(z/x)^{s-1}}{\Gamma(s)(1-x)}\, dx.$$

The measure is positive and finite; its support is $[0, z]$. The perturbation matrix is again rank one. Proposition 56.7 gives geometric limit $\log(z/4)$. Multiply row i by $b^i d_{n-1+i}^s$ and column j by b^j. Each rational term becomes $a^k b^{i+j-k} d_{n-1+i}^s / k^s$, an integer because $k \leq i+j \leq n-1+i$. The coefficient of X is also integral. Taking logarithms of the determinant multiplier gives $\log b + 3s/2$ on the quadratic scale. Adding $\log(z/4)$ proves the asserted cancellation of b. □

An exact rational calculation, included with this section's supporting code, checks the affine identity and primitive integer coefficients for $1 \le n \le 6$. Some small cases are

target	n	$[X]P_n$	$[1]P_n$
$\zeta(3)$	2	7	-8
$\zeta(3)$	3	4105	-4887
$\zeta(5)$	2	31	-32
$\zeta(5)$	3	4458601	-4622103
G	2	10	-9
G	3	61794	-56575

These computations illustrate the exact construction; the theorems above hold for every size and do not rely on these finite checks.

56.6 What would actually be needed for transcendence

The difference between irrationality and transcendence is not repaired by producing a small integer polynomial alone. The height and degree must be tracked together with its value and its nonvanishing.

Proposition 56.10 (A height-aware quadratic-scale criterion)**.** *Let θ be algebraic of degree d with primitive minimal polynomial having leading coefficient $a \neq 0$ and conjugates $\theta = \theta_1, \dots, \theta_d$. For $P \in \mathbb{Z}[X]$ of degree m and height $H \ge 1$, if $P(\theta) \neq 0$, then*

$$|P(\theta)| \ge |a|^{-m}\big((m+1)H\big)^{-(d-1)} \prod_{j=2}^{d} \max(1, |\theta_j|)^{-m}.$$

Consequently, suppose a sequence satisfies

$$m_n = o(n^2), \quad \log H(P_n) \le \beta n^2 + o(n^2), \quad 0 < |P_n(\theta)| \le \exp(-\alpha n^2 + o(n^2)),$$

where $\alpha > 0$ and $\beta \ge 0$. Then θ cannot have algebraic degree d whenever $\alpha > (d-1)\beta$.

Proof. The resultant of the minimal polynomial and P is the nonzero integer $a^m \prod_{j=1}^{d} P(\theta_j)$, so its absolute value is at least one. For each other conjugate, $|P(\theta_j)| \le (m+1)H \max(1, |\theta_j|)^m$. Division proves the inequality. Taking logarithms gives

$$\log |P(\theta)| \ge -(d-1)\log H - O_\theta(m + \log(m+1)),$$

which contradicts the specified upper bound in the indicated range. □

To exclude all algebraic degrees with positive quadratic height cost, the ratio α/β would need to be unbounded across a family of constructions. Alternatively, $\beta = 0$ and $\alpha > 0$ suffice, subject to the stated degree and nonvanishing hypotheses. None of our tail determinants provides either alternative. At the two zeta targets they do not even provide exponentially small normalized values, by Theorem 56.4. At G, Theorem 56.6 already rules out a positive quadratic exponent with degree space $O(n)$.

For an explicit check on this logical distinction, write $(1+\sqrt{2})^k = p_k + q_k\sqrt{2}$, where p_k, q_k are positive integers. Its conjugate is $(1-\sqrt{2})^k = p_k - q_k\sqrt{2}$. With $k = n^2$ the degree-one integer polynomials $Q_n(X) = q_{n^2}X - p_{n^2}$ satisfy

$$0 < |Q_n(\sqrt{2})| = (\sqrt{2}-1)^{n^2}, \qquad \log H(Q_n) = n^2 \log(1+\sqrt{2}) + O(1).$$

Thus even quadratic exponential smallness of nonzero integer linear forms is compatible with algebraic irrationality. Here $\alpha = \beta$, exactly at the degree-two threshold of Proposition 56.10.

Nor does a linear-independence theorem for additional constants settle an individual transcendence question. Lai proves that $1, \zeta(3), \zeta(4)$ are linearly independent over $\mathbb{Q}$ [264]. For comparison, $1, \alpha, \pi^4$ are linearly independent over $\mathbb{Q}$ for *every* algebraic irrational α: a nonzero coefficient of π^4 would make π^4 algebraic, whereas a zero coefficient reduces to irrationality of α. Since $\zeta(4) = \pi^4/90$, the form of this independence result is compatible with algebraic irrationality of the middle entry. No individual transcendence conclusion follows from it.

56.7 The actual transfer boundary and the remaining tasks

The earlier positivity and finite-witness methods transfer successfully to strict nonvanishing of our determinants. They do not transfer the required denominator savings. The obstruction here identifies the exact failure: all polynomial row changes leave the target dependence as one evaluation functional, and an integer leading coefficient forces the value to exceed the reciprocal Christoffel kernel. For zeta tails this reciprocal is only a power of the polynomial degree. For Catalan tails it is exponential in that degree, although the raw determinant is quadratic exponential.

The recent OpenAI Catalan construction illustrates a genuine escape from these hypotheses, rather than a counterexample. Its paper explicitly uses mixed signed moments, cancellation of an auxiliary $\zeta(2)$ contribution, and a separate arithmetic nonvanishing certificate at prime scales [267, Sections 1.2 and 4]. Those are not rational basis changes of the single positive tail functional studied here. In particular, our positive rank-one theorem cannot be applied to that mixed signed matrix. Fauzan's polynomial determinants likewise must not be replaced by these affine raw-tail determinants without proving a new identity.

For individual transcendence of G, $\zeta(5)$ and $\zeta(3)$, the unresolved task in this attempted approach is therefore specific: construct a family outside the rank-constrained tail class, certify that its integer-polynomial value is nonzero, and simultaneously obtain height, degree and decay bounds meeting Proposition 56.10 for every algebraic degree. Neither functional independence of a generating function nor finite numerical checks provide those three estimates. This section supplies a proved method obstruction and an exact arithmetic ledger; it records no new resolution of these individual transcendence questions.

57 Arithmetic information missing from ergodic and analytic recovery

This section tests two named targets: the Khinchin constant K and the period-doubling Feigenbaum constant δ. Neither constant is proved irrational or transcendental here. Instead, two precise transfers identify information that recovery arguments do not supply. Full finite-cylinder statistics need not control the unbounded logarithmic observable defining K. Even a uniformly effective, normalized analytic contraction, with a uniform spectral gap and a unique solution, need not permit an algorithmic decision of the arithmetic type of that solution. The latter assertion is proved for an explicit family with rational Taylor coefficients; it is not an undecidability assertion about either named constant.

57.1 Khinchin's constant: reconstruction of the law versus its logarithmic moment

Let μ be the Gauss probability measure on $(0,1)$,

$$d\mu(x) = \frac{dx}{(1+x)\log 2}, \qquad p_k = \mu\left(\frac{1}{k+1} < x < \frac{1}{k}\right) = \log_2 \frac{(k+1)^2}{k(k+2)} \quad (k \geq 1).$$

Writing the continued fraction of an irrational $x \in (0,1)$ as $[0; a_1, a_2, \ldots]$, the classical almost-everywhere geometric-mean theorem identifies

$$L := \log K = \sum_{k\geq 1} p_k \log k, \qquad \lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \log a_j = L \quad \text{for } \mu\text{-almost every } x. \tag{57.1}$$

Bailey, Borwein and Crandall give the product representation, accelerated evaluation methods, and a discussion of the unresolved arithmetic question [270]. Their finite searches for algebraic relations do not constitute a proof of irrationality. No such proof is supplied by the sources verified in this investigation.

For completeness, even the exact one-digit probabilities exhibit a warning against an arithmetic inference from the recovered probability law.

Proposition 57.1. *Every p_k is transcendental, whereas $\sum_{k\geq 1} p_k = 1$. Moreover, for every integer $B \geq 2$,*

$$0 \leq L - \sum_{k\leq B} p_k \log k \leq \frac{\log B + 1}{B \log 2}. \tag{57.2}$$

Proof. Put $r_k = (k+1)^2/[k(k+2)]$. This is rational and lies strictly between 1 and 2. If $p_k = a/b$ were rational in lowest terms, the equality $2^a = r_k^b$ and the 2-adic valuation would imply $b \mid a$, hence $b = 1$, contradicting $0 < p_k < 1$. If p_k were algebraic irrational, the Gelfond–Schneider theorem would make $2^{p_k} = r_k$ transcendental, also a contradiction. This application of the classical theorem has no novelty claim. Telescoping gives

$$\prod_{k=1}^{B} r_k = \frac{2(B+1)}{B+2} \longrightarrow 2,$$

and therefore the probabilities sum to one. Finally,

$$p_k = \frac{\log(1 + 1/[k(k+2)])}{\log 2} \leq \frac{1}{k^2 \log 2}.$$

The function $(\log t)/t^2$ decreases for $t \geq 2$, so its integral on $[B, \infty)$ bounds the tail and gives (57.2). □

The missing observation condition can be characterized exactly. The next statement is the elementary uniform-integrability principle specialized to the continued-fraction observable; its role is to expose the required additional hypothesis, rather than to claim a new general convergence theorem.

Proposition 57.2 (Exact tail requirement)**.** *Let (a_j) be any sequence of positive integers such that*

$$\frac{1}{n} \#\{j \leq n : a_j = k\} \longrightarrow p_k \quad \textit{for every } k \geq 1.$$

Then $n^{-1} \sum_{j\leq n} \log a_j \to L$ if and only if

$$\lim_{B\to\infty} \limsup_{n\to\infty} \frac{1}{n} \sum_{\substack{j\leq n \\ a_j > B}} \log a_j = 0. \tag{57.3}$$

Proof. For fixed B the truncated average converges to $L_B = \sum_{k\le B} p_k \log k$. If the full averages converge to L, their difference from the truncated averages converges to $L - L_B$; (57.2) then proves (57.3). Conversely, nonnegativity gives $\liminf_n n^{-1} \sum_{j\le n} \log a_j \ge L_B$. The limsup is at most L_B plus the limsup of the tail in (57.3). Let B tend to infinity in these two bounds. □

Observing every finite word, rather than only individual digits, still does not remove this obstruction. A Gauss-normal continued fraction means here that each finite word w of positive integers has limiting occurrence frequency $\mu(C_w)$, where C_w is its continued-fraction cylinder.

Theorem 57.3 (A computable exceptional orbit with the full cylinder law)**.** *There is a computable Liouville number $x \in (0,1)$ whose continued fraction is Gauss-normal and satisfies*

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \log a_j = +\infty.$$

Thus the entire collection of limiting finite-cylinder frequencies does not determine the logarithmic orbit average in (57.1)*.*

Proof. Start with a computable Gauss-normal digit sequence (b_i); the explicit construction of Adler, Keane and Smorodinsky supplies one [271]. Retain all these digits in their order. Immediately after the original digit b_{2^j}, insert one digit A_j. If the already constructed prefix before this insertion has length m_j and convergent denominator q_{m_j}, choose an integer

$$A_j \ge \max\{2, q_{m_j}^{j+2}\}.$$

This recursively defines a computable sequence of positive integers. Its infinite continued fraction is computable: its rational convergents are effective and their denominators have a uniform Fibonacci lower bound.

There are $O(\log n)$ insertions among the first n digits. For a fixed word of length r, deleting one inserted digit can change at most $2r$ candidate occurrences near that digit. After deleting all insertions, the prefix becomes a prefix of (b_i) of length $n - O(\log n)$. The discrepancy of its word count from the original prefix count is $O(r \log n)$, including the terminal boundary. Dividing by n proves that every finite word has the same limiting frequency as in (b_i).

At the convergent immediately before A_j, the standard continued-fraction estimate gives

$$0 < \left| x - \frac{p_{m_j}}{q_{m_j}} \right| < \frac{1}{A_j q_{m_j}^2} \le q_{m_j}^{-j-4}.$$

The denominators tend to infinity. For each positive integer exponent these inequalities give infinitely many sufficiently accurate rational approximations, so x is Liouville and hence transcendental. Finally the recurrence for convergent denominators gives $q_m \ge 2^{(m-1)/2}$: indeed $q_{m+2} \ge 2q_m$, with the usual initial values. At the prefix ending in A_j the logarithmic average is at least

$$\frac{\log A_j}{m_j + 1} \ge \frac{(j+2)(m_j - 1)}{2(m_j + 1)} \log 2 \longrightarrow +\infty.$$

This proves the assertion. □

The construction is an explicit application of sparse insertion and classical continued-fraction estimates; no priority claim for normal Liouville constructions is intended. It also does not contradict the almost-everywhere theorem: the orbit constructed is exceptional. For the named constant K, the precise failed transfer is now visible. Identifying the invariant distribution and proving an effective real tail estimate does not yield algebraic-height-controlled

rational approximations to its exponentiated logarithmic moment. In particular, the weights p_k are themselves transcendental, so Baker's theorem for finite linear forms with algebraic coefficients cannot simply be applied to $\sum p_k \log k$. Neither the transcendence of the individual weights nor their exact identification proves an arithmetic classification of K.

57.2 Feigenbaum's constant: uniqueness and a spectral gap are insufficient

The target here is the period-doubling parameter-scaling constant $\delta = 4.669201609\ldots$. For the principal period-doubling cascade of the logistic family, let r_n be its superstable parameter of period 2^n. With this convention,

$$\delta = \lim_{n\to\infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n}.$$

The equivalent renormalization description identifies the expanding scaling eigenvalue at the universal fixed point. Rigorous fixed-point methods originate in Lanford's work [272]; high-precision calculations and digit statistics are discussed in [273, 274, 275]. The verified sources do not establish its irrationality or transcendence. One must distinguish δ from the spatial-scaling constant $\alpha = -2.5029\ldots$: the polynomial-time theorem of Hertling and Spandl computes the fixed function and $\alpha = 1/g(1)$, and is not a stated polynomial-time theorem for δ [276, Theorem 2].

Could an arithmetic conclusion follow from effective analytic recovery, uniqueness and hyperbolicity alone? The following negative transfer is considerably more restrictive than encoding an arbitrary computable real as the constant term of a function. Both the constant and linear Taylor coefficients are fixed; all remaining coefficients are nonnegative rationals with a uniform algorithm and uniform analytic bounds.

Theorem 57.4 (Normalized analytic recovery does not decide arithmetic type)**.** *There is a uniformly specified family of entire functions $F_e(z) = \sum_{n\ge 0} c_{e,n} z^n$, indexed by finite machine descriptions, with the following properties.*

1. *Every coefficient is a nonnegative rational, and $c_{e,0} = 1/2$, $c_{e,1} = 1/8$. Given e and n in unary, $c_{e,n}$ is computable exactly in time polynomial in the description length of e and in n.*
2. *$\sup_{|z|\le 2} |F_e(z)| < 1$, $F_e([0,1]) \subset [1/2, 3/4]$, and $1/8 \le F_e'(x) < 1/4$ on $[0,1]$.*
3. *The unique fixed point $x_e \in [0,1]$ is uniformly computable to error 2^{-p} in time polynomial in p and the description length of e.*
4. *Each x_e is either $4/7$ or transcendental. There is no algorithm that, given these coefficient-program descriptions with all the preceding uniform guarantees, always decides which alternative holds.*

Proof. Fix a universal machine model and pad the computation so that any first halting step N satisfies $N \ge 2$. Put

$$F_0(z) = \frac{1}{2} + \frac{z}{8}, \qquad \epsilon_N = 2^{-2N-6}.$$

If machine e never halts, define $F_e = F_0$. If it first halts at step N, define

$$F_e(z) = F_0(z) + \epsilon_N z^N e^z. \tag{57.4}$$

These are mathematical definitions, not an algorithm required to decide whether halting occurs. To compute the coefficient of degree $n \geq 2$, simulate the machine for just n steps. If it has not halted, output zero. If its first halting step is $N \leq n$, output

$$c_{e,n} = \frac{2^{-2N-6}}{(n-N)!}.$$

This is correct even if the machine halts later than n. Its rational output has $O(n \log(n+2))$ bits. A universal simulation for n steps, factorial calculation and exact integer arithmetic all have polynomial cost in the stated input lengths. Thus the family has uniform coefficient algorithms, without using a halting oracle.

For $|z| \leq 2$, in the halting case,

$$|F_e(z)| \leq \frac{3}{4} + \epsilon_N 2^N e^2 = \frac{3}{4} + 2^{-N-6} e^2 < 1.$$

On $[0,1]$, its values lie between $1/2$ and $5/8 + \epsilon_N e < 3/4$. Also

$$F_e'(x) = \frac{1}{8} + \epsilon_N e^x (N x^{N-1} + x^N) \leq \frac{1}{8} + e(N+1)2^{-2N-6} < \frac{1}{4}.$$

For the final bound use $e < 3$ and $(N+1)4^{-N} \leq 3/16$ for $N \geq 2$. The nonhalting case satisfies the same bounds directly. The contraction theorem now gives a unique $x_e \in [1/2, 3/4]$.

We verify the claimed complexity rather than infer it from existence. Nonnegativity of the coefficients and $F_e(2) < 1$ give the uniform truncation estimate

$$\sup_{|x| \leq 1} \left| F_e(x) - \sum_{n=0}^{m} c_{e,n} x^n \right| \leq 2^{-m-1} \sum_{n>m} 2^n c_{e,n} < 2^{-m-1}.$$

At a dyadic argument with $O(p)$ bits, the polynomial of degree $p+5$ can be evaluated by rational arithmetic with polynomial bit cost and then rounded to error at most 2^{-p-4}. Including the truncation error, the total evaluation error is less than $\eta = 2^{-p-3}$. Start at $1/2$ and perform $p+2$ such approximate iterations, retaining $O(p)$ bits at each step. The computed iterates remain in $[0,1]$ because the exact image is in $[1/2, 3/4]$ and $\eta \leq 1/16$ for $p \geq 1$. Their distance from x_e is at most

$$4^{-p-2} + \frac{\eta}{1 - 1/4} < 2^{-p}.$$

Only polynomially many rational operations on polynomially many bits are used; each coefficient is obtained by the preceding bounded simulation.

If e never halts, its fixed point solves $x = 1/2 + x/8$ and equals $4/7$. If e halts at N and its fixed point were algebraic, then

$$e^{x_e} = \frac{(7/8)x_e - 1/2}{\epsilon_N x_e^N}$$

would be algebraic. This contradicts the Hermite–Lindemann theorem, since $x_e \geq 1/2$ is a nonzero algebraic number. Thus the fixed point is transcendental in the halting case. A decision procedure for the two arithmetic alternatives, on this uniformly generated family of promised inputs, would decide the halting problem [277]. □

The representation in this theorem matters. The input is an effective program for rational Taylor coefficients, with fixed analytic estimates; it is not a supplied elementary expression announcing in advance the halting time N. Every particular function happens to have one of the elementary forms above, but there is no uniform way to select the correct form by deciding whether the machine will halt. This is a limitation on a general arithmetic decision procedure, not on approximation: all of the fixed points remain efficiently computable real numbers.

Corollary 57.5 (Finite jets, convex minimizers and expanding eigenvalues)**.** *The family of Theorem 57.4 has the following additional properties.*

1. *For every finite Taylor order and every positive tolerance there are two functions satisfying its uniform analytic bounds, agreeing to that order and within the tolerance in the uniform norm on $|z| \le 2$, whose fixed points are respectively rational and transcendental.*

2. *The entire functions*
$$V_e(x) = \frac{x^2}{2} - \int_0^x F_e(t)\,dt$$
have rational Taylor coefficients, $V_e(0) = 0$, $V_e'(0) = -1/2$, and $3/4 < V_e'' \le 7/8$ on $[0,1]$. Their unique minimizers on this interval have the same undecidable rational/transcendental dichotomy.

3. *The entire maps*
$$\mathcal{R}_e(u,v) = \big(F_e(u), (4+u)v\big)$$
have a unique fixed point in $[0,1]\times\mathbb{R}$. It is hyperbolic, with one eigenvalue in $[1/8, 1/4)$ and a simple expanding eigenvalue $\lambda_e \in [9/2, 19/4]$. These expanding eigenvalues are uniformly polynomial-time computable, but their rational/transcendental dichotomy is undecidable on the same input representation.

Proof. The perturbation $\epsilon_N z^N e^z$ has vanishing coefficients below degree N and satisfies
$$\sup_{|z|\le 2} |\epsilon_N z^N e^z| \le e^2 2^{-N-6} \longrightarrow 0.$$
Compare it with F_0 and choose N larger than the prescribed order. Both functions can be realized by machines in the construction. For the second assertion, $V_e' = x - F_e(x)$ and $V_e'' = 1 - F_e'$. The derivative changes sign across the unique zero x_e, which lies in the interior; the asserted uniform convexity proves that it is the unique minimizer. Integration preserves rational Taylor coefficients. For the third assertion a fixed point must have $u = x_e$ and $v = (4 + x_e)v$, hence $v = 0$. At that point the derivative matrix is
$$D\mathcal{R}_e(x_e, 0) = \begin{pmatrix} F_e'(x_e) & 0 \\ 0 & 4 + x_e \end{pmatrix}.$$
Its expanding eigenvalue is $\lambda_e = 4 + x_e$, so rational translation preserves the dichotomy and the computational bounds. □

The corollary gives a precise obstruction to the attempted Feigenbaum transfer: rational coefficient access, analytic control, a unique recovered object and a simple expanding eigenvalue, even with uniform quantitative gaps, cannot by themselves provide an arithmetic classification algorithm. The maps $\mathcal{R}_e$ are model maps, not Feigenbaum renormalization operators. The theorem therefore neither resolves δ nor rules out a proof using its specific renormalization identities. Such a proof would need additional arithmetic structure—for example an identity governed by a transcendence theorem or approximants with controlled algebraic heights—which neither a spectral enclosure nor an arbitrarily long digit calculation supplies. These model consequences form one obstruction result; they are not counted as additional named constants.

58 What has been recovered, and what would constitute a further advance

The completed classifications have a common shape: analytic recovery is supplemented by an arithmetic invariant that is computable or rigid. In the quantile theorem it is a finite cyclotomic degree bound; in the logarithmic-root theorem it is the algebraic zero certificate; in the Mahler families it is a finite rational functional equation. Removing that invariant destroys the inference to arithmetic type.

Earlier failed transfer	Recovered ingredient	Exact result here
Logarithmic logic to arbitrary real inversion	Baker splitting at algebraic inputs plus a rational derivative	All isolated roots and their arithmetic types are decided in Theorem 51.1.
Normalization of an inverse problem by a transcendental period	An algebraic correction and its full Galois orbit	Finite order bound $N \leq 8r^2$ in Theorem 53.1.
Finite algebraic manipulation to a connection constant	Finite monodromy under translation	The common logarithmic coordinate is eliminated in Theorem 55.1.
Finite-dimensional compactness to a computable search	Factorial coercivity in numerator degree	Every fixed positive rational Laplace kernel has a finite integer candidate reduction, Theorem 55.4.
Small positive moment determinants to small integer forms	Rank and the evaluation kernel	Every rational integer normalization obeys Theorem 56.1; raw zeta tails lose exponential gain.
Exact limiting observations to the arithmetic of a limit	Uniform integrability and arithmetic structure	Theorems 57.3 and 57.4 supply explicit countermodels to the unrestricted transfer.

A narrower question genuinely decided. For a given rational polynomial arcsine tilt, the question of which algebraic probability levels have algebraic quantiles has an exact terminating answer here. The same is true for every positive half-integral beta law and every integer-degree Student or Fisher law. These are full classifications of stated natural families. They are not merely examples of a number that was defined to be transcendental, and they do not rely on floating-point tests for algebraicity. They are also not fifteen separate historical conjectures.

The remaining novelty question. The underlying transcendence theorems are classical. Dhillon–Saha's 2025 work explicitly studies incomplete beta values by logarithmic methods in a parameter range containing our half-integral case; the available public slides confirm that overlap. The normalized inverse classification, its real-root argument, and the finite cyclotomic bound need a line-by-line comparison with the full published article and the wider inverse-special-function literature before a priority claim is made. The entropy and Mahler statements likewise include explicitly credited antecedents. A complete proof and a new historical contribution are different assessments.

A quantitatively meaningful next target. The Hankel obstruction gives a concrete design constraint: a positive perturbation of small rank cannot create the desired integer-polynomial decay unless its inverse-Gram evaluation determinant is correspondingly large. A next construction must escape that condition by a genuinely different family, such as mixed signed functionals or a growing arithmetic perturbation rank, and must preserve nonvanishing. Even then, the height criterion in Proposition 56.10 has to exclude every algebraic degree before

a transcendence conclusion follows. The externally reported recent irrationality constructions illustrate some escapes; they do not automatically furnish the missing transcendence bounds.

For γ and δ, growing denominator kernels and provable common-factor savings are outside the fixed-lattice obstruction, but no such successful saving is established here. For B_1 and K, a new theorem controlling the infinite arithmetic combination is needed, not just faster evaluation. For the gamma fifths, rank information must be replaced by a statement identifying an individual value. For δ_F, the special renormalization identities must contribute arithmetic information beyond hyperbolicity. These are the explicit unfinished mathematical steps in the fifteen attempts.

58.1 Verification record

The beta classification and polynomial-tilt extension were independently checked, including the conjugacy argument, the bound $8r^2$, and the asymmetric beta and Fisher exceptions. The logarithmic zero and binary entropy proofs were checked separately from their numerical examples. The Laguerre degree and coefficient bounds, the Mahler pole bounds and algebraic loci, the Christoffel normalization bounds, and the computable contraction and normal-Liouville constructions received independent written-proof reviews. Exact rational checks verify the logarithmic isolators and the small Hankel coefficient tables; those checks do not replace the all-parameter proofs.

This document is a single self-contained LaTeX source with an embedded bibliography. It uses no external figures, source inclusions, or bibliography database. Its cross-references and citations are checked during assembly, and the final PDF is compiled and inspected for layout. None of these checks establishes historical novelty or formal verification of a theorem by a proof assistant.

Part VIII
Arithmetic Certificates

Editorial integration note. The five-atom entropy construction remains valid; subsequent parts give a three-atom example for algebraic irrational bit entropy and a four-atom example for rational bit entropy with irrational probabilities. The full-degree mixed Catalan pencil here is not asserted to be positive: later parts construct and distinguish positive alternatives. Exact moment and degree costs are sharpened in Part IX.

Original part framing.

Reading and verification note. This is a standalone research manuscript: its proofs do not require access to the earlier 223-page or 447-page compilations. The results added here are claims with supplied mathematical proofs, not formal machine certificates or an assertion of journal-level acceptance. Exact computations and separate proof reviews test specific identities and hypotheses. Bibliographic priority is not established merely by proving a statement. The foundational inputs are credited to Baker, Gelfond–Schneider, Wüstholz, Nishioka, and Adamczewski–Faverjon. Previously proved project results are identified as such, rather than counted again. The current investigation and its source checks were completed on 16–17 October 2026.

59 What survives a failed transfer

The preceding investigation supplied the forward criterion for algebraic-level quantiles of polynomially tilted arcsine laws and the bound $N \leq 8r^2$ for possible cyclotomic orders. It also identified several invalid inferences: rapid convergence does not control integer denominators, linear independence does not exclude nonlinear relations, and finite observations do not automatically determine arithmetic type. Rather than repeat these failures, we ask for the extra structure that makes each transfer valid.

The present manuscript develops four main positive directions. First, cyclotomic necessity can be reversed: every exception pattern allowed by Galois conjugacy is constructible, and finite moment constraints do not obstruct it. Second, a logarithmic value with algebraic coefficients admits a finite norm certificate, which classifies entropy in every finite alphabet. Third, additive functional equations permit polynomial dependence to be reduced to a rational coboundary, turning a relation ideal into finite linear algebra. Fourth, an exact differential on an elliptic curve reduces the quantile problem to finitely many algebraic torsion tests.

Earlier available result	Additional result here
Forward quantile test and finite order bound	All admissible finite exception sets, their minimum realization degree, positive beta mixtures, and arbitrarily many exact moment constraints (Theorems 60.2, 61.1).
Binary entropy classification	An exact classifier for every finite algebraic-weight logarithmic expression in bits, with explicit nonbinary counterexamples (Theorems 62.2, 62.7).
Linear Mahler relation locus	The complete polynomial relation ideal and its exact transcendence degree, including arbitrary finite collections of integer dilations (Theorem 65.2, Corollary 65.3).
Unperturbed elliptic division	A finite exception classifier after adding a nonzero algebraic exact-differential correction (Theorem 63.2).
Fixed-denominator and rank-one obstructions	Actual reduced denominators for unrestricted odd Taylor schedules; growing-pole content; full-rank signed odd-zeta pencils (Theorems 66.1, 66.3, 67.2).

The point of the inverse construction. For an admissible set S of s probability levels, the minimum polynomial tilt degree is exactly $s+1$. If the first M moments must also stay fixed, our constructive degree bound becomes $2s+M+2$. This latter bound is not asserted to be optimal. The realization does not say that an arbitrary subset of rational levels is possible: complete real cyclotomic conjugacy blocks are necessary and sufficient.

The point of the complete ideal. A linear relation test cannot exclude, for example, a quadratic relation among numbers. Theorem 65.2 proves that every polynomial relation in its specified additive cyclotomic class is generated by finitely many computable affine linear ones. The upgrade therefore supplies information that the previous linear locus did not contain. The deep value and purity theorems used for this passage are existing results, not new inputs proved here.

What is not counted as a new result. The bound $N \le 8r^2$, classical elliptic division, the previously covered Student progressions, and the earlier transcendence proof for $\log \mathrm{Ein}(1)$ are not new closures. The known algebraic independence of sparse Lambert blocks is attributed to Bundschuh–Väänänen. Recent external work on $\zeta(5)$ and Catalan's constant is not assigned to this project. Our fifteen named-target attempts are recorded at the end, including the exact point where each still stops.

60 Inverse realization of arithmetic quantile patterns

The cyclotomic divisibility criterion answers whether a specified polynomial tilt has an algebraic quantile. Here we solve the inverse problem: which finite sets of algebraic probability levels can be exactly the levels with algebraic quantiles? Every set allowed by Galois conjugacy can be realized, at the smallest possible polynomial degree, by a strictly positive rational tilt arbitrarily close to the arcsine law. The realization can additionally be chosen as a positive rational mixture of familiar half-integral beta distributions. These are self-posed inverse questions, not resolutions of named historical conjectures. The logarithmic transcendence input is classical [244]; prior incomplete-beta applications are credited below [279, 280].

60.1 Notation and the necessary Galois constraint

Let μ_0 be the arcsine probability measure on $(0,1)$, with

$$d\mu_0(x) = \frac{dx}{\pi\sqrt{x(1-x)}}, \qquad F_0(x) = \frac{2}{\pi}\arcsin\sqrt{x}.$$

For a nonzero rational polynomial w nonnegative on $[0,1]$, put $c_w = \int w\,d\mu_0 \in \mathbb{Q}_{>0}$ and $d\mu_w = (w/c_w)d\mu_0$. The rationality of c_w follows from

$$\int x^j\,d\mu_0(x) = 4^{-j}\binom{2j}{j}.$$

Write F_w and Q_w for its CDF and quantile function. Since a nonzero polynomial has only finitely many zeros, F_w is strictly increasing and every interior quantile is unique. Define the *algebraic-level exception set*

$$\mathcal{E}(w) = \{q \in \overline{\mathbb{Q}} \cap (0,1) : Q_w(q) \in \overline{\mathbb{Q}}\}. \tag{60.1}$$

This definition makes no assertion about transcendental probability levels.

For each integer $N \geq 3$, define the finite primitive-angle block and its corresponding set of quantile locations by

$$\mathcal{C}_N = \{2j/N : 1 \leq j < N/2,\ \gcd(j,N) = 1\}, \tag{60.2}$$

$$\mathcal{X}_N = \{\sin^2(\pi j/N) : 1 \leq j < N/2,\ \gcd(j,N) = 1\}. \tag{60.3}$$

Each has $h_N = \varphi(N)/2$ elements. The sets for different N are disjoint: N is the order of $e^{i\pi q}$ for a level q, and the map $q \mapsto \sin^2(\pi q/2)$ is injective on $(0,1)$. Let

$$\Psi_N(X) = \prod_{x \in \mathcal{X}_N}(X - x) \in \mathbb{Q}[X]. \tag{60.4}$$

This is an irreducible monic polynomial of degree h_N. Indeed, if ζ is a primitive N-th root of unity, then $x = (2 - \zeta - \zeta^{-1})/4$ has exactly these conjugates, and $[\mathbb{Q}(\zeta + \zeta^{-1}) : \mathbb{Q}] = \varphi(N)/2$: ζ is nonreal and quadratic over this real field. The usual cyclotomic construction therefore computes Ψ_N exactly. In particular,

$$\Psi_3 = X - \tfrac{3}{4}, \quad \Psi_4 = X - \tfrac{1}{2}, \quad \Psi_6 = X - \tfrac{1}{4}, \quad \Psi_5 = X^2 - \tfrac{5}{4}X + \tfrac{5}{16}.$$

Use the rational differential operator

$$\mathcal{L}(P)(X) = \frac{(1-2X)P(X) + 2X(1-X)P'(X)}{2}. \tag{60.5}$$

For every polynomial P,

$$\frac{\mathcal{L}(P)(x)}{\sqrt{x(1-x)}} = \frac{d}{dx}\bigl(\sqrt{x(1-x)}P(x)\bigr), \qquad \int \mathcal{L}(P)\,d\mu_0 = 0. \tag{60.6}$$

The endpoint terms vanish in the last integral. If $P \neq 0$ has degree d and leading coefficient a_d, then $\mathcal{L}(P)$ has degree $d+1$ and leading coefficient $-(d+1)a_d$.

Lemma 60.1 (Arithmetic necessity and exact cyclotomic blocks)**.** *For nonconstant w, there is a unique nonzero polynomial $P_w \in \mathbb{Q}[X]$ such that*

$$\frac{w}{c_w} = 1 + \mathcal{L}(P_w), \qquad F_w(x) = F_0(x) + \frac{\sqrt{x(1-x)}P_w(x)}{\pi}. \tag{60.7}$$

It satisfies $\deg P_w = \deg w - 1$. *Moreover,*

$$\mathcal{E}(w) = \bigcup_{\substack{N\geq 3\\ \Psi_N \mid P_w}} \mathcal{C}_N. \tag{60.8}$$

This is a finite disjoint union, and its exact cardinality is

$$|\mathcal{E}(w)| = \sum_{\substack{N\geq 3\\ \Psi_N \mid P_w}} \frac{\varphi(N)}{2} \leq \deg P_w.$$

All algebraic irrational levels have transcendental quantiles.

Proof. Put $x = \sin^2\theta$ and expand $w(\sin^2\theta)$ as a rational linear combination of $\cos(2j\theta)$ for $0 \leq j \leq \deg(w)$. Its constant coefficient is c_w. Integrate and use $\sin(2j\theta) = \sin(2\theta)U_{j-1}(\cos(2\theta))$, where $U_{j-1} \in \mathbb{Z}[X]$. This gives the CDF formula with $P_w \in \mathbb{Q}[X]$. Differentiation gives the first formula. The kernel of $\mathcal{L}$ on polynomials is zero: (60.6) makes $\sqrt{x(1-x)}P(x)$ constant, and its endpoint limits force that constant to be zero. Nonconstancy of w then gives $P_w \neq 0$. The leading-coefficient calculation following (60.6) proves the degree claim.

Suppose that $q, x \in \overline{\mathbb{Q}} \cap (0,1)$ and $F_w(x) = q$. For the principal logarithms, set

$$z = 1 - 2x + 2i\sqrt{x(1-x)}, \qquad \log(-1) = i\pi.$$

The point z is algebraic on the upper unit semicircle, and (60.7) yields

$$\log z - q\log(-1) = -i\sqrt{x(1-x)}P_w(x).$$

The right side is algebraic. Baker's theorem says a nonzero linear form with algebraic coefficients in logarithms of nonzero algebraic numbers is transcendental [244]. Consequently $P_w(x) = 0$ and $z = e^{i\pi q}$. Gelfond–Schneider forces $q \in \mathbb{Q}$: an algebraic irrational exponent of -1 cannot have an algebraic value. Thus $x = \sin^2(\pi q/2)$. Conversely, for rational q, this algebraic x is the quantile exactly when $P_w(x) = 0$.

Let N be the order of $e^{i\pi q}$. The last equality holds if and only if the minimal polynomial Ψ_N divides P_w. Every root of this irreducible factor gives the corresponding level in $\mathcal{C}_N$. Distinct factors have disjoint roots, which proves (60.8) and the cardinality assertion. □

For direct comparison with the preceding divisibility test, if $d = \deg P_w$ then

$$A_{P_w}(Z) = (4Z)^d P_w\left(\frac{2 - Z - Z^{-1}}{4}\right) = \sum_{k=0}^{d} p_k(-1)^k(4Z)^{d-k}(Z-1)^{2k}.$$

This has degree $2d$ and nonzero constant coefficient $(-1)^d p_d$. For $N \geq 3$, $\Psi_N \mid P_w$ is equivalent to $\Phi_N \mid A_{P_w}$. The factors $N = 1, 2$ correspond to endpoints and are excluded. The inverse problem below uses the real factors Ψ_N to keep the degree count sharp.

60.2 Realization, the smallest possible degree, and closeness

Theorem 60.2 (Complete inverse realization with minimal degree)**.** *A finite set $S \subset (0,1)$ is equal to $\mathcal{E}(w)$ for a nonconstant rational polynomial $w \geq 0$ on $[0,1]$ if and only if*

$$S = \bigcup_{N\in\mathcal{N}} \mathcal{C}_N \tag{60.9}$$

for a finite set $\mathcal{N} \subset \{3, 4, \dots\}$. The empty set is allowed. Set $s = |S|$ and

$$H_S(X) = \prod_{N \in \mathcal{N}} \Psi_N(X), \qquad H_\varnothing = 1.$$

Then $\deg H_S = s$. For every sufficiently small nonzero rational ε, the explicit polynomial

$$w_\varepsilon = 1 + \varepsilon \mathcal{L}(H_S) \tag{60.10}$$

is strictly positive on $[0, 1]$, is already normalized by $c_{w_\varepsilon} = 1$, and satisfies

$$\mathcal{E}(w_\varepsilon) = S, \qquad \deg w_\varepsilon = s + 1.$$

Every nonconstant polynomial realization of S has degree at least $s + 1$. Thus the displayed degree is minimal, including $s = 0$. At minimal degree its correction polynomial is necessarily a nonzero rational multiple of H_S.

For every fixed integer $k \geq 0$, these realizations can be made arbitrarily close to the arcsine law in the $C^k[0, 1]$ norm of the density ratio w_ε, in total variation, in uniform CDF distance, and in uniform quantile distance.

Proof. Necessity is Lemma 60.1. Conversely, let S have (60.9). Since all roots of H_S are exactly the locations in the chosen disjoint cyclotomic orbits, its degree is s. The polynomial $\mathcal{L}(H_S)$ is nonzero. Choose a positive rational bound $B \geq \|\mathcal{L}(H_S)\|_\infty$; for example, one plus the sum of the absolute values of its rational coefficients suffices. Any rational ε with $0 < |\varepsilon| < 1/(2B)$ gives $w_\varepsilon > 1/2$. Equation (60.6) gives normalization exactly, and its integrated version gives correction $P = \varepsilon H_S$. Lemma 60.1 proves that its exception set is exactly S. The degree calculation gives $s + 1$.

For any other nonconstant realization, all the distinct factors of H_S divide its correction polynomial. Therefore $\deg P_w \geq s$ and $\deg w = \deg P_w + 1 \geq s + 1$. At equality the quotient P_w/H_S is a nonzero rational constant. For $S = \varnothing$, a constant correction gives degree one and no exceptions. A constant tilt cannot realize a finite S, because it is the arcsine law and has all rational interior levels as exceptions.

For quantitative closeness, set $L = \mathcal{L}(H_S)$. Directly,

$$\|w_\varepsilon - 1\|_{C^k} = |\varepsilon| \|L\|_{C^k}, \tag{60.11}$$

$$\|\mu_{w_\varepsilon} - \mu_0\|_{\mathrm{TV}} = \frac{|\varepsilon|}{2} \int |L| \, d\mu_0 \leq \frac{|\varepsilon|}{2} \|L\|_\infty, \tag{60.12}$$

$$\|F_{w_\varepsilon} - F_0\|_\infty \leq \frac{|\varepsilon|}{2\pi} \|H_S\|_\infty. \tag{60.13}$$

Here total variation is the supremum of the differences on measurable sets. The inverse arcsine CDF is $Q_0(q) = \sin^2(\pi q/2)$ and is $\pi/2$-Lipschitz on $[0, 1]$. For $x = Q_{w_\varepsilon}(q)$, the CDF estimate gives $|F_0(x) - q| \leq \|F_{w_\varepsilon} - F_0\|_\infty$. Applying Q_0 proves

$$\sup_{0 \leq q \leq 1} |Q_{w_\varepsilon}(q) - Q_0(q)| \leq \frac{|\varepsilon|}{4} \|H_S\|_\infty. \tag{60.14}$$

All these distances tend to zero with a sequence of nonzero rational $\varepsilon \to 0$. □

The normalized realizations of minimal degree admit an exact further description. Put $m_S = \min_{[0,1]} \mathcal{L}(H_S)$ and $M_S = \max_{[0,1]} \mathcal{L}(H_S)$. Then $m_S < 0 < M_S$, because this nonzero polynomial has zero arcsine mean. All normalized minimal-degree realizations are $1 + \varepsilon \mathcal{L}(H_S)$, with rational $\varepsilon \neq 0$. They are strictly positive exactly when

$$-\frac{1}{M_S} < \varepsilon < -\frac{1}{m_S};$$

nonnegativity allows the two endpoints. The extrema and hence the endpoints are effectively computable real algebraic numbers. This follows by evaluating the rational polynomial at endpoints and at the finitely many real roots of its derivative.

The theorem separates a genuine obstruction from an artificial one: a single level $2/5$ cannot occur alone, since it forces the other level $4/5$ in $\mathcal{C}_5$; but the complete pair can occur arbitrarily close to the arcsine distribution. The level $1/3$ can occur alone because $\mathcal{C}_6 = \{1/3\}$. More generally, an arbitrary proposed finite list is decided by computing the orders of $e^{i\pi q}$ and checking closure under the corresponding blocks. The bound on the number of exceptions is attained by every realization above: its correction has exactly s simple interior roots, and all of them give algebraic quantiles.

60.3 Realization inside positive rational beta mixtures

Theorem 60.3 (A positive mixture realization)**.** *Every finite set S in Theorem 60.2 can be realized by a convex mixture with strictly positive rational weights of the $s+2$ distributions*

$$\mathrm{Beta}(k+1/2, s+3/2-k), \qquad 0 \le k \le s+1.$$

The normalized tilt has the minimal degree $s+1$ and can be made arbitrarily close to the arcsine law in all the senses stated in Theorem 60.2. No minimality assertion is made about the number of mixture components.

Proof. Put $\ell = s+1$. The Bernstein polynomials $b_{k,\ell}(x) = \binom{\ell}{k} x^k (1-x)^{\ell-k}$ form a basis over $\mathbb{Q}$ for polynomials of degree at most ℓ, and $\sum_{k=0}^{\ell} b_{k,\ell} = 1$. Thus the constructed tilt has an exact expansion

$$w_\varepsilon(x) = \sum_{k=0}^{\ell} (1+\varepsilon a_k) b_{k,\ell}(x), \qquad a_k \in \mathbb{Q}.$$

Make $\varepsilon \neq 0$ rational sufficiently small that all $1+\varepsilon a_k$ are strictly positive. Define

$$K_{k,\ell-k} = \frac{\pi}{B(k+1/2, \ell-k+1/2)} = \frac{4^\ell k!(\ell-k)!\ell!}{(2k)!(2\ell-2k)!} \in \mathbb{Q}_{>0},$$

and

$$\lambda_k = (1+\varepsilon a_k) \frac{\binom{\ell}{k}}{K_{k,\ell-k}}.$$

These are positive rational numbers. The identity between densities is exactly

$$w_\varepsilon(x) \frac{1}{\pi\sqrt{x(1-x)}} = \sum_{k=0}^{\ell} \lambda_k \frac{x^{k-1/2}(1-x)^{\ell-k-1/2}}{B(k+1/2, \ell-k+1/2)}.$$

Integrating and using $c_{w_\varepsilon} = 1$ gives $\sum_k \lambda_k = 1$. The degree, exception set, and closeness were already proved. Beta normalization and the CDF convention are standard [281]. □

Example 60.4 (Algebraic exceptions created by mixing)*.* Take the following rational mixture, in the displayed order:

$$\frac{1}{512}(185, 41, 131, 155)$$

on

$$\mathrm{Beta}(1/2, 7/2), \quad \mathrm{Beta}(3/2, 5/2), \quad \mathrm{Beta}(5/2, 3/2), \quad \mathrm{Beta}(7/2, 1/2).$$

Its algebraic-level algebraic quantiles are exactly

$$Q(2/5) = \frac{5-\sqrt{5}}{8}, \qquad Q(4/5) = \frac{5+\sqrt{5}}{8}. \tag{60.15}$$

Each of the four component distributions has a transcendental quantile at both of these levels.

Proof. Use $P = \Psi_5 = X^2 - 5X/4 + 5/16$ and

$$w = 1 + \mathcal{L}(P) = \frac{37}{32} - \frac{35}{16}X + 5X^2 - 3X^3.$$

Its degree-three Bernstein coefficients are $(37/32, 41/96, 131/96, 31/32)$, all positive. The conversion in Theorem 60.3 gives exactly the four weights above. Lemma 60.1 gives the full exception set $\mathcal{C}_5 = \{2/5, 4/5\}$ and the two roots of Ψ_5 give (60.15).

For completeness, the four component correction polynomials, in the same order, are

$$2+\frac{4}{3}(1-X)+\frac{16}{15}(1-X)^2, \quad \frac{2}{3}(4X-1)(3-2X), \quad -\frac{2}{3}(3-4X)(1+2X), \quad -2-\frac{4}{3}X-\frac{16}{15}X^2.$$

The first and last have no interior zeros; the middle two have only $1/4$ and $3/4$, corresponding respectively to levels $1/3$ and $2/3$. Thus neither $2/5$ nor $4/5$ is an algebraic-quantile level of any component, by the same logarithmic criterion. This verifies the final assertion individually. □

60.4 An attempted global classification of the beta quarter exception

A separate question is whether the positive integer solutions of

$$I_{1/4}(m + 1/2, n + 1/2) = 1/3 \tag{60.16}$$

consist only of $(m, n) = (1, 2)$. The construction theorems do not settle this more rigid two-parameter question. We record an exact integer recurrence which isolates the remaining arithmetic issue; we do not infer a global classification from a finite search.

Let $N = m + n$ and define

$$E_{m,n} = \int_0^{\pi/6} \sin^{2m}\theta \cos^{2n}\theta \, d\theta - \frac{1}{3}\int_0^{\pi/2} \sin^{2m}\theta \cos^{2n}\theta \, d\theta, \qquad R_{m,n} = \frac{2N!4^N}{\sqrt{3}}E_{m,n}.$$

Then $R_{0,0} = 0$, and integration by parts gives the integer recurrences

$$R_{0,n} = 2(2n-1)R_{0,n-1} + (n-1)!3^{n-1}, \qquad n \geq 1, \tag{60.17}$$

$$R_{m,n} = 2(2m-1)R_{m-1,n} - (m+n-1)!3^n, \qquad m \geq 1. \tag{60.18}$$

To verify the second, differentiate $\sin^{2m-1}\theta \cos^{2n+1}\theta$. Its derivative is

$$(2m-1)\sin^{2m-2}\theta\cos^{2n}\theta - 2N\sin^{2m}\theta\cos^{2n}\theta.$$

The endpoint contribution at $\pi/6$ is $\sqrt{3}\,3^n/4^N$; the complete integral has zero endpoint contribution. This gives (60.18). For (60.17), differentiate $\sin\theta\cos^{2n-1}\theta$ instead, whose endpoint contribution is $\sqrt{3}\,3^{n-1}/4^n$. Thus every $R_{m,n}$ is an explicitly computed integer and (60.16) is equivalent to $R_{m,n} = 0$.

The normalized incomplete beta recurrences [281, Equations 8.17.20–8.17.21] show that the left side of (60.16) is strictly decreasing in m and strictly increasing in n. Therefore there is at most one solution in each fixed row or column. Direct calculation gives $R_{1,2} = 0$, and consequently this is the only solution with $m = 1$, or with $n = 2$. These statements do not exclude solutions with other pairs of indices. The missing step is a proof that (60.17)–(60.18) never vanishes again for positive indices, or a counterexample. No such proof or counterexample is claimed here.

60.5 Prior art and scope of the additional results

Baker's logarithmic value theorem and cyclotomic conjugacy are classical. Dhillon–Saha study incomplete beta values [279], and Saha's primary slides explicitly cover rational noninteger parameters whose sum is a positive integer using logarithmic linear forms [280, slide 4/4]. That covers the arithmetic method used for individual half-integral beta components above. The preceding paper in this project already proved the exact cyclotomic test for polynomial tilts; it is recalled here solely to prove the inverse theorems.

The additional statements are the complete converse realization, the exact minimal degree, arbitrary closeness, and realization inside positive rational beta mixtures. They form one connected inverse-realization theory, not a list of separately solved famous conjectures. A primary-source search did not identify an explicit prior formulation of this inverse problem, but the full published Dhillon–Saha theorem statements were not accessible in the legal open sources checked. No historical-priority claim is made.

61 Finite moments can conceal any admissible arithmetic pattern

The inverse realization theorem can be made compatible with any prescribed finite list of moments of the arcsine law. This is a stronger statement than small perturbation: the moment equalities below are exact. It identifies a specific limitation of transferring statistical identifiability to arithmetic identifiability.

Theorem 61.1 (Moment-preserving inverse realization)**.** *Let S be any finite union of the blocks $\mathcal{C}_N$ in (60.2), including the empty union, and put $s = |S|$. For every integer $M \geq 0$, every integer $k \geq 0$, and every $\eta > 0$, there exists a nonconstant $w \in \mathbb{Q}[X]$, strictly positive on $[0,1]$, such that*

$$\int w \, d\mu_0 = 1, \qquad \mathcal{E}(w) = S, \tag{61.1}$$

$$\int x^j w(x) \, d\mu_0(x) = 4^{-j} \binom{2j}{j} \qquad (1 \leq j \leq M), \tag{61.2}$$

$$\|w - 1\|_{C^k[0,1]} < \eta, \qquad \deg w \leq 2s + M + 2. \tag{61.3}$$

The construction is effective for rational positive η, and $w\mu_0$ can additionally be expressed as a strictly positive rational mixture of finitely many half-integral beta laws.

Proof. Let $H = H_S$ be as in Theorem 60.2. Consider the positive measure

$$d\nu(x) = H(x)^2 \sqrt{x(1-x)} \, dx.$$

Its moment matrices are positive definite: a nonzero polynomial cannot vanish on an interval, and the density is positive except at finitely many points. Its moments divided by π are rational, by the half-integral beta integral. Gram–Schmidt therefore gives monic orthogonal polynomials $R_j \in \mathbb{Q}[X]$ of every degree j. Consecutive polynomials R_M, R_{M+1} have no common complex root. Indeed, their three-term recurrence has nonzero, positive backward coefficient; a common root would propagate to $R_0 = 1$.

Put $D = 2s + M + 1$ and consider

$$B_t = H^2(R_{M+1} + tR_M), \qquad t \in \mathbb{Q}.$$

This polynomial has degree D for every t. We choose t so that its only factors Ψ_N are those already dividing H. There are only finitely many candidates: a factor requires $\varphi(N)/2 \le D$, which implies $N \le 8D^2$. For completeness, $\varphi(N)^2 \ge N/2$ follows by multiplying the contributions $p^{a-2}(p-1)^2$ of prime powers $p^a \| N$; the only factor less than 1 can be $1/2$ from $p^a = 2$. For any root α of a candidate Ψ_N not dividing H, the equation $R_{M+1}(\alpha)+tR_M(\alpha) = 0$ forbids at most one value of t, since the two orthogonal polynomials have no common root. Thus finitely many choices are forbidden. Enumerating rational t and using exact polynomial divisibility tests finds an allowed choice after finitely many steps.

Choose a sufficiently small nonzero rational ϵ and set

$$P = \epsilon B_t, \qquad w = 1 + \mathcal{L}(P).$$

The derivative identity (60.6) gives normalization; Lemma 60.1 gives the exact exception set S. For $1 \le j \le M$, integration by parts gives

$$\begin{aligned}\int x^j (w-1)\, d\mu_0 &= -\frac{j}{\pi}\int_0^1 x^{j-1}\sqrt{x(1-x)}P(x)\, dx \\ &= -\frac{j\epsilon}{\pi}\int_0^1 x^{j-1}(R_{M+1} + tR_M)\, d\nu = 0.\end{aligned}$$

Boundary terms vanish and orthogonality proves the last equality. Shrinking $|\epsilon|$ ensures positivity and the stated C^k bound; these can be enforced by rational coefficient bounds. The degree is $D+1$ by the leading coefficient calculation for $\mathcal{L}$. Finally express w in the Bernstein basis of degree $D+1$. The constant polynomial 1 has all Bernstein coefficients equal to 1, so all coefficients remain positive for small $|\epsilon|$. Multiplying these basis functions by $d\mu_0$ and normalizing gives the claimed rational beta mixture, as in Theorem 60.3. □

Example 61.2 (The mean does not determine median arithmetic). Set

$$P(x) = \frac{1}{16}\left(x^2 - \frac{5}{16}\right), \qquad w(x) = 1 + \frac{1}{16}\left(-3x^3 + \frac{5}{2}x^2 + \frac{5}{16}x - \frac{5}{32}\right).$$

The sum of the absolute coefficients of $w-1$ is $191/512 < 1$, so $w > 0$ on $[0,1]$. Its integral against μ_0 is 1. Moreover $\int xw\, d\mu_0 = 1/2$, because

$$\frac{\int_0^1 x^2\sqrt{x(1-x)}\, dx}{\int_0^1 \sqrt{x(1-x)}\, dx} = \frac{5}{16}.$$

The positive root $\sqrt{5}/4$ of P has the negative conjugate $-\sqrt{5}/4$. It cannot be a root of any Ψ_N, all of whose conjugates lie in $(0,1)$. Hence $\mathcal{E}(w)$ is empty. This law has the same algebraic mean as the arcsine law but a transcendental median; indeed every algebraic interior probability has a transcendental quantile.

Corollary 61.3 (No finite-moment classifier on this class)**.** *For each fixed M, the first M exact moments do not determine the algebraic-level exception set of a strictly positive rational polynomial tilt of the arcsine law. This remains true in every C^k neighborhood of the constant weight* 1 *and within positive rational finite beta mixtures.*

Proof. Apply Theorem 61.1 to two different admissible sets, for example $S = \varnothing$ and $S = \{1/2\} = \mathcal{C}_4$. The resulting laws have identical specified moments and different exception sets. □

The degree bound is a constructive upper bound, not a claim of minimality under the moment constraints. The result does not prevent classification from the full polynomial density, which is precisely what the cyclotomic algorithm accomplishes.

62 Exact arithmetic of finite entropy: a norm algorithm and counterexamples

The binary entropy phenomenon does not extend unchanged to larger alphabets. We give an exact algorithm for deciding the algebraicity of finite information quantities at algebraic data, then exhibit five algebraic probabilities whose Shannon entropy in bits is an algebraic irrational number. Even rational probabilities with rational entropy need not be dyadic. These are precise corrections to tempting extensions, not resolutions of historically named conjectures.

The relevant prior result is Blakaj–Wolf's Theorem 2 [282]: entropy in nats at algebraic spectra is zero or transcendental, whereas entropy in an algebraic base at rational spectra is rational or transcendental. Their second hypothesis is rationality of the spectrum, not algebraicity. Our example respects both statements. Boche–Schaefer–Poor prove the binary rational-input exception theorem [283, Appendix B, Theorem 4]. Our previous binary algebraic-input classification therefore cannot be extrapolated solely from its rational-input special case. The results below use classical Baker theory; priority of the explicit algorithm or examples beyond these checked primary sources is not certified.

62.1 A finite exact test for algebraic information values

Write $\mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}$. All logarithms in this subsection are real logarithms of positive numbers. Algebraic inputs are specified by defining polynomials and isolating data, so exact field operations, minimal polynomials, number-field norms, and rational linear algebra are effective.

Lemma 62.1 (Detecting a rational power of two)**.** *Let $A > 0$ be algebraic, let $d = [\mathbb{Q}(A) : \mathbb{Q}]$, and let $n_A = |N_{\mathbb{Q}(A)/\mathbb{Q}}(A)| \in \mathbb{Q}_{>0}$. Then*

$$A \in 2^{\mathbb{Q}} \quad \Longleftrightarrow \quad n_A = 2^m \text{ for some } m \in \mathbb{Z} \quad \text{and} \quad A^d = 2^m.$$

These conditions are decidable by finite exact algebraic arithmetic. When they hold, $\log_2 A = m/d$.

Proof. If $A = 2^{a/b}$ with $a \in \mathbb{Z}$ and $b \geq 1$, then every conjugate of A has modulus A, since its b-th power is 2^a. Consequently $n_A = A^d = 2^{ad/b}$. A positive rational number that is a rational power of 2 is an integral power of 2: clear the exponent's denominator and compare prime valuations. Thus $n_A = 2^m$ as required. Conversely, positivity and $A^d = 2^m$ give $A = 2^{m/d}$. To implement the test, reduce the rational number n_A to lowest terms, check that its numerator and denominator are powers of 2, and check the stated algebraic equality. This also covers $A = 1$. □

We will use the following standard coefficient-splitting consequence of Baker's theorem [231, 62]. If algebraic numbers $\theta_1, \ldots, \theta_s$ are linearly independent over $\mathbb{Q}$, and each L_j is a rational linear combination of real logarithms of positive algebraic numbers, then

$$\sum_j \theta_j L_j = 0 \quad \Longrightarrow \quad L_j = 0 \quad \text{for all } j.$$

Indeed, expand the L_j in a $\mathbb{Q}$-basis of the finitely many logarithms involved. Baker's theorem says that this basis remains linearly independent over $\overline{\mathbb{Q}}$; the remaining coefficient equations split by independence of the θ_j. This is an argument for correctness and does not require an algorithm for finding a basis of logarithms.

Theorem 62.2 (Exact arithmetic classification)**.** *Given finitely many $\lambda_i \in \mathbb{A}$ and $\alpha_i \in \mathbb{A}_{>0}$, there is a terminating algorithm which determines whether*

$$V = \frac{\sum_i \lambda_i \log \alpha_i}{\log 2} \tag{62.1}$$

is algebraic or transcendental. In the algebraic case it returns V as an exact algebraic number and determines whether it is rational.

More explicitly, choose a $\mathbb{Q}$-basis $\theta_1, \ldots, \theta_s$ of the span of the λ_i, write $\lambda_i = \sum_j r_{ji}\theta_j$ with $r_{ji} \in \mathbb{Q}$, and choose $D \geq 1$ with $Dr_{ji} \in \mathbb{Z}$. Put

$$A_j = \prod_i \alpha_i^{Dr_{ji}}. \tag{62.2}$$

Then V is algebraic if and only if every A_j is a rational power of 2. If the norm test of Lemma 62.1 returns $\log_2 A_j = m_j/d_j$, then

$$V = \frac{1}{D} \sum_j \theta_j \frac{m_j}{d_j}. \tag{62.3}$$

If all coefficients are zero, the algorithm simply returns $V = 0$.

Proof. The sufficiency and formula follow from $DV \log 2 = \sum_j \theta_j \log A_j$. Suppose conversely that V is algebraic. First V belongs to the $\mathbb{Q}$-span of the θ_j. Otherwise these numbers together with V are independent over $\mathbb{Q}$, and coefficient splitting in this equality forces $D \log 2 = 0$, a contradiction. Write $V = \sum_j s_j\theta_j$, $s_j \in \mathbb{Q}$. Coefficient splitting now gives

$$\log A_j - Ds_j \log 2 = 0 \qquad (1 \leq j \leq s).$$

Hence every $A_j \in 2^{\mathbb{Q}}$.

All the θ_j, rational coefficients, and products A_j are computed in a common number field. The norm test uses only exact algebraic operations and finitely many checks on rational integers; no search for unknown logarithmic relations or unbounded approximation is needed. Formula (62.3) gives exact algebraic output. Testing whether its minimal polynomial has degree one decides rationality. If any norm test fails, the proved equivalence certifies transcendence. No running-time bound is asserted. □

The equivalent multiplicative-lattice criterion is also explicit. For fixed positive algebraic $\alpha_1, \ldots, \alpha_n$, set

$$\Lambda = \{(k_1, \ldots, k_n, k_0) \in \mathbb{Z}^{n+1} : \alpha_1^{k_1} \cdots \alpha_n^{k_n} 2^{k_0} = 1\}.$$

Baker's theorem gives, for any proposed algebraic v,

$$V = v \quad \Longleftrightarrow \quad (\lambda_1, \ldots, \lambda_n, -v) \in \operatorname{span}_{\overline{\mathbb{Q}}} \Lambda.$$

To verify this equivalence, the rational kernel of the row of real logarithms is exactly $\mathbb{Q}\Lambda$ after clearing denominators, and Baker says its algebraic kernel is its scalar extension. The norm algorithm avoids having to compute the whole lattice when only the arithmetic nature of the single value (62.1) is sought.

Corollary 62.3 (Finite information quantities)**.** *At exact algebraic finite probability data, algebraicity or transcendence in bits is decidable for Shannon entropy, conditional entropy, mutual information, and finite Kullback–Leibler divergence. The algebraic value, when present, is exactly recoverable. The same holds for any specified algebraic linear combination of these quantities.*

Proof. Delete zero-mass terms. Positive joint and marginal probabilities are positive algebraic numbers. Each indicated quantity is a finite sum of algebraic coefficients times logarithms of positive algebraic probabilities or their ratios, divided by $\log 2$. For relative entropy, a positive first-distribution mass paired with zero second-distribution mass is detected exactly and gives $+\infty$; otherwise the finite-value theorem applies. Thus the support cases, as well as all hypotheses of Theorem 62.2, are decidable. □

This is a pointwise arithmetic decision procedure with an exact algebraic input description. It does not make a positive-dimensional entropy level set semialgebraic, and does not contradict Blakaj–Wolf's non-semialgebraicity theorem.

62.2 Rational probabilities: the full valuation criterion

Proposition 62.4. *Let $p_i = a_i/N > 0$, where $a_i \in \mathbb{Z}_{>0}$ and $\sum_i a_i = N$. No coprimality hypothesis on this representation is needed. Then $H_2(p) = -\sum_i p_i \log_2 p_i$ is algebraic if and only if, for every odd prime ℓ,*

$$N v_\ell(N) = \sum_i a_i v_\ell(a_i). \tag{62.4}$$

Only primes dividing $N \prod_i a_i$ need be tested. When these conditions hold, the entropy is the rational number

$$H_2(p) = v_2(N) - \frac{1}{N} \sum_i a_i v_2(a_i). \tag{62.5}$$

If one condition fails, the entropy is transcendental.

Proof. The identity

$$N H_2(p) \log 2 = \log \left(\frac{N^N}{\prod_i a_i^{a_i}} \right)$$

together with Theorem 62.2 and Lemma 62.1, here applied to a positive rational number, shows that algebraicity is equivalent to the rational number in parentheses being an integral power of 2. Prime valuations give exactly (62.4) and (62.5). This refines the known rational-or-transcendental dichotomy by an explicit exact test; the dichotomy itself is already in [282]. □

Example 62.5 (Odd-prime cancellation with four outcomes). The rational distribution

$$p = \left(\frac{3}{8}, \frac{1}{4}, \frac{1}{3}, \frac{1}{24} \right)$$

has entropy $7/4$ bits, despite its non-dyadic probabilities. In the denominator-24 representation its numerators are $(9, 6, 8, 1)$. The only odd prime to check is 3, and

$$24 v_3(24) = 24 = 9 v_3(9) + 6 v_3(6).$$

The 2-adic formula gives $3 - (6 + 24)/24 = 7/4$. Thus neither cancellation of odd primes nor rational entropy requires each probability to be a negative integral power of 2. We make no claim that four is the least possible alphabet size for such a rational counterexample.

62.3 A five-outcome algebraic irrational entropy

The construction uses a double root of a positive Laurent polynomial. This mechanism is useful independently of the particular example.

Proposition 62.6 (Double-root construction)**.** *Let $k_i, m_i \in \mathbb{Z}$ and $u > 0$ be algebraic. Suppose*

$$F(t) = \sum_{i=1}^{n} 2^{-k_i} t^{m_i}, \qquad F(u) = 1, \qquad F'(u) = 0.$$

Then $p_i = 2^{-k_i} u^{m_i}$ form a positive algebraic probability vector, and

$$H_2(p) = \sum_i k_i p_i. \tag{62.6}$$

In particular this entropy is algebraic. If every coefficient of a positive Laurent polynomial F is dyadic rational, its monomials can be split into finitely many terms of the displayed form.

Proof. Normalization is $F(u) = 1$, and differentiating gives $\sum_i m_i p_i = uF'(u) = 0$. Therefore

$$-\sum_i p_i \log_2 p_i = \sum_i k_i p_i - \log_2 u \sum_i m_i p_i = \sum_i k_i p_i.$$

Every positive dyadic rational is a finite sum of integral powers of 2, giving the last assertion. Notice that the double-root identity cancels the coefficient of $\log u$ exactly; a numerical critical point would not supply this arithmetic certificate. □

Theorem 62.7 (Explicit algebraic irrational entropy)**.** *Put $u = 1 + \sqrt{3}$. The five probabilities*

$$p = \left(\frac{u^2}{16}, \frac{1}{2u}, \frac{2}{u^2}, \frac{1}{4u^2}, \frac{1}{u^3} \right) \tag{62.7}$$

are positive algebraic irrational numbers with sum 1, and

$$H_2(p) = \frac{6\sqrt{3} - 3}{4}. \tag{62.8}$$

Thus the rational-or-transcendental dichotomy fails for finite algebraic probability vectors in base 2.

Proof. The polynomial identity

$$(t^2 - 2t - 2)^2 (t + 4) = t^5 - 16t^3 + 8t^2 + 36t + 16 \tag{62.9}$$

shows that

$$F(t) = \frac{t^2}{16} + \frac{1}{2t} + \frac{2}{t^2} + \frac{1}{4t^2} + \frac{1}{t^3}$$

satisfies $F(u) = 1$ and $F'(u) = 0$; indeed $F(t) - 1 = (t^2 - 2t - 2)^2 (t + 4)/(16t^3)$. Apply Proposition 62.6 with

$$(k_i) = (4, 1, -1, 2, 0), \qquad (m_i) = (2, -1, -2, -2, -3).$$

The resulting entropy is

$$4p_1 + p_2 - p_3 + 2p_4 = \frac{u^2}{4} + \frac{1}{2u} - \frac{3}{2u^2} = \frac{6u - 9}{4} = \frac{6\sqrt{3} - 3}{4},$$

where $u^2 = 2u + 2$ was used. For a separate normalization check, writing $s = \sqrt{3}$ gives

$$p = \left(\frac{2+s}{8}, \frac{s-1}{4}, 2-s, \frac{2-s}{8}, \frac{3s-5}{4}\right).$$

Every entry is positive and irrational, and their rational and s coefficients sum to 1 and 0, respectively. The entropy in (62.8) is algebraic irrational since s is irrational. □

Corollary 62.8 (Rational entropy without rational probabilities)**.** *Replace the second and third atoms of* (62.7) *by four equal subatoms each, leaving the other three atoms intact. The resulting* 11*-outcome probability vector has only algebraic irrational entries and has rational entropy* 11/4 *bits.*

Proof. Splitting an atom of mass a into four equal atoms increases the entropy by $2a$. Therefore the new entropy is

$$\frac{6\sqrt{3}-3}{4} + 2\frac{\sqrt{3}-1}{4} + 2(2-\sqrt{3}) = \frac{11}{4}.$$

Division by 4 preserves irrationality of the affected entries. □

These examples answer two distinct extrapolation questions negatively: algebraic entropy does not force rational entropy, and rational entropy does not force rational probabilities. Example 62.5 separately shows that rational probabilities with rational entropy need not be dyadic. No smallest-alphabet assertion is needed for any conclusion. The same examples are spectra of diagonal algebraic density matrices, so they also apply to von Neumann entropy in bits.

62.4 Three named constants: exact outcomes and limits

$e + \pi$ and $e\pi$. These remain unresolved here, including their irrationality. Waldschmidt's primary notes record the classical gap [235]; a 2026 preprint still treats $e + \pi$ as an open irrationality target [237]. The norm algorithm requires algebraic logarithm bases and algebraic coefficients. Assuming $e + \pi = a \in \overline{\mathbb{Q}}$ gives $\log(a - \pi) = 1$, whose base is e, not an algebraic number. Assuming $e\pi = a$ gives $\log(a/\pi) = 1$ with the same defect. Replacing π by a branch of $\log(-1)/i$ introduces nested logarithms and does not repair this hypothesis. The new entropy construction supplies no theorem about such expressions.

There is also a precise obstruction to transferring an E-function value theorem: $e = e^1$ is an algebraic-point E-value, but a joint E-function system supplying π at an algebraic point has not been provided. Writing $\sin \pi = 0$ instead evaluates at the unknown transcendental point π, outside the algebraic-point specialization theorem. The elementary disjunction that at least one of $e + \pi, e\pi$ is transcendental follows from the quadratic with these two coefficients and is classical; it is not an individual closure.

$\log \mathrm{Ein}(1)$. This constant is already proved transcendental in the earlier E-function section of this project, so it is not counted as a new result of this turn. The applicable input is the stronger established statement

$$\mathrm{Ein}(1) \text{ is transcendental over } \mathcal{F}_{\exp} = \overline{\mathbb{Q}}(e^a : a \in \overline{\mathbb{Q}}).$$

For completeness, its mechanism is the Kolchin–Ostrowski primitive criterion followed by the Siegel–Shidlovskii specialization theorem [285, 284]. For any finite set of algebraic exponents, choose a rational basis $b_1, \ldots, b_r$ of their span, rescaled so that the exponents and 1 are integral combinations. In the abstract field $\mathbb{C}(z, w_1, \ldots, w_r)$, $w_i = e^{b_i z}$, the derivative $(1 - e^{-z})/z$ has

a nonzero simple pole at the formal divisor $z = 0$. A rational primitive cannot have such a derivative: a pole of order m gives derivative order $m + 1$, while a regular function has regular derivative, with the w-derivation unable to cancel the leading term. The primitive criterion gives transcendence of $\mathrm{Ein}(z)$ over that exponential field. The vector consisting of 1, these exponentials, e^{-z}, and $\mathrm{Ein}(z)$ is an E-function solution of a homogeneous system whose only finite possible singularity is 0. Specialization at 1 therefore preserves the transcendence degree [284, Theorem 1.1], proving the stated finite-field independence. A relation over $\mathcal{F}_{\exp}$ uses only finitely many exponents, so the whole-field claim follows.

Since $\mathrm{Ein}(1) > 0$, its real logarithm is defined. If it were algebraic b, then $\mathrm{Ein}(1) = e^b$ would belong to $\mathcal{F}_{\exp}$, a contradiction. This proof genuinely applies to this constant, but neither creates a new proof of algebraic independence of its logarithms nor solves $e + \pi$ or $e\pi$.

63 From roots of unity to elliptic torsion: another finite classifier

The cyclotomic test has a genus-one counterpart. Here a finite torsion test replaces cyclotomic divisibility. The transcendence input is the classical analytic subgroup theorem, not a new transcendence theorem. We give the full reduction and a terminating algorithm for a specified nonzero correction. Earlier project manuscripts already treated unperturbed elliptic division; the additional object here is the exact algebraic correction and its finite exceptional set.

Lemma 63.1 (Two consequences of the analytic subgroup theorem)**.** *Let E be an elliptic curve over $\overline{\mathbb{Q}}$ and use a nonzero invariant differential defined over $\overline{\mathbb{Q}}$ to identify its Lie algebra with $\mathbb{C}$.*

(i) If $u_1, \dots, u_m$ are elliptic logarithms of algebraic points and $c_1, \dots, c_m$ are algebraic, then $\sum c_j u_j$ is either zero or transcendental.

(ii) If E is real, u, v are nonzero real elliptic logarithms of algebraic points, and u/v is algebraic, then $u/v \in \mathbb{Q}$.

Proof. We use the logarithmic form of the analytic subgroup theorem: the smallest $\overline{\mathbb{Q}}$-linear Lie subspace containing a logarithm of an algebraic point is the Lie algebra of an algebraic subgroup [286, 287]. For (i), suppose $a = \sum c_j u_j$ is nonzero and algebraic, and apply the theorem in $\mathbb{G}_a \times E^m$ to $(a, u_1, \dots, u_m)$, in the hyperplane $t - \sum c_j z_j = 0$. A connected algebraic subgroup of $\mathbb{G}_a \times E^m$ that projects nontrivially to $\mathbb{G}_a$ contains $\mathbb{G}_a \times \{0\}$. Indeed, its affine part maps trivially to the abelian variety, whereas a homomorphism from an abelian variety to $\mathbb{G}_a$ is zero. This contradicts containment of its Lie algebra in the displayed hyperplane.

For (ii), put $q = u/v$. The analytic subgroup theorem in E^2 applied to (u, v) places it in the Lie algebra of a one-dimensional algebraic subgroup lying on $z_1 = qz_2$. Both projections are nonzero isogenies, so their tangent ratio belongs to $\mathrm{End}^0(E)$. In characteristic zero this algebra is $\mathbb{Q}$ or an imaginary quadratic field; its intersection with $\mathbb{R}$ is $\mathbb{Q}$. Thus the real number q is rational. This is the usual elliptic form of the Gelfond–Schneider phenomenon. □

Consider the smooth projective curve

$$E:\ y^2 = 1 - x^4, \qquad O = (-1, 0),$$

with the indicated point as group identity. Its real locus is the oval formed by the upper and lower branches for $-1 \le x \le 1$. The differential dx/y is regular and nonzero. Define

$$I(x) = \int_{-1}^{x} \frac{dt}{\sqrt{1 - t^4}}, \qquad \Omega = I(1) > 0.$$

On the upper branch, the elliptic logarithm of (x, y) is $I(x)$. The full real period is 2Ω and $(1, 0)$ is a point of order two: the involution $(x, y) \mapsto (x, -y)$ is negation with this choice of origin.

Let $P \in \mathbb{Q}[X] \setminus \{0\}$ and choose any nonzero rational ϵ sufficiently small that

$$w(x) = 1 + \epsilon\big((1 - x^4)P'(x) - 2x^3P(x)\big) > 0 \qquad (-1 \le x \le 1). \tag{63.1}$$

Such a choice is effective by a coefficient bound. The probability density and its CDF are

$$f(x) = \frac{w(x)}{\Omega\sqrt{1 - x^4}}, \qquad F(x) = \frac{I(x) + \epsilon\sqrt{1 - x^4}P(x)}{\Omega}. \tag{63.2}$$

Differentiation proves this formula, and the correction vanishes at both endpoints. Let $Q = F^{-1}$ on $(0, 1)$.

Theorem 63.2 (Effective elliptic exception classifier)**.** *For algebraic $q \in (0, 1)$, the quantile $Q(q)$ is algebraic if and only if there is a real root $x \in (-1, 1)$ of P for which*

$$T_x = (x, \sqrt{1 - x^4}) \in E(\overline{\mathbb{Q}}) \text{ is torsion}, \qquad q = I(x)/\Omega.$$

Every such q is rational. The entire list of exceptions, including their exact rational levels and algebraic quantiles, is computable by a terminating algorithm. There are at most $\deg P$ *exceptions; for nonzero constant P there are none.*

Proof. If $x = Q(q)$ and x, q are algebraic, (63.2) gives

$$I(x) - q\Omega = -\epsilon\sqrt{1 - x^4}P(x).$$

Both $I(x)$ and Ω are elliptic logarithms of algebraic points. The right side is algebraic, so Lemma 63.1(i) forces it to vanish. Hence $P(x) = 0$ and $I(x) = q\Omega$. Part (ii) of the lemma gives $q \in \mathbb{Q}$. Since 2Ω is a period, this makes T_x torsion. Conversely, a torsion point on the upper real arc has $I(x)/(2\Omega) \in \mathbb{Q}$. If $P(x) = 0$, the CDF formula gives the stated algebraic quantile. Strict increase of F ensures that different roots give different probability levels.

It remains to justify effectiveness, rather than merely recognize torsion by an unbounded search. Isolate the finitely many real roots of P in $(-1, 1)$. For each one, work in the explicit number field $K = \mathbb{Q}(x, \sqrt{1 - x^4})$ and use any algebraic Weierstrass model of (E, O). Choose two primes of good reduction of K with distinct residue characteristics p and ℓ. They can be found by enumeration since there are only finitely many primes of bad reduction. Compute the finite integers

$$A = \#E(k_{\mathfrak{p}}), \qquad B = \#E(k_{\mathfrak{l}}), \qquad D = AB.$$

If T_x has finite order n, its prime-to-p part injects under the first reduction, and its p-primary part injects under the second reduction. Consequently $n \mid D$. The reduction fact follows from the formal group: multiplication by an integer prime to the residue characteristic is invertible on the reduction kernel. Thus

$$T_x \text{ is torsion} \quad \Longleftrightarrow [D]T_x = O.$$

This is a finite exact algebraic computation. If it succeeds, factor D and test its divisors to find the exact order n. The real arc coordinate is then $I(x)/(2\Omega) = j/n$ with $0 < j < n/2$ and $\gcd(j, n) = 1$. Certified real approximations to the convergent integrals separate these finitely many candidates and recover the unique integer j, yielding the exact answer $q = 2j/n$. The approximation is used only to select from an already proved finite rational list. This completes the algorithm. □

Example 63.3. With $P = 1$ and sufficiently small nonzero rational ϵ, every algebraic-level quantile is transcendental. With $P = x$, the only algebraic-level algebraic quantile is $Q(1/2) = 0$. Indeed $I(0) = \Omega/2$ by symmetry, and P has no other root. These statements concern the perturbed laws in (63.2); for the unperturbed law $\epsilon = 0$, every rational-level quantile is algebraic by elliptic division.

This transfer removes the special dependence on roots of unity: the finite candidates are now zeros of a correction intersected with an elliptic torsion subgroup. It does not decide arbitrary elliptic constants or arbitrary real numbers. Nor is a polynomial complexity bound for the algorithm asserted.

64 Even-power rational densities and arithmetic transfer

We give a complete arithmetic classification for algebraic-probability quantiles of the rational densities proportional to $1/(1 + x^{2m})$. The mechanism is a sign-definite logarithmic part of the primitive, separated from its argument part by Baker's theorem. This is a different family from the earlier Student densities. No global priority claim is made for this application. We explicitly record the overlap of a related fractional Student calculation with the supplied earlier paper, then give three separate attempts on named constants that remain unresolved.

64.1 Separating modulus and phase in a rational primitive

We use Baker's theorem that logarithms of nonzero algebraic numbers which are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$; arbitrary determinations of the logarithms are allowed [294].

Lemma 64.1 (Real-logarithm and phase separation)**.** *Suppose $b_j > 0$ are algebraic, $\ell_k \in i\mathbb{R}$ are logarithms of nonzero algebraic numbers, and a_j, c_k are real algebraic numbers. If*

$$B + A = 0, \qquad B = \sum_j a_j \log b_j, \qquad A = \frac{1}{i} \sum_k c_k \ell_k,$$

then $B = A = 0$.

Proof. Choose a $\mathbb{Q}$-basis from each of the two finite lists $\{\log b_j\}$ and $\{\ell_k\}$, omitting zero elements. The union of these bases is $\mathbb{Q}$-linearly independent: its real and imaginary parts must vanish separately. All its members are logarithms of algebraic numbers. Baker's theorem makes the union $\overline{\mathbb{Q}}$-linearly independent as well. Expressing the hypothesized relation in that union forces every coefficient to vanish; in particular its two separate sums vanish. □

Theorem 64.2 (Algebraic quantiles of even-power rational laws)**.** *For an integer $m \geq 2$, let $Q_m(q)$ be the q-quantile of the probability density*

$$f_m(x) = \frac{m \sin(\pi/(2m))}{\pi} \frac{1}{1 + x^{2m}}, \qquad x \in \mathbb{R}.$$

For every algebraic $q \in (0, 1)$,

$$Q_m(q) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad q = \frac{1}{2}.$$

The algebraic quantile in that case is 0. Thus every noncentral algebraic-probability quantile is transcendental, including all noncentral rational-probability quantiles.

Proof. Write $Z_m = \int_{\mathbb{R}}(1+t^{2m})^{-1}dt = \pi\csc(\pi/(2m))/m$, by the substitution $u = t^{2m}$ and Euler's beta integral. Put $J_m(x) = \int_0^x (1+t^{2m})^{-1}dt$. The quantile equation is $J_m(x) = (q-1/2)Z_m$.

The simple poles ζ of the integrand satisfy $\zeta^{2m} = -1$. Their residues are $-\zeta/(2m)$. None lies on the real axis. Define $L_\zeta(x)$ as the continuous logarithm of $(x-\zeta)/(-\zeta)$ along the real segment from 0 to x, starting with value 0. Partial fractions, integrated with this normalization, give

$$J_m(x) = -\frac{1}{2m}\sum_{\zeta^{2m}=-1} \zeta L_\zeta(x).$$

For an upper-half-plane pole $\zeta = c + is$, $s > 0$, put $D_c(x) = x^2 - 2cx + 1 > 0$ and write $L_\zeta(x) = \frac{1}{2}\log D_c(x) + i\theta_\zeta(x)$. The lower pole is its complex conjugate, with conjugate logarithm. Their joint contribution is

$$-\frac{c}{2m}\log D_c(x) + \frac{s}{m}\theta_\zeta(x).$$

Consequently $J_m(x) = B_m(x) + A_m(x)$, where A_m is the sum of the argument terms, and pairing the upper poles with real parts c and $-c$ gives

$$B_m(x) = \sum_{\substack{\zeta^{2m}=-1 \\ \operatorname{Im}\zeta>0,\ \operatorname{Re}\zeta=c>0}} \frac{c}{2m}\log\frac{x^2+2cx+1}{x^2-2cx+1}. \tag{64.1}$$

An upper pole of real part zero contributes nothing to this sum. Since $m \geq 2$, at least one pole in the displayed sum exists. Every summand is positive for $x > 0$ and negative for $x < 0$. Therefore $B_m(x) = 0$ if and only if $x = 0$.

Now suppose that x and q are algebraic. All the positive numbers $D_c(x)$ are algebraic. Furthermore

$$\exp(i\theta_\zeta(x)) = \frac{(x-\zeta)/(-\zeta)}{\sqrt{D_c(x)}}$$

is algebraic and has modulus one, so $i\theta_\zeta(x)$ is a purely imaginary logarithm of an algebraic number. Since Z_m/π is real algebraic and $i\pi$ is a logarithm of -1, the quantile equation is a relation of exactly the type in Lemma 64.1: its real-logarithm part is $B_m(x)$, while $A_m(x) - (q-1/2)Z_m$ is an algebraic linear combination of purely imaginary logarithms divided by i. The lemma forces $B_m(x) = 0$. Thus $x = 0$ and $q = 1/2$. The converse follows from symmetry. □

For $m = 1$ the modulus part disappears: this is the Cauchy law, whose rational-probability quantiles are algebraic. The strict sign in (64.1) is the additional obstruction when $m \geq 2$. In particular the quartic density $\sqrt{2}/(\pi(1+x^4))$ has transcendental lower and upper quartiles. The theorem is a positive, uniform classification, not an inference from numerical approximations.

64.2 A recovered Student classification, not an additional discovery

The same investigation recovers the following result already contained in the supplied arc-length sequel [295]. For $\nu_0 \in \{1/3, 1/2, 2/3\}$ and algebraic $q \in (0,1)$, a Student quantile with ν_0 degrees of freedom is algebraic exactly when q is rational; with $\nu_0 + 2n$ degrees of freedom, $n \geq 1$, it is algebraic exactly when $q = 1/2$. We do not count either assertion as new.

Here is the short mechanism, to identify the overlap precisely. Set $a = (\nu+1)/2$, $I_a(s) = \int_0^s (1+t^2)^{-a}dt$ and $L_a = I_a(+\infty)$. The three starting exponents $2/3, 3/4, 5/6$ are first-kind elliptic integrals, by the respective substitutions

$$(X,Y) = ((1+s^2)^{1/3}, s), \quad s = \frac{2t}{1-t^2}, \quad (u,v) = ((1+s^2)^{-1/3}, s/\sqrt{1+s^2}),$$

on $Y^2 = X^3 - 1$, $y^2 = 1 - t^4$, and $v^2 = 1 - u^3$. Real algebraic fractions of these elliptic integrals are rational, by Schneider's theorem [288, 289]. For every increment one has

$$I_{a+1}(s) = \frac{s}{2a(1+s^2)^a} + \frac{2a-1}{2a} I_a(s).$$

Iteration gives $I_{a+n} = C_n I_a + D_n$, $L_{a+n} = C_n L_a$, where $C_n > 0$ is rational and $D_n(s)$ is nonzero algebraic of the sign of s at nonzero algebraic s. An algebraic-probability quantile would make $I_a(s) - (2q-1)L_a = -D_n(s)/C_n$. A linear form over the algebraic numbers in elliptic logarithms cannot be nonzero algebraic, by the analytic subgroup theorem [290]; hence $s = 0$. This is precisely the elliptic-division and exact-differential mechanism of the earlier sequel's elliptic-arc theorem and algebraic-parameter beta theorem (its source labels `dd-elliptic-alg` and `bs-algebraic-ratios`).

64.3 Three named constants: different transfers and their exact stopping points

The following attempts do not establish irrationality of Khinchin's K, the period-doubling Feigenbaum δ, or the Glaisher–Kinkelin A. They test whether the finite-dimensional arithmetic structures just used are actually present in natural representations of those constants.

Khinchin's constant: an algebraic-integrand test. The Gauss product formula [270] gives

$$(\log 2)\log K = \sum_{k\ge 1} \log k \log \frac{(k+1)^2}{k(k+2)}.$$

One can turn this into a positive Stieltjes integral, but its density does not become an algebraic differential.

Proposition 64.3. *The meromorphic function*

$$H(z) = \sum_{k=2}^{\infty} \frac{\log(k/(k-1))}{k+z}$$

satisfies $\int_0^1 H(t)\,dt = (\log 2)\log K$ *and is not algebraic over* $\mathbb{C}(z)$.

Proof. The series converges locally uniformly away from the negative integers $-2, -3, \ldots$, since its tail is $O(k^{-2})$ on every compact set. It has a simple pole at each $-k$, with nonzero residue $\log(k/(k-1))$. On $[0,1]$ summation by parts yields

$$H(t) = \sum_{k\ge 1} \log k \left(\frac{1}{k+t} - \frac{1}{k+1+t} \right).$$

The discarded boundary term is $O((\log k)/k)$. Integration, or Tonelli's theorem on the nonnegative summands, gives the claimed formula. An algebraic function over $\mathbb{C}(z)$ has only finitely many finite singularities: these lie among the zeros of the leading coefficient and discriminant of a defining polynomial. The infinitely many poles of H exclude algebraicity. □

This rules out interpreting this particular Stieltjes density as a first-kind algebraic differential in the same parameter. It does not rule out some other, presently unavailable representation of its definite integral. The unresolved step is a finite algebraic-group or height-controlled identity for the value, not recovery of H.

Feigenbaum's constant: bounded heights, but no degree control. Let r_n be the principal superstable logistic parameter of period 2^n, and put

$$d_n = \frac{r_n - r_{n-1}}{r_{n+1} - r_n}, \qquad \delta = \lim_{n\to\infty} d_n.$$

The existence and numerical evaluation of this scaling limit are distinct from an arithmetic classification [275]. Here h denotes the absolute logarithmic Weil height.

Proposition 64.4. *The approximants d_n are algebraic and satisfy $h(d_n) \le 14\log 2$. The construction supplies the degree bound*

$$[\mathbb{Q}(d_n) : \mathbb{Q}] \le 2^{7\cdot 2^{n-1}} \qquad (n \ge 1).$$

If a sequence of distinct algebraic approximants $\alpha_j \to \xi$ has bounded heights and

$$\frac{-\log|\xi - \alpha_j|}{[\mathbb{Q}(\alpha_j) : \mathbb{Q}]} \longrightarrow +\infty, \tag{64.2}$$

then ξ is transcendental. The bounds above do not verify (64.2) for δ.

Proof. The logistic map is affinely conjugate to $z \mapsto z^2 + c_n$, where $c_n = r_n(2 - r_n)/4$ and the critical point 0 has period 2^n. The polynomial $f_c^{\circ m}(0)$ is monic of degree 2^{m-1} in c. Hence c_n is an algebraic integer of degree at most 2^{2^n-1}. Every conjugate of c_n has modulus at most 2: if $|c| > 2$, the orbit of 0 escapes, since after its first step successive moduli grow by at least the factor $|c| - 1 > 1$. Therefore $h(c_n) \le \log 2$.

The relation $r_n = 1 \pm \sqrt{1 - 4c_n}$ and the standard height inequalities $h(ab) \le h(a) + h(b)$, $h(a + b) \le h(a) + h(b) + \log 2$, and $h(a^m) = |m|h(a)$ give $h(r_n) \le \log 8$. Applying the same inequalities to the two differences and their quotient gives $h(d_n) \le 4\log 8 + 2\log 2 = 14\log 2$. Also $[\mathbb{Q}(r_n) : \mathbb{Q}] \le 2^{2^n}$, so the degree of the field generated by r_{n-1}, r_n, r_{n+1} is at most the product of these three bounds, which is the stated bound for d_n.

Finally, if ξ were algebraic of degree D and $h(\alpha_j) \le C$, the product formula applied to the nonzero algebraic number $\xi - \alpha_j$ would give

$$|\xi - \alpha_j| \ge \exp\left(-D[\mathbb{Q}(\alpha_j) : \mathbb{Q}](h(\xi) + C + \log 2)\right).$$

Here we used $|z| \ge \exp(-\deg(z)h(z))$ for nonzero algebraic z, which follows by applying the height definition to z^{-1}. This contradicts (64.2). □

Thus even these uniformly bounded heights are not the missing arithmetic estimate by themselves. One would need approximation relative to the actual degrees, as in (64.2), or a different arithmetic identity. A geometric convergence estimate and the available large degree upper bound do not establish that condition. No lower bound for the actual degrees, and no impossibility theorem for a sharper approach, is being asserted.

Glaisher–Kinkelin's constant: the prime-product transfer. Use $\log A = 1/12 - \zeta'(-1)$, a normalization also used in the recent automatic-product paper [292]. Differentiating the zeta functional equation [293, 25.4.2] at -1 gives the exact reduction

$$12\log A = \log(2\pi) + \gamma - \frac{\zeta'(2)}{\zeta(2)} = \log(2\pi) + \gamma + \sum_p \frac{\log p}{p^2 - 1}. \tag{64.3}$$

Indeed the logarithmic derivative at -1 is $\log(2\pi) - 1 + \gamma - \zeta'(2)/\zeta(2)$ and $\zeta(-1) = -1/12$; the absolutely convergent prime sum is the logarithmic derivative of the Euler product at 2.

This is a concrete link to prime observation, but not a finite logarithmic certificate. To make the limitation quantitative, put

$$S_y = \sum_{p \le y} \frac{\log p}{p^2 - 1}, \qquad D_y = \operatorname{lcm}_{p \le y}(p^2 - 1), \qquad B_y = \prod_{p \le y} p^{D_y/(p^2-1)}.$$

Then $e^{S_y} = B_y^{1/D_y}$ is algebraic, has degree at most D_y, and has height exactly S_y, since B_y is a positive integer and heights scale under powers. These heights remain bounded. Nevertheless for $y \ge 2$ the immediate effective estimate is only

$$0 \le \sum_{p} \frac{\log p}{p^2 - 1} - S_y \le \frac{4}{3} \frac{\log y + 1}{y},$$

by comparison with $\sum_{m>y} (\log m)/m^2$ (take integer y). The denominator-clearing degrees are not controlled by a fixed number field. Moreover (64.3) still contains γ and π. Neither this product identity nor the presence of automatic sequences in other product evaluations proves that A is algebraic, irrational, or transcendental. The missing step is an arithmetic relation or independence theorem for the actual combined value in (64.3); the elliptic quantile theorem supplies none.

65 Complete relation ideals for cyclotomic Mahler sums

The earlier linear classification of sparse Lambert values admits a stronger, finite form: in the additive cyclotomic setting, every polynomial relation between algebraic-point values follows from a computable affine linear relation. This statement also survives mixing multiplicatively independent dilations. Its transcendence inputs are classical Mahler theory and the purity theorem of Adamczewski–Faverjon; the finite certificate and the applications below are proved explicitly. These results do not establish transcendence of the Erdős–Borwein constant or of the full reciprocal Fibonacci sum, or irrationality of the prime Mertens constant. The last subsection isolates the remaining obstructions through different decompositions of those three constants.

65.1 From algebraic dependence to an additive coboundary

We write $\overline{\mathbb{Q}}$ for the algebraic numbers and $\mu_N = \{z : z^N = 1\}$. A rational additive coboundary for the dilation $z \mapsto z^d$ is a function $g(z) - g(z^d)$, with $g \in \mathbb{C}(z)$. The following elementary argument is the additive part of the classical Mahler algebraic-independence mechanism; compare Kubota's theorem and the applications of Nishioka's theory in [296, 299].

Lemma 65.1 (The additive dependence criterion)**.** *Let $d \ge 2$, let $R_1, \dots, R_s \in \mathbb{C}(z)$ be regular at zero, and let analytic germs f_i satisfy*

$$f_i(z^d) = f_i(z) - R_i(z).$$

The functions $f_1, \dots, f_s$ are algebraically dependent over $\mathbb{C}(z)$ if and only if there are constants $c_1, \dots, c_s$, not all zero, and $g \in \mathbb{C}(z)$ such that

$$\sum_i c_i R_i(z) = g(z) - g(z^d).$$

Proof. Suppose there is a polynomial relation. In its nonzero ideal in $\mathbb{C}(z)[Y_1, \dots, Y_s]$, choose a polynomial P whose leading monomial is least in graded lexicographic order, and normalize the leading coefficient to one. The polynomial

$$\mathcal{T}P(z, Y) = P(z^d, Y_1 - R_1(z), \dots, Y_s - R_s(z))$$

is another relation with the same leading monomial and coefficient. Consequently $\mathcal{T}P - P = 0$; otherwise its leading monomial would be smaller. Write $m = \deg_Y P$, necessarily positive, and let P_j be its homogeneous part of degree j. The degree- m identity says that every coefficient of P_m is fixed by $z \mapsto z^d$. A rational function fixed by this substitution is constant: comparison of the first nonconstant Laurent coefficient at zero proves this directly. Thus $P_m \in \mathbb{C}[Y]$.

In degree $m-1$, the same identity gives

$$\sum_i R_i(z) \frac{\partial P_m}{\partial Y_i}(Y) = P_{m-1}(z^d, Y) - P_{m-1}(z, Y).$$

At least one coefficient vector of the gradient on the left is nonzero, since the characteristic is zero and $m > 0$. Comparing that monomial coefficient gives the required constants and a rational coboundary (with the sign of its rational function reversed).

Conversely, a rational g in the displayed coboundary cannot have a pole at zero: a pole of order $a > 0$ would give a pole of order da in $g(z^d)$, which neither other term could cancel. Thus $\sum_i c_i f_i - g$ is analytic at zero and invariant under $z \mapsto z^d$. Its first nonconstant Taylor coefficient would contradict invariance. It is constant, giving a linear relation over $\mathbb{C}(z)$. □

For clarity, the arithmetic transfer used below is a stated external input. For a one-variable linear Mahler system over $\overline{\mathbb{Q}}(z)$, analytic at zero, the transcendence degree of its values at a regular algebraic $0 < |\alpha| < 1$ equals the functional transcendence degree. This is Nishioka's theorem, also [297, Corollary 2.5 and equation (2.4), preprint numbering]. For several systems whose integer dilations are pairwise multiplicatively independent, the ideal of all value relations is generated by the ideals within the individual systems [297, Theorem 2.8, preprint numbering]. The older regular-singular version already suffices here [298, Theorem 2.4]. No unproved generalization of Loxton–van der Poorten's early multivariate claims is used.

65.2 One finite linear system computes the entire ideal

Theorem 65.2 (Complete cyclotomic relation certificate)**.** *Let K be a number field and $R_1, \dots, R_s \in K(z)$ satisfy $R_i(0) = 0$. Suppose their finite poles lie in μ_N, have orders at most m, and their polynomial parts have degree at most B, with a constant or zero part assigned degree zero. Put*

$$D(z) = (1 - z^N)^m, \qquad M = mN + \lfloor B/d \rfloor, \qquad f_i(z) = \sum_{k \geq 0} R_i(z^{d^k}), \qquad d \geq 2.$$

If all kernels are polynomials, take $m = 0, N = 1$. Solve the finite homogeneous linear system over K

$$A(z)D(z^d) - A(z^d)D(z) = D(z)D(z^d) \sum_{i=1}^{s} c_i R_i(z), \qquad A(0) = 0, \quad \deg A \leq M, \tag{65.1}$$

in $c \in K^s$ and $A \in K[z]$. Let $V \subset K^s$ be the projection of its solution space to c, let $r = \dim_K V$, and choose a basis $c^{(1)}, \dots, c^{(r)}$, with corresponding polynomials $A_1, \dots, A_r$.

For every algebraic $0 < |\alpha| < 1$, the entire ideal of polynomial relations over $\overline{\mathbb{Q}}$ between $f_1(\alpha), \dots, f_s(\alpha)$ is

$$\left(\sum_{i=1}^{s} c_i^{(j)} X_i - \frac{A_j(\alpha)}{D(\alpha)} \ : \ 1 \leq j \leq r \right). \tag{65.2}$$

In particular its transcendence degree is exactly $s - r$. The normalized A associated with any $c \in V$ is unique.

Proof. The defining sums converge locally uniformly in the disk and satisfy $f_i(z) - f_i(z^d) = R_i(z)$. We first verify that (65.1) captures every rational coboundary, including those initially having complex coefficients. Normalize its rational solution g by $g(0) = 0$; regularity at zero follows from the last paragraph of Lemma 65.1.

A pole of g outside μ_N forces poles at every successive d-th root of that point: these roots remain outside μ_N, where the right side of the coboundary equation is regular. Their number is unbounded, impossible for a rational function. All finite poles therefore lie in μ_N. If a pole had maximal order larger than m, the same backward propagation, now of that pole order, would again give infinitely many poles. Thus Dg is polynomial. If the polynomial part of g has positive degree b, that of $g - g(z^d)$ has degree exactly db, so $b \le \lfloor B/d \rfloor$. It follows that $g = A/D$ with $\deg A \le M$. This gives exactly (65.1). Conversely every solution of that identity gives a rational coboundary. A normalized solution is unique because a rational function invariant under $z \mapsto z^d$ is constant.

The coefficient equations are a finite linear system over K. Its complex solution space is the scalar extension of its solution space over K. In particular the space of all complex coefficient vectors giving rational coboundaries is $V \otimes_K \mathbb{C}$. Extend the chosen basis of V to the rows of an invertible matrix $S \in \mathrm{GL}_s(K)$, and put $h = Sf$. Its first r coordinates are the rational functions A_j/D, by normalized analytic uniqueness. The remaining $s - r$ coordinates are algebraically independent over $\mathbb{C}(z)$. Indeed, a relation among them would, by Lemma 65.1, produce a nonzero coboundary vector in the span of the complementary rows of S, contradicting their direct-sum decomposition with V.

Apply the stated one-variable value theorem to these remaining coordinates together with the constant function 1. Their system matrix has identity diagonal and the rational kernels in its first column. It and its inverse have poles only on the unit circle, and determinant one. The forward orbit of α is entirely inside the punctured disk, so α is regular; the one-variable pair is admissible. Thus these $s - r$ values remain algebraically independent. In the coordinates $Y = SX$, all relations are therefore generated by $Y_j - A_j(\alpha)/D(\alpha)$, $1 \le j \le r$: division by these linear polynomials leaves a polynomial in algebraically independent coordinates. Changing back to X proves (65.2). □

The theorem makes the arithmetic output exact, rather than merely giving a lower bound for the number of independent values. Its additive-dependence principle and its value theorem are classical; general rational Mahler solving and discrete residues are also established subjects [253, 254]. The specific contribution here is the explicit bounded matrix that computes the complete relation ideal for this cyclotomic input class. This is not a claim to a new general Mahler transcendence theorem.

Corollary 65.3 (Arbitrary finite collections of integer dilations)**.** *For each i, let $d_i \ge 2$ and let R_i be as in Theorem 65.2, allowing different cyclotomic pole sets and bounds. At a common algebraic $0 < |\alpha| < 1$, all algebraic relations between*

$$f_i(\alpha) = \sum_{k \ge 0} R_i(\alpha^{d_i^k})$$

are generated by affine linear relations, computable as follows. Partition the dilations into classes of multiplicative dependence. For each class choose a common power $D_0 = d_i^{e_i}$, replace R_i by $\sum_{k=0}^{e_i-1} R_i(z^{d_i^k})$, and apply (65.1) with dilation D_0 within that class. The union of the resulting class relations generates the full ideal; the transcendence degrees of the classes add.

Proof. Multiplicative dependence of positive integers greater than one is an equivalence relation: their prime-exponent vectors are rationally proportional. Each finite class therefore admits

the stated common power, effectively obtainable from those integer vectors. Iterating the functional equation gives precisely the replacement kernel. Its poles are still roots of unity, with finite explicitly computable orders, and it vanishes at zero. Theorem 65.2 computes the ideal for each class. Common powers for distinct classes are pairwise multiplicatively independent. The purity theorem stated above now gives exactly the sum of these extended ideals.

All its regularity hypotheses hold as in the preceding proof. Even if one uses only the older regular-singular purity theorem, our systems qualify: for a vector f the analytic triangular matrix $Y(z) = \left(\begin{smallmatrix} 1 & 0 \\ f(z) & I \end{smallmatrix}\right)$ satisfies $Y(z^{D_0}) = A(z)Y(z)$ and has determinant one, providing an analytic gauge to the identity system. Its coefficients are algebraic. Finally purity is a statement about the entire ideal, so no additional nonlinear mixed relations remain. □

65.3 Residues, Fibonacci contrasts, and infinite independent families

Theorem 65.4 (The complete residue-family ideal)**.** *For $d \geq 2$, define*

$$U_{d,j}(z) = \sum_{k \geq 0} \frac{z^{jd^k}}{1 - z^{d^{k+1}}}, \qquad 1 \leq j \leq d-1.$$

At every algebraic $0 < |\alpha| < 1$, their relation ideal is

$$\left(X_1 + \cdots + X_{d-1} - \frac{\alpha}{1-\alpha} \right),$$

and their transcendence degree is $d-2$. For pairwise multiplicatively independent $d_1, \ldots, d_t$, the union of these families, evaluated at arbitrary respective algebraic disk points $\alpha_1, \ldots, \alpha_t$, has only these t relations and transcendence degree $\sum_{i=1}^{t}(d_i - 2)$.

Proof. For a coefficient vector c, its kernel is $R(z) = (\sum_{j=1}^{d-1} c_j z^j)/(1 - z^d)$. The pole argument in Theorem 65.2 shows that a normalized rational coboundary solution has only simple poles in μ_d and is bounded at infinity. A pole at $\zeta \in \mu_d \setminus \{1\}$ is impossible: every d-th root of ζ is outside μ_d, where $g(z)$ and $R(z)$ are regular but $g(z^d)$ would have a pole. Hence $g(z) = cz/(1-z)$. Since

$$\frac{z}{1-z} - \frac{z^d}{1-z^d} = \frac{z + \cdots + z^{d-1}}{1 - z^d},$$

the coboundary space consists exactly of constant vectors. The complete-ideal theorem proves the first assertion. Purity proves the second, with the same regularity verification even when the evaluation points differ. The case $d = 2$ is interpreted literally: the single value is algebraic and contributes zero to the dimension. □

For example, $U_{3,1}(1/2), U_{4,1}(1/2), U_{4,2}(1/2)$ are algebraically independent. Bases 3 and 4 are multiplicatively independent although the latter is a power of 2. By contrast one cannot invoke purity between bases 2 and 4; their relations must first be computed at a common power.

Theorem 65.5 (All periodic Fibonacci residue relations)**.** *Let $F_1 = F_2 = 1$, and for $h \geq 1$ put*

$$V_{h,r} = \sum_{\substack{k \geq 0 \\ k \equiv r \pmod h}} \frac{1}{F_{2^k}}, \qquad 0 \leq r < h.$$

The entire algebraic relation ideal of this h-tuple over $\overline{\mathbb{Q}}$ is

$$\left(\sum_{r=0}^{h-1} X_r - \frac{7 - \sqrt{5}}{2} \right).$$

In particular its transcendence degree is $h-1$. Moreover the countable family

$$W_{m,r} = V_{2^{m+1},r} - V_{2^{m+1},r+2^m}, \qquad m \geq 0, \quad 0 \leq r < 2^m, \tag{65.3}$$

is algebraically independent over $\overline{\mathbb{Q}}$.

Proof. For a vector $a = (a_0, \ldots, a_{h-1})$, extend its entries periodically and define

$$H_a(z) = \sum_{k \geq 0} a_k \frac{z^{2^k}}{1 - z^{2^{k+1}}}.$$

For dilation $d = 2^h$, its kernel is

$$\sum_{r=0}^{h-1} \frac{a_r z^{2^r}}{1 - z^{2^{r+1}}} = \frac{\sum_{j=1}^{d-1} a_{v_2(j)} z^j}{1 - z^d}.$$

Each index j has the unique form $2^r(2s+1)$, proving the equality. The preceding proof shows that this kernel is a rational coboundary precisely when all a_r are equal. Theorem 65.2 applied to the h coordinate kernels therefore gives precisely one affine linear relation among their values and transcendence degree $h-1$.

Let $q = (\sqrt{5}-1)/2$. For every periodic a, Binet's formula gives, with $b_k = a_{k+1}$,

$$\sum_{k \geq 0} \frac{a_k}{F_{2^k}} = a_0 + \sqrt{5}\, H_b(q^2).$$

This is an invertible affine transformation of the h coordinate values: the shift of the residues is a permutation. The only relation has constant coefficients $a_r = 1$. In that case $H_b(z) = z/(1-z)$, whence the sum is $1 + \sqrt{5}q = (7-\sqrt{5})/2$. This proves the full ideal assertion.

For the second assertion, fix $M \geq 1$. Refinement gives $V_{h,r} = V_{2h,r} + V_{2h,r+h}$. At each node this sum, together with its difference in (65.3), recovers the two children by division by two. Thus the 2^M values $V_{2^M,r}$ are an invertible rational linear transform of their total sum and the $2^M - 1$ differences with $m < M$. Their transcendence degree is $2^M - 1$, and the total sum is algebraic. Consequently those $2^M - 1$ differences are algebraically independent. Every finite subfamily of (65.3) lies in one such level, completing the proof. □

The total-sum identity is classical, and substantial algebraic independence theorems for sparse Fibonacci/Lucas sums, including periodic weights, predate this work [300, 299, 301]. We do not assert priority for sparse Fibonacci transcendence. The advance over the previous project result is the full polynomial relation ideal, and the explicit compatible basis of independent contrasts at all dyadic refinement levels. These are consequences of the complete additive certificate, not separate historical conjecture closures.

65.4 Three named constants: transfers and the remaining gaps

Erdős–Borwein: independent blocks do not control their sum. For $b \geq 2$ and a prime p, let

$$E_b = \sum_{n \geq 1} \frac{1}{b^n - 1}, \qquad A_{p,m}(b) = \sum_{k \geq 0} \frac{1}{b^{mp^k} - 1} \quad (p \nmid m).$$

Then

$$E_b = \sum_{\substack{m \geq 1 \\ p \nmid m}} A_{p,m}(b), \tag{65.4}$$

and the countable family on the right is algebraically independent. This independence is already a special case of [299, Theorem 1]; it is not a new result. A short certificate proof helps explain its relation to our framework. Write $G_p(z) = \sum_{k\ge 0} z^{p^k}/(1-z^{p^k})$. For a finite set of indices m prime to p, the kernels of $G_p(z^m)$ are $z^m/(1-z^m)$. If a combination is a rational coboundary, the normalized rational solution has simple poles in μ_N, where N is the least common multiple of the indices, and is bounded at infinity. Since $p \nmid N$, each point of μ_N has $p-1$ of its p-th roots outside μ_N. Such a pole would be impossible by backward propagation. The solution is therefore constant. The kernels are linearly independent: a primitive root of the largest remaining index is a pole of its kernel alone, and induction removes all indices (with $m = 1$ treated last). Theorem 65.2 proves their algebraic independence at $z = 1/b$. The unique factorization $n = mp^k$, $p \nmid m$, proves (65.4) by positive rearrangement.

There is a genuine logical obstruction to drawing a conclusion about E_b from this independence alone. There exist positive, algebraically independent real numbers $t_1, t_2, \dots$ whose sum is 1. To prove this, put $r_0 = 1$, and recursively choose $t_n \in (r_{n-1}/3, 2r_{n-1}/3)$ transcendental over $\overline{\mathbb{Q}}(t_1, \dots, t_{n-1})$, then set $r_n = r_{n-1} - t_n$. Such a choice exists because the algebraic closure of this finitely generated countable field is countable. Independence follows inductively and $0 < r_n < (2/3)^n$, proving the sum assertion. Thus finite-block Mahler independence cannot by itself prove transcendence of E_2. An arithmetic estimate controlling the infinite tail relative to algebraic approximation is still missing. Duverney–Tachiya explicitly distinguish this transcendence question from known irrationality; Campbell's 2026 binary-pattern result does not close it [302, 248].

The full reciprocal Fibonacci sum: a theta–Lambert split. Set $L(z) = \sum_{n\ge 1} z^n/(1-z^n)$, $q = (\sqrt{5}-1)/2$, and $\theta_2(Q) = \sum_{n\in\mathbb{Z}} Q^{(n+1/2)^2}$ for real $0 < Q < 1$. Then the exact classical decomposition is

$$\sum_{n\ge 1} \frac{1}{F_n} = \sqrt{5}\left(L(q^2) - L(q^4) + \frac{1}{4}\theta_2(q^2)^2\right). \tag{65.5}$$

Indeed, Binet's formula gives $1/F_n = \sqrt{5}q^n/(1-(-1)^n q^{2n})$. The even-index contribution divided by $\sqrt{5}$ is $L(q^2) - L(q^4)$, using $t/(1-t) - t^2/(1-t^2) = t/(1-t^2)$. For the odd indices, Jacobi's two-square identity yields

$$\frac{1}{4}\theta_2(q^2)^2 = \sum_{\substack{m\ge 1\\ m \text{ odd}}} \frac{q^m}{1+q^{2m}}.$$

For completeness, the change of variables $(a, b) = (u+v+1, u-v)$ is a bijection from $\mathbb{Z}^2$ onto the pairs of opposite parity, and $a^2 + b^2 = 2(u+1/2)^2 + 2(v+1/2)^2$. Hence the theta square is $\sum_{n \text{ odd}} r_2(n)q^n$. Jacobi's formula $r_2(n) = 4\sum_{d\mid n} \chi_4(d)$, where $\chi_4(2j) = 0$ and $\chi_4(2j+1) = (-1)^j$, turns it into $4\sum_{m \text{ odd}} \sum_{j\ge 0}(-1)^j q^{m(2j+1)}$, proving the displayed identity by absolute convergence. See [303] for the classical two-square formula.

The odd-index sum is already known to be transcendental [304, slide 58]. This does not exclude its algebraic cancellation with the Lambert difference in (65.5). Precisely, proving that

$$L(q^2) - L(q^4) + \tfrac{1}{4}\theta_2(q^2)^2 \notin \overline{\mathbb{Q}}$$

would close the original target; the finite additive certificate does not control this mixed theta–Lambert relation. The full-sum transcendence question is explicitly left unresolved in [304, slide 59]. Thus this route identifies the missing mixed relation instead of reusing the earlier no-Mahler argument.

Prime Mertens: a zeta-value representation with an exponential tail. Let μ be the Möbius function and γ Euler's constant. The prime Mertens constant has the absolutely convergent forms

$$B_1 - \gamma = \sum_{m\ge 2} \frac{\mu(m)}{m} \log \zeta(m), \tag{65.6}$$

$$B_1 = \gamma - \frac{1}{2}\log\zeta(2) + \sum_{\substack{n\ge 3\\ n \text{ odd}}} \frac{\mu(n)}{n} \log \frac{\zeta(n)}{\sqrt{\zeta(2n)}}. \tag{65.7}$$

The tail in (65.6) after $m = M \ge 1$ has absolute value at most $3 \cdot 2^{-M}/(M+1)$. These representations belong to the classical computational theory of prime constants; compare [250]. To verify them directly, write $P(s) = \sum_p p^{-s}$. Euler's product gives $\log\zeta(m) = \sum_{k\ge 1} P(mk)/k$. Absolute convergence allows rearrangement, and the coefficient of $P(n)$ in the first series is

$$\frac{1}{n}\sum_{\substack{m\mid n\\ m\ge 2}} \mu(m) = -\frac{1}{n} \quad (n \ge 2).$$

It is therefore $\sum_p (\log(1-1/p) + 1/p) = B_1 - \gamma$. Pairing the terms n and $2n$ for odd n, using $\mu(2n) = -\mu(n)$ and $\mu(4n) = 0$, proves (65.7). Finally

$$0 < \log\zeta(m) < \zeta(m) - 1 \le 2^{-m} + \int_2^\infty x^{-m}\,dx \le 3\cdot 2^{-m} \quad (m \ge 2)$$

implies the asserted tail bound by summing a geometric series.

Although $\zeta(2n)$ is an algebraic multiple of π^{2n}, neither the arithmetic nature of the required odd zeta values nor their infinite interaction with γ is controlled here. Baker's theorem for a finite combination of logarithms of algebraic numbers has no such hypotheses. No verified irrationality proof for B_1 was located in the primary-source check. The faster tail in this representation is consequently a computational gain, not a proof that B_1 is irrational.

66 Growing Laplace kernels and actual arithmetic cancellation

The named targets here are Euler's constant γ and the Euler–Gompertz constant $\delta = \int_0^\infty e^{-t}/(1+t)\,dt$. The published 2025 work of Wolfs–Van Assche improves rational approximations to both constants without proving their irrationality [305]. The conditional application in Murabayashi–Yoshida [306] also retains an unproved input. No verified individual resolution was found in our current primary-source check, and none is claimed below. Known transcendence of $\mathrm{Ein}(1) = \gamma + \delta/e$ is a different statement.

We now leave the fixed-denominator integer-polynomial cone of the preceding investigation. We examine growing pole orders, primitive integer forms, and integer-valued rather than integer-coefficient polynomials. The main new internal results are an obstruction for *every* odd Taylor-order schedule using the actual reduced denominator, and a modular description of every prime occurring in the content of the classical Gompertz Padé forms. The Padé family itself is classical; no priority claim is made for the modular formulation. The proofs distinguish a proved exclusion from the higher-valuation estimate still missing for a positive irrationality argument.

66.1 Euler's constant: all odd Taylor schedules fail after primitive clearance

For integers $n \geq 1$ and $M \geq 1$, put

$$S_M(n) = \sum_{k=1}^{M} \frac{(-1)^{k-1} n^k}{k\, k!} = \frac{a_M(n)}{b_M(n)}, \qquad \gcd(a_M(n), b_M(n)) = 1, \quad b_M(n) > 0.$$

The denominator here is the actual reduced denominator, not a chosen factorial or least common multiple. Write $\delta_n = \int_0^\infty e^{-t}/(t+n)\, dt$.

Theorem 66.1 (Primitive odd-Taylor obstruction)**.** *For every choice of odd positive integers $M = M(n)$,*

$$b_M(n)\big[S_M(n) - \log n - \gamma\big] \longrightarrow +\infty \qquad (n \to \infty).$$

In particular the primitive integer linear forms $a_M(n) - b_M(n)\gamma - b_M(n)\log n$ are strictly positive, but never approach zero along an unbounded sequence of growing arguments and odd truncation orders. No growth restriction on $M(n)$ is imposed.

Proof. Taylor's integral remainder for e^{-t}, integrated after division by t, gives

$$\mathrm{Ein}(n) = S_M(n) + (-1)^M E_M(n), \qquad E_M(n) = \frac{1}{M!} \int_0^n \frac{1}{t} \int_0^t e^{-s}(t-s)^M\, ds\, dt > 0.$$

Together with $\mathrm{Ein}(n) = \gamma + \log n + e^{-n}\delta_n$, this proves for odd M that

$$S_M(n) - \log n - \gamma = E_M(n) + e^{-n}\delta_n > 0. \tag{66.1}$$

First suppose $n \geq 4$ and $M \leq 3n/2$. Restrict the outer integral to $n - n/(M+1) \leq t \leq n$ and the inner one to $0 \leq s \leq 1$. These ranges are compatible since $t \geq n/2 \geq 2$. They give

$$E_M(n) \geq \frac{e^{-1} n^M}{(M+1)M!} \left(1 - \frac{1}{M+1} - \frac{1}{n}\right)^M \geq \frac{e^{-11} n^M}{(M+1)M!}.$$

Indeed, with $u = 1/(M+1) + 1/n$, one has $u \leq 3/4$, $Mu \leq 5/2$, and $\log(1-u) \geq -u/(1-u)$. The last lower bound tends to infinity uniformly along all such schedules. To verify this, the arithmetic–geometric mean inequality $M! \leq ((M+1)/2)^M$ and $M + 1 \leq 7n/4$ imply

$$\frac{n^M}{(M+1)M!} \geq \frac{(8/7)^M}{M+1}.$$

This tends to infinity whenever $M \to \infty$; for M in any fixed finite set the original expression tends to infinity with n. A subsequence argument combines these alternatives. Since $b_M(n) \geq 1$, (66.1) proves the desired assertion in this range.

It remains to handle $M > 3n/2$. If $M/2 < p \leq M$ is prime and $p \nmid n$, the $k = p$ term of $S_M(n)$ has p-adic valuation -2. Every other summand has valuation at least -1: there is no other multiple of p among $1, \ldots, M$, and $v_p(k!) \leq 1$ there. The uniquely smallest valuation cannot cancel. Therefore

$$v_p\big(S_M(n)\big) = -2, \qquad p^2 \mid b_M(n). \tag{66.2}$$

Writing $\vartheta(x) = \sum_{p \leq x} \log p$, the distinct primes in this interval that divide n contribute at most $\log n$. Hence

$$\log b_M(n) \geq 2\big(\vartheta(M) - \vartheta(M/2)\big) - 2\log n = M + o(M) - 2\log n,$$

by the prime number theorem. Jensen's inequality applied to an exponential random variable of mean one gives $\delta_n \geq 1/(n+1)$. Thus

$$\log\big(b_M(n) e^{-n}\delta_n\big) \geq M - n + o(M) - 2\log n - \log(n+1) \geq M/3 + o(M) - 3\log(n+1) \longrightarrow +\infty.$$

Equation (66.1) completes the proof, including schedules alternating between the two ranges. □

This is an actual reduced-denominator statement. Its forms have coefficient gcd one because $a_M(n), b_M(n)$ are coprime. It does not prove a barrier for arbitrary approximants to γ: even truncation orders permit cancellation between the two positive terms in (66.1), and combinations at several arguments might cancel logarithms or introduce new common factors. No such cancellation estimate is supplied here. The theorem closes this specific growing-argument odd-Taylor route, including orders chosen after inspecting all previous coefficients.

66.2 Gompertz: a growing-pole Padé family and its exact content support

Define integers P_n, Q_n by

$$P_0 = 0, \quad P_1 = 1, \qquad Q_0 = 1, \quad Q_1 = 2, \qquad U_n = 2nU_{n-1} - (n-1)^2 U_{n-2} \quad (n \geq 2). \tag{66.3}$$

The recurrence applies separately to P and Q. Let $g_n = \gcd(P_n, Q_n) > 0$ for $n \geq 1$.

Proposition 66.2 (Exact positive error and growing poles)**.** *For every $n \geq 0$,*

$$Q_n = n! L_n(-1) = \sum_{j=0}^{n} \binom{n}{j}^2 j!, \tag{66.4}$$

$$Q_n \delta - P_n = n! I_n > 0, \qquad I_n = \int_0^\infty \frac{e^{-t} t^n}{(1+t)^{n+1}}\, dt. \tag{66.5}$$

Here L_n is the ordinary Laguerre polynomial. In particular P_n/Q_n is a rational approximation with a certified nonzero error, and its primitive integer form is exactly $(Q_n/g_n)\delta - P_n/g_n = n!I_n/g_n$.

Proof. The Rodrigues formula $L_n(t) = e^t (d/dt)^n (e^{-t}t^n)/n!$ and n integrations by parts give

$$\int_0^\infty \frac{e^{-t} L_n(t)}{1+t} dt = \int_0^\infty \frac{e^{-t} t^n}{(1+t)^{n+1}} dt.$$

All boundary terms vanish: the derivatives of $e^{-t}t^n$ of order less than n vanish at zero, and exponential decay handles infinity. The polynomial $A_n(t) = n! L_n(t)$ has integer coefficients. Consequently

$$Q_n = A_n(-1), \qquad P_n = \int_0^\infty e^{-t} \frac{A_n(-1) - A_n(t)}{1+t} dt$$

are integers, since the quotient is an integer polynomial and $\int_0^\infty e^{-t} t^k dt = k!$. They have the stated initial values. The Laguerre recurrence is $A_n = (2n-1-t)A_{n-1} - (n-1)^2 A_{n-2}$. At $t = -1$ this gives the recurrence for Q; integration after $1/(1+t)$ gives the same recurrence for the remainders when $n \geq 2$, because $\int e^{-t} A_{n-1}(t) dt = 0$. It follows for P as well. The finite Laguerre series, with its index reversed, gives (66.4). Positivity of I_n proves nonvanishing. □

The integral and recurrence are classical Stieltjes–Laguerre approximants, discussed alongside the newer constructions in [305, 306]; they are not new approximants. The following arithmetic reduction directly addresses their actual common factors.

Theorem 66.3 (Prime support of the primitive content)**.** *For every $n \geq 1$ and prime p,*

$$p \mid g_n \quad \Longleftrightarrow \quad p \leq n \ \text{ and } \ Q_r \equiv 0 \pmod{p}, \qquad r = n \bmod p \in \{0, \ldots, p-1\}.$$

Consequently 3 and 5 never divide g_n; $2 \mid g_n$ exactly when $n \geq 3$ is odd; and $7 \mid g_n$ exactly when $n \geq 7$ and $n \equiv 2 \pmod 7$.

Proof. The recurrence gives the exact determinant

$$P_nQ_{n-1} - P_{n-1}Q_n = ((n-1)!)^2 \qquad (n \geq 1). \tag{66.6}$$

This follows from its value one at $n = 1$ and multiplication by $(n-1)^2$ at each next step. Thus $p > n$ cannot divide g_n.

Modulo p, every summand in (66.4) with $j \geq p$ vanishes. If $n = kp + r$, the identity $(1+X)^n \equiv (1+X^p)^k(1+X)^r \pmod p$ shows that $\binom{n}{j} \equiv \binom{r}{j}$ for $0 \leq j < p$. It follows that

$$Q_n \equiv Q_r \pmod p. \tag{66.7}$$

For $k \geq 1$, the recurrence at $kp+1$ gives $P_{kp+1} \equiv 2P_{kp} \pmod p$, because $(kp)^2 \equiv 0$. Induction through the rest of the block, using the same recurrence coefficients as at indices $1, \ldots, p-1$, yields

$$P_{kp+r} \equiv P_{kp}Q_r \pmod p \qquad (0 \leq r < p).$$

Hence, for $p \leq n$, $Q_r \equiv 0$ implies that both P_n and Q_n vanish modulo p, and the reverse implication follows from (66.7). The small-prime assertions follow by evaluating $Q_0, \ldots, Q_{p-1}$ modulo p: modulo 3 these are $1, 2, 1$; modulo 5 they are $1, 2, 2, 4, 4$; and modulo 7 they are $1, 2, 0, 6, 6, 6, 6$. Modulo 2 the two values are $1, 0$. □

This theorem describes prime divisibility, not all prime powers. For example direct recurrence evaluation gives $g_9 = 14$, $g_{16} = 49$, and $g_{23} = 266$, so even a prime occurring only in one residue class can occur to higher order. Replacing the missing valuation analysis by an observed small gcd would be an unjustified arithmetic step.

Proposition 66.4 (Error scale and the exact remaining saving)**.** *As $n \to \infty$,*

$$I_n \sim \sqrt{\pi}e^{1/2}n^{-1/4}e^{-2\sqrt{n}}, \qquad \log\left|\delta - \frac{P_n}{Q_n}\right| = -4\sqrt{n} + O(\log n). \tag{66.8}$$

Therefore along any subsequence the primitive positive forms tend to zero if and only if

$$\frac{g_n}{n!\, n^{-1/4}e^{-2\sqrt{n}}} \longrightarrow +\infty. \tag{66.9}$$

Such a subsequence would prove δ irrational, but no estimate establishing (66.9) is proved here.

Proof. Put $\lambda = \sqrt{n}$ and $t = \lambda u$. On every compact interval of positive u the integrand becomes

$$\frac{1}{u + 1/\lambda} \exp\left[-\lambda(u + u^{-1}) + \frac{1}{2u^2} + O(\lambda^{-1})\right] du.$$

The phase $u + u^{-1}$ has its unique minimum 2 at $u = 1$, with second derivative 2. Taylor expansion there, followed by $v = \sqrt{\lambda}(u-1)$, gives the Gaussian factor $e^{1/2}\sqrt{\pi/\lambda}$ multiplying $e^{-2\lambda}$. Here the usual dominated convergence step can be justified by a fixed small neighborhood of 1, where the phase exceeds $2 + c(u-1)^2$, and then by a compact annulus where its excess has a positive minimum.

To check the noncompact tails rather than assume uniformity of the expansion, use $\log(1+1/t) \geq 1/(t+1)$ and $f(t) = t + n/(t+1)$. For large n, on $0 \leq t \leq \lambda/2$, $f(t) \geq f(\lambda/2) \geq (12/5)\lambda$. On $t \geq 2\lambda$, $f'(t) \geq 3/4$ and $f(2\lambda) \geq (12/5)\lambda$. The two tail integrals are thus bounded by a polynomial in λ times $e^{-(12/5)\lambda}$. They are negligible, proving the first asymptotic.

For the second, the series $L_n(-1) = \sum_{k=0}^{n} \binom{n}{k}/k!$ satisfies

$$L_n(-1) \leq \sum_{k \geq 0} \frac{n^k}{(k!)^2} \leq e^{2\sqrt{n}}.$$

The last inequality follows by retaining diagonal terms in the square of $\sum_{k\ge0}(\sqrt{n})^k/k!$. The single term with $k=\lfloor\sqrt{n}\rfloor$, using $\binom{n}{k}\ge n^k e^{-O(k^2/n)}/k!$ and Stirling's formula, gives $\log L_n(-1)\ge 2\sqrt{n}-O(\log n)$. Since the rational approximation error equals $I_n/L_n(-1)$, the stated logarithmic estimate follows. Finally (66.5) and (66.8) give the exact subsequence equivalence. A nonzero integer form in a rational number of fixed denominator is bounded away from zero, which proves the irrationality implication if the missing estimate were available. □

The determinant bound (66.6) alone is much too weak for this purpose. The prime-support theorem removes certain possible factors exactly, but does not establish a sufficiently large saving or rule one out. Thus the growing-pole attempt has reached a purely integer-content problem with certified nonvanishing and a sharp error scale; it has not resolved the named constant.

66.3 Integer-valued squared-binomial kernels: an explicit escaping attempt

The next family deliberately violates the earlier integer-coefficient hypothesis. Let

$$B_n(t)=\binom{t}{n}^2=\left(\frac{t(t-1)\cdots(t-n+1)}{n!}\right)^2\qquad(n\ge1).$$

It is nonnegative on the real line and integer-valued on every integer, but usually has nonintegral coefficients. Put

$$J_n=\int_0^\infty\frac{e^{-t}B_n(t)}{1+t}dt,\quad M_n=\int_0^\infty e^{-t}B_n(t)dt,\quad T_n=\int_0^\infty e^{-t}B_n(t)\log t\,dt.$$

Proposition 66.5 (Exact arithmetic and failure of this integer-valued family)**.** *There are rationals a_n,A_n with*

$$J_n=\delta+a_n,\qquad T_n=A_n-M_n\gamma,\qquad M_n\in\mathbb{Q}_{>0}.$$

The canonical integer forms obtained by multiplying these identities by $(n!)^2$ have integer coefficients. Nevertheless $J_n\to+\infty$ and $T_n/M_n\to+\infty$. Thus neither primitive clearance of the first family nor reduction of the rational candidate A_n/M_n in the second gives an approximation proving irrationality of the named constant.

Proof. Since $B_n(-1)=1$, polynomial division by $1+t$ proves the first identity with rational remainder; its coefficients have denominators dividing $(n!)^2$. If $B_n=\sum c_kt^k$, the moment identity

$$\int_0^\infty e^{-t}t^k\log t\,dt=k!(H_k-\gamma)$$

gives $M_n=\sum c_kk!$ and $A_n=\sum c_kk!H_k$. Here $H_0=0$ and $k!H_k\in\mathbb{Z}$ for $k\ge1$, proving all the integrality assertions.

For $3n\le t\le3n+1$, the falling product is increasing and $B_n(t)\ge\binom{3n}{n}^2$. The largest term of the Binomial$(3n,1/3)$ distribution occurs at n; since it is at least $1/(3n+1)$,

$$\binom{3n}{n}\ge\frac{(27/4)^n}{3n+1}.$$

Consequently

$$J_n\ge\frac{e^{-1}}{(3n+2)(3n+1)^2}\left(\frac{729}{16e^3}\right)^n\longrightarrow+\infty.\tag{66.10}$$

The base exceeds one since $e < 3$. If $a_n = u_n/v_n$ is in lowest terms, the primitive form is exactly $v_n\delta + u_n = v_n J_n$, whose coefficient gcd is one. Its positive value is at least J_n, so actual common-factor division cannot rescue this family.

For $0 \le t \le n$, write $k \le t \le k+1$ with $0 \le k \le n-1$. Bounding each factor in the falling product gives

$$\left|\binom{t}{n}\right| \le \frac{(k+1)!(n-1-k)!}{n!} \le 1.$$

Thus $\int_0^n e^{-t} B_n(t)dt \le 1$. For $0 \le t \le 1$ the sharper bound $|\binom{t}{n}| \le t/n$ gives a negative logarithmic contribution of absolute value at most $1/(9n^2)$. On $t \ge n$ the logarithm is at least $\log n$, and on $1 \le t \le n$ it is nonnegative. Hence

$$\frac{T_n}{M_n} \ge (\log n)\left(1 - \frac{1}{M_n}\right) - \frac{1}{9n^2 M_n}.$$

The same interval used in (66.10) shows $M_n \to \infty$ exponentially. The right side tends to infinity. Since $A_n/M_n = \gamma + T_n/M_n$, reducing its numerator and denominator does not alter this divergence or convert it into an approximation to γ. □

These computations address three genuine escapes from the previous fixed-kernel theorem. Growing pole order produces a useful classical Padé family, whose actual gcd now has an exact prime-support rule. Growing Taylor arguments still fail at every odd order even after primitive reduction. Integer-valued squared-binomial numerators are allowed, but their mass moves far into the exponential tail and their primitive forms fail for an explicit reason. None of these statements rules out other mixed kernels, cancellations across distinct logarithms, or a sufficiently strong new valuation theorem for g_n.

67 Beyond rank one: rational moment pencils and mixed Catalan moments

We investigate four specified targets: transcendence of G, $\zeta(3)$ and $\zeta(5)$, and irrationality of $\zeta(7)$. None is resolved here. The advance over the preceding raw-tail obstruction is constructive: we give full-rank rational moment pencils for every odd zeta value, prove their real-root and central-gap structure, and obtain rational two-sided approximants with a proved exponential error. We also exhibit an explicit full-degree mixed Catalan family that cancels its auxiliary period. Arithmetic estimates needed to turn these constructions into the named resolutions are stated separately.

The starting point is the actual rational functional in Fauzan's 2026 paper, Sections 2.1–2.4, rather than the rank-one tail functional [307]. Its $\zeta(5)$ determinant has growing degree and a positive value at the target; positivity of the value does not mean that its coefficient matrix is positive. The extension below makes the alternating signature explicit. Classical ingredients, including Hermite's formula, symmetric matrix pencils, and Gaussian quadrature, are identified as such; we claim no priority for those ingredients.

67.1 A positive rational functional for every odd weight

Fix an odd integer $s \geq 3$ and put $\theta_s = \zeta(s)$, $H_j^{(s)} = \sum_{v=1}^{j} v^{-s}$. On rational functions with only simple poles at negative integer squares, define a $\mathbb{Q}$-linear functional by

$$\mu_{s,X}(t^e) = (-1)^e B_{2e+2} \frac{(2e+s)!}{(s-1)!(2e+2)!}, \tag{67.1}$$

$$\mu_{s,X}\left(\frac{1}{t+j^2}\right) = j^{s-1}(X - H_j^{(s)}) - \frac{1}{s-1} + \frac{1}{2j}. \tag{67.2}$$

Polynomial division and simple partial fractions define the functional uniquely. The Bernoulli convention is $B_1 = -1/2$.

Theorem 67.1 (Odd-weight positive representation)**.** *For every rational function R in this domain,*

$$\mu_{s,\theta_s}(R) = \int_0^\infty R(y^2) w_s(y)\, dy, \qquad w_s(y) = \frac{2(2\pi)^{s-1}}{(s-1)!}\, y^s \sum_{\ell \geq 1} \ell^{s-1} e^{-2\pi\ell y} > 0.$$

The measure is finite, has all moments, and is strictly positive on $(0,\infty)$. This includes $s = 3, 5, 7$, with no arithmetic hypothesis on their zeta values.

Proof. Let $f(y) = (e^{2\pi y} - 1)^{-1}$. Since $s-1$ is even, $w_s(y) = 2y^s f^{(s-1)}(y)/(s-1)!$. The expansion at zero shows $w_s(y) \to 1/\pi$, and at infinity it is $O_s(y^s e^{-2\pi y})$. Termwise integration, justified by nonnegativity, gives

$$m_e := \int_0^\infty y^{2e} w_s(y) dy = \frac{2(2e+s)!}{(s-1)!(2\pi)^{2e+2}} \zeta(2e+2).$$

Euler's even-zeta formula gives (67.1).

For $a > 0$ one has the elementary identity

$$\frac{d^{s-1}}{dy^{s-1}} \frac{y^s}{y^2+a^2} = (s-1)! a^{s-1} \operatorname{Im}(a - iy)^{-s}.$$

Polynomial division leaves a polynomial of degree $s-2$ and remainder

$$(-1)^{(s-1)/2} a^{s-1} \frac{y}{y^2+a^2}.$$

Differentiate its two simple fractions. Integrating by parts $s-1$ times introduces no sign. All boundary terms vanish: at zero each product is $O(y)$, since the derivative orders sum to $s-2$, and at infinity the derivatives of f decay exponentially. Hermite's formula for Hurwitz zeta [308, 25.11.29] yields

$$\int_0^\infty \frac{w_s(y)}{y^2+a^2} dy = a^{s-1}\zeta(s,a) - \frac{1}{2a} - \frac{1}{s-1}.$$

At $a = j$ use $\zeta(s,j) = \theta_s - H_j^{(s)} + j^{-s}$ to obtain (67.2). Linearity proves the assertion for every R. □

67.2 Full rank, real roots, and the exact central gap

Let $0 \le N < K$, $h = K - N$, $r \ge 0$ be integers, and set

$$D_N(t) = \prod_{j=1}^{N}(t + j^2), \quad E(t) = \prod_{j=N+1}^{K}(t + j^2), \quad W(t) = D_N(t)^r.$$

Define the $h \times h$ rational affine pencil and its polynomial by

$$A_{s,N,K,r}(X) = \left[\mu_{s,X}\left(\frac{W(t)t^{i+j}}{E(t)}\right)\right]_{0\le i,j<h}, \qquad P_{s,N,K,r}(X) = \det A_{s,N,K,r}(X).$$

Theorem 67.2 (Signature and real-root structure)**.** *The polynomial $P_{s,N,K,r} \in \mathbb{Q}[X]$ has degree exactly h, is positive at θ_s, and has only real roots, counted with multiplicity. Its leading coefficient is*

$$(-1)^{h(h-1)/2} \prod_{j=N+1}^{K} j^{s-1} D_N(-j^2)^r.$$

If rN is even, precisely $\lceil h/2 \rceil$ roots are below θ_s and $\lfloor h/2 \rfloor$ are above it. If rN is odd, the counts are reversed. No root equals θ_s.

In particular, for $K = 40n$, $N = 3n$ and $r = s$ the degree is $37n$; there are exactly $\lfloor 37n/2 \rfloor$ roots below $\zeta(s)$ and $\lceil 37n/2 \rceil$ above. Thus the target lies in the central gap of an explicitly defined rational polynomial for every odd $s \ge 3$. For $s = 5$ this is the determinant family in Fauzan's construction, before its scalar normalizations.

Proof. At the target the matrix $M = A(\theta_s)$ is positive definite: its quadratic form at a nonzero real polynomial q of degree below h is $\int_0^\infty W(y^2)q(y^2)^2 w_s(y)/E(y^2)dy > 0$. Write $A(X) = M + (X - \theta_s)B$. Partial fractions show

$$B = V \operatorname{diag}(\alpha_j) V^{\mathsf{T}}, \qquad V_{ij} = (-j^2)^i, \qquad \alpha_j = \frac{j^{s-1}W(-j^2)}{E'(-j^2)}.$$

Every α_j is nonzero and V is square invertible. The Vandermonde identity gives the stated determinant of B. Moreover $\operatorname{sgn}(\alpha_j) = (-1)^{rN+j-N-1}$, so Sylvester's law of inertia gives the stated alternating positive and negative counts for B.

The real symmetric matrix $T = M^{-1/2}BM^{-1/2}$ is invertible. If its eigenvalues are $\gamma_1, \dots, \gamma_h$, then

$$P(X) = \det M \prod_{i=1}^{h}(1 + (X - \theta_s)\gamma_i).$$

Its roots are $\theta_s - 1/\gamma_i$. Positive eigenvalues give roots below the target and negative eigenvalues give roots above it. Congruence preserves inertia, proving all assertions. Finally, sN has the parity of n and $37n$ has the parity of n, giving the central counts. □

Corollary 67.3 (Certified algebraic positivity interval)**.** *If both signatures in Theorem 67.2 occur, let $a < \theta_s < b$ be its two adjacent roots. Then*

$$A(X) > 0 \quad \Longleftrightarrow \quad a < X < b.$$

Both endpoints are real algebraic numbers that can be isolated using the explicit rational pencil, without inserting a numerical value of θ_s. On this interval $\log\det A(X)$ is strictly concave and has exactly one critical point. That point is algebraic.

For principal subspaces of a fixed pencil, positivity intervals decrease under inclusion. The eigenvalues of the corresponding relative matrices $M^{-1/2}BM^{-1/2}$ interlace. These assertions do not claim nesting when the denominator E or the numerator W is changed.

Proof. All factors $1+(X-\theta_s)\gamma_i$ must be positive, giving exactly the stated interval. Standard exact real-root isolation of rational polynomials and rational positive-definiteness tests identify it; the root counts in Theorem 67.2 specify its position. Twice differentiating gives

$$\frac{d^2}{dX^2}\log\det A(X) = -\operatorname{tr}\big((A(X)^{-1/2}BA(X)^{-1/2})^2\big) < 0.$$

The first derivative tends to $+\infty$ at a and $-\infty$ at b. The unique critical point is a root of the rational polynomial P'. Restriction of quadratic forms proves interval nesting. The relative operator on a subspace is an orthogonal compression in the M inner product; the usual min–max argument proves eigenvalue interlacing. □

This is a genuine escape from the rank-one obstruction: the target coefficient has full rank $37n$, and its two signatures both grow. It supplies algebraic enclosures, not a transcendence conclusion.

67.3 Rational two-sided approximants with exponential error

Let ν_s be the pushforward of $w_s(y)dy$ under $t=y^2$. Its moments m_e are the rational numbers in (67.1). Write $S_s(z)=\int(t+z)^{-1}d\nu_s(t)$ for $z>0$. Let q_n be the monic degree-n orthogonal polynomial for ν_s and r_n that for $t\,d\nu_s$. Their coefficients are rational, because the defining positive definite moment systems have rational entries. Define

$$L_n(z)=\frac{1}{q_n(-z)}\int\frac{q_n(-z)-q_n(t)}{t+z}\,d\nu_s(t), \qquad U_n(z)=\frac{m_0-L_n^{t\nu_s}(z)}{z}.$$

The integrands after division are polynomials. These are rational functions over $\mathbb{Q}$, not integrals that still contain an unknown zeta value. This is the classical Gaussian/Radau construction, here applied to the odd-zeta functional.

Theorem 67.4 (Explicit rational enclosure family)**.** *For $n\ge 1$, $j\ge 1$, put*

$$b_{s,j}=H_j^{(s)}+\frac{j^{1-s}}{s-1}-\frac{1}{2j^s}, \quad \ell_{n,j}=b_{s,j}+j^{1-s}L_n(j^2), \quad u_{n,j}=b_{s,j}+j^{1-s}U_n(j^2).$$

Then $\ell_{n,j},u_{n,j}\in\mathbb{Q}$ and

$$0<\theta_s-\ell_{n,j}\le m_{2n}j^{-s-4n-1},$$
$$0<u_{n,j}-\theta_s\le m_{2n+1}j^{-s-4n-3}.$$

For each fixed j, the lower bounds increase and the upper bounds decrease with n. If $n(j)=\lfloor \pi j/2\rfloor$, then for all sufficiently large j,

$$0<u_{n(j),j}-\ell_{n(j),j}\le C_s j^{-1/2}e^{-2\pi j}.$$

All quantities entering the rational endpoints are obtained from finite linear systems with explicit Bernoulli moments.

Proof. Orthogonal polynomials for a positive measure of infinite support in $(0,\infty)$ have n simple positive zeros. For completeness, if fewer than n zeros in this interval changed sign, multiplying by the product of their linear factors would give a polynomial of degree below n whose product with the orthogonal polynomial has one sign and nonzero integral, a contradiction. This also

excludes other zeros by degree counting. Hence $|q_n(-z)|, |r_n(-z)| \geq z^n$. Polynomial division and orthogonality give the exact positive errors

$$S_s(z) - L_n(z) = \frac{1}{q_n(-z)^2} \int \frac{q_n(t)^2}{t+z} d\nu_s(t) > 0,$$

$$U_n(z) - S_s(z) = \frac{1}{z r_n(-z)^2} \int \frac{t r_n(t)^2}{t+z} d\nu_s(t) > 0.$$

For the first identity, subtract the definition of L_n from S_s; the difference between $q_n(t)$ and $q_n(t)^2/q_n(-z)$, divided by $t+z$, is q_n times a polynomial of degree below n. The second follows by applying the first to $t\nu_s$ and using $zS_s(z) = m_0 - \int t/(t+z) d\nu_s$. The minimal-norm property of monic orthogonal polynomials bounds their squared norms by m_{2n} and m_{2n+1}, respectively. Thus the errors are at most $m_{2n} z^{-2n-1}$ and $m_{2n+1} z^{-2n-2}$. Equation (67.2) proves the stated bounds for θ_s.

For monotonicity one may use the exact variational formula

$$L_n(z) = \max_{\deg p < n} \int \big(2p(t) - (t+z)p(t)^2\big) d\nu_s(t).$$

Completing the square proves it: the optimizer is $(1 - q_n(t)/q_n(-z))/(t+z)$. Nested polynomial spaces give monotonicity; apply the same argument to $t\nu_s$ for the upper bound.

Finally $m_k = 2(2k+s)!\zeta(2k+2)/((s-1)!(2\pi)^{2k+2})$. For $n = \lfloor \pi j/2 \rfloor$ the integers $4n+s$ and $4n+s+2$ differ from $2\pi j$ by bounded amounts depending only on s. Stirling's inequality applied to the two error bounds therefore gives $O_s(j^{-1/2} e^{-2\pi j})$ for each. Their sum proves the final assertion. □

Proposition 67.5 (A certified height bound, not an irrationality proof)**.** *Write the endpoints in Theorem 67.4 as reduced fractions. Their positive denominators have logarithms at most*

$$O_s(n^2 \log(n+j+2) + j \log(j+2)).$$

For $n \asymp j$ this is $O_s(j^2 \log j)$, whereas the certified error is only $e^{-2\pi j + O_s(\log j)}$.

Proof. The identity $B_m = \sum_{k=0}^{m} (-1)^k k! \left\{ {m \atop k} \right\} / (k+1)$, obtained by expanding $z/(e^z - 1)$, shows that the denominator of B_m divides $\operatorname{lcm}(1, \ldots, m+1)$. Thus all m_e, $e \leq 2n+1$, have a common denominator $D = (s-1)! \operatorname{lcm}(1, \ldots, 4n+5)$. The factorial formula bounds their logarithmic magnitudes by $O_s(n \log(n+2))$; the elementary bound $\log D = O_s(n \log(n+2))$ suffices.

The variational formula gives an explicit rational expression $L_n(z) = b^{\mathsf{T}} A^{-1} b$, with $A_{ik} = m_{i+k+1} + z m_{i+k}$ and $b_i = m_i$, $0 \leq i, k < n$. At $z = j^2$, DA and Db are integer arrays of logarithmic size $O_s(n \log(n+j+2))$. The adjugate formula and the elementary determinant bound $|\det C| \leq n! \max |C_{ik}|^n$ yield the asserted quadratic bound for numerator and denominator. The same argument applies to $t\nu_s$. Adding $b_{s,j}$ introduces at most $s \log(j!) + O_s(\log j)$, and multiplication by powers of j has the same stated bound. Reduction of a rational fraction cannot increase its denominator. □

This proves rapid rational approximation for $s = 3, 5, 7$ by one uniform construction. It does not prove irrationality of $\zeta(7)$: the estimates do not show that denominator times error tends to zero. They also do not prove that the actual denominators grow at the upper-bound rate.

67.4 An explicit full-degree mixed Catalan family

The different escape mechanism in OpenAI's Catalan paper is cancellation between two moment kernels [309, Section 2.1]. Put $f(t) = \sqrt{1-t^2}$ and define

$$M(i,j) = \int_{-1}^{1}\int_{0}^{1} \frac{|t|t^i u^j}{f(t)(1-tu)}\,du\,dt, \qquad Z(i,j) = \int_{0}^{1}\int_{0}^{1} \frac{t^i u^j}{1-tu}\,du\,dt.$$

For $c_k = 4^{-k}\binom{2k}{k}$, interpreted as zero unless k is a nonnegative integer, the paper's exact moment evaluation is

$$M(i,j) = M^0(i,j) + 4Gc_{(i-j)/2} + \tfrac{3}{2}\zeta(2)c_{(j-i-1)/2}, \qquad Z(i,j) = Z^0(i,j) + [i=j]\zeta(2),$$

where $M^0(i,j), Z^0(i,j)$ are rational. The following new choice of rows is much simpler than the paper's arithmetically optimized rows; it is used only to isolate what cancellation and rank growth accomplish.

Proposition 67.6 (Full degree after exact auxiliary-period cancellation)**.** *For $n \geq 1$ and $0 \leq r < n$ set*

$$P_r(t) = t^r, \qquad D_r(t) = \sum_{\substack{k\geq 0\\ r+1+2k<n}} c_k t^{r+1+2k}.$$

Extend M, Z linearly in their first argument. The matrix $F_n = [M(P_r,j) - \frac{3}{2}Z(D_r,j)]_{0\leq r,j<n}$ has entries in $\mathbb{Q}+\mathbb{Q}G$. Replacing G by X gives a rational matrix $F_n(X)$ such that $\det F_n(X)$ has degree exactly n and leading coefficient 4^n.

The actual value is not always positive. In dimension three,

$$\det F_3(X) = 64(X-7/8)(X^2 - 27X/32 - 9/128), \qquad \det F_3(G) < 0.$$

Thus full rank and exact period cancellation hold uniformly, but they do not supply the positive-Gram nonvanishing mechanism.

Proof. Since $f(t)^{-1} = \sum_{k\geq 0} c_k t^{2k}$, $tP_r/f - D_r$ vanishes to order at least n at zero. The coefficient of $\zeta(2)$ in column $j < n$ is $\frac{3}{2}[t^j](tP_r/f - D_r) = 0$. The X coefficient is $4c_{(r-j)/2}$, a lower triangular matrix with diagonal entries 4. Its determinant is 4^n, proving exact degree.

For an entirely rational check of the dimension-three formula, the matrix is

$$F_3(X) = \begin{pmatrix} 4X-3/2 & 3/2 & 5/4 \\ 7/8 & 4X-11/4 & 7/8 \\ 2X+1 & 2 & 4X-2 \end{pmatrix}.$$

Direct expansion gives the displayed polynomial. The alternating series gives $7/8 < G < 23/25$. The lower inequality follows already from $1 - 1/9 = 8/9$; for the upper inequality use the partial sum through $j = 8$, which is less than $23/25$ by rational arithmetic. The quadratic $X^2 - 27X/32 - 9/128$ is increasing on $[7/8, 23/25]$ and its value at $23/25$ is negative. The determinant is consequently negative. □

The optimized OpenAI construction includes separate arithmetic nonvanishing and prime-divisibility arguments; our elementary rows do not inherit them. The proposition gives a reproducible high-rank family and an exact demonstration of the additional issue that remains.

67.5 Four target verdicts and the quantitative missing inputs

Transcendence of $\zeta(5)$. The new signature theorem and rational enclosures apply. The external Fauzan estimates, however, are explicit about their current reach: for $K \in 40\mathbb{N}$, $\deg Q_K = (37/40)K$,

$$0 < Q_K(\zeta(5)) \le e^{-(79/1600)K^2}, \qquad \log H(Q_K) \le (511067/40000)K^2 + o(K^2)$$

[307, Appendices C.2–C.3]. A resultant lower bound for an algebraic target of degree d requires a decay-to-height ratio exceeding $d-1$. Here that ratio is only $1975/511067 < 0.004$. It excludes degree one but supplies no degree-two exclusion. The real-root factorization does not change these numerical estimates.

Transcendence of $\zeta(3)$. Theorems 67.1–67.4 apply with $s=3$ and give full rank and a strict exponential rational enclosure. Apéry's irrationality is external and classical. Neither these bounds nor Lai's 2026 linear independence of $1, \zeta(3), \zeta(4)$ supplies a degree-uniform transcendence estimate [310]. In particular, the heights in Proposition 67.5 do not make its rational errors small enough to exclude algebraic irrationality.

Irrationality of $\zeta(7)$. Positivity, exact degree and the central-root counts now follow from the same proved theorem, rather than from analogy with $s=5$. The arithmetic side remains unproved. A September 2026 claim by Anand must not be confused with an individual formalized result: the related Lean repository states that at least one of $\zeta(7), \zeta(9), \dots, \zeta(21)$ is irrational, which does not specify $\zeta(7)$ [311]. A primary issue report identifies an inconsistent outer-prime integral in Anand's proposed transfer [312]. We do not use that preprint's claimed negative margin. Its last piece can be checked independently without a numerical integration. For the printed scalar normalization, $K < p \le 2h < 2K$ implies

$$v_p(S_K) = -2\sum_{i=1}^{h-1} \left\lfloor \frac{2i}{p} \right\rfloor = -(2h-p-1).$$

The other factorials are p-units, and the paper explicitly chooses $\gamma_p^{\mathrm{out}} = 0$ for $p > K$. Hence the scaled outer integrand on $1 < y \le 37/20$ is $37/20 - y$, not the negative expression in its Table 5. Its integral is $289/800$, whereas the table's last row integrates to $-629/50$; the discrepancy is exactly $10353/800$. This verifies a specific defect in the proposed transfer, without claiming that this single correction settles all other estimates in that preprint. For our own constructions the exact missing assertion is an integral normalization Q_n with nonzero value, degree $O(n)$ and $\log|Q_n(\zeta(7))| \le -\delta n^2$ for some $\delta > 0$. The positive representation proves nonvanishing of the unnormalized determinant; it does not supply this denominator estimate.

Transcendence of G. The mixed family in Proposition 67.6 cancels $\zeta(2)$ and has degree n. It supplies neither a uniform nonvanishing theorem nor height and decay estimates. The later OpenAI irrationality result is an external input, not a transcendence theorem. A successful transfer would need all three estimates for the same integer polynomials; full degree by itself is insufficient.

For clarity, the resultant criterion used above follows from one line of arithmetic. If θ has degree d, minimal-polynomial leading coefficient a, and conjugates θ_j, then for a nonzero value of an integer polynomial Q of degree m and height H,

$$1 \le \left| a^m \prod_{j=1}^{d} Q(\theta_j) \right|, \qquad \log|Q(\theta)| \ge -(d-1)\log H - O_\theta(m + \log(m+1)).$$

Thus a fixed positive height cost cannot prove transcendence merely by increasing determinant size. One needs improved decay-to-height ratios excluding every degree, or a different arithmetic principle. The new results here concern the structure and approximation power of full-rank families; none is counted as a new resolution of the four specified individual questions.

68 Fifteen named targets and the remaining arithmetic step

The table records fifteen separately examined targets, not fifteen new theorems. Fourteen are not resolved by this investigation. The fifteenth is a previously proved project result, included because the new transfer had to be checked against it. Entries concern the precise arithmetic question in the second column; known irrationality does not settle a transcendence target.

No.	Target	Attempt and certified output	Remaining step
1	γ irrationality	Actual reduced-denominator analysis excludes every odd Taylor-order schedule; integer-valued square kernels also checked.	A different family with primitive, nonzero integer forms tending to zero.
2	Gompertz δ irrationality	Growing-pole Padé errors and every prime in the gcd content are determined exactly.	Higher prime-power valuations and enough total cancellation.
3	$e+\pi$ irrationality	Norm and E-function transfers are tested; both fail a concrete algebraic-input hypothesis.	Control involving both exponential and period values.
4	$e\pi$ irrationality	The analogous multiplicative substitution introduces the same uncontrolled logarithm base.	An individual nonvanishing result, not the classical disjunction.
5	Catalan G transcendence	Auxiliary $\zeta(2)$ is removed in a mixed moment family; the determinant has full degree.	Uniform nonvanishing and all-degree value/height control.
6	$\zeta(3)$ transcendence	Positive rational moments, exact real-root counts, and strict exponentially shrinking rational enclosures.	Arithmetic height bounds strong enough for every algebraic degree.
7	$\zeta(5)$ transcendence	The actual full-rank pencil is analyzed; the printed decay/height ratio is computed.	Its stated bounds do not exclude even algebraic degree two.
8	$\zeta(7)$ irrationality	The positive odd-weight functional and enclosures are proved directly; an external denominator calculation is checked.	A valid integral normalization with sufficiently small value.
9	Erdős–Borwein sum transcendence	Exact decomposition into countably algebraically independent sparse blocks, with prior result credited.	A finite-to-infinite arithmetic transfer with quantitative tails.
10	Full reciprocal Fibonacci sum transcendence	Exact theta–Lambert decomposition; sparse contrast ideals are completely classified.	Exclude cancellation between the full theta and Lambert values.
11	Prime Mertens constant irrationality	Möbius–zeta expansion with an explicit exponential tail.	Control the infinite combination involving γ and odd zeta values.
12	Khinchin constant irrationality	A positive Stieltjes representation has infinitely many poles, explaining failure of the finite-kernel transfer.	An arithmetic theorem for this infinite representation.

No.	Target	Attempt and certified output	Remaining step
13	Feigenbaum constant irrationality	Algebraic period-doubling approximants have bounded absolute height but increasing uncontrolled degree.	Degree-sensitive approximation separation for the limit.
14	Glaisher constant irrationality	Functional-equation and prime-logarithm reductions are made explicit.	Control an exponential involving uncontrolled constants and an infinite prime sum.
15	$\log \mathrm{Ein}(1)$ transcendence	Previously proved via independence over finite exponential fields; the argument is restated.	No new closure counted; outside the algebraic-input norm algorithm.

The named-target proofs and formulas are in Sections 62, 66, 67, 65, and 64. No numerical precision, count of examples, or number of pages is used as evidence of irrationality or transcendence.

68.1 A quantitative gate for a genuine further advance

The determinant and approximation branches now give concrete objects to improve. For a transcendental target candidate ξ, an argument through integer polynomials requires both a nonzero value and a value bound that defeats algebraic separation at every fixed degree. For example, suppose nonzero integer polynomials P_n satisfy

$$\deg P_n = O(n), \qquad \log H(P_n) \le An^2 + o(n^2), \qquad 0 < |P_n(\xi)| \le \exp(-Bn^2 + o(n^2)).$$

For an algebraic ξ of degree d, a resultant estimate yields

$$\log |P_n(\xi)| \ge -(d-1)\log H(P_n) - O_\xi(n + \log n).$$

Indeed, the resultant with the minimal polynomial of ξ is a nonzero integer, and the other conjugate factors are bounded by $(\deg P_n + 1)H(P_n)\max(1, |\xi_j|)^{\deg P_n}$. Thus $B > (d-1)A$ excludes that degree. A single finite ratio B/A does not exclude every degree. This is why real-rootedness and fast analytic error alone do not close the transcendence targets.

The analogous linear-form gate is even more concrete: after full reduction to coprime integer coefficients, one needs a nonzero form that tends to zero. The content theorem for Gompertz identifies where prime factors occur, but does not yet provide the factorial-scale total content needed for that gate. The odd-Taylor theorem excludes one broad family outright.

68.2 Proved boundaries and research priorities

The strongest completed conclusions in this manuscript concern exact arithmetic structure, not a new famous constant. Finite moment data can leave the complete algebraic-quantile pattern undetermined; full algebraic density data can determine it. Finite algebraic logarithmic data admit a terminating entropy classifier; nested logarithms and transcendental inputs are outside its proof. Additive cyclotomic Mahler values have a computable complete ideal; an infinite sum of independent blocks is outside that finite certificate. These are proved distinctions rather than heuristic cautions.

The most direct next mathematical questions are correspondingly specific: optimize the degree under moment constraints; classify which algebraic entropy values are realizable at a fixed alphabet size; extend the finite relation certificate to nonadditive systems without merely

invoking an existing general algorithm; and obtain a genuinely stronger primitive content or determinant-height estimate for one named constant. None of these extensions is silently assumed in the present conclusions.

Validation performed. The exact entropy identities were checked independently in $\mathbb{Q}(\sqrt{3})$; beta-mixture weights and the moment example were checked with rational arithmetic. The determinant branch has exact rational Padé and mixed-moment checks. The Gompertz recurrence and its prime-support consequences were checked at finite indices in addition to their general proofs. Separate proof reviews examined the main construction, norm criterion, analytic subgroup reduction, additive dependence criterion, and determinant estimates. These checks are not a formal verification of the manuscript or a substitute for external expert refereeing.

Part IX
Sharp Thresholds and Primitive Forms

Editorial integration note. The several minimum-alphabet questions have different hypotheses. Parts X–XII strengthen the rational-entropy branch, primitive-content estimates, and formal Gamma analysis while preserving these distinctions. The present prime-grid isolation bound is extended to unrestricted auxiliary denominators in Part X.

Original part framing.

Scope and verification. This standalone manuscript supplies the statements, hypotheses, and proofs used in its arguments. It does not require the earlier 223-page or 447-page compilations. Separate proof reviews and exact algebraic checks are not a formal machine verification, and a supplied proof does not establish historical priority. The depth theorems of Baker, Gelfond–Schneider, Wüstholz, and the CM torsion literature are credited as external inputs. The Wilson–Apéry connection already appears in Prévost's 2017 work and is not claimed as a discovery here. Source checks and this investigation date from 16–17 October 2026. Publication level and priority remain matters for further expert review.

69 From examples and convergence to exact costs

The preceding manuscript established exact arithmetic classifiers for specified quantile and entropy families, together with finite Mahler relation certificates. It also located failures in several attempts to pass from rapid analytic convergence to arithmetic conclusions. The question here is more demanding: which complexity costs are unavoidable, what is the actual reduced denominator, and can a failed finite transfer be repaired within its proposed class?

Three and four are sharp entropy thresholds. Among positive algebraic probability vectors, the earlier five-atom example is reduced to the exact minimum three:

$$p = \left(\frac{1+\sqrt{3}}{4},\ 2-\sqrt{3},\ \frac{3\sqrt{3}-5}{4} \right), \qquad H_2(p) = \frac{3(\sqrt{3}-1)}{2}.$$

Theorem 70.4 proves both existence and minimality. A different optimization problem has answer four: rational probabilities with rational entropy can first include a non-dyadic entry at four atoms (Corollary 70.8). The Galois criterion in Theorem 70.12 detects transcendence from a distinct simultaneous positive conjugate, and Corollary 70.14 transfers it to quantum spectra.

Exact moment costs and elliptic degree costs. Preserving the first M arcsine moments requires weight degree at least $M+1$. At that degree the exception pattern is forced to be the grid $k/(M+1)$; deleting every exception requires exactly $M+2$ when $M \geq 1$ (Theorem 71.2). Theorem 71.3 computes the precise optimum for every admissible finite pattern. In the elliptic family, the cost depends on all Galois conjugates, including nonreal ones. Real CM cyclotomy after a quadratic twist gives $n \leq 8d^2$, and the first noncentral exception has correction degree four and weight degree seven (Theorems 72.2 and 72.3).

Primitive arithmetic, rather than an unreduced denominator. Theorem 73.1 determines the entire gcd of the classical Laguerre numerator and denominator, not just its prime support. The residue tree and its local analytic branches describe all prime powers.

For Euler's constant, Theorem 73.5 reduces the attempted even Taylor escape to at most one crossover order per large argument, with a necessary exponential cancellation. For each fixed bound, the arguments admitting any bounded primitive even-order form have natural density zero (Corollary 73.6); an infinite sparse exceptional subsequence is not excluded. For the odd-zeta trial family, Theorem 74.4 proves width $\exp(-8\log(1+\sqrt{2})n + O_s(\log n))$ together with $\log \operatorname{den} = O_s(n)$. The constants still matter: an order-n denominator bound by itself is not an irrationality criterion.

Two complete limits on proposed transfers. Theorem 75.2 excludes rational linear identities among all finite collections of full Lambert dilates and their derivatives. The same obstruction survives the theta perturbation in the reciprocal Fibonacci function. Separately, Theorem 76.1 determines the exact number of beta factors needed for a specified formal gamma isolation. It explicitly concerns the defined identity system; it does not assert that actual gamma values have no additional relations.

These are uniform results with supplied proofs. They are not obtained by counting separate numerical examples as separate conjecture solutions. The final target ledger records what remains unproved for each named constant.

70 Sharp alphabet thresholds and Galois rigidity of entropy

Throughout this section the entropy is measured in bits:

$$H_2(p) = -\sum_{i=1}^{n} p_i \log_2 p_i, \qquad p_i > 0, \qquad \sum_i p_i = 1.$$

Zero entries, if present, are deleted; alphabet size means the number of positive entries. The arithmetic dichotomy for rational probability vectors follows from classical transcendence theory and is explicitly recorded by Blakaj–Wolf [313, Theorem 2]. Here we prove two sharp small-alphabet statements, a construction in every real quadratic field, and a Galois obstruction which transfers to quantum entropy. These are not claims about the irrationality of a historically named constant. The priority of the specific refinements below has not been certified beyond the primary literature checked here.

70.1 Two elementary consequences of Baker's theorem

We use the following form of Baker's theorem [231]: real logarithms of positive algebraic numbers which are linearly independent over $\mathbb{Q}$ remain linearly independent over $\overline{\mathbb{Q}}$. Two consequences will be used repeatedly.

Lemma 70.1 (Splitting and transporting logarithmic relations)**.** *Suppose that $\theta_1, \dots, \theta_d$ are algebraic numbers independent over $\mathbb{Q}$, and each L_j is a rational linear combination of real logarithms of positive algebraic numbers. Then*

$$\sum_j \theta_j L_j = 0 \quad \Longrightarrow \quad L_j = 0 \text{ for all } j.$$

Equivalently, if $\alpha_i > 0$ are algebraic, the algebraic linear relations among $\log \alpha_1, \dots, \log \alpha_m$ are the $\overline{\mathbb{Q}}$-span of their rational linear relations. Every rational relation, after clearing denominators, is the logarithm of an identity $\prod_i \alpha_i^{k_i} = 1$, with $k_i \in \mathbb{Z}$.

Proof. Choose a $\mathbb{Q}$-basis of the finite-dimensional space generated by the logarithms. Baker's theorem makes this basis independent over $\overline{\mathbb{Q}}$. Comparing its coefficients proves both statements; in the first statement compare also the coefficients of the independent θ_j. Positivity makes exponentiation and taking real logarithms mutually inverse in the last assertion. □

Lemma 70.2 (Positive rational powers of two)**.** *If $r_i \in \mathbb{Q}$ and $\sum_{i=1}^{n} 2^{r_i} = 1$, then every r_i is an integer. In particular, if $2^r + 2^s = 1$, then $r = s = -1$.*

Proof. Take a common positive denominator d and put $v = 2^{1/d}$. Its minimal polynomial $X^d - 2$ is irreducible by Eisenstein's criterion, so $1, v, \dots, v^{d-1}$ are independent over $\mathbb{Q}$. Write each integer dr_i as $da_i + b_i$ with $0 \le b_i < d$. Then $2^{r_i} = 2^{a_i} v^{b_i}$. In the claimed sum every coefficient of a nonconstant basis vector is nonnegative, and is strictly positive if some b_i equals that vector's exponent. Independence forces all $b_i = 0$. For two terms, write the negative integral exponents as $-m, -n$, with $1 \le m \le n$. The equality $2^{n-m} + 1 = 2^n$ forces $m = n = 1$. □

70.2 The least alphabet for algebraic irrational entropy is three

Proposition 70.3 (Binary lower bound)**.** *For algebraic $0 < p < 1$,*

$$H_2(p, 1-p) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad p = \tfrac{1}{2}.$$

The algebraic value is then 1.

Proof. First suppose p is irrational and the entropy h is algebraic. The identity

$$-\log(1-p) - p \log \frac{p}{1-p} = h \log 2$$

and Lemma 70.1 imply $h \in \mathbb{Q} + \mathbb{Q}p$: otherwise $1, p, h$ would be independent and would force $\log 2 = 0$. Write $h = a + bp$ with $a, b \in \mathbb{Q}$. Splitting the coefficients of $1, p$ now gives

$$1 - p = 2^{-a}, \qquad \frac{p}{1-p} = 2^{-b}.$$

Thus both p and $1 - p$ are rational powers of two, contradicting irrationality by Lemma 70.2.

For rational $p = a/N$, take positive coprime integers $a, N - a$. The identity

$$N H_2(p, 1-p) \log 2 = \log \frac{N^N}{a^a (N-a)^{N-a}}$$

and Baker's theorem show that algebraicity forces the positive rational number in the logarithm to be an integral power of two. For each odd prime ℓ this gives

$$N v_\ell(N) = a v_\ell(a) + (N-a) v_\ell(N-a).$$

If $\ell \mid N$, coprimality implies it divides neither numerator, a contradiction. If $\ell \nmid N$, the equation prohibits it from dividing either numerator. Hence $N, a, N - a$ are all powers of two. Coprimality gives $a = N - a = 1$ and $N = 2$. The converse is immediate. □

Theorem 70.4 (Sharp three-atom construction)**.** *The least number of positive algebraic probabilities whose entropy in bits is algebraic irrational is exactly three. An explicit vector is*

$$p = \left(\frac{1+\sqrt{3}}{4},\ 2 - \sqrt{3},\ \frac{3\sqrt{3} - 5}{4} \right), \qquad H_2(p) = \frac{3(\sqrt{3} - 1)}{2}. \tag{70.1}$$

Every entry of this vector is irrational.

Proof. Put $u = 1 + \sqrt{3}$, so that $u^2 - 2u - 2 = 0$. The identity

$$(t^2 - 2t - 2)^2 = t^4 - 4t^3 + 8t + 4$$

gives, for $F(t) = t/4 + 2/t^2 + 1/t^3$,

$$F(t) - 1 = \frac{(t^2 - 2t - 2)^2}{4t^3}, \qquad F(u) = 1, \qquad F'(u) = 0.$$

Consequently the positive probabilities $(p_1, p_2, p_3) = (u/4, 2/u^2, 1/u^3)$ sum to one and satisfy $p_1 - 2p_2 - 3p_3 = 0$. Expanding their logarithms therefore gives

$$H_2(p) = 2p_1 - p_2 - (p_1 - 2p_2 - 3p_3)\log_2 u = \frac{u}{2} - \frac{2}{u^2} = \frac{3(\sqrt{3} - 1)}{2}.$$

The displayed radical formulas follow by rationalization. Positivity uses $5/3 < \sqrt{3} < 2$; the entropy is algebraic irrational. One atom has zero entropy, and Proposition 70.3 rules out two atoms. □

This improves the preceding five-atom construction to the exact minimum. The conclusion is a sharp arithmetic threshold, not a numerical search result.

Corollary 70.5 (Five irrational atoms with rational entropy)**.** *Put $A = (1+\sqrt{3})/4$, $B = 2-\sqrt{3}$ and $C = (3\sqrt{3} - 5)/4$. The five positive irrational probabilities*

$$(A, B/2, B/4, B/4, C)$$

have entropy $3/2$ bits. Thus rational entropy can occur with every probability irrational using five atoms. We do not assert that five is the minimum for this different property.

Proof. Splitting an atom of mass B according to a conditional distribution of entropy h increases total entropy by Bh, as follows directly by expanding the logarithms. Here the split proportions are $(1/2, 1/4, 1/4)$ and their entropy is $3/2$. Hence the new entropy is

$$\frac{3(\sqrt{3} - 1)}{2} + \frac{3}{2}(2 - \sqrt{3}) = \frac{3}{2}.$$

Every displayed atom is a nonzero rational multiple of an irrational entry of (70.1). □

70.3 Complete rational classification for three atoms

Lemma 70.6 (Rational valuation test)**.** *Let $a_1, \dots, a_n$ be positive integers with sum N. For $p_i = a_i/N$, algebraicity of $H_2(p)$ is equivalent to*

$$N v_\ell(N) = \sum_i a_i v_\ell(a_i) \quad \text{for every odd prime } \ell. \tag{70.2}$$

If it holds, the entropy is rational.

Proof. Use $N H_2(p) \log 2 = \log R$, where $R = N^N / \prod_i a_i^{a_i} \in \mathbb{Q}_{>0}$. Baker's theorem implies that algebraicity of the ratio $\log R / \log 2$ forces it to be rational. If a positive rational number is a rational power of two, comparison of prime valuations after clearing the exponent's denominator shows it is an integral power of two. Thus the condition is precisely that all odd valuations of R vanish. Sufficiency is the same identity read backwards. □

Theorem 70.7 (Rational ternary classification)**.** *For positive rational p_1, p_2, p_3 with sum one,*

$$H_2(p_1, p_2, p_3) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad (p_1, p_2, p_3) \text{ is a permutation of } (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The algebraic entropy is $3/2$; every other positive rational ternary distribution has transcendental entropy in bits.

Proof. Write $p = (a/N, b/N, c/N)$ with a, b, c positive integers, $a+b+c = N$ and $\gcd(a, b, c) = 1$. Assume the valuation criterion of Lemma 70.6. An odd prime not dividing N cannot divide any of a, b, c, by (70.2).

Fix an odd prime $\ell \mid N$. At least one numerator is not divisible by ℓ, by primitivity. At least two are not divisible by ℓ: otherwise reducing their sum modulo ℓ would contradict $\ell \mid N$. At least one numerator is divisible by ℓ, since the left side of (70.2) is positive. Exactly one numerator therefore contains ℓ. If that numerator is a, put $t = v_\ell(N)$ and $s = v_\ell(a)$. The valuation identity reads

$$\frac{a}{N} = \frac{t}{s}, \qquad s > t \geq 1.$$

The reduced denominator of a/N is at most s, and $s \leq \log_\ell a < \log_3 N < \sqrt{N}$. The last strict inequality holds for every $N > 0$ for which the logarithm is positive: indeed $\log N/\sqrt{N} \leq 2/e < \log 3$.

If odd primes occurred in two different numerators, the reduced denominators d_1, d_2 of those two probabilities would both be smaller than $\sqrt{N}$. Since the third probability is one minus their sum, its denominator divides $\operatorname{lcm}(d_1, d_2)$. Primitivity says that N is the least common denominator of all three probabilities, hence

$$N = \operatorname{lcm}(d_1, d_2) \leq d_1 d_2 < N,$$

a contradiction. Therefore all odd primes, if any exist, occur in only one numerator. Relabel it a, so that $b = 2^u$ and $c = 2^v$.

We now exclude this remaining possibility. Write $a = 2^h A$ and $N = 2^k B$, where A, B are odd. They are greater than one in the present case. The odd valuation identities give $A^a = B^N$. Reduce $a/N = t/s$, with $\gcd(s, t) = 1$ and $s > t$. Then $A^t = B^s$, so unique factorization gives an odd integer $C > 1$ with

$$A = C^s, \qquad B = C^t.$$

The equality $sa = tN$ becomes

$$sC^{s-t} = t2^{k-h}.$$

The left side is greater than t, so $j = k - h > 0$. Coprimality forces $s = 2^l$ for some $l \geq 1$; t is odd. Taking the 2-adic valuation gives $j = l$, leaving

$$t = C^d, \qquad d = s - t = 2^l - t > 0 \text{ odd}, \qquad C^d + d = 2^l. \tag{70.3}$$

Furthermore

$$b + c = N - a = 2^h C^t d.$$

If $u = v$, the left side is a power of two, impossible since $C > 1$. Otherwise comparison of odd parts of the sum of two powers of two gives an integer $w \geq 1$ with

$$C^t d = 1 + 2^w. \tag{70.4}$$

If $l \geq 3$, the oddness of C, d, t and (70.3) imply $t \equiv C \pmod 8$ and $d \equiv -C \pmod 8$. Hence $C^t d \equiv -C^2 \equiv 7 \pmod 8$, whereas $1 + 2^w$ is congruent to 3, 5 or 1 modulo 8. If $l = 2$, equation

$C^d + d = 4$ allows only $C = 3, d = 1$, which would require $27 = 1 + 2^w$, also impossible. If $l = 1$, already $C^d + d \geq 4 > 2$. Thus odd primes do not occur.

It follows that N, a, b, c are powers of two, so all three probabilities are negative integral powers of two. Order their exponents as $1 \leq r_1 \leq r_2 \leq r_3$. If $r_1 \geq 2$, their sum is at most $3/4$. Thus $r_1 = 1$, and the remaining two terms sum to $1/2$. Lemma 70.2, or direct subtraction, gives $r_2 = r_3 = 2$. The converse follows by evaluating the entropy. □

Corollary 70.8 (A second sharp alphabet threshold)**.** *The least alphabet size for a rational probability vector with rational entropy and at least one non-dyadic probability is four. An example is*

$$p = (\tfrac{3}{8}, \tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{24}), \qquad H_2(p) = \tfrac{7}{4}.$$

Proof. Proposition 70.3 and Theorem 70.7 exclude smaller alphabets. For the displayed denominator-24 numerators $(9, 6, 8, 1)$, the odd prime condition is $24 = 9 \cdot 2 + 6 \cdot 1$ at 3. The 2-adic valuation gives the entropy $3 - (6 + 24)/24 = 7/4$. The example was available in the preceding manuscript; its minimality is the additional conclusion proved here. □

70.4 Algebraic irrational entropy in every real quadratic field

Proposition 70.9 (Dyadic critical-point construction)**.** *Suppose $u > 0$ is algebraic and $F(t) = \sum_i 2^{-k_i} t^{m_i}$, with $k_i, m_i \in \mathbb{Z}$, satisfies $F(u) = 1$ and $F'(u) = 0$. Then the positive probabilities $p_i = 2^{-k_i} u^{m_i}$ have algebraic entropy $H_2(p) = \sum_i k_i p_i$. The same conclusion holds after splitting each positive dyadic coefficient of a Laurent polynomial into a finite sum of powers of two.*

Proof. The derivative identity is $\sum_i m_i p_i = 0$. Substitute $\log_2 p_i = -k_i + m_i \log_2 u$ into the entropy. A positive dyadic rational has a finite binary expansion, allowing the indicated split. □

Theorem 70.10 (Every real quadratic field occurs)**.** *Let $D > 1$ be a squarefree integer. There is a finite positive probability vector with all entries in $\mathbb{Q}(\sqrt{D})$ whose entropy in bits generates $\mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$. The construction is effective using rational arithmetic and real-algebraic comparisons.*

Proof. Choose a positive dyadic rational q such that $1 < q^2 D < 3$, and set

$$b = q^2 D - 1 \in (0, 2), \qquad u = 1 + q\sqrt{D}.$$

Thus $u^2 - 2u - b = 0$. The Laurent polynomial

$$F(t) = \frac{t}{4} + \frac{1 - b/2}{t} + \frac{b}{t^2} + \frac{b^2}{4t^3}$$

has strictly positive dyadic coefficients and satisfies

$$F(t) - 1 = \frac{(t^2 - 2t - b)^2}{4t^3}.$$

Split its coefficients into powers of two, leaving the first monomial $t/4$ intact. Proposition 70.9 gives a positive probability vector with entropy $h \in \mathbb{Q}(\sqrt{D})$ and an irrational atom $u/4$. If h is irrational, the construction is done. If h is rational, split this atom into two equal atoms. Such a split increases entropy by $u/4$, so the new entropy is irrational and still belongs to $\mathbb{Q}(\sqrt{D})$. Any irrational element of a quadratic field generates that field. Choosing q, splitting the dyadic coefficients and testing the rationality of h are all effective. □

Proposition 70.11 (A genuine limit on the single-parameter construction)**.** *Let $F \in \mathbb{Q}[t, t^{-1}]$ have positive coefficients, with some nonzero exponent. If an algebraic $u > 0$ satisfies $F(u) = 1$ and $F'(u) = 0$, then u has no other positive real conjugate over $\mathbb{Q}$.*

Proof. For $G(x) = F(e^x)$ one has $G''(x) = \sum_m c_m m^2 e^{mx} > 0$. Consequently G has at most one critical point. Any positive conjugate v of u also satisfies $F(v) = 1$ and $F'(v) = 0$, and would give a second critical point $\log v$ unless $v = u$. □

Thus this construction does not establish that every positive algebraic number is itself an entropy value, nor does it directly extend to a prescribed totally positive primitive element of a higher-degree field.

70.5 Galois rigidity and quantum transfer

Theorem 70.12 (Positive-conjugate obstruction)**.** *Let $p = (p_1, \dots, p_n)$ be a positive algebraic probability vector. If $H_2(p)$ is algebraic and a field embedding σ of a number field containing its entries has $\sigma(p_i) > 0$ for every i, then*

$$\sigma(p_i) = p_i \quad \textit{for all } i.$$

Equivalently, a distinct simultaneous positive real conjugate of p certifies transcendence of its entropy in bits.

Proof. Write $h = H_2(p) \in \overline{\mathbb{Q}}$. The vector $(p_1, \dots, p_n, h)$ is an algebraic relation among $\log p_1, \dots, \log p_n, \log 2$. By Lemma 70.1, it is an algebraic linear combination of integer relation vectors k satisfying

$$\prod_i p_i^{k_i} 2^{k_0} = 1.$$

Apply σ to the bases in this multiplicative identity. Since $q_i = \sigma(p_i) > 0$ and $\sigma(2) = 2$, taking real logarithms gives the same integer relation among $\log q_1, \dots, \log q_n, \log 2$. Use the same algebraic linear combination of these relations, without conjugating its coefficients. It follows that

$$\sum_i p_i \log q_i + h \log 2 = 0.$$

Subtracting the original entropy identity yields $\sum_i p_i \log(p_i/q_i) = 0$. Both p and q are positive probability vectors, since $\sum q_i = \sigma(1) = 1$. The strict equality case of the elementary relative-entropy inequality now gives $p = q$. Explicitly,

$$\sum_i p_i \log \frac{q_i}{p_i} \le \log \left(\sum_i p_i \frac{q_i}{p_i} \right) = 0,$$

and strict concavity of the logarithm makes equality possible only when every ratio q_i/p_i is the same; normalization makes it one. □

Corollary 70.13 (Totally positive probabilities)**.** *If every entry of a positive algebraic probability vector is totally positive over $\mathbb{Q}$, and at least one entry is irrational, its entropy in bits is transcendental.*

Proof. Use a normal closure of the common number field. Every automorphism sends every entry to a positive real number. Were the entropy algebraic, Theorem 70.12 would make all entries fixed by the entire Galois group, hence rational. □

Corollary 70.14 (Rational characteristic polynomials)**.** *Let ρ be a finite-dimensional density matrix whose characteristic polynomial has rational coefficients. If any eigenvalue of ρ is irrational, its von Neumann entropy in bits is transcendental. Consequently its entropy is algebraic if and only if all nonzero eigenvalues are rational and satisfy the odd-prime valuation test in Lemma 70.6; in that case it is rational. This applies, in particular, to rational density matrices.*

Proof. The eigenvalues are nonnegative algebraic numbers and sum to one. After deleting zeros, every algebraic conjugate of any remaining eigenvalue is a nonzero root of the rational characteristic polynomial, hence another positive eigenvalue. Thus every remaining eigenvalue is totally positive. Apply Corollary 70.13, and then Lemma 70.6 in the rational-spectrum case. □

For example, the rational positive definite density matrix

$$\rho = \frac{1}{8}\begin{pmatrix} 5 & 1 \\ 1 & 3 \end{pmatrix}$$

has eigenvalues $(4 \pm \sqrt{2})/8$, so its entropy in bits is transcendental. The Galois statement also applies in arbitrary rank and does not require a formula for the eigenvalues. In contrast, the diagonal density matrix with spectrum (70.1) has algebraic irrational entropy; its characteristic polynomial cannot have rational coefficients. This distinction identifies an arithmetic hypothesis that survives the passage from classical probabilities to quantum spectra.

71 Sharp degree costs under exact moment constraints

The inverse arcsine construction from the preceding manuscript can be sharpened from an existence theorem to an exact minimum-degree algorithm. Its first two degree thresholds have closed forms.

On $(0,1)$, let $\rho_0(x) = 1/(\pi\sqrt{x(1-x)})$. For $P \in \mathbb{Q}[x]$, define

$$\mathcal{L}(P) = \frac{(1-2x)P + 2x(1-x)P'}{2}, \qquad w_P = 1 + \mathcal{L}(P),$$

and suppose $P \neq 0$ and $w_P > 0$ on $[0,1]$. Then $w_P\rho_0$ is a probability density, with CDF

$$F_P(x) = \frac{2\arcsin\sqrt{x} + \sqrt{x(1-x)}P(x)}{\pi}.$$

Every fixed nonzero rational polynomial becomes admissible after multiplication by a sufficiently small nonzero rational scalar. For a polynomial of degree d,

$$\deg w_P = d + 1, \tag{71.1}$$

because its leading coefficient a becomes $-(d+1)a$ in degree $d+1$ under $\mathcal{L}$.

We use the established arithmetic classifier from the preceding manuscript: for algebraic $q \in (0,1)$,

$$F_P^{-1}(q) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad q \in \mathbb{Q} \quad \text{and} \quad P\left(\sin^2\frac{\pi q}{2}\right) = 0. \tag{71.2}$$

Its transcendence input is Baker's theorem for algebraic linear forms in logarithms, together with Gelfond–Schneider in the irrational-algebraic exponent case. In outline, write the arcsine term as the logarithm of an algebraic number when x is algebraic. The CDF equation makes

its difference from an algebraic multiple of $\log(-1)$ algebraic. Baker's theorem forces this difference to vanish; hence $P(x) = 0$. Gelfond–Schneider then forces the algebraic level q to be rational. The converse is immediate. We recall this established classifier solely to specify the input to the new degree problem; see also the classical logarithmic results [327, 328].

Lemma 71.1 (Moment constraints are orthogonality constraints)**.** *For $M \geq 0$, the density $w_P \rho_0$ has the same first M raw moments as ρ_0 if and only if*

$$\int_0^1 x^j P(x) \sqrt{x(1-x)}\, dx = 0 \qquad (0 \leq j < M). \tag{71.3}$$

These are homogeneous rational linear conditions on the coefficients of P.

Proof. The difference of CDFs is $\sqrt{x(1-x)}P(x)/\pi$, which vanishes at both endpoints. Integration by parts therefore gives, for $k \geq 1$,

$$\int_0^1 x^k (w_P - 1)\rho_0\, dx = -\frac{k}{\pi} \int_0^1 x^{k-1} P(x) \sqrt{x(1-x)}\, dx.$$

This proves the equivalence. Each monomial integral is a beta integral $B(j + 3/2, 3/2)$, a rational multiple of π. Thus division by π makes every constraint rational. □

Theorem 71.2 (Two sharp thresholds)**.** *Fix $M \geq 0$. Among nonconstant positive rational polynomial tilts preserving the first M arcsine moments, the minimum weight degree is exactly $M + 1$. At this degree the correction is uniquely determined up to a nonzero scalar:*

$$P(x) = c\, U_M(2x - 1),$$

where U_M is the Chebyshev polynomial of the second kind. Its full algebraic-quantile exception set is

$$G_M = \{k/(M+1) : 1 \leq k \leq M\}. \tag{71.4}$$

For $M \geq 1$, requiring the exception set to be empty raises the exact minimum weight degree by one, to

$$\boxed{M + 2}. \tag{71.5}$$

Both minima can be attained arbitrarily close to the unperturbed weight 1 in any prescribed finite C^k norm.

Proof. The positive weight $\sqrt{x(1-x)}$ makes its polynomial inner product positive definite. A nonzero polynomial of degree less than M cannot be orthogonal to every polynomial of degree less than M, since it would be orthogonal to itself. At degree M, the orthogonal complement is one-dimensional. The change $u = 2x - 1$ identifies its generator with $U_M(u)$, whose roots are $\cos(k\pi/(M+1))$, $1 \leq k \leq M$. Equations (71.1) and (71.2) prove the first minimum and the exact grid (71.4).

For $M \geq 1$ this grid is nonempty, so an empty exception set requires correction degree at least $M + 1$. The polynomials

$$P_t(x) = U_{M+1}(2x - 1) + t U_M(2x - 1), \qquad t \in \mathbb{Q},$$

all satisfy (71.3). Consecutive Chebyshev polynomials have no common zero, as follows from their three-term recurrence and $U_0 = 1$. Therefore each candidate algebraic root α excludes at most one parameter t. Only finitely many candidates need be considered: if a rational polynomial of degree $M + 1$ vanishes at $\sin^2(\pi q/2)$ with rational q, the order N of $e^{i\pi q}$ satisfies $\varphi(N) \leq 2(M + 1)$, and hence $N \leq 8(M + 1)^2$. The last inequality follows from $N \leq 2\varphi(N)^2$. Avoid the finitely many forbidden rational values of t. The resulting polynomial has no exception, proving (71.5). Finally multiply the chosen correction by an arbitrarily small nonzero rational scalar. Orthogonality, degree, and its root set stay unchanged, whereas positivity and the asserted norm closeness follow. □

71.1 An algorithm for the exact optimum of every admissible pattern

For $N \geq 3$, put

$$C_N = \{2j/N : 0 < j < N/2,\ \gcd(j, N) = 1\}, \qquad \Psi_N(X) = \prod_{q \in C_N} \left(X - \sin^2 \frac{\pi q}{2} \right).$$

Classical real cyclotomy gives distinct irreducible monic polynomials $\Psi_N \in \mathbb{Q}[X]$, of degree $\varphi(N)/2$. The admissible finite exception sets are exactly finite unions of the disjoint blocks C_N. For such a set S, write

$$H_S = \prod_{C_N \subset S} \Psi_N, \qquad s = \deg H_S = |S|; \qquad H_\varnothing = 1.$$

Theorem 71.3 (Exact minimum-degree algorithm with prescribed moments)**.** *Given $M \geq 0$ and an admissible finite exception set S, the following finite algorithm computes the exact minimum weight degree and constructs a strictly positive rational polynomial tilt attaining it while preserving the first M arcsine moments.*

For $d = 0, 1, \ldots, 2s + M + 1$, compute by rational linear algebra

$$V_d = \{P \in \mathbb{Q}[x] : \deg P \leq d,\ H_S \mid P,\ P \text{ satisfies } (71.3)\}.$$

If $V_d = \{0\}$, reject this degree. Otherwise form the monic greatest common divisor G_d of any rational basis of V_d. Accept the first degree for which every real cyclotomic factor Ψ_N of G_d also divides H_S. The exact minimum weight degree is then $d + 1$.

All tests are finite: at degree $d \geq 1$, it is enough to test $3 \leq N \leq 8d^2$ with $\varphi(N) \leq 2d$. At $d = 0$ there are no such factors. An attaining polynomial is obtained by enumerating rational coefficient combinations of the basis until none is divisible by an unselected Ψ_N, and then scaling by a sufficiently small nonzero rational number.

Proof. By (71.2), a correction with all exceptions in S and containing S must be a nonzero member of V_d and have no real cyclotomic factor outside H_S. If such a factor divides G_d, it divides every member of V_d, so degree d is impossible.

Conversely suppose no unwanted cyclotomic factor divides G_d. For each of the finitely many unwanted factors of degree at most d, its divisibility condition cuts out a proper rational linear subspace of V_d. A finite-dimensional vector space over the infinite field $\mathbb{Q}$ is not the union of finitely many proper linear subspaces. Thus there is a nonzero $P \in V_d$ avoiding all these factors. Rational enumeration finds one, and the classifier gives its exact exception set. This proves both the test and its minimality. Positivity follows after scaling without changing the constraints or exceptions.

It remains to prove the stated finite stopping bound. Let R_j be the monic orthogonal polynomial of degree j for the positive measure

$$H_S(x)^2 \sqrt{x(1-x)}\, dx \quad (0 < x < 1).$$

Its moments divided by π are rational, so each $R_j \in \mathbb{Q}[x]$. Positive definiteness gives its usual three-term recurrence with nonzero subdiagonal coefficients; consequently R_M and R_{M+1} are coprime. The polynomial

$$P_t = H_S^2(R_{M+1} + tR_M), \qquad t \in \mathbb{Q},$$

has degree $2s + M + 1$ and satisfies every constraint in (71.3). Outside the selected factors of H_S, each possible cyclotomic root forbids at most one t, by coprimality. Avoiding finitely many values therefore gives exactly S. The search must terminate within the asserted range. □

The result computes an exact optimum, not merely an upper bound. For instance, if only the first moment is fixed and no algebraic quantile exception is allowed, the sharp weight degree is three: $P = x^2 - 5/16$ has the required orthogonality, and its only root in $(0,1)$ is $\sqrt{5}/4$. Its other conjugate is negative, whereas every conjugate of a real cyclotomic quantile coordinate lies in $(0,1)$. It therefore has no cyclotomic factor. Scaling this P makes a positive tilt with mean exactly $1/2$ and a transcendental quantile at every algebraic level, including the median.

72 Inverse elliptic exceptions, exact degree cost, and a sharp first threshold

The preceding project paper gave a finite torsion test for a fixed elliptic correction. Here we solve the inverse problem, account for nonreal Galois conjugates in its exact degree cost, and identify the first degree at which a noncentral exception can occur. The uniform torsion bound used below is a consequence of classical real cyclotomy for CM curves; it is not a new bound for CM torsion.

72.1 The family and the previously established classifier

Let

$$E : y^2 = 1 - x^4, \quad O = (-1, 0), \qquad I(x) = \int_{-1}^{x} \frac{dt}{\sqrt{1-t^4}}, \quad \Omega = I(1).$$

The smooth projective curve is elliptic with identity O. Its upper real arc has elliptic logarithm $I(x)$, and its real period is 2Ω. For a nonzero rational polynomial P and a sufficiently small nonzero rational ϵ, set

$$L_E(P)(x) = (1-x^4)P'(x) - 2x^3P(x), \tag{72.1}$$
$$w_P(x) = 1 + \epsilon L_E(P)(x) > 0 \quad (-1 \le x \le 1),$$
$$F_P(x) = \frac{I(x) + \epsilon\sqrt{1-x^4}P(x)}{\Omega}. \tag{72.2}$$

This is the CDF of the density $w_P(x)/(\Omega\sqrt{1-x^4})$. For $q \in \mathbb{Q} \cap (0,1)$, denote by ξ_q the unique point in $(-1,1)$ satisfying $I(\xi_q) = q\Omega$. Elliptic division implies that ξ_q is algebraic. The order of $T_q = (\xi_q, \sqrt{1-\xi_q^4})$ is

$$n(q) = \operatorname{den}(q/2). \tag{72.3}$$

Here den means the denominator in lowest terms.

We recall explicitly the previously proved classification:

$$\{q \in \overline{\mathbb{Q}} \cap (0,1) : F_P^{-1}(q) \in \overline{\mathbb{Q}}\} = \{q \in \mathbb{Q} \cap (0,1) : P(\xi_q) = 0\}. \tag{72.4}$$

Indeed, if q and $x = F_P^{-1}(q)$ are algebraic, then $I(x) - q\Omega = -\epsilon\sqrt{1-x^4}P(x)$ is algebraic. The analytic subgroup theorem implies that an algebraic linear combination of elliptic logarithms of algebraic points is either zero or transcendental, so $P(x) = 0$. The real elliptic Gelfond–Schneider theorem then gives $I(x)/\Omega \in \mathbb{Q}$. Both consequences follow from the analytic subgroup theorem in $\mathbb{G}_a \times E^2$ and E^2, respectively [286, 314]. Conversely $P(\xi_q) = 0$ gives $F_P(\xi_q) = q$. We use (72.4) as an established input, not as an additional discovery of this section.

72.2 Galois packets and the inverse problem

Let $m_q \in \mathbb{Q}[X]$ be the monic minimal polynomial of ξ_q. Define its real packet by

$$\mathcal{C}(q) = \{q' \in \mathbb{Q} \cap (0,1) : m_q(\xi_{q'}) = 0\}. \tag{72.5}$$

These are finite equivalence classes: two packets are equal exactly when their minimal polynomials agree. They need not have cardinality $\deg m_q$, since some conjugates can be nonreal or lie outside the probability interval in the x coordinate.

Theorem 72.1 (Exact inverse elliptic realization)**.** *A finite set $S \subset \overline{\mathbb{Q}} \cap (0,1)$ occurs as the full exception set on the left side of (72.4) for a nonzero rational polynomial correction if and only if S is a finite union of packets $\mathcal{C}(q)$. Write*

$$H_S = \prod_{\mathcal{C} \subset S} m_{\mathcal{C}}, \qquad D(S) = \deg H_S = \sum_{\mathcal{C} \subset S} \deg m_{\mathcal{C}},$$

where one minimal polynomial is included for each distinct packet; for $S = \varnothing$, take $H_S = 1$ and $D(S) = 0$. Then the exact minimum correction degree is $D(S)$, and the exact minimum weight degree is

$$\boxed{\deg w_P = D(S) + 3}. \tag{72.6}$$

The minimum is attained by $P = H_S$. The resulting weight can be made arbitrarily close to 1 in every prescribed finite C^k norm by choosing ϵ sufficiently small and nonzero.

Proof. If an exception q occurs, m_q divides P. Every real root $x \in (-1,1)$ of m_q is the coordinate of a torsion point: choose an embedding carrying ξ_q to x and extend it to the coordinates of T_q. Its image has coordinates (x, y), with $y = \pm\sqrt{1-x^4}$, and still has order $n(q)$. Changing the sign of y is group inversion. Thus the upper-arc point is torsion, its level $I(x)/\Omega$ is rational, and the entire packet occurs. Distinct minimal polynomials are coprime. Consequently $H_S \mid P$ and $\deg P \geq D(S)$.

Conversely the roots of H_S in the interval give exactly the selected packets, so (72.4) proves exact realization. The polynomial $L_E(H_S)$ is bounded on the compact interval; taking $0 < |\epsilon| < 1/(1 + \max |L_E(H_S)|)$ ensures strict positivity. For a polynomial P of degree d and leading coefficient $a \neq 0$, $L_E(P)$ has leading term $-(d+2)ax^{d+3}$. Hence w_P has degree exactly $d+3$, including when $d = 0$. This proves both minimality assertions. Every derivative of $w_P - 1$ is proportional to ϵ, which proves the last claim. □

72.3 A degree-only bound: real cyclotomy survives the transfer

The real cyclotomy theorem of Aoki and Bourdon–Clark–Stankewicz states that if an elliptic curve over a real number field F has complex multiplication by $\mathbb{Q}(i)$ and an F-rational point of order n, then

$$\mathbb{Q}(\zeta_n)^+ \subset F, \qquad \mathbb{Q}(\zeta_n) \subset F(i).$$

Here $\mathbb{Q}(\zeta_n)^+$ is the maximal real cyclotomic subfield. We use the version in [315, Theorem 4.12]. For curves with CM by the maximal order $\mathbb{Z}[i]$, of discriminant -4, the same paper's square-root SPY bound gives, for odd $n \geq 3$,

$$\varphi(n)^2 \leq 4[F : \mathbb{Q}] \tag{72.7}$$

because the real field does not contain $\mathbb{Q}(i)$ and $\gcd(n, -4) = 1$ [315, Theorem 4.19].

Theorem 72.2 (Elliptic degree certificate)**.** *For $q \in \mathbb{Q} \cap (0,1)$, put $d = [\mathbb{Q}(\xi_q) : \mathbb{Q}]$ and $n = n(q)$. Then*

$$\mathbb{Q}(\zeta_n)^+ \subset \mathbb{Q}(\xi_q), \qquad \varphi(n) \le 2d, \qquad \boxed{n \le 8d^2}. \tag{72.8}$$

For the odd part n_{odd} of n, one also has $\varphi(n_{\mathrm{odd}})^2 \le 4d$ whenever $n_{\mathrm{odd}} > 1$. Consequently every exceptional level for a correction P of positive degree D belongs to the finite list

$$q = 2j/n, \quad 3 \le n \le 8D^2, \quad 0 < j < n/2, \quad \gcd(j,n) = 1, \quad \varphi(n) \le 2D. \tag{72.9}$$

Thus elliptic division and exact substitution into P give a second terminating classifier, with an a priori order cutoff depending only on the correction degree.

Proof. The explicit change of variables

$$U = \frac{2(1-x)}{1+x}, \qquad V = \frac{4y}{(1+x)^2} \tag{72.10}$$

is a rational isomorphism from (E, O) to $E' : V^2 = U^3 + 4U$, carrying O to infinity. It identifies the coordinate fields $\mathbb{Q}(x) = \mathbb{Q}(U)$. This Weierstrass curve has CM by $\mathbb{Z}[i]$ via $(U, V) \mapsto (-U, iV)$. Let $U_0 = 2(1-\xi_q)/(1+\xi_q) > 0$, set $F = \mathbb{Q}(\xi_q)$ and $a = U_0^3 + 4U_0 > 0$. The quadratic twist

$$aZ^2 = U^3 + 4U$$

is defined over the real number field F, has the same geometric CM, and contains the F-rational point $(U_0, 1)$ of order n. The order is unchanged by its isomorphism to E' over $F(\sqrt{a})$. Applying real cyclotomy to this twist gives the two field inclusions and $\varphi(n) \le [F(i) : \mathbb{Q}] = 2d$.

For completeness, $n \le 2\varphi(n)^2$ follows directly from

$$\frac{\varphi(n)^2}{n} = \prod_{p^a \| n} p^{a-2}(p-1)^2 \ge \frac{1}{2}.$$

Each odd-prime factor is at least one, and the factor at 2 is at least $1/2$. This proves the stated quadratic cutoff. Multiplying the twisted point by the 2-part of n produces an F-rational point of order n_{odd}, to which (72.7) applies. Finally a root of P has degree at most D, so (72.9) follows from (72.3). Division points on an elliptic curve over a number field are computable by division polynomials and exact algebraic arithmetic. Selecting the upper real point with logarithm $2j\Omega/n$ uses certified real isolation among an already finite algebraic list. No unrestricted numerical zero test is required. □

72.4 A sharp first noncentral exception

The packet cost is genuinely different from the number of real exceptions. The following result determines its first nontrivial case.

Theorem 72.3 (Degree seven is the first noncentral weight)**.** *In the family (72.2), every correction of degree at most three has exception set either $\varnothing$ or $\{1/2\}$. A noncentral exception first becomes possible at correction degree four, or equivalently weight degree seven. More precisely,*

$$P_+(x) = x^4 - 2x^3 - 2x + 1$$

has exactly the exception $q = 2/3$, and $P_-(x) = P_+(-x)$ has exactly the exception $q = 1/3$. Each is degree-optimal for its singleton set. For $P = P_+$ the explicit choice $\epsilon = 1/64$ is valid.

Proof. First determine the real points of order three on E'. Its third division polynomial is

$$\psi_3(U) = 3U^4 + 24U^2 - 16.$$

After (72.10),

$$(1+x)^4\psi_3\left(\frac{2(1-x)}{1+x}\right) = 128(x^4 - 2x^3 - 2x + 1).$$

Writing $t = x + x^{-1}$ transforms $P_+(x) = 0$ into $t^2 - 2t - 2 = 0$. For $t = 1 + \sqrt{3} > 2$ there are two positive reciprocal real roots, exactly one in $(0, 1)$; for $t = 1 - \sqrt{3} \in (-2, 2)$ the two roots are nonreal. The polynomial is irreducible over $\mathbb{Q}$: t has degree two and $t^2 - 4 = 2\sqrt{3}$ is not a square in $\mathbb{Q}(\sqrt{3})$, since $(a + b\sqrt{3})^2 = 2\sqrt{3}$ with $a, b \in \mathbb{Q}$ would give $a^2 + 3b^2 = 0$ and $2ab = 2$. The unique root in $(-1, 1)$ is the upper real point of order three, whose arc coordinate is $q = 2/3$. Its negative has level $1/3$ by $I(-x) = \Omega - I(x)$, and P_- is its minimal polynomial.

Now suppose ξ_q has degree $d \le 3$ and its point has order n. The odd-part bound in Theorem 72.2 yields $\varphi(n_{\mathrm{odd}})^2 \le 12$ if the odd part is nontrivial, so $n_{\mathrm{odd}} = 3$. In that case a multiple of the point has order three. Its x coordinate belongs to $\mathbb{Q}(\xi_q)$: the x coordinate of multiplication by an integer is a rational function of x on this degree-two quotient by group inversion. That coordinate is the degree-four number just found, a contradiction to $d \le 3$. Thus n is a power of two. Real cyclotomy gives $\varphi(n) \le 6$, hence $n \le 8$. Order one and order two give arc endpoints; order four gives only $\xi_{1/2} = 0$ on the open upper arc.

For order eight, doubling gives one of the two order-four points $(U, V) = (2, \pm 4)$ on E'. The doubling formula

$$U([2]T) = \frac{(U(T)^2 - 4)^2}{4U(T)(U(T)^2 + 4)}$$

therefore gives, after the same substitution, the equation

$$x^4 + 2x^2 - 1 = 0.$$

Its real roots are $\pm\sqrt{\sqrt{2} - 1}$, both of degree four: $\sqrt{2} - 1$ has norm -1 in $\mathbb{Q}(\sqrt{2})$ and hence is not a square in that real quadratic field. This excludes order eight when $d \le 3$ and proves the lower threshold.

Finally

$$L_E(P_+) = -6x^7 + 10x^6 + 6x^4 + 2x^3 - 6x^2 - 2.$$

The sum of the absolute values of its coefficients is 32, so $1 + L_E(P_+)/64 \ge 1/2$ on $[-1, 1]$. Classification (72.4) proves the claimed singleton exception, and Theorem 72.1 proves exact optimality. □

Corollary 72.4 (The two quartiles form one forced packet)**.** *The set $\{1/4, 3/4\}$ is one real Galois packet, with minimal polynomial $P_8 = x^4 + 2x^2 - 1$ and minimum weight degree seven. It can be the full exception set, although its two quantiles are opposites and the median of the perturbed law is transcendental. Neither singleton $\{1/4\}$ nor $\{3/4\}$ is realizable in this family.*

Proof. The order-eight calculation in the theorem identifies the two interior real roots as $\xi_{1/4} = -\sqrt{\sqrt{2} - 1}$ and $\xi_{3/4} = \sqrt{\sqrt{2} - 1}$, with irreducible polynomial P_8. Its other two roots are nonreal, so the real packet is exactly the stated pair. The inverse theorem gives the degree and the impossibility of splitting the packet. Taking $P = P_8$ gives exactly these exceptions; since $P_8(0) = -1$, the classifier excludes an algebraic median. For example $\epsilon = 1/48$ is positive and valid: the coefficient absolute-value sum of $L_E(P_8) = -6x^7 - 8x^5 + 6x^3 + 4x$ is 24. □

For comparison, the singleton median has minimal polynomial x, so it costs one correction degree and four weight degrees. The singleton $\{2/3\}$ costs four correction degrees and seven weight degrees because its other Galois conjugates must also be present in any rational polynomial, even though they do not give real probability levels. This distinction is absent from the elementary count of visible real exceptions. The results here solve an inverse and minimal-degree problem for a specified elliptic probability family; they do not settle the arithmetic of unrelated classical constants.

73 Primitive Laguerre content and the single remaining Taylor crossover

This section returns to two named targets: Euler's constant γ and the Euler–Gompertz constant $\delta = \int_0^\infty e^{-t}/(1+t)\,dt$. The classical Laguerre approximants themselves are not new [316]. Nor is their congruence-preserving property: for parameter one, it was explicitly recorded by Bala in 2022 [317]. Our improvement within the present investigation is to determine their *entire primitive content*, including all prime powers, and to identify the local analytic objects controlling those powers. For γ we isolate the one even truncation order per argument at which the previous odd-order obstruction can possibly be evaded. Neither result proves either constant irrational.

73.1 The complete content of a shifted Laguerre family

For a positive integer A, put

$$\delta_A = \int_0^\infty \frac{e^{-t}}{1+At}\,dt, \qquad Q_n(A) = \sum_{j=0}^{n} j!\binom{n}{j}^2 A^j = A^n n! L_n(-1/A).$$

Define $P_0(A) = 0$, $P_1(A) = 1$, and apply the recurrence

$$U_n = (1 + (2n-1)A)U_{n-1} - (n-1)^2 A^2 U_{n-2} \qquad (n \geq 2) \tag{73.1}$$

to both P_n and Q_n. Write $g_n(A) = \gcd(P_n(A), Q_n(A))$.

Theorem 73.1 (Exact primitive content)**.** *For every $A, n \geq 1$,*

$$\gcd(Q_n(A), Q_{n-1}(A)) = 1, \tag{73.2}$$

$$g_n(A) = \gcd(Q_n(A), ((n-1)!)^2). \tag{73.3}$$

In particular, for every prime p,

$$v_p(g_n(A)) = \min\{v_p(Q_n(A)),\, 2v_p((n-1)!)\}.$$

There is no unknown numerator cancellation left in this formula.

Proof. If $p \mid A$, the defining sum gives $Q_n(A) \equiv 1 \pmod p$. If $p \nmid A$, write $n = kp + r$, $0 \leq r < p$. Terms with $j \geq p$ vanish modulo p, and Lucas's elementary binomial congruence gives $Q_n(A) \equiv Q_r(A) \pmod p$. In particular $Q_{kp}(A) \equiv 1$. If p divided two consecutive terms, their residues would give two consecutive zero terms among $Q_0(A), \ldots, Q_{p-1}(A)$. For indices $2 \leq j < p$ the backward coefficient $(j-1)^2A^2$ in (73.1) is nonzero modulo p. Backward propagation would therefore give $Q_0(A) \equiv 0$, a contradiction. This proves (73.2).

The recurrence and the initial determinant one imply

$$P_n(A)Q_{n-1}(A) - P_{n-1}(A)Q_n(A) = A^{2n-2}((n-1)!)^2.$$

Reduction modulo $Q_n(A)$, followed by $\gcd(Q_n(A), Q_{n-1}(A)) = \gcd(Q_n(A), A) = 1$, gives (73.3). Taking valuations proves the last assertion. □

The exact positive remainder is

$$Q_n(A)\delta_A - P_n(A) = A^{2n} n! \int_0^\infty \frac{e^{-t}t^n}{(1+At)^{n+1}}\,dt > 0. \tag{73.4}$$

For completeness, take $B_n(t) = A^n n! L_n(t)$. The polynomial $[B_n(-1/A) - B_n(t)]/(1+At)$ has integer coefficients: division of each difference $(-1/A)^j - t^j$ leaves powers A^{n-j} with nonnegative exponents. Its integral against e^{-t} is therefore an integer. The Laguerre recurrence identifies this integer with $P_n(A)$. Rodrigues's formula and n integrations by parts give (73.4). This also supplies nonvanishing independently of the arithmetic nature of δ_A.

73.2 All prime powers as a finite residue tree

Lemma 73.2 (Congruence preservation)**.** *For every positive integer m and every $n \geq 0$,*

$$Q_{n+m}(A) \equiv Q_n(A) \pmod{m}.$$

Proof. It is enough to work modulo p^a. For nonnegative integers u, v with $u \equiv v \pmod{p^a}$, Vandermonde's identity implies

$$v_p\left(\binom{u}{j} - \binom{v}{j}\right) \geq a - \lfloor \log_p j \rfloor \qquad (j \geq 1),$$

where a negative right side is harmless. One first proves this for a shift by p^a, using $v_p\binom{p^a}{h} = a - v_p(h)$ for $1 \leq h \leq p^a$, and then telescopes. Since $v_p(j!) \geq \lfloor \log_p j \rfloor$, the difference $j!A^j[\binom{u}{j}^2 - \binom{v}{j}^2]$ is divisible by p^a. Summing the finitely many nonzero terms proves the claim. □

Let

$$R_{p,a}(A) = \{0 \leq r < p^a : Q_r(A) \equiv 0 \pmod{p^a}\}.$$

Starting from $R_{p,1}(A)$, construct $R_{p,a+1}(A)$ by testing the p children $r + bp^a$, $0 \leq b < p$, of every retained residue. Congruence preservation proves that no other residue can survive. Theorem 73.1 now gives the exact finite formula

$$v_p(g_n(A)) = \sum_{a=1}^{2v_p((n-1)!)} \mathbf{1}_{\{n \bmod p^a \in R_{p,a}(A)\}}. \tag{73.5}$$

Thus the primitive coefficients can be determined through finitely many modular computations without computing the numerator $P_n(A)$. This is an exact algorithm, not an asymptotic upper bound on $g_n(A)$.

73.3 Simple roots give exact p-adic distance formulas

For an odd prime p, define the continuous interpolation

$$\mathcal{Q}_{p,A}(x) = \sum_{j=0}^{\infty} j!A^j \binom{x}{j}^2 \qquad (x \in \mathbb{Z}_p).$$

The series converges uniformly, since $\binom{x}{j} \in \mathbb{Z}_p$ and $v_p(j!) \to \infty$. At nonnegative integer arguments it equals $Q_n(A)$. If $0 \le r < p$, put

$$d_r(A) = 2\sum_{j=1}^{r} j!A^j \binom{r}{j}^2 (H_r - H_{r-j}), \qquad H_0 = 0.$$

This number is p-integral.

Theorem 73.3 (A complete branch above each simple residue root)**.** *Suppose p is odd, $p \nmid A$, $0 \le r < p$, and*

$$Q_r(A) \equiv 0 \pmod{p}, \qquad d_r(A) \not\equiv 0 \pmod{p}.$$

There is a unique $\alpha_{p,r,A} \in r + p\mathbb{Z}_p$ with $\mathcal{Q}_{p,A}(\alpha_{p,r,A}) = 0$, and for every $x \in r + p\mathbb{Z}_p$,

$$v_p(\mathcal{Q}_{p,A}(x)) = v_p(x - \alpha_{p,r,A}). \tag{73.6}$$

In particular the contribution of this residue disk to the actual content is exactly

$$\min\{v_p(n - \alpha_{p,r,A}),\, 2v_p((n-1)!)\}.$$

Proof. Work on $x = r + py$. In the falling factorial $(r+py)_j$, at least $\lfloor j/p \rfloor$ factors have all coefficients divisible by p. Consequently every coefficient of $(r+py)_j^2 A^j/j!$ has valuation at least $2\lfloor j/p \rfloor - v_p(j!)$, which tends to infinity for odd p. The series is therefore a restricted power series in y.

We compute it modulo p^2, coefficient by coefficient. For $j < p$, Taylor expansion at r gives $Q_r(A) + pd_r(A)y$ modulo p^2; the terms $r < j < p$ have a double zero at r and do not contribute. Terms with $j \ge 2p$ have all coefficients divisible by p^2, since $2\lfloor j/p \rfloor - v_p(j!) \ge 2$ in that range. For $j = p + s$, $0 \le s \le r$, there is exactly one factor congruent to zero at r in $(r+py)_j$. Wilson's theorem gives

$$\frac{(r+py)_{p+s}^2 A^{p+s}}{(p+s)!} \equiv -py^2 A\, \frac{(r)_s^2 A^s}{s!} \pmod{p^2}.$$

If $r < s < p$, there are two such factors and the term vanishes modulo p^2. Summation yields the restricted-series identity

$$\mathcal{Q}_{p,A}(r+py) \equiv Q_r(A) + pd_r(A)y - pAQ_r(A)y^2 \pmod{p^2}. \tag{73.7}$$

The root assumption therefore gives

$$\mathcal{Q}_{p,A}(r+py)/p = c + d_r(A)y + pG(y), \qquad c = Q_r(A)/p, \quad G \in \mathbb{Z}_p\langle y \rangle.$$

Since $d_r(A)$ is a unit, Hensel's lemma gives a unique zero $\beta \in \mathbb{Z}_p$. Moreover $G(y) - G(\beta)$ is divisible by $y - \beta$ with an integral restricted-series quotient. Thus

$$\mathcal{Q}_{p,A}(r+py)/p = (y-\beta)\big[d_r(A) + p\widetilde{G}(y)\big],$$

and the bracket is a unit. Taking $\alpha_{p,r,A} = r + p\beta$ proves (73.6). □

The simplicity hypothesis is essential to the proved statement; it has not been established for every prime and every root. Without it, the finite tree (73.5) remains valid. For $A = 1$, the prime 2 needs no analytic argument: periodicity modulo 4, together with $(Q_0, Q_1, Q_2, Q_3) \equiv (1, 2, 3, 2) \pmod 4$, gives $v_2(Q_n) = 1$ for odd n and 0 for even n. For $p = 7$, the only root is $r = 2$ and $d_2 = 10 \equiv 3$; its unique lift modulo 49 is 16. For $p = 11$, the roots $4, 7$ have derivatives $4, 6$ modulo 11. Every higher valuation in these three disks is therefore governed by one explicitly computable p-adic zero.

73.4 A precise bridge to the left-factorial problem

Write $!p = \sum_{j=0}^{p-1} j!$. The familiar congruence $\binom{p-1}{j} \equiv (-1)^j \pmod p$ gives

$$Q_{p-1}(1) \equiv !p \pmod p.$$

Connections between this Laguerre sequence and Kurepa's hypothesis already occur in the literature [318]; we record the following consequence for the *actual reduced approximants*.

Proposition 73.4. *Kurepa's assertion $p \nmid !p$ for every odd prime p is equivalent to*

$$\gcd(g_n(1), n+1) = \begin{cases} 1, & n \text{ even or } n = 1, \\ 2, & n \geq 3 \text{ odd.} \end{cases}$$

Proof. If an odd prime p divides $n+1$ and $p \leq n$, then $n \equiv p-1 \pmod p$ and $p \mid g_n(1)$ if and only if $p \mid !p$, by Theorem 73.1 and the prime-residue computation. If $p > n$, it cannot divide $g_n(1)$. This proves the forward implication, including the exact factor 2 established above. Conversely, if $p \mid !p$ for an odd prime, choose $n = 2p - 1$. Then $p \mid g_n(1)$ and $p \mid n+1$, contradicting the displayed formula. □

This equivalence does not settle Kurepa's assertion, whose continuing computational and theoretical difficulty is discussed in Harvey's 2026 ICM account [319]. It is a warning against replacing a genuine local arithmetic problem by an unproved universal "no exceptional residue" assumption.

73.5 What the exact content does and does not yield

The saddle point in (73.4), after writing $c = 1/A$ and $t = \sqrt{nc}\, u$, gives

$$Q_n(A)\delta_A - P_n(A) \sim \sqrt{\pi} e^{1/(2A)} A^{n-3/4} n! n^{-1/4} e^{-2\sqrt{n/A}}.$$

The phase $u + u^{-1}$ has a unique nondegenerate minimum at $u = 1$; the same elementary Laplace calculation as in the parameter-one case proves this formula. Accordingly the primitive positive form tends to zero along a subsequence exactly when

$$\log g_n(A) - \log(n! A^n) + 2\sqrt{n/A} + \tfrac{1}{4} \log n \longrightarrow +\infty. \tag{73.8}$$

Equations (73.5) and (73.6) express the entire left side arithmetically, without the unknown constant δ_A. But neither an upper bound ruling out (73.8) nor a lower bound establishing it has been proved here. A fixed-prime Hensel description is not a bound for the sum of contributions from all primes up to n. This is the precise global estimate still missing.

73.6 Euler's constant: only one even crossover can escape

Put

$$S_M(n) = \sum_{k=1}^{M} \frac{(-1)^{k-1} n^k}{k\,k!} = \frac{a_M(n)}{b_M(n)}, \qquad \gcd(a_M(n), b_M(n)) = 1, \qquad b_M(n) > 0.$$

Let

$$E_M(n) = \frac{1}{M!} \int_0^n \frac{1}{t} \int_0^t e^{-s} (t-s)^M \, ds\, dt, \quad T_n = e^{-n} \int_0^\infty \frac{e^{-t}}{n+t}\, dt.$$

The exact remainder identity is

$$S_M(n) - \log n - \gamma = T_n + (-1)^{M+1} E_M(n).$$

Odd orders have two errors of the same sign and were excluded in the previous investigation. Even orders require a different argument. Let $\kappa > e$ be the unique solution of

$$\kappa(\log \kappa - 1) = 1 \qquad (\kappa \approx 3.59112).$$

Theorem 73.5 (A unique possible even crossover)**.** *For every sufficiently large integer n, at most one even integer $M \geq 2n$ satisfies*

$$\tfrac{1}{2} T_n \leq E_M(n) \leq 2T_n. \tag{73.9}$$

As $n \to \infty$, for every sequence of positive even integers $M = M(n)$ outside this possible exceptional choice,

$$b_M(n)\,|S_M(n) - \log n - \gamma| \longrightarrow \infty.$$

Any orders satisfying (73.9) necessarily obey

$$M = \kappa n - \frac{\log n}{2 \log \kappa} + O(1). \tag{73.10}$$

If an even-order sequence actually has primitive forms tending to zero, then, in addition,

$$\left| \frac{E_M(n)}{T_n} - 1 \right| \leq \exp\bigl(-[\kappa - 1 + o(1)]n\bigr). \tag{73.11}$$

Proof. The integral defining E_M gives the exact inequality

$$\frac{E_{M+2}(n)}{E_M(n)} \leq \frac{n^2}{(M+1)(M+2)} < \frac{1}{4} \qquad (M \geq 2n).$$

Two even orders cannot both satisfy (73.9). Outside that window, $|T_n - E_M(n)| \geq T_n/2$.

For $M \leq 2n$, restricting the outer integral to $n - n/(M+1) \leq t \leq n$ and the inner integral to $0 \leq s \leq 1$ gives, for $n \geq 4$,

$$E_M(n) \geq e^{-13} \frac{n^M}{(M+1)M!}.$$

This tends to infinity uniformly over positive orders $M \leq 2n$. For $M \leq \sqrt{n}$ use $M! \leq M^M$; for $\sqrt{n} < M \leq 2n$ use Stirling's bound and $en/M \geq e/2 > 1$. Since $T_n \to 0$ and $b_M(n) \geq 1$, these orders are excluded.

For $M \geq 2n$, every prime $M/2 < p \leq M$ with $p \nmid n$ occurs to exact valuation -2 in the sum $S_M(n)$: its $k = p$ term has valuation -2 and every other term has valuation at least -1. Thus the *actual reduced denominator* satisfies

$$\log b_M(n) \geq 2\bigl(\vartheta(M) - \vartheta(M/2)\bigr) - 2\log n = M + o(M) - 2\log n.$$

Also $T_n \ge e^{-n}/(n+1)$ by Jensen's inequality. Hence $b_M(n)T_n \to \infty$ uniformly for $M \ge 2n$, proving the assertion outside (73.9).

It remains to locate that window. Uniformly for M/n in a compact subset of $(0,\infty)$, boundary Laplace asymptotics give

$$E_M(n) \sim \frac{n^M}{M!\,M(1+M/n)}. \tag{73.12}$$

Indeed, set $t = n - v$ in the defining double integral. After division by $n^{M-1}/M!$, the integrand tends to $\exp[-cv - (1+c)s]$ when $M/n \to c > 0$; its integral is $1/[c(1+c)]$. On $t \ge n/2$, a fixed exponential in $v+s$ dominates uniformly for c in a compact interval, while the remaining range is exponentially negligible. This proves the stated uniform asymptotic.

Since $T_n \sim e^{-n}/n$, Stirling's formula gives, with $c = M/n$,

$$\log \frac{E_M(n)}{T_n} = n\{1 + c(1 - \log c)\} - \tfrac{1}{2}\log n - \tfrac{3}{2}\log c - \log(1+c) - \tfrac{1}{2}\log(2\pi) + o(1).$$

First compare orders near $2n$ and $4n$ and use monotonicity to confine the window to $2 < M/n < 4$. The displayed formula then forces $M/n \to \kappa$. Expansion at κ, whose derivative is $-\log\kappa$, proves (73.10). Finally, if $b_M|T_n - E_M| \to 0$, divide by $b_M T_n$ and apply the denominator lower bound and (73.10) to obtain (73.11). □

Corollary 73.6 (Density-one exclusion of bounded even-order forms)**.** *For every fixed $B > 0$, the set*

$$\{n \ge 1 : \text{ there is an even } M \ge 2 \text{ with } b_M(n)|S_M(n) - \log n - \gamma| \le B\}$$

has natural density zero. More precisely, the set of arguments n at which the factor-two window (73.9) contains an even order has natural density

$$\frac{\log 2}{\log \kappa}.$$

Proof. Define

$$C_\kappa = \tfrac{3}{2}\log\kappa + \log(1+\kappa) + \tfrac{1}{2}\log(2\pi), \qquad c_n = \kappa n - \frac{\log n}{2\log\kappa} - \frac{C_\kappa}{\log\kappa}.$$

The expansion in the proof of Theorem 73.5 yields, uniformly for bounded $M - c_n$,

$$\log(E_M(n)/T_n) = -\log\kappa\,(M - c_n) + o(1). \tag{73.13}$$

The number κ is transcendental: if it were algebraic, $\log\kappa = 1 + 1/\kappa$ would be a nonzero algebraic number whose exponential is algebraic, contrary to the Hermite–Lindemann theorem. In particular κ is irrational.

The sequence c_n is equidistributed modulo 2. Indeed, for any nonzero integer h, its Weyl sum is a constant of modulus one times

$$\sum_{n \le N} e^{\pi i h \kappa n} n^{-\pi i h/(2\log\kappa)}.$$

The geometric partial sums are bounded because $e^{\pi i h\kappa} \ne 1$. Partial summation, using a derivative of magnitude $O_h(1/n)$ for the second factor, bounds the displayed sum by $O_h(\log N)$. Weyl's criterion proves the assertion.

Write $a = \log 2/\log\kappa < 1$. Equations (73.10) and (73.13) show that the existence of a window order is equivalent, apart from a boundary of width $o(1)$, to $\operatorname{dist}(c_n, 2\mathbb{Z}) \le a$.

Equidistribution modulo 2 gives density $2a/2 = a$; the shrinking boundary contributes density zero.

If the primitive magnitude is at most B, the uniform exclusions in Theorem 73.5 first place M in the window. There $b_M(n)T_n \to \infty$ uniformly, so $|E_M/T_n - 1| \le B/(b_M T_n) \to 0$ uniformly over these choices. Equation (73.13) now forces $\operatorname{dist}(c_n, 2\mathbb{Z}) \to 0$ on the set in question. For every $\varepsilon > 0$, all sufficiently large members therefore lie in $\operatorname{dist}(c_n, 2\mathbb{Z}) \le \varepsilon$, a set of density ε when $0 < \varepsilon < 1$. Letting $\varepsilon \downarrow 0$ proves density zero. □

This is a genuine narrowing of the failed Taylor attack: arbitrary even schedules are reduced to at most one crossover per argument, and that crossover would need exponentially precise cancellation. The density statement does not rule out an infinite sparse sequence of successful crossovers. Further, the forms still contain $\log n$, so a successful proof for γ alone would also have to eliminate that logarithmic term. No estimate accomplishing either remaining task is asserted here.

74 Wilson trial functions, primitive determinants, and four zeta targets

The previous construction produced positive odd-zeta moment pencils and rational enclosures by solving growing moment systems. The corresponding quadratic logarithmic height bound was too expensive for arithmetic applications. Here we replace those linear systems by one explicit polynomial family. This gives a proved improvement of the height estimate from $O_s(n^2 \log n)$ to $O_s(n)$, together with a stronger exponential enclosure rate. It does not establish the transcendence of G, $\zeta(3)$ or $\zeta(5)$, or the irrationality of $\zeta(7)$.

There is an important attribution correction. The positive Stieltjes representation of the Hurwitz zeta function, including positive real weights beyond odd integers, is already in Prévost's work [320, Theorems 1–3]. Wilson-polynomial remainder Padé approximation is also developed by Prévost and Rivoal [321]. Neither ingredient is claimed as a new discovery here. Our statement below is an explicit comparison and transfer for the particular moment-enclosure construction used in this project; historical priority for this precise formulation is not asserted.

74.1 An exact Wilson specialization

For an odd integer $s \ge 3$, write $\theta_s = \zeta(s)$ and

$$d\nu_s(t) = w_s(y)\,dy, \qquad t = y^2, \qquad w_s(y) = \frac{2(2\pi)^{s-1}}{(s-1)!} y^s \sum_{\ell \ge 1} \ell^{s-1} e^{-2\pi \ell y}.$$

Its moments are

$$m_{s,e} = \int t^e\, d\nu_s(t) = (-1)^e B_{2e+2} \frac{(2e+s)!}{(s-1)!(2e+2)!}. \tag{74.1}$$

Put $S_s(z) = \int (t+z)^{-1} d\nu_s(t)$. Hermite's formula gives, for integers $j \ge 1$,

$$\theta_s = b_{s,j} + j^{1-s} S_s(j^2), \qquad b_{s,j} = \sum_{v=1}^{j} v^{-s} + \frac{j^{1-s}}{s-1} - \frac{1}{2j^s}. \tag{74.2}$$

Define the rational polynomial

$$R_n(t) = \sum_{k=0}^{n} \frac{(-1)^k}{(k!)^2} \binom{n}{k} \binom{n+k+1}{k} \prod_{r=0}^{k-1} (t + r^2), \qquad n \ge 0. \tag{74.3}$$

The empty product equals one.

Lemma 74.1 (Wilson identity and exact norm)**.** *The polynomials R_n are orthogonal for ν_3, and*

$$\int R_n(t)^2\, d\nu_3(t) = \frac{n+1}{4}.$$

If q_n is their monic normalization, then

$$tq_n = q_{n+1} + \beta_n q_n + \lambda_n q_{n-1},$$

where

$$\beta_n = \frac{(n+1)^2(2n^2+4n+1)}{(2n+1)(2n+3)}, \qquad \lambda_n = \frac{n^3(n+1)^3}{4(2n+1)^2} \quad (n \geq 1).$$

Proof. The gamma identities $|\Gamma(iy)|^2 = \pi/(y\sinh \pi y)$ and $|\Gamma(1+iy)|^2 = \pi y/\sinh \pi y$ show that

$$w_3(y) = \pi^2 y^3 \frac{\cosh \pi y}{\sinh^3 \pi y} = \frac{1}{4\pi}\left|\frac{\Gamma(iy)\Gamma(1+iy)^3}{\Gamma(2iy)}\right|^2.$$

Thus this is the Wilson weight with parameters $(0,1,1,1)$. The hypergeometric definition of Wilson polynomials gives $R_n(t) = W_n(t;0,1,1,1)/(n!)^3$ [322, 323]. The limiting parameter zero causes no extra mass: take the first parameter $a \downarrow 0$ in the positive-parameter orthogonality. Near $y = 0$ the factor $|\Gamma(a+iy)/\Gamma(2iy)|^2$ is bounded by a constant times $y^2/(a^2+y^2)$, hence is uniformly bounded; at infinity a common exponential majorant times a polynomial applies. Dominated convergence therefore proves the asserted limiting orthogonality.

The Wilson recurrence at these parameters gives the displayed β_n, λ_n. Since $m_{3,0} = 1/4$, the monic squared norms are

$$h_n = \frac{1}{4}\prod_{k=1}^{n} \lambda_k = \frac{(n!)^5((n+1)!)^3}{4((2n+1)!)^2}.$$

The leading coefficient of R_n is $(-1)^n(2n+1)!/((n+1)!(n!)^3)$. Multiplication by its square gives $(n+1)/4$. □

Lemma 74.2 (A uniform transfer of the squared norm)**.** *For fixed odd $s \geq 3$, there is an effective constant $C_s > 0$ such that*

$$0 < \int R_n^2\, d\nu_s \leq C_s(n+1)^{s-2}.$$

For $k = (s-3)/2$, the more explicit estimate

$$\int R_n^2\, d\nu_s \leq \frac{n+1}{2}\left((1+4\pi^2)[1+4(n+k+2)^2]\right)^k$$

is valid.

Proof. For $q \in (0,1)$ and integer $m \geq 2$,

$$\sum_{\ell \geq 1} \ell^m q^\ell \leq \frac{m!\,q}{(1-q)^{m+1}}, \qquad \sum_{\ell \geq 1} \ell^2 q^\ell = \frac{q(1+q)}{(1-q)^3}.$$

The first inequality follows from $\ell^m \leq m!\binom{\ell+m-1}{m}$ and its generating function. Using $y/(1-e^{-2\pi y}) \leq y + (2\pi)^{-1}$, it follows that

$$\frac{w_s(y)}{w_3(y)} \leq 2(1+2\pi y)^{s-3} \leq 2(1+4\pi^2)^k(1+y^2)^k.$$

In the orthonormal Wilson basis, multiplication by t is the Jacobi matrix with diagonal β_n and off-diagonals $\sqrt{\lambda_n}$. Its finite section through index $n+k$ has operator norm at most $4(n+k+2)^2$, by the displayed recurrence and the maximum row-sum bound. A path contributing to its (n,n) entry in any power at most k never uses an index above $n+k$. Consequently

$$\int (1+t)^k R_n(t)^2 \, d\nu_3(t) \le [1+4(n+k+2)^2]^k \frac{n+1}{4}.$$

The weight comparison proves the result. Strict positivity follows because R_n is nonzero and $w_s > 0$. □

74.2 Explicit rational intervals without moment-matrix inversion

For $n, j \ge 1$ let $a_{n,j} = R_n(-j^2) > 0$, and put

$$p_{n,j}(t) = \frac{1 - R_n(t)/a_{n,j}}{t + j^2} \in \mathbb{Q}[t].$$

Positivity of $a_{n,j}$ follows directly from (74.3): every term with $k \le j$ is positive and every term with $k > j$ is zero. For a polynomial p, define

$$\mathcal{L}_s(p; z) = \int [2p - (t+z)p^2] d\nu_s(t),$$
$$\mathcal{U}_s(p; z) = \frac{m_{s,0} - \int [2tp - t(t+z)p^2] d\nu_s(t)}{z}.$$

All these integrals are finite rational combinations of the explicit moments (74.1).

Lemma 74.3 (Integer-valued arithmetic and Bernoulli evaluation)**.** *Let* $\mathrm{Int}(\mathbb{Z})$ *denote the rational polynomials mapping* $\mathbb{Z}$ *into* $\mathbb{Z}$*, let* $L_d = \mathrm{lcm}(1, \dots, d)$ *with* $L_0 = 1$*, and let* $\mathcal{B} : \mathbb{Q}[x] \longrightarrow \mathbb{Q}$ *be defined by* $\mathcal{B}(x^m) = B_m$ *with* $B_1 = -1/2$*. For* $f \in \mathrm{Int}(\mathbb{Z})$ *of degree at most* d*:*

1. $L_d f'$ *is integer-valued;*
2. *if* $a \in \mathbb{Z}$ *and* $f(a) = 0$*, then* $L_d f(x)/(x-a)$ *is integer-valued;*
3. $L_{d+1}\mathcal{B}(f) \in \mathbb{Z}$*.*

Moreover, for every polynomial $F \in \mathbb{Q}[t]$*,*

$$\int F(t) d\nu_s(t) = \frac{\mathcal{B}((x^s F(-x^2))^{(s-2)})}{(s-1)!}.$$

Proof. An integer-valued polynomial has integer coefficients in the Newton basis $\binom{x}{k}$, because these coefficients are its finite differences at zero. The identities

$$\frac{d}{dx}\binom{x}{k} = \sum_{j=1}^{k} \frac{(-1)^{j-1}}{j}\binom{x}{k-j}, \qquad \frac{1}{x-a}\binom{x-a}{k} = \frac{1}{k}\binom{x-a-1}{k-1}$$

prove the first two assertions; for the second, expand at a and use $f(a) = 0$ to remove the constant term. The formal identity $\mathcal{B}((1+z)^x) = \log(1+z)/z$ gives $\mathcal{B}(\binom{x}{k}) = (-1)^k/(k+1)$, proving the third assertion. Finally the displayed integral identity follows on each monomial $F(t) = t^e$ from (74.1), and then by linearity. □

Theorem 74.4 (A quantitative Wilson trial enclosure)**.** *Set*

$$\ell_{s,n,j} = b_{s,j} + j^{1-s}\mathcal{L}_s(p_{n,j}; j^2), \qquad u_{s,n,j} = b_{s,j} + j^{1-s}\mathcal{U}_s(p_{n,j}; j^2).$$

Then both endpoints are rational and

$$\ell_{s,n,j} < \zeta(s) < u_{s,n,j}, \qquad u_{s,n,j} - \ell_{s,n,j} = \frac{\displaystyle\int R_n(t)^2 d\nu_s(t)}{j^{s+1} R_n(-j^2)^2}.$$

On the diagonal $j = n$ one has

$$0 < u_{s,n,n} - \ell_{s,n,n} \le \exp\bigl(-8\log(1+\sqrt{2})n + O_s(\log n)\bigr). \tag{74.4}$$

The positive reduced denominators of both endpoints satisfy

$$\log \operatorname{den}(\ell_{s,n,n}), \quad \log \operatorname{den}(u_{s,n,n}) \le O_s(n). \tag{74.5}$$

In particular the exponent in the enclosure rate is about $7.05099n$. The construction involves explicit polynomial arithmetic and moment summation, with no growing linear system to invert.

Proof. Completing the square gives, for every polynomial p,

$$S_s(z) - \mathcal{L}_s(p; z) = \int \frac{[1-(t+z)p(t)]^2}{t+z}\, d\nu_s(t),$$

$$\mathcal{U}_s(p; z) - S_s(z) = \frac{1}{z}\int \frac{t[1-(t+z)p(t)]^2}{t+z}\, d\nu_s(t).$$

For the specified p, their sum is $\int R_n^2 d\nu_s/(z a_{n,j}^2)$; both terms are strictly positive. Equation (74.2) proves the exact interval and width.

Let $A_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$ be the classical Apéry integer. Direct factorial simplification in (74.3) gives

$$R_n(-n^2) = \sum_{k=0}^n \frac{n(n+k+1)}{(n+1)(n+k)} \binom{n}{k}^2 \binom{n+k}{k}^2.$$

Hence $nA_n/(n+1) \le R_n(-n^2) \le A_n$. Uniform elementary factorial estimates show that the logarithm of the kth summand of A_n equals $nf(k/n) + O(\log(n+1))$, where

$$f(x) = 2[(1+x)\log(1+x) - (1-x)\log(1-x) - 2x\log x].$$

The endpoint terms use $0\log 0 = 0$. Its maximum occurs at $x = 1/\sqrt{2}$, where $f(x) = 4\log(1+\sqrt{2})$. Bounding the sum by $(n+1)$ times its maximum term, and using the integer nearest $n/\sqrt{2}$ for a lower bound, proves

$$\log R_n(-n^2) = 4\log(1+\sqrt{2})n + O(\log n).$$

Together with Lemma 74.2, this proves (74.4).

For the denominator bound, first observe that

$$R_n(-x^2) = \sum_{k=0}^n \binom{n}{k}\binom{n+k+1}{k}\binom{x}{k}\binom{x+k-1}{k} \in \operatorname{Int}(\mathbb{Z}).$$

The identity follows from pairing the factors $x-r$ and $x+r$. In particular $a_n = R_n(-n^2)$ is a positive integer. The numerator $a_n - R_n(-x^2)$ vanishes at both $x = n$ and $x = -n$. Apply Lemma 74.3 twice to conclude that

$$a_n L_{2n}^2 p_{n,n}(-x^2) \in \operatorname{Int}(\mathbb{Z}).$$

Consequently, multiplying either of the polynomials $2p-(t+n^2)p^2$ and $2tp-t(t+n^2)p^2$ by $E_n=a_n^2L_{2n}^4$ makes it integer-valued after $t=-x^2$. Their degrees in t are at most $2n-1$ and $2n$, respectively. The Bernoulli-evaluation formula in Lemma 74.3, followed by its derivative and denominator assertions, proves that

$$E_n(s-1)!L_{4n+s+1}^{s-1}$$

clears their two moment integrals. Here the undifferentiated polynomial $x^sF(-x^2)$ has degree at most $4n+s$, and there are $s-2$ derivatives followed by one Bernoulli evaluation. The same integer also clears $m_{s,0}$. Thus one explicit common denominator for the two endpoints is

$$Q_{s,n}=2(s-1)n^{s+1}L_n^sa_n^2L_{2n}^4(s-1)!L_{4n+s+1}^{s-1}. \tag{74.6}$$

The elementary Chebyshev bound $\log L_m=O(m)$ suffices here. It follows, for example, by bounding the product of primes between m and $2m$ by $\binom{2m}{m}\le 4^m$, summing dyadic intervals, and then summing prime powers. The proved exponential size of a_n now gives $\log Q_{s,n}=O_s(n)$. Reduction cannot increase a denominator, completing the proof. □

For an explicit asymptotic accounting, put $\psi(m)=\log L_m$. Equation (74.6) gives

$$\log Q_{s,n}=2\log a_n+4\psi(2n)+(s-1)\psi(4n+s+1)+s\psi(n)+O_s(\log n).$$

If the prime number theorem $\psi(x)=x+o(x)$ is used, its leading coefficient is $8\log(1+\sqrt{2})+5s+4$. This upper bound is larger than the decay coefficient $8\log(1+\sqrt{2})$; it proves no positive irrationality margin. It is not a lower bound for the actual reduced denominators.

The old estimate $e^{-2\pi j+O_s(\log j)}$ used moment-polynomial degree $\lfloor \pi j/2\rfloor$ and a generic linear-system height bound. The new statement uses degree j and the exact explicit norm comparison. It is a strict improvement over those proved estimates, not an assertion that the previous actual reduced denominators had quadratic growth. For $s=3$ the lower trial is the actual Gaussian Padé approximant; classical cancellations can improve its arithmetic further. For $s\ge 5$ it is a variational trial and need not be the Padé optimizer.

74.3 Why changing a basis cannot change a primitive determinant

For $0\neq P\in\mathbb{Q}[X]$, denote by $\operatorname{prim}(P)$ its unique primitive integer multiple with positive leading coefficient.

Proposition 74.5 (Primitive invariance)**.** *Let $A(X)$ be a square matrix over $\mathbb{Q}[X]$ with nonzero determinant. For any constant matrices $C,D\in \mathrm{GL}_h(\mathbb{Q})$,*

$$\operatorname{prim}(\det(CA(X)D))=\operatorname{prim}(\det A(X)).$$

Consequently every rational change from monomials to an integer-valued, Newton, orthogonal, or Lagrange basis leaves the primitive integer polynomial, its height, and its value at a fixed target exactly unchanged. This applies both to congruences of the odd-zeta moment pencils and to independent row and column changes in a mixed Catalan matrix.

Proof. The determinants differ by the nonzero rational scalar $\det C\det D$. Primitive normalization removes precisely such a scalar. All conclusions are then identities, not asymptotic comparisons. □

This does not make a good basis useless: it can expose large common factors and improve a previously crude bound. It does rule out the claim that smaller entry denominators, by

themselves, have created a new primitive arithmetic certificate. A target-dependent orthogonal basis is outside the proposition and may have coefficients involving the very constant whose arithmetic nature is being investigated; it cannot be treated as a rational integerization step.

There is an analogous invariant for the variational approximants above: expressing the same polynomial p in a different rational basis does not change either endpoint. Replacing the optimizer by $p_{n,j}$, however, changes the approximation problem, which is why the constructive transfer in Theorem 74.4 is not merely a basis relabeling.

74.4 Four individual target conclusions

The transcendence of $\zeta(3)$. The Wilson identification supplies exact norms, explicit approximants, and a stronger estimate than our previous generic moment calculation. Its links with Apéry's sequence and remainder Padé approximation are classical, not new irrationality or transcendence results [320, 321]. Our present height and width bounds do not exclude any algebraic degree beyond previously known results.

The transcendence of $\zeta(5)$. Theorem 74.4 applies with $s = 5$ and quantitatively improves the proved enclosure/height tradeoff in this construction, including removal of its factorial denominator loss. Proposition 74.5 shows that reexpressing the older full-rank pencil in a rational orthogonal or integer-valued basis cannot itself improve the primitive polynomial's actual height. A new common-divisor estimate can improve its certified bound; none sufficient for a transcendence conclusion has been proved here.

The irrationality of $\zeta(7)$. Theorem 74.4 gives an explicit pair of rational bounds at rate $e^{-7.05099n+O(\log n)}$. The proved logarithmic denominator bound is now $O(n)$. The explicit constants in (74.6) still do not imply the required smallness after integerization. The missing step is a substantially stronger estimate for the actual reduced denominators or a different certificate whose denominator and error bounds have a strict positive exponential margin. No claim of irrationality follows from the faster numerical convergence alone.

The transcendence of Catalan's constant G. The previous elementary mixed Catalan determinant cancels its auxiliary $\zeta(2)$ term and has full degree. Rational row/column transformations preserve its primitive polynomial by Proposition 74.5. They therefore cannot remove its missing uniform nonvanishing theorem or create a better primitive height by scalar bookkeeping. A genuinely different kernel or polynomial family is required for this proposed escape; no such transcendence certificate is supplied here.

75 Logarithmic boundary signatures and the finite-to-infinite barrier

Let

$$L(z) = \sum_{n\ge1} \frac{z^n}{1-z^n} = \sum_{n\ge1} \tau(n) z^n, \qquad |z| < 1, \qquad \mathcal{D} = z\frac{d}{dz}.$$

Here $\tau(n)$ counts positive divisors. The classical Lambert identity is recorded in [324]. The previously obtained complete relation certificates apply to finite additive Mahler systems. They do not automatically extend to the full Lambert series. We now give a direct obstruction that survives arbitrary finite rational differential filters and arbitrary integer dilations. This is a functional result, not a transcendence theorem for $L(1/2)$. In particular, the already known

failure of an ordinary Mahler equation for L, also a consequence of the recent classification of Bell–Smertnig [325], is not counted as a new result.

75.1 A greatest-common-divisor matrix detects every linear filter

Lemma 75.1 (Radial logarithmic signature)**.** *If ζ is a primitive N-th root of unity, $m \geq 1$, and $r \geq 0$, then, as $t \downarrow 0$,*

$$\mathcal{D}^r L(z^m)|_{z=\zeta e^{-t}} = \frac{r!\,\gcd(N,m)}{Nm}\,\frac{\log(1/t)}{t^{r+1}} + O_{N,m,r}(t^{-r-1}). \tag{75.1}$$

Proof. Put $f(u) = (e^u - 1)^{-1}$. Its Laurent expansion at zero gives

$$(-1)^r f^{(r)}(u) = r!u^{-r-1} + O_r(1) \quad (0 < u \leq 1),$$

and the left side is $O_r(e^{-u})$ for $u \geq 1$. Termwise differentiation, justified on compact subintervals of $t > 0$, and separation at $n = \lfloor 1/t \rfloor$ therefore give

$$\left(-\frac{d}{dt}\right)^r L(e^{-t}) = r!t^{-r-1}\sum_{n\leq 1/t}\frac{1}{n} + O_r(t^{-r-1}) = r!t^{-r-1}\log(1/t) + O_r(t^{-r-1}).$$

Indeed the small-argument error sums as $\sum_{n\leq 1/t} n^r$, and the large-argument tail is bounded by $\sum_{n>1/t} n^r e^{-nt} = O_r(t^{-r-1})$.

For a primitive Q-th root ξ, the summands of $L(\xi e^{-u})$ whose index is divisible by Q sum to $L(e^{-Qu})$. In each remaining residue class the function $\xi^a e^{-v}/(1-\xi^a e^{-v})$, together with all its derivatives, is bounded near $v = 0$ and decays exponentially at infinity. Its differentiated contribution is thus $O_{Q,r}(u^{-r-1})$, by the same estimates. The preceding display consequently has leading coefficient $r!/Q$ at ξ. Finally take $\xi = \zeta^m$, $Q = N/\gcd(N,m)$, and $u = mt$. Since $\mathcal{D} = -d/dt$ on the chosen radial path, the factor from the derivatives is m^r, which proves (75.1). □

Theorem 75.2 (No finite differential–dilation relation)**.** *For any finite set M of distinct positive integers and any $R \geq 0$, the functions*

$$1, \qquad \mathcal{D}^r L(z^m) \quad (m \in M,\ 0 \leq r \leq R)$$

are linearly independent over $\mathbb{C}(z)$. More strongly, every combination

$$G(z) = \sum_{m\in M}\sum_{r=0}^{R} a_{m,r}(z)\mathcal{D}^r L(z^m), \qquad a_{m,r} \in \mathbb{C}(z),$$

with at least one nonzero coefficient has logarithmic singularities at a dense subset of the unit circle. It admits no meromorphic continuation through any open arc of that circle.

Proof. Take the largest R for which some $a_{m,R} \neq 0$. At every root of unity ζ of order N that is not a pole of a coefficient, Lemma 75.1 gives

$$G(\zeta e^{-t}) = \frac{R!}{N}\left(\sum_{m\in M}\frac{\gcd(N,m)}{m}a_{m,R}(\zeta)\right)t^{-R-1}\log(1/t) + O(t^{-R-1}). \tag{75.2}$$

Replacing $a_{m,R}(\zeta e^{-t})$ by its value at ζ costs only $O(t^{-R}\log(1/t))$, which is absorbed by the error term. Lower derivative orders are absorbed as well.

For each $d \in M$ define the rational function

$$Q_d(z) = \sum_{m \in M} \frac{\gcd(d,m)}{m} a_{m,R}(z).$$

The matrix $(\gcd(d,m))_{d,m \in M}$ is nonsingular. To verify this without a determinant formula, use

$$\gcd(d,m) = \sum_{e \mid d,\ e \mid m} \varphi(e).$$

This represents it as $E\,\mathrm{diag}(\varphi(e))E^T$, where $E_{d,e} = 1_{e \mid d}$ and the columns run through all divisors of members of M. The submatrix consisting of the columns $e \in M$, ordered increasingly, is lower triangular with diagonal one. Thus E has full row rank, and the matrix is positive definite. Some Q_d is therefore nonzero.

Let ℓ range over primes not dividing any member of M, and set $N = d\ell$. Then $\gcd(N,m) = \gcd(d,m)$ for all $m \in M$. Apart from the finitely many zeros and poles of Q_d and of the coefficients, every primitive $d\ell$-th root has a nonzero logarithmic coefficient in (75.2). Such primitive roots are dense on the circle as $\ell \to \infty$: in any fixed arc, the number of eligible numerators coprime to d grows proportionally to ℓ, whereas at most d are multiples of ℓ.

A nonzero term $t^{-R-1}\log(1/t)$, with error $O(t^{-R-1})$, cannot be the radial expansion of a meromorphic function at that point. This proves the continuation assertion. In particular G cannot equal a rational function. This is exactly the claimed linear independence, including the function 1. □

Corollary 75.3 (Every finite Lambert filter retains the obstruction)**.** *Let $A \subset \mathbb{Z}_{>0}$ be finite, let $(c_a)_{a \in A} \subset \mathbb{C}$ be nonzero, and suppose H is analytic in the unit disk and satisfies*

$$\mathcal{D}^r H(\zeta e^{-t}) = O_{\zeta,r}(t^{-r-1}) \quad (t \downarrow 0)$$

for every root of unity ζ and every $r \geq 0$. Set

$$F(z) = \sum_{a \in A} c_a L(z^a) + H(z).$$

Then 1 and all finitely many selected functions $\mathcal{D}^r F(z^m)$, for distinct positive m and nonnegative r, are linearly independent over $\mathbb{C}(z)$. Every nonzero rational differential combination of these dilates has the same natural-boundary conclusion as in Theorem 75.2.

Proof. At the highest derivative order, the Lambert coefficients become the finite Dirichlet convolution

$$b_n(z) = \sum_{am=n} c_a a_{m,R}(z).$$

This convolution cannot vanish identically. If a_0 is the least index with $c_{a_0} \neq 0$, and m_0 is the least index with $a_{m_0,R} \neq 0$, then $b_{a_0 m_0} = c_{a_0} a_{m_0,R} \neq 0$: every other contributing pair would need one index smaller than its respective minimum. The terms involving $H(z^m)$ have size $O(t^{-R-1})$, so they do not change the logarithmic coefficients. The proof of Theorem 75.2 applies to the nonzero vector (b_n). □

75.2 Two familiar constants: what the obstruction now excludes

For the Erdős–Borwein constant $E_2 = L(1/2)$, the new conclusion is exact: no choice of finitely many integer dilations, finitely many derivatives, and rational coefficient functions puts the full Lambert function into a nontrivial linear relation. Consequently enlarging the earlier additive

system by finitely many such derivatives or dilates cannot repair the attempted transfer. An ordinary finite-dimensional linear Mahler system containing L would already give a rational linear dependence of finitely many iterates and is therefore excluded.

This obstruction concerns the proposed method, not the arithmetic nature of E_2. Known value results are stronger than mere irrationality in a different direction: Duverney–Tachiya prove that $1, L(1/b), \dots, L(1/b^h)$ are linearly independent over $\mathbb{Q}$, for integer $|b| > 1$ [326, Corollary 1.1]. Their result does not give algebraic independence or transcendence of E_2, and neither does the functional theorem above.

For the full reciprocal Fibonacci sum, put

$$\mathscr{F}(z) = \sum_{n\geq 1} \frac{z^n}{1-(-1)^n z^{2n}}, \qquad q = \frac{\sqrt{5}-1}{2}.$$

Binet's formula gives $\sum_{n\geq 1} F_n^{-1} = \sqrt{5}\,\mathscr{F}(q)$. The theta–Lambert decomposition is

$$\mathscr{F}(z) = L(z^2) - L(z^4) + T(z), \qquad T(z) = \frac{1}{4}\sum_{u,v\in\mathbb{Z}} z^{2(u+1/2)^2+2(v+1/2)^2}. \tag{75.3}$$

All exponents in T are positive integers. The identity follows by separating even and odd n and applying Jacobi's two-square identity to the odd part. It is the same classical decomposition used in the preceding manuscript, rather than a new identity.

The theta perturbation satisfies the hypothesis of Corollary 75.3. Indeed, with $A_{u,v} = 2(u+1/2)^2 + 2(v+1/2)^2$, absolute values give

$$\left|\mathcal{D}^r T(\zeta e^{-t})\right| \leq \frac{1}{4}\sum_{u,v\in\mathbb{Z}} A_{u,v}^r e^{-tA_{u,v}} = O_r(t^{-r-1}).$$

The final estimate follows by comparison with a two-dimensional Gaussian integral, or by bounding $x^r e^{-x} \leq C_r e^{-x/2}$ and then separating the two one-dimensional Gaussian sums. Thus:

Corollary 75.4 (The full Fibonacci function has no finite repair)**.** *The functions* 1 *and any finite collection* $\mathcal{D}^r\mathscr{F}(z^m)$ *with distinct pairs* (m,r) *are linearly independent over* $\mathbb{C}(z)$*. In particular, no finite rational differential–dilation filter removes the Lambert part of* (75.3) *to leave a function with only the indicated theta-type radial growth.*

The remaining numerical target is still $\mathscr{F}(q) \notin \overline{\mathbb{Q}}$. Functional linear independence at a variable does not prove this assertion at the single algebraic point q; the required finite Mahler system is precisely what the corollary rules out. Nonlinear, modular, or genuinely infinite-dimensional methods are not excluded.

75.3 Prime Mertens: exact independence of every finite cutoff

The standard prime correction representation is

$$B_1 - \gamma = \sum_p c_p, \qquad c_p = \log\left(1-\frac{1}{p}\right) + \frac{1}{p}.$$

See [329] for the classical constant and its computation. We test a distinct proposed repair: cancel the finite logarithmic part by combining several prime cutoffs. The following is an explicit application of Baker's classical theorem, not a new transcendence theorem of comparable generality.

Proposition 75.5 (No finite algebraic cancellation of prime corrections)**.** *For every finite set of distinct primes S, the numbers 1 and c_p, $p \in S$, are linearly independent over $\overline{\mathbb{Q}}$. Consequently the same holds for 1 and the successive prime cutoffs $C_k = \sum_{j=1}^{k} c_{p_j}$, $1 \le k \le n$. Moreover the reduced denominator of $A_k = \sum_{j=1}^{k} p_j^{-1}$ is exactly $P_k = \prod_{j=1}^{k} p_j$.*

Proof. The rational numbers $(p-1)/p$, $p \in S$, are multiplicatively independent. In a product of their integer powers equal to one, let p be the largest prime with a nonzero exponent. Its valuation in every smaller factor is zero, whereas its valuation in its own factor is minus that exponent, a contradiction. Thus their real logarithms are linearly independent over $\mathbb{Q}$. Baker's theorem makes them linearly independent over $\overline{\mathbb{Q}}$, and its inhomogeneous form says that a nonzero algebraic-coefficient linear form in these logarithms is transcendental [327, 328]. A relation among $1, c_p$ would make such a form algebraic. It must therefore vanish, forcing every coefficient to be zero. Passing to the C_k is an invertible triangular rational change of coordinates.

Finally, the numerator of A_k over denominator P_k is $\sum_{p \le p_k} P_k/p$. Modulo any prime $p \mid P_k$, only the term P_k/p is nonzero. No factor of P_k can cancel, proving the denominator assertion. □

For completeness the prime number theorem makes the mismatch of the unaccelerated cutoff scales explicit. If

$$T(x) = -\sum_{p>x} c_p,$$

then $T(x) \sim (2x \log x)^{-1}$, since $-c_p = 1/(2p^2) + O(p^{-3})$ and partial summation gives $\sum_{p>x} p^{-2} \sim (x \log x)^{-1}$. Meanwhile $\log \prod_{p \le x} p \sim x$. Hence

$$\left(\prod_{p \le x} p \right) T(x) \longrightarrow \infty.$$

The logarithms in these approximants do not disappear; the proposition proves that finite algebraic recombinations cannot make them disappear. The display is therefore an audit of this specific cutoff strategy, not a universal lower bound on every possible approximation to B_1. It leaves both the infinite prime tail and its interaction with γ unresolved, so it supplies no irrationality claim for B_1.

76 The exact multiplicative depth of a prime-grid gamma isolation

The five individual targets in this section are

$$\Gamma(1/5), \quad \Gamma(2/5), \quad \Gamma(1/7), \quad \Gamma(2/7), \quad \Gamma(3/7).$$

None is proved irrational or transcendental here. The earlier project argument established that reflection identities and the known transcendence-degree bound for the fifth-denominator values do not isolate either individual value. We now replace that qualitative obstruction by a sharp calculation: in a precisely specified reflection-and-recurrence model, isolating a fifth-denominator gamma power needs at least three beta factors, and isolating a seventh-denominator gamma power needs at least five. The bounds are attained by explicit formulas. The result is a theorem about a certificate model, not a lower bound for all possible transcendence proofs.

76.1 What the classical period theorems do and do not supply

The analytic subgroup theorem and its applications to Fermat Jacobians give very strong *linear* information about beta values. Wolfart–Wüstholz determine the algebraic-coefficient linear relations among rational beta values through the classical Deligne–Koblitz–Ogus relations [330, 331]. This is different from determining all polynomial relations among gamma values. The positive-characteristic gamma analogue does have substantially stronger known algebraic independence results; transferring those results to Euler's gamma function is not legitimate. The distinction is explicit in [331, Sections 1.4.3–1.4.4].

For fifth denominators, the known lower bound

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(\pi, \Gamma(1/5), \Gamma(2/5)) \geq 2$$

does not identify which gamma value must supply transcendence; see [332, Section 3.3.2]. Sheth's 2026 paper explicitly records that the transcendence of $\Gamma(1/5)$ remains open [333, discussion after Conjecture 4.2]. Our primary-source search did not locate a resolution of the other four specified individual questions. Thus they remain unresolved *in this investigation*; the statement is not a claim to have exhaustively certified the entire current literature.

76.2 A formal model with a verifiable interpretation

Let $p \geq 5$ be prime and $h = (p-1)/2$. Work in the Laurent polynomial ring

$$\mathcal{R}_p = \overline{\mathbb{Q}}[T^{\pm 1}, X_1^{\pm 1}, \ldots, X_h^{\pm 1}].$$

The variable T represents π, and X_j represents $\Gamma(j/p)$. Define formal gamma values for $1 \leq a < p$ by

$$\mathcal{G}_a = \begin{cases} X_a, & a \leq h, \\ \dfrac{T}{\sin(\pi a/p) X_{p-a}}, & a > h. \end{cases} \tag{76.1}$$

The sine coefficients are nonzero algebraic numbers. Also set $\mathcal{G}_p = 1$, and, for $p < a < 2p$, set $\mathcal{G}_a = ((a-p)/p)\mathcal{G}_{a-p}$. Finally put

$$\mathcal{B}_{a,b} = \frac{\mathcal{G}_a \mathcal{G}_b}{\mathcal{G}_{a+b}} \qquad (1 \leq a, b < p). \tag{76.2}$$

Every identity in this ring becomes a true identity of gamma and beta values after the indicated substitution. The converse is not assumed: an arithmetic identity not implied by these formal rules could exist. Standard gamma and beta identities used here are [334, Sections 5.5 and 5.12].

For $m \in \mathbb{Z}$ and $1 \leq j \leq h$, define $\lambda_p(j, m)$ to be the smallest value of $\sum_{a,b} |n_{a,b}|$ in a formal identity

$$X_j^m = cT^s \prod_{1 \leq a,b < p} \mathcal{B}_{a,b}^{n_{a,b}}, \qquad c \in \overline{\mathbb{Q}}^{\times}, \quad s, n_{a,b} \in \mathbb{Z}. \tag{76.3}$$

If no identity exists, set $\lambda_p(j, m) = \infty$. Thus powers of π and algebraic factors have zero cost; beta factors and their reciprocals each have unit cost, with multiplicity. This convention is part of the definition. Arbitrary rational powers and beta values with other denominators are not included.

Theorem 76.1 (Sharp gamma-isolation depth)**.** *For every prime $p \geq 5$, $1 \leq j \leq (p-1)/2$, and $m \in \mathbb{Z}$,*

$$\boxed{\lambda_p(j, m) = \begin{cases} |m/p|(p-2), & p \mid m, \\ \infty, & p \nmid m. \end{cases}}$$

In particular, the first nonzero isolatable power is the pth power, and its exact beta-factor cost is $p-2$.

Proof. First discard the power of T and the algebraic coefficient of each Laurent monomial and retain its exponent vector in $\mathbb{Z}^h$. Write $e_1,\dots,e_h$ for the standard basis and define, with indices reduced modulo p,

$$v(0)=0, \qquad v(a)=\begin{cases} e_a, & 1\le a\le h,\\ -e_{p-a}, & h<a<p. \end{cases}$$

Then the exponent vector of a beta factor is

$$b(a,b)=v(a)+v(b)-v(a+b). \tag{76.4}$$

The homomorphism

$$\chi:\mathbb{Z}^h\longrightarrow \mathbb{Z}/p\mathbb{Z}, \qquad \chi(e_a)=a \pmod p$$

satisfies $\chi(v(a))=a$ and consequently annihilates every $b(a,b)$. If (76.3) holds, then $mj=0$ modulo p. Since p is prime and $j\neq 0$ modulo p, necessarily $p\mid m$.

For the sharp metric lower bound, let t be the inverse of j modulo p, and let $r_t(a)\in\{1,\dots,p-1\}$ be the least positive residue of ta for $a\not\equiv 0 \pmod p$. Define the real linear functional L_t on $\mathbb{R}^h$ by

$$L_t(e_a)=1-\frac{2r_t(a)}{p}.$$

The identity $r_t(p-a)=p-r_t(a)$ gives

$$L_t(v(a))=1-\frac{2r_t(a)}{p} \quad (a\neq 0), \qquad L_t(v(0))=0.$$

If $a+b\neq 0$ modulo p, the difference $r_t(a)+r_t(b)-r_t(a+b)$ is either 0 or p. Hence

$$L_t(b(a,b))\in\{-1,1\}.$$

If $a+b=0$ modulo p, the same expression is 0. Thus $|L_t(b(a,b))|\le 1$ in every case, whereas $L_t(e_j)=1-2/p$. Applying L_t to the exponent-vector identity in (76.3) yields

$$\sum_{a,b}|n_{a,b}|\ge |m|\frac{p-2}{p}. \tag{76.5}$$

For the matching construction, put $r_k\in\{1,\dots,p-1\}$ equal to the least positive residue of kj, for $1\le k\le p-1$. Thus $r_1=j$ and $r_{p-1}=p-j$. For $1\le k\le p-2$, write

$$r_k+j=r_{k+1}+p\varepsilon_k, \qquad \varepsilon_k\in\{0,1\}.$$

Recurrence and telescoping give

$$\prod_{k=1}^{p-2}\mathcal{B}_{r_k,j}=\frac{\mathcal{G}_j^{\,p-1}}{\mathcal{G}_{p-j}}\prod_{k=1}^{p-2}\left(\frac{p}{r_{k+1}}\right)^{\varepsilon_k}.$$

Reflection then proves the formal and actual identity

$$\boxed{\Gamma(j/p)^p=\pi C_{p,j}\prod_{k=1}^{p-2}B(r_k/p,j/p), \qquad C_{p,j}=\frac{1}{\sin(\pi j/p)}\prod_{k=1}^{p-2}\left(\frac{r_{k+1}}{p}\right)^{\varepsilon_k}.} \tag{76.6}$$

The coefficient $C_{p,j}$ is positive algebraic. This uses exactly $p-2$ beta factors. Raising the identity to an integer power, or taking its reciprocal, attains (76.5) for all m divisible by p. The case $m=0$ is the empty product. □

Remark 76.2 (The model boundary is essential). Theorem 76.1 does not prove that every possible expression for $\Gamma(j/p)$ requires $p-2$ periods. It treats integer products of the specified beta values under the stated formal relations. It neither assumes nor proves the Rohrlich–Lang conjecture. A theorem about nonlinear period products, an identity involving other denominators, or a direct Diophantine approximation argument could supply genuinely additional information.

76.3 A finite-degree non-detection theorem

The lower bound also describes exactly one way in which low-degree period tests fail to see whether a specified gamma coordinate has become algebraic. For an integer $d \geq 0$, let $\mathcal{V}_d \subset \mathcal{R}_p$ be the $\overline{\mathbb{Q}}[T^{\pm 1}]$-module spanned by products of at most d formal beta factors or reciprocals. It contains constants. Equivalently, its exponent support in the X variables is contained in the radius-d word ball for the generators (76.4).

Theorem 76.3 (Low-degree specialization preserves the formal tests)**.** *Fix j and any nonzero algebraic number α. If $2d < p-2$, then substitution $X_j = \alpha$ is injective on $\mathcal{V}_d$. In particular, this is true for linear beta tests when $p \geq 5$, and for tests using at most two beta factors per monomial when $p \geq 7$.*

Proof. Two distinct X-exponent vectors u, w in this word ball can coalesce after substitution only if $u - w = me_j$ with $m \neq 0$. Their difference has a representation using at most $2d$ beta generators or their negatives. The congruence and lower-bound arguments in the proof of Theorem 76.1 give $p \mid m$ and $2d \geq |m/p|(p-2) \geq p-2$, a contradiction. Thus distinct X-monomials remain distinct after the substitution. Multiplication of their coefficients by the nonzero algebraic powers of α cannot make a nonzero coefficient vanish. The coefficients remain Laurent polynomials in the independent variable T, proving injectivity. □

For a numerical interpretation one may set T equal to the actual π, set X_j to any positive algebraic α, and choose the other X variables positive and algebraically independent over $\overline{\mathbb{Q}}(\pi)$. Then every nonzero formal test in $\mathcal{V}_d$ remains nonzero. Such tuples exist by countability. They are not alternative gamma functions and do not match every analytic property of gamma. They show the insufficiency of precisely this collection of formal finite-degree tests. In the fifth case they can also satisfy the known transcendence-degree lower bound quoted above.

76.4 The five named targets and their optimal certificates

For $p = 5$, write $B_{a,b} = B(a/5, b/5)$. Formula (76.6) gives

$$\Gamma(1/5)^5 = \frac{\pi}{\sin(\pi/5)} B_{1,1} B_{2,1} B_{3,1}, \tag{76.7}$$

$$\Gamma(2/5)^5 = \frac{\pi}{5\sin(2\pi/5)} B_{2,2} B_{4,2} B_{1,2}. \tag{76.8}$$

Each uses three factors, and three is minimal in the model. For example, reflection also reduces $B_{3,1} = [\sin(\pi/5)/\sin(2\pi/5)]B_{1,1}$. Thus the first identity is a cubic product of two beta periods, not a new linear-period relation. Transcendence of each factor separately would not decide transcendence of the product after multiplication by π.

For $p = 7$, write $B_{a,b} = B(a/7, b/7)$. The three optimal identities are

$$\Gamma(1/7)^7 = \frac{\pi}{\sin(\pi/7)} B_{1,1}B_{2,1}B_{3,1}B_{4,1}B_{5,1}, \tag{76.9}$$

$$\Gamma(2/7)^7 = \frac{\pi}{7\sin(2\pi/7)} B_{2,2}B_{4,2}B_{6,2}B_{1,2}B_{3,2}, \tag{76.10}$$

$$\Gamma(3/7)^7 = \frac{2\pi}{49\sin(3\pi/7)} B_{3,3}B_{6,3}B_{2,3}B_{5,3}B_{1,3}. \tag{76.11}$$

Each uses five factors, and five is minimal. In all five identities the equivalence

$$\Gamma(j/p) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \pi \prod_{k=1}^{p-2} B(r_k/p, j/p) \in \overline{\mathbb{Q}}$$

is exact, because the omitted coefficient is nonzero algebraic and an algebraic pth power has an algebraic pth root. This equivalence relocates the missing statement; it does not prove it. For irrationality, even this final algebraicity equivalence is too coarse: an algebraic irrational gamma value would remain an unresolved rationality target.

Target	First power	Minimal beta cost	Outcome here
$\Gamma(1/5)$	5	3	individual question unresolved
$\Gamma(2/5)$	5	3	individual question unresolved
$\Gamma(1/7)$	7	5	individual question unresolved
$\Gamma(2/7)$	7	5	individual question unresolved
$\Gamma(3/7)$	7	5	individual question unresolved

The exact formulas and vector calculations were checked with rational arithmetic for every prime $5 \le p \le 97$ and every $1 \le j \le (p-1)/2$. These checks supplement the uniform proof; they are not substitutes for it. For $p = 5, 7$, an exhaustive search in the finite exponent word balls independently confirmed the costs 3 and 5.

76.5 What can be recycled next

The elliptic quantile classifier from the preceding manuscript uses a one-dimensional algebraic group: a linear relation between elliptic logarithms can force an endomorphism relation and then torsion. A higher-dimensional Fermat Jacobian supplies beta periods, but normalizing or solving a quantile problem does not turn the products in (76.7)–(76.11) into linear forms. Taking logarithms is not a remedy: the resulting logarithms are of beta values not known to be algebraic, so Baker's theorem on logarithms of algebraic numbers cannot be invoked.

The useful new output is therefore an exact choice of the next missing arithmetic statement. A direct attack on fifth gamma values must go beyond the tested linear beta information; the seventh case still escapes even the degree-two formal tests of Theorem 76.3. The proof gives a reusable lower-bound certificate for avoiding repeated attempts to extract one gamma coordinate from inadequate multiplicative data. Historical priority for this certificate formulation has not been established.

77 Named targets and the remaining arithmetic step

The following compact ledger separates the fourteen named-constant targets from the family theorems proved above. The target is stated explicitly: an irrationality result does not settle a transcendence target. "Unresolved here" describes this investigation and is not a claim to have

exhaustively classified every external preprint. Kurepa's left-factorial conjecture is a fifteenth, related target, rather than an additional constant.

Target	Transfer attempted	Exact remaining step
γ: irrationality	Even Taylor schedules and actual denominator clearance	Control the remaining sparse crossovers and eliminate the auxiliary $\log n$ term; see Theorem 73.5.
δ: irrationality	Exact primitive Laguerre content, prime-power trees	Establish a positive asymptotic margin after division by the exact gcd; see (73.8).
G: transcendence	Primitive full-rank determinant and basis changes	A different kernel or a new uniform nonvanishing and height estimate is required; rational basis changes preserve the primitive polynomial.
$\zeta(3)$: transcendence	Wilson trial functions and exact norms	Exclude algebraic degrees beyond known irrationality; the Wilson–Apéry connection is prior work.
$\zeta(5)$: transcendence	Exponential intervals with $O(n)$ logarithmic denominator bound	Prove a genuinely stronger arithmetic certificate; the bounds supplied here do not establish transcendence.
$\zeta(7)$: irrationality	The same explicit interval construction	Improve the reduced-denominator constants or the certificate enough to obtain a strict positive integerization margin.
E_2: transcendence	Full Lambert dilation and derivative closure	Every proposed finite rational linear closure is excluded by Theorem 75.2; an arithmetic value method outside this class is needed.
$\sum_{n\geq 1} F_n^{-1}$: transcendence	Remove the Lambert component and retain the theta component	No nonzero finite differential–dilation filter does this; the mixed value at the algebraic point remains uncontrolled.
B_1: irrationality	Algebraic recombinations of prime cutoffs	Finite logarithmic cancellation is excluded by Proposition 75.5; control of the infinite tail and its interaction with γ is absent.
$\Gamma(1/5)$: transcendence	Optimal prime-grid beta certificate	The exact formal cost is three factors for the fifth power; the required actual period relation remains unproved.
$\Gamma(2/5)$: transcendence	Its separate optimal beta certificate	The same formal cost does not establish the arithmetic nature of this gamma value.
$\Gamma(1/7)$: transcendence	Optimal prime-grid beta certificate	Five factors isolate the seventh power formally; the needed numerical algebraic-independence input is missing.
$\Gamma(2/7)$: transcendence	Its separate optimal beta certificate	The explicit identity does not supply the missing period-value transcendence step.
$\Gamma(3/7)$: transcendence	Its separate optimal beta certificate	The same limitation persists despite the proved optimal formal depth.
Kurepa conjecture	Transfer through shifted Laguerre residue classes	The exact congruence bridge in Proposition 73.4 does not prove global nonvanishing.

No new named-constant conjecture is closed by these attempts. What changes is the content of the next required estimate. A vague denominator problem becomes an exact gcd and local residue problem; a numerical entropy example becomes a minimum-alphabet theorem; an inverse quantile construction becomes an exact degree algorithm; and a failed additive transfer becomes a theorem excluding every finite rational differential–dilation repair in the stated class.

What has and has not been verified. The manuscript includes proofs of the stated family classifications, thresholds, and obstructions. Separate reviews checked sensitive steps, including

CM twisting hypotheses, exact valuation identities, the rational ternary entropy argument, and denominator clearance. Small exact computations corroborate algebraic identities and instances; they do not replace general proofs. The assembled text is a single TeX source with an embedded bibliography and no external input files. There is no Lean or comparable formal certification. Historical novelty, completeness of the literature comparison, and the appropriate journal level require further assessment.

Part X

Sparse Certificates and Information Constants

Editorial integration note. The unresolved ternary entropy question is narrowed in Parts XI and XII. The latter excludes cubic probability fields. The three-jet functional result not yet obtained in this part appears with a precise derivative condition in Part XI. The factorial-threshold cancellation count here is different from the final linear-threshold count.

Original part framing.

Scope and verification. This is a standalone successor to the earlier arithmetic-threshold manuscript. Each use of a previous construction is restated, and the additional conclusions are identified explicitly. The proofs use classical external theorems, including Baker's logarithmic theorem and orthogonal-polynomial identities. Independent proof reviews and exact rational checks are not a formal verification of this manuscript. The 2026 external sources report proofs of irrationality for $\zeta(5)$ and Catalan's constant [342, 343, 344]; their complete proofs and Lean dependencies are not independently verified here. Our corresponding targets are transcendence. Publication significance and historical priority require further expert review.

78 The additional conclusions and their arithmetic limits

Rapid approximation, exact finite reconstruction, and functional independence do not automatically give irrationality of a distinguished constant. The preceding manuscript exposed several such gaps. This successor asks which remaining assertions can actually be proved, whether an unsuccessful transfer can be excluded throughout a larger class, and whether the same obstruction yields arithmetic theorems in another field.

From local content to global sparsity. The previous Laguerre content formula determined the exact common divisor at a fixed index. Theorem 82.3 now proves

$$\#\{2 \le n \le X : \log g_n \ge \varepsilon n \log n\} = O_{A,\varepsilon}(\log X).$$

This controls the sum of all local contributions without assuming that every p-adic root is simple. It excludes bounded primitive linear forms outside a logarithmically sparse set, but does not eliminate an infinite exceptional subsequence. A different failed route to Euler's constant is controlled by nonzero curvature: Theorem 83.1 bounds the number of indices admitting a bounded primitive even-Taylor crossover form by $O_B(X^{1/3})$. The logarithmic term in those forms remains present.

From a transcendental entropy to an independent family. Theorem 79.7 supplies the rational matrices

$$\rho_r = \frac{1}{2} I_2 + 2^{-(r+3)/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \qquad (r \text{ an odd prime}).$$

Their entropies and 1 are linearly independent over $\overline{\mathbb{Q}}$. This is stronger than proving each entropy transcendental separately, but is not algebraic independence. The channel theorem also identifies exactly how totally positive algebraic inputs can lose only an algebraic amount of entropy.

A sharp finite-data obstruction. Theorem 80.1 determines

$$d(M) = M + 1.$$

Here $d(M)$ is the minimum common alphabet size for two positive algebraic distributions with the same first M power sums and different arithmetic types of Shannon entropy: rational versus transcendental. This is not the earlier polynomial-density degree problem. Theorem 81.1 gives a separate four-atom distribution whose entries are all irrational algebraic numbers and whose entropy is exactly one bit. The minimum for this latter all-irrational problem remains either three or four.

Analytic tuning and nonlinear transfer. The Wilson shift theorem gives an exact exponential rate and shows that equal degree and shift uniquely maximize it per total budget. The Catalan construction produces strict rational intervals without an auxiliary zeta value. These facts improve the construction but do not supply a transcendence certificate. The Lambert results instead address functions: they exclude nonlinear relations between distinct differential–dilation jets, including mixed Lambert–Fibonacci pairs, and exclude algebraic replacement by finite-level modular functions. Appending the prime-harmonic function to such a pair gives an algebraically independent triple over $\mathbb{C}(q)$ (Theorem 85.10). Evaluation at one algebraic point is a further arithmetic step which is not established here.

Unrestricted gamma identities still have an exact cost. For a reduced rational argument a/q, Theorem 86.2 determines the least number of beta factors in an integer beta word representing $\Gamma(a/q)^m$, modulo algebraic factors and rational powers of π, throughout the standard identity calculus:

$$\ell(m[a/q]) = \begin{cases} |m|(q-2)/q, & q \mid 2m, \\ \infty, & q \nmid 2m. \end{cases}$$

The auxiliary arguments have no denominator restriction. This closes a possible formal shortcut from the earlier prime-grid model. It does not classify unknown numerical gamma relations outside the standard identities.

The final ledger records what each named target gains and which step is still missing. A new example, an obstruction theorem, an external recent result, and a solution of a famous conjecture are kept distinct throughout.

79 Arithmetic information loss and independent entropy constants

Throughout this section $H_2(p) = -\sum_i p_i \log p_i / \log 2$. All probability vectors have strictly positive entries; zero rows of a channel are deleted. A rational channel is a matrix $A = (A_{yi}) \in \mathbb{Q}_{\geq 0}^{m \times n}$ with $\sum_y A_{yi} = 1$. We study the signed entropy difference

$$L_A(p) = H_2(p) - H_2(Ap).$$

It is a conditional entropy when A is a deterministic coarse graining, but it need not be nonnegative for a general channel.

The prior entropy obstruction extends to a complete structural classification for channels. Its additional input is the equality case of relative-entropy data processing, proved below. We also obtain an explicit infinite family of rational two-dimensional density matrices whose

entropies, together with 1, are linearly independent over $\overline{\mathbb{Q}}$. Linear independence is not algebraic independence. Baker's logarithmic theorem is the classical arithmetic input [231]; the arithmetic statements here are distinct from the semialgebraic-level-set results of [335, 336]. No exhaustive priority claim is made.

79.1 Transport of logarithmic relations

Lemma 79.1 (Relations and multiplicative valuations)**.** *For positive algebraic numbers $a_1, \dots, a_N$, every algebraic linear relation among their real logarithms is an algebraic linear combination of rational linear relations. Consequently:*

1. *If $\sigma(a_j) > 0$ for all j and $\sum_j c_j \log a_j = 0$, where c_j are algebraic, then $\sum_j c_j \log \sigma(a_j) = 0$. The coefficients c_j are* not *conjugated.*
2. *If v is an additive valuation of a number field containing the a_j, then $\sum_j c_j v(a_j) = 0$.*

Proof. Choose a rational basis of the span of the logarithms. Baker's theorem makes it independent over $\overline{\mathbb{Q}}$, so the two relation spaces have the stated relation. A rational relation, after clearing denominators, is equivalent to $\prod_j a_j^{k_j} = 1$ with integers k_j. Apply σ or v to this identity and then take the same algebraic linear combination of the resulting relations. Positivity justifies real logarithms in the first case. □

79.2 A Galois criterion for exact information preservation

Join input symbols i, j by an edge when some row y of A has $A_{yi}A_{yj} > 0$. Let $\mathcal{C}(A)$ be the connected components of this graph. Every nonzero output row belongs to exactly one component; write Y_C for those rows associated with C.

Theorem 79.2 (Galois conjugates and channel sufficiency)**.** *Let p be a positive algebraic probability vector. If $L_A(p)$ is algebraic and $q = \sigma(p)$ is a simultaneous positive real conjugate, then*

$$\frac{p_i}{q_i} = \frac{p_j}{q_j} \qquad (i, j \in C,\ C \in \mathcal{C}(A)).$$

Equivalently, the channel loses no relative entropy between p and q:

$$D(p\|q) = D(Ap\|Aq).$$

Here D uses natural logarithms. In particular, if the overlap graph is connected and p has a distinct positive conjugate, $L_A(p)$ is transcendental.

Proof. Put $u = Ap$ and $v = Aq = \sigma(u)$. If $h = L_A(p)$ is algebraic, Lemma 79.1 applied to

$$-\sum_i p_i \log p_i + \sum_y u_y \log u_y - h \log 2 = 0$$

allows p_i, u_y inside the logarithms to be replaced by q_i, v_y, while retaining all coefficients. Subtraction yields $D(p\|q) - D(u\|v) = 0$.

For completeness, for each output y define posterior vectors on its positive support by

$$r_{i|y} = \frac{A_{yi}p_i}{u_y}, \qquad s_{i|y} = \frac{A_{yi}q_i}{v_y}.$$

A direct expansion gives

$$D(p\|q) - D(u\|v) = \sum_y u_y D(r_{\cdot|y}\|s_{\cdot|y}) \geq 0.$$

The elementary logarithm inequality gives equality only when every pair of posterior vectors agrees. This says $p_i/q_i = u_y/v_y$ whenever $A_{yi} > 0$, which is equivalent to constancy on each connected component. For a connected graph normalization forces the constant to be one. □

Call an algebraic number totally positive when all its conjugates over $\mathbb{Q}$ are positive real numbers. The next theorem gives a finite exact arithmetic test, not merely a sufficient obstruction.

Theorem 79.3 (Complete classification for totally positive input)**.** *Suppose every p_i is totally positive. For $C \in \mathcal{C}(A)$ put*

$$P_C = \sum_{i \in C} p_i, \qquad r_i = \frac{p_i}{P_C} \ (i \in C), \qquad s_y = \sum_{i \in C} A_{yi} r_i \ (y \in Y_C).$$

Then $L_A(p)$ is algebraic if and only if both of the following hold:

1. *Every r_i is rational.*
2. *For every odd prime ℓ,*

$$\sum_C P_C b_{C,\ell} = 0, \qquad b_{C,\ell} = -\sum_{i \in C} r_i v_\ell(r_i) + \sum_{y \in Y_C} s_y v_\ell(s_y).$$

Only finitely many primes occur. When these conditions hold,

$$L_A(p) = \sum_C P_C b_{C,2}.$$

For a connected overlap graph, algebraicity therefore forces p to be rational, and the algebraic entropy difference is rational.

Proof. Take a normal closure of the field generated by the p_i. Every automorphism gives a positive conjugate. Theorem 79.2 therefore implies $p_i/P_C = \sigma(p_i/P_C)$ for every automorphism, so each r_i is rational. The s_y are then rational as well.

Both input and output have component masses P_C. Applying the entropy chain rule separately to input and output cancels their common mass entropy and gives

$$L_A(p) = \sum_C P_C\{H_2(r_C) - H_2(s_C)\} = \sum_\ell \left(\sum_C P_C b_{C,\ell} \right) \frac{\log \ell}{\log 2}.$$

The logarithms of distinct primes are rationally independent by unique factorization, hence linearly independent over $\overline{\mathbb{Q}}$ by Baker's theorem. Algebraicity of the last display is therefore exactly cancellation of its odd-prime coefficients. The displayed value follows, proving necessity and sufficiency. □

Corollary 79.4 (Conditional entropy)**.** *For a positive joint law $p(x, y)$ whose entries are totally positive algebraic numbers, algebraicity of $H_2(X \mid Y)$ forces every conditional probability $p(x \mid y)$ to be rational. After this structural test, the odd-prime test of Theorem 79.3 is necessary and sufficient.*

Proof. Apply the theorem to the deterministic map $(x, y) \mapsto y$. Its overlap components are exactly the fibres with fixed y. □

The conclusion concerns conditional probabilities, not the joint probabilities themselves. For example, set $a = (2+\sqrt{2})/4$ and take the two columns of a positive 3×2 joint law to be

$$a(1/2, 1/4, 1/4), \qquad (1-a)(1/4, 3/8, 3/8).$$

All entries are totally positive. This example illustrates the structural test, but it does not pass the odd-prime test: its conditional entropy has a nonzero $\log 3$ coefficient and is transcendental. To obtain an algebraic irrational conditional entropy instead, use a first column with conditional distribution $(1/2, 1/4, 1/8, 1/8)$ and a second with $(1/4, 1/4, 1/4, 1/4)$, with masses $a, 1-a$. The resulting strictly positive joint law has

$$H_2(X \mid Y) = \frac{7}{4}a + 2(1-a) = 2 - \frac{a}{4}.$$

Thus disconnected channels genuinely permit algebraic irrational entropy differences, even with totally positive input.

Corollary 79.5 (A cone of transcendental entropy differences)**.** *Let $p^{(j)}$ be finitely many positive algebraic laws, let A_j be rational channels, and let $c_j > 0$ be real algebraic numbers. Suppose a common field embedding makes all $q^{(j)} = \sigma(p^{(j)})$ positive. If*

$$\sum_j c_j L_{A_j}(p^{(j)})$$

is algebraic, then every A_j preserves relative entropy between $p^{(j)}$ and $q^{(j)}$. Consequently, if just one pair has a strict data-processing inequality, that weighted sum is transcendental.

Proof. Transport the combined logarithmic relation while retaining the coefficients c_j. Subtraction gives a sum of nonnegative data-processing gaps, weighted by the positive c_j, equal to zero. Each gap must vanish. □

79.3 Correlations with rational marginals

Theorem 79.6 (Marginally invisible algebraic correlations)**.** *Let $p(x, y)$ be a positive algebraic joint law. If a distinct positive conjugate $q = \sigma(p)$ has the same two marginals as p, then $I_2(X;Y)$ is transcendental. More generally, for a positive law $p(x, y, z)$, a distinct positive conjugate with the same XZ and YZ marginals certifies transcendence of $I_2(X;Y \mid Z)$. In particular, totally positive algebraic joint entries and rational specified marginals have this property whenever the joint law has an irrational entry.*

Proof. The logarithmic formulas are

$$I_2(X;Y) = \frac{1}{\log 2} \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p_X(x)p_Y(y)}$$

and

$$I_2(X;Y \mid Z) = \frac{1}{\log 2} \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y,z)p_Z(z)}{p_{XZ}(x,z)p_{YZ}(y,z)}.$$

If the relevant value were algebraic, transporting bases and subtracting would give $D(p\|q) = 0$: all marginal logarithms cancel by hypothesis. Thus $p = q$, a contradiction. For the final claim, choose an automorphism moving an irrational entry; total positivity makes the conjugate law positive and rational marginals are fixed. □

79.4 An infinite linearly independent family of quantum entropies

Theorem 79.7 (Independent entropy constants in dimension two)**.** *For each odd prime r, define the rational density matrix*

$$\rho_r = \frac{1}{2} I_2 + 2^{-(r+3)/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad h_r = -\operatorname{tr}(\rho_r \log_2 \rho_r).$$

Then 1 *and all the h_r are linearly independent over $\overline{\mathbb{Q}}$. In particular every nonzero finite algebraic linear combination of the h_r is transcendental.*

Proof. The two eigenvalues are

$$\lambda_r^{\pm} = \frac{1}{2} \pm \frac{\sqrt{2}}{2^{(r+3)/2}}, \qquad \lambda_r^{+} \lambda_r^{-} = \frac{2^r - 1}{2^{r+2}}.$$

They are positive and lie in $K = \mathbb{Q}(\sqrt{2})$. For distinct primes r, s,

$$\gcd(2^r - 1, 2^s - 1) = 2^{\gcd(r,s)} - 1 = 1.$$

Fix a rational prime q dividing $2^r - 1$. It is odd. At a prime ideal $\mathfrak{q}$ above q where one of $\lambda_r^{\pm}$ has positive valuation, the other has zero valuation: their sum is 1, and both are integral away from 2. Existence follows by taking the valuation of their product. For every other prime index s, both $\lambda_s^{\pm}$ have zero $\mathfrak{q}$-valuation, by the coprimality just proved. Also $v_{\mathfrak{q}}(2) = 0$.

Suppose a finite relation $b + \sum_r c_r h_r = 0$ holds with algebraic coefficients. Multiplying by $\log 2$ gives

$$b \log 2 - \sum_r c_r (\lambda_r^{+} \log \lambda_r^{+} + \lambda_r^{-} \log \lambda_r^{-}) = 0.$$

Apply the valuation consequence of Lemma 79.1 at the ideal chosen for index r. All terms disappear except $-c_r \lambda_r^{\pm} v_{\mathfrak{q}}(\lambda_r^{\pm})$. Its two non-coefficient factors are nonzero, so $c_r = 0$. Repeat for every index, then obtain $b = 0$. □

This proof needs neither an unproved independence conjecture nor Zsigmondy's primitive-divisor theorem: the restriction to odd prime indices supplies pairwise coprime norms by an elementary identity. It does not prove algebraic independence of the entropy values.

Corollary 79.8 (Higher-order correlations)**.** *For an odd prime r put $t_r = \sqrt{2}/2^{(r+1)/2}$. For any $d \geq 2$ define a law on $\{-1, 1\}^d$ by*

$$p_{r,d}(x_1, \ldots, x_d) = 2^{-d} \left(1 + t_r \prod_{i=1}^{d} x_i \right).$$

Every proper marginal is uniform and rational, while every atom is totally positive algebraic. Its total correlation and its conditional mutual information between the first two coordinates given the remaining coordinates both equal $1 - h_r$. Consequently 1 *and these correlation constants, as r runs through odd primes, are linearly independent over $\overline{\mathbb{Q}}$.*

Proof. Summing over any missing coordinate cancels the parity term. Each of the two parity classes has 2^{d-1} atoms and mass $(1 \pm t_r)/2 = \lambda_r^{\pm}$, so the joint entropy is $d - 1 + h_r$. Uniformity of proper marginals gives total correlation $d - (d - 1 + h_r) = 1 - h_r$ and, using the entropy formula for conditional mutual information, the same value for the stated conditional mutual information. For $d = 2$, the conditioning variable is empty. The linear-independence assertion follows immediately from Theorem 79.7. □

These are explicit arithmetic conclusions for information-theoretic constants. They do not classify Euler's constant, Gompertz's constant, or any gamma or zeta value. Their contribution is an exact channel criterion and an infinite independent family, rather than a claimed resolution of a classical named transcendence problem.

80 A sharp limit on arithmetic inference from finitely many moments

For a positive probability vector $p = (p_1, \dots, p_n)$, write $s_k(p) = \sum_i p_i^k$. The normalization $s_1 = 1$ is included in the moment count. For an integer $M \geq 1$, let $d(M)$ be the least common alphabet size of two positive algebraic probability vectors with identical $s_1, \dots, s_M$, one having rational Shannon entropy in bits and the other having transcendental Shannon entropy in bits.

Theorem 80.1 (Exact moment threshold)**.** *For every integer $M \geq 1$,*

$$d(M) = M + 1.$$

The rational-entropy vector may be chosen dyadic. The other vector may be chosen arbitrarily close to it, with its $M + 1$ entries all distinct, positive, and conjugate roots of one irreducible polynomial over $\mathbb{Q}$. Thus its diagonal density matrix has rational characteristic polynomial. The pair has identical integer-order Rényi entropies of orders $2, \dots, M$.

Proof. First suppose $n \leq M$. Newton's identities express every elementary symmetric polynomial in n variables as a rational polynomial in $s_1, \dots, s_n$. Equality of these power sums therefore determines the monic polynomial $\prod_i (X - p_i)$, hence the unordered vector and its Shannon entropy. This proves $d(M) \geq M + 1$.

For the converse put $n = M + 1$. Use the dyadic vector

$$p^{(0)} = (2^{-1}, 2^{-2}, \dots, 2^{-(n-2)}, 2^{-(n-1)}, 2^{-(n-1)})$$

when $n \geq 3$, and $p^{(0)} = (1/2, 1/2)$ when $n = 2$. Its entries sum to one, and its entropy is rational because the negative base-two logarithm of every entry is an integer. Set

$$f(X) = \prod_{i=1}^{n} (X - p_i^{(0)}) \in \mathbb{Z}[1/2][X], \qquad a = 2^{-(n-1)}.$$

The root a is double; the remaining roots, if present, are simple and greater than a. Factor $f(X) = (X - a)^2 g(X)$, where

$$\operatorname{sign} g(a) = (-1)^{n-2}.$$

Consequently, for sufficiently small nonzero τ with sign $(-1)^{n-1}$, the polynomial $f(X) + \tau$ has n distinct positive real roots. To see this directly, the simple roots persist under a small real perturbation, and near a the equation is $(X - a)^2 = -\tau / g(X)$ with positive right-hand side. Its two roots lie on opposite sides of a and approach a as $\tau \to 0$. They stay positive after shrinking the perturbation. Equivalently, the same count follows by the intermediate value theorem on two small intervals adjacent to a and on disjoint intervals around the simple roots. The degree then exhausts the possible roots.

Fix an odd rational prime ℓ. Choose positive integers b with $\ell \nmid b$, tending to infinity, and put

$$\tau = \frac{(-1)^{n-1}}{\ell b}.$$

We next prove that $f+\tau$ is irreducible. Write $f(X)=X^n+c_{n-1}X^{n-1}+\cdots+c_0$. All c_j are integral at ℓ, since their denominators are powers of two. The reciprocal polynomial

$$R(X)=\ell X^n\{f(1/X)+\tau\}$$

has constant coefficient ℓ, intermediate coefficients $\ell c_{n-1},\dots,\ell c_1$, and leading coefficient $\ell c_0+(-1)^{n-1}/b$. Its leading coefficient is an ℓ-adic unit; every other coefficient is divisible by ℓ; and its constant coefficient is not divisible by ℓ^2. After division by the leading unit, Eisenstein's criterion over $\mathbb{Z}_\ell$ applies. Hence R is irreducible over $\mathbb{Q}_\ell$, and therefore over $\mathbb{Q}$. Reciprocation proves irreducibility of $f+\tau$ as well. Clearing denominators prime to ℓ gives the same elementary Eisenstein argument over the integers if preferred.

Let $p^{(b)}$ be the roots of $f+\tau$, ordered to converge to $p^{(0)}$. The coefficient of X^{n-1} is unchanged, so their sum is one. In fact all coefficients except the constant coefficient are unchanged. Newton's identities give

$$s_k(p^{(b)})=s_k(p^{(0)})\qquad(1\le k\le n-1=M).$$

Each root has degree $n\ge 2$, and all its conjugates are among these positive roots. Thus the entries are totally positive and irrational. The positive-conjugate entropy criterion (the one-output case of Theorem 79.2) proves that $H_2(p^{(b)})$ is transcendental. Explicitly, the one-output channel has output entropy zero and a connected overlap graph; an automorphism moving one root provides the distinct positive conjugate required by that theorem.

Finally, the integer-order Rényi entropy is $\mathcal{R}_k(p)=(1-k)^{-1}\log_2 s_k(p)$ for $k\ge 2$. The asserted equalities follow from the matching power sums. Convergence of the roots supplies arbitrary closeness. □

Example 80.2 (The first nontrivial spectral moment). For $M=2$ the dyadic vector $(1/2,1/4,1/4)$ has Shannon entropy $3/2$ and second power sum $3/8$. The three roots of

$$X^3-X^2+\frac{5}{16}X-\frac{23}{768}$$

are positive, sum to one, and have the same second power sum $3/8$, but their Shannon entropy is transcendental. This polynomial is $(X-1/2)(X-1/4)^2+1/768$. Its sign is negative at 0, positive at $1/4$, negative at $5/12$, and positive at $1/2$, so it has three positive roots in the intervening intervals. The reciprocal Eisenstein argument uses $\ell=3$ and $b=256$.

Corollary 80.3 (No uniform finite-moment arithmetic classifier)**.** *No fixed finite list of initial power sums can decide, for all finite-dimensional algebraic probability vectors or spectra, whether the Shannon entropy in bits is rational or transcendental. For each M, even exact knowledge of $s_1,\dots,s_M$ fails at dimension $M+1$, and this dimension is optimal for such a pair.*

Proof. The two vectors in Theorem 80.1 produce exactly the same supplied data and require different answers. □

This is a limitation of the specified observation data, not of all algorithms. For a known dimension n, the first n power sums do determine the spectrum. The construction uses rational characteristic polynomials, but does not assert that the displayed diagonal matrices have rational entries. It does not resolve a classical named irrationality conjecture, and historical novelty of the sharp arithmetic formulation has not been established by an exhaustive literature review.

81 Four irrational probabilities with one bit of entropy

For a probability vector with positive entries, write

$$H_b(p) = -\sum_i p_i \log_b p_i, \qquad b > 1.$$

The preceding manuscript established that algebraic irrational entropy in bits first occurs on three atoms. It also constructed five positive irrational algebraic probabilities whose entropy is rational, without claiming the minimum for that different problem. We reduce the latter construction to four atoms. Its exact minimum is still either three or four. The other results in this section classify the binary problem for all algebraic logarithm bases, and give an exact critical-value reduction for the remaining ternary problem. These statements use classical Baker theory, rather than a new transcendence theorem.

81.1 An exact four-atom certificate

Theorem 81.1 (Four irrational atoms, exactly one bit)**.** *Let $\varphi = (1+\sqrt{5})/2$. The four positive algebraic irrational numbers*

$$p = \left(\frac{\varphi}{2}, \frac{\varphi^{-5}}{2}, \frac{\varphi^{-4}}{2}, \frac{\varphi^{-4}}{2}\right) = \left(\frac{1+\sqrt{5}}{4}, \frac{5\sqrt{5}-11}{4}, \frac{7-3\sqrt{5}}{4}, \frac{7-3\sqrt{5}}{4}\right)$$

sum to one and satisfy $H_2(p) = 1$.

Proof. The equality $\varphi^2 = \varphi + 1$ gives

$$\varphi^{-4} = 5 - 3\varphi, \qquad \varphi^{-5} = 5\varphi - 8.$$

Consequently

$$\varphi + \varphi^{-5} + 2\varphi^{-4} = 2, \qquad \varphi - 5\varphi^{-5} - 8\varphi^{-4} = 0.$$

Positivity follows either from the original expressions or from $125 > 121$ and $49 > 45$. The nonzero $\sqrt{5}$ coefficients show that every entry is irrational. For $m = (1, -5, -4, -4)$, the two displayed identities say $\sum_i p_i = 1$ and $\sum_i m_i p_i = 0$. Substitution into $\log_2 p_i = -1 + m_i \log_2 \varphi$ gives

$$H_2(p) = 1 - \log_2 \varphi \sum_i m_i p_i = 1.$$

□

A polynomial certificate explains the simultaneous cancellation. Put

$$F(t) = \frac{t + t^{-5} + 2t^{-4}}{2}.$$

Then

$$F(t) - 1 = \frac{(t^2 - t - 1)^2 (t^2 + 1)}{2t^5}. \tag{81.1}$$

Thus $F(\varphi) = 1$ and $F'(\varphi) = 0$. The four summands of F, counting the repeated $t^{-4}/2$ separately, are exactly the four probabilities. This is a verifiable algebraic identity, not a high-precision numerical entropy calculation.

Corollary 81.2. *Let $m_{\mathrm{irr,rat}}$ be the smallest number of positive algebraic irrational probabilities with rational entropy in bits. Then*

$$3 \le m_{\mathrm{irr,rat}} \le 4.$$

Proof. One positive probability equals one and hence is rational. The binary case is excluded by Theorem 81.3 below, since the base 2 is not an algebraic unit. Theorem 81.1 supplies the upper bound. □

81.2 The complete binary classification for an algebraic base

We use Baker's theorem in the following standard form: a set of real logarithms of positive algebraic numbers which is linearly independent over $\mathbb{Q}$ remains linearly independent over $\overline{\mathbb{Q}}$ [231]. Equivalently, every algebraic-coefficient logarithmic relation is a linear combination of rational-coefficient logarithmic relations. The rational-spectrum entropy dichotomy is also recorded by Blakaj–Wolf [337, Theorem 2].

Theorem 81.3 (Binary algebraic-base classification)**.** *Let $b > 1$ be algebraic and let $0 < p < 1$ be algebraic irrational. Then $H_b(p, 1-p)$ is algebraic if and only if there exist distinct positive rational numbers u, v such that*

$$p = b^{-u}, \qquad 1 - p = b^{-v}.$$

In that case

$$H_b(p, 1-p) = up + v(1-p)$$

is algebraic irrational, and b is an algebraic unit.

Proof. Suppose $h = H_b(p, 1-p)$ is algebraic. If h were outside the rational span of $1, p$, Baker's theorem applied after splitting the coefficients with respect to the independent numbers $1, p, h$ would force $\log b = 0$. Thus $h = a + cp$ for $a, c \in \mathbb{Q}$. In the identity

$$p \log p + (1-p)\log(1-p) + (a + cp)\log b = 0,$$

splitting the independent algebraic coefficients $1, p$ gives

$$\log(1-p) + a \log b = 0, \qquad \log p - \log(1-p) + c \log b = 0.$$

Hence $1 - p = b^{-a}$ and $p = b^{-(a+c)}$. Positivity and $b > 1$ imply $a > 0$ and $a + c > 0$. Set $v = a$ and $u = a + c$. Equality $u = v$ would force $p = 1/2$, contrary to the hypothesis. The converse follows by direct substitution, and the displayed affine expression in the irrational number p is irrational because $u - v \neq 0$.

Choose a positive integer d such that $m = du$ and $n = dv$ are distinct positive integers, and put $\alpha = b^{-1/d}$. The identity $\alpha^m + \alpha^n = 1$ shows that α is a root of the monic polynomial $X^{\max(m,n)} + X^{\min(m,n)} - 1$. Its reciprocal is also an algebraic integer, since the reciprocal polynomial is monic up to sign. Thus α, and therefore $b = \alpha^{-d}$, is an algebraic unit. □

Corollary 81.4 (A complete obstruction for nonunit bases)**.** *If an algebraic $b > 1$ is not an algebraic unit, every binary distribution with irrational algebraic probabilities has transcendental entropy in base b. This includes every rational base $b > 1$.*

Proof. The first assertion is the contrapositive of the theorem. A rational algebraic unit is 1 or -1, so no rational $b > 1$ is a unit. □

For completeness, the rational-probability case has a different exact criterion and must not be confused with Theorem 81.3. If $p = a/N$ with $0 < a < N$, define the positive algebraic number

$$A = \left(\frac{N^N}{a^a (N-a)^{N-a}} \right)^{1/N}.$$

Then $H_b(p, 1-p) = \log A / \log b$ is algebraic if and only if $A = b^r$ for some rational r; in that case the entropy equals r. Indeed an algebraic irrational ratio would contradict the Gelfond–Schneider theorem. The individual probabilities need not be rational powers of b in this rational-probability case.

Example 81.5 (Golden and plastic bases). For $b = \varphi$, the probabilities $(\varphi^{-1}, \varphi^{-2})$ have entropy

$$H_\varphi = \frac{5 - \sqrt{5}}{2}.$$

Let $\rho > 1$ be the real root of $X^3 - X - 1$. Then

$$\rho^{-2} + \rho^{-3} = 1, \qquad \rho^{-1} + \rho^{-5} = 1,$$

the second identity following from $\rho^5 = \rho^4 + 1$. These produce two different binary distributions with algebraic irrational entropy in base ρ. The irrationality of their probabilities can be checked without a numerical approximation: if a positive power of ρ were rational, its nonreal conjugates would have the same modulus as ρ, whereas their product of moduli is $1/\rho$ and $\rho > 1$. These examples show that the base-2 minimum alphabet of three is sensitive to the choice of algebraic base.

81.3 An exact reduction of the unresolved ternary case

Lemma 81.6 (Arithmetic dimension and logarithmic rank)**.** *Let $p_1, \dots, p_n > 0$ be algebraic with $\sum_i p_i = 1$, and let $b > 1$ be algebraic. Define*

$$t = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{p_1, \dots, p_n\}, \qquad r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\log p_1, \dots, \log p_n, \log b\}.$$

If $h = H_b(p)$ is algebraic, then

$$h \in \operatorname{span}_{\mathbb{Q}}\{p_1, \dots, p_n\}, \qquad t + r \le n + 1.$$

Proof. If h were outside the indicated rational span, split the entropy identity with respect to a basis of that span together with h. Baker's theorem would force $\log b = 0$, a contradiction. Now choose a rational basis $\theta_1, \dots, \theta_t$ of the probability span and expand every p_i and h in that basis. Splitting the entropy identity produces t rational logarithmic relations. Their coefficient rows are independent because the submatrix formed by the p_i coordinates has row rank t. The rational relation space among the $n + 1$ logarithms has dimension $n + 1 - r$, proving the inequality. □

Theorem 81.7 (Three-monomial critical-value reduction)**.** *Suppose an algebraic ternary probability vector p has at least one irrational entry and $H_2(p) = h \in \mathbb{Q}$. Then there exist a positive algebraic number w and integers m_1, m_2, m_3, not all zero, such that*

$$p_i = 2^{-h} w^{m_i}, \qquad \sum_{i=1}^{3} w^{m_i} = 2^h, \qquad \sum_{i=1}^{3} m_i w^{m_i} = 0. \tag{81.2}$$

Moreover $\log w$ and $\log 2$ are rationally independent. Conversely, any such algebraic data with $h \in \mathbb{Q}$ give a probability vector of entropy h. The rational span of the probabilities and the rational span of their logarithms together with $\log 2$ both have dimension two in the nonrational case.

Proof. In Lemma 81.6, nonrationality gives $t \geq 2$. If $r = 1$, each probability is a rational power of two. Such positive powers summing to one must all have integral exponents: for a common exponent denominator d, the independence of $1, 2^{1/d}, \dots, 2^{(d-1)/d}$ follows from Eisenstein's criterion for $X^d - 2$, and every nonconstant basis coefficient in the sum is nonnegative. Hence $r = 1$ would make every probability rational. Thus $r \geq 2$, and $t + r \leq 4$ forces $t = r = 2$.

Choose a positive algebraic u such that $\log 2, \log u$ form a rational basis of the logarithm span. Write $\log p_i = a_i \log 2 + b_i \log u$ with $a_i, b_i \in \mathbb{Q}$. Baker's theorem gives

$$\sum_i b_i p_i = 0, \qquad \sum_i a_i p_i = -h.$$

The kernel of the rational map $(x_i) \mapsto \sum_i x_i p_i$ is one-dimensional, and $b = (b_i)$ is a nonzero vector in it. As $\sum_i (a_i + h) p_i = 0$, there is a rational λ with $a_i + h = \lambda b_i$. Therefore

$$p_i = 2^{-h} (2^{\lambda} u)^{b_i}.$$

Take a common positive denominator d of the b_i, put $w = (2^{\lambda} u)^{1/d}$, and set $m_i = d b_i$. This gives (81.2). Rational independence is preserved by this change of logarithm basis. The converse is obtained by substituting $\log_2 p_i = -h + m_i \log_2 w$. □

The last theorem converts the missing lower bound into a precise arithmetic problem about a stationary value of three Laurent monomials. It does not prove that such data are impossible. In particular, the four-atom construction proves an improved upper bound, not the assertion $m_{\mathrm{irr,rat}} = 4$. No claim that a previously named irrationality conjecture is closed is made in this section.

82 A global sparsity theorem for primitive Laguerre content

The earlier analysis gave an exact formula for the common divisor of the numerator and denominator of each Laguerre approximation to the Euler–Gompertz constant. It did not control that common divisor globally as the index varied. Here we prove a quantitative global restriction: factorial-sized common divisors occur at only logarithmically many indices. The proof uses positive Jacobi matrices and the elementary arithmetic of least common multiples, rather than a uniform hypothesis about the simplicity of every local p-adic root.

82.1 The fixed family and the arithmetic input

Fix a positive integer A, and set

$$\delta_A = \int_0^{\infty} \frac{e^{-t}}{1 + At}\, dt, \qquad Q_n = \sum_{j=0}^{n} j! \binom{n}{j}^2 A^j.$$

Let $P_0 = 0$, $P_1 = 1$, $Q_0 = 1$, $Q_1 = 1 + A$, and apply

$$U_n = (1 + (2n-1)A) U_{n-1} - (n-1)^2 A^2 U_{n-2} \qquad (n \geq 2) \tag{82.1}$$

to both sequences. We write

$$g_n = \gcd(P_n, Q_n), \qquad q_n = Q_n / g_n, \qquad \ell_n = (Q_n \delta_A - P_n)/g_n.$$

Thus q_n is the actual reduced denominator and ℓ_n is the corresponding primitive positive linear form. These classical approximants are reviewed in [338]; recent continued-fraction work on the same special function is discussed in [339].

For self-containment, we record the two arithmetic facts needed below:

$$\gcd(Q_n, Q_{n-1}) = 1, \qquad g_n = \gcd(Q_n, ((n-1)!)^2). \tag{82.2}$$

Indeed, primes dividing A never divide Q_n. For a prime p not dividing A, Lucas's congruence gives $Q_{kp+r} \equiv Q_r \pmod{p}$ for $0 \le r < p$. Two consecutive zero terms would propagate backwards under (82.1) to $Q_0 \equiv 0$, whereas $Q_0 = 1$. The determinant identity

$$P_n Q_{n-1} - P_{n-1} Q_n = A^{2n-2}((n-1)!)^2$$

then proves the second equality in (82.2) by reduction modulo Q_n. These facts were established in the preceding manuscript; the global estimates below are the additional conclusions.

82.2 Separated indices are controlled by positive continuants

For $1 \le n < m$, let $C_{n,m}$ be the determinant of the symmetric matrix whose rows and columns have indices $k = n+2, \dots, m$, with diagonal entries $1 + (2k-1)A$ and off-diagonal entry $-kA$ between k and $k+1$. The empty determinant, when $m = n+1$, is one.

Lemma 82.1 (A uniform overlap bound)**.** *For every $1 \le n < m$,*

$$\gcd(Q_n, Q_m) \mid C_{n,m}, \tag{82.3}$$

$$1 \le C_{n,m} \le ((2A+1)m)^{m-n-1}. \tag{82.4}$$

Proof. Expansion of the determinant along its last row gives the same three-term recurrence as (82.1), starting with $C_{n,n+1} = 1$ and $C_{n,n+2} = 1 + (2n+3)A$. Consequently

$$Q_m \equiv C_{n,m} Q_{n+1} \pmod{Q_n}.$$

Since Q_n and Q_{n+1} are coprime, this congruence proves (82.3).

The matrix defining $C_{n,m}$ is positive definite. In an interior row, its diagonal exceeds the sum of the absolute values of the two off-diagonal entries by exactly one. At boundary rows the excess is larger. Equivalently, its quadratic form is a sum of positive multiples of squared adjacent differences, positive boundary terms, and the Euclidean squared norm. Its determinant is therefore positive. Hadamard's inequality for positive definite matrices bounds the determinant by the product of the diagonal entries. Each is at most $(2A+1)m$, proving (82.4). □

Lemma 82.2 (Packing primitive contents)**.** *If $1 \le n_1 < \cdots < n_h = N$, then*

$$\prod_{i=1}^{h} g_{n_i} \le ((N-1)!)^2 ((2A+1)N)^{\sum_{i<j}(n_j - n_i - 1)}. \tag{82.5}$$

Proof. For any positive integers $a_1, \dots, a_h$,

$$\prod_i a_i \;\mid\; \operatorname{lcm}(a_1, \dots, a_h) \prod_{i<j} \gcd(a_i, a_j).$$

To verify this, order their valuations at a fixed prime as $v_1 \le \cdots \le v_h$. The valuation on the right is $v_h + \sum_{i=1}^{h-1}(h-i)v_i$, which is at least $\sum_i v_i$. Apply this observation with $a_i = g_{n_i}$. By (82.2), their least common multiple divides $((N-1)!)^2$. Each pairwise gcd divides $\gcd(Q_{n_i}, Q_{n_j})$, which is bounded by Lemma 82.1. □

The use of positive continuants matters. A formal polynomial remainder would not suffice if it vanished at one of the selected integer indices. Positive definiteness excludes that possibility uniformly for every gap and every positive integer A.

82.3 Only logarithmically many factorial-sized cancellations

Theorem 82.3 (Global large-content sparsity)**.** *For every fixed positive integer A and every $\varepsilon > 0$,*

$$\#\{2 \le n \le X : \ \log g_n \ge \varepsilon n \log n\} = O_{A,\varepsilon}(\log X). \tag{82.6}$$

More precisely, there are constants $h \ge 1$, $\eta > 0$, and N_0 depending only on A, ε, such that every interval $[N, (1+\eta)N]$ with $N \ge N_0$ contains at most $h-1$ members of this set.

Proof. It is enough to prove the assertion for $0 < \varepsilon \le 1$. Choose an integer h with $h\varepsilon > 2$, and then choose $\eta > 0$ so small that

$$2(1+\eta) + \binom{h}{2}\eta < h\varepsilon. \tag{82.7}$$

Suppose that h exceptional indices $n_1 < \cdots < n_h$ lie in $[N, (1+\eta)N]$. Their defining lower bounds give

$$\sum_{i=1}^{h} \log g_{n_i} \ge h\varepsilon N \log N.$$

On the other hand, Lemma 82.2, the bound $n_j - n_i - 1 \le \eta N$, and Stirling's elementary upper bound yield

$$\sum_{i=1}^{h} \log g_{n_i} \le \left(2(1+\eta) + \binom{h}{2}\eta\right) N \log N + O_{A,h,\eta}(N).$$

This contradicts (82.7) for all sufficiently large N. Covering $[N_0, X]$ by $O_\eta(\log X)$ consecutive multiplicative intervals proves (82.6). □

This is an unconditional bound for the sum of all local prime-power contributions at almost every index, with an explicit order of magnitude for the exceptional counting function. It does not assert the pointwise estimate $\log g_n = o(n \log n)$, because an infinite exceptional subsequence is still permitted.

82.4 Consequences for irrationality certificates

The exact positive remainder is

$$R_n := Q_n \delta_A - P_n = A^{2n} n! \int_0^\infty \frac{e^{-t} t^n}{(1+At)^{n+1}}\, dt > 0. \tag{82.8}$$

It follows either by Rodrigues's formula and integration by parts, or directly from the Laguerre construction. Only the coarse estimates

$$\log R_n = n \log n + O_A(n), \qquad \log Q_n = n \log n + O_A(n) \tag{82.9}$$

are needed here. For the first upper bound, use $t^n/(1+At)^n \le A^{-n}$ in (82.8). For its lower bound, restrict the integral to $\sqrt{n} \le t \le \sqrt{n} + 1$. On that interval

$$\left(\frac{At}{1+At}\right)^n \ge \exp(-C_A\sqrt{n}), \qquad \frac{e^{-t}}{1+At} \ge \frac{\exp(-\sqrt{n}-1)}{1+A(\sqrt{n}+1)},$$

and Stirling's formula completes the estimate. For Q_n, its top summand gives $Q_n \ge n! A^n$, while

$$\frac{Q_n}{n! A^n} = L_n(-1/A) = \sum_{k=0}^{n} \binom{n}{k} \frac{A^{-k}}{k!} \le \sum_{k=0}^{\infty} \frac{(n/A)^k}{(k!)^2} \le e^{2\sqrt{n/A}}.$$

Corollary 82.4 (Primitive forms are large outside a logarithmic set)**.** *Fix $A \geq 1$ and $0 < \varepsilon < 1$. Apart from $O_{A,\varepsilon}(\log X)$ indices $n \leq X$ and a fixed finite set, both q_n and ℓ_n lie in*

$$[\exp((1-\varepsilon)n\log n),\, \exp((1+\varepsilon)n\log n)]\,.$$

In particular, for every fixed $B > 0$,

$$\#\{n \leq X :\ 0 < \ell_n \leq B\} = O_{A,B}(\log X). \tag{82.10}$$

Proof. Apply Theorem 82.3 with $\varepsilon/2$, and combine $\log \ell_n = \log R_n - \log g_n$ and $\log q_n = \log Q_n - \log g_n$ with (82.9). The upper bounds follow from $g_n \geq 1$. □

There is also a concrete lacunarity constraint on every possible successful subsequence. Set

$$\lambda_* := \frac{1+\sqrt{37}}{6} > 1.$$

Theorem 82.5 (Lacunarity of bounded primitive forms)**.** *Fix $A \geq 1$ and $B > 0$. If $a < b < c$ are indices with $\ell_a, \ell_b, \ell_c \leq B$, then*

$$3a + b \leq 3c + O_{A,B}(c/\log c). \tag{82.11}$$

If $n_1 < n_2 < \cdots$ is an infinite sequence of such indices, then for every $1 < \lambda < \lambda_*$ there is $C > 0$ such that $n_j \geq C\lambda^j$. Consequently

$$\limsup_{X\to\infty} \frac{\#\{n \leq X : \ell_n \leq B\}}{\log X} \leq \frac{1}{\log \lambda_*} < 6.028.$$

The displayed constant is a proved upper bound, not a claim of optimality.

Proof. Boundedness of ℓ_n and (82.9) imply $\log g_n \geq n\log n - O_{A,B}(n)$. Apply Lemma 82.2 to a, b, c. The sum of the three gap exponents is $(b-a-1)+(c-a-1)+(c-b-1) = 2(c-a)-3$. Thus

$$a\log a + b\log b + c\log c \leq (4c-2a)\log c + O_{A,B}(c).$$

For $0 < u \leq c$, $u\log(u/c) \geq -c/e$. Replacing the logarithms on the left by $\log c$ therefore changes the bound by only $O(c)$, giving (82.11).

For every $\tau > 0$, the consecutive terms of the selected subsequence eventually satisfy

$$(3+\tau)n_{j+2} \geq 3n_j + n_{j+1}.$$

The positive root λ_τ of $(3+\tau)\lambda^2 - \lambda - 3 = 0$ tends to λ_* as $\tau \downarrow 0$. Induction from two sufficiently late initial terms gives $n_j \geq C_\tau \lambda_\tau^j$. The asserted growth and counting bound follow. □

82.5 What has and has not been closed

For $A = 1$, Theorems 82.3 and 82.5 close a definite global subproblem left by the preceding content formula: large cancellation cannot occur at a positive-density set, or even at more than logarithmically many indices up to X. Any vanishing primitive-form proof of the irrationality of δ using this fixed classical family must select a geometrically sparse subsequence. The results neither exclude such a subsequence nor construct one, and therefore do not prove δ irrational.

The earlier Kurepa equivalence is not resolved by a sparse-index bound. For any fixed prime p with $p \mid \sum_{j=0}^{p-1} j!$, congruence preservation makes p divide the content along a whole

residue class, but that contributes only a fixed local valuation at typical indices. This is entirely compatible with Theorem 82.3. No conclusion about the universal nonvanishing of the left factorial is drawn.

For Euler's constant γ, the separate Taylor-crossover route still requires exponentially accurate cancellation and removal of the remaining $\log n$ term. The present theorem concerns the Laguerre family for δ_A; transferring it to the Taylor family without a suitable common recurrence and content divisor would be unjustified.

83 Curvature turns a Taylor density-zero obstruction into a counting bound

The even Taylor truncations considered in the preceding manuscript could succeed only at an exponentially accurate crossover. Their possible bounded primitive forms were shown to have density zero. Here a lattice-point argument gives a power bound for their number, without an irrationality-measure estimate for the crossover slope.

For positive integers n, M, write

$$S_M(n) = \sum_{k=1}^{M} \frac{(-1)^{k-1} n^k}{k\, k!} = \frac{a_M(n)}{b_M(n)}, \qquad \gcd(a_M(n), b_M(n)) = 1, \quad b_M(n) > 0.$$

For real $x > 0$ and $u > 0$, define

$$E_u(x) = \frac{1}{\Gamma(u+1)} \int_0^x \frac{1}{t} \int_0^t e^{-s} (t-s)^u \, ds \, dt,$$
$$T_x = e^{-x} \int_0^\infty \frac{e^{-t}}{x+t} \, dt.$$

Taylor's integral remainder gives, for even integers M,

$$S_M(n) - \log n - \gamma = T_n - E_M(n). \tag{83.1}$$

Let $\kappa > e$ be the unique solution of $\kappa(\log \kappa - 1) = 1$.

Theorem 83.1 (A quantitative bound for bounded crossover forms)**.** *For every fixed $B > 0$,*

$$\#\{n \le X : \exists M \ge 2 \text{ even with } b_M(n) | S_M(n) - \log n - \gamma| \le B\} = O_B(X^{1/3}). \tag{83.2}$$

This statement neither proves that the set is finite nor removes the $\log n$ *term from the displayed forms.*

83.1 An exact smooth crossover with controlled curvature

Set $F(x, c) = \log(E_{cx}(x)/T_x)$, with c restricted to a compact subinterval of $(0, \infty)$. The following differentiated estimate is the analytic input.

Lemma 83.2 (A differentiated boundary expansion)**.** *Uniformly for c in such a compact interval,*

$$F(x, c) = x f(c) - \tfrac{1}{2} \log x + g(c) + r(x, c), \tag{83.3}$$

where

$$f(c) = 1 + c(1 - \log c), \qquad g(c) = -\tfrac{3}{2} \log c - \log(1 + c) - \tfrac{1}{2} \log(2\pi),$$

and, for $0 \le a, b \le 2$,

$$\partial_x^a \partial_c^b r(x, c) = O(x^{-1-a}). \tag{83.4}$$

Proof. The change of variables $t = x - v$ gives

$$E_{cx}(x) = \frac{x^{cx-1}}{\Gamma(cx+1)} J(x,c), \qquad J(x,c) = \int_{\substack{v,s\geq 0\\ v+s\leq x}} \frac{e^{-s}(1-(v+s)/x)^{cx}}{1-v/x}\, dv\, ds.$$

The limiting integral is

$$J_0(c) = \int_0^\infty \int_0^\infty e^{-cv-(1+c)s}\, dv\, ds = \frac{1}{c(1+c)}.$$

We verify, with the same range of derivatives,

$$\partial_x^a \partial_c^b (J(x,c) - J_0(c)) = O(x^{-1-a}). \tag{83.5}$$

On $v + s \leq x/2$, put $y = v + s$ and use

$$cx \log(1 - y/x) = -cy - \frac{cy^2}{2x} + O(y^3/x^2).$$

Differentiation at fixed c shows that the first and second x derivatives of the exponent beyond $-cy$ are bounded by $C(y^2 + y^3)x^{-2}$ and $C(y^2 + y^4)x^{-3}$, respectively, on this range. The reciprocal factor $1/(1 - v/x)$ has analogous bounds with powers of v. Up to two c derivatives only introduce additional polynomial factors in y. The inequalities $\log(1 - y/x) \leq -y/x$ and $c \geq c_0 > 0$ provide an integrable dominating function, a fixed polynomial in $v + s$ times $e^{-c_0(v+s)}e^{-s}$. The mean-value estimate for the exponential then proves (83.5) on the inner range.

On the complementary range, write $z = 1 - y/x$. The denominator is at most z^{-1}, while every derivative of order under consideration introduces only finitely many additional powers of z^{-1}, x, and $|\log z|$. Split $z^{cx} = z^{cx/2}z^{cx/2}$. The first factor is at most $e^{-c_0 x/4}$ when $y \geq x/2$; for all sufficiently large x, the second absorbs all the stated singular powers and logarithms. Thus this range, its differentiated integrals, and the corresponding tails of J_0 are exponentially small. More explicitly, extend the integrand by zero to the fixed quadrant $v, s \geq 0$. For sufficiently large x, its required derivatives vanish at $v + s = x$, since they are bounded there by a fixed power of x times $z^{cx-C}(1 + |\log z|)^C$. Differentiation over the fixed quadrant is therefore legitimate. Contributions at the artificial splitting boundary $v + s = x/2$ cancel, or can individually be absorbed in the exponentially small error. This proves (83.5).

Also

$$T_x = \frac{e^{-x}}{x} K(x), \qquad K(x) = \int_0^\infty \frac{e^{-t}}{1 + t/x}\, dt = 1 + O(x^{-1}),$$

where the first two differentiated remainders are $O(x^{-2})$ and $O(x^{-3})$ by differentiation under the integral. Finally, the usual differentiated Stirling expansion for $\log\Gamma(cx+1)$ has an $O(x^{-1})$ remainder with the derivative bounds in (83.4); this also follows directly by differentiating Binet's integral formula. Combining these estimates, and using that $J_0(c)$ is bounded away from zero on the chosen compact interval, proves the lemma. □

Lemma 83.3 (The exact crossover curve)**.** *For all sufficiently large real x, there is a unique $u(x)$ in $(2x, 4x)$ such that $E_{u(x)}(x) = T_x$. It is smooth and satisfies*

$$u(x) = \kappa x - \frac{\log x}{2\log\kappa} + O(1), \tag{83.6}$$

$$u''(x) = \frac{1}{2\log\kappa\, x^2} + O\left(\frac{(\log x)^2}{x^3}\right). \tag{83.7}$$

Proof. In Lemma 83.2, $f(2) > 0$, $f(4) < 0$, and $f'(c) = -\log c < 0$ on $[2,4]$. Thus F has exactly one zero there for all sufficiently large x, and its derivative with respect to the real order cx is bounded away from zero. The implicit function theorem applies.

Write $u(x) = \kappa x + s(x)$. Taylor expansion of (83.3) first yields $s(x) = O(\log x)$ and then the identity

$$0 = f'(\kappa)s - \tfrac{1}{2}\log x + g(\kappa) + \mathcal{R}(x,s), \qquad \mathcal{R}(x,s) = O((1+s^2)/x).$$

For $|s| \le C\log x$, the differentiated estimates from Lemma 83.2 imply

$$\begin{aligned} \mathcal{R}_x &= O((1+s^2)/x^2), & \mathcal{R}_s &= O((1+|s|)/x), & & \\ \mathcal{R}_{xx} &= O((1+s^2)/x^3), & \mathcal{R}_{xs} &= O((1+|s|)/x^2), & \mathcal{R}_{ss} &= O(x^{-1}). \end{aligned}$$

Implicit differentiation gives

$$s'(x) = \frac{1}{2f'(\kappa)x} + O((\log x)^2/x^2), \qquad s''(x) = -\frac{1}{2f'(\kappa)x^2} + O((\log x)^2/x^3).$$

Since $f'(\kappa) = -\log\kappa$, these are the asserted formulas. □

83.2 Bounded arithmetic forms must track the curve exponentially

Lemma 83.4. *For every $B > 0$ there are constants $c_B > 0$ and n_B such that, whenever $n \ge n_B$ and an even $M \ge 2$ satisfies the inequality in (83.2), one has*

$$|M - u(n)| \le e^{-c_B n}. \tag{83.8}$$

Proof. We give the arithmetic reduction explicitly. For $M \le 2n$, restricting the defining integral for $E_M(n)$ to $n - n/(M+1) \le t \le n$ and $0 \le s \le 1$ gives

$$E_M(n) \ge e^{-13}\frac{n^M}{(M+1)M!} \qquad (n \ge 4).$$

This tends to infinity uniformly for $2 \le M \le 2n$: use $M! \le M^M$ when $M \le \sqrt{n}$, and Stirling's bound with $en/M \ge e/2 > 1$ on the remaining range. Since $T_n \to 0$ and $b_M(n) \ge 1$, bounded forms cannot have such orders once n is large.

For $M \ge 2n$, every prime $M/2 < p \le M$ with $p \nmid n$ occurs at exact valuation -2 in $S_M(n)$: its $k = p$ summand has that valuation and every other summand has valuation at least -1. Therefore, with $\vartheta(y) = \sum_{p \le y}\log p$,

$$\log b_M(n) \ge 2\{\vartheta(M) - \vartheta(M/2)\} - 2\log n = M + o(M) - 2\log n.$$

Jensen's inequality gives $T_n \ge e^{-n}/(n+1)$, so $b_M(n)T_n \to \infty$ uniformly for $M \ge 2n$. It follows from (83.1) that $|E_M(n)/T_n - 1| \to 0$ uniformly over bounded forms.

The real-order function $E_u(n)$ decreases for $u \ge 2n$ when n is large: its logarithmic derivative is at most $\log n - \psi(u+1) < 0$. Lemma 83.2 at $c = 2, 4$ thus confines M/n to $[2,4]$, and the same lemma forces $M/n \to \kappa$. The denominator estimate now improves the relative-error bound to

$$\left|\frac{E_M(n)}{T_n} - 1\right| \le \exp(-[\kappa - 1 + o(1)]n).$$

The derivative of $\log(E_u(n)/T_n)$ with respect to u is $-\log\kappa + o(1)$ near its zero, by Lemma 83.2. The mean-value theorem proves (83.8), with any fixed $c_B < \kappa - 1$ after increasing the threshold n_B. □

83.3 Integer triangles cannot follow nonzero curvature too often

Proof of Theorem 83.1. On each interval $[N, 2N]$, Lemma 83.3 gives constants $0 < c_1 < c_2$ independent of N such that

$$c_1 N^{-2} \le u''(x) \le c_2 N^{-2}.$$

Choose a fixed small $\eta > 0$ and set $H = \eta N^{2/3}$. Suppose that three successful integers $n_1 < n_2 < n_3$ lie in an interval of length at most H, and let M_i be their associated even orders. By Lemma 83.4, $|M_i - u(n_i)| \le e^{-c_B N}$.

The integer determinant

$$D = (n_2 - n_1)(M_3 - M_1) - (n_3 - n_1)(M_2 - M_1)$$

differs from

$$D_u = (n_2 - n_1)(n_3 - n_1)(n_3 - n_2)u[n_1, n_2, n_3]$$

by at most $4He^{-c_B N}$, where the divided difference satisfies $u[n_1, n_2, n_3] = u''(\xi)/2$ for an intermediate point ξ. Consequently $0 < D_u \le c_2 H^3/(2N^2)$. Choose η so that this upper bound is below $1/2$. For large N, we then have $|D| < 1$, hence $D = 0$. But the positive integer gaps give $D_u \ge c_1 N^{-2}$, whereas $4He^{-c_B N} = o(N^{-2})$; this contradicts $D = 0$.

There are therefore at most two successful integers in each interval of length H. Partitioning $[N, 2N]$ gives $O(N/H + 1) = O(N^{1/3})$ successful integers. Summation over dyadic intervals proves (83.2). □

The curvature argument is independent of whether κ has a known finite irrationality measure. It strengthens the previous density-zero conclusion to an explicit counting estimate for this specific failed approximation route. It leaves open whether the exceptional set is infinite and whether any of its forms actually tends to zero; it does not settle the irrationality of γ.

84 Shift optimization and positive Catalan certificates

This section continues the Wilson trial construction, rather than counting its earlier exponential-height estimate as a second result. It determines the sharp exponential effect of changing the shift, computes the leading constant in the transferred squared norms, and removes an auxiliary-constant obstruction in the Catalan application. None of these conclusions proves a new irrationality or transcendence statement about the named constants.

The starting Stieltjes representation is due to Prévost [340]; Wilson remainder Padé approximations were developed further by Prévost and Rivoal [341]. Recent context is particularly important here: Fauzan's September 2026 preprint reports a proof of the irrationality of $\zeta(5)$ by normalized Hankel determinants [342]; Lai's October 2026 preprint uses the determinant method to prove the linear independence of $1, \zeta(3), \zeta(4)$ [343]. Its revised version also records OpenAI's reported Lean-formalized irrationality proof for Catalan's constant [344]. Their complete proofs and Lean dependencies are not independently verified here. These are external results, not discoveries of this project. Our $\zeta(5)$ and Catalan targets below are transcendence.

84.1 The explicit family and its squared norms

For integers $s \ge 2$, let

$$d\nu_s(t) = w_s(y)\, dy, \qquad t = y^2, \qquad w_s(y) = \frac{2(2\pi)^{s-1} y^s}{(s-1)!} \sum_{\ell \ge 1} \ell^{s-1} e^{-2\pi \ell y}.$$

Its polynomial moments and Stieltjes transform are

$$m_{s,e} = (-1)^e B_{2e+2} \frac{(2e+s)!}{(s-1)!(2e+2)!}, \tag{84.1}$$

$$S_s(z) = \int_0^\infty \frac{d\nu_s(t)}{t+z}. \tag{84.2}$$

For $a > 0$ the Hurwitz formula reads

$$\zeta(s,a) = \frac{1}{2a^s} + \frac{a^{1-s}}{s-1} + a^{1-s} S_s(a^2). \tag{84.3}$$

Define

$$R_n(t) = \sum_{k=0}^{n} \frac{(-1)^k}{(k!)^2} \binom{n}{k} \binom{n+k+1}{k} \prod_{r=0}^{k-1} (t + r^2). \tag{84.4}$$

The classical Wilson specialization $(0,1,1,1)$ gives

$$\int R_n^2 \, d\nu_3 = \frac{n+1}{4}. \tag{84.5}$$

For the associated monic orthogonal polynomials, the recurrence coefficients are

$$\beta_n = \frac{(n+1)^2(2n^2+4n+1)}{(2n+1)(2n+3)}, \qquad \lambda_n = \frac{n^3(n+1)^3}{4(2n+1)^2}.$$

Thus multiplication by t in the orthonormal basis has diagonal $\beta_n = n^2/2 + O(n)$ and adjacent entries $\sqrt{\lambda_n} = n^2/4 + O(n)$.

Theorem 84.1 (Exact leading transferred norm)**.** *For fixed odd $s \geq 3$, put*

$$\kappa_s = \frac{\pi^{s-3}}{2(s-1)!} \binom{s-3}{(s-3)/2}.$$

Then

$$H_{s,n} := \int R_n(t)^2 \, d\nu_s(t) = \kappa_s n^{s-2} (1 + O_s(n^{-1})). \tag{84.6}$$

For $s = 2$ one has the useful exact comparison

$$0 < H_{2,n} \leq \frac{n+1}{4}. \tag{84.7}$$

Proof. Write $s = 2k+3$. Directly from the defining densities,

$$\frac{w_s(y)}{w_3(y)} - \frac{2(2\pi)^{2k}}{(2k+2)!} y^{2k}$$

is bounded on $(0,\infty)$. At infinity this follows by separating the $\ell = 1$ terms, whose remaining relative error decays exponentially. At zero both densities tend to $1/\pi$; this limit follows from the elementary Riemann-sum limit $u^{m+1} \sum_{\ell \geq 1} \ell^m e^{-u\ell} \to m!$. Consequently the error in replacing $H_{s,n}$ by $2(2\pi)^{2k}/(2k+2)!$ times $\int t^k R_n^2 d\nu_3$ is $O_s(n)$.

Let J denote the orthonormal Jacobi matrix. The identity

$$\int t^k R_n^2 d\nu_3 = \frac{n+1}{4} (J^k)_{nn}$$

involves paths of length k staying within k of n. Each of the finitely many coefficients on such paths, divided by n^2, tends to $1/2$ on the diagonal and $1/4$ off the diagonal, with error $O_k(n^{-1})$. Therefore

$$(J^k)_{nn} = n^{2k}[z^0]\left(\frac{1}{2} + \frac{z+z^{-1}}{4}\right)^k + O_k(n^{2k-1}) = \frac{1}{4^k}\binom{2k}{k}n^{2k} + O_k(n^{2k-1}).$$

The case $k = 0$ is supplied exactly by (84.5); for $k \geq 1$ these formulas prove (84.6). Finally geometric-series summation gives

$$\frac{w_2(y)}{w_3(y)} = \frac{\tanh(\pi y)}{\pi y} \leq 1,$$

which proves (84.7). □

84.2 The optimal shift at a fixed total budget

For $\rho > 0$ define

$$v = \sqrt{1+\rho^2}, \qquad \Phi(\rho) = 2\,\operatorname{arsinh}\rho + 2\rho\,\operatorname{arsinh}(1/\rho), \qquad C(\rho) = \frac{(v+\rho)^2}{4\pi^{3/2}\rho^{3/2}v^{1/2}}.$$

Theorem 84.2 (Uniform shift asymptotics)**.** *Let j, n be positive integers and suppose $\rho = j/n$ ranges over a fixed compact subinterval of $(0,\infty)$. Uniformly in this range,*

$$R_n(-j^2) = C(\rho)n^{-3/2}e^{n\Phi(\rho)}(1 + O(n^{-1})). \tag{84.8}$$

For the Wilson trial intervals for $\zeta(s)$, $s \geq 3$ odd, the exact interval width is

$$W_{s,n,j} = \frac{H_{s,n}}{j^{s+1}R_n(-j^2)^2},$$

and hence

$$W_{s,n,j} = \frac{\kappa_s}{\rho^{s+1}C(\rho)^2}e^{-2n\Phi(\rho)}(1 + O_s(n^{-1})). \tag{84.9}$$

Among fixed positive shift ratios, the exponential decay rate per total budget $n + j$ is uniquely maximized by $j/n = 1$. The maximum is $4\log(1+\sqrt{2})$.

Proof. All nonzero terms in

$$R_n(-j^2) = \sum_{k=0}^{\min(n,j)} \binom{n}{k}\binom{n+k+1}{k}\binom{j}{k}\binom{j+k-1}{k}$$

are positive. For $x = k/n$ in the interior of the summation range, uniform Stirling expansion gives a term $n^{-2}g_\rho(x)e^{nf_\rho(x)}(1 + O(n^{-1}))$, where

$$\begin{aligned} f_\rho(x) &= (1+x)\log(1+x) - (1-x)\log(1-x) \\ &\quad + (\rho+x)\log(\rho+x) - (\rho-x)\log(\rho-x) - 4x\log x, \\ g_\rho(x) &= \frac{\rho(1+x)}{4\pi^2x^2}\sqrt{\frac{1+x}{(1-x)(\rho+x)(\rho-x)}}. \end{aligned}$$

The unique maximum satisfies $(1-x^2)(\rho^2-x^2) = x^4$, so

$$x_* = \frac{\rho}{\sqrt{1+\rho^2}}, \qquad f_\rho(x_*) = \Phi(\rho), \qquad f_\rho''(x_*) = -\frac{2(1+\rho^2)^{5/2}}{\rho^3}.$$

For ratios in a compact set this saddle remains uniformly separated from both endpoints. Strict concavity makes the complementary part of the sum exponentially smaller. Taylor expansion on a shrinking neighborhood of the saddle, followed by Gaussian summation, gives

$$R_n(-j^2) = n^{-3/2} g_\rho(x_*) \sqrt{\frac{2\pi}{|f_\rho''(x_*)|}} e^{n\Phi(\rho)} (1 + O(n^{-1})).$$

The Gaussian odd term cancels; the even remainder is $O(n^{-1})$. The prefactor simplifies to $C(\rho)$, proving (84.8). Combining it with Theorem 84.1 proves (84.9).

The exponential rate per total budget is $F(\rho) = 2\Phi(\rho)/(1+\rho)$. Since $\Phi'(\rho) = 2\,\mathrm{arsinh}(1/\rho)$,

$$F'(\rho) = \frac{4}{(1+\rho)^2} \big(\mathrm{arsinh}(1/\rho) - \mathrm{arsinh}\,\rho\big).$$

This is positive below 1 and negative above 1, proving the strict optimality assertion. □

Remark 84.3 (What parameter tuning cannot repair). For integer shifts the previously proved common denominator is

$$Q_{s,n,j} = 2(s-1) j^{s+1} L_j^s R_n(-j^2)^2 L_{2n}^4 (s-1)! L_{4n+s+1}^{s-1},$$

where $L_m = \mathrm{lcm}(1, \dots, m)$. The prime number theorem and Theorem 84.2 give, for $j/n \to \rho > 0$,

$$\frac{\log Q_{s,n,j}}{n} = 2\Phi(\rho) + 4s + 4 + s\rho + o(1).$$

The two occurrences of $2\Phi(\rho)$ cancel when this denominator bound is compared with the interval width. No fixed positive shift ratio makes this particular estimate certify irrationality. This is a limitation of the common-denominator estimate, not a lower bound on the reduced denominator and not an impossibility theorem for the actual approximants.

84.3 An exact cancellation diagnostic

For $z = j^2$, set $a = R_n(-z)$ and $q = (a - R_n(t))/(t+z)$. Let μ_s denote integration against ν_s. The lower variational trial is exactly

$$\mu_s\left(2\frac{q}{a} - (t+z)\frac{q^2}{a^2}\right) = \frac{\mu_s(q)}{a} + \frac{\mu_s(R_n q)}{a^2}. \tag{84.10}$$

Indeed $(t+z)q = a - R_n$ proves the identity by multiplication. For $s = 3$, orthogonality makes the second term vanish, because $\deg q < n$. This explains the loss of one power of a in the classical Gaussian Padé case. For $s = 5, 7$ the exact rational number $\mu_s(R_n q)$ is the obstruction to this same cancellation; one cannot delete it on the basis of orthogonality for a different weight. If an integer M clears both moments in (84.10), the remaining cancellation is determined exactly by

$$\gcd(aM\mu_s(q) + M\mu_s(R_n q),\ Ma^2).$$

This identity separates orthogonality cancellation from genuine arithmetic common factors.

84.4 Strict Catalan intervals with linear logarithmic height

Put $G = \sum_{v\ge 0} (-1)^v/(2v+1)^2$. For any polynomial p and $z > 0$ define rational moment expressions

$$\mathcal{L}_s(p; z) = \int (2p - (t+z)p^2)\, d\nu_s,$$

$$\mathcal{U}_s(p; z) = \frac{m_{s,0} - \int (2tp - t(t+z)p^2)\, d\nu_s}{z}.$$

For rational $c>0$ with $R_n(-c^2)\neq 0$, put

$$p_{n,c}(t)=\frac{1-R_n(t)/R_n(-c^2)}{t+c^2},\quad L_{n,c}=\mathcal{L}_2(p_{n,c};c^2),\quad U_{n,c}=\mathcal{U}_2(p_{n,c};c^2).$$

Completing squares proves

$$L_{n,c}<S_2(c^2)<U_{n,c},\qquad U_{n,c}-L_{n,c}=\frac{H_{2,n}}{c^2R_n(-c^2)^2}. \tag{84.11}$$

Theorem 84.4 (A direct Catalan transfer)**.** *Let $n\geq 1$, $a=n+1/4$, $b=n+3/4$, and*

$$B_n=\sum_{v=0}^{n-1}\left(\frac{1}{(4v+1)^2}-\frac{1}{(4v+3)^2}\right)+\frac{1}{16}\left(\frac{1}{2a^2}+\frac{1}{a}-\frac{1}{2b^2}-\frac{1}{b}\right).$$

Then the rational numbers

$$\ell_n^G=B_n+\frac{1}{16}\left(\frac{L_{n,a}}{a}-\frac{U_{n,b}}{b}\right),\quad u_n^G=B_n+\frac{1}{16}\left(\frac{U_{n,a}}{a}-\frac{L_{n,b}}{b}\right)$$

satisfy strict inequalities $\ell_n^G<G<u_n^G$. Moreover

$$u_n^G-\ell_n^G=\frac{H_{2,n}}{16}\left(\frac{1}{a^3R_n(-a^2)^2}+\frac{1}{b^3R_n(-b^2)^2}\right), \tag{84.12}$$

$$0<u_n^G-\ell_n^G\leq\exp(-8\log(1+\sqrt{2})n+O(\log n)), \tag{84.13}$$

$$\log\operatorname{den}(\ell_n^G),\quad \log\operatorname{den}(u_n^G)=O(n). \tag{84.14}$$

No $\zeta(2)$ term or unproved determinant nonvanishing statement occurs in these certificates.

Proof. The tail identity for the defining series and (84.3) give

$$G=B_n+\frac{1}{16}\left(\frac{S_2(a^2)}{a}-\frac{S_2(b^2)}{b}\right).$$

All summands defining $R_n(-a^2)$ and $R_n(-b^2)$ are positive: the indices $k\leq n$ satisfy $k-1<a,b$. Thus the polynomials in the statement are defined. Formula (84.11) proves the strict interval and its exact width. The saddle analysis in Theorem 84.2 also applies to real shifts $c>n$ with $c=n+O(1)$: all terms remain positive, the same gamma-function Stirling expansion holds, and the saddle remains interior. Consequently $\log R_n(-c^2)=4\log(1+\sqrt{2})n+O(\log n)$. Equation (84.7) proves (84.13).

Here are explicit arithmetic details for (84.14). For integers x and $k\geq 0$,

$$2^{2k+v_2(k!)}\binom{x/4}{k}\in\mathbb{Z}.$$

For each odd prime, the numerator $\prod_{r=0}^{k-1}(x-4r)$ contains at least as many multiples of each prime power as $k!$; this proves the assertion after taking valuations. The same bound holds for $\binom{x/4+k-1}{k}$. Since $v_2(k!)\leq k$, the polynomial

$$F_n(x)=2^{6n}R_n(-x^2/16)$$

is integer-valued on $\mathbb{Z}$, of degree $2n$. Write $k_1=4n+1$, $k_3=4n+3$, and $A_i=F_n(k_i)$, positive integers. For $i=1,3$, the polynomial $A_i-F_n(x)$ vanishes at both $x=k_i$ and $x=-k_i$. The elementary divided-difference property of integer-valued polynomials implies

$$A_iL_{2n}^2p_{n,k_i/4}(-x^2/16)\in\operatorname{Int}(\mathbb{Z}).$$

For completeness, if an integer-valued polynomial f of degree at most d vanishes at an integer c, expansion in $\binom{x-c}{r}$ gives $L_d f(x)/(x-c) \in \mathrm{Int}(\mathbb{Z})$, since $\binom{x-c}{r}/(x-c) = r^{-1}\binom{x-c-1}{r-1}$. Applying this twice proves the displayed assertion.

Replacing x by $4x$ preserves integer-valuedness. Therefore, with $E = A_1^2 A_3^2 L_{2n}^4$, multiplying either moment polynomial appearing in $L_{n,c}$ or $U_{n,c}$ by $16E$ makes its substitution $t = -x^2$ integer-valued. The factor 16 clears $c^2 = k_i^2/16$. For the Bernoulli functional $\mathcal{B}(x^r) = B_r$, one has

$$\mu_2(P) = \mathcal{B}(x^2 P(-x^2)), \qquad L_{d+1}\mathcal{B}(f) \in \mathbb{Z} \quad (f \in \mathrm{Int}(\mathbb{Z}),\ \deg f \le d).$$

The last fact follows from $\mathcal{B}(\binom{x}{r}) = (-1)^r/(r+1)$, obtained by applying $\mathcal{B}$ to $(1+z)^x$. All relevant degrees are at most $4n+2$. An explicit, deliberately nonminimal, common denominator for the two Catalan endpoints is consequently

$$Q_n^G = 2^{12}(k_1 k_3)^3 L_{4n+3}^3 A_1^2 A_3^2 L_{2n}^4. \tag{84.15}$$

The two harmonic sums are cleared by L_{4n+3}^2; the remaining powers of a, b are covered by $2^{12}(k_1k_3)^3$. Since $\log A_i = (6\log 2 + 4\log(1+\sqrt{2}))n + O(\log n)$ and $\log L_m = O(m)$, this proves (84.14). □

Arithmetic scope of the four attacks. For $\zeta(3)$, the exact defect (84.10) vanishes, recovering the classical orthogonality cancellation but providing no exclusion of algebraic degrees greater than one. For $\zeta(5)$ and $\zeta(7)$, the shift theorem proves that a change of the degree-to-shift ratio does not improve the exponential rate at a fixed total budget; the common-denominator estimate still has a strictly adverse margin. For Catalan's constant the direct construction replaces the earlier mixed-determinant nonvanishing problem by strict positive integrals, but (84.15) does not even independently recover irrationality, let alone establish transcendence. The remaining arithmetic question in each case concerns the actual cancellation, not numerical convergence.

Exact finite checks, not asymptotic proof. The accompanying rational-arithmetic script constructs every coefficient and Bernoulli moment with exact fractions. It verifies (84.10), the $s = 3$ orthogonality cancellation, and the exact interval widths. For diagonal degree $n = 50$, the ratios $\log \mathrm{den}(\ell_{s,n,n})/n$ were approximately $6.1096, 11.3195, 13.4737$ for $s = 3, 5, 7$, while $-\log W_{s,n,n}/n$ was approximately $7.0171, 7.0064, 7.0058$. These data support no new irrationality claim. In particular, finite data do not prove a lower asymptotic bound for denominators. The Catalan script additionally checks the exact interval formula and that the displayed common denominator clears both endpoints, for $n = 1, 2, 3, 5, 10, 15, 20$. It encloses the target using a separate Euler transformation with a rigorous rational tail bound 2^{-400}.

85 Nonlinear independence beyond the Lambert obstruction

The previous manuscript excluded rational linear differential–dilation relations for the full divisor Lambert series. We now retain the constant beside its leading logarithm. This supplies an algebraic independence argument that the leading logarithm alone does not give. The conclusions concern functions, and do not establish transcendence of their values at an algebraic point.

Put

$$L(q) = \sum_{n\ge1} \frac{q^n}{1-q^n} = \sum_{n\ge1} d(n) q^n, \qquad D = q\frac{d}{dq}, \qquad F_{m,r}(q) = D^r[L(q^m)] \quad (m \ge 1,\ r \ge 0).$$

In particular, differentiation is taken *after* composition. This convention accounts for the factor m^{-1}, rather than m^{-r-1}, in the next formula.

85.1 Two terms at a rational cusp

Lemma 85.1 (A logarithmic cusp with a power-saving error)**.** *Let ξ be a primitive N-th root of unity and put $g=\gcd(N,m)$, $N_m=N/g$. For fixed N,m,r, as $t\downarrow 0$,*

$$F_{m,r}(\xi e^{-t})=\frac{r!}{mN_m\,t^{r+1}}\left(\log\frac{1}{t}+\gamma-2\log N_m-\log m+H_r\right)+O_{N,m,r}(t^{-r-1/2}),\qquad \text{(NL1)}$$

where $H_0=0$.

Proof. For a primitive N-th root ξ, define

$$Z_\xi(s)=\sum_{u,v\ge 1}\frac{\xi^{uv}}{(uv)^s}=N^{-2s}\sum_{a,b=1}^{N}\xi^{ab}\zeta(s,a/N)\zeta(s,b/N)\qquad(\Re s>1).$$

Here $\zeta(s,x)$ is the Hurwitz zeta function. Its expansion at 1 is $\zeta(s,x)=(s-1)^{-1}-\psi(x)+O(s-1)$. The elementary root sums

$$\sum_{a,b=1}^{N}\xi^{ab}=N,\qquad \sum_{a,b=1}^{N}\xi^{ab}\psi(a/N)=N\psi(1)=-N\gamma$$

give

$$Z_\xi(s)=\frac{1/N}{(s-1)^2}+\frac{(2\gamma-2\log N)/N}{s-1}+O(1).$$

Mellin inversion yields

$$L(\xi e^{-t})=\frac{1}{2\pi i}\int_{(c)}\Gamma(s)Z_\xi(s)t^{-s}\,ds,\qquad c>1.$$

After applying $(-\partial_t)^r$, replace $\Gamma(s)$ by $\Gamma(s+r)$ and t^{-s} by t^{-s-r}. Move the contour to $\Re s=1/2$. The finite Hurwitz-zeta expression has polynomial growth on vertical strips, while the gamma factor decays exponentially. Thus the new integral is $O(t^{-r-1/2})$, and the only crossed pole is $s=1$. Since $\Gamma(r+1)=r!$ and $\psi(r+1)=H_r-\gamma$, its residue is

$$\frac{r!}{Nt^{r+1}}\left(\log(1/t)+\gamma-2\log N+H_r\right).$$

Finally apply this result to the primitive N_m-th root ξ^m, replace t by mt, and multiply by m^r. □

The expansion belongs to the classical theory of Lambert series at roots of unity; much more extensive transseries are available in [345]. The use made of its second term below is the additional step in the present research program.

85.2 Every pair of distinct jets is algebraically independent

Theorem 85.2 (Pairwise algebraic independence)**.** *If $(m,r)\neq(n,s)$, then*

$$\boxed{F_{m,r}(q),\ F_{n,s}(q)\ \text{are algebraically independent over}\ \mathbb{C}(q).}$$

Consequently $L(q)$ satisfies no nonzero first-order algebraic differential equation over $\mathbb{C}(q)$, and, for every $m>1$, there is no nonzero polynomial $P(q,X,Y)\in\mathbb{C}(q)[X,Y]$ with $P(q,L(q),L(q^m))=0$.

Proof. Suppose a nonzero polynomial $P(q, X, Y)$ gives a relation. Choose a prime N not dividing mn and a primitive N-th root ξ at which all coefficients are finite and every coefficient needed below is nonzero. Such a choice is possible because a nonzero rational function has finitely many zeros and poles.

First suppose $r \neq s$. Give X, Y weights $r+1, s+1$, and let W be the maximum weight of a monomial occurring in P. Among its weight-W monomials choose one of maximum ordinary degree. This monomial is unique: the equations

$$(r+1)i + (s+1)j = W, \qquad i + j = d$$

uniquely determine (i, j). Write $u = \log(1/t)$. Formula (NL1) shows that, after multiplication by t^W, the terms of weight W give a polynomial in u whose leading coefficient is nonzero. All smaller weights tend to zero; the power-saving remainder in (NL1) remains $o(1)$ after multiplication by any fixed power of u. A nonzero polynomial in u cannot tend to zero as $u \to +\infty$, a contradiction.

Now suppose $r = s$, so $m \neq n$. Let P_d be the highest ordinary homogeneous part of P, chosen so that $P_d(\xi; X, Y) \neq 0$. Set

$$A = \frac{r!}{Nm}, \quad B = \frac{r!}{Nn}, \quad C = \gamma - 2\log N + H_r.$$

Multiplication of the alleged relation by $t^{(r+1)d}$ gives

$$P_d(\xi; A(u + C - \log m), B(u + C - \log n)) + o(1) = 0. \tag{NL2}$$

The displayed polynomial in u is nonzero. Indeed, its two linear arguments describe an affine line not passing through the origin, because their normalized intercepts differ by $\log(n/m) \neq 0$. A nonzero homogeneous polynomial in two variables cannot vanish identically on such a line: its zero set is a finite union of lines through the origin. This contradicts (NL2). □

Remark 85.3 (The scope of pairwise independence). The theorem does not prove joint algebraic independence of three or more jets, nor differential transcendence of L. It excludes algebraic differential equations involving one derivative, but does not exclude equations of order two or higher. Radial asymptotics have previously been used to prove algebraic independence for Mahler functions [346]; our series is the full Lambert series, not a sparse Mahler series. The related work of Pupyrev and Bézivin [347, 348] must also be distinguished: Bézivin differentiates $\sum a^n q^n/(1-q^n)$ with respect to a, whereas D here differentiates the nome q. A comprehensive priority claim for Theorem 85.2 is not made.

85.3 Algebraic exclusion of finite-level modular substitutions

Let $\mathcal{K}_{\mathrm{mod}}$ be the field of meromorphic functions on the upper half-plane generated by $q = e^{2\pi i\tau}$, the quasimodular Eisenstein series $E_2(\tau)$, and all integer-weight meromorphic modular forms of finite congruence level which are meromorphic at every cusp. Each individual element uses only finitely many generators. We regard functions of q on the upper half-plane by composition. This convention also accommodates forms with a period greater than one.

Lemma 85.4 (Logarithm-free leading monomials)**.** *At every rational cusp a/c, a nonzero $h \in \mathcal{K}_{\mathrm{mod}}$, evaluated at $\tau = a/c + it/(2\pi)$, has an asymptotic*

$$h(\tau) = Ce^{-A/t}t^b(1 + o(1)), \qquad C \neq 0, \quad A \in \mathbb{R}, \quad b \in \mathbb{Z}. \tag{NL3}$$

The conclusion includes coefficients that are exponentially small at the cusp.

Proof. Take a common congruence subgroup for the finitely many generators in an expression for h, and choose $\begin{pmatrix} a & b_0 \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. The inverse-transformed variable is exactly

$$\tau' = -d/c + \frac{2\pi i}{c^2 t}.$$

A modular form of weight k becomes a constant times t^{-k} times a convergent Laurent series in

$$Q = \exp(-\alpha/t), \qquad \alpha > 0,$$

after a common cusp width is chosen; phases are absorbed into the coefficients. The transformation formula for E_2 adds only a multiple of t^{-1}. The remaining generator $q = e^{2\pi i a/c} e^{-t}$ is analytic in t. These standard transformations are reviewed in [349].

After denominators are cleared, the numerator and denominator of h are convergent Laurent series in Q, bounded below in their Q-exponents, whose coefficients are meromorphic germs in t with a common bound on pole order. They arise from finite polynomials in the generators, so these bounds are uniform in the Q-index. A nonzero such expression has a first nonzero Q-coefficient. That coefficient has a first nonzero Laurent term Ct^b; the later Q-terms are exponentially smaller. If all Q-coefficients vanished identically as germs, normal convergence would make the original expression identically zero. Taking the quotient of the two leading monomials proves (NL3). □

Lemma 85.5 (A logarithm cannot be algebraic over a cusp field)**.** *Suppose every nonzero element of a coefficient field has* (NL3) *at a fixed cusp. If*

$$f(t) \sim C_0 t^{-w} \log(1/t), \qquad C_0 \neq 0,$$

then f is transcendental over that field.

Proof. In a nonzero sum $\sum_j a_j f^j$, compare first the smallest exponential exponent A_j, then the smallest power $b_j - wj$, and finally the largest j. The selected term uniquely dominates the remaining terms: exponentials dominate powers, and powers dominate powers of a logarithm. □

Theorem 85.6 (Nonlinear modular obstruction for every rational filter)**.** *Let*

$$F(q) = \sum_{m \in S} \sum_{r=0}^{R} a_{m,r}(q) F_{m,r}(q), \qquad a_{m,r} \in \mathbb{C}(q),$$

where S is a finite set of distinct positive integers and the coefficients are not all zero. Then F is transcendental over $\mathcal{K}_{\mathrm{mod}}$.

Proof. Take the largest $r = R_0$ with a nonzero coefficient and set

$$A_d(q) = \sum_{m \in S} \frac{\gcd(d, m)}{m} a_{m,R_0}(q), \qquad d \in S.$$

The gcd matrix on S is invertible. Indeed,

$$\gcd(d, m) = \sum_{e \mid d,\ e \mid m} \varphi(e)$$

expresses it as a positive Gram matrix; the divisibility incidence matrix has full row rank, since its columns indexed by S form a triangular matrix with diagonal 1, after increasing ordering. Hence some A_d is a nonzero rational function.

Choose a prime ℓ not dividing any member of S, put $N = d\ell$, and choose a primitive N-th root ξ avoiding the finitely many relevant poles and zeros. Then $\gcd(N, m) = \gcd(d, m)$. Formula (NL1) implies

$$F(\xi e^{-t}) \sim \frac{R_0! A_d(\xi)}{N}\, t^{-R_0-1} \log(1/t).$$

Apply Lemmas 85.4 and 85.5. □

This strengthens the earlier linear exclusion to all polynomial relations between one such filter and finitely many modular generators. It also excludes adjoining finitely many algebraic roots of modular expressions, by transitivity of algebraic extensions. It does not exclude identities involving integrals, nonholomorphic completions, mock modular functions, or arbitrary solutions of nonlinear differential equations.

85.4 Fibonacci sums and a prime-harmonic transfer

Define

$$\mathcal{F}(q) = \sum_{n \ge 1} \frac{q^n}{1 - (-1)^n q^{2n}}, \qquad \vartheta(q) = \sum_{k \in \mathbb{Z}} q^{k^2}.$$

Splitting even and odd n gives the exact identity

$$\mathcal{F}(q) = L(q^2) - L(q^4) + \frac{\vartheta(q)^2 - \vartheta(-q)^2}{8}. \tag{NL4}$$

For the odd part, the coefficient of q^j is zero for even j and $\sum_{d \mid j} \chi_4(d)$ for odd j; the two-square theta identity gives the last term in (NL4). The theta squares are integer-weight modular forms of finite level. Consequently Theorem 85.6 shows that $\mathcal{F}$ is transcendental over $\mathcal{K}_{\mathrm{mod}}$. At $q = e^{-t}$, the theta term is $O(t^{-1})$, and

$$\mathcal{F}(e^{-t}) \sim \frac{1}{4t} \log(1/t). \tag{NL5}$$

Lemma 85.7 (The Fibonacci cusp intercept)**.** *Let N be an odd prime, ξ a primitive N-th root, and $N \nmid m$. Set $\mathcal{F}_{m,r}(q) = D^r[\mathcal{F}(q^m)]$ and $\epsilon_N = (-1)^{(N-1)/2}$. Then*

$$\mathcal{F}_{m,r}(\xi e^{-t}) = \frac{r!}{4Nm\, t^{r+1}} \left(\log(1/t) + \gamma - 2\log N - \log m + H_r + \epsilon_N \pi/2\right) + O_{N,m,r}(t^{-r-1/2}).$$

Proof. Poisson summation on residue classes modulo N gives

$$\vartheta(\xi e^{-t}) = \frac{\sqrt{\pi}}{N\sqrt{t}} \sum_{a=0}^{N-1} \xi^{a^2} + O(t^{-1/2} e^{-c_N/t}).$$

The square of the quadratic Gauss sum is $\epsilon_N N$, independently of the primitive numerator in ξ. For $\vartheta(-\xi e^{-t})$, use residue classes modulo $2N$; the zero-frequency Gauss sum vanishes, since the terms indexed by a and $a + N$ cancel. Consequently

$$\frac{\vartheta(\xi e^{-t})^2 - \vartheta(-\xi e^{-t})^2}{8} = \frac{\epsilon_N \pi}{8Nt} + O(t^{-1} e^{-c'_N/t}).$$

The differentiated Poisson expansions have the same exponential saving, with harmless additional powers of t^{-1}. Combine this with (NL4) and Lemma 85.1. The constants involving $\log 2$ cancel in $L(q^{2m}) - L(q^{4m})$, giving the stated intercept. □

Theorem 85.8 (Mixed Lambert–Fibonacci pairs)**.** *Any two distinct members of the family*

$$\{F_{m,r} : m \geq 1, r \geq 0\} \ \cup \ \{\mathcal{F}_{m,r} : m \geq 1, r \geq 0\}$$

are algebraically independent over $\mathbb{C}(q)$*. In particular, the full Fibonacci function satisfies no first-order algebraic differential equation over* $\mathbb{C}(q)$*.*

Proof. For different derivative orders, repeat the unique maximum-weight, maximum-degree argument of Theorem 85.2. For equal derivative orders, the normalized intercepts of the two cusp expansions differ by a nonzero $\log(m/n)$ for distinct dilations in the same family. For a mixed pair the difference has the form $\log(m/n) + \epsilon_N \pi/2$. At least one choice $\epsilon_N = 1$ or -1 makes this nonzero. Choose infinitely many primes in the corresponding class modulo 4, avoiding the finitely many coefficient zeros and poles. The affine-line argument of Theorem 85.2 applies. □

Now introduce the ordinary prime-harmonic generating function

$$\mathcal{P}(q) = \sum_{p \text{ prime}} \frac{q^p}{p}.$$

This is not the prime zeta function $\sum_p p^{-s}$. The classical Mertens theorem gives

$$\mathcal{P}(e^{-t}) = \log\log(1/t) + B_1 + o(1). \tag{NL6}$$

For completeness, let $A(x) = \sum_{p \leq x} 1/p$. Partial summation gives $\mathcal{P}(e^{-t}) = \int_{2t}^{\infty} e^{-u} A(u/t)\, du$. Insert $A(x) = \log\log x + B_1 + o(1)$. On $u \in [t^{1/2}, t^{-1/2}]$, the difference $\log\log(u/t) - \log\log(1/t)$ tends pointwise to zero and is bounded by an integrable function after multiplication by e^{-u}; the lower and upper tails contribute $o(1)$. This proves (NL6). Quantitative versions of the classical input are in [350].

Theorem 85.9 (Two logarithmic scales over the modular field)**.** *Both pairs*

$$\boxed{(L(q), \mathcal{P}(q)), \qquad (\mathcal{F}(q), \mathcal{P}(q))}$$

are algebraically independent over $\mathcal{K}_{\mathrm{mod}}$*.*

Proof. Write $u = \log(1/t)$, and let f be either L or $\mathcal{F}$. Then $f(e^{-t}) \sim Ct^{-1}u$, $C \neq 0$, while $\mathcal{P}(e^{-t}) \sim \log u$. In a nonzero polynomial $\sum_{i,j} a_{ij} f^i \mathcal{P}^j$, use Lemma 85.4 and compare, in order, the smallest exponential exponent A_{ij}, the smallest power $b_{ij} - i$, the largest i, and the largest j. The selected term has a unique leading asymptotic

$$C_{ij} e^{-A_{ij}/t} t^{b_{ij}-i} u^i (\log u)^j.$$

It cannot be canceled. This proves the theorem. □

Theorem 85.10 (A third independent function from primes)**.** *Let* f, g *be any two distinct members of the family in Theorem 85.8. Then*

$$\boxed{f(q),\ g(q),\ \mathcal{P}(q) \text{ are algebraically independent over } \mathbb{C}(q).}$$

This assertion has coefficient field $\mathbb{C}(q)$*, rather than* $\mathcal{K}_{\mathrm{mod}}$*.*

Proof. For a primitive root ξ of odd prime order N, the Mertens theorem in arithmetic progressions, followed by the same partial summation as above, gives

$$\mathcal{P}(\xi e^{-t}) = -\frac{1}{N-1}\log\log(1/t) + C_\xi + o(1). \tag{NL7}$$

Indeed, each nonzero residue class has prime-harmonic sum $(N-1)^{-1}\log\log x + C_a + o(1)$, while $\sum_{a=1}^{N-1}\xi^a = -1$; the prime N contributes only a bounded term. The arithmetic-progression form of Mertens' theorem is recorded, for example, in [352].

Suppose $P(q, f, g, \mathcal{P}) = 0$. Choose a prime cusp as in Theorem 85.8, avoiding the finitely many nonzero rational coefficient zeros and poles. Let $u = \log(1/t)$. If the derivative orders of f, g differ, take the largest weighted degree in their variables and then the largest ordinary degree in those variables. The selected pair of exponents is unique. Its coefficient is a nonzero polynomial in $\mathcal{P}(\xi e^{-t})$, hence has a nonzero leading power of $\log u$, by (NL7). Other pairs have smaller powers of u, and lower weights have a power saving in t, so they cannot cancel this term.

For equal derivative orders, take the highest homogeneous part in the two variables corresponding to f, g, regarding its coefficients as polynomials in a third indeterminate Z. Substitution of the affine cusp line is injective on homogeneous polynomials in the first two variables, over $\mathbb{C}[Z]$. It therefore produces a nonzero polynomial $Q(u, Z)$. Taking $Z = \mathcal{P}(\xi e^{-t})$, the largest power of u in Q, and then the largest power of Z in its coefficient, has a unique nonzero leading term $cu^j(\log u)^k$. The error from the two Lambert–Fibonacci cusp expansions is $O(t^{1/2})$ times fixed powers of u and $\log u$, hence tends to zero. This is again a contradiction. □

Remark 85.11 (What did and did not transfer to the named constants). The Erdős–Borwein constant is $L(1/2)$, and, if $\phi = (1+\sqrt{5})/2$, the Fibonacci reciprocal sum is $\sqrt{5}\,\mathcal{F}(\phi^{-1})$. The results above exclude complete families of functional algebraic reductions, including nonlinear modular substitutions. They supply no specialization theorem at $1/2$ or ϕ^{-1}. Bell–Smertnig's characterization [351] already rules out the ordinary linear Mahler framework for L. For B_1, (NL6) is a renormalized boundary limit. Algebraic independence of the functions does not determine the arithmetic nature of its constant term. Thus the transcendence of the two reciprocal-sum constants and the irrationality of B_1 remain unresolved in this work.

86 Gamma isolation with unrestricted auxiliary denominators

The preceding manuscript computed a beta-factor cost in a fixed prime reflection model. A natural repair of an unsuccessful isolation argument is to introduce gamma values with new denominators and use Gauss multiplication. We show that this repair does not reduce that cost. Moreover, we determine the complete cost functions for monomials in the fifth and seventh gamma values. These are theorems about the standard identity calculus. They are not proofs of the transcendence of the five individual gamma values under investigation.

86.1 The standard identity calculus

All gamma and beta arguments in this section are positive rational numbers. Algebraic nonzero factors and rational powers of π have zero cost. The identities allowed are recurrence, reflection, Gauss multiplication, and their finite products, quotients, and rational powers. Rational powers can be taken as positive real powers; any choices of algebraic roots can be absorbed into the algebraic factor.

Here is a precise formal description. Let $\mathcal{D}$ be the rational vector space on symbols $[x]$, $x \in \mathbb{Q}/\mathbb{Z}$, modulo

$$[0] = [1/2] = 0, \qquad [-x] = -[x], \tag{86.1}$$

$$\sum_{k=0}^{d-1} [x + k/d] = [dx] \qquad (d \geq 1). \tag{86.2}$$

A symbol $[x]$ represents $\Gamma(x)$ modulo algebraic factors and rational powers of π. Recurrence removes integer shifts, reflection gives (86.1), and Gauss multiplication gives (86.2). When a representative is an integer, its symbol is zero. This also handles the limiting multiplication identity at $x = 0$. For $x, y \in \mathbb{Q}/\mathbb{Z}$ write

$$b(x, y) = [x] + [y] - [x + y].$$

It represents $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x + y)$ after choosing positive representatives. The choices affect only algebraic factors.

For $v \in \mathcal{D}$ define

$$\ell(v) = \min \left\{ \sum_i |n_i| : v = \sum_i n_i b(x_i, y_i), \quad n_i \in \mathbb{Z} \right\}, \tag{86.3}$$

with $\ell(v) = \infty$ when no such representation exists. The arguments x_i, y_i have no denominator restriction. The beta factors in the final word have integer exponents; allowing arbitrary rational exponents in that final word would define a different problem. The equalities in (86.3) may nevertheless use rational combinations of all standard gamma identities.

Lemma 86.1 (Sawtooth certificates, including the endpoints)**.** *Set*

$$s(x) = \begin{cases} 1 - 2\{x\}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}. \end{cases}$$

Then

$$\sum_{k=0}^{d-1} s(x + k/d) = s(dx), \qquad s(-x) = -s(x), \qquad s(1/2) = 0. \tag{86.4}$$

Also $s(x) + s(y) - s(x + y) \in \{-1, 0, 1\}$. Suppose a formal identity involves only finitely many standard relations, and let N be a common denominator of every argument appearing in that proof. For every integer t coprime to N, substitution $[x] \mapsto s(tx)$ respects every relation in the proof.

Proof. Put $u = \{dx\}$. If $u \neq 0$, the fractional parts of $x + k/d$ are $(u + j)/d$, $0 \leq j < d$, in some order. Their sum is $(d - 1)/2 + u$, so the sum of their sawtooth values is $1 - 2u$. If $u = 0$, the fractional parts are $0, 1/d, \ldots, (d - 1)/d$. The zero term contributes zero and the remaining terms sum to zero. This proves the distribution identity also at the discontinuities. The oddness and the assertion about $1/2$ follow directly. If neither x, y, nor $x + y$ is an integer, the beta defect is 1 or -1 according as $\{x\} + \{y\} < 1$ or > 1; the omitted cases have defect zero.

For a distribution relation with multiplier d, its arguments have differences $1/d$, so $d \mid N$. Since $(t, d) = 1$, multiplication by t permutes the fractions k/d modulo one. Consequently

$$\sum_{k=0}^{d-1} s(tx + tk/d) = \sum_{k=0}^{d-1} s(tx + k/d) = s(tdx).$$

The other relations were already checked. A rational linear combination of these relations is respected as well. □

A certificate need not extend to a real-valued functional on all of $\mathcal{D}$ for one fixed t. The proof above is deliberately finite: for any proposed finite identity, a suitable unit t is chosen after all its auxiliary denominators have been included in N. This avoids making an unjustified global invariance assertion.

Theorem 86.2 (Exact isolation cost in the full standard calculus)**.** *Let $x = a/q$ be reduced with $q \geq 3$, and let $m \in \mathbb{Z}$. Then*

$$\boxed{\ell(m[x]) = \begin{cases} |m|(q-2)/q, & q \mid 2m, \\ \infty, & q \nmid 2m. \end{cases}} \tag{86.5}$$

This remains true even when arbitrarily large auxiliary denominators and every standard gamma multiplication identity are permitted.

Proof. Consider any finite representation of $m[x]$ by an integer beta word, including a finite proof from the standard relations. Enlarge N so $q \mid N$. Choose t coprime to N with $ta \equiv 1 \pmod{q}$. Such a choice exists: impose the inverse of a modulo q, and impose $t \equiv 1$ modulo each prime dividing N but not q, then use the Chinese remainder theorem. Lemma 86.1 gives

$$m(1 - 2/q) = \sum_i n_i \epsilon_i, \qquad \epsilon_i \in \{-1, 0, 1\}.$$

The right side is an integer. Hence $q \mid 2m$, and its absolute value is at most $\sum_i |n_i|$. This proves both the obstruction and the lower bound, including after saturation by rational powers of standard identities.

If q is odd, telescoping gives

$$\sum_{k=1}^{q-2} b(kx, x) = (q-1)[x] - [(q-1)x] = q[x].$$

This is a word of length $q - 2$, and $q \mid 2m$ then means $q \mid m$. If q is even, a is odd and $(q/2)x = 1/2$ modulo one. Thus

$$\sum_{k=1}^{q/2-1} b(kx, x) = (q/2)[x] - [(q/2)x] = (q/2)[x].$$

This word has length $q/2 - 1$, and $q/2 \mid m$. Repetition and inversion attain the stated bound for every allowed m. □

For example, the least admissible powers at reduced denominators $4, 5, 6, 7, 8, 9, 10$ are respectively $2, 5, 3, 7, 4, 9, 5$, with exact costs $1, 3, 2, 5, 3, 7, 4$. In particular, introducing denominators $10, 15, 35$, or larger ones cannot shorten the three-factor certificate for a fifth power or the five-factor certificate for a seventh power.

86.2 Complete monomial cost at fifths and sevenths

The single-coordinate theorem has a stronger lattice counterpart at the two denominators relevant here. Put

$$v_5(A, B) = A[1/5] + B[2/5], \qquad v_7(A, B, C) = A[1/7] + B[2/7] + C[3/7].$$

All exponents below are integers and may be negative.

Theorem 86.3 (The entire fifth-gamma lattice)**.** *One has*

$$\ell(v_5(A,B)) = \begin{cases} (|2A-B| + |A+2B|)/5, & B \equiv 2A \pmod 5, \\ \infty, & B \not\equiv 2A \pmod 5. \end{cases} \tag{86.6}$$

The finite value is also $\max\{|3A+B|, |A-3B|\}/5$.

Proof. Applying the finite certificates with $t \equiv 1, 2 \pmod 5$ gives respectively

$$(3A+B)/5, \qquad (A-3B)/5.$$

Each must be an integer and its absolute value is a lower bound for any word length, regardless of its auxiliary denominators. Their integrality is equivalent to the displayed congruence.

Modulo (86.1), the two beta symbols $b(1/5, 1/5)$ and $b(1/5, 2/5)$ have vectors $u = (2, -1)$ and $v = (1, 2)$ in the ordered coordinates $([1/5], [2/5])$. Under the congruence hypothesis,

$$(A, B) = ru + sv, \qquad r = (2A-B)/5, \quad s = (A+2B)/5$$

with $r, s \in \mathbb{Z}$. The corresponding word has length $|r| + |s|$. Its two certificate values are $r+s$ and $r-s$, and $\max\{|r+s|, |r-s|\} = |r| + |s|$. Thus the lower bound is attained. □

Theorem 86.4 (The entire seventh-gamma lattice)**.** *If* $A + 2B + 3C \not\equiv 0 \pmod 7$*, then* $\ell(v_7(A,B,C)) = \infty$*. Otherwise*

$$\boxed{\ell(v_7(A,B,C)) = \frac{1}{7}\max\{|5A+3B+C|,\ |3A-B-5C|,\ |A-5B+3C|\}.} \tag{86.7}$$

An optimal word can be computed by taking a median of four explicit integers.

Proof. The certificate values for $t \equiv 1, 2, 3 \pmod 7$ are precisely the three linear forms divided by seven in (86.7). Their integrality is equivalent to $A + 2B + 3C \equiv 0 \pmod 7$. For example, $5A + 3B + C \equiv 5(A + 2B + 3C) \pmod 7$. Their absolute values bound every word from below.

The four beta symbols with argument pairs $(1,1), (1,2), (1,3), (2,2)$ divided by seven have coordinate vectors

$$u = (2, -1, 0), \quad v = (1, 1, -1), \quad w = (1, 0, 2), \quad z = (0, 2, 1), \qquad u + z = v + w.$$

Set

$$r = \frac{2A - 3B - C}{7}, \qquad s = \frac{2A + 4B - C}{7}, \qquad h = \frac{A + 2B + 3C}{7}.$$

These are integers under the congruence condition, and $(A, B, C) = ru + sv + hw$. For every integer d the same vector is

$$(r+d)u + (s-d)v + (h-d)w + dz.$$

Its word length is

$$|d + r| + |d - s| + |d - h| + |d|.$$

Choose d between the two middle order statistics of $-r, s, h, 0$. Such an integer exists. The minimum is the sum of the two largest of these numbers minus the sum of the two smallest. Equivalently it is

$$\max\{|r + s + h|,\ |r + s - h|,\ |r - s + h|\}.$$

Indeed these are, up to sign, the three ways to subtract the sum of one pair from the sum of its complementary pair. Substitution gives exactly (86.7), proving sharpness. □

These formulas solve an integer optimization problem for every exponent vector, rather than testing only a bounded list of candidate identities. The unrestricted-denominator lower bounds are essential: enumerating beta words at denominators five and seven alone would not establish this conclusion.

86.3 What the formal theorem does and does not decide

There is a natural map from $\mathcal{D}$ to the multiplicative classes of actual gamma values modulo algebraic factors and rational powers of π. The standard identities show this map is well defined; its injectivity is not known. Rohrlich's conjectural description of multiplicative gamma relations is exactly the missing type of input [353]. Under that conjecture, the cost formulas above also govern actual numerical beta-product identities. Without it, an unexpected numerical relation may identify distinct formal classes. In particular, a formal infinite cost for $\Gamma(1/5)$ does not prove that this number is transcendental. This is not a semantic distinction: an algebraic gamma value would itself give a zero-cost numerical identity that is invisible to the standard calculus.

For comparison there is a genuine, but classical, nonlinear arithmetic subsystem inside the seventh-gamma problem. Let

$$X = \Gamma(1/7), \quad Y = \Gamma(2/7), \quad Z = \Gamma(3/7), \qquad B_7 = B(1/7, 2/7) = XY/Z, \qquad Q_7 = e^{\pi\sqrt{7}}.$$

Reflection gives the exact identity

$$\mathrm{CS}(-7) = \frac{\sin(\pi/7)\sin(2\pi/7)}{\sin(3\pi/7)}\,\frac{B_7^2}{\pi}. \tag{86.8}$$

The classical Chowla–Selberg and Nesterenko results imply that π, $\mathrm{CS}(-7)$, and Q_7 are algebraically independent; see the explicit formulation and recent quantitative refinements of Cohen and Zudilin [354].

Proposition 86.5 (A completely classified classical CM subsystem)**.** *The numbers π, B_7, Q_7 are algebraically independent over $\overline{\mathbb{Q}}$. Consequently, for $R \in \overline{\mathbb{Q}}(U, V, W)$, its evaluation $R(\pi, B_7, Q_7)$ is algebraic if and only if R is a constant rational function. Every nonzero denominator remains nonzero on evaluation.*

Proof. By (86.8), adjoining B_7 or adjoining $\mathrm{CS}(-7)$ to $\overline{\mathbb{Q}}(\pi, Q_7)$ gives fields that are algebraic over each other. Their transcendence degree is three, proving the first assertion. If a nonconstant rational function evaluated to an algebraic number a, its numerator minus a times its denominator would give a nonzero polynomial relation. The same independence excludes a zero denominator. □

Proposition 86.5 is stated to mark the usable arithmetic input, not as a newly proved transcendence theorem. It classifies nonlinear expressions in a distinguished beta period. The corresponding vector $(1, 1, -1)$ has exact beta cost one by Theorem 86.4; each isolated seventh power has cost five. Knowing the arithmetic of one CM beta period does not classify products of the other beta periods in a five-factor isolating word.

For the five named individual targets $\Gamma(1/5), \Gamma(2/5), \Gamma(1/7), \Gamma(2/7), \Gamma(3/7)$, this section therefore establishes no irrationality or transcendence closure. Its new increment over the preceding manuscript is the unrestricted-denominator theorem, its even and composite cases, and the complete optimal monomial formulas at fifths and sevenths. The primary 2026 discussion of gamma-value transcendence by Sheth [355] also makes clear that geometric transcendence does not supply the missing individual-value result.

87 Fourteen constant targets and one related conjecture

The named targets in this section are the arithmetic questions carried forward from the previous manuscript. The word "unresolved" below describes what the present arguments do

not prove. It is not a claim that no relevant external work exists. In particular, irrationality of $\zeta(5)$ and Catalan's constant is reported in the cited 2026 external sources; neither is counted as a discovery of this manuscript, and their complete proofs are not independently verified here.

Target	Additional result and remaining obstruction
γ: irrationality or transcendence	Curvature bounds the indices admitting a bounded primitive even-Taylor crossover form by $O_B(X^{1/3})$. An infinite exceptional set and the logarithmic term in the form remain. No value conclusion.
δ: irrationality	Factorial-size primitive content occurs at only $O(\log X)$ indices up to X. Bounded primitive forms must be sparse and lacunary. An infinite exceptional certificate subsequence is not excluded.
G: transcendence	Direct positive rational intervals avoid the auxiliary $\zeta(2)$ term and determinant nonvanishing gap. Their proved denominator bound does not yield a transcendence criterion.
$\zeta(3)$: transcendence	The exact cancellation diagnostic recovers the classical Gaussian orthogonality cancellation. It does not exclude higher algebraic degrees.
$\zeta(5)$: transcendence	Exact norm and shift asymptotics; reduced-denominator experiments. The reported external irrationality result is not strengthened to transcendence here.
$\zeta(7)$: irrationality	The degree-to-shift ratio is optimized exactly at a fixed total budget. The common-denominator estimate retains a negative arithmetic margin; a sufficient actual cancellation estimate is absent.
Erdős–Borwein constant: transcendence	Distinct Lambert jets are algebraically independent as functions, and finite rational filters remain transcendental over the indicated modular field. No specialization theorem establishes the value claim.
Reciprocal Fibonacci sum: transcendence	A theta–Lambert decomposition transfers the nonlinear modular obstruction. Functional independence still does not imply the arithmetic nature of the specified sum.
Prime Mertens constant B_1: irrationality	The prime-harmonic generating function has a distinct logarithmic scale, giving functional exclusion and independence. Its constant term is not shown irrational.
$\Gamma(1/5)$: transcendence	Exact unrestricted beta-word cost, including the full fifth-gamma monomial lattice. A formal isolation identity does not determine the numerical value.
$\Gamma(2/5)$: transcendence	The same complete fifth-gamma cost applies after every permitted auxiliary denominator. No new value theorem.
$\Gamma(1/7)$: transcendence	Exact unrestricted isolation and the complete seventh-gamma monomial cost. Numerical algebraic relations remain beyond this calculus.
$\Gamma(2/7)$: transcendence	The same seventh-gamma cost and its classical CM comparison apply. No individual value conclusion.
$\Gamma(3/7)$: transcendence	The same seventh-gamma cost closes the formal shortcut, without proving transcendence of this individual gamma value.
Kurepa's left-factorial conjecture	The previous local Laguerre-content connection can be compared with the new global sparsity theorem. Sparse large content is compatible with either outcome of the conjecture, so no resolution follows.

What is newly closed inside the stated models

The proved conclusions include a logarithmic counting bound for large primitive cancellation; a complete totally positive channel-loss classification; an infinite algebraically linearly independent family of entropy constants; the exact finite-moment threshold $d(M) = M + 1$; the fixed-budget Wilson shift optimum; and the unrestricted standard gamma-calculus cost. The explicit

all-irrational four-atom entropy construction and direct Catalan intervals are constructive additions. The nonlinear Lambert conclusions strengthen the previous functional obstruction and include independent triples with the prime-harmonic function. These statements have their precise hypotheses and proofs in the corresponding sections. They are not renamed solutions of the named-constant questions in the table.

Verification and remaining research obligations

Exact rational arithmetic checks accompany the Laguerre content, Wilson and Catalan interval formulas, entropy identities, and finite gamma costs. Such checks can detect a faulty formula but do not prove an asymptotic assertion; those assertions are justified in the text. The main remaining obligations for a named-constant breakthrough are an arithmetic specialization theorem, an adequate primitive-height bound, or elimination of a surviving exceptional subsequence. None is silently substituted by convergence, functional independence, or a finite numerical search.

Part XI

Spectral Entropy and Constant Transfer

Editorial integration note. The lower bound of degree three for a remaining ternary counterexample is superseded in Part XII by degree at least four, or odd degree at least five for odd entropy denominator. The final part also distinguishes the invariant Catalan space from its convergent full-space repair and passes from formal Gamma signatures to a specified actual beta-period span.

Original part framing.

Scope. This standalone successor restates the constructions needed for its proofs. It distinguishes totally real numbers (all conjugates real) from totally positive numbers (all conjugates positive). Classical external inputs, independent proof reviews, and exact finite checks are identified; the manuscript is not formally verified. The cited 2026 external sources report irrationality proofs for $\zeta(5)$ and Catalan's constant [359, 360, 344]. Their full proofs and formal verification dependencies are not certified here. The corresponding targets in this manuscript are transcendence, and those external results are not counted as ours.

88 The new conclusions and their precise scope

The starting point is an arithmetic mismatch. Rapid convergence, positivity of a moment matrix, and algebraic independence of functions can all be proved without deciding the arithmetic nature of a specified constant. We treat the mismatch as a source of questions with exact answers, and then return to the named targets to state what is still missing. The claims below are additional conclusions of this research program; none is a declaration of historical priority.

An exact arithmetic classifier for entropy powers. For a positive probability vector p put $E(p) = \prod_i p_i^{-p_i}$. Theorem 89.3 proves, when every p_i is totally positive and $n \geq 2$, that for nonzero real algebraic c,

$$E(p)^c \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad p_1, \ldots, p_n, c \in \mathbb{Q}.$$

A symmetric relative-entropy identity forces every positive conjugate vector to coincide with the original one. The result applies to any density matrix with rational characteristic polynomial. It also makes Laplacian integrality the exact criterion for algebraicity of the effective rank of a finite graph. The final part of that section strengthens the earlier linear independence of entropy constants to an exclusion of algebraic-exponent products of their effective ranks. It does not assert algebraic independence of those ranks.

A sharp threshold over all totally real fields. Theorem 90.6 excludes rational entropy in bits for every nonrational ternary probability vector whose entries are totally real. Finite valuations first force a three-monomial generator to be an algebraic unit with odd exponents. A count of real embeddings then excludes every totally real probability field, without restricting its degree. The explicit golden-ratio four-atom example gives the matching upper bound:

$$\min\{\text{alphabet size with all entries irrational and totally real, and rational entropy}\} = 4.$$

The statement concerns rational entropy; it does not classify the entropy as algebraic or transcendental in every case. Without the totally real hypothesis, the minimum remains three or four here.

A higher functional transfer, with a finite-order limit. Theorem 91.3 proves algebraic independence over $\mathbb{C}(q)$ of three distinct Lambert–Fibonacci differential–dilation jets, provided their derivative orders are not all equal, together with the prime-harmonic series. In particular the Lambert and Fibonacci functions satisfy no algebraic differential equation of order at most two. The proof varies prime cusp denominators to separate two polynomial gradings. An explicit order-three relation in the leading cusp model explains why that proof does not establish full differential transcendence. These are function theorems; the named values require a specialization argument that is not provided.

Positivity and the cost of higher algebraic degrees. Theorem 92.2 converts a small primitive real-rooted polynomial value and a single off-axis estimate into exclusion of a specified algebraic degree. The exact Pell family in Theorem 92.3 reaches the quadratic boundary while retaining positive Hankel matrices, simple roots, and primitive coefficients. For Catalan's constant, Theorem 92.5 constructs a positive Gram matrix $A_m + 16GI_m$ with A_m rational. Paired interpolation cancels π^2 without losing determinant degree. Quantitative bounds for primitive value and off-axis growth remain necessary.

Two broad arithmetic transfer classes can be measured exactly. The Laguerre parameter, unlike the index, has only simple local roots. Theorem 93.2 therefore determines the complete distribution of the primitive common divisor over parameter residues, with independent truncated geometric valuations. Uniform exponential-tail bounds follow. They describe parameter averages and do not settle the fixed Euler–Gompertz parameter $A = 1$. For gamma values, Theorem 94.1 identifies exactly which prime-denominator monomials can be reduced to elliptic complex-multiplication periods using all standard gamma identities, all auxiliary rational denominators, and rational powers. This is an identity-calculus theorem; unknown numerical gamma relations remain outside its conclusion.

89 Algebraic powers of effective rank

For a positive probability vector $p = (p_i)_{i=1}^n$, define its natural-log entropy and its effective number of outcomes by

$$S(p) = -\sum_i p_i \log p_i, \qquad E(p) = e^{S(p)} = \prod_i p_i^{-p_i}.$$

Thus $E(p) = 2^{H_2(p)}$. The quantity E is also the effective rank in the sense of Roy and Vetterli [356] of a density matrix when p is its nonzero spectrum. It is not the Gaussian-normalized entropy power used for continuous distributions. The preceding manuscript proved transcendence of H_2 under a rational-characteristic-polynomial hypothesis. Here we classify *all real algebraic powers of* E under the more general hypothesis of totally positive algebraic probabilities. The proof uses Baker's classical logarithmic theorem [358]; historical priority of the resulting arithmetic formulation has not been established by an exhaustive literature review.

89.1 A logarithmic coordinate lemma

Lemma 89.1 (When an algebraic linear combination is an algebraic logarithm)**.** *Let $u_1, \dots, u_t$ be real logarithms of positive algebraic numbers, linearly independent over $\mathbb{Q}$. For algebraic*

real numbers $b_1, \dots, b_t$ one has

$$\exp\left(\sum_j b_j u_j\right) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad b_j \in \mathbb{Q} \text{ for every } j.$$

Furthermore, algebraic linear relations among logarithms are the algebraic span of their rational relations, and therefore remain valid after conjugating their algebraic coefficients alone.

Proof. Rational coefficients give a product of rational powers of algebraic numbers. Conversely, if the exponential is a positive algebraic number a, consider the logarithms $u_1, \dots, u_t, \log a$. If $\log a$ were independent of the u_j over $\mathbb{Q}$, Baker's theorem would prohibit the displayed algebraic relation. Hence $\log a = \sum_j r_j u_j$ with rational r_j. Baker's theorem applied to the independent u_j gives $b_j = r_j$. The final assertion follows by choosing a rational basis of any finite logarithmic span and comparing coefficients. □

Theorem 89.2 (Positive-conjugate rigidity for algebraic powers)**.** *Let $p = (p_1, \dots, p_n)$ be a positive algebraic probability vector, $n \geq 2$, and let $c \neq 0$ be real algebraic. Suppose $E(p)^c$ is algebraic. For every field embedding σ for which all $q_i = \sigma(p_i)$ are positive, one has*

$$q_i = p_i \quad \text{for every } i, \qquad \sigma(c) = c.$$

Here the embedding is taken on a number field containing p and c.

Proof. Put $A = E(p)^c > 0$, which is algebraic by assumption. Then

$$\log A + c \sum_i p_i \log p_i = 0.$$

Baker's rational-generation property allows conjugation of the algebraic coefficients alone, retaining every logarithm. Writing $d = \sigma(c)$, this gives

$$\log A + d \sum_i q_i \log p_i = 0.$$

An extension of σ to the algebraic closure may be used for this coefficient operation; the base A is not conjugated. Subtract the two identities. Since $0 < p_i < 1$ and $q_i > 0$, both $\sum_i p_i \log p_i$ and $\sum_i q_i \log p_i$ are negative. Thus

$$\frac{d}{c} = \frac{\sum_i p_i \log p_i}{\sum_i q_i \log p_i} > 0.$$

In particular d is real and has the same sign as c.

Set $r_i = cp_i - dq_i$. We have $\sum_i r_i \log p_i = 0$. Again by Baker's rational-generation property, this relation is an algebraic linear combination of rational relations among the $\log p_i$. Every rational relation, after clearing denominators, is a multiplicative identity in the p_i. Conjugating its bases by σ and using positivity yields the same rational relation among the $\log q_i$. Taking the original algebraic linear combination, without conjugating the r_i, therefore gives $\sum_i r_i \log q_i = 0$. Subtraction now yields

$$cD(p\|q) + dD(q\|p) = 0.$$

Divide by c. Both divergences are nonnegative and $d/c > 0$, so each divergence is zero. Their strict equality condition gives $p = q$. The earlier negative sums then give $d = c$. □

Theorem 89.3 (Exact algebraic-power classifier)**.** *Let $p_1, \dots, p_n > 0$, $n \geq 2$, sum to one, and suppose every p_i is totally positive over $\mathbb{Q}$. For every nonzero real algebraic number c,*

$$\boxed{E(p)^c \in \overline{\mathbb{Q}} \iff p_1, \dots, p_n \in \mathbb{Q} \text{ and } c \in \mathbb{Q}.}$$

In particular, one irrational entry forces every nonzero real algebraic power of the effective number of outcomes to be transcendental.

Proof. Take a normal field containing all entries and c. Total positivity makes every conjugate vector positive. If $E(p)^c$ is algebraic, Theorem 89.2 shows that every automorphism fixes each p_i and c, hence they are rational. Conversely, rational p_i and rational c give a finite product of rational powers of rational numbers, which is algebraic. □

Corollary 89.4 (Every algebraic logarithm base)**.** *Under the irrational-entry hypothesis of Theorem 89.3, the entropy $S(p)/\log b$ is transcendental for every real algebraic base $b > 1$.*

Proof. If $a = S(p)/\log b$ were algebraic, then $a > 0$ and $E(p)^{1/a} = b$ would contradict Theorem 89.3. □

Corollary 89.5 (Rational spectral data)**.** *Let ρ be a density matrix with rational characteristic polynomial and at least two positive eigenvalues. If any eigenvalue is irrational, then*

$$\exp\bigl(c[-\operatorname{tr}(\rho \log \rho)]\bigr)$$

is transcendental for every nonzero real algebraic c. If all nonzero eigenvalues are rational, this expression is algebraic exactly for rational c. This includes every rational-entry density matrix. For a pure state the expression is always 1 and is an explicit exception.

Proof. Delete zero eigenvalues. Every conjugate of each remaining eigenvalue is another positive root of the rational characteristic polynomial. Thus the eigenvalues are totally positive, and Theorem 89.3 applies. □

89.2 An exact criterion for graph constants

For a finite simple graph G with $m \geq 1$ edges, let L_G be its combinatorial Laplacian and put

$$\rho_G = \frac{L_G}{2m}, \qquad E_G = \exp\{-\operatorname{tr}(\rho_G \log \rho_G)\}.$$

This graph density matrix and its entropy are classical [357]. A graph is called Laplacian integral when every eigenvalue of L_G is an integer.

Corollary 89.6 (Graph integrality is the exact arithmetic criterion)**.** *Suppose* $\operatorname{rank} L_G \geq 2$. *For every nonzero real algebraic c,*

$$E_G^c \in \overline{\mathbb{Q}} \iff G \text{ is Laplacian integral and } c \in \mathbb{Q}.$$

For the path P_n, $E_{P_n}^c$ is transcendental for every $n \geq 4$ and every such c. For cycles C_n, $n \geq 3$, one has

$$E_{C_n}^c \in \overline{\mathbb{Q}} \iff n \in \{3, 4, 6\} \text{ and } c \in \mathbb{Q}.$$

The rank-one exception consists of one edge and any number of isolated vertices; for these graphs $E_G^c = 1$.

Proof. The matrix ρ_G is rational and positive semidefinite, with trace one. The eigenvalues of the integer matrix L_G are algebraic integers. They are rational exactly when they are integers, so the first equivalence follows from Corollary 89.5.

The path eigenvalues are $2 - 2\cos(k\pi/n)$ for $0 \le k < n$. For $n \ge 4$, the first positive eigenvalue lies strictly between zero and one, hence is not an integer. The cycle eigenvalues are $2 - 2\cos(2\pi k/n)$, $0 \le k < n$. At $n = 3, 4, 6$ all are integers. For $n = 5$, the eigenvalue $(5 - \sqrt{5})/2$ is irrational. For $n \ge 7$, the first positive eigenvalue lies strictly between zero and one. These spectral formulas follow directly by applying the Laplacian to the usual discrete cosine vectors for a path and Fourier vectors for a cycle. Finally, Laplacian rank equals the number of vertices minus the number of connected components, so rank one means exactly one two-vertex component and isolated vertices. □

The criterion propagates through standard graph operations. A disjoint union, or a Cartesian product of nonempty graphs, is Laplacian integral exactly when both factors are: the respective spectra are the multiset union and all pairwise sums, and each factor has a zero eigenvalue. Also a graph is Laplacian integral exactly when its complement is, since on the orthogonal complement of the all-ones vector the two Laplacians sum to nI. Thus one nonintegral example yields whole families of canonical transcendental effective-rank constants, subject only to the explicit rank and nonempty-edge conditions above.

Corollary 89.7 (Positive rational mixtures of spectral entropies)**.** *Let $p^{(1)}, \dots, p^{(k)}$ satisfy the hypotheses of Theorem 89.3, and let $w_j > 0$ be rational. If any entry of any $p^{(j)}$ is irrational, then*

$$\exp\left(c \sum_{j=1}^{k} w_j S(p^{(j)})\right)$$

is transcendental for every nonzero real algebraic c. If all entries are rational, it is algebraic exactly when c is rational.

Proof. Suppose the displayed exponential is algebraic, and let $q^{(j)}$ be a simultaneous conjugate of the input vectors. Put $d = \sigma(c)$. Coefficient-only conjugation gives the same logarithm as a weighted sum of cross-entropies. Their positivity implies $d/c > 0$. Subtracting these two relations and then transporting their bases as in Theorem 89.2 gives

$$\sum_j w_j \{cD(p^{(j)} \| q^{(j)}) + dD(q^{(j)} \| p^{(j)})\} = 0.$$

All summands have the same nonnegative sign after division by c, so $q^{(j)} = p^{(j)}$ for every j, and then $d = c$. Total positivity allows every automorphism, making all entries and c rational. Conversely rational data give an algebraic product. □

The total-positivity hypothesis cannot simply be dropped. Let $t = (\sqrt{5} - 1)/2$ and $p = (t, t^2)$. Then $t + t^2 = 1$ and

$$E(p) = t^{-(2-t)}, \qquad E(p)^{1/(2-t)} = t^{-1} \in \overline{\mathbb{Q}}.$$

Thus a positive algebraic vector with irrational entries can have an algebraic nonzero algebraic power of its effective number of outcomes. Its nonpositive conjugates prevent the Galois argument.

89.3 Algebraic-exponent independence of an explicit family

For each odd prime r, retain the preceding manuscript's rational matrix

$$\rho_r = \frac{1}{2} I_2 + 2^{-(r+3)/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad h_r = H_2(\rho_r), \qquad E_r = 2^{h_r}.$$

The previously proved linear independence of $1, h_3, h_5, \ldots$ does not by itself decide algebraicity of products of the E_r. The following strengthens it in that direction.

Theorem 89.8 (No algebraic-exponent product is algebraic)**.** *For any finitely supported real algebraic coefficients c_r, not all zero,*

$$\prod_r E_r^{c_r} = 2^{\sum_r c_r h_r}$$

is transcendental. No positivity assumption on the c_r is needed. Consequently the E_r are multiplicatively independent even modulo positive algebraic numbers and with arbitrary real algebraic exponents. This is not algebraic independence of the E_r.

Proof. The two eigenvalues are

$$\lambda_r^{\pm} = \frac{1}{2} \pm \frac{\sqrt{2}}{2^{(r+3)/2}}, \qquad \lambda_r^{+} \lambda_r^{-} = \frac{2^r - 1}{2^{r+2}}.$$

For distinct odd primes r, s, the positive integers $2^r - 1$ and $2^s - 1$ are coprime. Choose an odd prime $q \mid 2^r - 1$. In $K = \mathbb{Q}(\sqrt{2})$ there is a prime ideal $\mathfrak{q}_+$ at which one eigenvalue, say λ_r^+, has positive valuation and the other has zero valuation. The sum of the eigenvalues is 1, so they cannot both have positive valuation. Their product proves existence, and their denominators are powers of two. At the conjugate prime ideal $\mathfrak{q}_-$, λ_r^- has positive valuation and λ_r^+ has zero valuation. These ideals are distinct. All eigenvalues with other indices have valuation zero at both ideals.

Suppose $A = \prod_r E_r^{c_r}$ were algebraic. Then

$$\log A + \sum_r c_r (\lambda_r^+ \log \lambda_r^+ + \lambda_r^- \log \lambda_r^-) = 0.$$

Every algebraic logarithmic relation is an algebraic combination of rational ones. Each rational relation yields a multiplicative identity and hence an identity of prime-ideal valuations. Work in a number field containing K and A, and extend each of $\mathfrak{q}_+$ and $\mathfrak{q}_-$ to that field. Evaluating the relation at the first extended valuation gives $c_r \lambda_r^+ \in \mathbb{Q}$; at the second it gives $c_r \lambda_r^- \in \mathbb{Q}$. Valuations of algebraic numbers are integers after normalization, and ramification only multiplies the relevant nonzero valuations by positive integers. If $c_r \neq 0$, division gives $\lambda_r^+ / \lambda_r^- \in \mathbb{Q}$, which is false because their sum is 1 and each is irrational. Thus every $c_r = 0$, a contradiction. □

Together with the preceding manuscript's result, this provides explicit constants for which every nonzero algebraic linear combination $L = \sum c_r h_r$ is transcendental and, additionally, 2^L is transcendental. No algebraic independence of L and 2^L is asserted. In particular these statements do not resolve algebraic independence questions for general Gelfond–Schneider numbers or for classical gamma and zeta constants.

90 A unit obstruction and the exact totally-real entropy threshold

The preceding manuscript reduced the remaining three-atom problem to a stationary value of three Laurent monomials. That reduction can be strengthened substantially by using finite valuations. We prove that a nonrational algebraic ternary vector with rational entropy would have to arise from an algebraic unit with three odd exponents. This excludes all such vectors in totally real number fields, of arbitrary degree. Combined with the explicit four-atom example already obtained, the least alphabet with totally real irrational algebraic probabilities is exactly four. The corresponding problem over all number fields remains open in this work.

Throughout, $H_2(p) = -\sum_i p_i \log_2 p_i$, all probabilities are positive, and a nonrational vector means that at least one entry is irrational. An algebraic unit is an algebraic number whose value and reciprocal are both algebraic integers.

90.1 The earlier reduction, with primitive exponents

Lemma 90.1 (Three-monomial reduction)**.** *Suppose $p = (p_1, p_2, p_3)$ is an algebraic, nonrational probability vector and $h = H_2(p) \in \mathbb{Q}$. There are a positive algebraic number w and integers m_1, m_2, m_3 such that*

$$p_i = 2^{-h} w^{m_i}, \qquad \sum_i w^{m_i} = 2^h, \qquad \sum_i m_i w^{m_i} = 0, \qquad \gcd(m_1, m_2, m_3) = 1. \tag{90.1}$$

The logarithms $\log w$ and $\log 2$ are rationally independent. The exponents are pairwise distinct, and their minimum is negative while their maximum is positive.

Proof. We include the reduction to make the new obstruction self-contained. Baker's logarithmic independence theorem [358] implies that algebraic-coefficient relations among logarithms of positive algebraic numbers are generated by rational-coefficient relations. Put

$$t = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{p_1, p_2, p_3\}, \quad r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\log p_1, \log p_2, \log p_3, \log 2\}.$$

Splitting the entropy relation with respect to a rational basis of the probabilities, which also spans h because $\sum_i p_i = 1$ and $h \in \mathbb{Q}$, gives t independent rational relations among the four logarithms. Thus $t + r \le 4$. Nonrationality gives $t \ge 2$. If $r = 1$, all the p_i are rational powers of two. Positive rational powers of two summing to one must have integral exponents: clear a common denominator d, use the independence of $1, 2^{1/d}, \dots, 2^{(d-1)/d}$ supplied by Eisenstein's criterion for $X^d - 2$, and observe that all coefficients are nonnegative. Hence $r = 1$ would make all entries rational. Therefore $t = r = 2$.

Choose a positive algebraic u with $\log 2, \log u$ a rational basis of the logarithm span, and write $\log p_i = a_i \log 2 + b_i \log u$ with rational a_i, b_i. Baker's theorem then gives $\sum_i b_i p_i = 0$ and $\sum_i a_i p_i = -h$. The rational kernel of $(x_i) \mapsto \sum_i x_i p_i$ is one-dimensional; as $b \neq 0$, it follows that $a_i + h = \lambda b_i$ for one rational λ. Clearing a common denominator in the b_i gives (90.1), except possibly for primitivity. Replacing w by a positive integral power and dividing all m_i by their common gcd makes the exponents primitive.

If, for example, $m_1 = m_2 = a$ and $m_3 = b \neq a$, stationarity and normalization imply $a(p_1 + p_2) + bp_3 = 0$, while $p_1 = p_2$. These two linear equations make every probability rational. If all three exponents coincide, stationarity makes that exponent zero, contrary to logarithmic independence. Thus the exponents are distinct. Their weighted mean zero, with positive weights, forces both a positive and a negative exponent. □

90.2 Finite valuations force a unit and odd exponents

Theorem 90.2 (Unit, parity, and norm restrictions)**.** *Under the hypotheses of Lemma 90.1, the normalized generator w is an algebraic unit, each m_i is odd, and*

$$1 \le h < \log_2 3. \tag{90.2}$$

Let $K = \mathbb{Q}(p_1, p_2, p_3)$ and $d = [K : \mathbb{Q}]$. Then

$$\left|N_{K/\mathbb{Q}}(p_i)\right| = 2^{-hd} \quad (i = 1, 2, 3), \qquad hd \in \mathbb{Z}. \tag{90.3}$$

All three probabilities are irrational.

Proof. Put $C = 2^h$ and $L = \mathbb{Q}(w)$. Since $h > 0$ is rational, C is an algebraic integer: if $h = a/b$ in lowest terms, then C satisfies $X^b - 2^a = 0$. Also $C = \sum_i w^{m_i} \in L$.

Let v be any finite additive valuation of L. If $v(w) > 0$, the term with the unique smallest exponent has strictly the smallest valuation in $\sum_i w^{m_i}$, and that valuation is negative because $\min_i m_i < 0$. The ultrametric inequality with a unique minimum therefore gives $v(C) < 0$, contrary to integrality. If $v(w) < 0$, the unique largest exponent gives the same contradiction. Thus $v(w) = 0$ at every finite place. Equivalently, w is an algebraic unit.

Choose a prime of L above 2, reduce the two sum identities in (90.1) in its residue field, and write x for the nonzero residue of w. Since $h > 0$, the residue of C is zero. Thus

$$\sum_i x^{m_i} = 0, \qquad \sum_{i:\ m_i\ \mathrm{mod}\, 2} x^{m_i} = 0.$$

If exactly one exponent is odd, the second sum is a single nonzero term. If exactly two are odd, subtracting the second sum from the first gives a single nonzero term. Both are impossible. Primitivity excludes three even exponents, so all three exponents are odd. Consequently stationarity also gives

$$C = \sum_i (1 - m_i) w^{m_i} \in 2\mathcal{O}_L,$$

because each coefficient $1 - m_i$ is even and every power of w is an algebraic integer. At a valuation normalized by $v(2) = 1$, this says $h = v(C) \ge 1$. The strict entropy upper bound in (90.2) follows from $H_2(p) \le \log_2 3$, with equality only for the rational uniform vector.

Primitivity supplies integers c_i with $\sum_i c_i m_i = 1$; hence

$$w = \prod_i (C p_i)^{c_i}, \qquad L = K(C).$$

Every conjugate of C has absolute value C, while the absolute norm of a unit is one. Taking norms of $p_i = C^{-1} w^{m_i}$ from L to $\mathbb{Q}$, and then using norm transitivity, proves $|N_{K/\mathbb{Q}}(p_i)| = 2^{-hd}$. This is rational, so hd is an integer, by uniqueness of prime factorization applied to a rational power of two.

Finally, if p_i were rational and positive, its norm identity would give $p_i = 2^{-h}$, so $w^{m_i} = 1$. As $w > 0$ and $w \ne 1$, this forces $m_i = 0$, contradicting oddness. □

The lower bound $h \ge 1$ is a new exclusion region for the remaining ternary problem. It is not a statement that a nonrational ternary vector of entropy exactly one bit exists.

Proposition 90.3 (The generator has very few real conjugates)**.** *Write $h = a/b$ in lowest positive terms. The field $L = \mathbb{Q}(w)$ from Theorem 90.2 has exactly one real embedding if b is odd and exactly two real embeddings if b is even. In the latter case the two real conjugates of w are w and $-w$. In particular, w is not totally real.*

Proof. Set $F(x) = \sum_i x^{m_i}$. On the positive real axis,

$$\frac{d^2}{dy^2} F(e^y) = \sum_i m_i^2 e^{m_i y} > 0.$$

Thus F has exactly one positive stationary point, namely w: its first derivative in y is strictly increasing and tends to opposite infinities at the two ends, because the exponents have both signs. Since all m_i are odd, F is odd on the nonzero real axis. Every real conjugate of w is a stationary point of F, and its F-value is a real conjugate of C. Therefore it can only be w or $-w$.

The number $C = 2^{a/b}$ has degree b: Bezout's identity for a, b shows $\mathbb{Q}(C) = \mathbb{Q}(2^{1/b})$, and $X^b - 2$ is Eisenstein. If b is odd, C has just one real conjugate, which is positive; only w is possible. If b is even, b divides $[L : \mathbb{Q}]$, so that degree is even. Its real embeddings consequently have an even count; as the count is positive and at most two, it is exactly two.

If w were totally real, all its conjugates would be among $w, -w$. Then w^2 would be rational, and also a positive rational algebraic unit. This forces $w^2 = 1$, contradicting the rational independence of $\log w, \log 2$. □

90.3 Closing the quadratic-field case

Theorem 90.4 (No nonrational quadratic ternary vector has rational entropy)**.** *Let K be a real quadratic number field and let $p_1, p_2, p_3 \in K$ be positive with sum one. If at least one p_i is irrational, then $H_2(p_1, p_2, p_3) \notin \mathbb{Q}$. More generally, a nonrational ternary vector in any totally real field cannot have entropy in $\frac{1}{2}\mathbb{Z}$.*

Proof. For the more general statement, suppose $h \in \frac{1}{2}\mathbb{Z}$ and let K_0 be the totally real field generated by the probabilities. Theorem 90.2 gives a unit w with

$$\mathbb{Q}(w) = K_0(2^h) \subseteq K_0(\sqrt{2}).$$

The compositum of totally real fields is totally real. This makes w totally real, contradicting Proposition 90.3.

For a nonrational ternary vector in a real quadratic field, its probability field has degree two. Formula (90.3) gives $2h \in \mathbb{Z}$, and (90.2) leaves only $h = 1$ or $h = 3/2$. Both have just been excluded. □

90.4 Extension to every totally real number field

The quadratic argument extends to arbitrary totally real fields. The needed radical-degree fact has a short proof using the constant term of a minimal polynomial; no unproved binomial irreducibility assumption is needed.

Lemma 90.5 (A rational radical over a totally real field)**.** *Let K be a totally real number field and let $C = 2^{a/b}$, where a, b are coprime positive integers. Then*

$$[K(C) : K] = \begin{cases} b/2, & b \text{ is even and } \sqrt{2} \in K, \\ b, & \text{otherwise.} \end{cases}$$

In the first case the minimal polynomial is $X^{b/2} - 2^{(a-1)/2}\sqrt{2}$; in the second it is $X^b - 2^a$.

Proof. Let $e = [K(C) : K]$, and consider the minimal polynomial of C over K at the given real embedding of K. Its e roots belong to $\{C\zeta : \zeta^b = 1\}$. Its nonzero constant coefficient is real and has absolute value C^e, so $C^e \in K$. A positive rational power $2^{ae/b}$ has degree $b/\gcd(b, e)$ over $\mathbb{Q}$, by Eisenstein and Bezout as in Proposition 90.3. If this degree exceeds two, some conjugates are nonreal. Since every element of K is totally real, we obtain $b/\gcd(b, e) \le 2$.

As $1 \le e \le b$, this forces $e = b$ when b is odd, and $e \in \{b/2, b\}$ when b is even. The case $e = b/2$ puts $C^e = 2^{a/2}$ in K; as a is odd, it forces $\sqrt{2} \in K$. Conversely, if $\sqrt{2} \in K$, the displayed polynomial of degree $b/2$ vanishes at C, so $e \le b/2$ and equality follows. The degree-b case has the other displayed minimal polynomial. □

Theorem 90.6 (Irrational ternary entropy over totally real fields)**.** *Let p_1, p_2, p_3 be positive totally real algebraic numbers with sum one. If at least one entry is irrational, then*

$$H_2(p_1, p_2, p_3) \notin \mathbb{Q}.$$

Proof. Suppose instead that $h = a/b \in \mathbb{Q}$, in lowest positive terms. The probability field $K = \mathbb{Q}(p_1, p_2, p_3)$ is totally real; write $d = [K : \mathbb{Q}] > 1$. The previous results give $L = K(C) = \mathbb{Q}(w)$, with $C = 2^{a/b}$, and a real-embedding count $r_1(L)$ equal to one for odd b and two for even b.

If b is odd, Lemma 90.5 makes $X^b - 2^a$ the minimal polynomial over K. Under every real embedding of K it has exactly one real root. Counting extensions of embeddings therefore gives $r_1(L) = d$, contradicting $r_1(L) = 1$.

If b is even and $\sqrt{2} \notin K$, the same minimal polynomial has exactly two real roots under each embedding of K. Thus $r_1(L) = 2d$, contradicting $r_1(L) = 2$.

It remains that b is even and $\sqrt{2} \in K$. Put $n = b/2$. The minimal polynomial is

$$X^n - 2^{(a-1)/2}\sqrt{2}.$$

Exactly half of the d real embeddings of K send $\sqrt{2}$ to $\sqrt{2}$, and half send it to $-\sqrt{2}$. If n is odd, each embedding has one real extension; if n is even, the first half have two real extensions and the second half have none. In both cases $r_1(L) = d$. Hence $d = 2$. The norm restriction $hd \in \mathbb{Z}$ forces $b \mid 2$, so $b = 2$ and $L = K(C) = K$ is totally real. This contradicts the fact that w is not totally real, proved in Proposition 90.3. □

Corollary 90.7 (The exact minimum for totally real probabilities)**.** *Among probability vectors all of whose entries are positive totally real algebraic irrational numbers, the least number of entries allowing rational entropy in bits is exactly four. The minimum is already attained within the quadratic field $\mathbb{Q}(\sqrt{5})$.*

Proof. One atom is necessarily rational. For two irrational algebraic probabilities, rational entropy in base 2 is impossible by Baker's theorem: splitting the entropy relation with respect to $1, p$ forces p and $1 - p$ to be rational powers of two, and positivity in the Eisenstein basis used above makes them rational. Three atoms are excluded by Theorem 90.6.

For the upper bound put $\varphi = (1 + \sqrt{5})/2$ and use the previously constructed vector

$$\left(\frac{\varphi}{2}, \frac{\varphi^{-5}}{2}, \frac{\varphi^{-4}}{2}, \frac{\varphi^{-4}}{2}\right).$$

All its entries are positive irrational elements of $\mathbb{Q}(\sqrt{5})$. The identities

$$\varphi + \varphi^{-5} + 2\varphi^{-4} = 2, \qquad \varphi - 5\varphi^{-5} - 8\varphi^{-4} = 0$$

show that they sum to one and have entropy exactly one bit. □

The corollary minimizes over all totally real probability vectors; it does not assert that four atoms suffice in each prescribed number field. The global minimum over unrestricted algebraic probabilities remains either three or four. Any ternary counterexample would now need a probability field that is not totally real, of degree at least three, rational entropy in $[1, \log_2 3)$ with denominator dividing that degree, and the unit and signature restrictions proved above.

Literature and scope. The logarithmic input is classical Baker theory. The rational-spectrum rational-or-transcendental dichotomy appears explicitly in Blakaj–Wolf [337, Theorem 2]. The finite-place unit obstruction, parity restriction, and totally-real threshold above are additions to this research program; the primary-source comparison does not establish their historical priority. No arithmetic classification of Euler's constant, the Gompertz constant, or an odd zeta value follows from this entropy theorem.

91 A third Lambert–Fibonacci jet and a fourth independent function

The preceding two-jet argument used one rational cusp at a time. We now vary the denominator of the cusp. The residue $1/N$ separates ordinary polynomial degree from differential weight. This proves a three-jet theorem, together with a fourth independent function supplied by the prime-harmonic series.

Write

$$L(q) = \sum_{n\ge 1} \frac{q^n}{1-q^n}, \qquad \mathcal{F}(q) = \sum_{n\ge 1} \frac{q^n}{1-(-1)^n q^{2n}}, \qquad \mathcal{P}(q) = \sum_{p \text{ prime}} \frac{q^p}{p}, \qquad D = q\frac{d}{dq}.$$

For $\eta \in \{0,1\}$, define

$$J_{m,r,\eta}(q) = \begin{cases} D^r[L(q^m)], & \eta = 0, \\ D^r[\mathcal{F}(q^m)], & \eta = 1. \end{cases}$$

Here $m \ge 1,\ r \ge 0$, and differentiation follows composition.

91.1 The fixed-cusp input

For a prime $N \equiv 1 \pmod 4$, a primitive N-th root ξ, and $N \nmid m$, the cusp expansions are

$$J_{m,r,\eta}(\xi e^{-t}) = \frac{a_{m,r,\eta}}{N} t^{-w_r}\bigl(U + \gamma - 2\log N + \beta_{m,r,\eta}\bigr) + O_{N,m,r}(t^{-w_r+1/2}), \tag{TL1}$$

where

$$U = \log(1/t), \quad w_r = r+1, \quad a_{m,r,\eta} = \frac{r!}{4^\eta m}, \quad \beta_{m,r,\eta} = H_r - \log m + \eta\pi/2.$$

We recall the derivation to make the use of the remainder explicit. For ξ of order N, the Mellin transform of L is

$$Z_\xi(s) = N^{-2s} \sum_{a,b=1}^{N} \xi^{ab}\zeta(s, a/N)\zeta(s, b/N) = \frac{1/N}{(s-1)^2} + \frac{(2\gamma - 2\log N)/N}{s-1} + O(1).$$

In the inverse Mellin integral for the r-th radial derivative, the factor is $\Gamma(s+r)t^{-s-r}$. Shifting to $\Re s = 1/2$ gives the power-saving remainder in (TL1). Its residue contains $+H_r$. These are standard Lambert cusp expansions; see [345] for a much more extensive treatment.

For the Fibonacci function use the exact identity

$$\mathcal{F}(q) = L(q^2) - L(q^4) + \frac{\vartheta(q)^2 - \vartheta(-q)^2}{8}, \qquad \vartheta(q) = \sum_{k\in\mathbb{Z}} q^{k^2}.$$

At a primitive odd-prime cusp, Poisson summation and quadratic Gauss sums give

$$\vartheta(\xi e^{-t})^2 = \frac{(-1)^{(N-1)/2}\pi}{Nt} + O(t^{-1}e^{-c_N/t}), \qquad \vartheta(-\xi e^{-t})^2 = O(t^{-1}e^{-c_N/t}).$$

The square of the Gauss sum is independent of the primitive numerator. Differentiation preserves an exponential saving. For $N \equiv 1 \pmod 4$, this gives the extra intercept $\pi/2$ in (TL1); the two $\log 2$ contributions cancel.

The Mertens theorem in arithmetic progressions, followed by partial summation, gives at the same cusp

$$\mathcal{P}(\xi e^{-t}) = -\frac{1}{N-1}\log U + C_\xi + o(1). \tag{TL2}$$

Indeed, every nonzero class modulo N contributes $(N-1)^{-1}\log\log x + M(N,a) + o(1)$, and $\sum_{a=1}^{N-1}\xi^a = -1$; the prime N contributes only a bounded term. An explicit statement of the classical input is [352, Eq. (1)].

Lemma 91.1 (Distinct cusp intercepts)**.** *The real numbers $\beta_{m,r,\eta}$ attached to distinct triples (m,r,η) are distinct.*

Proof. An equality would imply

$$H_r - H_s = \log(m/n) - (\eta - \eta')\pi/2.$$

If $m/n \neq 1$, the real logarithm $\log(m/n)$ and $\log(-1) = i\pi$ are linearly independent over $\mathbb{Q}$. Baker's theorem [358] implies that $1, \log(m/n), i\pi$ are linearly independent over $\overline{\mathbb{Q}}$, contradicting the displayed equality. If $m/n = 1$, irrationality of π forces $\eta = \eta'$; then strict increase of H_r forces $r = s$. Thus the two triples coincide. □

The use of Baker's theorem here is only to distinguish the known intercepts; it does not transfer to the values of L or $\mathcal{F}$.

91.2 An injectivity lemma with two gradings

Lemma 91.2. *Let w_1, w_2, w_3 be positive integers, not all equal, and let $\beta_1, \beta_2, \beta_3$ be distinct complex numbers. On the vector space of polynomials homogeneous of ordinary degree d and weighted degree W, the substitution*

$$(X_1, X_2, X_3) \longmapsto (V + \beta_1, V + \beta_2, V + \beta_3)$$

is injective. The assertion remains valid when the coefficients belong to any purely transcendental extension of $\mathbb{C}$.

Proof. The exponent vectors of the allowed monomials satisfy

$$e_1 + e_2 + e_3 = d, \qquad w_1e_1 + w_2e_2 + w_3e_3 = W.$$

The integer kernel of these two independent equations has rank one. Hence, after choosing one exponent vector e^0, every other vector is $e^0 + k\nu$, for a fixed nonzero integer vector ν. Factoring out the image of the first monomial, a proposed relation becomes a Laurent polynomial in

$$R(V) = \prod_{i=1}^{3}(V + \beta_i)^{\nu_i}.$$

The rational function R is nonconstant, since its distinct zeros and poles have orders ν_i, not all zero. It is therefore transcendental over the coefficient field, which does not contain V. A nonzero Laurent polynomial in $R(V)$ cannot vanish. □

Theorem 91.3 (Three jets, and four functions with primes)**.** *Choose three distinct triples (m_i, r_i, η_i), $1 \le i \le 3$, whose derivative orders r_i are not all equal. Then*

$$\boxed{J_{m_1,r_1,\eta_1}(q),\ J_{m_2,r_2,\eta_2}(q),\ J_{m_3,r_3,\eta_3}(q),\ \mathcal{P}(q) \text{ are algebraically independent over } \mathbb{C}(q).}$$

In particular, either of the triples

$$L,\ DL,\ D^2L \qquad \text{and} \qquad \mathcal{F},\ D\mathcal{F},\ D^2\mathcal{F}$$

is algebraically independent over $\mathbb{C}(q)$. Thus neither L nor $\mathcal{F}$ satisfies a nonzero algebraic differential equation of order at most two over $\mathbb{C}(q)$.

Proof. Suppose

$$P(q, X_1, X_2, X_3, Z) \in \mathbb{C}[q, X_1, X_2, X_3, Z]$$

is a nonzero polynomial giving an identity after the indicated substitution; rational denominators have been cleared. Let M bound its degree in q. Give X_i weight $w_i = r_i + 1$, and give Z weight zero. Let W be the largest weight occurring, and decompose that part by ordinary degree in the three X-variables:

$$P_W = \sum_d P_{W,d}.$$

Set $a_i = a_{m_i,r_i,\eta_i}$, $\beta_i = \beta_{m_i,r_i,\eta_i}$, and define

$$Q_d(q, V, Z) = P_{W,d}(q, a_1(V + \beta_1), a_2(V + \beta_2), a_3(V + \beta_3), Z).$$

Take any sufficiently large prime $N \equiv 1 \pmod 4$ not dividing $m_1 m_2 m_3$, and any primitive N-th root ξ. For this fixed N, put $V = U + \gamma - 2\log N$. The assumed identity and (TL1) imply

$$\sum_d N^{-d} Q_d(\xi, V, \mathcal{P}(\xi e^{-t})) + o(1) = 0. \tag{TL3}$$

All error terms really are $o(1)$: after multiplication by t^W, the jet error has a factor $t^{1/2}$, lower weights have a positive integer power of t, and the remaining factors grow only as fixed powers of U and $\log U$. Replacing the polynomial coefficients at $q = \xi e^{-t}$ by their values at ξ contributes only $O(t)$ times fixed powers of U and $\log U$, again $o(1)$.

The polynomial

$$Q_{N,\xi}(V, Z) = \sum_d N^{-d} Q_d(\xi, V, Z)$$

must be zero. Otherwise, its highest power of V, followed by the highest power of Z in that coefficient, would give a unique nonzero leading term $cV^j(\log V)^k$ after (TL2) is inserted. Such a term cannot satisfy (TL3). No coefficient extraction from the $o(1)$ prime remainder is needed; only its leading $\log V$ growth is used.

Now use the *exact root count*. Every coefficient of $V^j Z^k$ in $\sum_d N^{-d} Q_d(q, V, Z)$ is a polynomial in q of degree at most M. It vanishes at all $N - 1$ primitive N-th roots. For $N - 1 > M$, it vanishes identically. Consequently

$$\sum_d N^{-d} Q_d(q, V, Z) = 0$$

as a polynomial identity for infinitely many primes N. Regarding this as a polynomial in $1/N$, one obtains $Q_d(q, V, Z) = 0$ for every d. By Lemmas 91.1 and 91.2, over the coefficient field $\mathbb{C}(q, Z)$, each $P_{W,d}$ must vanish. This contradicts the definition of W. □

Remark 91.4 (A genuine increment and its scope). The previous manuscript established independence of two jets and, after adjoining $\mathcal{P}$, three functions. Theorem 91.3 increases these counts to three jets and four functions whenever the derivative orders are not all equal. It does not establish full differential transcendence or independence of every finite collection of jets. It also says nothing new about the transcendence of $L(1/2)$, the Fibonacci reciprocal sum, or the irrationality of B_1. A complete historical priority check for this three-jet extension has not been completed.

91.3 An explicit obstruction at the next order

The next proposition explains why the leading cusp family alone cannot prove full differential transcendence.

Proposition 91.5 (An order-three identity of the cusp model)**.** *Define*

$$\begin{aligned}\mathscr{R}(A,B,C,E) = A^2E^2 - 23ABCE - 44B^2C^2 \\ + 36B^3E + 25AC^3.\end{aligned}$$

It is homogeneous of ordinary degree 4 *and weighted degree* 10 *for weights* $1,2,3,4$*, and*

$$\mathscr{R}(axv,\ ax^2(v+1),\ ax^3(2v+3),\ ax^4(6v+11)) = 0 \tag{TL4}$$

identically in a, x, v*. Nevertheless,*

$$\mathscr{R}(L, DL, D^2L, D^3L) = -5q^4 + O(q^5),$$

so (TL4) *is not a differential equation for* L*.*

Proof. For the four arguments in (TL4), elimination of x uses

$$Ax^2 - 3Bx + C = 0, \qquad 4Bx^2 - 5Cx + E = 0.$$

The resultant is

$$(AE - 4BC)^2 - (12B^2 - 5AC)(5C^2 - 3BE),$$

which expands to $\mathscr{R}$. The two homogeneity assertions are immediate. Finally $D^rL = q + O(q^2)$ for every r, and the sum of the five coefficients in $\mathscr{R}$ is -5. □

To exclude higher-order differential equations one must use additional information beyond the two displayed cusp terms. The proposition is a barrier to this proof mechanism, not a barrier theorem for all possible methods.

91.4 Why the stronger functional theorem does not decide a value

Proposition 91.6 (Specialization can destroy all finite-jet information)**.** *Let* $\alpha \in \overline{\mathbb{Q}}$ *lie inside the common analytic domain of functions* $f_0, \dots, f_s$ *which are algebraically independent over* $\mathbb{C}(q)$*. For arbitrary prescribed algebraic values* $c_0, \dots, c_s$*, the functions*

$$g_i(q) = (q - \alpha)f_i(q) + c_i$$

are still algebraically independent over $\mathbb{C}(q)$*, but* $g_i(\alpha) = c_i$*. If the original functions have algebraic Taylor coefficients, so do the new ones.*

More specifically, if $f, Df, \dots, D^Rf$ *are independent, then the first* $R+1$ *jets of* $g = (q-\alpha)^{R+1}f$ *remain independent over* $\mathbb{C}(q)$*, while* $g(\alpha) = Dg(\alpha) = \cdots = D^Rg(\alpha) = 0$*.*

Proof. The first change of variables is invertible over $\mathbb{C}(q)$. In the second statement, Leibniz' rule gives an invertible triangular linear transformation of the jets, whose diagonal is $(q-\alpha)^{R+1}$. The vanishing at α follows from the zero multiplicity. □

This elementary proposition identifies the missing bridge: functional independence, rational Taylor coefficients, and even independent finite jets do not constitute a specialization theorem. The arithmetic targets remain separate from the new functional result.

92 Positive determinant certificates and the algebraic-degree barrier

The preceding research program supplied rapidly shrinking rational intervals, but not a sufficient arithmetic margin. We now examine the stronger polynomial certificates supplied by positive affine matrix pencils. Two conclusions are obtained. A single off-axis determinant controls the passage from irrationality to exclusion of higher algebraic degrees. A separate interpolation construction removes the auxiliary π^2 from a Catalan moment matrix while retaining positivity and full degree. Neither conclusion is a new solution of a named-constant problem.

The external motivation is Fauzan's 2026 determinant construction and Lai's subsequent independence argument [359, 360]. For this investigation we read Fauzan's actual manuscript, including Sections 2–4 and Appendix C, rather than relying only on its abstract. Its complete proof and formal verification are not independently certified here. The relative-matrix factorization used below already appears in its Appendix C; our purpose is to extract an explicit degree-sensitive criterion and test how sharp its obstruction is.

92.1 A definite affine pencil has only real zeros

Lemma 92.1 (Spectrum and exact degree)**.** *Let A, B be rational symmetric $h \times h$ matrices, and suppose $G = A + \theta B$ is positive definite for a real number θ. Put $P(X) = c\det(A + XB)$, where $c \neq 0$ is rational. If $C = G^{-1/2}BG^{-1/2}$ has nonzero eigenvalues $\lambda_1, \dots, \lambda_r$, then*

$$P(X) = P(\theta) \prod_{j=1}^{r} (1 + (X - \theta)\lambda_j).$$

Consequently $\deg P = \operatorname{rank} B = r$*, every zero of P is real, and the number of zeros below or above θ, counted with multiplicity, is the positive or negative inertia of B, respectively. Moreover*

$$\log|P(\theta + i)| = \log|P(\theta)| + \frac{1}{2}\sum_{j=1}^{r} \log(1 + \lambda_j^2). \tag{92.1}$$

Proof. Factor $A + XB = G^{1/2}(I + (X - \theta)C)G^{1/2}$ and apply the spectral theorem to the real symmetric matrix C. Congruence preserves inertia. The zero corresponding to λ_j is $\theta - 1/\lambda_j$. Taking absolute values at $X = \theta + i$ proves (92.1). □

In particular the off-axis quantity is an entire spectral budget, not just the largest eigenvalue. One may also compute it through positive matrices:

$$|\det(G + iB)|^2 = \det G \, \det(G + BG^{-1}B).$$

Both matrices on the right are positive definite. This identity follows from $\det(I + C^2)$ and does not require a choice of eigenvectors.

Theorem 92.2 (A single off-axis test for algebraic degree)**.** *Let θ be real. Suppose $P_n \in \mathbb{Z}[X]$ have only real zeros, $P_n(\theta) \neq 0$, and $\deg P_n = o(T_n)$ for some $T_n \to \infty$. Assume*

$$\limsup_n T_n^{-1} \log |P_n(\theta)| \leq -\tau < 0, \qquad \limsup_n T_n^{-1} \log |P_n(\theta + i)| \leq \eta.$$

Then θ cannot be algebraic of degree d whenever

$$\tau > (d-1)\eta. \tag{92.2}$$

In particular, $\eta \leq 0$ would imply transcendence.

Proof. For a real u and arbitrary complex z, Cauchy–Schwarz gives

$$|z - \theta - u| \leq \sqrt{1 + |z-\theta|^2}\sqrt{1+u^2}.$$

Factor a real-rooted polynomial P into its linear factors and put $u = r - \theta$ for each root r. Thus, if $h = \deg P$,

$$|P(z)| \leq (1 + |z-\theta|^2)^{h/2} |P(\theta + i)|. \tag{92.3}$$

Suppose now that θ has degree d with primitive minimal polynomial f of leading coefficient a, and conjugates $\theta = \theta_1, \ldots, \theta_d$. Since $P_n(\theta) \neq 0$, f and P_n are coprime. Their resultant is a nonzero integer:

$$1 \leq \left| a^{\deg P_n} \prod_{j=1}^{d} P_n(\theta_j) \right|.$$

Apply (92.3) to the $d-1$ other factors. The logarithm of the right side is at most

$$\log |P_n(\theta)| + (d-1) \log |P_n(\theta + i)| + O_\theta(\deg P_n).$$

After division by T_n, condition (92.2) makes its upper limit negative, a contradiction. □

This is a specialized resultant criterion, not a replacement for classical transcendence theory. Its practical advantage for a positive pencil is that the missing arithmetic information is reduced to one explicit off-axis determinant, rather than separate bounds at unknown conjugates.

92.2 The barrier is sharp even for positive Hankel pencils

Theorem 92.3 (An exact quadratic counterexample)**.** *Put $\lambda = 1 + \sqrt{2}$ and define positive integers p_m, q_m by*

$$p_m + q_m \sqrt{2} = \lambda^{2m+1} \qquad (m \geq 0).$$

For $n \geq 1$, the primitive squarefree integer polynomial

$$P_n(X) = \prod_{m=n}^{2n-1} (q_m X - p_m)$$

has degree n, all its zeros are real and less than $\sqrt{2}$, and

$$P_n(\sqrt{2}) = \lambda^{-3n^2} > 0, \tag{92.4}$$

$$\log |P_n(\sqrt{2} + i)| = 3n^2 \log \lambda + O(n), \tag{92.5}$$

$$N_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(P_n(\sqrt{2})) = (-1)^n. \tag{92.6}$$

Up to a nonzero rational scalar, P_n is the determinant of a rational affine Hankel pencil which is positive definite at $\sqrt{2}$. Thus positivity, full degree, simple real zeros, and quadratic exponential decay do not by themselves exclude degree two.

Proof. Conjugation gives $p_m - q_m\sqrt{2} = -\lambda^{-(2m+1)}$, so $p_m^2 - 2q_m^2 = -1$, $\gcd(p_m, q_m) = 1$, and $q_m\sqrt{2} - p_m = \lambda^{-(2m+1)}$. The roots p_m/q_m increase strictly to $\sqrt{2}$. Each factor is primitive, so Gauss's lemma makes their product primitive. Summing the exponents gives

$$\sum_{m=n}^{2n-1} (2m+1) = 3n^2,$$

which proves (92.4); the norm of each factor is -1. Also $q_m = \lambda^{2m+1}(1 + O(\lambda^{-4m-2}))/(2\sqrt{2})$, and

$$|q_m(\sqrt{2} + i) - p_m| = \sqrt{q_m^2 + \lambda^{-4m-2}}.$$

Multiplication proves (92.5).

Choose distinct rational nodes $x_m = m$ and form

$$H_n(X)_{ab} = \sum_{m=n}^{2n-1} (q_m X - p_m) x_m^{a+b}, \qquad 0 \le a, b < n.$$

Writing $V_{am} = x_m^a$ gives $H_n(X) = V \operatorname{diag}(q_m X - p_m) V^T$. The Vandermonde matrix is invertible. Thus $H_n(\sqrt{2})$ is positive definite and $\det H_n(X) = (\det V)^2 P_n(X)$. Primitive normalization removes the rational scalar. Finally (92.4) and (92.5) give $\tau = \eta = 3\log\lambda$ on the scale $T_n = n^2$, exactly the boundary of (92.2) for $d = 2$. □

Proposition 92.4 (Amplification does not improve this ratio)**.** *Suppose finitely many primitive real-rooted polynomial families $P_{j,n}$ have limits*

$$-T_n^{-1} \log|P_{j,n}(\theta)| \to \tau_j > 0, \qquad T_n^{-1} \log|P_{j,n}(\theta + i)| \to \eta_j > 0.$$

For fixed nonnegative integers a_j, not all zero, their product $R_n = \prod_j P_{j,n}^{a_j}$ is primitive, and its ratio of these two rates is

$$\frac{\sum_j a_j \tau_j}{\sum_j a_j \eta_j} \le \max_j \frac{\tau_j}{\eta_j}.$$

At each fixed index, the same weighted-average identity holds for arbitrary nonnegative powers. Consequently powers, direct sums of determinant pencils, and Kronecker products do not improve this ratio merely by amplification.

Proof. Gauss's lemma proves primitivity of the product; its two logarithms add exactly. The ratio is a weighted average with positive weights $a_j\eta_j$. A direct sum multiplies determinants. The determinant of a Kronecker product is a product of powers of the original determinants, so the same exact observation applies. □

92.3 What the published quintic-zeta bounds currently certify

For the normalization in Fauzan's Appendix C, put $\xi = \zeta(5)$. The cited bounds have degree $37K/40$, value at most $\exp(-79K^2/1600)$, and relative matrix norm at most $\exp(6933K/500)$ for all sufficiently large admissible K [359, Appendix C]. Lemma 92.1 therefore gives

$$\tau = \frac{79}{1600}, \qquad \eta = \frac{37}{40}\frac{6933}{500} - \frac{79}{1600} = \frac{511067}{40000}, \qquad \frac{\tau}{\eta} = \frac{1975}{511067} < 0.003865.$$

These particular estimates certify degree one exclusion, but not degree two exclusion. This is not a lower bound on the true off-axis growth.

There is also a concrete content target. If the displayed integer polynomial factors as $Q_K = c_K P_K$ with P_K primitive and

$$\liminf K^{-2} \log c_K \geq \chi,$$

then the same estimates give $\tau_P \geq \tau + \chi$ and $\eta_P \leq \eta - \chi$. Thus a sufficient additional content bound for degree-d exclusion is

$$\chi > \frac{(d-1)\eta - \tau}{d}. \tag{92.7}$$

For degree two this asks for $\chi > 127273/20000 = 6.36365$. No such content estimate is proved here. Passing to the primitive polynomial is useful only when its quantitative common divisor is actually controlled; a formal renormalization does not establish (92.7).

92.4 A full-degree Catalan pencil with no auxiliary period

Use the positive measure

$$d\nu_2(t) = 4\pi y^2 \sum_{\ell \geq 1} \ell e^{-2\pi \ell y}\, dy, \qquad t = y^2,$$

whose moments are $\mu_2(t^e) = (-1)^e B_{2e+2}$ and whose Stieltjes transform satisfies

$$\mu_2((t+c^2)^{-1}) = c\zeta(2,c) - 1 - \frac{1}{2c}.$$

These are classical Hurwitz moment identities [361]. Let $G = \sum_{k \geq 0} (-1)^k/(2k+1)^2$, and set, for $j \geq 0$,

$$a_j = j + \frac{1}{4}, \qquad b_j = j + \frac{3}{4}, \qquad K_j(t) = \frac{1}{a_j(t+a_j^2)} - \frac{1}{b_j(t+b_j^2)}.$$

Notice that $K_j(t) > 0$ on $[0,\infty)$, since

$$K_j(t) = \frac{(b_j - a_j)(t + a_j^2 + a_j b_j + b_j^2)}{a_j b_j (t+a_j^2)(t+b_j^2)}.$$

Theorem 92.5 (Paired interpolation cancels the nuisance period)**.** *For every integer $m \geq 1$, let $f_0, \ldots, f_{m-1} \in \mathbb{Q}[t]$ be the unique polynomials of degree less than $2m$ satisfying*

$$f_i(-a_j^2) = f_i(-b_j^2) = \delta_{ij} \qquad (0 \leq i, j < m).$$

Put $W_m = \sum_{j=0}^{m-1} K_j$. There is an explicitly computable rational symmetric matrix A_m such that

$$\left(\int W_m(t) f_i(t) f_j(t) d\nu_2(t) \right)_{i,j<m} = A_m + 16 G I_m. \tag{92.8}$$

The matrix on the left is positive definite. Hence

$$D_m(X) = \det(A_m + 16 X I_m)$$

has exact degree m, only real zeros, and every zero is strictly less than G. The coefficient of G has full rank m even though the coefficient of the auxiliary π^2 vanishes identically.

Proof. Interpolation at the $2m$ distinct rational nodes gives the f_i uniquely. Explicitly f_i is the sum of the two cardinal Lagrange polynomials for its paired nodes. The classical identities

$$\zeta(2, 1/4) = \pi^2 + 8G, \qquad \zeta(2, 3/4) = \pi^2 - 8G$$

follow by separating the odd reciprocal squares into their two classes modulo four. The Hurwitz shift relation therefore gives

$$\mu_2((t+a_j^2)^{-1}) = a_j\pi^2 + 8a_jG + r_j^+, \quad \mu_2((t+b_j^2)^{-1}) = b_j\pi^2 - 8b_jG + r_j^-,$$

with explicit rationals

$$r_j^+ = -16a_j\sum_{k=0}^{j-1}(4k+1)^{-2} - 1 - (2a_j)^{-1}, \quad r_j^- = -16b_j\sum_{k=0}^{j-1}(4k+3)^{-2} - 1 - (2b_j)^{-1}.$$

For a polynomial F, division by $t+c^2$ leaves remainder $F(-c^2)$; the quotient has rational moment under μ_2. Thus the coefficients of π^2 and G in $\mu_2(K_jf_if_k)$ are respectively

$$f_i(-a_j^2)f_k(-a_j^2) - f_i(-b_j^2)f_k(-b_j^2) = 0,$$

and

$$8(f_i(-a_j^2)f_k(-a_j^2) + f_i(-b_j^2)f_k(-b_j^2)) = 16\delta_{ij}\delta_{kj}.$$

Summation proves (92.8), including rationality and computability of A_m. The f_i are linearly independent by evaluation at the nodes. Since $W_m > 0$ and ν_2 has positive density, their Gram matrix is positive definite. Lemma 92.1 with $B = 16I_m$ proves every assertion about D_m. □

For $m = 1$, the polynomial is $16X - 88/9$, whose primitive multiple is $18X - 11$. For larger m the construction is a rationally computable algebraic lower certificate. We have not proved a useful asymptotic smallness estimate for its primitive value at G, nor the off-axis estimate needed in Theorem 92.2. The construction closes the auxiliary-period and nonvanishing problems for this family, not the transcendence problem.

92.5 Six individual target outcomes

1. $\zeta(3)$ transcendence: the positive-pencil test applies, but the preceding orthogonal-polynomial cancellation does not supply a sufficient off-axis budget. No new algebraic degree is excluded.

2. $\zeta(5)$ transcendence: the actual published coefficient and relative-matrix bounds were inspected. Their exact degree ratio and the sufficient new content target are displayed above. No higher degree exclusion follows from those estimates.

3. $\zeta(7)$ irrationality: positive odd-zeta moment pencils have real-rooted determinant polynomials, but no sufficient primitive-value decay estimate is established.

4. $\zeta(9)$ irrationality: the same spectral criterion applies; positivity alone cannot replace the missing local denominator estimate.

5. $\zeta(11)$ irrationality: likewise, no individual irrationality conclusion follows. An existing disjunctive statement about several odd zeta values is not an individual proof.

6. Catalan transcendence: Theorem 92.5 removes the auxiliary period without loss of determinant degree. Primitive height and off-axis growth remain to be controlled. The reported external irrationality result is not counted as our result.

Historical priority is not claimed for the specialized resultant criterion, spectral facts, or interpolation formulation. The exact Pell example demonstrates why the identified arithmetic barrier is substantive even under all the displayed positivity hypotheses.

93 An exact parameter law for primitive Gompertz cancellation

The preceding index-sparsity theorem leaves open a geometrically sparse sequence of unusually large common divisors for a fixed Euler–Gompertz constant. We now change the variable in the local arithmetic analysis. Roots as a function of the integer index need not have proved simple derivatives. In contrast, roots as a function of the Laguerre parameter are always simple. This distinction gives an unconditional distribution law for the complete primitive content. The resulting statement is about parameter averages; its limitation at a prescribed parameter, especially $A = 1$, is stated explicitly.

For positive integers A, n, use the classical family

$$\delta_A = \int_0^\infty \frac{e^{-t}}{1+At}\,dt, \qquad Q_n(A) = \sum_{j=0}^{n} j!\binom{n}{j}^2 A^j = n!A^n L_n(-1/A).$$

Let $P_0 = 0$, $P_1 = 1$, and give both P_n, Q_n the recurrence

$$U_n = (1 + (2n-1)A)U_{n-1} - (n-1)^2A^2U_{n-2}.$$

The earlier exact-content calculation gives

$$g_n(A) := \gcd(P_n(A), Q_n(A)) = \gcd(Q_n(A), D_n), \qquad D_n = ((n-1)!)^2. \tag{93.1}$$

This follows from adjacent-denominator coprimality and the determinant $P_nQ_{n-1} - P_{n-1}Q_n = A^{2n-2}((n-1)!)^2$. In particular $g_n(A)$ is a periodic function of A modulo D_n. The Laguerre approximations themselves are classical [338, 339].

93.1 Every parameter root is simple

Lemma 93.1. *Fix $n \geq 1$ and a prime p, and put $r = n \bmod p$, $0 \leq r < p$. As polynomials over $\mathbb{F}_p$,*

$$Q_n(A) = Q_r(A).$$

The polynomial $Q_r(A)$ has degree r, constant term one, and has no repeated root in an algebraic closure of $\mathbb{F}_p$. If

$$\rho_p(n) = \#\{a \in \mathbb{F}_p : Q_n(a) = 0\},$$

then $0 \leq \rho_p(n) \leq r \leq p-1$, and for every $k \geq 1$ there are exactly $\rho_p(n)$ roots of $Q_n(A)$ modulo p^k.

Proof. The terms with $j \geq p$ vanish modulo p because of $j!$. For $j < p$, Lucas's congruence gives $\binom{n}{j} \equiv \binom{r}{j}$, which proves the polynomial identity. Since $r!$ is nonzero modulo p, its degree is exactly r.

For $r = 0$ there are no roots. For $1 \leq r < p$, the identity $Q_r(A) = r!A^rL_r(-1/A)$ reduces simplicity to that of $L_r(x)$ at nonzero points. The classical Laguerre differential equation [362, Table 18.8.1] is

$$xL_r'' + (1-x)L_r' + rL_r = 0;$$

it remains valid over $\mathbb{F}_p$, since the denominators in L_r are invertible. Also $L_r(0) = 1$. If a nonzero root had multiplicity $m \geq 2$, then $2 \leq m \leq r < p$. The term of degree $m-2$ in the differential equation would have nonzero coefficient $x_0m(m-1)c$, where c is the leading local coefficient. The other terms have higher vanishing order, a contradiction. The case $p = 2$ is automatic since $r \leq 1$.

Thus every root modulo p has a nonzero parameter derivative. Taylor expansion shows that a root a modulo p^k has exactly one lift $a + tp^k$ modulo p^{k+1}: the choice of t mod p solves a linear equation with nonzero coefficient $Q_n'(a)$. Induction proves the final assertion. □

For completeness, the same reduction proves the adjacent-denominator coprimality used in (93.1). If a prime p divides both $Q_n(A)$ and $Q_{n-1}(A)$, then $p \nmid A$ because each $Q_k(A)$ is congruent to one modulo A. Put $r = n \bmod p$. The case $r = 0$ is impossible since $Q_n \equiv 1 \pmod p$. Otherwise the common root is a root of Q_r and Q_{r-1} modulo p. If $r = 1$, this contradicts $Q_0 = 1$. If $r \geq 2$, the recurrence, whose coefficient $(r-1)^2A^2$ is nonzero modulo p, forces the root to be a root of Q_{r-2}. Descent again reaches $Q_0 = 1$. Thus $\gcd(Q_n(A), Q_{n-1}(A)) = 1$. The displayed adjacent determinant, together with $\gcd(Q_n(A), A) = 1$, now gives exactly $\gcd(P_n(A), Q_n(A)) = \gcd(Q_n(A), D_n)$.

The squarefreeness input is an elementary consequence of the classical differential equation, rather than a new theorem about Laguerre zeros. Its use here eliminates the exceptional-root hypothesis that remained in the index-variable analysis.

93.2 Independent truncated geometric valuations

For $p \mid D_n$, write $e_p = 2v_p((n-1)!)$. Let A be uniform on any complete residue system modulo D_n, and set $V_p = v_p(g_n(A))$.

Theorem 93.2 (The exact parameter distribution)**.** *The random variables V_p, for $p \mid D_n$, are independent, and their distributions are determined by*

$$\mathbb{P}(V_p \geq k) = \frac{\rho_p(n)}{p^k} \qquad (1 \leq k \leq e_p). \tag{93.2}$$

In particular,

$$\mathbb{E} \log g_n(A) = \sum_{p \mid D_n} \frac{\rho_p(n) \log p}{p-1}(1 - p^{-e_p}) \leq \vartheta(n-1), \tag{93.3}$$

$$\mathbb{E}[g_n(A)^s] = \prod_{p \mid D_n} \left[1 + \frac{\rho_p(n)(p^s-1)}{p} \sum_{j=0}^{e_p-1} p^{(s-1)j}\right] \qquad (s > 0), \tag{93.4}$$

where $\vartheta(y) = \sum_{p \leq y} \log p$.

Proof. The Chinese remainder theorem makes the residues of A modulo the distinct p^{e_p} independent and uniform. Lemma 93.1 counts exactly $\rho_p(n)p^{e_p-k}$ roots modulo p^{e_p} of the congruence $Q_n(A) \equiv 0 \pmod{p^k}$, proving (93.2). Sum the tail probabilities to get $\mathbb{E}V_p$, and use

$$p^{sV_p} = 1 + \sum_{k=1}^{e_p} (p^{sk} - p^{s(k-1)}) \mathbf{1}_{\{V_p \geq k\}}$$

to obtain (93.4). The inequality in (93.3) uses $\rho_p(n) \leq p - 1$. □

Corollary 93.3 (Exponential moment bound)**.** *For every fixed $0 < s < 1$,*

$$\log \mathbb{E}[g_n(A)^s] \leq s\vartheta(n-1) + O_s(n^s) = sn + o_s(n). \tag{93.5}$$

Consequently, for every real T,

$$\mathbb{P}\{\log g_n(A) \geq \vartheta(n-1) + T\} \leq \exp(-sT + O_s(n^s)). \tag{93.6}$$

Proof. Extend the geometric sum in (93.4) to infinity and use $\rho_p(n) \leq p - 1$. The resulting factor is at most

$$1 + \frac{(p-1)(p^s-1)}{p - p^s} \leq \frac{p^s}{1 - p^{s-1}}.$$

After taking logarithms, the additional error is at most

$$\sum_{p\le n-1} -\log(1-p^{s-1}) \le C_s \sum_{j=2}^{n-1} j^{s-1} = O_s(n^s).$$

The prime number theorem gives $\vartheta(n-1) = n + o(n)$. Markov's inequality proves (93.6). □

93.3 Uniform exclusion for most finite parameter choices

Theorem 93.4 (A finite simultaneous parameter bound)**.** *For every $\eta > 0$, there are positive constants C, c and M_0 such that, for all $N \ge M \ge M_0$, the fraction of residues $A \bmod D_N$ for which at least one $n \in [M, N]$ satisfies $\log g_n(A) > (1+\eta)n$ is at most Ce^{-cM}. The same proportion among $1 \le A \le H$ is at most*

$$Ce^{-cM} + \frac{D_N}{H}. \tag{93.7}$$

Proof. Since $D_n \mid D_N$ for $n \le N$, the marginal distribution at every such n is the one in Theorem 93.2. Fix $0 < s < 1$. Corollary 93.3 gives

$$\mathbb{P}\{\log g_n(A) > (1+\eta)n\} \le e^{-s\eta n/2}$$

for sufficiently large n. Sum this geometric tail over $M \le n \le N$; independence between different indices is not needed. Finally partition $[1, H]$ into complete blocks of length D_N and a remaining block of length less than D_N. □

This finite statement does not require a probabilistic interpretation of the constants themselves. It is an exact counting assertion about ordinary positive integer parameters and all approximation orders in a specified finite range.

We can also quantify the parameter rarity of bounded primitive linear forms, uniformly over positive integer A. Write

$$R_n(A) = Q_n(A)\delta_A - P_n(A), \qquad \ell_n(A) = R_n(A)/g_n(A).$$

The positive remainder identity gives

$$R_n(A) = n!A^n \int_0^\infty \frac{e^{-t}}{1+At}\left(\frac{At}{1+At}\right)^n dt.$$

Restricting to $\sqrt{n} \le t \le \sqrt{n}+1$, using $A \ge 1$, and using $\log(1+y) \le y$, we obtain the uniform lower bound

$$R_n(A) \ge \frac{n!e^{-2\sqrt{n}-1}}{\sqrt{n}+2}. \tag{93.8}$$

Thus for fixed $B > 0$ and $0 < s < 1$, the fraction of $A \in \{1, \dots, D_N\}$ for which $\ell_n(A) \le B$ for at least one $M \le n \le N$ is at most

$$\exp(-sM\log M + O_{s,B}(M)) \tag{93.9}$$

as $M \to \infty$, uniformly in $N \ge M$. Indeed, boundedness forces $\log g_n(A) \ge n\log n - n - O_B(\sqrt{n} + \log n)$. Apply (93.5) and Markov's inequality at each n, then sum the rapidly decreasing bounds. The corresponding proportion in $1 \le A \le H$ has the additional error D_N/H. Here periodic-block counting is applied to the necessary gcd-threshold superset; the real-valued forms $\ell_n(A)$ themselves are not asserted to be periodic in A.

93.4 The almost-sure statement and its fixed-parameter limitation

Equip the profinite integers $\widehat{\mathbb{Z}}$ with Haar probability measure, and define $g_n(A) = \gcd(Q_n(A), D_n)$ through the residue of the polynomial $Q_n(A)$ modulo D_n. For every $\eta > 0$, the exceptional probabilities $\mathbb{P}\{\log g_n(A) > (1+\eta)n\}$ are summable by Corollary 93.3. The first Borel–Cantelli lemma therefore gives, after intersecting the full-measure sets for $\eta = 1/j$, $j = 1, 2, \ldots$,

$$\limsup_{n\to\infty} \frac{\log g_n(A)}{n} \le 1 \quad \text{for Haar-almost every } A \in \widehat{\mathbb{Z}}. \tag{93.10}$$

This is an all-large-orders conclusion for almost every profinite parameter, stronger than an index-density statement for a fixed parameter.

It cannot be specialized to $A = 1$. Every singleton, and hence the whole embedded set of ordinary integers, has Haar measure zero in $\widehat{\mathbb{Z}}$. A full-measure theorem is logically compatible with all prescribed integer parameters belonging to its exceptional set. Accordingly (93.10) does not close the irrationality of δ, any specified δ_A, or Euler's constant γ. The new arithmetic result is the exact parameter law and its quantitative finite simultaneous consequences; the fixed-fibre arithmetic estimate remains an identifiable missing step.

93.5 Why curvature and a transcendental slope cannot finish the Euler attack

The prior Euler–Taylor analysis gave an exact crossover curve with positive curvature asymptotic to a/x^2 and a transcendental leading slope. It is tempting to infer that exponentially accurate integer crossovers must eventually stop. The following countermodel shows that those two pieces of information alone cannot imply this.

Proposition 93.5 (Exponentially close lattice points despite curvature)**.** *Fix $a > 0$, $b \in \mathbb{R}$, an arbitrary nonempty open interval $I \subset \mathbb{R}$, and any function $\Psi : \mathbb{N} \to (0, \infty)$. There are uncountably many transcendental $\alpha \in I$ for which*

$$|\alpha n - a \log n - b - 2m| < e^{-\Psi(n)} \tag{93.11}$$

holds for infinitely many pairs $(n, m) \in \mathbb{N}\times\mathbb{Z}$. Every corresponding curve $f_\alpha(x) = \alpha x - a\log x - b$ has the exact strictly positive curvature $f''_\alpha(x) = a/x^2$.

Proof. For each fixed integer n, the centers

$$\frac{2m + a\log n + b}{n} \qquad (m \in \mathbb{Z})$$

form a grid of spacing $2/n$. Inside any nondegenerate compact interval, a sufficiently large n therefore gives at least two well-separated centers in its interior. Around each chosen center take a closed interval of radius

$$r_n = \frac{e^{-\Psi(n)}}{4n}.$$

Choosing n large enough makes these intervals disjoint, contained in the parent interval, and of arbitrarily small diameter. Every α in either child interval satisfies (93.11) at that n.

Start with a compact interval inside I. At each level carry out this construction for every current interval, choosing all new indices larger than every previous index and all diameters below 2^{-j} at level j. The resulting binary Cantor construction has uncountably many distinct limit points. Every limit point lies in infinitely many selected intervals and hence satisfies (93.11) infinitely often. Removing the countable set of algebraic numbers leaves uncountably many transcendental limit points. The derivative formula is immediate. □

This proposition does not replace the actual slope κ by a counterexample or assert that Euler's exact crossover has infinitely many exceptional indices. It establishes a precise limitation of an attempted inference: positive x^{-2} curvature, transcendence of the slope, and even the exact leading linear–logarithmic shape are compatible with arbitrarily accurate, infinitely sparse lattice contacts. A proof excluding all Euler crossovers must use additional arithmetic information about the particular slope and the exact integral defining the crossover, or a different approximation family.

94 Exactly which prime-denominator gamma monomials enter the elliptic CM sector

The preceding manuscript solved an unrestricted beta-word optimization problem. We now ask a different question: can one isolate the five individual fifth and seventh gamma values by adjoining arbitrarily many elliptic CM periods and using their Chowla–Selberg formulas? We determine the entire subspace accessible to this multiplicative strategy. The result permits auxiliary gamma values of arbitrary rational denominators, all standard gamma identities, and rational powers. The obstruction is a Dirichlet-character component, rather than beta-word length.

94.1 The precise identity calculus

Let $\mathcal{D}$ be the rational vector space generated by $[x]$ for $x \in \mathbb{Q}/\mathbb{Z}$, modulo

$$[0] = [1/2] = 0, \qquad [-x] = -[x], \qquad \sum_{k=0}^{m-1} [x + k/m] = [mx] \quad (m \geq 1). \tag{94.1}$$

The interpretation is the class of $\Gamma(x)$ modulo nonzero algebraic factors and rational powers of π. Recurrence, reflection and Gauss multiplication give these relations. Taking the quotient as a rational vector space allows rational powers of the standard identities. All asserted identities in this calculus have finite proofs.

For a negative fundamental discriminant D, put $d = |D|$ and let χ_D be its primitive quadratic Dirichlet character. Define

$$c_D = \sum_{a=1}^{d-1} \chi_D(a)[a/d], \qquad \mathcal{E} = \operatorname{span}_{\mathbb{Q}}\{c_D : D < 0 \text{ fundamental}\} \subset \mathcal{D}. \tag{94.2}$$

Only finite linear combinations occur in this span. This is precisely the multiplicative sector generated by the standard Chowla–Selberg expressions for elliptic CM periods, after ignoring algebraic factors and rational powers of π. Indeed, if h_D and w_D are the class number and number of roots of unity, a nonzero period of a CM elliptic curve in the corresponding isogeny class is, up to a nonzero algebraic factor,

$$\sqrt{\pi}\left(\prod_{a=1}^{d-1} \Gamma(a/d)^{\chi_D(a)}\right)^{w_D/(4h_D)}.$$

We use this classical formula only to identify the sector, not as a new transcendence theorem; see Gross [363]. Periods of nonmaximal orders introduce no additional sector because CM elliptic curves with the same imaginary quadratic field are isogenous over $\overline{\mathbb{Q}}$.

For an odd prime p, put $h = (p-1)/2$ and

$$W_p = \operatorname{span}_{\mathbb{Q}}\{[1/p], \ldots, [h/p]\}.$$

The theorem below also proves these h symbols are linearly independent in the full standard calculus.

Theorem 94.1 (Complete elliptic-CM recognition at prime denominators)**.** *Let $\mathcal{P}$ be any finite set of odd primes. An element*

$$v = \sum_{p\in\mathcal{P}} \sum_{a=1}^{(p-1)/2} e_{p,a}[a/p], \qquad e_{p,a} \in \mathbb{Q},$$

belongs to $\mathcal{E}$ if and only if the following conditions hold separately for every $p \in \mathcal{P}$:

1. *if $p \equiv 1 \pmod 4$, then $e_{p,a} = 0$ for every a;*
2. *if $p \equiv 3 \pmod 4$, there is $r_p \in \mathbb{Q}$ such that $e_{p,a} = r_p \left(\frac{a}{p}\right)$ for every a.*

Consequently

$$\dim_{\mathbb{Q}} \left(\left(\sum_{p\in\mathcal{P}} W_p \right) \cap \mathcal{E} \right) = \#\{p \in \mathcal{P} : p \equiv 3 \pmod 4\}.$$

All auxiliary rational denominators and all relations (94.1) are allowed in this assertion.

94.2 Finite-proof character certificates

Define the centered sawtooth

$$s(x) = \begin{cases} 1 - 2\{x\}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}. \end{cases}$$

Its endpoint convention is essential. Direct summation gives

$$s(-x) = -s(x), \quad s(1/2) = 0, \quad \sum_{k=0}^{m-1} s(x + k/m) = s(mx). \tag{94.3}$$

For completeness, if $u = \{mx\} \neq 0$, the fractional parts on the left are $(u + j)/m$, $0 \le j < m$, in some order, and their sawtooth sum is $1 - 2u$. If $u = 0$, their fractional parts are $0, 1/m, \ldots, (m-1)/m$ and the sum is zero.

Given any finite proof in $\mathcal{D}$, let N contain every denominator appearing in it. For $t \in (\mathbb{Z}/N\mathbb{Z})^\times$, the substitution

$$[x] \longmapsto s(tx)$$

respects every relation in that proof. For multiplication by m, one has $m \mid N$, and t permutes the shifts k/m modulo one; thus (94.3) applies. This is a finite-proof certificate, not an assertion that one fixed t acts on every denominator globally.

For the CM symbol it gives

$$\sum_{a=1}^{d-1} \chi_D(a) s(ta/d) = K_D \chi_D(t), \qquad K_D = \sum_{a=1}^{d-1} \chi_D(a) s(a/d) = 2L(0, \chi_D) \neq 0. \tag{94.4}$$

The first equality follows by the substitution $a \mapsto ta$ modulo d. The formula for K_D follows from $L(0,\chi) = -d^{-1} \sum_{a=1}^{d-1} a\chi(a)$. For an odd primitive character, $L(0,\chi)$ is nonzero by the Dirichlet functional equation and $L(1,\overline{\chi}) \neq 0$. These are classical Dirichlet L-function facts; see [364]. Thus every elliptic CM generator has a single quadratic-character profile, independently of how large a common denominator is chosen.

Lemma 94.2 (No invisible prime-denominator direction)**.** *For an odd prime p, the map*

$$(e_1,\dots,e_h)\longmapsto \Phi_e(t)=\sum_{a=1}^{h} e_a s(ta/p),\quad t\in(\mathbb{Z}/p\mathbb{Z})^\times,$$

is an isomorphism from $\mathbb{C}^h$ onto the space of odd functions on $(\mathbb{Z}/p\mathbb{Z})^\times$. It preserves the character decomposition: the inverse image of the line generated by a quadratic odd character is the same character on the half-residue representatives.

Proof. Extend the coefficients oddly to the nonzero residues: $E(a)=e_a$ for $1\le a\le h$, and $E(-a)=-e_a$. Then

$$\Phi_e(t)=\frac{1}{2}\sum_{a\in(\mathbb{Z}/p\mathbb{Z})^\times} E(a)s(ta/p).$$

For an odd Dirichlet character χ modulo p, its Fourier coefficient equals

$$\sum_t \Phi_e(t)\overline{\chi(t)}=\frac{1}{2}\left(\sum_a E(a)\chi(a)\right)\left(\sum_t s(t/p)\overline{\chi(t)}\right).$$

The second factor is $2L(0,\overline{\chi})\neq 0$: every odd character modulo the prime p is primitive. Fourier inversion therefore gives the claimed isomorphism on the odd subspace. For a quadratic character inversion does not change its character line, proving the final assertion. □

Proof of Theorem 94.1. Suppose $v=\sum_D r_D c_D$ in $\mathcal{D}$, and choose N containing all prime denominators, all $|D|$ on the right, and all auxiliary denominators in a finite proof of this identity. Its profiles satisfy

$$\sum_{p\in\mathcal{P}} \Phi_{e_p}(t)=\sum_D r_D K_D\chi_D(t)\qquad (t\in(\mathbb{Z}/N\mathbb{Z})^\times). \tag{94.5}$$

All characters are inflated to this common group. The odd characters on the left have primitive conductor exactly p; characters belonging to distinct primes have distinct conductors. The characters on the right are primitive odd quadratic characters of conductor $|D|$. Distinct primitive characters remain distinct when inflated to a common multiple of their conductors, and characters of a finite abelian group are linearly independent.

There is no odd quadratic character of conductor p when $p\equiv 1 \pmod 4$. When $p\equiv 3 \pmod 4$, the unique one is $\chi_{-p}(a)=(a/p)$. Equation (94.5) and Lemma 94.2 therefore force precisely the coefficient conditions in the theorem. This argument also shows that a linear combination of the prime-denominator symbols can vanish in $\mathcal{D}$ only when every coefficient vanishes.

Conversely, for $p\equiv 3 \pmod 4$, reflection and oddness give

$$c_{-p}=\sum_{a=1}^{p-1}\left(\frac{a}{p}\right)[a/p]=2\sum_{a=1}^{h}\left(\frac{a}{p}\right)[a/p].$$

Each permitted direction thus belongs to $\mathcal{E}$ using only reflection. Summing these identities proves sufficiency and the dimension formula. □

94.3 The five targets and the exact remaining arithmetic gap

For the particular joint attack, write

$$X_5=\Gamma(1/5),\quad Y_5=\Gamma(2/5),\qquad X_7=\Gamma(1/7),\quad Y_7=\Gamma(2/7),\quad Z_7=\Gamma(3/7).$$

The theorem gives the complete criterion

$$\begin{aligned} X_5^a Y_5^b X_7^A Y_7^B Z_7^C \text{ is standard-reducible to elliptic CM products} \\ \iff \quad a = b = 0, \quad A = B = -C. \end{aligned} \tag{94.6}$$

for rational exponents, with algebraic factors and rational powers of π ignored. Thus mixing the fifth and seventh denominators does not cancel their nonquadratic character components. Nor can adding CM elliptic curves of arbitrarily large discriminant cancel them.

The surviving seventh direction is

$$B_7 = X_7 Y_7 / Z_7 = B(1/7, 2/7).$$

Its actual arithmetic is classical: reflection gives

$$\mathrm{CS}(-7) = \frac{\sin(\pi/7)\sin(2\pi/7)}{\sin(3\pi/7)} \frac{B_7^2}{\pi},$$

and the Chowla–Selberg–Nesterenko input implies algebraic independence of $\pi, B_7, e^{\pi\sqrt{7}}$; see Cohen–Zudilin [365]. This is the same classical CM direction used in the preceding manuscript. The new classification proves that it is the only one available in the entire joint prime-denominator span by the stated standard-calculus strategy.

More generally, for $p \equiv 3 \pmod 4$ define

$$P_p = \prod_{a=1}^{(p-1)/2} \Gamma(a/p)^{\left(\frac{a}{p}\right)}.$$

Reflection identifies a rational power of P_p, up to an algebraic factor and a rational power of π, with $\mathrm{CS}(-p)$. Therefore π and P_p are algebraically independent by the same classical CM theorem. This last transcendence assertion is a consequence of known results, not a newly settled conjecture.

The restriction to the standard calculus cannot be removed for arbitrary gamma monomials with the results used here. Rohrlich's conjecture would make it legitimate to pass from a numerical multiplicative algebraic relation to such a formal relation, and would then promote (94.6) to the corresponding assertion about actual numerical CM-product identities. We do not assume that conjecture. Wüstholz's theorem controls algebraic *linear* relations between periods; applying it to unrestricted products of beta periods is not justified. The tensor products that arise in those gamma-isolating words are exactly where the attempted value transfer still stops.

For orientation, a known result of Vasil'ev and Grinspan already says that at least two of $\pi, \Gamma(1/5), \Gamma(2/5)$ are algebraically independent; see the discussion and original references in Waldschmidt [366]. Our formal classification neither improves that transcendence-degree bound nor selects which individual gamma value is transcendental. The five individual irrationality or transcendence targets remain unclosed in this investigation.

95 Sixteen named targets and the remaining arithmetic steps

The entries below state outcomes of this manuscript. An unresolved entry means that our arguments do not prove its target conclusion. It does not erase external results or establish a complete survey of the literature. In particular, the reported external irrationality results for G and $\zeta(5)$ are distinct from our transcendence targets. The sixteen targets are counted as values, not as multiple renamings of the same conditional criterion.

Named target	Additional conclusion and remaining step
Euler's constant γ: irrationality	An explicit transcendental-slope construction admits infinitely many exponentially close lattice points despite curvature. Thus that abstract curvature route cannot settle the fixed Euler target. No irrationality conclusion follows.
Euler–Gompertz $\delta = \delta_1$: irrationality	The exact parameter law and exponential tails control primitive cancellation for most residues. They do not exclude exceptional behavior at the prescribed parameter $A = 1$.
Catalan G: transcendence	The rational pencil $A_m + 16XI_m$ has full degree and all zeros below G; the π^2 coefficient vanishes exactly. Adequate primitive-value and off-axis growth estimates are not proved.
$\zeta(3)$: transcendence	The single off-axis degree criterion and sharp positive-Hankel counterexample identify the required additional arithmetic margin. No higher algebraic degree is newly excluded.
$\zeta(5)$: transcendence	The inspected external determinant estimates have rate ratio 1975/511067. A sufficient quadratic-exclusion content rate is $\chi > 127273/20000$, but no such bound is established.
$\zeta(7)$: irrationality	The positive-pencil criterion applies to the proposed certificate framework; a sufficient primitive-value decay estimate is missing.
$\zeta(9)$: irrationality	The same degree criterion requires arithmetic denominator control that positivity and amplification do not supply.
$\zeta(11)$: irrationality	No individual value conclusion follows. Independence or a disjunction involving several odd zeta values does not prove this target.
Erdős–Borwein $\sum_{n\geq 1}(2^n-1)^{-1}$: transcendence	The Lambert function has no algebraic differential equation of order at most two and belongs to an independent four-function family. Specialization at $q = 1/2$ is not resolved.
Reciprocal Fibonacci sum $\sum_{n\geq 1} F_n^{-1}$: transcendence	The mixed Lambert–Fibonacci jet theorem strengthens functional independence. The corresponding arithmetic specialization is unproved.
Prime Mertens B_1: irrationality	The prime-harmonic function adds a fourth independent function to three jets. Its constant term is not thereby shown irrational.
$\Gamma(1/5)$: transcendence	The fifth-denominator space has zero intersection with the entire elliptic CM sector in the standard identity calculus. Numerical transcendence requires information beyond that calculus.
$\Gamma(2/5)$: transcendence	The same complete fifth-denominator obstruction holds with all auxiliary denominators and all elliptic CM periods. No new individual value theorem.
$\Gamma(1/7)$: transcendence	The seventh-denominator CM intersection is exactly the Legendre-symbol line. This coordinate is outside it; the formal obstruction does not prove the value transcendental.
$\Gamma(2/7)$: transcendence	The same exact one-dimensional CM recognition applies. Unknown numerical multiplicative relations are not excluded.
$\Gamma(3/7)$: transcendence	The same complete formal classification applies, without an individual transcendence conclusion.

The associated Kurepa left-factorial question is likewise not resolved. The parameter-average law does not decide the special residue patterns needed for that conjecture.

What is proved, and what is not counted as a closure

The exact totally-real alphabet threshold, the totally-positive algebraic-power classifier, the three-jet/four-function independence theorem, the parameter distribution law, the gamma CM recognition theorem, and the full-degree auxiliary-free Catalan pencil have complete proofs in their stated settings. The determinant criterion and its sharp Pell example clarify

an algebraic-degree obstruction. These are structural results. The number of newly resolved famous constant conjectures is zero.

Verification and next obligations

Independent proof reviews accompany the sections. Exact checks cover paired Catalan interpolation and primitive determinants, Pell/Hankel identities, Laguerre residue distributions, gamma character profiles, and the Lambert cusp-model obstruction. Finite computations verify formulas and examples; asymptotic conclusions rely on the proofs. The principal remaining obligations are a suitable arithmetic specialization theorem, a quantitative primitive determinant estimate, or a fixed-parameter argument that goes beyond an average statement. Neither functional independence nor rapid approximation is used as a substitute for any of those obligations. Historical priority and publication significance require further expert comparison.

Part XII

Quadratic Exceptions and Constant Transfer

Editorial integration note. This is the final source manuscript in the edition. Its exact entropy classifications, fixed-parameter cancellation bounds, Catalan repair, and actual beta-period calculation provide the latest conclusions in their respective branches. Its sixteen-target ledger states the remaining arithmetic gaps explicitly.

Original part framing.

Reading convention. This is a standalone mathematical successor to the earlier research manuscripts, not a merger of their full contents. Definitions and inputs needed for the present proofs are included. Totally positive means that *every* conjugate is positive; positivity of only the displayed real embedding is weaker. Algebraic powers of positive numbers always use the real logarithm. Classical external theorems, proofs supplied here, finite checks, and unresolved transfers have different roles. Independent proof reviews and exact computations do not amount to formal verification.

96 Arithmetic classifications and the limits of transfer

The main distinction is between a number whose conjugates satisfy positivity constraints and a named constant approached by algebraic numbers. The former setting allows logarithmic relations to be transported and compared; in the latter, roots of approximating polynomials are not known conjugates of the target. This distinction explains both the rigidity results and the remaining barriers.

Relative entropy has a genuine quadratic exception. For $D(p\|r) = \sum_i p_i \log(p_i/r_i)$, with r rational and p totally positive algebraic, Theorem 97.2 proves that an algebraic value of $\exp(cD(p\|r))$, for $c \neq 0$ real algebraic and $p \neq r$, forces

$$[\mathbb{Q}(p_1, \ldots, p_n) : \mathbb{Q}] \leq 2.$$

It gives an explicit multiplicative test for all quadratic exceptions. The proof compares two independent transports of logarithmic relations. Three distinct positive conjugate laws would require three nonzero coefficients to have pairwise opposite signs. The resulting degree barrier is sharp: binary laws have no nonrational exception, whereas a concrete three-symbol law has one even for the uniform reference. Thus the exact threshold in this specified problem is three, not an unrestricted statement about all entropy problems.

Conditional laws and Markov sources have exact classifiers. For a totally positive algebraic joint law, an algebraic value of $\exp(cH(X \mid Y))$ forces every conditional probability to be rational. The remaining issue is an exact rational-line test on prime-valuation coefficients (Theorem 97.9). Irrational marginals and irrational successful exponents can occur. For a strictly positive stochastic matrix with totally positive algebraic entries, the stationary constraints remove that freedom: Theorem 97.11 gives algebraicity precisely for rational transition probabilities and rational exponent. These are arithmetic classifications of explicitly specified entropy values.

The cubic ternary case closes inside a larger open problem. Theorem 98.5 excludes rational bit entropy for every positive ternary algebraic probability vector whose probability field has degree three, including cubic fields with a pair of complex embeddings. A new restriction in the unit normal form says that the greatest common divisor of exponent differences is a power of two. For odd entropy denominator, the probability field equals the unit generator field. The cubic case then reduces to two elementary unit bounds. This does not exclude every ternary example over an arbitrary number field: a remaining example would need degree at least four, or odd degree at least five when the entropy denominator is odd.

Classical period theory upgrades a formal obstruction. Theorem 99.3 computes an actual $\overline{\mathbb{Q}}$-linear basis of size fourteen for 1 together with the fifth and seventh beta-period spans. Its input is the classical completeness theorem for beta values, not an assumption of Rohrlich's conjecture for all Gamma monomials. Twelve displayed transcendental ratios and an infinite family of exact linear-fractional tests follow. These are explicit consequences of classical transcendence theory; they are not twelve previously open constant problems solved. No individual fifth or seventh Gamma value is classified here.

Fixing the Gompertz parameter changes what can be averaged. For each fixed positive integer A, Theorem 100.1 proves $\rho_p(A) = O(p^{2/3})$ for the number of index residues at which a prime divides the classical Laguerre denominator. Theorem 100.2 then gives a limiting distribution and finite exponential moments for a prime-support upper bound on normalized logarithmic content. This applies in particular to $A = 1$. A separate positive-continuant packing argument gives the stronger content estimate

$$\frac{1}{X}\sum_{n\le X}\frac{\log g_n(A)}{n} = O_A(X^{-1/3}\log(2X)),$$

and only $O_{A,\varepsilon}((\log(2X))^4)$ indices up to X have $\log g_n(A) \ge \varepsilon n$ (Theorem 100.4). On a single set of density one the reduced denominator and primitive form both retain their first two factorial growth terms. These results do not rule out every exceptional index. Theorem 100.5 shows that a bounded primitive form would require cancellation involving a growing number of primes.

Catalan convergence can be repaired without an arithmetic conclusion. A fixed quadratic polynomial invariant removes the auxiliary period and produces positive Hankel pencils of fixed determinant degree. Theorem 101.3 proves that their roots stay uniformly below Catalan's constant. An explicit moment-null density provides a quantitative gap. The full paired polynomial space contains an additional sector omitted by that invariant construction. Theorem 101.7 proves that restoring the full space makes all roots converge to G from below at every fixed determinant degree. The unproved step is a sufficiently strong comparison with primitive integer coefficient height; analytic convergence alone supplies no irrationality or transcendence certificate.

A Lambert optimization is a limit of specified estimates. Theorem 102.2 identifies the unique optimum of the published common-shift denominator budget and proves $\Lambda(x) > 2/5$ uniformly. It therefore cannot reach the $1/3$ threshold for degree-three exclusion. Independent shifts in the proved central cone do not improve this optimum. The result concerns those sufficient clearing bounds, not a lower bound for the actual primitive denominators and not a universal obstruction to other Lambert constructions. The external October 2026

nonquadraticity announcement is attributed separately and is not counted as a result of this manuscript.

The sections below contain proofs and their assumptions. The final ledger records what they imply, and fail to imply, for each named constant. Historical priority remains a separate question from the validity of the supplied arguments.

97 Relative entropy: a degree-two barrier and its exact exceptions

All logarithms in this section are real natural logarithms. For strictly positive probability vectors write

$$D(p\|r) = \sum_i p_i \log \frac{p_i}{r_i}, \qquad H(p) = -\sum_i p_i \log p_i.$$

An algebraic number is *totally positive* if every one of its conjugates over $\mathbb{Q}$ is positive. This hypothesis is imposed on the individual probabilities, not merely on their displayed values. The conclusions below concern arithmetic values, rather than the semialgebraicity of entropy level sets studied in [335, 367]. They use classical logarithmic transcendence [231]. Their historical priority has not been established by an exhaustive literature review.

97.1 Two independent transports of logarithmic relations

Lemma 97.1 (Baker transport and algebraic logarithms)**.** *Let $\beta_1, \dots, \beta_N$ be positive algebraic numbers. Every algebraic linear relation among the $\log \beta_i$ is an algebraic linear combination of their rational relations. Consequently the following operations are valid independently:*

1. *conjugate the algebraic coefficients of a relation, keeping all logarithmic bases fixed;*
2. *conjugate its bases by an embedding under which all of them remain positive, keeping all coefficients fixed.*

If $u_1, \dots, u_s$ are rationally independent logarithms of positive algebraic numbers and b_j are real algebraic, then

$$\exp\Big(\sum_j b_j u_j\Big) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad b_j \in \mathbb{Q} \text{ for every } j.$$

Proof. Choose a rational basis of the span of the logarithms. Baker's linear independence theorem says that this basis remains independent over $\overline{\mathbb{Q}}$. Comparing its coefficients proves the rational-generation assertion and coefficient transport. A rational relation, after clearing denominators, is a multiplicative identity among the bases. Conjugation preserves this identity, and positivity permits taking real logarithms; this proves base transport. For the final assertion, if the exponential is algebraic $A > 0$, Baker's theorem forces $\log A$ into the rational span of the u_j. Comparing coefficients gives $b_j \in \mathbb{Q}$. The converse is a product of rational powers of algebraic numbers. □

Theorem 97.2 (Exact rational-reference classification)**.** *Let r be a strictly positive rational probability vector, let p be a strictly positive probability vector with totally positive algebraic entries, and put $K = \mathbb{Q}(p_1, \dots, p_n)$. For nonzero real algebraic c, the algebraicity of $\exp(cD(p\|r))$ is classified as follows.*

1. *If $p = r$, the value is 1 for every c.*

2. *If $p \neq r$ and $K = \mathbb{Q}$, it is algebraic exactly when $c \in \mathbb{Q}$.*

3. *If $[K : \mathbb{Q}] \geq 3$, it is transcendental for every such c.*

4. *Suppose $K = \mathbb{Q}(t)$, where $t = \sqrt{d}$ and $d > 1$ is a squarefree integer. Write $p_i = m_i + tv_i$ with $m_i, v_i \in \mathbb{Q}$. The value is algebraic exactly when $c = a + bt \in K$, with $a, b \in \mathbb{Q}$, and*

$$\prod_i \left(\frac{p_i}{r_i}\right)^{av_i + bm_i} = 1. \tag{97.1}$$

All powers in this test are positive real powers with rational exponents. Equivalently, clear a common denominator to obtain an equality of algebraic numbers with integer exponents.

In case (4), every successful nonzero exponent satisfies $(a + bt)(a - bt) < 0$.

Proof. The zero case follows from strict positivity of relative entropy. Assume henceforth $p \neq r$, so $D(p\|r) > 0$, and suppose $A = \exp(cD(p\|r))$ is algebraic. Work in a finite normal field L containing p and c. Put $\beta_i = p_i/r_i$. For $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ set $p^\sigma = (\sigma p_i)_i$ and $c_\sigma = \sigma c$. Coefficient transport in

$$\log A - \sum_i cp_i \log \beta_i = 0$$

gives, without changing the base A,

$$\sum_i c_\sigma p_i^\sigma \log \beta_i = \log A.$$

Subtract the equations for σ and τ. Base transport then gives the same resulting relation with bases p_i^σ/r_i and, separately, with bases p_i^τ/r_i. Subtracting these two relations yields the key identity

$$c_\sigma D(p^\sigma\|p^\tau) + c_\tau D(p^\tau\|p^\sigma) = 0. \tag{97.2}$$

Total positivity makes all these divergences well defined.

If $p^\sigma = p$, comparison with the identity embedding in the fixed-base equation gives $c_\sigma = c$, because $D(p\|r) > 0$. Thus the stabilizer of p fixes c, and $c \in K$. If $p^\sigma \neq p$, equation (97.2) with the identity embedding gives

$$\frac{c_\sigma}{c} = -\frac{D(p\|p^\sigma)}{D(p^\sigma\|p)} < 0.$$

In particular all conjugates of c are real. There cannot be three distinct conjugate vectors: their three pairwise identities would require three nonzero real numbers to have pairwise opposite signs. The number of distinct conjugate vectors is $[K : \mathbb{Q}]$. Therefore $[K : \mathbb{Q}] \leq 2$. If $K = \mathbb{Q}$, the already proved inclusion $c \in K$ gives the necessary rationality of c; its sufficiency is immediate from rational powers of rational numbers.

It remains to make the quadratic case exact. For $c = a + bt$,

$$cD(p\|r) = \sum_i (am_i + bdv_i) \log \beta_i + t \sum_i (av_i + bm_i) \log \beta_i.$$

The exponential of the first sum is algebraic. The exponential of the second sum before multiplication by t is also algebraic, since all its coefficients are rational. If that sum is nonzero, Lemma 97.1, applied to its single nonzero logarithm, shows that its exponential after multiplication by the irrational algebraic t is transcendental. Thus algebraicity is equivalent to the second sum vanishing, precisely (97.1). Finally the two distinct conjugate vectors in (97.2) force c and its quadratic conjugate to have opposite signs. □

Corollary 97.3 (Three positive conjugates suffice)**.** *Let p be a positive algebraic probability vector and r a positive rational probability vector. If p has three distinct conjugate vectors whose entries are all positive, then $\exp(cD(p\|r))$ is transcendental for every nonzero real algebraic c, even when other conjugate vectors are not positive.*

Proof. The coefficient and base transports in the proof of Theorem 97.2 apply to any chosen positive conjugate vectors. If there were an algebraic exponential, the pairwise identity (97.2) would force three corresponding nonzero conjugate coefficients to have pairwise negative real ratios, which is impossible. □

Example 97.4 (A cubic family with no exceptional reference)*.* For $j = 1, 2, 3$ put

$$p_j = \frac{3 + 2\cos(2\pi j/7)}{8}.$$

These are the positive roots of $512X^3 - 512X^2 + 152X - 13$, and their sum is one. The polynomial is irreducible modulo 3 (its cubic reduction has no root), hence irreducible over $\mathbb{Q}$. Thus $[\mathbb{Q}(p_1, p_2, p_3) : \mathbb{Q}] \geq 3$, and every p_j is totally positive. For every strictly positive rational reference r and every nonzero real algebraic c,

$$\prod_{j=1}^{3} (p_j/r_j)^{cp_j} \quad \text{is transcendental.}$$

The rational reference may vary freely; no quadratic escape is possible for this one fixed cubic law.

Corollary 97.5 (The exceptional exponent space is one rational line)**.** *In the nonrational quadratic case of Theorem 97.2, the set*

$$\{c \in \overline{\mathbb{Q}} \cap \mathbb{R} : \exp(cD(p\|r)) \in \overline{\mathbb{Q}}\}$$

is either $\{0\}$ or $c_0\mathbb{Q}$ for one nonzero $c_0 \in K$ of negative norm. In particular a positive-norm quadratic exponent can never be an exception.

Proof. The test (97.1) is the kernel over $\mathbb{Q}$ of the real linear expression

$$(a, b) \longmapsto a \sum_i v_i \log \beta_i + b \sum_i m_i \log \beta_i.$$

Its two coefficients are not both zero, since otherwise $D(p\|r) = 0$. The kernel has rational dimension at most one. The norm assertion follows from Theorem 97.2. □

Example 97.6 (A genuine quadratic escape at a uniform reference)*.* Let

$$t = \frac{\sqrt{17}}{36}, \qquad p = \left(\frac{1}{4} + t, \frac{1}{4} - t, \frac{1}{2}\right), \qquad r = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

All entries of p are totally positive, and its multiset is Galois invariant. Moreover $p_1 p_2 = 4/81$, so

$$(3p_1)(3p_2)(3/2)^2 = 1, \qquad D(p\|r) = t \log \frac{p_1}{p_2}.$$

It follows that

$$\boxed{\exp(D(p\|r)/t) = \frac{9 + \sqrt{17}}{9 - \sqrt{17}} \in \overline{\mathbb{Q}}.}$$

Thus even a rational-characteristic-polynomial spectrum and a uniform reference do not imply transcendence of every algebraic power of the exponential relative entropy. The degree-two exception in Theorem 97.2 is necessary.

Corollary 97.7 (Binary rigidity and the sharp three-symbol threshold)**.** *Let $p = (p_1, 1 - p_1)$ have totally positive algebraic entries and let r be a strictly positive rational binary law. If $p \neq r$, then*

$$\exp(cD(p\|r)) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad p_1 \in \mathbb{Q} \text{ and } c \in \mathbb{Q}$$

for nonzero real algebraic c. More generally, a nonrational quadratic law has no exceptional exponent whenever its rational reference lies on the affine line joining the law to its conjugate. Consequently the minimum alphabet size allowing a nonrational totally positive law, a rational reference, and a nonzero algebraic exponent with algebraic exponential relative entropy is exactly 3.

Proof. The degree barrier reduces the issue to the quadratic case $p = m + tv$, $q = m - tv$, $c = a + bt$. A successful exponent has negative norm, hence $b \neq 0$ and $|a/b| < t$. Therefore $z = m + (a/b)v$ is a strictly positive rational probability vector on the open segment joining p and q. Dividing (97.1) by b gives

$$\sum_i z_i \log p_i = \sum_i z_i \log r_i.$$

Clear denominators in this rational logarithmic relation and conjugate its bases. Since z, r are rational, this also gives $\sum_i z_i \log q_i = \sum_i z_i \log r_i$. On the positive part of the affine line through p, q, the function $x \mapsto \sum_i z_i \log x_i$ is strictly concave: its second derivative along the nonzero direction v is $-\sum_i z_i v_i^2/x_i^2 < 0$. Three distinct points on this line cannot have the same value. If r lies on the line, p, q, r would be three such points, a contradiction. Every binary probability vector lies on the same affine line, proving binary rigidity. One symbol has no nonrational law, and Example 97.6 supplies the three-symbol exception. □

Corollary 97.8 (The Galois-averaged reference has no exceptions)**.** *Let p be a nonrational probability vector whose entries are totally positive algebraic numbers. In a finite normal field containing its entries, define*

$$m_i = \frac{1}{|G|} \sum_{\sigma \in G} \sigma(p_i), \qquad G = \mathrm{Gal}(L/\mathbb{Q}).$$

Then m is a strictly positive rational probability vector and

$$\exp(cD(p\|m)) \text{ is transcendental}$$

for every nonzero real algebraic c. If the multiset of entries of p is Galois invariant, one also has

$$D(p\|m) = H(m) - H(p) > 0.$$

Thus every nonzero real algebraic power of $\exp(H(m) - H(p))$ is transcendental in this spectral case.

Proof. The average is fixed by G, positive, and normalized. If $[\mathbb{Q}(p) : \mathbb{Q}] \geq 3$, use Theorem 97.2. If this degree is two, m is the midpoint of the two conjugate vectors and Corollary 97.7 applies. Nonrationality excludes degree one and ensures $p \neq m$. For a Galois-invariant multiset, m_i is constant on each conjugacy orbit of entries. Conjugating p permutes its entries within these orbits, so $\sum_i p_i \log m_i$ is unchanged. Averaging gives $\sum_i p_i \log m_i = \sum_i m_i \log m_i$. Substitution into the definition of relative entropy proves the entropy identity. □

97.2 Conditional entropy and stationary Markov sources

For a strictly positive joint law $p = (p_{xy})$ put $P_y = \sum_x p_{xy}$ and $r_{x|y} = p_{xy}/P_y$. Let

$$F(p) = H(X \mid Y) = -\sum_{x,y} p_{xy} \log r_{x|y}.$$

The X-alphabet is assumed to have at least two symbols.

Theorem 97.9 (Exact conditional effective-number classifier)**.** *Suppose all p_{xy} are totally positive algebraic numbers. For nonzero real algebraic c, the number $\exp(cF(p))$ is algebraic if and only if both of the following hold:*

1. *Every conditional probability $r_{x|y}$ is rational.*
2. *For every rational prime ℓ,*

$$cB_\ell \in \mathbb{Q}, \qquad B_\ell = -\sum_y P_y \sum_x r_{x|y} v_\ell(r_{x|y}). \tag{97.3}$$

Here $v_\ell(\ell) = 1$ and only finitely many B_ℓ are nonzero.

If a conditional probability is irrational, every nonzero real algebraic power of $\exp(F(p))$ is transcendental. If all conditional probabilities are rational, the successful exponents, together with zero, form either $\{0\}$ or one rational line $\beta^{-1}\mathbb{Q}$: the latter occurs exactly when the nonzero B_ℓ span a one-dimensional rational vector space $\beta\mathbb{Q}$.

Proof. Suppose $A = \exp(cF(p))$ is algebraic and choose a normal field containing p, c. For an automorphism σ put $q = \sigma p$, $Q_y = \sum_x q_{xy}$, $s_{x|y} = q_{xy}/Q_y$, and $d = \sigma c$. Coefficient transport gives

$$\log A = -c \sum_{xy} p_{xy} \log r_{x|y} = -d \sum_{xy} q_{xy} \log r_{x|y}.$$

Both sums without the prefactors are strictly negative, since $0 < r_{x|y} < 1$. Hence $d/c > 0$ is real. The coefficient vector $cp - dq$ annihilates the logarithms of r, and base transport shows that it also annihilates the logarithms of s. Subtracting the two relations yields

$$c \sum_y P_y D(r_{\cdot|y} \| s_{\cdot|y}) + d \sum_y Q_y D(s_{\cdot|y} \| r_{\cdot|y}) = 0.$$

All weights are positive after division by c. Thus every conditional law equals its σ-conjugate. Since this holds for all σ, each $r_{x|y}$ is rational.

For rational conditionals, prime factorization gives $F(p) = \sum_\ell B_\ell \log \ell$ with the coefficients in (97.3). Distinct prime logarithms are rationally independent. Lemma 97.1 proves both necessity and sufficiency of that test. Finally, since $F(p) > 0$, at least one B_ℓ is nonzero. A common nonzero multiplier makes all the B_ℓ rational precisely when their rational span has dimension one; its successful multipliers are then exactly $\beta^{-1}\mathbb{Q}$. □

Example 97.10 (Irrational marginals and an irrational successful exponent)*.* Take two conditional laws on four symbols,

$$r^{(1)} = (1/2, 1/4, 1/8, 1/8), \qquad r^{(2)} = (1/4, 1/4, 1/4, 1/4),$$

and marginal probabilities $P_1 = a = (2 + \sqrt{2})/4$, $P_2 = 1 - a$. The joint entries $P_y r_x^{(y)}$ are totally positive algebraic, and

$$F(p) = (2 - a/4) \log 2, \qquad \exp\left(\frac{F(p)}{2 - a/4}\right) = 2.$$

The marginal and the successful exponent are irrational. Consequently conditional entropy cannot inherit the stronger claim that all entries and the exponent must be rational.

Theorem 97.11 (Exact classifier for Markov entropy rates)**.** *Let T be an $n \times n$ strictly positive stochastic matrix, $n \geq 2$, whose entries are totally positive algebraic numbers. Let π be its stationary probability vector and let*

$$h(T) = -\sum_{i,j} \pi_i T_{ij} \log T_{ij}$$

be the entropy rate of its stationary Markov chain. For every nonzero real algebraic c,

$$\boxed{\exp(ch(T)) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad T_{ij} \in \mathbb{Q} \text{ for all } i, j \text{ and } c \in \mathbb{Q}.}$$

If an entry of T is irrational, $h(T)/\log b$ is transcendental for every algebraic real base $b > 1$.

Proof. Strict positivity gives a unique stationary vector with positive entries. Solving the stationary linear equations shows that π belongs to the number field generated by the entries of T. Under every embedding, the conjugate matrix remains strictly positive and stochastic, and the conjugate vector is its unique normalized stationary vector. Thus each π_i and each joint probability $\pi_i T_{ij}$ is totally positive. Apply Theorem 97.9 to consecutive states of the chain: algebraicity forces every transition probability T_{ij} to be rational. Then the stationary equations have rational coefficients and a unique solution, so π is rational. Hence $h(T)$ is a nonzero rational linear combination of prime logarithms. Its positive exponential is algebraic, and Lemma 97.1 forces c to be rational. The converse follows from rational powers of rational numbers. If $h(T)/\log b = a$ were algebraic, it would be positive and $\exp(h(T)/a) = b$ would contradict the theorem. □

Example 97.12 (An asymmetric algebraic Markov source). For $a = (2 + \sqrt{2})/4$, put

$$T = \begin{pmatrix} a & 1-a \\ 1/3 & 2/3 \end{pmatrix}, \qquad \pi = \left(\frac{1/3}{4/3 - a}, \frac{1-a}{4/3-a} \right).$$

Both conjugate transition matrices are strictly positive. Every nonzero real algebraic power of $\exp(h(T))$ is transcendental, although the rate involves an irrational stationary mixture of two different row entropies.

97.3 Scope of the transfer

The degree barrier for relative entropy is not an unrestricted relative-entropy rigidity claim: rational p already makes $\exp(D(p\|q))$ algebraic for every positive algebraic reference q. The reference is rational in Theorem 97.2, and its quadratic exceptions are real. The conditional theorem concerns classical, strictly positive joint laws. It does not assert a corresponding classification for quantum conditional entropy or general hidden-Markov entropy rates. None of these value classifications establishes algebraic independence of the resulting transcendental constants or resolves a previously named open conjecture.

98 A two-primary obstruction and the cubic entropy case

The previous manuscript excluded rational bit entropy for nonrational ternary probabilities in totally real number fields. Here we exclude *every cubic probability field*, including fields with a complex pair of embeddings. The new ingredient is an arithmetic restriction on the differences of the three Laurent exponents: their greatest common divisor is a power of two. This also identifies the probability field with its unit-generator field whenever the entropy denominator is odd. No unrestricted three-versus-four alphabet theorem is claimed.

Write $H_2(p) = -\sum_i p_i \log_2 p_i$. Probabilities are strictly positive, and "nonrational vector" means that at least one entry is irrational. The probability field is $K = \mathbb{Q}(p_1, p_2, p_3)$.

98.1 The inherited normal form

We state and prove the previously obtained reduction to specify exactly which inputs the new argument uses. Baker’s logarithmic independence theorem is the transcendence-theoretic input [231]. The arithmetic of entropy values is also discussed in [335]; the following field-degree question is separate from the non-semialgebraicity questions treated there.

Lemma 98.1 (Inherited unit normal form)**.** *Suppose $p = (p_1, p_2, p_3)$ is a nonrational algebraic probability vector and $h = H_2(p) \in \mathbb{Q}$. Write $h = a/b$ in lowest positive terms, put $C = 2^h$, and let $d = [K : \mathbb{Q}]$. There is a positive algebraic unit $w \neq 1$ and pairwise distinct odd integers m_1, m_2, m_3 such that*

$$p_i = C^{-1} w^{m_i}, \qquad \sum_i w^{m_i} = C, \qquad \sum_i m_i w^{m_i} = 0, \qquad \gcd(m_1, m_2, m_3) = 1. \tag{98.1}$$

There are exponents of both signs. With $L = \mathbb{Q}(w)$ one has

$$L = K(C), \qquad 1 \leq h < \log_2 3, \qquad hd \in \mathbb{Z}, \qquad r_1(L) = \begin{cases} 1 & b \text{ odd}, \\ 2 & b \text{ even}. \end{cases} \tag{98.2}$$

Here r_1 counts real embeddings. Every p_i is irrational.

Proof. Let t be the rational dimension of the span of the probabilities, and let r be that of $\log p_1, \log p_2, \log p_3, \log 2$. Splitting the entropy relation against a rational basis of the probabilities, and applying Baker’s theorem, gives t independent rational relations among the four logarithms. Thus $t + r \leq 4$. Nonrationality gives $t \geq 2$. If $r = 1$, the probabilities are rational powers of two. Clearing a common denominator of the exponents and using the independence of $1, 2^{1/q}, \ldots, 2^{(q-1)/q}$, by Eisenstein for $X^q - 2$, shows that a positive sum of such powers can equal one only if all the powers have integral exponents. This contradicts nonrationality. Hence $t = r = 2$.

Write $\log p_i = \alpha_i \log 2 + \beta_i \log u$ in a rational basis, with rational α_i, β_i and positive algebraic u. Baker’s theorem gives $\sum \beta_i p_i = 0$ and $\sum \alpha_i p_i = -h$. The rational kernel of $(x_i) \mapsto \sum x_i p_i$ is one-dimensional, so $\alpha_i + h = \lambda \beta_i$ for one rational λ. Clearing denominators and making the exponent vector primitive gives (98.1), initially without the unit and parity assertions. The logarithms of w and 2 are rationally independent. Equal exponents would give two equal probabilities and, using stationarity and normalization, make all three probabilities rational. Thus the exponents are distinct. Their positive weighted mean zero forces both signs.

The number C is integral. At a finite valuation, $v(w) > 0$ makes the unique smallest exponent contribute a strictly negative, uniquely minimal valuation to $\sum w^{m_i}$; $v(w) < 0$ does the same with the largest exponent. Both contradict integrality of C. Thus w is a unit. Reducing normalization and stationarity at a prime above two shows that either all three exponents are odd or all three are even: otherwise subtraction isolates one nonzero residue term. Primitivity excludes the even case. Consequently $C = \sum (1 - m_i) w^{m_i} \in 2\mathcal{O}_L$, so $h \geq 1$. The entropy bound is strict since the uniform vector is rational.

Bezout for the primitive exponents gives $w \in K(C)$, and the sum identity gives $C \in \mathbb{Q}(w)$; hence $L = K(C)$. Taking absolute norms in $p_i = C^{-1} w^{m_i}$ and descending from L to K gives $|N_{K/\mathbb{Q}}(p_i)| = 2^{-hd}$. Rationality of this number forces $hd \in \mathbb{Z}$. A rational positive p_i would then equal C^{-1}, forcing $m_i = 0$, contrary to oddness.

Finally $F(x) = \sum x^{m_i}$ has a unique positive stationary point, because $y \mapsto F(e^y)$ is strictly convex and its derivative tends to opposite infinities at the two ends. Oddness of the exponents makes its only possible real stationary points $w, -w$. Every real conjugate of w must be one of these. The field $\mathbb{Q}(C) = \mathbb{Q}(2^{1/b})$ has degree b by Eisenstein and Bezout. If b is odd, its only

real conjugate of C is positive, so only w occurs. If b is even, $[L:\mathbb{Q}]$ is even, and the positive real-embedding count, at most two, must also be even. This proves the signature assertion. □

98.2 The new restriction on exponent differences

Theorem 98.2 (Two-primary exponent spacing)**.** *Under Lemma 98.1, set* $g = \gcd(m_2 - m_1, m_3 - m_1) > 0$. *Then*

$$g = 2^s \quad (s \geq 1), \qquad [L:K] \text{ is a power of } 2. \tag{98.3}$$

If the reduced denominator b *of* h *is odd, then*

$$L = K, \qquad r_1(K) = 1, \qquad [K:\mathbb{Q}] \text{ is odd.} \tag{98.4}$$

Proof. All exponents are odd, so g is even. Suppose an odd prime ℓ divides g. Then $m_i \equiv r \pmod{\ell}$ for a common residue r, which is nonzero because the exponent vector is primitive. Write $m_i = r + \ell n_i$ with integers n_i. Stationarity yields

$$0 = rC + \ell \sum_i n_i w^{m_i}.$$

The sum is an algebraic integer, while rC is a unit at every prime above ℓ: indeed $C^b = 2^a$ and $\ell \nmid 2r$. Reduction at such a prime is a contradiction. Therefore g has no odd prime divisor.

The ratios $p_2/p_1 = w^{m_2-m_1}$ and $p_3/p_1 = w^{m_3-m_1}$ belong to K. Bezout for these two differences gives $w^g \in K$. The tower adjoining successively $w^{2^{s-1}}, w^{2^{s-2}}, \ldots, w$ has every step of degree at most two. Since $K \subseteq L = \mathbb{Q}(w)$, this proves that $[L:K]$ is a power of two.

If b is odd, Lemma 98.1 gives $r_1(L) = 1$. The identity $[L:\mathbb{Q}] = r_1(L) + 2r_2(L)$ makes $[L:\mathbb{Q}]$ odd. Its factor $[L:K]$, being a power of two, must be one. The remaining assertions follow. □

The exclusion concerns the gcd of the *differences*, not merely the gcd of the exponents. For example, every primitive triple of odd exponents all congruent to 1 modulo 3 is excluded immediately, without evaluating a stationary point.

98.3 Two elementary cubic unit bounds

Lemma 98.3 (A small cubic unit exclusion)**.** *There is no algebraic unit* u *of degree three with exactly one real conjugate and* $1 < u < 2^{1/3}$.

Proof. Write the other conjugates as $z, \bar{z}$. Positivity of u and the unit property give $u|z|^2 = 1$. The minimal polynomial therefore has the form $X^3 - sX^2 + tX - 1$, with $s, t \in \mathbb{Z}$, and

$$s = u + 2\operatorname{Re} z, \qquad t = su - u^2 + u^{-1}.$$

Put $\theta = 2^{1/3}$. Since $1 < u < \theta < 4/3$, $-1 < u - 2/\sqrt{u} \leq s \leq u + 2/\sqrt{u} < 4$, whence $s \in \{0, 1, 2, 3\}$. For these four integers set $f_s(x) = sx - x^2 + x^{-1}$. For $x > 1$,

$$f_s'(x) \leq 3 - 2x - x^{-2} = -\frac{(x-1)^2(2x+1)}{x^2} < 0,$$

so $f_s(u) < f_s(1) = s$. On the other hand,

$$f_s(u) - (s-1) = 1 - u^2 + u^{-1} + s(u-1) > 1 - \theta^2 + \theta^{-1} = 1 - \frac{\theta^2}{2} > 0.$$

Thus the integer t lies strictly between $s-1$ and s, impossible. □

Lemma 98.4 (A pure cubic unit exclusion)**.** *In $E = \mathbb{Q}(\theta)$, $\theta = 2^{1/3}$, there is no algebraic unit u satisfying $1 < u < 2\theta$.*

Proof. We use the standard integral basis $\mathcal{O}_E = \mathbb{Z}[\theta]$, with a verification recalled here to keep the arithmetic input explicit; see also [368]. The power-basis discriminant is -108, so its index can have only the prime divisors two and three. Locally at two, $X^3 - 2$ is Eisenstein, and θ is a uniformizer generating the full ring of integers. Locally at three the generator $1 + \theta$ satisfies the Eisenstein polynomial $X^3 - 3X^2 + 3X - 3$. Thus neither prime divides the index, as claimed.

Write $u = A + B\theta + D\theta^2$ with integers A, B, D. Let $\zeta = e^{2\pi i/3}$ and $z = A + B\theta\zeta + D\theta^2\zeta^2$; the third conjugate is $\bar{z}$. As u is a positive unit, $u|z|^2 = 1$. Fourier inversion gives

$$3A = u + z + \bar{z}, \quad 3B\theta = u + \zeta^2 z + \zeta\bar{z}, \quad 3D\theta^2 = u + \zeta z + \zeta^2\bar{z}.$$

Each right side is real and belongs to $[u - 2/\sqrt{u}, u + 2/\sqrt{u}]$. The lower endpoint is greater than -1, so integrality gives $A, B, D \geq 0$. The function $x + 2/\sqrt{x}$ is strictly increasing for $x > 1$. At $x = 2\theta = \theta^4$ its value is 3θ. Consequently

$$A < \theta < 2, \qquad B < 1, \qquad D < \theta^{-1} < 1.$$

It follows that $B = D = 0$ and $A \in \{0, 1\}$, contradicting $u > 1$. □

These two lemmas are elementary unit bounds, not claims of new results about the smallest Pisot number or the fundamental unit of a pure cubic field. Their role here is to eliminate the two possible rational entropy values in a cubic probability field.

98.4 Closing the cubic probability-field case

Theorem 98.5 (No cubic ternary rational entropy)**.** *Let p_1, p_2, p_3 be positive algebraic numbers with sum one. If $[\mathbb{Q}(p_1, p_2, p_3) : \mathbb{Q}] = 3$, then*

$$H_2(p_1, p_2, p_3) \notin \mathbb{Q}.$$

No total-reality assumption is required.

Proof. Suppose h is rational. The normal form gives $3h \in \mathbb{Z}$ and $1 \leq h < \log_2 3 < 5/3$, so $h = 1$ or $h = 4/3$. Both denominators are odd. Theorem 98.2 gives $L = K$, of signature $(1, 1)$. Replacing w by w^{-1} and all exponents by their negatives if necessary, assume $w > 1$.

First suppose $h = 1$, so $C = 2$. Two positive exponents would make their two monomials alone exceed two. There is therefore one positive odd exponent a, with the other two equal to $-b, -c$, where $1 \leq b < c$ are odd. If $a = 1$ and $b = 1$, then $w + w^{-1} + w^{-c} > 2$. If $a = 1$ and $b \geq 3$, then $c \geq 5$ and

$$F(\sqrt{2}) = \sqrt{2} + 2^{-b/2} + 2^{-c/2} \leq \frac{11}{4\sqrt{2}} < 2.$$

But w is the global minimum of F on the positive real axis by the strict convexity argument in Lemma 98.1; hence $F(w) \leq F(\sqrt{2}) < 2$, another contradiction. Thus $a \geq 3$, and the normalization gives $w^a < 2$, whence $1 < w < 2^{1/3}$. This contradicts Lemma 98.3.

Now suppose $h = 4/3$. Since $C = 2^{4/3} \in L = K$ and both $\mathbb{Q}(C)$ and K have degree three, $K = \mathbb{Q}(2^{1/3})$. At least one exponent a is positive, so $w \leq w^a < C = 2\theta$. This contradicts Lemma 98.4. □

Corollary 98.6 (Degree barrier for the unresolved ternary case)**.** *If a nonrational algebraic ternary probability vector has rational bit entropy, its probability field has degree at least four. If the reduced entropy denominator is odd, the probability field has exactly one real embedding and odd degree at least five. In particular, a one-bit example would require degree at least five.*

Proof. Degree one is excluded by nonrationality and degree three by Theorem 98.5. For completeness, degree two is also impossible: the real quadratic probability field K is totally real, the norm restriction leaves $h = 1$ or $3/2$, and $L = K(2^h) \subseteq K(\sqrt{2})$ is totally real. Yet Lemma 98.1 permits at most the real conjugates $w, -w$ for w. Total reality would force w^2 to be rational; as a positive rational algebraic unit it equals one, a contradiction. For odd denominator use Theorem 98.2 and exclude the remaining odd degrees one and three. □

Scope of the advance. The cubic theorem rules out a family not covered by the earlier totally-real theorem. It does not exclude non-totally-real fields of degree four or higher, does not establish transcendence of these entropy values, and does not classify algebraic irrational entropy. The unrestricted least alphabet for entirely irrational algebraic probabilities and rational bit entropy remains either three or four. The exact value four for totally real probabilities belongs to the previous manuscript and is not recounted as a new result here.

99 Actual arithmetic separation of fifth and seventh beta periods

The previous formal Gamma calculus can be upgraded to a statement about *actual numbers* on a restricted, but infinite, family. The upgrade uses the classical Wolfart–Wüstholz theorem; it does not establish the Rohrlich conjecture for arbitrary Gamma monomials. The resulting rank calculation and linear-fractional classification below are explicit specializations of classical period theory, not claimed solutions of previously open individual Gamma-value problems.

99.1 The precise classical input

Write $u \sim v$ if $u/v \in \overline{\mathbb{Q}}^{\times}$. We use the following two parts of the beta-value theorem of Wolfart and Wüstholz [369]. The first is also stated as Theorem 5 in [370].

1. For positive rational a_j, b_j, algebraic linear dependence of the numbers $B(a_j, b_j)$ implies that two of them have algebraic ratio. Equivalently, representatives of distinct $\sim$-classes of positive rational beta values are linearly independent over $\overline{\mathbb{Q}}$.

2. An algebraic ratio of two rational beta values is accounted for by the classical Gamma identities: recurrence, reflection, and Gauss multiplication, with algebraic factors and rational powers admitted. In particular, any invariant of these identities must assign the same value to the two beta monomials.

The second assertion is the beta-value case of the classical Gauss–Legendre completeness statement in [369]; the first assertion alone would not justify our signature test. This completeness is known for these periods of Fermat Jacobians. It is not known for arbitrary Gamma monomials. We also use Schneider's classical theorem that $B(a, b)$ is transcendental when $a, b \in \mathbb{Q} \setminus \mathbb{Z}$ and $a + b \notin \mathbb{Z}_{\leq 0}$; see Theorem 4 of [370] and [371].

99.2 A signature that survives unrestricted auxiliary denominators

Put

$$s(x) = \begin{cases} 1 - 2\{x\}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}. \end{cases}$$

For $m \geq 2$, $1 \leq a, b < m$, and $t \in (\mathbb{Z}/m\mathbb{Z})^{\times}$, define

$$\sigma_{m;a,b}(t) = s(ta/m) + s(tb/m) - s(t(a+b)/m).$$

If $a + b = m$ the signature is identically zero. Otherwise it takes its values in $\{1, -1\}$, and is odd under $t \mapsto -t$.

Here is the verification that these are legitimate obstructions, even when a proposed identity introduces other denominators. The elementary identity

$$\sum_{j=0}^{h-1} s(x + j/h) = s(hx), \qquad h \geq 1,$$

holds at endpoints as well as away from them. When hx is not an integer it follows by summing fractional parts; when hx is an integer, the nonintegral residues pair as $u, 1-u$ and the integral residue has value zero. Also $s(x+1) = s(x)$ and $s(-x) = -s(x)$. For a finite Gamma-identity derivation choose L divisible by every denominator and every multiplication index in the derivation. For each t prime to L, assign $\Gamma(x)$ the additive value $s(tx)$ and assign algebraic factors and powers of π the value zero. Recurrence and reflection preserve this assignment. In a multiplication identity, $j \mapsto tj$ permutes the residues modulo its multiplication index, so the displayed distribution identity proves preservation too. Reduction of units modulo a divisor of L is surjective. Thus unequal signatures cannot become equal by introducing auxiliary denominators.

Lemma 99.1 (Actual separation from unequal signatures)**.** *Let $\beta = B(a/m, b/m)$ and $\beta' = B(c/n, d/n)$ with $1 \leq a, b < m$ and $1 \leq c, d < n$. Inflate the two signatures to the units modulo* $\operatorname{lcm}(m, n)$*. If the inflated signatures differ, then β/β' is transcendental.*

Proof. An algebraic ratio would, by the classical beta completeness theorem, admit a standard Gamma-identity derivation. The finite-derivation invariant just proved would then make the signatures equal. This is a contradiction. Since the ratio is nonzero, failure to be algebraic is precisely transcendence. □

Theorem 99.2 (Separation of coprime conductors)**.** *For $m \geq 2$ let*

$$V_m = \operatorname{span}_{\overline{\mathbb{Q}}}\{B(a/m, b/m) : 1 \leq a, b < m\}.$$

If $m_1, \ldots, m_k \geq 2$ are pairwise coprime, then the sum of the quotient spaces $V_{m_i}/\overline{\mathbb{Q}}\pi$ is direct inside $(V_{m_1} + \cdots + V_{m_k})/\overline{\mathbb{Q}}\pi$. In particular,

$$V_m \cap V_n = \overline{\mathbb{Q}}\pi \quad ((m,n) = 1), \qquad \dim(V_{m_1} + \cdots + V_{m_k}) = 1 + \sum_{i=1}^{k} (\dim V_{m_i} - 1).$$

Moreover 1 is linearly independent of this sum.

Proof. Each V_m contains π by reflection. Its beta values with zero signature are algebraic multiples of π. A nonzero signature cannot equal the zero signature. A signature from conductor m cannot equal a nonzero signature from a coprime conductor n: by the Chinese

remainder theorem, such an equality would equate a function of an arbitrary unit modulo m with a function of an independent arbitrary unit modulo n. Both functions would be constant. Their oddness would force them to be zero, contrary to their values ± 1. Choose representatives of the algebraic-ratio classes within each conductor. Lemma 99.1 shows that the only class shared by two conductors is the class of π. The Wolfart–Wüstholz linear independence theorem now proves all stated assertions about the sum and its dimension. Finally include $B(1,1)=1$ among the beta values. Schneider's theorem says that none of the other classes is its class; the same linear independence theorem includes 1. $\square$

99.3 A complete fourteen-dimensional calculation

Set

$$X=\Gamma(1/5),\quad Y=\Gamma(2/5),\qquad R=\Gamma(1/7),\quad S=\Gamma(2/7),\quad T=\Gamma(3/7),$$

and define the actual positive numbers

$$A=\frac{X^2}{Y},\quad B=\frac{XY^2}{\pi},\qquad U=\frac{R^2}{S},\quad V=\frac{RS}{T},\quad W=\frac{RT^2}{\pi},\quad Z=\frac{S^2T}{\pi}.$$

Here $A=B(1/5,1/5)$ and $\sin(2\pi/5)B=B(1/5,2/5)$; $U=B(1/7,1/7)$, $V=B(1/7,2/7)$, $\sin(3\pi/7)W=B(1/7,3/7)$, and $\sin(3\pi/7)Z=B(2/7,2/7)$. These sine factors are nonzero algebraic numbers.

Theorem 99.3 (Exact mixed beta-period basis)**.** *The following ordered list is linearly independent over $\overline{\mathbb{Q}}$:*

$$\mathcal{M}=\left(1,\pi,A,B,\frac{\pi}{A},\frac{\pi}{B},U,V,W,Z,\frac{\pi}{U},\frac{\pi}{V},\frac{\pi}{W},\frac{\pi}{Z}\right).$$

It is a basis of $\overline{\mathbb{Q}}+V_5+V_7$. Consequently

$$\dim V_5=5,\qquad \dim V_7=9,\qquad \dim(\overline{\mathbb{Q}}+V_5+V_7)=14.$$

Proof. The signatures of A,B on $t=1,2,3,4$ are respectively

$$(1,1,-1,-1),\qquad (1,-1,1,-1).$$

The signatures of U,V,W,Z on $t=1,\dots,6$ are respectively

U	1	1	1	−1	−1	−1
V	1	1	−1	1	−1	−1
W	1	−1	1	−1	1	−1
Z	1	−1	−1	1	1	−1

The signature of π/Q is the negative of that of Q. These four fifth signatures and eight seventh signatures are distinct and nonzero. Each reciprocal is, up to an algebraic factor, a positive rational beta value: for nonintegral $a,b,a+b$ with $0<a,b<1$,

$$B(a,b)B(1-a,1-b)=\frac{\pi\sin(\pi(a+b))}{(1-a-b)\sin(\pi a)\sin(\pi b)}.$$

The expression remains positive when $a+b>1$ because both the numerator sine and $1-a-b$ change sign. The values π and 1 are represented by $B(1/2,1/2)$ and $B(1,1)$ respectively. Lemma 99.1, the coprime-conductor argument, and Schneider's theorem therefore separate all fourteen algebraic-ratio classes. Their linear independence follows from Wolfart–Wüstholz.

For completeness the spanning calculation uses only recurrence and reflection. To reduce $B(a/p,b/p)$, write a Gamma argument as r/p modulo integers, replace $\Gamma(r/p)$ for $r>(p-1)/2$

by an algebraic multiple of $\pi/\Gamma((p-r)/p)$, and collect exponents. For $p=5$ the resulting classes are exactly

$$\pi,\ A,\ B,\ \pi/A,\ \pi/B,$$

and for $p=7$ they are exactly

$$\pi,\ U,\ V,\ W,\ Z,\ \pi/U,\ \pi/V,\ \pi/W,\ \pi/Z.$$

An exhaustive integer-exponent enumeration of the 16 and 36 pairs is supplied with the manuscript; alternatively the displayed reduction checks this finite assertion directly. This proves spanning and all dimensions. □

99.4 An exact algebraicity test on an infinite explicit family

For $c=(c_0,\dots,c_{13})\in\overline{\mathbb{Q}}^{14}$ write $L_c=\sum_j c_j\mathcal{M}_j$.

Corollary 99.4 (Linear-fractional arithmetic classification)**.** *For $d\neq 0$, $L_d\neq 0$, and*

$$\frac{L_c}{L_d}\in\overline{\mathbb{Q}} \quad\Longleftrightarrow\quad c=\alpha d \text{ for some } \alpha\in\overline{\mathbb{Q}}.$$

In particular, L_c is algebraic if and only if $c_1=\cdots=c_{13}=0$. Every ratio of two distinct entries of $\mathcal{M}$ is transcendental.

Proof. The denominator cannot vanish by Theorem 99.3. If the ratio is $\alpha\in\overline{\mathbb{Q}}$, then $L_{c-\alpha d}=0$, so linear independence forces $c=\alpha d$. The converse is immediate, and taking $d=(1,0,\dots,0)$ proves the algebraicity statement. □

This yields actual monomial exclusions that do not assert algebraic independence of the five individual Gamma values. Examples include

$$\frac{\pi X}{Y^3},\quad \frac{X^4}{\pi Y^2},\quad \frac{X^2Y^4}{\pi^3},\quad \frac{RT}{S^2},\quad \frac{\pi S}{T^3},\quad \frac{\pi R^2}{S^3T},$$
$$\frac{R^4}{\pi S^2},\quad \frac{R^2S^2}{\pi T^2},\quad \frac{R^2T^4}{\pi^3},\quad \frac{S^4T^2}{\pi^3},\quad \frac{X^2S}{YR^2},\quad \frac{XY^2T}{\pi RS}.$$

They are respectively A/B, $A/(\pi/A)$, $B/(\pi/B)$, U/V, V/W, U/Z, $U/(\pi/U)$, $V/(\pi/V)$, $W/(\pi/W)$, $Z/(\pi/Z)$, A/U, and B/V. They are twelve explicit consequences of one classical period theorem and the finite signature certificate, not twelve newly closed famous conjectures.

For a polynomial version, replace π with a formal variable P in the fourteen Laurent monomials defining $\mathcal{M}$. Regard P,X,Y,R,S,T as independent variables. Evaluation at the actual six constants is injective on their $\overline{\mathbb{Q}}$-linear span. Multiplication by one common Laurent monomial turns this into an explicit finite-support polynomial family having no nonzero vanishing member. This is a complete statement for that support, and extends to ratios of two such expressions by Corollary 99.4.

99.5 The remaining boundary

Linear independence of this fourteen-element list is compatible with nonlinear relations among its entries. Indeed the exact identity

$$UZ=VW$$

already holds. The argument therefore cannot be promoted to algebraic independence of the beta periods. It also does not decide the irrationality or transcendence of any one of X, Y, R, S, T separately. The new contribution to the present manuscript is an explicit actual period-space calculation, its coprime-conductor separation proof, and the resulting complete linear-fractional criterion. The deep transcendence input and its general beta-value completeness are classical, and no claim of first historical discovery is made for these specializations.

100 Fixed-parameter modular scarcity and the prime cost of cancellation

The parameter law of the previous paper does not specialize to $A = 1$. Here the parameter remains fixed throughout. We instead bound the number of *index residues* at which a prime divides the classical Laguerre denominator. No assertion about simplicity of these index roots is required. This yields a uniform bound on mean logarithmic content at scale n, together with an obstruction to producing a small primitive form using only a few primes. It does not establish the irrationality of the Euler–Gompertz constant.

Fix a positive integer A and define

$$\delta_A = \int_0^\infty \frac{e^{-t}}{1+At}\,dt, \qquad Q_n = Q_n(A) = \sum_{j=0}^{n} j!\binom{n}{j}^2 A^j,$$

Let $P_0 = 0$, $P_1 = 1$, and give both sequences the recurrence

$$U_n = (1 + (2n-1)A)U_{n-1} - (n-1)^2A^2U_{n-2}.$$

For $n \geq 1$, put

$$g_n = \gcd(Q_n, ((n-1)!)^2), \quad \ell_n = \frac{Q_n\delta_A - P_n}{g_n} > 0.$$

These definitions have the following exact arithmetic and analytic identities:

$$g_n = \gcd(P_n, Q_n), \qquad Q_n\delta_A - P_n = n!A^n \int_0^\infty \frac{e^{-t}}{1+At}\left(\frac{At}{1+At}\right)^n dt. \tag{100.1}$$

In particular,

$$\log(Q_n\delta_A - P_n) = n\log n + O_A(n), \qquad v_p(g_n) \leq \frac{2n}{p-1}\mathbf{1}_{\{p|Q_n\}}. \tag{100.2}$$

The approximations themselves and their real and p-adic settings are classical; see [372, 373]. The estimates below are consequences of their recurrence, rather than new evaluations of δ_A.

Here are short verifications to make the section self-contained. The binomial formula implies $Q_{n+p} \equiv Q_n \pmod p$ for every prime p, by deleting terms with $j \geq p$ and reducing the other binomial coefficients. If a prime divided two consecutive Q_n, it would not divide A; reducing indices modulo p and using the recurrence backwards would reach $Q_0 = 1$, a contradiction. Thus $\gcd(Q_n, Q_{n-1}) = 1$. The recurrence also gives

$$P_nQ_{n-1} - P_{n-1}Q_n = A^{2n-2}((n-1)!)^2.$$

Since $Q_n \equiv 1 \pmod A$, taking the gcd with Q_n proves the first identity in (100.1).

For the second, put $F_n(t) = n!A^nL_n(t)$, so $F_n(-1/A) = Q_n$. The polynomial quotient $(Q_n - F_n(t))/(1+At)$ has integral P_n against $e^{-t}dt$: the Laguerre recurrence gives exactly the stated recurrence and initial values after integration, since $\int_0^\infty e^{-t}L_m(t)dt = 0$ for $m \geq 1$.

Rodrigues's identity $n!e^{-t}L_n(t) = (d/dt)^n(e^{-t}t^n)$ and n integrations by parts now give the integral in (100.1); all boundary terms vanish. Its upper bound is $n!A^n\delta_A$, while restricting the integral to $[\sqrt{n}, \sqrt{n}+1]$ gives a lower bound $n!A^n \exp(-O_A(\sqrt{n}+\log n))$. Stirling's estimate proves the logarithmic assertion in (100.2). The valuation bound there follows directly from $g_n \mid ((n-1)!)^2$ and Legendre's formula.

100.1 Short shifts constrain the index root set

For a prime p, put

$$R_p(A) = \{r \in \mathbb{F}_p : Q_r(A) = 0 \pmod p\}, \qquad \rho_p(A) = |R_p(A)|,$$

where r in Q_r is represented by $0 \le r < p$. The binomial formula shows that $Q_{n+p} \equiv Q_n \pmod p$. If $p \mid A$, then $R_p(A)$ is empty.

Theorem 100.1 (Uniform scarcity of index roots)**.** *For $p \nmid A$ and $1 \le d < p$,*

$$|R_p(A) \cap (R_p(A) - d)| \le d - 1. \tag{100.3}$$

Consequently, for every integer $1 \le h \le p$,

$$h\rho_p(A)^2 \le p\rho_p(A) + \frac{p(h-1)(h-2)}{3}. \tag{100.4}$$

In particular, with the absolute constant $C_0 = (1+\sqrt{7/3})/2$,

$$\rho_p(A) \le C_0 p^{2/3}. \tag{100.5}$$

Proof. Consecutive Q_n cannot both vanish modulo p. Indeed, after reducing the indices modulo p, the recurrence

$$Q_k = (1 + (2k-1)A)Q_{k-1} - (k-1)^2A^2Q_{k-2}$$

allows descent to $Q_0 = 1$; an index congruent to zero already has value one. Define polynomials in x by

$$C_1(x) = 1, \quad C_2(x) = 1 + (2x+3)A,$$

and, for $d \ge 3$, by

$$C_d(x) = (1 + (2x + 2d - 1)A)C_{d-1}(x) - (x+d-1)^2A^2C_{d-2}(x).$$

Reduction of the recurrence modulo Q_n gives $Q_{n+d} \equiv C_d(n)Q_{n+1} \pmod{Q_n}$. The coefficient of x^{d-1} in C_d is dA^{d-1}: its recurrence is $c_d = 2Ac_{d-1} - A^2c_{d-2}$ with $c_1 = 1, c_2 = 2A$. Thus C_d has degree exactly $d-1$ over $\mathbb{F}_p$ when $d < p$. If both r and $r+d$ are index roots, then $Q_{r+1} \ne 0$ and $C_d(r) = 0$. This proves (100.3), including shifts that wrap around the residue circle, by the periodicity already noted.

For each $x \in \mathbb{F}_p$ let $t_x = \sum_{j=1}^{h} \mathbf{1}_{R_p(A)}(x+j)$. Then $\sum_x t_x = h\rho_p(A)$ and

$$\begin{aligned}\sum_x t_x^2 &= h\rho_p(A) + 2\sum_{d=1}^{h-1}(h-d)|R_p(A) \cap (R_p(A)-d)| \\ &\le h\rho_p(A) + \frac{h(h-1)(h-2)}{3}.\end{aligned}$$

Cauchy–Schwarz gives (100.4). Take $h = \lceil p^{1/3} \rceil$. Since $p/h \le p^{2/3}$ and $(h-1)(h-2)/h \le h - 1 \le p^{1/3}$, the resulting quadratic inequality is

$$\rho_p(A)^2 \le p^{2/3}\rho_p(A) + \tfrac{1}{3}p^{4/3}.$$

Its positive root is (100.5). □

This argument concerns zeros in the discrete index, not zeros in the Laguerre parameter. In particular it neither asserts nor needs squarefreeness of a polynomial interpolating the index sequence. The method is an elementary short-shift counting argument. Its application here is proved in full; priority for this particular application has not been established by the literature search.

100.2 A limiting law for a prime-support upper bound

Define

$$W_A(n) = 2 \sum_{\substack{p \le n \\ p \mid Q_n(A)}} \frac{\log p}{p-1}, \qquad b_p = \frac{2\log p}{p-1}, \qquad a_p = \frac{\rho_p(A)}{p}.$$

In particular $\log g_n/n \le W_A(n)$. Let B_p be independent Bernoulli random variables with $\mathbb{P}(B_p = 1) = a_p$.

Theorem 100.2 (Fixed-parameter linear-scale control)**.** *The nonnegative random series $W_A = \sum_p b_p B_p$ converges almost surely and in L^1. As $N \to \infty$, the empirical distribution of $W_A(n)$ for $1 \le n \le N$ converges weakly to that of W_A, and its first moment converges to*

$$\mu_A = 2 \sum_p \frac{\rho_p(A) \log p}{p(p-1)} < \infty. \tag{100.6}$$

The constants μ_A are bounded by an absolute constant uniformly in positive integer A. Thus

$$\limsup_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \frac{\log g_n(A)}{n} \le \mu_A. \tag{100.7}$$

The moment generating function exists for every real t and equals

$$M_A(t) = \mathbb{E} e^{tW_A} = \prod_p \left(1 + a_p(e^{tb_p} - 1)\right). \tag{100.8}$$

There are absolute positive constants c, L_0 such that, for $L \ge L_0$,

$$\limsup_{N\to\infty} \frac{1}{N} \#\{n \le N : \log g_n(A) \ge Ln\} \le \exp(-ce^{L/2}). \tag{100.9}$$

Proof. The root bound implies $a_p \le C_0 p^{-1/3}$, and hence $\sum_p a_p b_p < \infty$, even if the sum is enlarged to all integers at least two. Finite expectation proves convergence of the positive random series and also convergence in L^1.

Fix Y. By the Chinese remainder theorem, the divisibility events $p \mid Q_n(A)$ for $p \le Y$ have independent limiting frequencies a_p as n runs through integers. This is a finite periodic statement. The omission of terms $p > n$ changes this truncated assertion at only finitely many indices. To pass to all primes, use the elementary count

$$\#\{n \le N : p \mid Q_n(A)\} \le (N/p + 1)\rho_p(A).$$

For an absolute C, uniformly in A, this gives

$$\begin{aligned} \frac{1}{N} \sum_{n \le N} \sum_{\substack{Y < p \le n \\ p \mid Q_n}} b_p &\le C \sum_{p > Y} \frac{\log p}{p^{4/3}} + \frac{C}{N} \sum_{p \le N} \frac{\log p}{p^{1/3}} \\ &\le CY^{-1/3} \log(2Y) + CN^{-1/3} \log(2N). \end{aligned}$$

The final inequalities follow by comparison with sums over integers. Truncation therefore proves convergence against bounded Lipschitz functions, hence weak convergence, and also convergence of the first moment. This proves (100.6) and (100.7).

For fixed t, the tail of the logarithm of the product in (100.8) is absolutely convergent because $e^{tb_p} - 1 = O_t(b_p)$. Independence and monotone convergence for $t \geq 0$ give the claimed identity; bounded convergence covers $t < 0$. We record a uniform estimate, for $t \geq 2$:

$$\log M_A(t) \leq 2t \log t + Ct. \tag{100.10}$$

For $p \leq 24t$, bound the logarithm of each factor by tb_p and use the classical elementary estimate $\sum_{p \leq x} \log p/(p-1) = \log x + O(1)$. For $p > 24t$, one has $tb_p \leq (\log p)/6$; consequently

$$\log(1 + a_p(e^{tb_p} - 1)) \leq a_p t b_p e^{tb_p} \leq Ct \frac{\log p}{p^{7/6}}.$$

Its tail sum is $O(t^{5/6} \log(2t)) = O(t)$. For completeness, the elementary prime-sum estimate follows from $\log(m!) = \sum_{p \leq m} v_p(m!) \log p$ and $v_p(m!) = m/(p-1) + O(\log m/\log p)$, together with the elementary Chebyshev bound $\pi(m) = O(m/\log m)$ and Stirling's estimate.

Weak convergence gives, for the closed set $[L, \infty)$, an upper limit bounded by $\mathbb{P}(W_A \geq L)$. Markov's inequality and (100.10), optimized at $t = \exp((L - C - 2)/2)$, bound this probability by $\exp(-ce^{L/2})$ after enlarging L_0. □

The order of limits in (100.9) matters: L is fixed before N tends to infinity. Substituting $L = \log N$ into this statement is not justified. In particular, the law does not eliminate the sparse exceptional indices left by the earlier $O(\log N)$ theorem. Unlike the preceding parameter law, however, all conclusions here hold with $A = 1$ fixed.

100.3 Sublinear content with a quantitative exceptional set

The support law gives more information about the distribution of prime divisibility than about the content itself. The latter admits a stronger estimate by varying the number of indices in a continuant packing argument. This refinement also holds for every fixed A.

Lemma 100.3 (Positive continuant packing)**.** *If $1 \leq n_1 < \cdots < n_h \leq M$, then*

$$\sum_{i=1}^{h} \log g_{n_i} \leq 2 \log((M-1)!) + \log((2A+1)M) \sum_{i<j} (n_j - n_i - 1). \tag{100.11}$$

Proof. For $n < m$, the polynomial value $C_{m-n}(n)$ used in the proof of Theorem 100.1 is the determinant of a symmetric tridiagonal matrix indexed by $k = n+2, \ldots, m$. Its diagonal entries are $1 + (2k-1)A$, with off-diagonal entry $-kA$ between k and $k+1$. The empty determinant is one. Expansion at the last row verifies the defining recurrence. In an interior row the diagonal exceeds the sum of the absolute off-diagonal entries by one; in boundary rows it exceeds that sum by more. The matrix is therefore positive definite. Hadamard's determinant inequality gives

$$1 \leq C_{m-n}(n) \leq ((2A+1)m)^{m-n-1}.$$

The recurrence congruence and adjacent coprimality imply $\gcd(Q_n, Q_m) \mid C_{m-n}(n)$.

For positive integers $a_1, \ldots, a_h$ one has

$$\prod_i a_i \mid \operatorname{lcm}(a_1, \ldots, a_h) \prod_{i<j} \gcd(a_i, a_j).$$

Indeed, ordering their valuations at a prime as $v_1 \le \cdots \le v_h$, the right side has valuation $v_h + \sum_{i<h}(h-i)v_i \ge \sum_i v_i$. Apply this to $a_i = g_{n_i}$. Their least common multiple divides $((M-1)!)^2$, and each pairwise gcd divides the corresponding positive continuant. Taking logarithms proves the claim. □

Theorem 100.4 (Quantitative negligible content)**.** *For each fixed positive integer A,*

$$\frac{1}{X}\sum_{n\le X}\frac{\log g_n(A)}{n} = O_A(X^{-1/3}\log(2X)). \tag{100.12}$$

For every fixed $\varepsilon > 0$,

$$\#\{n \le X : \log g_n(A) \ge \varepsilon n\} = O_{A,\varepsilon}((\log(2X))^4). \tag{100.13}$$

Consequently, outside a set of natural density zero,

$$\log(Q_n/g_n) = n\log n + (\log A - 1)n + o(n), \tag{100.14}$$

$$\log \ell_n = n\log n + (\log A - 1)n + o(n). \tag{100.15}$$

These are density statements; exceptional infinite subsequences remain possible.

Proof. Work first in $[N, 2N]$, with $N \ge 2$. An interval of span at most H containing h selected indices satisfies, by Lemma 100.3,

$$\sum_i \log g_{n_i} \le C_A(N + h^2H)\log(2N) \tag{100.16}$$

for a fixed constant $C_A \ge 1$. Partition $[N, 2N]$ into intervals containing at most $\lceil N^{1/3}\rceil$ consecutive integers and apply this inequality to all indices in each interval. Each sum is $O_A(N\log(2N))$, and there are $O(N^{2/3})$ intervals. Hence

$$\sum_{N\le n\le 2N}\log g_n = O_A(N^{5/3}\log(2N)), \qquad \sum_{N\le n\le 2N}\frac{\log g_n}{n} = O_A(N^{2/3}\log(2N)).$$

Dyadic summation proves (100.12).

For the sharper exceptional count, put $L = \log(2N)$, $h = \lceil 4C_AL/\varepsilon\rceil$, and

$$H = \left\lfloor \frac{\varepsilon N}{4C_AhL}\right\rfloor.$$

For sufficiently large N these are positive integers, and $H \asymp_{A,\varepsilon} N/L^2$. If an interval of span at most H contained h exceptional indices, their logarithmic contents would sum to at least $h\varepsilon N$. But (100.16) is at most

$$C_ANL + C_Ah^2HL \le \tfrac{1}{4}h\varepsilon N + \tfrac{1}{4}h\varepsilon N,$$

a contradiction. Covering $[N, 2N]$ by $O_{A,\varepsilon}(L^2)$ such intervals gives $O_{A,\varepsilon}(L^3)$ exceptional indices. Summing over dyadic intervals proves (100.13).

For clarity, a single density-one set can support the stated $o(n)$. For each integer $j \ge 1$, let $E_j = \{n : \log g_n \ge n/j\}$; each has density zero by (100.13). Choose increasing cutoffs T_j so large that $|E_j \cap [1, X]| \le X/j$ for all $X \ge T_j$. They may also be chosen so that $T_j \ge jT_{j-1}$. The union of $E_j \cap [T_j, T_{j+1})$ has density zero: for $T_j \le X < T_{j+1}$, its part below T_j is at most $T_{j-1} + T_j/(j-1)$, and its current part is at most X/j. Outside this union $\log g_n = o(n)$.

Finally, the binomial formula and the exact remainder give

$$\log Q_n = \log(n!) + n\log A + O_A(\sqrt{n}), \qquad \log(Q_n\delta_A - P_n) = \log(n!) + n\log A + O_A(\sqrt{n} + \log n).$$

For the first estimate use $1 \le Q_n/(n!A^n) \le \sum_{j\ge 0}(n/A)^j/(j!)^2 \le e^{2\sqrt{n/A}}$; the second was proved after (100.1). Subtracting $\log g_n = o(n)$ and applying Stirling proves (100.14)–(100.15). □

The new threshold in (100.13) is linear in n. The earlier logarithmic exceptional count concerned the larger threshold $\varepsilon n \log n$. These different thresholds must not be conflated. In particular, even the present stronger result does not exclude all indices capable of giving an irrationality certificate for δ_A.

100.4 A small primitive form requires many growing primes

For $Y \geq 2$, write $g_{n,\leq Y} = \prod_{p \leq Y} p^{v_p(g_n)}$ and $g_{n,>Y} = g_n / g_{n,\leq Y}$.

Theorem 100.5 (Necessary prime complexity)**.** *For every $Y \geq 2$ and $n \geq 2$,*

$$\log g_{n,\leq Y} \leq 2n \log Y + O(n), \tag{100.17}$$

with an absolute implied constant. Fix $B > 0$ and $0 < \theta < 1/2$. At every sufficiently large index with $\ell_n \leq B$,

$$\log g_{n,>n^\theta} \geq (1 - 2\theta) n \log n - O_{A,B}(n), \tag{100.18}$$

$$\#\{p > n^\theta : p \mid g_n\} \geq \left(\frac{1 - 2\theta}{2} + o_{A,B,\theta}(1) \right) n^\theta. \tag{100.19}$$

More sharply, there are positive constants $c_{A,B}, c'_{A,B}$ such that any such sufficiently large index satisfies

$$\#\{p > c_{A,B}\sqrt{n} : p \mid g_n\} \geq c'_{A,B} \frac{\sqrt{n}}{\log n}. \tag{100.20}$$

Proof. By (100.2) and the prime-sum estimate used above,

$$\log g_{n,\leq Y} \leq 2n \sum_{p \leq Y} \frac{\log p}{p - 1} \leq 2n \log Y + O(n).$$

If $\ell_n \leq B$, then (100.2) gives $\log g_n \geq n \log n - O_{A,B}(n)$. Subtracting (100.17) at $Y = n^\theta$ proves (100.18). Every contributing prime is at most $n - 1$, and, when $p > n^\theta$, its contribution is at most

$$v_p(g_n) \log p \leq \frac{2n \log n}{n^\theta - 1} = (2 + o(1)) n^{1-\theta} \log n.$$

Division proves (100.19).

Finally set $Y = c\sqrt{n}$. The two preceding estimates give $\log g_{n,>Y} \geq (-2 \log c - C_{A,B})n$ for a fixed constant $C_{A,B}$. Choose $c > 0$ small enough to make the parenthesis at least one. Each prime exceeding Y contributes at most $2n \log n / (Y - 1) = O_{A,B}(\sqrt{n} \log n)$, so at least $c'_{A,B}\sqrt{n}/\log n$ such primes are needed. □

Thus finitely many exceptional primes, no matter how their higher valuations behave, cannot produce bounded primitive forms indefinitely. The remaining arithmetic difficulty is simultaneous cancellation at a growing collection of primes. A bound on the number of roots modulo each prime controls averages, but the Chinese remainder theorem alone does not prevent one exceptionally small positive index from meeting many of those congruences. Neither the Kurepa conjecture nor the irrationality of δ follows. In particular, the known congruence $Q_{p-1}(1) \equiv \sum_{j=0}^{p-1} j! \pmod{p}$ is a boundary residue identity, not an equivalence between all the index-root conditions used here and Kurepa's conjecture.

101 Polynomial period elimination creates a uniform Catalan barrier

The previous paired interpolation construction removed π^2 from a positive Catalan pencil. A tempting improvement is to replace the changing interpolation space by powers of a single polynomial invariant, thereby obtaining genuine Hankel matrices of arbitrarily large size while the determinant degree remains fixed. We construct this improvement, then prove that the restricted invariant space cannot produce arbitrarily small primitive integer polynomial values. The obstruction is quantitative: in the simplest case every determinant root stays an explicit positive distance below Catalan's constant, independently of matrix size. We then identify the omitted polynomial sector and prove that the full paired spaces restore convergence at every fixed determinant degree. The remaining primitive-height estimate is not supplied.

The signed density with all moments zero used below is classical in Stieltjes moment theory; see Kleiber [374, Section 4] and López-García [375, Section 2]. Our application concerns the specific period-eliminating pencil and its uniform arithmetic failure. No historical priority claim is made for the general moment mechanism.

101.1 A moment-null density with an exact negative-axis transform

Lemma 101.1. *Let $p \geq 4$ be an even integer, $c > 0$, $\alpha = \pi/p$, and*

$$g_c(u) = e^{-cu^{1/p}} \sin(c \tan\alpha \, u^{1/p}) \qquad (u > 0).$$

All its nonnegative integer moments vanish. Moreover, for $s < 0$,

$$\int_0^\infty \frac{g_c(u)}{u - s}\, du = \pi \exp\{-c \sec\alpha \, (-s)^{1/p}\} > 0. \tag{101.1}$$

Proof. Put $z = c(1 - i\tan\alpha)$. Substitution $u = y^p$ gives

$$\int_0^\infty u^k g_c(u)\, du = p \operatorname{Im} \frac{\Gamma(p(k+1))}{z^{p(k+1)}} = 0,$$

because $\arg z = -\pi/p$. All integrals are absolutely convergent. For the transform, write $-s = v^p$ and integrate

$$\frac{p y^{p-1} e^{-zy}}{y^p + v^p}$$

over the sector between arguments zero and α. The exponential decays uniformly on its large circular arcs. The only pole on the rotated boundary is $y = ve^{i\alpha}$, with positive real residue $e^{-c\sec\alpha v}$. Rotate the boundary with a small indentation at that pole. The principal-value integral on the ray is real: its integrand, parametrized by $y = re^{i\alpha}$, is

$$\frac{p r^{p-1} e^{-c\sec\alpha r}}{r^p - v^p}.$$

The half-residue contribution to the imaginary part of the original integral is $\pi e^{-c\sec\alpha v}$. This proves (101.1). Equivalently one may rotate to angles below α and use $(x - i0)^{-1} = \mathrm{PV}(1/x) + i\pi\delta_0$; the same sign and coefficient result. □

Theorem 101.2 (Bounded evaluations and a fixed-degree gap)**.** *Let $w(u) > 0$ be continuous on $(0, \infty)$, with*

$$w(u) \asymp u^{-1/2} \quad (u \downarrow 0), \qquad \log w(u) = -au^{1/p} + O(\log u) \quad (u \to \infty),$$

where $a > 0$ and $p \geq 4$ is even. Fix distinct negative points $s_1, \dots, s_k$ and positive weights ω_j. Write

$$H_n = \left(\int_0^\infty u^{i+j} w(u)\, du \right)_{0 \leq i,j < n}, \quad B_n = \left(\sum_{j=1}^{k} \omega_j s_j^{i+\ell} \right)_{0 \leq i,\ell < n}.$$

For $c > a/2$, set

$$M_j = \frac{e^{2c \sec(\pi/p)(-s_j)^{1/p}}}{\pi^2} \int_0^\infty \frac{g_c(u)^2}{(u - s_j)^2 w(u)}\, du, \qquad K = \sum_{j=1}^{k} \omega_j M_j.$$

These constants are finite and positive, and $B_n \leq K H_n$ for every n. Consequently the polynomial

$$Q_n(X) = \det(H_n + (X - \theta) B_n)$$

has degree k for $n \geq k$, all its zeros are real, and each zero is at most $\theta - K^{-1}$. If Q_n is a nonzero scalar multiple of an integer polynomial P_n of degree k, then

$$|P_n(\theta)| \geq K^{-k}. \tag{101.2}$$

This remains true for the primitive integer normalization.

Proof. Near zero, $g_c(u) = O(u^{1/p})$, so the integrand defining M_j is $O(u^{2/p+1/2})$. At infinity it is bounded by a polynomial factor times $e^{-(2c-a)u^{1/p}}$. Thus M_j is finite. For any polynomial f, the vanishing moments give

$$\int_0^\infty \frac{f(u) - f(s_j)}{u - s_j} g_c(u)\, du = 0.$$

Use (101.1) and Cauchy–Schwarz to obtain

$$|f(s_j)|^2 \leq M_j \int_0^\infty |f(u)|^2 w(u)\, du.$$

Summing with weights proves the matrix inequality. The positive matrix $H_n^{-1/2} B_n H_n^{-1/2}$ has rank k for $n \geq k$, by the Vandermonde rank of the distinct s_j, and its nonzero eigenvalues lie in $(0, K]$. The determinant factors accordingly, with roots $\theta - 1/\lambda_j \leq \theta - 1/K$. Finally, the leading coefficient of a nonzero integer polynomial has absolute value at least one. Multiplying its k distances from θ proves (101.2). □

For $c > a$, the quotient g_c/w is bounded, and is not identically zero. Thus $w + \varepsilon g_c$ is positive for all sufficiently small $|\varepsilon|$ and has exactly the same moments as w. This gives an explicit proof of Stieltjes indeterminacy. The bounded-evaluation argument is stronger for the present purpose because it provides a uniform gap and requires only $c > a/2$.

101.2 A genuine Hankel family for Catalan

Use the classical positive Hurwitz moment measure [376]

$$d\nu_2(t) = 4\pi y^2 \sum_{\ell \geq 1} \ell e^{-2\pi \ell y}\, dy, \quad t = y^2,$$

for which

$$\int t^e\, d\nu_2(t) = (-1)^e B_{2e+2}, \qquad \int \frac{d\nu_2(t)}{t + c^2} = c\zeta(2, c) - 1 - \frac{1}{2c}.$$

Let $G = \sum_{n\ge 0}(-1)^n/(2n+1)^2$. Suppose that rational pairs $a_j < b_j$, $1 \le j \le k$, satisfy

$$a_j \in \tfrac{1}{4} + \mathbb{Z}_{\ge 0}, \quad b_j \in \tfrac{3}{4} + \mathbb{Z}_{\ge 0}, \quad a_j^2 + b_j^2 = C,$$

and are distinct. Put

$$U(t) = t(t+C), \quad s_j = -a_j^2 b_j^2 < 0, \quad W(t) = \sum_{j=1}^{k} \left(\frac{1}{a_j(t+a_j^2)} - \frac{1}{b_j(t+b_j^2)} \right).$$

The s_j are distinct, because their sum-of-squares and product would otherwise determine the same unordered positive pair. Each summand of W is strictly positive on $[0,\infty)$.

Theorem 101.3 (Full degree, followed by saturation)**.** *There are rational numbers r_ℓ, computable by polynomial division and Bernoulli numbers, such that for every $\ell \ge 0$,*

$$\int U(t)^\ell W(t)\, d\nu_2(t) = r_\ell + 16G \sum_{j=1}^{k} s_j^\ell.$$

Thus the rational Hankel pencil

$$T_n(X) = \left(r_{i+\ell} + 16X \sum_{j=1}^{k} s_j^{i+\ell} \right)_{0\le i,\ell<n}$$

is positive definite at $X = G$, and $\det T_n$ has degree exactly k whenever $n \ge k$. For every fixed pole set there is an $\varepsilon > 0$, independent of n, such that every zero of $\det T_n$ is at most $G - \varepsilon$. For its primitive integer normalization P_n,

$$|P_n(G)| \ge \varepsilon^k > 0.$$

In particular increasing the matrix size at a fixed pole set cannot supply an irrationality or transcendence criterion based on values tending to zero.

Proof. One has $U(-a_j^2) = U(-b_j^2) = s_j$. Polynomial division therefore gives the same remainder s_j^ℓ at the paired poles. The two Hurwitz transforms, divided by a_j and b_j, have π^2 coefficients 1 and 1, and G coefficients 8 and -8. Their difference has coefficient $16 s_j^\ell$ in G and zero coefficient in π^2; all remaining terms are rational.

Push the positive measure $W(t)d\nu_2(t)$ forward by $u = U(t)$, and call its density $w(u)$. Since $U'(t) = 2t + C > 0$, this density is positive and continuous. At zero, $W(t)$ tends to a positive constant and the t-density of ν_2 is asymptotic to $(2\pi\sqrt{t})^{-1}$, so $w(u) \asymp u^{-1/2}$. At infinity, $W(t) \asymp t^{-1}$ and

$$w(u) \asymp u^{-3/4} e^{-2\pi u^{1/4}}.$$

Theorem 101.2 applies with $p = 4$, $a = 2\pi$, $\theta = G$, and weights $\omega_j = 16$. All assertions follow. □

Proposition 101.4 (Arbitrarily many poles on a common circle)**.** *For every k there is a pole set as above with at least k pairs. It can be chosen with $\log C = O((\log(k+1))^2)$.*

Proof. For an integer $h \ge 0$, let $N_j = 10^{j+2} + 1$ and form the 2^h Gaussian integers

$$(1+3i)\prod_{j=1}^{h}(N_j + 4\epsilon_j i), \qquad \epsilon_j \in \{-1, 1\}.$$

Their coordinates P, Q satisfy $P \equiv 1$, $Q \equiv 3 \pmod 4$. All their arguments lie in $(\pi/4, \pi/2)$: the total perturbation from $\arg(1+3i)$ is at most $\sum_j 4/10^{j+2} < 1/100$. They are distinct. Indeed, if $\theta_j = \arctan(4/N_j)$, then $\theta_j > 2/10^{j+2} > \sum_{l>j} \theta_l$, so a first differing sign cannot be canceled by all later terms. Thus $0 < P < Q$ and $a = P/4$, $b = Q/4$ give 2^h distinct permissible pairs, all with

$$C = \frac{10}{16} \prod_{j=1}^{h} (N_j^2 + 16).$$

Taking $h = \lceil \log_2 k \rceil$ proves both assertions. □

101.3 An explicit gap for the smallest Catalan invariant

Theorem 101.5. *Take $a = 1/4$, $b = 3/4$, $C = 5/8$, and $U(t) = t(t + 5/8)$. For every $n \geq 1$ the determinant polynomial in Theorem 101.3 is linear. If its root is x_n, then*

$$x_n \leq G - \varepsilon_0, \qquad \varepsilon_0 = \frac{39\pi^3(4-\pi)}{1600} e^{-2\sqrt{6}} > 0. \tag{101.3}$$

Its primitive integer normalization satisfies $|P_n(G)| \geq \varepsilon_0$ for every n.

Proof. Here $s = -9/256$, and

$$W(t) = \frac{8}{3} \frac{t + 13/16}{(t + 1/16)(t + 9/16)}, \quad U(t) - s = (t + 1/16)(t + 9/16).$$

Use Lemma 101.1 with $p = 4$ and any $c > \pi$. Write $E = \int_0^\infty g_c(u)^2/((u-s)^2 w(u))\, du$. On substituting $t = y^2$ and then $u = U(t)$, the geometric sum in the measure gives exactly

$$E = \frac{3}{2\pi} \int_0^\infty \frac{e^{-2cU(y^2)^{1/4}} \sin^2(cU(y^2)^{1/4})(2y^2 + 5/8)^2 \sinh^2(\pi y)}{(y^2 + 1/16)(y^2 + 9/16)(y^2 + 13/16)}\, dy.$$

Since $U(y^2)^{1/4} \geq y$, $\sin^2 \leq 1$, and $\sinh^2(\pi y) \leq e^{2\pi y}/4$, while

$$\frac{(2y^2 + 5/8)^2}{(y^2 + 1/16)(y^2 + 9/16)(y^2 + 13/16)} \leq \frac{100}{9} \frac{16}{13} = \frac{1600}{117},$$

we obtain $E \leq 100/(39\pi(c - \pi))$. The rational inequality uses $(x+z)^2/(xz) = 4 + (x-z)^2/(xz)$ with $x = y^2 + 1/16$ and $z = y^2 + 9/16$. Also $(-s)^{1/4} = \sqrt{3}/4$, so

$$M_s \leq \frac{100 e^{c\sqrt{6}/2}}{39\pi^3(c - \pi)}.$$

The gap in Theorem 101.2, with weight 16, is at least $1/(16M_s)$. Taking $c = 4$ proves (101.3) and the primitive integer bound. □

Exact rational computations for $n = 1, 2, 5, 10$ give respectively

$$x_n \approx 0.6111111111, \quad 0.6636009857, \quad 0.6837781755, \quad 0.6892645428.$$

These are finite diagnostics, not the proof of saturation. The uniform conclusion is (101.3). It does not apply to the previous moving-pole interpolation family, where the pole set changes with the matrix dimension.

101.4 The omitted sector repairs convergence, but not arithmetic height

The saturation theorem concerns polynomials in $U(t)$ only. The full space of paired polynomials is larger. For the common-circle pole set, write $D(u) = \prod_{j=1}^{k}(u - s_j)$. Polynomial division by the monic quadratic $t^2 + Ct - u$ gives a unique decomposition

$$f(t) = h(U(t)) + (2t + C)r(U(t)).$$

The involution $t \mapsto -C - t$ fixes the first summand and negates the second. Consequently the paired constraints are equivalent to $r(s_j) = 0$ for every j, and hence

$$\{f : f(-a_j^2) = f(-b_j^2) \text{ for all } j\} = \mathbb{Q}[U] \ \oplus \ (2t + C)D(U)\mathbb{Q}[U]. \tag{101.4}$$

The first omitted polynomial has degree $2k + 1$. Thus changing a basis inside $\mathbb{Q}[U]$ cannot repair the saturation, whereas adjoining the second summand changes the approximation problem itself.

Lemma 101.6 (No bounded negative-point evaluation combination)**.** *For the measure $d\mu(t) = W(t)d\nu_2(t)$ above, no nonzero finite linear combination of evaluations at distinct negative points is bounded on the polynomial space with its $L^2(\mu)$ norm.*

Proof. Suppose $\sum_j c_j f(-v_j^2)$ were bounded, with distinct $v_j > 0$. By extension to the Hilbert-space closure and the Riesz representation theorem, there would be $h \in L^2(\mu)$ such that

$$\int f(t)h(t)\, d\mu(t) = \sum_j c_j f(-v_j^2)$$

for every polynomial f. Complex coefficients, if present, can be handled with the corresponding conjugate convention for h. Symmetrize the measure under $t = y^2$ to an even measure $d\sigma(y)$ on $\mathbb{R}$. Its density has exponential tail $e^{-2\pi|y|}$ times a rational factor. Cauchy–Schwarz therefore makes

$$F(z) = \int_{\mathbb{R}} e^{izy} h(y^2)\, d\sigma(y)$$

holomorphic on $|\operatorname{Im} z| < \pi$. Its Taylor coefficients at zero are determined by the displayed moment identities and symmetry, giving $F(z) = \sum_j c_j \cosh(v_j z)$ near zero and hence throughout the strip. But $h(y^2)d\sigma(y)$ has finite total variation, so F is bounded on the real axis. A nonzero sum of $\cosh(v_j z)$ with distinct positive v_j is unbounded there, as is seen from its largest exponent. This is a contradiction. □

Theorem 101.7 (Full paired spaces recover every fixed determinant degree)**.** *Fix any k distinct permissible quarter-integer pairs $a_j < b_j$; the common-circle hypothesis is not required. Let W be their positive kernel sum and let*

$$V_N = \{f \in \mathbb{Q}[t] : \deg f \le N,\ f(-a_j^2) = f(-b_j^2) \text{ for every } j\},$$

assuming all $2k$ poles are distinct. For $N \ge 2k - 1$, choose any rational basis of V_N and form its Gram matrix in $L^2(W\nu_2)$. It has the form $A_N + 16GE_N^T E_N$, with A_N rational, where E_N evaluates the basis at $-a_1^2, \dots, -a_k^2$. Then $\det(A_N + 16XE_N^T E_N)$ has degree exactly k. All its roots approach G from below as $N \to \infty$. After division by the leading coefficient these polynomials converge coefficientwise to $(X - G)^k$.

Proof. The paired evaluation cancels the π^2 term and leaves coefficient $16E_N^T E_N$ exactly as before. Paired Lagrange interpolation shows that E_N has rank k once $N \ge 2k - 1$. Let $H_N = A_N + 16GE_N^T E_N > 0$ and $K_N = E_N H_N^{-1} E_N^T$. The matrices K_N increase in the positive

semidefinite order as the polynomial spaces increase; their quadratic forms are the squared norms of linear combinations of the evaluations.

We claim that the smallest eigenvalue of K_N tends to infinity. Otherwise take unit vectors c_N along a subsequence with $c_N^T K_N c_N$ bounded, and pass to a limit $c \neq 0$. Monotonicity shows that $c^T K_M c$ is bounded uniformly in M. Thus the functional $L(f) = \sum_j c_j f(-a_j^2)$ is bounded on the union of the paired spaces. Extend it to a bounded functional on $L^2(\mu)$. On all polynomials, the difference from $\sum_j c_j f(-a_j^2)$ annihilates the kernel of the surjective constraint map

$$f \longmapsto (f(-a_j^2) - f(-b_j^2))_{j=1}^k.$$

It is therefore a linear combination of these constraints. The bounded extension would then be a nonzero linear combination of negative-point evaluations on all polynomials: its coefficients at the two members of each pair cannot all vanish unless $c = 0$. This contradicts Lemma 101.6.

The matrix determinant lemma now gives

$$\frac{\det(A_N + 16X E_N^T E_N)}{\det H_N} = \det(I_k + 16(X - G)K_N).$$

Its roots are $G - 1/(16\lambda_j(K_N))$. All k eigenvalues tend to infinity, proving the root convergence and the final assertion. □

For the single pair $(1/4, 3/4)$, the full spaces give roots approximately 0.7047630881, 0.8206648111, and 0.8337339123 at polynomial degree 3, 10, and 12. The corresponding primitive coefficient heights have logarithms approximately 26.98, 136.72, and 177.88. These finite values illustrate both the repaired convergence and the unresolved arithmetic cost. The proof supplies no rate relating primitive height to the small normalized value; it therefore supplies no irrationality or transcendence certificate.

101.5 What this does and does not transfer to other constants

A rational positive reweighting of a Hurwitz moment measure retains an exponential tail in $\sqrt{t}$. If a polynomial invariant U of degree $d \geq 2$, strictly increasing on $[0, \infty)$ with $U(0) = 0$ and $U'(0) > 0$, replaces t, the pushed density instead has exponential tail in $u^{1/(2d)}$. The same construction applies with $p = 2d$. Whenever the resulting coefficient of a target constant is a fixed positive sum of evaluations at negative u-points, Theorem 101.2 prevents small primitive values at fixed rank. This is a stated-condition obstruction, not a claim that all possible zeta constructions have that form.

For G transcendence we have proved a complete uniform obstruction for the invariant Hankel route and an explicit primitive lower bound. The full paired spaces repair analytic convergence at every fixed determinant degree, but the primitive-height estimate remains open. For $\zeta(3)$ and $\zeta(5)$ transcendence, no higher algebraic degree is excluded; the required off-axis or content bound remains unproved. For $\zeta(7)$, $\zeta(9)$, and $\zeta(11)$ irrationality, no sufficient individual primitive-value estimate is obtained. The entropy rigidity theorem cannot be applied to an unknown constant merely because rational determinant polynomials have algebraic roots: those roots are approximants, not conjugates of the target. The number of newly resolved named-constant problems in this section is zero.

102 Lambert values: a sharp optimization limit for a published determinant budget

The preceding manuscript established functional independence of three Lambert–Fibonacci jets, with a fourth function supplied by prime sums. That result did not specialize to an arithmetic value. Here we examine an actual arithmetic determinant construction and optimize its shape. The outcome is a precise limitation: the displayed local bounds cannot exclude cubic values, even after optimizing the common shift. Separating the two shifts also gives no improvement in the central parameter region. This is a result about a particular certificate budget, not a theorem that Lambert values can be cubic.

102.1 The external arithmetic input and its status

Write

$$F(q) = \sum_{n\ge 1} \frac{1}{q^n - 1} = L(1/q), \qquad q > 1.$$

Lambrinoudis's October 2026 manuscript [377] announces that $1, F(q), F(q)^2$ are linearly independent over $\mathbb{Q}$ for integer $q \ge 2$. It supplies a general degree criterion with

$$\lambda = \frac{42963 + 428220/\pi^2}{202140} = 0.4271829289\ldots, \qquad e\lambda \log M(q) < \log q.$$

The source includes a claimed complete Lean formalization. We read the actual mathematical source, including its local denominator argument; we have not rebuilt that formalization or independently certified the entire external theorem. The announcement is not counted as our result. Its six-presentation refinement is stronger than the single-presentation shape optimization below.

We retain the source's notation for its positive determinant. For pole and moment shift k, matrix size h, the normalized analytic decay has scale

$$\mathcal{T}_{h,k} = G_{h,k} + h \sum_{j=k}^{k+h-1} j + \frac{h(h+1)(2h+1)}{6} + \frac{kh(h+1)}{2}, \qquad G_{h,k} = k\left(hk + \frac{h(h-1)}{2}\right).$$

The denominator budget comprises a pole at zero and cyclotomic poles. All statements below optimize these explicit bounds; they do not assert that the bounds equal the actual denominator orders.

102.2 The complete common-shift budget

Put $k/h \longrightarrow x \ge 0$, and normalize all cubic quantities by h^3. Define

$$S(x) = x^2 + 2x + \frac{5}{6}, \qquad B(x) = \frac{1}{8} + \frac{7/4 + (9/2)x + 3x^2}{\pi^2},$$

and

$$A(x) = \begin{cases} \dfrac{1}{18} + \dfrac{x}{3} + \dfrac{2x^2}{3} + \dfrac{x^3}{9}, & 0 \le x \le 1, \\ x^2 + \dfrac{1}{6}, & x \ge 1. \end{cases} \qquad C(x) = A(x) + B(x), \quad \Lambda(x) = \frac{C(x)}{S(x)}.$$

Proposition 102.1 (Evaluation of the common-shift budget)**.** *The analytic decay scale is $S(x)$. The zero-place and cyclotomic clearing estimates used in the source give the sufficient budgets $A(x)$ and $B(x)$, respectively. In particular $x = 1/20$ recovers the source's displayed $h = 60N, k = 3N$ constants.*

Proof. The formula for S follows directly from $\mathcal{T}_{h,k}$. For the zero place let $r \in [0,1]$ be the normalized column index and $j \geq 0$ the normalized selected node. The source's slotwise lower bound has continuous cost

$$w_x(j,r) = (x+2-2r)j - \int_x^{x+1} \min(j,v)\,dv.$$

The relaxed capacity bound is $j \geq r$ if $r < x$, and $j \geq (x+r)/2$ otherwise. For $0 \leq x \leq 1$, the minimizing nodes are

$$j(r) = \begin{cases} r, & 0 \leq r < x, \\ (x+r)/2, & x \leq r \leq (x+2)/3, \\ 2r-1, & (x+2)/3 \leq r \leq 1. \end{cases}$$

Indeed, before the pole window the cost is affine; inside the window it is $j^2/2 + (1-2r)j + x^2/2$. Its unconstrained minimizer is $2r-1$. The expression after the window is nondecreasing. Direct integration of these three pieces gives

$$x(x+1/2) - \int_0^1 w_x(j(r),r)\,dr = \frac{1}{18} + \frac{x}{3} + \frac{2x^2}{3} + \frac{x^3}{9}.$$

For $x \geq 1$ all capacity bounds are $j \geq r$, the minimum is $j = r$, and the integral is $x/2 - 1/6$, giving the second expression for A. Floors and endpoint rounding change the discrete sum by $O(h^2)$, which does not affect its cubic limit.

For completeness, the cyclotomic envelope can be written using $(u)_+ = \max(u,0)$ as

$$\begin{aligned} E_h(n) &= 3(h-n)_+ - (h-2n)_+ + \sum_{j\geq 1} (h-(2j+1)n)_+, \\ c_{h,k}(n) &= E_h(n) + 2(h+k-n)_+ - 2(k+h/2-n)_+ \\ &\qquad - 2(h-n)_+ + 2(h/2-n)_+. \end{aligned}$$

The source's filtration bound is used when $n \leq h/2$; its residue rank bound gives this envelope for $n > h/2$. The totient asymptotic and partial summation give

$$h^{-3} \sum_n \varphi(n)(Kh - mn)_+ \longrightarrow \frac{K^3}{\pi^2 m^2}.$$

Applying it to the displayed tents, and using $\sum_{j\geq 1}(2j+1)^{-2} = \pi^2/8 - 1$, gives

$$B(x) = \frac{1}{8} + \frac{7/4 + 2((1+x)^3 - (1/2+x)^3 - 1 + 1/8)}{\pi^2},$$

which simplifies to the stated formula. □

Theorem 102.2 (The common-shift optimization and its cubic barrier)**.** *The function Λ has a unique global minimum. Its minimizer x_* is the unique positive root of*

$$\frac{x^4}{9} + \frac{4x^3}{9} + \left(\frac{23}{18} + \frac{3}{2\pi^2}\right)x^2 + \left(\frac{3}{4} + \frac{3}{2\pi^2}\right)x + \frac{1}{4\pi^2} - \frac{1}{12} = 0.$$

Numerically,

$$x_* \simeq 0.05873628, \qquad \Lambda(x_*) \simeq 0.4271370864.$$

Moreover the elementary uniform bound

$$\boxed{\Lambda(x) > \frac{2}{5} \quad (x \geq 0)}$$

holds. Hence this entire common-shift budget cannot certify degree-three exclusion at an integer base, for which the criterion requires $\Lambda < 1/3$.

Proof. For $0 \le x \le 1$, the numerator of $\Lambda'(x)$ is exactly the quartic displayed above. Its constant term is negative, all its other coefficients are positive, and it is strictly increasing on $x \ge 0$. It has one positive zero, lying between zero and one. For $x \ge 1$, the numerator of Λ' is

$$\left(2 + \frac{3}{2\pi^2}\right) x^2 + \left(\frac{13}{12} + \frac{3}{2\pi^2}\right) x + \frac{1}{4\pi^2} - \frac{7}{12},$$

which is positive for $x \ge 1$. The two formulas agree at $x = 1$, so the minimum is global and unique.

For the uniform inequality use $\pi^2 < 10$. If $0 \le x \le 1$,

$$C(x) - \frac{2}{5} S(x) > \frac{1}{45} - \frac{x}{60} + \frac{17x^2}{30} + \frac{x^3}{9} \ge \frac{1}{180} > 0.$$

If $x \ge 1$, the same difference is greater than

$$\frac{9}{10} x^2 - \frac{7}{20} x + \frac{2}{15} > 0.$$

The numerical values are diagnostics of the exact unique-root characterization, not premises of the barrier proof. □

102.3 Allowing independent moment and pole shifts

It is natural to ask whether the common shift itself caused the loss. Let the moment shift divided by h tend to a, and the pole shift divided by h tend to b. The determinant is still a positive Gram determinant at the real Lambert value: only its positive scalar weight changes. Its analytic scale becomes

$$S(a, b) = ab + a + b + 5/6.$$

The augmented matrix in the external construction is symmetric in its two indices. Transposition and the monomial factors in its Vandermonde normalization therefore give the exact symmetry $Q_{a,b} = Q_{b,a}$ at the discrete integer-shift level.

Proposition 102.3 (Asymmetric shifts do not improve the central cone)**.** *In the parameter region*

$$a, b \ge 0, \qquad 2a \le b + 1, \qquad 2b \le a + 1,$$

put $s = a + b$ and $p = ab$. The same local bounds give

$$A(a, b) = \frac{1}{18} + \frac{s}{6} + \frac{s^2}{6} - \frac{s^3}{9} + \frac{sp}{2},$$
$$B(a, b) = \frac{1}{8} + \frac{7/4 + 9s/4 + 9s^2/4 + 3s^3/4 - (6 + 3s)p}{\pi^2}.$$

For fixed s, the ratio $(A + B)/(p + s + 5/6)$ is minimized at $a = b = s/2$. Thus independent shifts in this entire cone do not beat the global minimum in Theorem 102.2.

Proof. The zero-place cost is

$$w_{a,b}(j, r) = (a + 2 - 2r)j - \int_b^{b+1} \min(j, v)\, dv.$$

The minimizing nodes in the central cone are r for $r < b$, $(b + r)/2$ for $b \le r \le (2 + 2a - b)/3$, and $2r - 1 - a + b$ thereafter. Integrating these expressions and subtracting from the infinity normalization $a(b + 1/2)$ gives $A(a, b)$.

For the cyclotomic bound, the residue matrix is supported on at most $\tau_a(n)$ rows plus $\tau_b(n)$ columns, rather than twice one common support size. Here the normalized support function is $\tau_a(u) = \min(1, (1 + a - u)_+)$. In the present cone $|a - b| \le 1$, and the cap appearing in the pole envelope is

$$\min(1, \tau_a(u) + \tau_b(u)) = (1 + a - u)_+ + (1 + b - u)_+ - 2((1 + a + b)/2 - u)_+.$$

The same tent summation as before gives

$$B(a, b) = \frac{1}{8} + \frac{(1 + a)^3 + (1 + b)^3 - (1 + a + b)^3/4}{\pi^2},$$

which expands to the stated expression.

The cone implies $0 \le s \le 2$. For fixed s, the coefficient of p in $A + B$ is

$$\frac{s}{2} - \frac{6 + 3s}{\pi^2} < 0,$$

using $\pi^2 < 10$ and $s \le 2$. The numerator is positive, whereas the denominator increases with p. Thus the ratio strictly decreases with p. Its largest allowed p is $s^2/4$, attained at $a = b$; that point remains inside the cone. This proves the assertion. □

No assertion is made here that asymmetric shifts outside this cone are optimal or impossible to improve. Nor does the result cover all six bordered presentations or better local denominator estimates.

102.4 Arithmetic outcomes and the remaining step

The external nonquadraticity announcement is an important new literature input. Our contribution in this branch is the exact optimization and its obstruction, not another proof of that announcement. The optimized single-presentation rational-base cutoff would be approximately $1 - \Lambda(x_*) = 0.5728629136$; the source already reports a stronger six-presentation cutoff 0.573290, so the numerical optimization is not a new best irrationality region.

The transcendence of $L(1/2)$ is not proved. The full reciprocal Fibonacci sum is not classified as transcendental, and B_1 is not classified as irrational. Sparse Lambert series with exponents d^h have arithmetic specialization theorems [378], but they are different functions from the full Lambert series. The modular component of the Fibonacci decomposition likewise has classical transcendence results [379], which do not classify the full sum.

A cubic exclusion by this determinant route needs genuinely smaller local clearing costs or a different certificate family; changing the common shape, or introducing central-cone asymmetry, does not supply it.

103 Sixteen named targets: individual outcomes

This ledger records the conclusions of this manuscript, rather than declaring the complete current status of every target in the literature. A statement that a target is unclosed means that our arguments do not prove its listed conclusion. External announcements are not counted as our results. The twelve Gamma-ratio examples in Section 99 are not twelve additional historically open problems.

For clarity, $\gamma = \lim_{n\to\infty}(\sum_{j=1}^{n} j^{-1} - \log n)$, $\delta = \int_0^\infty e^{-t}/(1 + t)\, dt$, and $G = \sum_{n\ge0}(-1)^n/(2n+1)^2$. Write $L(z) = \sum_{n\ge1} z^n/(1-z^n)$ for $|z| < 1$, let $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$, and define the prime Mertens constant by $B_1 = \lim_{x\to\infty}(\sum_{p\le x} p^{-1} - \log\log x)$.

Named target	Result of this follow-up and remaining step
Euler's constant γ: irrationality	No new Euler-specific arithmetic certificate is obtained. Entropy rigidity cannot be applied to an unknown target merely by approximating it with algebraic numbers. The previous curvature obstruction is not removed. This target remains unclosed here.
Euler–Gompertz δ: irrationality	The fixed-A results include $A = 1$. The mean of $\log g_n/n$ tends to zero quantitatively, and only $O_\varepsilon((\log X)^4)$ indices violate a fixed linear threshold. A bounded primitive form would require many growing prime divisors. An exceptional infinite subsequence is not excluded, so no irrationality conclusion follows.
Catalan G: transcendence	The polynomial-invariant Hankel family has a uniform positive root gap. Restoring the full paired polynomial space makes all roots converge to G at every fixed determinant degree. Primitive height relative to the small normalized value remains uncontrolled.
$\zeta(3)$: transcendence	The moment-space analysis explains how a choice of invariant can destroy approximation. It gives no new exclusion of an algebraic degree for $\zeta(3)$; sufficient primitive-value and off-axis bounds are still absent.
$\zeta(5)$: transcendence	No primitive-content gain sufficient for a transcendence certificate is established. Statements about external irrationality results would not settle this stronger target and are not counted here.
$\zeta(7)$: irrationality	No individual integer linear forms with a sufficient primitive-value estimate are constructed. The general moment obstruction is not an irrationality proof for this value.
$\zeta(9)$: irrationality	The denominator and content estimate needed for this individual value remains missing. Positivity and fixed-degree root convergence do not replace it.
$\zeta(11)$: irrationality	No individual conclusion is obtained. A relation or a disjunction involving several odd zeta values would not by itself establish this target.
Erdős–Borwein $L(1/2)$: transcendence	The specified common-shift Lambert budget has a unique optimum and never reaches the cubic-exclusion threshold. The separate external nonquadraticity announcement is not our theorem and does not prove transcendence.
Reciprocal Fibonacci sum $\sum_{n\geq 1} F_n^{-1}$: transcendence	The functional transfers and classical modular-component results do not classify the full sum. The present Lambert budget optimization provides no sufficient specialization theorem.
Prime Mertens B_1: irrationality	No arithmetic classification of this constant is obtained. Functional independence of a generating function is not a value theorem for its limiting constant.
$\Gamma(1/5)$: transcendence	Writing $X = \Gamma(1/5)$ and $Y = \Gamma(2/5)$, the actual period-space calculation proves, for example, that $\pi X/Y^3$ is transcendental. It does not isolate the arithmetic nature of X.
$\Gamma(2/5)$: transcendence	The same exact span gives transcendence of X^2Y^4/π^3 and a complete linear-fractional test on the stated support. No conclusion for Y alone follows.
$\Gamma(1/7)$: transcendence	For $R = \Gamma(1/7)$, $S = \Gamma(2/7)$ and $T = \Gamma(3/7)$, $R^4/(\pi S^2)$ is among the explicit classical-period consequences. No individual conclusion for R is proved.
$\Gamma(2/7)$: transcendence	The ratio $\pi S/T^3$ is transcendental by the rank calculation. That does not decide the arithmetic nature of S separately.
$\Gamma(3/7)$: transcendence	The ratio R^2T^4/π^3 is transcendental. The nonlinear identity $UZ = VW$ also shows why the period basis is not an algebraically independent family. No individual conclusion for T follows.

The number of newly closed previously open named-constant conjectures in this ledger is therefore **zero**. The related Kurepa left-factorial conjecture is likewise not proved: its boundary congruence does not encode all index-root conditions. This accounting is compatible with proving exact classifications for the restricted entropy and period-space questions developed above.

104 Verification record and precise remaining problems

The mathematical claims were reviewed independently within the research workflow. The checks verify exact finite identities and sample cases; the proofs carry the universal quantifiers. In particular, the Gamma certificate enumerates all 16 fifth-denominator and 36 seventh-denominator beta pairs, checks their distinct signatures on units modulo 35, and verifies 9,300 exact distribution identities including endpoints. The twelve displayed ratios are checked as integer exponent differences. The entropy checks confirm the explicit examples and finite algebraic steps. The Gompertz, Catalan and Lambert computations are reproducible diagnostics of recurrences, moment identities and budget formulas. They do not establish irrationality by numerical approximation.

The use of the full Wolfart–Wüstholz beta completeness theorem is explicit. Its primary publisher abstract and later precise research statements were consulted; the paywalled full original article was not inspected in this workflow. The weaker pair-class independence statement alone would not justify the signature obstruction, so it is not substituted for completeness. The external Lambert manuscript and its mathematical source were inspected, while its announced Lean verification was not rebuilt. No result is described here as formally verified, and the literature search does not certify historical priority of our specializations or formulations.

Four concrete problems emerge from the proved results. First, can a nonrational ternary algebraic probability vector in a field of degree at least four have rational bit entropy? The cubic exclusion and two-primary spacing condition narrow that question but do not answer it. Second, can the full paired Catalan spaces be equipped with a primitive-height estimate strong enough to turn their repaired root convergence into arithmetic information? Third, can simultaneous Gompertz divisibility by a growing prime set be controlled at every relevant index, beyond the limiting distribution and polylogarithmic exceptional count proved here? Fourth, can the Lambert clearing budget be lowered by a new local estimate or a different presentation rather than by the shape changes already optimized? None of these missing estimates is supplied by renaming an approximating root as a conjugate of the target constant.

The complete relative-entropy, conditional-entropy and Markov classifications stand independently of these open transfers. The period calculation supplies a second form of exactness: within its specified fourteen-dimensional support, algebraicity of every linear-fractional expression is decided by proportionality of coefficient vectors. Beyond those hypotheses and supports, further arithmetic input is required.

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